

Egyptian Baccalaureate Mathematics

Student Edition • Open Workbook / Guided Practice • Part1

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Concept 1 | Binomial and Multinomial Theorems

English Student Edition • Focused formulas and guided practice

Formula opener

Binomial

$$\binom{n}{r} a^r b^{n-r}$$

Multinomial

$$\binom{n}{r} \binom{n-r}{s} a^r b^s c^{n-r-s}$$

Exponent check

$$r + (n - r) = n$$

Worked Solutions

Binomial and Multinomial Theorems

Q001 Challenge

For $(2x - 3)^5$, find the coefficient of x^3 and the sum of all coefficients without writing the full expansion:

Solution

1. Use the binomial term rule: $q = 3$ and $r = 2$, so the coefficient is 5C_2 times 72.
2. For the sum of coefficients, evaluate the expression at $x = 1$: -1 .

Correct option: A

Final answer

Coefficient = 720; sum = -1.

Worked Solutions

Binomial and Multinomial Theorems

Q002 Challenge

Find the coefficient of x^2y^3z in $(x + y + z)^6$, then find the coefficient of x^2yz^3 in $(x - 2y + z)^6$:

Solution

1. The first exponents are (2,3,1), so use $6!/(2!3!1!) = 60$.
2. The second exponents are (2,1,3); the factor (-2) is raised to the first power of y, giving -120.

Correct option: A

Final answer

First coefficient = 60; second coefficient = -120.

Worked Solutions

Binomial and Multinomial Theorems

Q003 Challenge

For $(x^2 + x + 1)^4$, find the greatest coefficient and verify it by identifying the coefficient of x^4 :

Solution

1. Expand and collect like powers: the central coefficient is 19.
2. The coefficient of x^4 is also 19; it is therefore the greatest.

Correct option: A

Final answer

Greatest coefficient = 19.

Worked Solutions

Binomial and Multinomial Theorems

Q004 Written

Expand $(x + 2)^5$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x + 2)^5 = x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$.

Final answer

$$x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$$

Q005 MCQ

Expand $(3x - 2)^4$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(3x - 2)^4 = 81x^4 - 216x^3 + 216x^2 - 96x + 16$.

Correct option: D**Final answer**

$$81x^4 - 216x^3 + 216x^2 - 96x + 16$$

Worked Solutions

Binomial and Multinomial Theorems

Q006 Written

Expand $(x + 2)^6$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x + 2)^6 = x^6 + 12x^5 + 60x^4 + 160x^3 + 240x^2 + 192x + 64$.

Final answer

$$x^6 + 12x^5 + 60x^4 + 160x^3 + 240x^2 + 192x + 64$$

Q007 MCQ

Expand $(x + 1)^4$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x + 1)^4 = x^4 + 4x^3 + 6x^2 + 4x + 1$.

Correct option: B**Final answer**

$$x^4 + 4x^3 + 6x^2 + 4x + 1$$

Worked Solutions

Binomial and Multinomial Theorems

Q008 MCQ

Expand $(a + b)^4$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$.

Correct option: C**Final answer**

$$a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

Q009 Written

Expand $(a - 2b)^4$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a - 2b)^4 = a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$.

Final answer

$$a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$$

Worked Solutions

Binomial and Multinomial Theorems

Q010 MCQ

Expand $(x - 2)^5$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x - 2)^5 = x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32$.

Correct option: A**Final answer**

$$x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32$$

Q011 Written

Expand $(3x + 2y)^5$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(3x + 2y)^5 = 243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 + 240xy^4 + 32y^5$.

Final answer

$$243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 + 240xy^4 + 32y^5$$

Worked Solutions

Binomial and Multinomial Theorems

Q012 **Written****Expand $(2a^2 - 1)^5$ using the Binomial Theorem.****Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2a^2 - 1)^5 = 32a^{10} - 80a^8 + 80a^6 - 40a^4 + 10a^2 - 1$.

Final answer

$$32a^{10} - 80a^8 + 80a^6 - 40a^4 + 10a^2 - 1$$

Q013 **Written****Expand $(x - 3)^6$ using the Binomial Theorem.****Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x - 3)^6 = x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$.

Final answer

$$x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$$

Worked Solutions

Binomial and Multinomial Theorems

Q014 Written

Expand $(2x - 3y)^6$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2x - 3y)^6 = 64x^6 - 576x^5y + 2160x^4y^2 - 4320x^3y^3 + 4860x^2y^4 - 2916xy^5 + 729y^6$.

Final answer

$$64x^6 - 576x^5y + 2160x^4y^2 - 4320x^3y^3 + 4860x^2y^4 - 2916xy^5 + 729y^6$$

Q015 Written

Expand $\left(x + \frac{1}{3}\right)^6$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $\left(x + \frac{1}{3}\right)^6 = x^6 + 2x^5 + \frac{5x^4}{3} + \frac{20x^3}{27} + \frac{5x^2}{27} + \frac{2x}{81} + \frac{1}{729}$.

Final answer

$$x^6 + 2x^5 + \frac{5x^4}{3} + \frac{20x^3}{27} + \frac{5x^2}{27} + \frac{2x}{81} + \frac{1}{729}$$

Worked Solutions

Binomial and Multinomial Theorems

Q016 MCQ

Expand $(x + 3)^3$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x + 3)^3 = x^3 + 9x^2 + 27x + 27$.

Correct option: B**Final answer**

$$x^3 + 9x^2 + 27x + 27$$

Worked Solutions

Binomial and Multinomial Theorems

Q017 MCQ

Expand $(a + 2b)^5$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a + 2b)^5 = a^5 + 10a^4b + 40a^3b^2 + 80a^2b^3 + 80ab^4 + 32b^5$.

Correct option: D**Final answer**

$$a^5 + 10a^4b + 40a^3b^2 + 80a^2b^3 + 80ab^4 + 32b^5$$

Worked Solutions

Binomial and Multinomial Theorems

Q018 MCQ

Expand $(x + 2)^2$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x + 2)^2 = x^2 + 4x + 4$.

Correct option: D**Final answer**

$$x^2 + 4x + 4$$

Q019 MCQ

Expand $(a - b)^3$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$.

Correct option: A**Final answer**

$$a^3 - 3a^2b + 3ab^2 - b^3$$

Worked Solutions

Binomial and Multinomial Theorems

Q020 MCQ

Expand $(2x + 1)^5$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2x + 1)^5 = 32x^5 + 80x^4 + 80x^3 + 40x^2 + 10x + 1$.

Correct option: B**Final answer**

$$32x^5 + 80x^4 + 80x^3 + 40x^2 + 10x + 1$$

Q021 Written

Expand $(3a - 2b)^5$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(3a - 2b)^5 = 243a^5 - 810a^4b + 1080a^3b^2 - 720a^2b^3 + 240ab^4 - 32b^5$.

Final answer

$$243a^5 - 810a^4b + 1080a^3b^2 - 720a^2b^3 + 240ab^4 - 32b^5$$

Worked Solutions

Binomial and Multinomial Theorems

Q022 MCQ

Expand $(2x - 1)^4$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2x - 1)^4 = 16x^4 - 32x^3 + 24x^2 - 8x + 1$.

Correct option: C**Final answer**

$$16x^4 - 32x^3 + 24x^2 - 8x + 1$$

Q023 MCQ

Expand $(x - 3)^4$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x - 3)^4 = x^4 - 12x^3 + 54x^2 - 108x + 81$.

Correct option: B**Final answer**

$$x^4 - 12x^3 + 54x^2 - 108x + 81$$

Worked Solutions

Binomial and Multinomial Theorems

Q024 MCQ

Expand $(2a + b)^3$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2a + b)^3 = 8a^3 + 12a^2b + 6ab^2 + b^3$.

Correct option: C**Final answer**

$$8a^3 + 12a^2b + 6ab^2 + b^3$$

Q025 MCQ

Expand $(2a^2 + 3)^4$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2a^2 + 3)^4 = 16a^8 + 96a^6 + 216a^4 + 216a^2 + 81$.

Correct option: C**Final answer**

$$16a^8 + 96a^6 + 216a^4 + 216a^2 + 81$$

Worked Solutions

Binomial and Multinomial Theorems

Q026 MCQ

Expand $(3a^2 - 2)^3$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(3a^2 - 2)^3 = 27a^6 - 54a^4 + 36a^2 - 8$.

Correct option: B**Final answer**

$$27a^6 - 54a^4 + 36a^2 - 8$$

Q027 Written

Find the coefficient of a^3b^2 in the expansion of $(a + b)^5$.**Solution**

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 2$.
2. Use 5C_2 times the first factor to power q and the second factor to power r .
3. The required coefficient is 10.

Final answer

10

Worked Solutions

Binomial and Multinomial Theorems

Q028 Written

Find the coefficient of x^4y^2 in the expansion of $(x - 2y)^6$.**Solution**

1. The target power uses $q = 4$ copies of the first term, so $r = n - q = 2$.
2. Use $6C2$ times the first factor to power q and the second factor to power r .
3. The required coefficient is 60.

Final answer

60

Q029 Written

Find the coefficient of x^4 in the expansion of $(2x^2 - 3)^7$.**Solution**

1. The target power uses $q = 2$ copies of the first term, so $r = n - q = 5$.
2. Use $7C5$ times the first factor to power q and the second factor to power r .
3. The required coefficient is -20412 .

Final answer -20412

Worked Solutions

Binomial and Multinomial Theorems

Q030 Written

Find the coefficient of x^6y in the expansion of $(2x - 3y)^7$.**Solution**

1. The target power uses $q = 6$ copies of the first term, so $r = n - q = 1$.
2. Use $7C1$ times the first factor to power q and the second factor to power r .
3. The required coefficient is -1344 .

Final answer -1344

Q031 MCQ

Find the coefficient of x^2 in the expansion of $(3x - 2)^5$:**Solution**

1. The target power uses $q = 2$ copies of the first term, so $r = n - q = 3$.
2. Use $5C3$ times the first factor to power q and the second factor to power r .
3. The required coefficient is -720 .

Correct option: B**Final answer** -720

Worked Solutions

Binomial and Multinomial Theorems

Q032 Written

Find the coefficient of a^6 in the expansion of $(3a^2 - 2)^8$.**Solution**

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 5$.
2. Use $8C5$ times the first factor to power q and the second factor to power r .
3. The required coefficient is -48384 .

Final answer -48384

Q033 MCQ

Find the coefficient of x^4y^4 in the expansion of $(2x - y)^8$:**Solution**

1. The target power uses $q = 4$ copies of the first term, so $r = n - q = 4$.
2. Use $8C4$ times the first factor to power q and the second factor to power r .
3. The required coefficient is 1120.

Correct option: A**Final answer**

1120

Worked Solutions

Binomial and Multinomial Theorems

Q034 MCQ

Find the coefficient of ab^3 in the expansion of $(3a + b)^4$:**Solution**

1. The target power uses $q = 1$ copies of the first term, so $r = n - q = 3$.
2. Use $4C3$ times the first factor to power q and the second factor to power r .
3. The required coefficient is 12.

Correct option: B**Final answer**

12

Q035 MCQ

Find the coefficient of x^3y^2 in the expansion of $(x - 4y)^5$:**Solution**

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 2$.
2. Use $5C2$ times the first factor to power q and the second factor to power r .
3. The required coefficient is 160.

Correct option: C**Final answer**

160

Worked Solutions

Binomial and Multinomial Theorems

Q036 MCQ

Find the coefficient of x^4 in the expansion of $(3x + 1)^5$:**Solution**

1. The target power uses $q = 4$ copies of the first term, so $r = n - q = 1$.
2. Use $5C1$ times the first factor to power q and the second factor to power r .
3. The required coefficient is 405.

Correct option: D**Final answer**

405

Q037 Written

Find the coefficient of a in the expansion of $(2a - 3)^4$.**Solution**

1. The target power uses $q = 1$ copies of the first term, so $r = n - q = 3$.
2. Use $4C3$ times the first factor to power q and the second factor to power r .
3. The required coefficient is -216 .

Final answer -216

Worked Solutions

Binomial and Multinomial Theorems

Q038 Written

Find the coefficient of x^3 in the expansion of $(3x + 2)^5$.**Solution**

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 2$.
2. Use 5C_2 times the first factor to power q and the second factor to power r .
3. The required coefficient is 1080.

Final answer

1080

Q039 Written

Find the coefficient of x^6 in the expansion of $(2x^2 - 1)^6$.**Solution**

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 3$.
2. Use 6C_3 times the first factor to power q and the second factor to power r .
3. The required coefficient is -160 .

Final answer -160

Worked Solutions

Binomial and Multinomial Theorems

Q040 MCQ

Find the coefficient of x^4 in the expansion of $(3x^2 - 1)^5$:**Solution**

1. The target power uses $q = 2$ copies of the first term, so $r = n - q = 3$.
2. Use $5C3$ times the first factor to power q and the second factor to power r .
3. The required coefficient is -90 .

Correct option: D**Final answer** -90

Q041 Written

Find the coefficient of x^6 in the expansion of $(3x^2 + 2)^6$.**Solution**

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 3$.
2. Use $6C3$ times the first factor to power q and the second factor to power r .
3. The required coefficient is 4320 .

Final answer 4320

Worked Solutions

Binomial and Multinomial Theorems

Q042 Written

If $(x + 1)^5 = a_0 + a_x + \dots$, find the sum of its coefficients.

Solution

1. Put $x = 1$; the sum of the coefficients is the value of the expansion at $x = 1$.
2. Therefore the required value is 32.

Final answer

32

Q043 MCQ

Find the coefficient of xyz in the expansion of $(x + y + z)^3$.

Solution

1. The target powers are $p = 1$, $q = 1$, $r = 1$, and $p + q + r = 3$.
2. Coefficient = $\frac{3!}{1! \times 1! \times 1!} \times (1)^1 \times (1)^1 \times (1)^1 = 6$.
3. Therefore, the required coefficient is 6.

Correct option: A

Final answer

6

Worked Solutions

Binomial and Multinomial Theorems

Q044 MCQ

Find the coefficient of a^4b^3c in the expansion of $(a + b + c)^8$:**Solution**

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\frac{n!}{p! \times q! \times r!} = 280$ on the signed factors.
3. The required coefficient is 280.

Correct option: C**Final answer**

280

Q045 MCQ

Find the coefficient of a^3bc^2 in the expansion of $(a + b + c)^6$:**Solution**

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\frac{n!}{p! \times q! \times r!} = 60$ on the signed factors.
3. The required coefficient is 60.

Correct option: A**Final answer**

60

Worked Solutions

Binomial and Multinomial Theorems

Q046 MCQ

Find the coefficient of $a^2b^2c^2$ in the expansion of $(a + b + c)^6$:**Solution**

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\frac{n!}{p! \times q! \times r!} = 90$ on the signed factors.
3. The required coefficient is 90.

Correct option: B**Final answer**

90

Q047 MCQ

Find the coefficient of xy^5z in the expansion of $(x + y + z)^7$:**Solution**

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\frac{n!}{p! \times q! \times r!} = 42$ on the signed factors.
3. The required coefficient is 42.

Correct option: A**Final answer**

42

Worked Solutions

Binomial and Multinomial Theorems

Q048 Written

Find the coefficient of x^4y^2z in the expansion of $(x + y + z)^7$.**Solution**

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\frac{n!}{p! \times q! \times r!} = 105$ on the signed factors.
3. The required coefficient is 105.

Final answer

105

Q049 Written

Find the coefficient of $x^2y^3z^2$ in the expansion of $(x + y + z)^7$.**Solution**

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\frac{n!}{p! \times q! \times r!} = 210$ on the signed factors.
3. The required coefficient is 210.

Final answer

210

Worked Solutions

Binomial and Multinomial Theorems

Q050 Written

Find the coefficient of x^6yz^2 in the expansion of $(x + y + z)^9$.**Solution**

1. The target exponents add to 9; compare this with $n = 9$.
2. Use $\frac{n!}{p! \times q! \times r!} = 252$ on the signed factors.
3. The required coefficient is 252.

Final answer

252

Q051 Written

Find the coefficient of $x^3y^3z^3$ in the expansion of $(x + y + z)^9$.**Solution**

1. The target exponents add to 9; compare this with $n = 9$.
2. Use $\frac{n!}{p! \times q! \times r!} = 1680$ on the signed factors.
3. The required coefficient is 1680.

Final answer

1680

Worked Solutions

Binomial and Multinomial Theorems

Q052 Written

Find the coefficient of $x^2y^4z^3$ in the expansion of $(x + y + z)^9$.**Solution**

1. The target exponents add to 9; compare this with $n = 9$.
2. Use $\frac{n!}{p! \times q! \times r!} = 1260$ on the signed factors.
3. The required coefficient is 1260.

Final answer

1260

Q053 MCQ

Find the coefficient of a^4b^3c in the expansion of $(a - 2b + 3c)^8$:**Solution**

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\frac{n!}{p! \times q! \times r!} = 280$ on the signed factors.
3. The required coefficient is -6720 .

Correct option: D**Final answer** -6720

Worked Solutions

Binomial and Multinomial Theorems

Q054 MCQ

Find the coefficient of $a^2b^2c^2$ in the expansion of $(a + 3b - 2c)^6$:**Solution**

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\frac{n!}{p! \times q! \times r!} = 90$ on the signed factors.
3. The required coefficient is 3240.

Correct option: C**Final answer**

3240

Q055 MCQ

Find the coefficient of ab^2c^3 in the expansion of $(a + 3b - 2c)^6$:**Solution**

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\frac{n!}{p! \times q! \times r!} = 60$ on the signed factors.
3. The required coefficient is -4320 .

Correct option: D**Final answer** -4320

Worked Solutions

Binomial and Multinomial Theorems

Q056 Written

Find the coefficient of $a^2b^3c^2$ in the expansion of $(a + 2b - 3c)^7$.**Solution**

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\frac{n!}{p! \times q! \times r!} = 210$ on the signed factors.
3. The required coefficient is 15120.

Final answer

15120

Q057 Written

Find the coefficient of abc^5 in the expansion of $(a + 2b - 3c)^7$.**Solution**

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\frac{n!}{p! \times q! \times r!} = 42$ on the signed factors.
3. The required coefficient is -20412 .

Final answer -20412

Worked Solutions

Binomial and Multinomial Theorems

Q058 Written

Find the coefficient of a^2b^4c in the expansion of $(a + 2b - 3c)^7$.**Solution**

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\frac{n!}{p! \times q! \times r!} = 105$ on the signed factors.
3. The required coefficient is -5040 .

Final answer -5040

Q059 Written

Find the coefficient of $a^2b^2c^4$ in the expansion of $(2a - b + 3c)^8$.**Solution**

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\frac{n!}{p! \times q! \times r!} = 420$ on the signed factors.
3. The required coefficient is 136080.

Final answer

136080

Worked Solutions

Binomial and Multinomial Theorems

Q060 Written

Find the coefficient of a^4bc^3 in the expansion of $(2a - b + 3c)^8$.**Solution**

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\frac{n!}{p! \times q! \times r!} = 280$ on the signed factors.
3. The required coefficient is -120960 .

Final answer -120960

Q061 Written

Find the coefficient of $a^3b^3c^2$ in the expansion of $(2a - b + 3c)^8$.**Solution**

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\frac{n!}{p! \times q! \times r!} = 560$ on the signed factors.
3. The required coefficient is -40320 .

Final answer -40320

Worked Solutions

Binomial and Multinomial Theorems

Q062 Written

Find the greatest coefficient in the expansion of $(x^2 + x + 1)^4$.

Solution

1. Expand and collect like powers: $x^8 + 4x^7 + 10x^6 + 16x^5 + 19x^4 + 16x^3 + 10x^2 + 4x + 1$.
2. The greatest coefficient is 19.

Final answer

19

Quiz MCQ 1 Concept Quiz · MCQ

The coefficient of x^4 in $(2x - 3)^5$ is:

Solution

1. $q = 4$ and $r = 1$, so the coefficient is -240 .

Correct option: A

Final answer -240

Worked Solutions

Binomial and Multinomial Theorems

Quiz MCQ 2 [Concept Quiz](#) · [MCQ](#)

The coefficient of $x^2y^2z^2$ in $(x + y + z)^6$ is:

Solution

1. Use $6!/(2!2!2!)$.

Correct option: C

Final answer

90

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

The coefficient of x^6 in $(3x^2 - 2)^6$ is:

Solution

1. Choose the first factor three times.

Correct option: A

Final answer

−4320

Worked Solutions

Binomial and Multinomial Theorems

Quiz MCQ 4 Concept Quiz · MCQ

The coefficient of x^5 in $(2x^2 + 1)^4$ is:

Solution

1. All powers of x are even, so the coefficient is zero.

Correct option: A

Final answer

0

Quiz WRITTEN 5 Concept Quiz · Written

Expand $(2x - 1)^4$ fully.

Solution

1. The expansion is $16x^4 - 32x^3 + 24x^2 - 8x + 1$.

Final answer

$16x^4 - 32x^3 + 24x^2 - 8x + 1$

Worked Solutions

Binomial and Multinomial Theorems

Quiz WRITTEN 6 Concept Quiz · Written

Find the coefficient of x^4 in $(3x^2 + 2)^5$. Show the exponent check.

Solution

1. The first-term exponent is $q = 2$ and $r = 3$, giving ${}^5C_3 \times 9 \times 8 = 720$.

Final answer

720

Quiz WRITTEN 7 Concept Quiz · Written

Find the coefficient of x^2y^3z in $(x - 2y + z)^6$.

Solution

1. Use the multinomial coefficient and the signed factor; the result is -480 .

Final answer

 -480

Worked Solutions

Binomial and Multinomial Theorems

Quiz WRITTEN 8 Concept Quiz · Written

Find the greatest coefficient in $(x^2 + x + 1)^4$. Show enough work to justify the maximum.

Solution

1. Expand: $x^8 + 4x^7 + 10x^6 + 16x^5 + 19x^4 + 16x^3 + 10x^2 + 4x + 1$; the greatest coefficient is 19.

Final answer

19

Answer key

Use the question number shown on each page.

Q001 A	Q018 D	Q036 D	Q054 C
Q002 A	Q019 A	Q037 -216	Q055 D
Q003 A	Q020 B	Q038 1080	Q056 15120
Q004 $x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$	Q021 $243a^5 - 810a^4b + 1080a^3b^2 - 720a^2b^3 + 240ab^4 - 32b^5$	Q039 -160	Q057 -20412
Q005 D	Q022 C	Q040 D	Q058 -5040
Q006 $x^6 + 12x^5 + 60x^4 + 160x^3 + 240x^2 + 192x + 64$	Q023 B	Q041 4320	Q059 136080
Q007 B	Q024 C	Q042 32	Q060 -120960
Q008 C	Q025 C	Q043 A	Q061 -40320
Q009 $a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$	Q026 B	Q044 C	Q062 19
Q010 A	Q027 10	Q045 A	Quiz MCQ 1 A
Q011 $243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 + 240xy^4 + 32y^5$	Q028 60	Q046 B	Quiz MCQ 2 C
Q012 $32a^{10} - 80a^8 + 80a^6 - 40a^4 + 10a^2 - 1$	Q029 -20412	Q047 A	Quiz MCQ 3 A
Q013 $x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$	Q030 -1344	Q048 105	Quiz MCQ 4 A
Q014 $64x^6 - 576x^5y + 2160x^4y^2 - 4320x^3y^3 + 4860x^2y^4 - 2916xy^5 + 729y^6$	Q031 B	Q049 210	Quiz WRITTEN 5 $16x^4 - 32x^3 + 24x^2 - 8x + 1$
Q015 $x^6 + 2x^5 + \frac{5x^4}{3} + \frac{20x^3}{27} + \frac{5x^2}{27} + \frac{2x}{81} + \frac{1}{729}$	Q032 -48384	Q050 252	Quiz WRITTEN 6 720
Q016 B	Q033 A	Q051 1680	Quiz WRITTEN 7 -480
Q017 D	Q034 B	Q052 1260	Quiz WRITTEN 8 19
	Q035 C	Q053 D	

Concept 2 | Polynomial Division

English Student Edition • Focused formulas and guided practice

Formula opener

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder

$$R = P(a) \quad \text{for divisor } x - a$$

Degree check

$$\deg R < \deg B$$

Worked Solutions

Polynomial Division

Q001 Challenge

Find the quotient and the remainder when $2x^3 + x^2 - 7x + 3$ is divided by $x^2 + x - 2$. Then verify the result using $A = B \times Q + R$:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 + x^2 - 7x + 3 = (x^2 + x - 2) \times (2x - 1) + (-2x + 1)$.
4. Therefore, quotient $Q = 2x - 1$ and remainder $R = -2x + 1$.

Correct option: B

Final answer

$$Q = 2x - 1, \quad R = -2x + 1$$

Worked Solutions

Polynomial Division

Q002 Challenge

Find the polynomial A when it is divided by $x^2 - 2x + 3$, the quotient is $3x - 1$, and the remainder is $x + 2$. Then divide your answer back to check it:

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (x^2 - 2x + 3) \times (3x - 1) + (x + 2)$.
3. Expand and collect like powers: $A = 3x^3 - 7x^2 + 12x - 1$.
4. Therefore, $A = 3x^3 - 7x^2 + 12x - 1$.

Correct option: A

Final answer

$$3x^3 - 7x^2 + 12x - 1$$

Worked Solutions

Polynomial Division

Q003 Challenge

When $x^3 + 5x^2 + 7x - 7$ is divided by B, the quotient is $x + 3$ and the remainder is $2x - 4$. Find B. Then verify the degree condition:

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{x^3+5x^2+7x-7-(2x-4)}{x+3}$.
3. Divide and simplify: $B = x^2 + 2x - 1$.
4. Check by multiplying B by Q and adding R to recover $A = x^3 + 5x^2 + 7x - 7$.

Correct option: D

Final answer

$$x^2 + 2x - 1$$

Worked Solutions

Polynomial Division

Q004 Challenge

A cuboid has volume $2x^3 + 9x^2 + 4x - 15$ and base area $x^2 + 2x - 3$. Write its height in terms of x . State the mathematical check Which of the following is correct?:

Solution

1. Use height = volume $\div$ base area: $h = \frac{2x^3+9x^2+4x-15}{x^2+2x-3}$.
2. Divide the polynomials and check that the remainder is zero.
3. Therefore, $h = 2x + 5$.

Correct option: C

Final answer

$$2x + 5$$

Worked Solutions

Polynomial Division

Q005 Challenge

Find the quotient and the remainder when $3x^4 - x^3 + 2x^2 - 4x + 1$ is divided by $x^2 - 2x + 2$. Decide whether the result is exact and justify from R:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^4 - x^3 + 2x^2 - 4x + 1 = (x^2 - 2x + 2) \times (3x^2 + 5x + 6) + (-2x - 11)$.
4. Therefore, quotient $Q = 3x^2 + 5x + 6$ and remainder $R = -2x - 11$.

Correct option: B

Final answer

$$Q = 3x^2 + 5x + 6, \quad R = -2x - 11$$

Worked Solutions

Polynomial Division

Q006 Challenge

Find the quotient and the remainder when $x^4 - 3x^2 + 2x + 6$ is divided by $x + 2$. Explain why a zero placeholder is needed:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^4 - 3x^2 + 2x + 6 = (x + 2) \times (x^3 - 2x^2 + x) + (6)$.
4. Therefore, quotient $Q = x^3 - 2x^2 + x$ and remainder $R = 6$.

Correct option: A

Final answer

$$Q = x^3 - 2x^2 + x, \quad R = 6$$

Worked Solutions

Polynomial Division

Q007 Written

Find the quotient and the remainder when $x^2 - 5x + 1$ is divided by $x - 2$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^2 - 5x + 1 = (x - 2) \times (x - 3) + (-5)$.
4. Therefore, quotient $Q = x - 3$ and remainder $R = -5$.

Final answer

$$Q = x - 3, \quad R = -5$$

Q008 Written

Find the quotient and the remainder when $x^3 - 4x + 1$ is divided by $x - 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 - 4x + 1 = (x - 3) \times (x^2 + 3x + 5) + (16)$.
4. Therefore, quotient $Q = x^2 + 3x + 5$ and remainder $R = 16$.

Final answer

$$Q = x^2 + 3x + 5, \quad R = 16$$

Worked Solutions

Polynomial Division

Q009 MCQ

Find the quotient and the remainder when $3x^3 + 2x^2 - 7x + 5$ is divided by $x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 + 2x^2 - 7x + 5 = (x - 2) \times (3x^2 + 8x + 9) + (23)$.
4. Therefore, quotient $Q = 3x^2 + 8x + 9$ and remainder $R = 23$.

Correct option: D**Final answer**

$$Q = 3x^2 + 8x + 9, \quad R = 23$$

Worked Solutions

Polynomial Division

Q010 MCQ

Find the quotient and the remainder when $x^3 + 6x^2 - 4$ is divided by $2x - 3$:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 + 6x^2 - 4 = (2x - 3) \times \left(\frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8} \right) + \left(\frac{103}{8} \right)$.
4. Therefore, quotient $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}$ and remainder $R = \frac{103}{8}$.

Correct option: B

Final answer

$$Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}, \quad R = \frac{103}{8}$$

Worked Solutions

Polynomial Division

Q011 MCQ

Find the quotient and the remainder when $3x^3 - 4x^2 + 2x + 7$ is divided by $x - 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 - 4x^2 + 2x + 7 = (x - 1) \times (3x^2 - x + 1) + (8)$.
4. Therefore, quotient $Q = 3x^2 - x + 1$ and remainder $R = 8$.

Correct option: B**Final answer**

$$Q = 3x^2 - x + 1, \quad R = 8$$

Q012 MCQ

Find the quotient and the remainder when $2x^4 + x^3 - 5x + 4$ is divided by $x + 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^4 + x^3 - 5x + 4 = (x + 2) \times (2x^3 - 3x^2 + 6x - 17) + (38)$.
4. Therefore, quotient $Q = 2x^3 - 3x^2 + 6x - 17$ and remainder $R = 38$.

Correct option: A**Final answer**

$$Q = 2x^3 - 3x^2 + 6x - 17, \quad R = 38$$

Worked Solutions

Polynomial Division

Q013 MCQ

Find the quotient and the remainder when $4x^4 - 3x^3 + x - 6$ is divided by $x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $4x^4 - 3x^3 + x - 6 = (x - 2) \times (4x^3 + 5x^2 + 10x + 21) + (36)$.
4. Therefore, quotient $Q = 4x^3 + 5x^2 + 10x + 21$ and remainder $R = 36$.

Correct option: D**Final answer**

$$Q = 4x^3 + 5x^2 + 10x + 21, \quad R = 36$$

Worked Solutions

Polynomial Division

Q014 MCQ

Find the quotient and the remainder when $2x^3 - 9x + 1$ is divided by $3x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 9x + 1 = (3x - 2) \times \left(\frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27} \right) + \left(-\frac{119}{27} \right)$.
4. Therefore, quotient $Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}$ and remainder $R = -\frac{119}{27}$.

Correct option: B**Final answer**

$$Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}, \quad R = -\frac{119}{27}$$

Worked Solutions

Polynomial Division

Q015 Written

Find the quotient and the remainder when $2x^3 - 5x^2 - 2$ is divided by $2x^2 - x - 1$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 5x^2 - 2 = (2x^2 - x - 1) \times (x - 2) + (-x - 4)$.
4. Therefore, quotient $Q = x - 2$ and remainder $R = -x - 4$.

Final answer

$$Q = x - 2, \quad R = -x - 4$$

Q016 Written

Find the quotient and the remainder when $2x^3 - 13x + 1$ is divided by $x^2 + 2x - 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 13x + 1 = (x^2 + 2x - 3) \times (2x - 4) + (x - 11)$.
4. Therefore, quotient $Q = 2x - 4$ and remainder $R = x - 11$.

Final answer

$$Q = 2x - 4, \quad R = x - 11$$

Worked Solutions

Polynomial Division

Q017 Written

Find the quotient and the remainder when $2x^3 - 3x^2 + 3x + 4$ is divided by $x^2 - 2x + 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 3x^2 + 3x + 4 = (x^2 - 2x + 3) \times (2x + 1) + (-x + 1)$.
4. Therefore, quotient $Q = 2x + 1$ and remainder $R = -x + 1$.

Final answer

$$Q = 2x + 1, \quad R = -x + 1$$

Q018 Written

Find the quotient and the remainder when $x^3 - 7x + 6$ is divided by $x^2 + 2x - 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 - 7x + 6 = (x^2 + 2x - 3) \times (x - 2) + (0)$.
4. Therefore, quotient $Q = x - 2$ and remainder $R = 0$.

Final answer

$$Q = x - 2, \quad R = 0$$

Worked Solutions

Polynomial Division

Q019 MCQ

Find the quotient and the remainder when $x^3 + 2x^2 - 5x + 4$ is divided by $x^2 + x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 + 2x^2 - 5x + 4 = (x^2 + x - 2) \times (x + 1) + (-4x + 6)$.
4. Therefore, quotient $Q = x + 1$ and remainder $R = -4x + 6$.

Correct option: D**Final answer**

$$Q = x + 1, \quad R = -4x + 6$$

Q020 MCQ

Find the quotient and the remainder when $x^4 - 3x^3 + 2x - 5$ is divided by $x^2 + 2x - 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^4 - 3x^3 + 2x - 5 = (x^2 + 2x - 1) \times (x^2 - 5x + 11) + (-25x + 6)$.
4. Therefore, quotient $Q = x^2 - 5x + 11$ and remainder $R = -25x + 6$.

Correct option: B**Final answer**

$$Q = x^2 - 5x + 11, \quad R = -25x + 6$$

Worked Solutions

Polynomial Division

Q021 MCQ

Find the quotient and the remainder when $2x^3 + x^2 - 4x + 5$ is divided by $x^2 - 2x + 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 + x^2 - 4x + 5 = (x^2 - 2x + 2) \times (2x + 5) + (2x - 5)$.
4. Therefore, quotient $Q = 2x + 5$ and remainder $R = 2x - 5$.

Correct option: D**Final answer**

$$Q = 2x + 5, \quad R = 2x - 5$$

Q022 MCQ

Find the quotient and the remainder when $x^4 - 2x^3 + 5x^2 - x + 3$ is divided by $x^2 + 3x - 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^4 - 2x^3 + 5x^2 - x + 3 = (x^2 + 3x - 1) \times (x^2 - 5x + 21) + (-69x + 24)$.
4. Therefore, quotient $Q = x^2 - 5x + 21$ and remainder $R = -69x + 24$.

Correct option: C**Final answer**

$$Q = x^2 - 5x + 21, \quad R = -69x + 24$$

Worked Solutions

Polynomial Division

Q023 MCQ

Find the quotient and the remainder when $3x^3 - 5x + 2$ is divided by $x^2 - x - 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 - 5x + 2 = (x^2 - x - 1) \times (3x + 3) + (x + 5)$.
4. Therefore, quotient $Q = 3x + 3$ and remainder $R = x + 5$.

Correct option: B**Final answer**

$$Q = 3x + 3, \quad R = x + 5$$

Q024 MCQ

Find the quotient and the remainder when $2x^4 - x^3 + 4x^2 - 7$ is divided by $x^2 + x + 3$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^4 - x^3 + 4x^2 - 7 = (x^2 + x + 3) \times (2x^2 - 3x + 1) + (8x - 10)$.
4. Therefore, quotient $Q = 2x^2 - 3x + 1$ and remainder $R = 8x - 10$.

Correct option: A**Final answer**

$$Q = 2x^2 - 3x + 1, \quad R = 8x - 10$$

Worked Solutions

Polynomial Division

Q025 Written

Find the quotient and the remainder when $6x^3 - 8x^2 + 5$ is divided by $2x^2 - 2x - 5$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $6x^3 - 8x^2 + 5 = (2x^2 - 2x - 5) \times (3x - 1) + (13x)$.
4. Therefore, quotient $Q = 3x - 1$ and remainder $R = 13x$.

Final answer

$$Q = 3x - 1, \quad R = 13x$$

Q026 Written

Find the quotient and the remainder when $3x^4 - 5x^3 + x - 1$ is divided by $-x^2 + 3x + 2$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^4 - 5x^3 + x - 1 = (-x^2 + 3x + 2) \times (-3x^2 - 4x - 18) + (63x + 35)$.
4. Therefore, quotient $Q = -3x^2 - 4x - 18$ and remainder $R = 63x + 35$.

Final answer

$$Q = -3x^2 - 4x - 18, \quad R = 63x + 35$$

Worked Solutions

Polynomial Division

Q027 MCQ

Find the quotient and the remainder when $4x^3 - x^2 + 8x - 3$ is divided by $2x^2 + x - 4$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $4x^3 - x^2 + 8x - 3 = (2x^2 + x - 4) \times \left(2x - \frac{3}{2}\right) + \left(\frac{35x}{2} - 9\right)$.
4. Therefore, quotient $Q = 2x - \frac{3}{2}$ and remainder $R = \frac{35x}{2} - 9$.

Correct option: D**Final answer**

$$Q = 2x - \frac{3}{2}, \quad R = \frac{35x}{2} - 9$$

Q028 MCQ

Find the quotient and the remainder when $x^3 - 8x + 2$ is divided by $x^2 - 4$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 - 8x + 2 = (x^2 - 4) \times (x) + (-4x + 2)$.
4. Therefore, quotient $Q = x$ and remainder $R = -4x + 2$.

Correct option: B**Final answer**

$$Q = x, \quad R = -4x + 2$$

Worked Solutions

Polynomial Division

Q029 MCQ

Find the quotient and the remainder when $5x^3 - 4x^2 + x - 6$ is divided by $2x^2 - x + 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $5x^3 - 4x^2 + x - 6 = (2x^2 - x + 1) \times \left(\frac{5x}{2} - \frac{3}{4}\right) + \left(-\frac{9x}{4} - \frac{21}{4}\right)$.
4. Therefore, quotient $Q = \frac{5x}{2} - \frac{3}{4}$ and remainder $R = -\frac{9x}{4} - \frac{21}{4}$.

Correct option: D**Final answer**

$$Q = \frac{5x}{2} - \frac{3}{4}, \quad R = -\frac{9x}{4} - \frac{21}{4}$$

Worked Solutions

Polynomial Division

Q030 MCQ

Find the quotient and the remainder when $3x^4 + 2x^3 - x^2 + 5$ is divided by $4x^2 + x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^4 + 2x^3 - x^2 + 5 = (4x^2 + x - 2) \times \left(\frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64} \right) + \left(\frac{37x}{64} + \frac{163}{32} \right)$.
4. Therefore, quotient $Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}$ and remainder $R = \frac{37x}{64} + \frac{163}{32}$.

Correct option: C**Final answer**

$$Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}, \quad R = \frac{37x}{64} + \frac{163}{32}$$

Worked Solutions

Polynomial Division

Q031 MCQ

Find the quotient and the remainder when $3x^3 + 4x^2 - x + 9$ is divided by $2x^2 - x + 3$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 + 4x^2 - x + 9 = (2x^2 - x + 3) \times \left(\frac{3x}{2} + \frac{11}{4}\right) + \left(-\frac{11x}{4} + \frac{3}{4}\right)$.
4. Therefore, quotient $Q = \frac{3x}{2} + \frac{11}{4}$ and remainder $R = -\frac{11x}{4} + \frac{3}{4}$.

Correct option: D**Final answer**

$$Q = \frac{3x}{2} + \frac{11}{4}, \quad R = -\frac{11x}{4} + \frac{3}{4}$$

Worked Solutions

Polynomial Division

Q032 MCQ

Find the quotient and the remainder when $6x^3 - 5x + 1$ is divided by $2x^2 + 3x - 2$:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $6x^3 - 5x + 1 = (2x^2 + 3x - 2) \times \left(3x - \frac{9}{2}\right) + \left(\frac{29x}{2} - 8\right)$.
4. Therefore, quotient $Q = 3x - \frac{9}{2}$ and remainder $R = \frac{29x}{2} - 8$.

Correct option: B

Final answer

$$Q = 3x - \frac{9}{2}, \quad R = \frac{29x}{2} - 8$$

Q033 Written

Find the quotient and the remainder when $-9a^2 - 9ax + 4x^2$ is divided by $-3a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-9a^2 - 9ax + 4x^2 = (-3a + x) \times (3a + 4x) + (0)$.
4. Therefore, quotient $Q = 3a + 4x$ and remainder $R = 0$.

Final answer

$$Q = 3a + 4x, \quad R = 0$$

Worked Solutions

Polynomial Division

Q034 Written

Find the quotient and the remainder when $-5a^3 + 5a^2x - 3ax^2 + x^3$ is divided by $-2a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-5a^3 + 5a^2x - 3ax^2 + x^3 = (-2a + x) \times (3a^2 - ax + x^2) + (a^3)$.
4. Therefore, quotient $Q = 3a^2 - ax + x^2$ and remainder $R = a^3$.

Final answer

$$Q = 3a^2 - ax + x^2, \quad R = a^3$$

Q035 Written

Find the quotient and the remainder when $-6a^2 - 5ax + 4x^2$ is divided by $-2a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-6a^2 - 5ax + 4x^2 = (-2a + x) \times (3a + 4x) + (0)$.
4. Therefore, quotient $Q = 3a + 4x$ and remainder $R = 0$.

Final answer

$$Q = 3a + 4x, \quad R = 0$$

Worked Solutions

Polynomial Division

Q036 Written

Find the quotient and the remainder when $-a^3 + a^2x + ax^2 + x^3$ is divided by $2a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-a^3 + a^2x + ax^2 + x^3 = (2a + x) \times (3a^2 - ax + x^2) + (-7a^3)$.
4. Therefore, quotient $Q = 3a^2 - ax + x^2$ and remainder $R = -7a^3$.

Final answer

$$Q = 3a^2 - ax + x^2, \quad R = -7a^3$$

Q037 Written

Find the quotient and the remainder when $3a^3 - 2ax^2 + x^3$ is divided by $a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3a^3 - 2ax^2 + x^3 = (a + x) \times (3a^2 - 3ax + x^2) + (0)$.
4. Therefore, quotient $Q = 3a^2 - 3ax + x^2$ and remainder $R = 0$.

Final answer

$$Q = 3a^2 - 3ax + x^2, \quad R = 0$$

Worked Solutions

Polynomial Division

Q038 MCQ

Find the quotient and the remainder when $-2a^3 + 7a^2x - 4ax^2 + x^3$ is divided by $-a + x$. Treat it as a polynomial in x ; other letters are constants:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-2a^3 + 7a^2x - 4ax^2 + x^3 = (-a + x) \times (4a^2 - 3ax + x^2) + (2a^3)$.
4. Therefore, quotient $Q = 4a^2 - 3ax + x^2$ and remainder $R = 2a^3$.

Correct option: D

Final answer

$$Q = 4a^2 - 3ax + x^2, \quad R = 2a^3$$

Q039 MCQ

Find the quotient and the remainder when $x^2 + 3xy - 4y^2$ is divided by $x + y$. Treat it as a polynomial in x ; other letters are constants:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^2 + 3xy - 4y^2 = (x + y) \times (x + 2y) + (-6y^2)$.
4. Therefore, quotient $Q = x + 2y$ and remainder $R = -6y^2$.

Correct option: B

Final answer

$$Q = x + 2y, \quad R = -6y^2$$

Worked Solutions

Polynomial Division

Q040 Written

Find the quotient and the remainder when $x^3 + 3x^2y - 2y^3$ is divided by $x + y$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 + 3x^2y - 2y^3 = (x + y) \times (x^2 + 2xy - 2y^2) + (0)$.
4. Therefore, quotient $Q = x^2 + 2xy - 2y^2$ and remainder $R = 0$.

Final answer

$$Q = x^2 + 2xy - 2y^2, \quad R = 0$$

Q041 Written

Find the polynomial A when it is divided by $x^2 - 2x - 1$, the quotient is $x + 3$, and the remainder is $-2x + 5$.

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (x^2 - 2x - 1) \times (x + 3) + (-2x + 5)$.
3. Expand and collect like powers: $A = x^3 + x^2 - 9x + 2$.
4. Therefore, $A = x^3 + x^2 - 9x + 2$.

Final answer

$$x^3 + x^2 - 9x + 2$$

Worked Solutions

Polynomial Division

Q042 Written

Find the polynomial **A** when it is divided by $x^2 - 2x - 1$, the quotient is $2x - 3$, and the remainder is $-2x$.

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (x^2 - 2x - 1) \times (2x - 3) + (-2x)$.
3. Expand and collect like powers: $A = 2x^3 - 7x^2 + 2x + 3$.
4. Therefore, $A = 2x^3 - 7x^2 + 2x + 3$.

Final answer

$$2x^3 - 7x^2 + 2x + 3$$

Q043 Written

Find the polynomial **A** when it is divided by $x^2 - 3x + 2$, the quotient is $2x + 6$, and the remainder is $2x - 3$.

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (x^2 - 3x + 2) \times (2x + 6) + (2x - 3)$.
3. Expand and collect like powers: $A = 2x^3 - 12x + 9$.
4. Therefore, $A = 2x^3 - 12x + 9$.

Final answer

$$2x^3 - 12x + 9$$

Worked Solutions

Polynomial Division

Q044 Written

Find the polynomial A when it is divided by $2x^2 - x + 2$, the quotient is $2x - 1$, and the remainder is 2.

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (2x^2 - x + 2) \times (2x - 1) + (2)$.
3. Expand and collect like powers: $A = 4x^3 - 4x^2 + 5x$.
4. Therefore, $A = 4x^3 - 4x^2 + 5x$.

Final answer

$$4x^3 - 4x^2 + 5x$$

Q045 Written

Find the polynomial A when it is divided by $2x^2 + x + 1$, the quotient is $3x + 2$, and the remainder is $-2x + 6$.

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (2x^2 + x + 1) \times (3x + 2) + (-2x + 6)$.
3. Expand and collect like powers: $A = 6x^3 + 7x^2 + 3x + 8$.
4. Therefore, $A = 6x^3 + 7x^2 + 3x + 8$.

Final answer

$$6x^3 + 7x^2 + 3x + 8$$

Worked Solutions

Polynomial Division

Q046 Written

When $4x^3 + 8x - 9$ is divided by B, the quotient is $2x - 3$ and the remainder is $x + 15$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{4x^3+8x-9-(x+15)}{2x-3}$.
3. Divide and simplify: $B = 2x^2 + 3x + 8$.
4. Check by multiplying B by Q and adding R to recover $A = 4x^3 + 8x - 9$.

Final answer

$$2x^2 + 3x + 8$$

Q047 Written

When $x^3 - x^2 + 3x + 1$ is divided by B, the quotient is $x^2 - 2x + 5$ and the remainder is -4 . Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{x^3-x^2+3x+1-(-4)}{x^2-2x+5}$.
3. Divide and simplify: $B = x + 1$.
4. Check by multiplying B by Q and adding R to recover $A = x^3 - x^2 + 3x + 1$.

Final answer

$$x + 1$$

Worked Solutions

Polynomial Division

Q048 Written

When $2x^3 + x^2 - 3x + 5$ is divided by B, the quotient is $x + 1$ and the remainder is $-5x + 2$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{2x^3+x^2-3x+5-(-5x+2)}{x+1}$.
3. Divide and simplify: $B = 2x^2 - x + 3$.
4. Check by multiplying B by Q and adding R to recover $A = 2x^3 + x^2 - 3x + 5$.

Final answer

$$2x^2 - x + 3$$

Q049 Written

When $6x^3 + 11x^2 - 2$ is divided by B, the quotient is $2x^2 + 3x - 1$ and the remainder is -1 . Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{6x^3+11x^2-2-(-1)}{2x^2+3x-1}$.
3. Divide and simplify: $B = 3x + 1$.
4. Check by multiplying B by Q and adding R to recover $A = 6x^3 + 11x^2 - 2$.

Final answer

$$3x + 1$$

Worked Solutions

Polynomial Division

Q050 Written

When $2x^3 + 4x^2 + 7$ is divided by B, the quotient is $x + 2$ and the remainder is $3x + 13$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{2x^3+4x^2+7-(3x+13)}{x+2}$.
3. Divide and simplify: $B = 2x^2 - 3$.
4. Check by multiplying B by Q and adding R to recover $A = 2x^3 + 4x^2 + 7$.

Final answer

$$2x^2 - 3$$

Q051 Written

When $x^4 + x^2 + 3x - 1$ is divided by B, the quotient is $x^2 + x + 1$ and the remainder is $3x - 2$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{x^4+x^2+3x-1-(3x-2)}{x^2+x+1}$.
3. Divide and simplify: $B = x^2 - x + 1$.
4. Check by multiplying B by Q and adding R to recover $A = x^4 + x^2 + 3x - 1$.

Final answer

$$x^2 - x + 1$$

Worked Solutions

Polynomial Division

Q052 Written

A cuboid has volume $4x^3 + 4x^2 - 9x - 9$ and base area $2x^2 - x - 3$. Write its height in terms of x .

Solution

1. Use height = volume $\div$ base area: $h = \frac{4x^3 + 4x^2 - 9x - 9}{2x^2 - x - 3}$.
2. Divide the polynomials and check that the remainder is zero.
3. Therefore, $h = 2x + 3$.

Final answer

$$2x + 3$$

Quiz MCQ 1 Concept Quiz · MCQ

For $x^2 - 5x + 1$ divided by $x - 2$, what is R ?:

Solution

1. Long division gives $R = -5$.

Correct option: A

Final answer

$$-5$$

Worked Solutions

Polynomial Division

Quiz MCQ 2 **Concept Quiz · MCQ**

In a polynomial in x , a is treated as... Which of the following is correct?:

Solution

1. Only powers of x determine the division order.

Correct option: B

Final answer

constant

Quiz MCQ 3 **Concept Quiz · MCQ**

To recover B from $A = B \times Q + R$, use... Which of the following is correct?:

Solution

1. Isolate B after moving R to the other side.

Correct option: C

Final answer

$$\frac{A - R}{Q}$$

Worked Solutions

Polynomial Division

Quiz MCQ 4 Concept Quiz · MCQ

If B has degree 2, a valid remainder can have degree...:

Solution

1. The remainder degree must be below 2.

Correct option: C

Final answer

1

Quiz WRITTEN 5 Concept Quiz · Written

Divide $3x^3 - 5x^2 + 2x - 7$ by $x - 2$. State Q, R, and verify $A = B \times Q + R$.

Solution

1. Long division gives $Q = 3x^2 + x + 4, R = 1$.
2. Substitution verifies $3x^3 - 5x^2 + 2x - 7 = (x - 2) \times (3x^2 + x + 4) + (1)$.

Final answer

$$Q = 3x^2 + x + 4, \quad R = 1$$

Worked Solutions

Polynomial Division

Quiz WRITTEN 6 Concept Quiz · Written

Recover A when $B = x^2 - x - 2$, $Q = 2x + 3$, $R = -x + 1$.

Solution

1. Use $A = B \times Q + R$.
2. $A = 2x^3 + x^2 - 8x - 5$.

Final answer

$$2x^3 + x^2 - 8x - 5$$

Quiz WRITTEN 7 Concept Quiz · Written

A cuboid has volume $4x^3 + 4x^2 - 9x - 9$ and base area $2x^2 - x - 3$. Find its height.

Solution

1. Divide volume by base area: $h = 2x + 3$.
2. The remainder is zero, so the height is polynomial.

Final answer

$$2x + 3$$

Worked Solutions

Polynomial Division

Quiz WRITTEN 8 Concept Quiz · Written

Divide $x^4 - 2x^3 + 3x - 1$ by $x^2 - x + 1$ and explain why the remainder is valid.

Solution

1. Long division gives $Q = x^2 - x - 2$ and $R = 2x + 1$.
2. The remainder degree is less than 2, so it is valid even though it is nonzero.

Final answer

$$Q = x^2 - x - 2, \quad R = 2x + 1$$

Answer key

Use the question number shown on each page.

Q001 B	Q016 $Q = 2x - 4$, $R = x - 11$	Q031 D	Q046 $2x^2 + 3x + 8$
Q002 A	Q017 $Q = 2x + 1$, $R = -x + 1$	Q032 B	Q047 $x + 1$
Q003 D	Q018 $Q = x - 2$, $R = 0$	Q033 $Q = 3a + 4x$, $R = 0$	Q048 $2x^2 - x + 3$
Q004 C	Q019 D	Q034 $Q = 3a^2 - ax + x^2$, $R = a^3$	Q049 $3x + 1$
Q005 B	Q020 B	Q035 $Q = 3a + 4x$, $R = 0$	Q050 $2x^2 - 3$
Q006 A	Q021 D	Q036 $Q = 3a^2 - ax + x^2$, $R = -7a^3$	Q051 $x^2 - x + 1$
Q007 $Q = x - 3$, $R = -5$	Q022 C	Q037 $Q = 3a^2 - 3ax + x^2$, $R = 0$	Q052 $2x + 3$
Q008 $Q = x^2 + 3x + 5$, $R = 16$	Q023 B	Q038 D	Quiz MCQ 1 A
Q009 D	Q024 A	Q039 B	Quiz MCQ 2 B
Q010 B	Q025 $Q = 3x - 1$, $R = 13x$	Q040 $Q = x^2 + 2xy - 2y^2$, $R = 0$	Quiz MCQ 3 C
Q011 B	Q026 $Q = -3x^2 - 4x - 18$, $R = 63x + 35$	Q041 $x^3 + x^2 - 9x + 2$	Quiz MCQ 4 C
Q012 A	Q027 D	Q042 $2x^3 - 7x^2 + 2x + 3$	Quiz WRITTEN 5 $Q = 3x^2 + x + 4$, $R = 1$
Q013 D	Q028 B	Q043 $2x^3 - 12x + 9$	Quiz WRITTEN 6 $2x^3 + x^2 - 8x - 5$
Q014 B	Q029 D	Q044 $4x^3 - 4x^2 + 5x$	Quiz WRITTEN 7 $2x + 3$
Q015 $Q = x - 2$, $R = -x - 4$	Q030 C	Q045 $6x^3 + 7x^2 + 3x + 8$	Quiz WRITTEN 8 $Q = x^2 - x - 2$, $R = 2x + 1$

Concept 3 | Rational Expressions

English Student Edition • Focused formulas and guided practice

Formula opener

Restriction

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Worked Solutions

Rational Expressions

Q001 Challenge

After simplifying this transfer chain before substitution, which result is correct for 2():

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{2x(x+3)}{(x-3)(x-1)}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{2x(x+3)}{(x-3)(x-1)}$$

Worked Solutions

Rational Expressions

Q002 Challenge

After simplifying this transfer chain before substitution, which result is correct for 3():

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3(x-3)(x+2)}{x(x+3)}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{3(x-3)(x+2)}{x(x+3)}$$

Worked Solutions

Rational Expressions

Q003 Challenge

After simplifying this transfer chain before substitution, which result is correct for 4():

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{8(x-4)(x+1)}{(x-3)(x-2)}$.
4. Retain the original denominator restrictions.

Correct option: C

Final answer

$$\frac{8(x-4)(x+1)}{(x-3)(x-2)}$$

Worked Solutions

Rational Expressions

Q004 Challenge

After simplifying this transfer chain before substitution, which result is correct for 8():?

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{8}{x}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{8}{x}$$

Worked Solutions

Rational Expressions

Q005 Challenge

After simplifying this transfer chain before substitution, which result is correct for 9():

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{9(x-1)(x^4+x^3+x^2+x+1)}{2}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{9(x-1)(x^4+x^3+x^2+x+1)}{2}$$

Worked Solutions

Rational Expressions

Q006 Challenge

After simplifying this transfer chain before substitution, which result is correct for 10():

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{10(x-4)(x-1)(x+4)}{x(x-3)(x-2)}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{10(x-4)(x-1)(x+4)}{x(x-3)(x-2)}$$

Worked Solutions

Rational Expressions

Q007 Written

Simplify $\frac{12ab^5}{9a^3b^2}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{4b^3}{3a^2}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{4b^3}{3a^2}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{4b^3}{3a^2}$$

Q008 Written

Simplify $\frac{6x^2y^3}{20x^3y}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{3y^2}{10x}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{3y^2}{10x}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{3y^2}{10x}$$

Worked Solutions

Rational Expressions

Q009 Written

Simplify $\frac{6a^2bc^2}{8ab^4c^2}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{3a}{4b^3}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{3a}{4b^3}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{3a}{4b^3}$$

Q010 Written

Simplify $\frac{2a^2b}{8a^3b^3}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{1}{4ab^2}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{1}{4ab^2}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{1}{4ab^2}$$

Worked Solutions

Rational Expressions

Q011 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{12ab^5}{9a^3b^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{4b^3}{a^2}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{4b^3}{a^2}$$

Q012 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{6x^2y^3}{20x^3y}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{9y^2}{10x}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{9y^2}{10x}$$

Worked Solutions

Rational Expressions

Q013 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{6a^2bc^2}{8ab^4c^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3a}{b^3}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{3a}{b^3}$$

Q014 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{2a^2b}{8a^3b^3}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{1}{2ab^2}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{1}{2ab^2}$$

Worked Solutions

Rational Expressions

Q015 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{18a^3b^2}{24ab^5}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3a^2}{2b^3}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{3a^2}{2b^3}$$

Q016 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{-15x^4y^3}{20x^2y}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= -\frac{9x^2y^2}{4}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$-\frac{9x^2y^2}{4}$$

Worked Solutions

Rational Expressions

Q017 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{45a^2b^4c}{60a^5bc^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3b^3}{a^3c}$.
4. Retain the original denominator restrictions.

Correct option: C

Final answer

$$\frac{3b^3}{a^3c}$$

Q018 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{14x^5y^2}{21x^2y^4}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{4x^3}{3y^2}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{4x^3}{3y^2}$$

Worked Solutions

Rational Expressions

Q019 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{32a^4}{48ab^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{2a^3}{b^2}$.
4. Retain the original denominator restrictions.

Correct option: A**Final answer**

$$\frac{2a^3}{b^2}$$

Q020 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{27x^3y^5}{36xy^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= 3x^2y^3$.
4. Retain the original denominator restrictions.

Correct option: B**Final answer**

$$3x^2y^3$$

Worked Solutions

Rational Expressions

Q021 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{-42a^2c^3}{56a^5c}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= -\frac{3c^2}{2a^3}$.
4. Retain the original denominator restrictions.

Correct option: C**Final answer**

$$-\frac{3c^2}{2a^3}$$

Q022 Written

Simplify $\frac{a^3b^2}{3c} \times \frac{9c^3}{a^4b^2}$.

Solution

1. Factor each polynomial: $\frac{a^3b^2}{3c}$ and $\frac{9c^3}{a^4b^2}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{3c^2}{a}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{3c^2}{a}$$

Worked Solutions

Rational Expressions

Q023 Written

Simplify $\frac{6bc^3}{5a^2} \times \frac{10a^4}{3b^3}$.

Solution

1. Factor each polynomial: $\frac{6bc^3}{5a^2}$ and $\frac{10a^4}{3b^3}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{4a^2c^3}{b^2}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{4a^2c^3}{b^2}$$

Q024 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{a^3b^2}{3c} \times \frac{9c^3}{a^4b^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{12c^2}{a}$.
4. Retain the original denominator restrictions.

Correct option: C

Final answer

$$\frac{12c^2}{a}$$

Worked Solutions

Rational Expressions

Q025 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{6bc^3}{5a^2} \times \frac{10a^4}{3b^3}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{8a^2c^3}{b^2}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{8a^2c^3}{b^2}$$

Q026 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{6a^2b}{5c} \times \frac{15c^2}{4ab^3}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{27ac}{2b^2}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{27ac}{2b^2}$$

Worked Solutions

Rational Expressions

Q027 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{9x^3y}{14a^2} \times \frac{21a^3}{6xy^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{9ax^2}{y}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{9ax^2}{y}$$

Q028 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{-8ab^2}{3c^2} \times \frac{12c^3}{-5a^2b}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{64bc}{5a}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{64bc}{5a}$$

Worked Solutions

Rational Expressions

Q029 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{10x^2y^3}{9ab} \times \frac{27a^2b^2}{25xy}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{18abxy^2}{5}$.
4. Retain the original denominator restrictions.

Correct option: C**Final answer**

$$\frac{18abxy^2}{5}$$

Q030 Written

Simplify $\frac{x^2-4}{3x^2+7x+2}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x-2}{3x+1}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x-2}{3x+1}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x-2}{3x+1}$$

Worked Solutions

Rational Expressions

Q031 Written

Simplify $\frac{2x^2-3x-5}{x^2+3x+2}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{2x-5}{x+2}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{2x-5}{x+2}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{2x-5}{x+2}$$

Q032 Written

Simplify $\frac{x^2+3x}{x^2-9}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x}{x-3}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x}{x-3}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x}{x-3}$$

Worked Solutions

Rational Expressions

Q033 Written

Simplify $\frac{x^2+x-2}{x^2-5x+4}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x+2}{x-4}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x+2}{x-4}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x+2}{x-4}$$

Q034 Written

Simplify $\frac{2x^2-4xy+2y^2}{x^3+3x^2y-4xy^2}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{2(x-y)}{x(x+4y)}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{2(x-y)}{x(x+4y)}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{2(x-y)}{x(x+4y)}$$

Worked Solutions

Rational Expressions

Q035 Written

Simplify $\frac{x^3-1}{x^2+2x-3}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x^2+x+1}{x+3}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x^2+x+1}{x+3}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x^2 + x + 1}{x + 3}$$

Q036 Written

Simplify $\frac{x^2-11x+24}{x^2-6x-16} \times \frac{x^2+2x}{x^2-6x+9}$.

Solution

1. Factor each polynomial: $\frac{(x-8)(x-3)}{(x-8)(x+2)}$ and $\frac{x(x+2)}{(x-3)^2}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{x}{x-3}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{x}{x-3}$$

Worked Solutions

Rational Expressions

Q037 Written

Simplify $\frac{x+1}{x^2-4} \times \frac{x^2-2x-8}{x^2+x}.$

Solution

1. Factor each polynomial: $\frac{x+1}{(x-2)(x+2)}$ and $\frac{(x-4)(x+2)}{x(x+1)}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{x-4}{x(x-2)}.$
4. Retain the original denominator restrictions.

Final answer

$$\frac{x-4}{x(x-2)}$$

Q038 Written

Simplify $\frac{x^2-9}{x^2-2x} \times \frac{x^2-7x+10}{x^2+x-12}.$

Solution

1. Factor each polynomial: $\frac{(x-3)(x+3)}{x(x-2)}$ and $\frac{(x-5)(x-2)}{(x-3)(x+4)}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{(x-5)(x+3)}{x(x+4)}.$
4. Retain the original denominator restrictions.

Final answer

$$\frac{(x-5)(x+3)}{x(x+4)}$$

Worked Solutions

Rational Expressions

Q039 Written

Simplify $\frac{x^2+x-6}{2x^2+x-6} \times \frac{x^2+6x+8}{3x^2-5x-2}$.

Solution

- Factor each polynomial: $\frac{(x-2)(x+3)}{(x+2)(2x-3)}$ and $\frac{(x+2)(x+4)}{(x-2)(3x+1)}$.
- Multiply the numerators and denominators, then cancel common factors across the product.
- Therefore the simplified expression is $= \frac{(x+3)(x+4)}{(2x-3)(3x+1)}$.
- Retain the original denominator restrictions.

Final answer

$$\frac{(x+3)(x+4)}{(2x-3)(3x+1)}$$

Q040 Written

Simplify $\frac{x^2-4x-5}{2x^2+5x+2} \div \frac{x^2-x-20}{2x^2-5x-3}$.

Solution

- Change division to multiplication by the reciprocal of the second fraction.
- Factor before cancelling, then multiply by the reciprocal: $\frac{(x-5)(x+1)}{(x+2)(2x+1)}$ and $\frac{(x-3)(2x+1)}{(x-5)(x+4)}$.
- Cancel only common factors; the result is $= \frac{(x-3)(x+1)}{(x+2)(x+4)}$.
- Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{(x-3)(x+1)}{(x+2)(x+4)}$$

Worked Solutions

Rational Expressions

Q041 Written

Simplify $\frac{x^2-9}{x^2-x-2} \div \frac{2x^2+5x-3}{x+1}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x-3)(x+3)}{(x-2)(x+1)}$ and $\frac{x+1}{(x+3)(2x-1)}$.
3. Cancel only common factors; the result is $= \frac{x-3}{(x-2)(2x-1)}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{x-3}{(x-2)(2x-1)}$$

Q042 Written

Simplify $\frac{2xy}{3ab} \div \frac{8x^3y}{9a^2b^3}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{2xy}{3ab}$ and $\frac{9a^2b^3}{8x^3y}$.
3. Cancel only common factors; the result is $= \frac{3ab^2}{4x^2}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{3ab^2}{4x^2}$$

Worked Solutions

Rational Expressions

Q043 Written

Simplify $\frac{2x^2-x-10}{x^2-2x-3} \div \frac{x^2+x-2}{x^2-6x+9}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x+2)(2x-5)}{(x-3)(x+1)}$ and $\frac{(x-3)^2}{(x-1)(x+2)}$.
3. Cancel only common factors; the result is $= \frac{(x-3)(2x-5)}{(x-1)(x+1)}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{(x-3)(2x-5)}{(x-1)(x+1)}$$

Q044 Written

Simplify $\frac{x^2-3x-10}{2x^2+3x-2} \div \frac{x^2-2x-15}{6x-3}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x-5)(x+2)}{(x+2)(2x-1)}$ and $\frac{3(2x-1)}{(x-5)(x+3)}$.
3. Cancel only common factors; the result is $= \frac{3}{x+3}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{3}{x+3}$$

Worked Solutions

Rational Expressions

Q045 Written

Simplify $\frac{x^2-y^2}{x^2-2xy+y^2} \div \frac{x^2+xy}{x-y}.$

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x-y)(x+y)}{(x-y)^2}$ and $\frac{x-y}{x(x+y)}.$
3. Cancel only common factors; the result is $= \frac{1}{x}.$
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{1}{x}$$

Q046 Written

Simplify $\frac{x^3+8}{x^2-x-6} \times \frac{x^2-6x+9}{x^2-2x+4} \div \frac{2x-6}{x^5-1}$ for $x = \sqrt{123}.$

Solution

1. Factor every polynomial and postpone substitution until the symbolic cancellation is complete.
2. The chain simplifies to $= \frac{(x-1)(x^4+x^3+x^2+x+1)}{2}.$
3. Substitute $x = \sqrt{123}$ only after simplifying: $= -\frac{1}{2} + \frac{15129\sqrt{123}}{2}.$
4. Check all original denominators and divisor-nonzero conditions.

Final answer

$$-\frac{1}{2} + \frac{15129\sqrt{123}}{2}$$

Worked Solutions

Rational Expressions

Q047 Challenge

Simplify $\left(\frac{x+1}{x-2} - \frac{2}{x+3} + \frac{3}{x+3} - \frac{1}{x-2}\right)$; state every excluded value.**Solution**

1. Group like denominators, then combine over the two-factor LCM.
2. The reduced expression is $= \frac{x^2 + 4x - 2}{(x-2)(x+3)}$.
3. The original denominators still control the domain.

Final answer

$$\frac{x^2 + 4x - 2}{(x-2)(x+3)}$$

Worked Solutions

Rational Expressions

Q048 Written

Simplify

$$\frac{x^2 - 4}{x + 1} + \frac{3}{x + 1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= x - 1$.
4. Retain all values excluded by the original denominators.

Final answer

$$x - 1$$

Worked Solutions

Rational Expressions

Q049 MCQ

Simplify

$$\frac{x^2 + 2x + 3}{x + 1} + \frac{2x - 1}{x + 1}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{x^2 + 4x + 2}{x + 1}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$\frac{x^2 + 4x + 2}{x + 1}$$

Worked Solutions

Rational Expressions

Q050 MCQ

Simplify

$$\frac{2x^2 - 5}{x^2 - 4} + \frac{x + 6}{x^2 - 4}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{2x^2 + x + 1}{(x - 2)(x + 2)}$.
5. Keep every value excluded by an original denominator.

Correct option: D**Final answer**

$$\frac{2x^2 + x + 1}{(x - 2)(x + 2)}$$

Worked Solutions

Rational Expressions

Q051 MCQ

Simplify

$$\frac{3x^2 + x - 2}{(x - 2)(x + 1)} + \frac{4x - 3}{(x - 2)(x + 1)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{3x^2 + 5x - 5}{(x - 2)(x + 1)}$.
5. Keep every value excluded by an original denominator.

Correct option: A**Final answer**

$$\frac{3x^2 + 5x - 5}{(x - 2)(x + 1)}$$

Worked Solutions

Rational Expressions

Q052 MCQ

Simplify

$$\frac{x^3 - 1}{x + 2} + \frac{2x^2 + x + 4}{x + 2};$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{x^3 + 2x^2 + x + 3}{x + 2}$.
5. Keep every value excluded by an original denominator.

Correct option: B**Final answer**

$$\frac{x^3 + 2x^2 + x + 3}{x + 2}$$

Worked Solutions

Rational Expressions

Q053 Written

Simplify

$$\frac{3}{x^2 + x} - \frac{-x + 3}{x^2 + x}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{1}{x + 1}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{1}{x + 1}$$

Worked Solutions

Rational Expressions

Q054 MCQ

Simplify

$$\frac{x^2 + 4x + 4}{x + 2} - \frac{x + 4}{x + 2}.$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{x(x + 3)}{x + 2}$.
5. Keep every value excluded by an original denominator.

Correct option: A**Final answer**

$$\frac{x(x + 3)}{x + 2}$$

Worked Solutions

Rational Expressions

Q055 MCQ

Simplify

$$\frac{3x^2 - 2x + 1}{x^2 - 1} - \frac{2x - 5}{x^2 - 1}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{3x^2 - 4x + 6}{(x - 1)(x + 1)}$.
5. Keep every value excluded by an original denominator.

Correct option: B**Final answer**

$$\frac{3x^2 - 4x + 6}{(x - 1)(x + 1)}$$

Worked Solutions

Rational Expressions

Q056 MCQ

Simplify

$$\frac{x^2 - 6x + 8}{(x - 1)(x + 2)} - \frac{x^2 - 3x + 2}{(x - 1)(x + 2)}.$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= -\frac{3(x - 2)}{(x - 1)(x + 2)}.$
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$-\frac{3(x - 2)}{(x - 1)(x + 2)}$$

Worked Solutions

Rational Expressions

Q057 MCQ

Simplify

$$\frac{2x^3 + x}{x - 3} - \frac{x^2 + 7}{x - 3}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{2x^3 - x^2 + x - 7}{x - 3}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$\frac{2x^3 - x^2 + x - 7}{x - 3}$$

Worked Solutions

Rational Expressions

Q058 Written

Simplify

$$\frac{x+4}{x^2-2x} - \frac{3}{x^2-3x+2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{x+2}{x(x-1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{x+2}{x(x-1)}$$

Q059 Written**Simplify**

$$\frac{x-1}{x^2+4x+3} + \frac{2}{x^2+5x+6}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{x}{(x+1)(x+2)}.$
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{x}{(x+1)(x+2)}$$

Worked Solutions

Rational Expressions

Q060 MCQ

Simplify

$$\frac{x-1}{(x+1)(x+3)} + \frac{2x+1}{(x+3)(x+5)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{3x^2 + 7x - 4}{(x+1)(x+3)(x+5)}$.
5. Keep every value excluded by an original denominator.

Correct option: B**Final answer**

$$\frac{3x^2 + 7x - 4}{(x+1)(x+3)(x+5)}$$

Worked Solutions

Rational Expressions

Q061 MCQ

Simplify

$$\frac{x+3}{(x+2)(x+4)} + \frac{x-2}{(x+4)(x+6)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{2x^2 + 9x + 14}{(x+2)(x+4)(x+6)}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$\frac{2x^2 + 9x + 14}{(x+2)(x+4)(x+6)}$$

Worked Solutions

Rational Expressions

Q062 MCQ

Simplify

$$\frac{3x + 2}{(x + 1)(x + 4)} + \frac{x + 5}{(x + 4)(x + 7)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{4x^2 + 29x + 19}{(x + 1)(x + 4)(x + 7)}$.
5. Keep every value excluded by an original denominator.

Correct option: D**Final answer**

$$\frac{4x^2 + 29x + 19}{(x + 1)(x + 4)(x + 7)}$$

Worked Solutions

Rational Expressions

Q063 MCQ

Simplify

$$\frac{x+1}{(x+2)(x+5)} + \frac{4x-3}{(x+5)(x+8)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{5x^2 + 14x + 2}{(x+2)(x+5)(x+8)}$.
5. Keep every value excluded by an original denominator.

Correct option: A**Final answer**

$$\frac{5x^2 + 14x + 2}{(x+2)(x+5)(x+8)}$$

Worked Solutions

Rational Expressions

Q064 MCQ

Simplify

$$\frac{2x + 1}{(x + 3)(x + 6)} + \frac{x + 4}{(x + 6)(x + 9)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{3x^2 + 26x + 21}{(x + 3)(x + 6)(x + 9)}$.
5. Keep every value excluded by an original denominator.

Correct option: B**Final answer**

$$\frac{3x^2 + 26x + 21}{(x + 3)(x + 6)(x + 9)}$$

Worked Solutions

Rational Expressions

Q065 MCQ

Simplify

$$\frac{x-2}{(x+1)(x+5)} + \frac{3x+1}{(x+5)(x+8)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{4x^2 + 10x - 15}{(x+1)(x+5)(x+8)}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$\frac{4x^2 + 10x - 15}{(x+1)(x+5)(x+8)}$$

Worked Solutions

Rational Expressions

Q066 Written

Simplify

$$\frac{x}{x-1} - \frac{2}{x^2-1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{x+2}{x+1}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{x+2}{x+1}$$

Q067 Written**Simplify**

$$\frac{x+2}{x} - \frac{x-2}{x-1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{3x-2}{x(x-1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{3x-2}{x(x-1)}$$

Worked Solutions

Rational Expressions

Q068 **Written****Simplify**

$$\frac{2}{x-1} + \frac{3}{x^2+5x-6}$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2x+15}{(x-1)(x+6)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2x+15}{(x-1)(x+6)}$$

Worked Solutions

Rational Expressions

Q069 MCQ

Simplify

$$\frac{x+2}{(x+1)(x+3)} - \frac{2}{x+1}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= -\frac{x+4}{(x+1)(x+3)}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$-\frac{x+4}{(x+1)(x+3)}$$

Worked Solutions

Rational Expressions

Q070 Written

Simplify

$$\frac{1}{x+2} + \frac{1}{x-2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2x}{(x-2)(x+2)}.$
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2x}{(x-2)(x+2)}$$

Q071 Written**Simplify**

$$\frac{x+4}{x^2+2x} - \frac{x+5}{x^2+x-2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= -\frac{2}{x(x-1)}.$
4. Retain all values excluded by the original denominators.

Final answer

$$-\frac{2}{x(x-1)}$$

Worked Solutions

Rational Expressions

Q072 Written

Simplify

$$\frac{x+8}{x^2+x-2} - \frac{x+5}{x^2-1}$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2}{(x+1)(x+2)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2}{(x+1)(x+2)}$$

Q073 Written**Simplify**

$$\frac{x-3}{x^2+x} + \frac{x}{x^2-1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2x^2 - 4x + 3}{x(x-1)(x+1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2x^2 - 4x + 3}{x(x-1)(x+1)}$$

Worked Solutions

Rational Expressions

Q074 **Written****Simplify**

$$\frac{4x}{x^2 - 1} - \frac{x - 1}{x^2 + x}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{3x - 1}{x(x - 1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{3x - 1}{x(x - 1)}$$

Q075 Written**Simplify**

$$\frac{1}{x^2 + 5x + 6} + \frac{1}{x^2 + 7x + 12}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2}{(x + 2)(x + 4)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2}{(x + 2)(x + 4)}$$

Worked Solutions

Rational Expressions

Q076 Written

Simplify

$$\frac{x-1}{x^2+3x+2} - \frac{x-3}{x^2+4x+3}$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{3}{(x+2)(x+3)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{3}{(x+2)(x+3)}$$

Worked Solutions

Rational Expressions

Q077 MCQ

A learner combines

 $\frac{2x-1}{(x-1)(x+2)} - \frac{x+4}{x+2}$. Correct the numerator line, then simplify the expression:

Solution

1. Keep one common denominator; never add the denominators.
2. Put the entire second numerator in parentheses before applying the sign.
3. After factoring and reducing, the correct result is $= -\frac{x^2+x-3}{(x-1)(x+2)}$.
4. Check the original denominator restrictions.

Correct option: B

Final answer

$$-\frac{x^2+x-3}{(x-1)(x+2)}$$

Worked Solutions

Rational Expressions

Q078 **Written****Simplify**

$$\frac{x^3 + 8x}{x^2 - x - 2} - \frac{8}{x - 2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{x^2 + 2x + 4}{x + 1}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{x^2 + 2x + 4}{x + 1}$$

Worked Solutions

Rational Expressions

Q079 Challenge

Simplify the single-inner fraction together with the reciprocal-pair quotient; state the full domain.

Solution

1. Identify the two family triggers before manipulating the expression.
2. Simplify each component inside-out, keeping the original restrictions.
3. Combine the reduced components: $= -\frac{x^2 - 4x + 2}{2(x - 2)}$.
4. Audit every original inner and outer denominator.

Final answer

$$-\frac{x^2 - 4x + 2}{2(x - 2)}$$

Worked Solutions

Rational Expressions

Q080 Written

Simplify

$$\frac{1}{1 - \frac{1}{x-1}}.$$

Solution

1. The inner denominator requires $x-1 \neq 0$.
2. Multiply the outer numerator and denominator by $x-1$.
3. The result is $\frac{x-1}{x-2}$.
4. The outer denominator also excludes $x = 2$.

Final answer

$$\frac{x-1}{x-2}$$

Q081 Written**Simplify**

$$\frac{1}{1 - \frac{x}{x+1}}.$$

Solution

1. The inner denominator requires $x+1 \neq 0$.
2. Clear the inner denominator $x+1$ in the outer fraction.
3. The denominator becomes 1, so the result is $x + 1$.
4. Retain $x \neq -1$ from the original expression.

Final answer

$$x + 1$$

Worked Solutions

Rational Expressions

Q082 MCQ

Simplify

$$\frac{1}{1 - \frac{x+2}{x+1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -x - 1$.
4. State every restriction from the original denominators.

Correct option: D

Final answer

$$-x - 1$$

Q083 MCQ

Simplify

$$\frac{1}{1 - \frac{x+3}{x+2}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -x - 2$.
4. State every restriction from the original denominators.

Correct option: A**Final answer**

$$-x - 2$$

Worked Solutions

Rational Expressions

Q084 MCQ

Simplify

$$\frac{1}{1 - \frac{2x+1}{x+1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -\frac{x+1}{x}$.
4. State every restriction from the original denominators.

Correct option: B

Final answer

$$-\frac{x+1}{x}$$

Q085 MCQ

Simplify

$$\frac{1}{1 - \frac{2x + 3}{x + 1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -\frac{x + 1}{x + 2}$.
4. State every restriction from the original denominators.

Correct option: C**Final answer**

$$-\frac{x + 1}{x + 2}$$

Worked Solutions

Rational Expressions

Q086 MCQ

Simplify

$$\frac{1}{1 - \frac{3x+1}{x+2}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -\frac{x+2}{2x-1}$.
4. State every restriction from the original denominators.

Correct option: D

Final answer

$$-\frac{x+2}{2x-1}$$

Q087 MCQ

Simplify

$$\frac{1}{1 - \frac{x+4}{x+3}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -x - 3$.
4. State every restriction from the original denominators.

Correct option: A**Final answer**

$$-x - 3$$

Worked Solutions

Rational Expressions

Q088 MCQ

Simplify

$$\frac{1}{1 - \frac{2x+5}{x+1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -\frac{x+1}{x+4}$.
4. State every restriction from the original denominators.

Correct option: B

Final answer

$$-\frac{x+1}{x+4}$$

Q089 Written**Simplify**

$$\frac{1}{1-x} + \frac{1}{1+x} \div \left(\frac{1}{1-x} - \frac{1}{1+x} \right).$$

Solution

1. Multiply both outer parts by $(1-x)(1+x)$.
2. The numerator pair becomes 2; the denominator pair becomes $2x$.
3. Reduce to $1/x$.
4. The original inner denominators and the outer denominator give $x \neq -1, 0, 1$.

Final answer

$$1/x$$

Worked Solutions

Rational Expressions

Q090 MCQ

Simplify

$$\frac{1}{x+1} + \frac{1}{x+4} \div \left(\frac{1}{x+1} - \frac{1}{x+4} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= \frac{2x+5}{3}$.
4. State every restriction from the original denominators.

Correct option: A

Final answer

$$\frac{2x+5}{3}$$

Q091 MCQ**Simplify**

$$\frac{1}{x+1} + \frac{1}{x+5} \div \left(\frac{1}{x+1} - \frac{1}{x+5} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= \frac{x+3}{2}$.
4. State every restriction from the original denominators.

Correct option: A**Final answer**

$$\frac{x+3}{2}$$

Worked Solutions

Rational Expressions

Q092 Written

Simplify

$$\frac{1}{x+2} + \frac{1}{x-2} \div \left(\frac{1}{x+2} - \frac{1}{x-2} \right).$$

Solution

1. Multiply numerator and denominator by $(x+2)(x-2)$.
2. The numerator pair becomes $2x$; the denominator pair becomes -4 .
3. The result is $-x/2$.
4. Keep $x \neq -2$ and $x \neq 2$.

Final answer

$$-\frac{x}{2}$$

Q093 MCQ**Simplify**

$$\frac{1}{x+1} + \frac{1}{x+3} \div \left(\frac{1}{x+1} + \frac{1}{x+3} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: = 1.
4. State every restriction from the original denominators.

Correct option: C**Final answer**

1

Worked Solutions

Rational Expressions

Q094 MCQ

Simplify

$$\frac{1}{x+2} + \frac{1}{x+5} \div \left(\frac{1}{x+2} + \frac{1}{x+5} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: = 1.
4. State every restriction from the original denominators.

Correct option: D

Final answer

1

Q095 MCQ

Simplify

$$\frac{1}{x+3} + \frac{1}{x+6} \div \left(\frac{1}{x+3} + \frac{1}{x+6} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: = 1.
4. State every restriction from the original denominators.

Correct option: B**Final answer**

1

Worked Solutions

Rational Expressions

Q096 Written

Simplify

$$\frac{1}{x+y} + \frac{1}{x-y} \div \left(\frac{1}{x+y} - \frac{1}{x-y} \right).$$

Solution

1. Clear both inner denominators using $(x+y)(x-y)$.
2. The numerator pair becomes $2x$; the denominator pair becomes $-2y$.
3. Reduce to $-x/y$.
4. Keep $y \neq 0$ and $x \neq \pm y$.

Final answer

$$-\frac{x}{y}$$

Q097 Written**Simplify**

$$\frac{1}{1 - \frac{1}{1 - \frac{1}{x}}}.$$

Solution

1. Start with the innermost $1 - 1/x$.
2. Substitute that result into the next reciprocal layer.
3. The outer result reduces to $1 - x$.
4. The original denominators exclude $x = 0$ and $x = 1$.

Final answer

$$1 - x$$

Worked Solutions

Rational Expressions

Q098 Written

Simplify

$$\frac{1}{1 - \frac{1}{1 + \frac{1}{x}}}.$$

Solution

1. Simplify $1 + 1/x$ before entering the next layer.
2. Substitute the result into $1 - 1/(...)$.
3. The final result is $x + 1$.
4. Retain $x \neq 0$ and $x \neq -1$.

Final answer

$$x + 1$$

Q099 Written**Simplify**

$$\left(1 + \frac{x-y}{x+y}\right) \div \left(1 - \frac{x-y}{x+y}\right).$$

Solution

1. Let $u = (x - y)/(x + y)$, then combine $1+u$ and $1-u$.
2. Clear $x+y$ in the resulting numerator and denominator.
3. The reduced result is $\frac{x}{y}$.
4. Require $y \neq 0$ and $x + y \neq 0$.

Final answer

$$\frac{x}{y}$$

Worked Solutions

Rational Expressions

Q100 Written

Simplify

$$\left(x - \frac{2}{x-1}\right) \div \left(\frac{x}{x-1} - 2\right).$$

Solution

1. Combine the polynomial pieces over $x-1$ in the numerator and denominator.
2. Form the outer quotient only after both pieces are simplified.
3. Cancel complete factors to obtain $-(x+1)$.
4. The original denominators retain $x \neq 1$ and $x \neq 2$.

Final answer

$$-(x+1)$$

Q101 Written

Simplify

$$\left(\frac{x+1}{x-1} - \frac{x-1}{x+1}\right) \div \left(\frac{x+1}{x-1} + \frac{x-1}{x+1}\right).$$

Solution

1. Use $(x-1)(x+1)$ inside both the numerator and denominator.
2. The difference gives $4x$; the sum gives $2(x^2+1)$.
3. The simplified result is $\frac{2x}{x^2+1}$.
4. The original factors require $x \neq \pm 1$.

Final answer

$$\frac{2x}{x^2+1}$$

Worked Solutions

Rational Expressions

Q102 Written

Simplify

$$\frac{1}{1 + \frac{1}{1 + \frac{1}{1 + x}}}.$$

Solution

1. Translate the three booster stages as an inside-out reciprocal chain.
2. Simplify the innermost stage $1 + 1/(1+x)$.
3. Continuing outward gives $\frac{x+2}{2x+3}$.
4. The stated model condition is $x > 0$.

Final answer

$$\frac{x+2}{2x+3}$$

Quiz MCQ 1 Concept Quiz · MCQ

Simplify $\frac{x^2-4}{3x^2+7x+2}$:

Solution

1. Factor both polynomials and cancel $x+2$.

Correct option: A

Final answer

$$\frac{x-2}{3x+1}$$

Worked Solutions

Rational Expressions

Quiz MCQ 2 [Concept Quiz · MCQ](#)**Simplify**

$$\frac{x+1}{x-2} + \frac{2}{x-2}:$$

Solution

1. The denominators match, so add the numerators only.

Correct option: A

Final answer

$$\frac{x+3}{x-2}$$

Quiz MCQ 3 [Concept Quiz · MCQ](#)**In**

$\frac{3}{x^2+x} - \frac{3-x}{x^2+x}$, the numerator after the minus sign is... Which of the following is correct?:

Solution

1. Parentheses preserve the sign of every term in the second numerator.

Correct option: A

Final answer

$$3 - (3 - x)$$

Worked Solutions

Rational Expressions

Quiz MCQ 4 Concept Quiz · MCQ

Which statement is safe when simplifying a compound fraction?:

Solution

1. A common multiplier preserves the quotient and exposes the structure.

Correct option: A

Final answer

multiply top and bottom by the same inner denominator

Quiz WRITTEN 5 Concept Quiz · Written

Simplify $\frac{x^2-9}{x^2-2x-3} \times \frac{x^2-4}{x^2+x-6}$ and state the original restrictions.

Solution

1. Factor all four polynomials and cancel common factors.
2. Final form: $= \frac{x+2}{x+1}$; retain original denominator exclusions.

Final answer

$$\frac{x+2}{x+1}$$

Worked Solutions

Rational Expressions

Quiz WRITTEN 6 Concept Quiz · Written

Simplify

$\frac{x^2 - 4}{x + 1} + \frac{3}{x + 1}$ and state the original restriction.

Solution

1. Use the common denominator $x+1$ and combine the numerators.
2. Final form: $= x - 1$; $x \neq -1$ remains excluded.

Final answer

$$x - 1$$

Quiz WRITTEN 7 Concept Quiz · Written

Simplify

$\frac{x^2 - y^2}{x^2 - 2xy + y^2} - \frac{x + y}{x - y}$ in x and y .

Solution

1. Factor $x^2 - y^2$, use the common $(x - y)$ structure, and apply the minus sign to $x + y$.
2. Final form: $= 0$.

Final answer

$$0$$

Worked Solutions

Rational Expressions

Quiz WRITTEN 8 Concept Quiz · Written

Explain why $(x^2 - 16)/(x - 4) = x + 4$ is not an identity for every real x .

Solution

1. The cancellation is valid only when $x - 4 \neq 0$.
2. At $x = 4$ the original expression is undefined, so the graph has a hole.

Final answer

$x \neq 4$; removable hole at $(4, 8)$

Answer key

Use the question number shown on each page.

Q001 A
Q002 B
Q003 C
Q004 D
Q005 A
Q006 B
Q007 $\frac{4b^3}{3a^2}$
Q008 $\frac{3y^2}{10x}$
Q009 $\frac{3a}{4b^3}$
Q010 $\frac{1}{4ab^2}$
Q011 B
Q012 A
Q013 A
Q014 B
Q015 A
Q016 B
Q017 C
Q018 D
Q019 A
Q020 B
Q021 C
Q022 $\frac{3c^2}{a}$
Q023 $\frac{4a^2c^3}{b^2}$

Q024 C
Q025 D
Q026 D
Q027 A
Q028 B
Q029 C
Q030 $\frac{x-2}{3x+1}$
Q031 $\frac{2x-5}{x+2}$
Q032 $\frac{x}{x-3}$
Q033 $\frac{x+2}{x-4}$
Q034 $\frac{2(x-y)}{x(x+4y)}$
Q035 $\frac{x^2+x+1}{x+3}$
Q036 $\frac{x}{x-3}$
Q037 $\frac{x-4}{x(x-2)}$
Q038 $\frac{(x-5)(x+3)}{x(x+4)}$
Q039 $\frac{(x+3)(x+4)}{(2x-3)(3x+1)}$
Q040 $\frac{(x-3)(x+1)}{(x+2)(x+4)}$
Q041 $\frac{x-3}{(x-2)(2x-1)}$
Q042 $\frac{3ab^2}{4x^2}$

Q043 $\frac{(x-3)(2x-5)}{(x-1)(x+1)}$
Q044 $\frac{3}{x+3}$
Q045 $\frac{1}{x}$
Q046 $-\frac{1}{2} + \frac{15129\sqrt{123}}{2}$
Q047 $\frac{x^2+4x-2}{(x-2)(x+3)}$
Q048 $x-1$
Q049 C
Q050 D
Q051 A
Q052 B
Q053 $\frac{1}{x+1}$
Q054 A
Q055 B
Q056 C
Q057 C
Q058 $\frac{x+2}{x(x-1)}$
Q059 $\frac{x}{(x+1)(x+2)}$
Q060 B
Q061 C
Q062 D
Q063 A
Q064 B
Q065 C

Q066 $\frac{x+2}{x+1}$
Q067 $\frac{3x-2}{x(x-1)}$
Q068 $\frac{2x+15}{(x-1)(x+6)}$
Q069 C
Q070 $\frac{2x}{(x-2)(x+2)}$
Q071 $-\frac{2}{x(x-1)}$
Q072 $\frac{2}{(x+1)(x+2)}$
Q073 $\frac{2x^2-4x+3}{x(x-1)(x+1)}$
Q074 $\frac{3x-1}{x(x-1)}$
Q075 $\frac{2}{(x+2)(x+4)}$
Q076 $\frac{3}{(x+2)(x+3)}$
Q077 B
Q078 $\frac{x^2+2x+4}{x+1}$
Q079 $-\frac{x^2-4x+2}{2(x-2)}$
Q080 $\frac{x-1}{x-2}$
Q081 $x+1$
Q082 D
Q083 A
Q084 B
Q085 C

Q086 D

Q087 A

Q088 B

Q089 $1/x$

Q090 A

Q091 A

Q092 $-\frac{x}{2}$

Q093 C

Q094 D

Q095 B

Q096 $-\frac{x}{y}$ Q097 $1 - x$ Q098 $x + 1$ Q099 $\frac{x}{y}$ Q100 $-(x + 1)$ Q101 $\frac{2x}{x^2 + 1}$ Q102 $\frac{x + 2}{2x + 3}$

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 A

Quiz WRITTEN 5 $\frac{x + 2}{x + 1}$ Quiz WRITTEN 6 $x - 1$

Quiz WRITTEN 7 0

Quiz WRITTEN 8 $x \neq 4$; removable hole at (4, 8)

Concept 4 | Identities and Proofs

English Student Edition • Focused formulas and guided practice

Formula opener

Proof route

LHS $\longrightarrow$ RHS

Square

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Difference

$$a^2 - b^2 = (a - b)(a + b)$$

Worked Solutions

Identities and Proofs

Q001 Challenge

$2x^2 + 3x + 1 = (x + 1)(ax + b) + c$. Find **a, b, c**.

Solution

1. Decide that this is a polynomial identity and expand the product.
2. Match the x^2 , x , and constant coefficients.
3. The constants are $a = 2$, $b = 1$, $c = 0$; substitute back.

Final answer

$$a = 2, b = 1, c = 0$$

Worked Solutions

Identities and Proofs

Q002 Written

$$x^2 + 6 = a(x + 1)^2 + b(x + 1) + c$$

Solution

1. Expand the shifted square: the right side is $ax^2 + (2a+b)x + (a+b+c)$.
2. Match coefficients with $x^2 + 0x + 6$: $a = 1$, $2a + b = 0$, $a + b + c = 6$.
3. Solve in order to obtain $a = 1$, $b = -2$, $c = 7$.
4. Substitution reproduces the original identity.

Final answer

$$a = 1, b = -2, c = 7$$

Q003 Written

$$x^2 - x - 3 = a(x - 1)^2 + b(x - 1) + c$$

Solution

1. Expand and collect the right side as $ax^2 + (-2a+b)x + (a-b+c)$.
2. Match with $x^2 - x - 3$ and solve $a = 1$, $b = 1$, $c = -3$.
3. Check all three coefficients in the original identity.

Final answer

$$a = 1, b = 1, c = -3$$

Worked Solutions

Identities and Proofs

Q004 MCQ

For the identity $x^2 + 6x + 5 = a(x + 2)^2 + b(x + 2) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2 , x , and constant coefficients.
3. Read off $a = 1$, $b = 2$, $c = -3$.
4. Substitute back to check.

Correct option: C**Final answer** $(1, 2, -3)$

Q005 MCQ

For the identity $2x^2 + 9x + 8 = a(x + 3)^2 + b(x + 3) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2 , x , and constant coefficients.
3. Read off $a = 2$, $b = -3$, $c = -1$.
4. Substitute back to check.

Correct option: C**Final answer** $(2, -3, -1)$

Worked Solutions

Identities and Proofs

Q006 MCQ

For the identity $3x^2 + 28x + 65 = a(x + 4)^2 + b(x + 4) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2 , x , and constant coefficients.
3. Read off $a = 3$, $b = 4$, $c = 1$.
4. Substitute back to check.

Correct option: D

Final answer

$(3, 4, 1)$

Q007 MCQ

For the identity $x^2 + 5x + 3 = a(x + 5)^2 + b(x + 5) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2 , x , and constant coefficients.
3. Read off $a = 1$, $b = -5$, $c = 3$.
4. Substitute back to check.

Correct option: A

Final answer

$(1, -5, 3)$

Worked Solutions

Identities and Proofs

Q008 Written

$$a(x + 1) - b(x + 5) = 8x$$

Solution

1. Collect coefficients: $(a - b)x + (a - 5b) = 8x$.
2. Solve $a - b = 8$ and $a - 5b = 0$.
3. Hence $a = 10$ and $b = 2$; the constant term checks as zero.

Final answer

$$a = 10, b = 2$$

Q009 MCQ

For the identity $a(x + 1) - b(x + 3) = x - 1$, the ordered tuple (a, b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a = 2, b = 1$.
4. Check both terms.

Correct option: A

Final answer

$$(2, 1)$$

Worked Solutions

Identities and Proofs

Q010 MCQ

For the identity $a(x + 2) - b(x + 4) = 5x + 14$, the ordered tuple (a, b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a = 3, b = -2$.
4. Check both terms.

Correct option: B

Final answer $(3, -2)$

Q011 MCQ

For the identity $a(x + 3) - b(x + 5) = x - 3$, the ordered tuple (a, b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a = 4, b = 3$.
4. Check both terms.

Correct option: D

Final answer $(4, 3)$

Worked Solutions

Identities and Proofs

Q012 MCQ

For the identity $a(x + 4) - b(x + 6) = 9x + 44$, the ordered tuple (a, b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a = 5$, $b = -4$.
4. Check both terms.

Correct option: D

Final answer

$(5, -4)$

Q013 Written

$$(x + a)(b + 3) = 4x + 12$$

Solution

1. Expand as $(b+3)x+a(b+3)$.
2. Match $b + 3 = 4$ and $a(b + 3) = 12$.
3. Therefore $b = 1$ and $a = 3$; c is not present in the printed identity and is not applicable.

Final answer

$a = 3, b = 1$

Worked Solutions

Identities and Proofs

Q014 MCQ

For the identity $(x + a)(b + 4) = 3x + 6$, the ordered tuple (a, b) is:

Solution

1. Expand as $(b+d)x+a(b+d)$.
2. Match the coefficient of x and the constant term.
3. Obtain $a = 2$, $b = -1$; no extra parameter is present.
4. Substitute back.

Correct option: B

Final answer

$(2, -1)$

Q015 MCQ

For the identity $(x + a)(b + 5) = 5x + 15$, the ordered tuple (a, b) is:

Solution

1. Expand as $(b+d)x+a(b+d)$.
2. Match the coefficient of x and the constant term.
3. Obtain $a = 3$, $b = 0$; no extra parameter is present.
4. Substitute back.

Correct option: C

Final answer

$(3, 0)$

Worked Solutions

Identities and Proofs

Q016 MCQ

For the identity $(x + a)(b + 6) = 7x + 28$, the ordered tuple (a, b) is:

Solution

1. Expand as $(b+6)x+a(b+6)$.
2. Match the coefficient of x and the constant term.
3. Obtain $a = 4$, $b = 1$; no extra parameter is present.
4. Substitute back.

Correct option: C

Final answer

$(4, 1)$

Q017 Written

$$2x^2 + 5x + 4 = (x + 1)(ax + b) + c$$

Solution

1. Expand the right side to $ax^2+(a+b)x+(b+c)$.
2. Match $(a, a + b, b + c) = (2, 5, 4)$.
3. Solve $a = 2$, $b = 3$, $c = 1$ and check the constant term.

Final answer

$a = 2$, $b = 3$, $c = 1$

Worked Solutions

Identities and Proofs

Q018 Written

$$(ax + b)(x + 1) = 4x^2 + cx + 1$$

Solution

1. Expand: $ax^2 + (a+b)x + b$.
2. Match the x^2 and constant coefficients to get $a = 4$ and $b = 1$.
3. Then $c = a + b = 5$.

Final answer

$$a = 4, b = 1, c = 5$$

Q019 Written

$$2x^2 - 7x + 8 = (x - 3)(ax + b) + c$$

Solution

1. Expand to $ax^2 + (b-3a)x + (-3b+c)$.
2. Match with $(2, -7, 8)$.
3. Solve $a = 2$, $b = -1$, $c = 5$ and substitute back.

Final answer

$$a = 2, b = -1, c = 5$$

Worked Solutions

Identities and Proofs

Q020 MCQ

For the identity $(ax - 1)(x + 1) + c = x^2 - 3$, the ordered tuple (a, c) is:

Solution

1. Expand the product to $ax^2 + (ah+b)x + bh + c$.
2. Match the x^2 , x , and constant coefficients.
3. Use $a = 1$ and $c = -2$, then verify the middle coefficient.
4. Check the full identity.

Correct option: B

Final answer

$(1, -2)$

Q021 MCQ

For the identity $(ax + 2)(x + 2) + c = 2x^2 + 6x + 3$, the ordered tuple (a, c) is:

Solution

1. Expand the product to $ax^2 + (ah+b)x + bh + c$.
2. Match the x^2 , x , and constant coefficients.
3. Use $a = 2$ and $c = -1$, then verify the middle coefficient.
4. Check the full identity.

Correct option: C

Final answer

$(2, -1)$

Worked Solutions

Identities and Proofs

Q022 MCQ

For the identity $(ax + 3)(x + 3) + c = 3x^2 + 12x + 9$, the ordered tuple (a, c) is:

Solution

1. Expand the product to $ax^2 + (ah+b)x + bh + c$.
2. Match the x^2 , x , and constant coefficients.
3. Use $a = 3$ and $c = 0$, then verify the middle coefficient.
4. Check the full identity.

Correct option: B

Final answer

$(3, 0)$

Q023 Written

$$\frac{3x + 1}{(x + 2)(x - 3)} = \frac{a}{x + 2} + \frac{b}{x - 3}$$

Solution

1. Exclude $x = -2$ and $x = 3$ from the original denominator.
2. Multiply by $(x+2)(x-3)$: $3x + 1 = a(x - 3) + b(x + 2)$.
3. Match coefficients: $a + b = 3$ and $-3a + 2b = 1$, giving $a = 1$, $b = 2$.
4. Substitute a and b back into the rational identity.

Final answer

$a = 1$, $b = 2$

Worked Solutions

Identities and Proofs

Q024 Written

$$\frac{x+8}{(x-2)(x+3)} = \frac{a}{x-2} + \frac{b}{x+3}$$

Solution

1. Exclude $x = 2$ and $x = -3$.
2. Clear denominators: $x + 8 = a(x + 3) + b(x - 2)$.
3. Match $a + b = 1$ and $3a - 2b = 8$, giving $a = 2$, $b = -1$.
4. Substitute and verify.

Final answer

$$a = 2, b = -1$$

Q025 Written

$$\frac{1}{(2x-1)(x+1)} = \frac{a}{2x-1} + \frac{b}{x+1}$$

Solution

1. Exclude $x = 1/2$ and $x = -1$.
2. Clear denominators: $1 = a(x + 1) + b(2x - 1)$.
3. Match $a + 2b = 0$ and $a - b = 1$, giving $a = 2/3$, $b = -1/3$.
4. Verify in the original decomposition.

Final answer

$$a = \frac{2}{3}, b = -\frac{1}{3}$$

Worked Solutions

Identities and Proofs

Q026 Written

$$\frac{a}{x-1} + \frac{b}{x-5} = \frac{5x-1}{(x-1)(x-5)}$$

Solution

1. Exclude $x = 1$ and $x = 5$.
2. Clear denominators: $a(x-5) + b(x-1) = 5x-1$.
3. Match $a+b=5$ and $-5a-b=-1$, giving $a=-1$, $b=6$.
4. Check the original rational identity.

Final answer

$$a = -1, b = 6$$

Q027 Written

$$\frac{3x-8}{3x^2+5x-2} = \frac{a}{x+2} + \frac{b}{3x-1}$$

Solution

1. Factor $3x^2 + 5x - 2 = (x+2)(3x-1)$.
2. Exclude $x = -2$ and $x = 1/3$, then clear denominators.
3. Match $3a+b=3$ and $-a+2b=-8$, giving $a=2$, $b=-3$.
4. Check the factored rational identity.

Final answer

$$a = 2, b = -3$$

Worked Solutions

Identities and Proofs

Q028 Written

$$f(t) = \frac{22t + 15}{8t^2 + 10t + 3} = \frac{a}{2t + 1} + \frac{b}{4t + 3}, \quad t > 0$$

Solution

1. Factor ($8t^2 + 10t + 3 = (2t + 1)(4t + 3)$).
2. The stated domain is ($t > 0$); the algebraic exclusions lie outside it.
3. Clear denominators: ($22t + 15 = a(4t + 3) + b(2t + 1)$).
4. Match ($4a + 2b = 22$) and ($3a + b = 15$) to get ($a = 4, b = 3$).

Final answer

$$a = 4, \quad b = 3$$

Q029 Written

Prove that $(x + y)^2 - (x - y)^2 = 4xy$.**Solution**

1. Expand both conjugate squares on the left.
2. Remove the parentheses: $x^2 + 2xy + y^2 - x^2 + 2xy - y^2$.
3. Collect like terms to obtain $4xy$, which is the right-hand side.

Final answer

$$4xy$$

Worked Solutions

Identities and Proofs

Q030 Written

Prove that $(2a + b)^2 - (a + 2b)^2 = 3(a^2 - b^2)$.**Solution**

1. Recognise a difference of squares: $U^2 - V^2 = (U - V)(U + V)$.
2. Set $U = 2a + b$ and $V = a + 2b$; then $U - V = a - b$ and $U + V = 3a + 3b$.
3. Multiply to get $(a - b)(3a + 3b) = 3(a^2 - b^2)$.

Final answer

$$3a^2 - 3b^2$$

Q031 MCQ

For the identity $-(x - 2y + 1)^2 + (x + 2y + 1)^2 = 2y(4x + 4)$, **the value of the simplified left-hand side is:****Solution**

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $8xy + 8y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: A**Final answer**

$$8xy + 8y$$

Worked Solutions

Identities and Proofs

Q032 MCQ

For the identity $-(x - 3y + 2)^2 + (x + 3y + 2)^2 = 3y(4x + 8)$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $12xy + 24y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D**Final answer** $12xy + 24y$

Q033 MCQ

For the identity $-(x - y + 3)^2 + (x + y + 3)^2 = y(4x + 12)$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $4xy + 12y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D**Final answer** $4xy + 12y$

Worked Solutions

Identities and Proofs

Q034 MCQ

For the identity $-(x - 2y + 4)^2 + (x + 2y + 4)^2 = 2y(4x + 16)$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $8xy + 32y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: A

Final answer

$$8xy + 32y$$

Q035 Written

Prove that $(x + y)^2 + (x - y)^2 = 2x^2 + 2y^2$.

Solution

1. Expand both squares on the left.
2. Add $x^2 + 2xy + y^2$ and $x^2 - 2xy + y^2$.
3. The mixed terms cancel, leaving $2x^2 + 2y^2$.

Final answer

$$2x^2 + 2y^2$$

Worked Solutions

Identities and Proofs

Q036 MCQ

For the identity $(x - 2y + 1)^2 + (x + 2y + 1)^2 = 8y^2 + 2(x + 1)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2 + 4x + 8y^2 + 2$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C**Final answer**

$$2x^2 + 4x + 8y^2 + 2$$

Q037 MCQ

For the identity $(x - 3y + 2)^2 + (x + 3y + 2)^2 = 18y^2 + 2(x + 2)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2 + 8x + 18y^2 + 8$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C**Final answer**

$$2x^2 + 8x + 18y^2 + 8$$

Worked Solutions

Identities and Proofs

Q038 MCQ

For the identity $(x - y + 3)^2 + (x + y + 3)^2 = 2y^2 + 2(x + 3)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2 + 12x + 2y^2 + 18$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C

Final answer

$$2x^2 + 12x + 2y^2 + 18$$

Q039 MCQ

For the identity $(x - 2y + 4)^2 + (x + 2y + 4)^2 = 8y^2 + 2(x + 4)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2 + 16x + 8y^2 + 32$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D

Final answer

$$2x^2 + 16x + 8y^2 + 32$$

Worked Solutions

Identities and Proofs

Q040 Written

Prove that $(a + 2b)^2 - 4ab = (a - 2b)^2 + 4ab$.**Solution**

1. Expand the left-hand side: $a^2 + 4ab + 4b^2 - 4ab = a^2 + 4b^2$.
2. Expand the right-hand side: $a^2 - 4ab + 4b^2 + 4ab = a^2 + 4b^2$.
3. Both sides simplify to the same expression.

Final answer

$$a^2 + 4b^2$$

Q041 MCQ

For the identity $-(x - 2y + 3)^2 + (x + 2y - 1)^2 = (8x + 8)(y - 1)$, **the value of the simplified left-hand side is:****Solution**

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $8xy - 8x + 8y - 8$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D**Final answer**

$$8xy - 8x + 8y - 8$$

Worked Solutions

Identities and Proofs

Q042 MCQ

For the identity $-(2x - 3y + 5)^2 + (2x + 3y - 1)^2 = (24x + 24)(y - 1)$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $24xy - 24x + 24y - 24$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C**Final answer**

$$24xy - 24x + 24y - 24$$

Q043 Written

Prove that $(x^2 + x + 1)(x^2 - x + 1) = x^4 + x^2 + 1$.

Solution

1. Group the factors as $[(x^2 + 1) - x][(x^2 + 1) + x]$.
2. Use $(U - V)(U + V) = U^2 - V^2$ with $U = x^2 + 1$ and $V = x$.
3. Simplify $(x^2 + 1)^2 - x^2$ to $x^4 + x^2 + 1$.

Final answer

$$x^4 + x^2 + 1$$

Worked Solutions

Identities and Proofs

Q044 Written

Prove that $(a^2 - ab + b^2)(a^2 + ab + b^2) = (a^2 + b^2)^2 - (ab)^2$.

Solution

1. Group the factors as $[(a^2 + b^2) - ab][(a^2 + b^2) + ab]$.
2. Apply $(U - V)(U + V) = U^2 - V^2$ with $U = a^2 + b^2$ and $V = ab$.
3. This gives $(a^2 + b^2)^2 - (ab)^2$, exactly the right-hand side.

Final answer

$$a^4 + a^2b^2 + b^4$$

Q045 Written

Prove that $(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$.

Solution

1. Expand the left-hand product to $a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$.
2. Expand the first right-hand square: $a^2c^2 - 2abcd + b^2d^2$.
3. Expand the second right-hand square: $a^2d^2 + 2abcd + b^2c^2$.
4. Add the two expansions; the $-2abcd$ and $+2abcd$ terms cancel, leaving the left-hand expression.

Final answer

$$a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$$

Worked Solutions

Identities and Proofs

Q046 Written

Prove that $(x^2 + 1)(y^2 + 1) = (xy + 1)^2 + (x - y)^2$.

Solution

1. Expand the left-hand side to $x^2y^2 + x^2 + y^2 + 1$.
2. Expand the right-hand side: $x^2y^2 + 2xy + 1 + x^2 - 2xy + y^2$.
3. The $+2xy$ and $-2xy$ terms cancel, leaving the same expression as the left.

Final answer

$$x^2y^2 + x^2 + y^2 + 1$$

Q047 Written

Prove that $(a^2 - b^2)(c^2 - d^2) = (ac - bd)^2 - (ad - bc)^2$.

Solution

1. Expand the first right-hand square: $a^2c^2 - 2abcd + b^2d^2$.
2. Expand the second right-hand square: $a^2d^2 - 2abcd + b^2c^2$.
3. Subtract; the mixed terms cancel, giving $a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2$.
4. Factor the result as $(a^2 - b^2)(c^2 - d^2)$.

Final answer

$$a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2$$

Worked Solutions

Identities and Proofs

Q048 Written

Prove that $(a + b)^3 - 3ab(a + b) = a^3 + b^3$.**Solution**

1. Expand $(a + b)^3$ to $a^3 + 3a^2b + 3ab^2 + b^3$.
2. Expand $3ab(a+b)$ to $3a^2b + 3ab^2$.
3. Subtract the matching cross terms; the result is $a^3 + b^3$.

Final answer

$$a^3 + b^3$$

Q049 Written

Prove that $(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = a^3 + b^3 + c^3 - 3abc$.**Solution**

1. Distribute $a+b+c$ across the second bracket.
2. Write all products in a fixed order: a^3, b^3, c^3 , then the mixed terms.
3. Each mixed term cancels with its opposite, while $-abc$ appears three times.
4. The remaining expression is $a^3 + b^3 + c^3 - 3abc$.

Final answer

$$a^3 - 3abc + b^3 + c^3$$

Worked Solutions

Identities and Proofs

Q050 Challenge

For the integrated identity $(x^2 + 1)(-(x - 2y + 1)^2 + (x + 2y + 1)^2) = 2y(4x + 4)(x^2 + 1)$, the value of the simplified left-hand side is:

Solution

1. Recognise the cubic or symmetric pattern before expanding.
2. Expand in descending degree and collect like terms.
3. The cross terms cancel, leaving $8x^3y + 8x^2y + 8xy + 8y$.
4. Therefore the proposed equality is proved.

Correct option: B

Final answer

$$8x^3y + 8x^2y + 8xy + 8y$$

Worked Solutions

Identities and Proofs

Q051 Written

Under the condition $x + y = 1$, prove that $x^2 + y^2 + 1 = 2(x + y - xy)$.

Solution

1. Use $x + y = 1$, so $(x + y)^2 = 1$.
2. Expand: $x^2 + 2xy + y^2 = 1$, hence $x^2 + y^2 + 1 = 2 - 2xy$.
3. Since $x + y = 1$, $2 - 2xy = 2(x + y - xy)$. Therefore the equality holds.

Final answer

$$x^2 + y^2 + 1 = 2(x + y - xy)$$

Q052 Written

Under the condition $x + y = 1$, prove that $x^2 + y^2 = 1 - 2xy$.

Solution

1. Square the condition: $(x + y)^2 = 1$.
2. Expand to $x^2 + 2xy + y^2 = 1$.
3. Rearrange to obtain $x^2 + y^2 = 1 - 2xy$.

Final answer

$$x^2 + y^2 = 1 - 2xy$$

Worked Solutions

Identities and Proofs

Q053 MCQ

Under the condition $x + y = 2$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = 2$.
2. $y = 2 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 4$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: C

Final answer

$$-2xy + 4$$

Q054 MCQ

Under the condition $x + y = 3$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = 3$.
2. $y = 3 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 9$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: D

Final answer

$$-2xy + 9$$

Worked Solutions

Identities and Proofs

Q055 MCQ

Under the condition $x + y = -1$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = -1$.
2. $y = -1 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 1$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: C

Final answer

$$-2xy + 1$$

Q056 MCQ

Under the condition $x + y = 4$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = 4$.
2. $y = 4 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 16$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: A

Final answer

$$-2xy + 16$$

Worked Solutions

Identities and Proofs

Q057 Written

Under the condition $x - y = 1$, prove that $x^2 + y^2 = 2xy + 1$.

Solution

1. Square the condition: $(x - y)^2 = 1$.
2. Expand to $x^2 - 2xy + y^2 = 1$.
3. Rearrange to obtain $x^2 + y^2 = 2xy + 1$.

Final answer

$$x^2 + y^2 = 2xy + 1$$

Q058 MCQ

Under the condition $x - y = 1$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = 1$.
2. $x = y + 1$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 1$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: B**Final answer**

$$2xy + 1$$

Worked Solutions

Identities and Proofs

Q059 MCQ

Under the condition $x - y = 2$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = 2$.
2. $x = y + 2$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 4$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: B

Final answer

$$2xy + 4$$

Q060 MCQ

Under the condition $x - y = -2$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = -2$.
2. $x = y - 2$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 4$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: C

Final answer

$$2xy + 4$$

Worked Solutions

Identities and Proofs

Q061 MCQ

Under the condition $x - y = 3$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = 3$.
2. $x = y + 3$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 9$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: D

Final answer

$$2xy + 9$$

Q062 Written

Under the condition $x + y = 1$, prove that $x^2 - x = y^2 - y$.

Solution

1. Move the right-hand side to the left: $x^2 - y^2 - x + y$.
2. Factor: $x^2 - y^2 - x + y = (x - y)(x + y - 1)$.
3. The condition makes $x + y - 1 = 0$, so the difference is zero.

Final answer

$$x^2 - x = y^2 - y$$

Worked Solutions

Identities and Proofs

Q063 Written

Under the condition $a + b + c = 0$, prove that $a^2 + b^2 + c^2 = -2(ab + bc + ca)$.

Solution

1. From $a + b + c = 0$, write $c = -a - b$.
2. Substitute into the left-hand side: $a^2 + b^2 + (-a - b)^2 = 2a^2 + 2ab + 2b^2$.
3. Substitute into the right-hand side: $-2[ab + b(-a - b) + (-a - b)a] = 2a^2 + 2ab + 2b^2$.
4. Both sides reduce to the same expression, so the equality is proved.

Final answer

$$a^2 + b^2 + c^2 = -2(ab + bc + ca)$$

Q064 Written

Under the condition $a + b + c = 0$, prove that $a^3 + b^3 + c^3 = 3abc$.

Solution

1. Use the identity $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$.
2. The condition gives $a + b + c = 0$, so the product on the right is zero.
3. Therefore $a^3 + b^3 + c^3 - 3abc = 0$, which proves $a^3 + b^3 + c^3 = 3abc$.

Final answer

$$a^3 + b^3 + c^3 = 3abc$$

Worked Solutions

Identities and Proofs

Q065 Written

Under the condition $a + b + c = 0$, prove that $a^2 + ca = b^2 + bc$.

Solution

1. Subtract the right-hand side: $a^2 + ca - b^2 - bc$.
2. Factor by grouping: $(a - b)(a + b) + (a - b)c = (a - b)(a + b + c)$.
3. Since $a + b + c = 0$, the difference is zero; hence the two sides are equal.

Final answer

$$a^2 + ca = b^2 + bc$$

Q066 Written

Under the condition $a + b + c = 0$, prove that $(a + b)(b + c)(c + a) + abc = 0$.

Solution

1. From $a + b + c = 0$, write $a + b = -c$, $b + c = -a$, and $c + a = -b$.
2. Then $(a + b)(b + c)(c + a) = (-c)(-a)(-b) = -abc$.
3. Adding abc gives zero, so the required equality is proved.

Final answer

$$(a + b)(b + c)(c + a) + abc = 0$$

Worked Solutions

Identities and Proofs

Q067 Written

Under the condition $a + b + c = 0$, $abc \neq 0$, prove that $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$.

Solution

1. Because $a + b + c = 0$, $b + c = -a$, $c + a = -b$, and $a + b = -c$.
2. The non-zero condition ensures that all three denominators are non-zero.
3. Therefore each fraction equals -1: $a/(b + c) = b/(c + a) = c/(a + b) = -1$.
4. Their sum is $-1 - 1 - 1 = -3$.

Final answer

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$$

Worked Solutions

Identities and Proofs

Q068 Challenge

Under the condition $x + y = 3$, the expression $x^2 + y^2 + 1$ is equal to:

Solution

1. Use $x + y = 3$ to replace each cyclic denominator.
2. The non-zero condition keeps every original denominator defined.
3. Each term simplifies and the total becomes $-2xy + 10$.
4. Thus the required constant sum is proved.

Correct option: C

Final answer

$$-2xy + 10$$

Worked Solutions

Identities and Proofs

Q069 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{ab}{a^2+b^2} = \frac{cd}{c^2+d^2}$.

Solution

1. Let $a/b = c/d = k$, so $a = bk$ and $c = dk$.
2. Substitute into the left-hand side: $ab/(a^2+b^2) = k/(k^2+1)$.
3. Substitute into the right-hand side: $cd/(c^2+d^2) = k/(k^2+1)$.
4. The two sides are equal.

Final answer

$$\frac{ab}{a^2 + b^2} = \frac{cd}{c^2 + d^2}$$

Q070 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: C

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions

Identities and Proofs

Q071 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{cd}{c^2 + d^2}$$

Q072 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: C

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions

Identities and Proofs

Q073 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions

Identities and Proofs

Q074 Challenge

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions

Identities and Proofs

Q075 Challenge

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: C

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions

Identities and Proofs

Q076 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{ab+cd}{ab-cd} = \frac{a^2+c^2}{a^2-c^2}$.

Solution

1. Let $a = bk$ and $c = dk$.
2. The left-hand side becomes $k(b^2 + d^2)/(b^2 - d^2)$.
3. The right-hand side becomes $k^2(b^2 + d^2)/[k^2(b^2 - d^2)]$, which is the same expression.
4. Hence the equality is proved.

Final answer

$$\frac{ab + cd}{ab - cd} = \frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions

Identities and Proofs

Q077 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions

Identities and Proofs

Q078 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions

Identities and Proofs

Q079 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions

Identities and Proofs

Q080 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions

Identities and Proofs

Q081 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{a+b}{a-b} = \frac{c+d}{c-d}$.

Solution

1. Write $a = bk$ and $c = dk$.
2. Then $(a + b)/(a - b) = (bk + b)/(bk - b) = (k + 1)/(k - 1)$.
3. Similarly $(c + d)/(c - d) = (dk + d)/(dk - d) = (k + 1)/(k - 1)$.
4. The equality follows.

Final answer

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Q082 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: B

Final answer

$$\frac{c+d}{c-d}$$

Worked Solutions

Identities and Proofs

Q083 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: B

Final answer

$$\frac{c+d}{c-d}$$

Q084 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{c+d}{c-d}$$

Worked Solutions

Identities and Proofs

Q085 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{c+d}{c-d}$$

Q086 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$.

Solution

1. Substitute $a = bk$ and $c = dk$.
2. The left-hand side becomes $(3k + k)/(3 + 1) = k$.
3. The right-hand side becomes $(2k - k)/(2 - 1) = k$.
4. Thus both sides are equal.

Final answer

$$\frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$$

Worked Solutions

Identities and Proofs

Q087 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{2a-c}{2b-d}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{2a - c}{2b - d}$$

Q088 Written

Under the condition $x/a = y/b$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b}, \quad \frac{a^2+x^2}{a^2-x^2} = \frac{b^2+y^2}{b^2-y^2}$.

Solution

1. Let $x = ak$ and $y = bk$.
2. The left-hand side is $(a^2 + a^2k^2)/(a^2 - a^2k^2) = (1 + k^2)/(1 - k^2)$.
3. The right-hand side reduces to the same value.
4. Hence the equality is proved.

Final answer

$$\frac{a^2 + x^2}{a^2 - x^2} = \frac{b^2 + y^2}{b^2 - y^2}$$

Worked Solutions

Identities and Proofs

Q089 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{a+b}{a-b} = \frac{c+d}{c-d}.$

Solution

1. Use the equal concentration condition to write $a = bk$ and $c = dk$.
2. The first ratio becomes $(k+1)/(k-1)$.
3. The second ratio becomes $(k+1)/(k-1)$.
4. The required equality follows, provided the denominators are non-zero.

Final answer

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Q090 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}, \quad \frac{x^2+y^2+z^2}{a^2+b^2+c^2} = \frac{xy+yz+zx}{ab+bc+ca}.$

Solution

1. Let the common ratio be k , so $x = ak$, $y = bk$ and $z = ck$.
2. Then $x^2 + y^2 + z^2 = k^2(a^2 + b^2 + c^2)$.
3. Also $xy + yz + zx = k^2(ab + bc + ca)$.
4. Both ratios therefore equal k^2 .

Final answer

$$\frac{x^2 + y^2 + z^2}{a^2 + b^2 + c^2} = \frac{xy + yz + zx}{ab + bc + ca}$$

Worked Solutions

Identities and Proofs

Q091 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $\frac{x+y+z}{a+b+c} = \frac{x}{a}$.

Solution

1. Let the common ratio be k .
2. Then $x + y + z = k(a + b + c)$.
3. Therefore $(x + y + z)/(a + b + c) = k = x/a$.

Final answer

$$\frac{x + y + z}{a + b + c} = \frac{x}{a}$$

Q092 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $\frac{x+2y+3z}{a+2b+3c} = \frac{x+y+z}{a+b+c}$.

Solution

1. Set $x = ak$, $y = bk$ and $z = ck$.
2. The left numerator is $k(a+2b+3c)$, so the left ratio is k .
3. The right ratio is also k because $x + y + z = k(a + b + c)$.
4. The two ratios are equal.

Final answer

$$\frac{x + 2y + 3z}{a + 2b + 3c} = \frac{x + y + z}{a + b + c}$$

Worked Solutions

Identities and Proofs

Q093 Written

Under the condition $x : y : z = 2 : 3 : 5$, prove or find the required expression: $x : y : z = 2 : 3 : 5$, $\frac{2xy+yz+3zx}{x^2+y^2+z^2}$.

Solution

1. Let $x = 2k$, $y = 3k$ and $z = 5k$.
2. The numerator is $(12 + 15 + 30)k^2 = 57k^2$.
3. The denominator is $(4 + 9 + 25)k^2 = 38k^2$.
4. Therefore the value is $57/38$.

Final answer

$$\frac{57}{38}$$

Q094 Written

Under the condition $a : b : c = 2 : 3 : 4$, prove or find the required expression: $a : b : c = 2 : 3 : 4$, $\frac{ab+bc+ca}{a^2+b^2+c^2}$.

Solution

1. Let $a = 2k$, $b = 3k$ and $c = 4k$.
2. The numerator is $(6 + 12 + 8)k^2 = 26k^2$.
3. The denominator is $(4 + 9 + 16)k^2 = 29k^2$.
4. Therefore the value is $26/29$.

Final answer

$$\frac{26}{29}$$

Worked Solutions

Identities and Proofs

Q095 Written

Under the condition $a : b : c = 3 : 5 : 2$, prove or find the required expression: $a : b : c = 3 : 5 : 2$, $\frac{a^2+b^2-c^2}{a^2-b^2+c^2}$.

Solution

1. Let $a = 3k$, $b = 5k$ and $c = 2k$.
2. The numerator is $(9 + 25 - 4)k^2 = 30k^2$.
3. The denominator is $(9 - 25 + 4)k^2 = -12k^2$.
4. Therefore the value is $-5/2$.

Final answer

$$-\frac{5}{2}$$

Q096 Written

Under the condition $x : y : z = 2 : -3 : 4$, prove or find the required expression: $x : y : z = 2 : -3 : 4$, $\frac{x^2+y^2+z^2}{xy+yz+zx}$.

Solution

1. Let $x = 2k$, $y = -3k$ and $z = 4k$.
2. The numerator is $(4 + 9 + 16)k^2 = 29k^2$.
3. The denominator is $(-6 - 12 + 8)k^2 = -10k^2$.
4. Therefore the value is $-29/10$.

Final answer

$$-\frac{29}{10}$$

Worked Solutions

Identities and Proofs

Q097 Written

Under the condition $x/7 = y/3 \neq 0$, prove or find the required expression: $\frac{x}{7} = \frac{y}{3} \neq 0$, $\frac{x^2+5y^2}{x^2+xy+y^2}$.

Solution

1. Let $x = 7k$ and $y = 3k$, where k is non-zero.
2. The numerator is $(49 + 45)k^2 = 94k^2$.
3. The denominator is $(49 + 21 + 9)k^2 = 79k^2$.
4. Therefore the value is $94/79$.

Final answer

$$\frac{94}{79}$$

Q098 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$.

Solution

1. Let $x = ak$, $y = bk$ and $z = ck$.
2. Then $x^2 + y^2 + z^2 = k^2(a^2 + b^2 + c^2)$.
3. Also $ax + by + cz = k(a^2 + b^2 + c^2)$.
4. Both sides equal $k^2(a^2 + b^2 + c^2)^2$.

Final answer

$$(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$$

Worked Solutions

Identities and Proofs

Quiz MCQ 1 Concept Quiz · MCQ

If $x^2 + 4x + 1 = (x + 1)(ax + b) + c$, then (a, b, c) is... Which of the following is correct?:

Solution

1. Expand to $ax^2 + (a + b)x + b + c$.
2. Match $a = 1$, $a + b = 4$, $b + c = 1$, so $b = 3$, $c = -2$.

Correct option: A

Final answer

$(1, 3, -2)$

Quiz MCQ 2 Concept Quiz · MCQ

For the identity $(x^2 - 2x + 2)(x^2 + 2x + 2) = -4x^2 + (x^2 + 2)^2$, the value of the simplified left-hand side is:

Solution

1. Rewrite the factors as $(U-V)(U+V)$.
2. Apply $U^2 - V^2$ before carrying out any long expansion.
3. Simplify to $x^4 + 4$.
4. This is exactly the stated right-hand side.

Correct option: A

Final answer

$x^4 + 4$

Worked Solutions

Identities and Proofs

Quiz MCQ 3 Concept Quiz · MCQ

Under the condition $a + b + c = 0$, the expression $3a^2 + 3ac$ is equal to:

Solution

1. Move the right-hand side to the left and factor the difference.
2. The factor containing the condition $a + b + c = 0$ is zero.
3. The remaining product is therefore 0, giving $3b^2 + 3bc$.
4. Hence the two sides are equal under the given condition.

Correct option: A

Final answer

$$3b^2 + 3bc$$

Quiz MCQ 4 Concept Quiz · MCQ

Under the condition $x : y : z = 2 : 3 : 5$, the required expression is equal to:

Solution

1. Write the variables as the stated multiples of k.
2. Substitute into the numerator and denominator.
3. Cancel the common factor and simplify to $\frac{3}{2}$.
4. Hence the required value is obtained.

Correct option: B

Final answer

$$\frac{3}{2}$$

Worked Solutions

Identities and Proofs

Quiz WRITTEN 5 Concept Quiz · Written

Find a, b, c if $x^2 - 3x + 5 = (x - 2)(ax + b) + c$.

Solution

1. Expand to $ax^2 + (b - 2a)x - 2b + c$.
2. Match $a = 1$, $b - 2 = -3$, $-2b + c = 5$.
3. Therefore $b = -1$, $c = 3$.

Final answer

$$a = 1, b = -1, c = 3$$

Quiz WRITTEN 6 Concept Quiz · Written

Prove that $(x^2 - 3x + 4)(x^2 + 3x + 4) = -9x^2 + (x^2 + 4)^2$.

Solution

1. Rewrite the factors as $(U - V)(U + V)$.
2. Apply $U^2 - V^2$ before carrying out any long expansion.
3. Simplify to $x^4 - x^2 + 16$.
4. This is exactly the stated right-hand side.

Final answer

$$x^4 - x^2 + 16$$

Worked Solutions

Identities and Proofs

Quiz WRITTEN 7 Concept Quiz · Written

Under the condition $a + b + c = 0$, prove that $4a^2 + 4ac = 4b^2 + 4bc$.

Solution

1. Move the right-hand side to the left and factor the difference.
2. The factor containing the condition $a + b + c = 0$ is zero.
3. The remaining product is therefore 0, giving $4b^2 + 4bc$.
4. Hence the two sides are equal under the given condition.

Final answer

$$4b^2 + 4bc$$

Quiz WRITTEN 8 Concept Quiz · Written

Under the condition $x/a = y/b = z/c$, prove that $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$.

Solution

1. Use the common multiplier from $x/a = y/b = z/c$.
2. Substitute and collect the common factor.
3. Both sides simplify to $(ax + by + cz)^2$.
4. Thus the identity is proved.

Final answer

$$(ax + by + cz)^2$$

Answer key

Use the question number shown on each page.

Q001 $a = 2, b = 1, c = 0$

Q002 $a = 1, b = -2, c = 7$

Q003 $a = 1, b = 1, c = -3$

Q004 C

Q005 C

Q006 D

Q007 A

Q008 $a = 10, b = 2$

Q009 A

Q010 B

Q011 D

Q012 D

Q013 $a = 3, b = 1$

Q014 B

Q015 C

Q016 C

Q017 $a = 2, b = 3, c = 1$

Q018 $a = 4, b = 1, c = 5$

Q019 $a = 2, b = -1, c = 5$

Q020 B

Q021 C

Q022 B

Q023 $a = 1, b = 2$

Q024 $a = 2, b = -1$

Q025 $a = \frac{2}{3}, b = -\frac{1}{3}$

Q026 $a = -1, b = 6$

Q027 $a = 2, b = -3$

Q028 $a = 4, b = 3$

Q029 $4xy$

Q030 $3a^2 - 3b^2$

Q031 A

Q032 D

Q033 D

Q034 A

Q035 $2x^2 + 2y^2$

Q036 C

Q037 C

Q038 C

Q039 D

Q040 $a^2 + 4b^2$

Q041 D

Q042 C

Q043 $x^4 + x^2 + 1$

Q044 $a^4 + a^2b^2 + b^4$

Q045 $a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$

Q046 $x^2y^2 + x^2 + y^2 + 1$

Q047 $a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2$

Q048 $a^3 + b^3$

Q049 $a^3 - 3abc + b^3 + c^3$

Q050 B

Q051 $x^2 + y^2 + 1 = 2(x + y - xy)$

Q052 $x^2 + y^2 = 1 - 2xy$

Q053 C

Q054 D

Q055 C

Q056 A

Q057 $x^2 + y^2 = 2xy + 1$

Q058 B

Q059 B

Q060 C

Q061 D

Q062 $x^2 - x = y^2 - y$

Q063 $a^2 + b^2 + c^2 = -2(ab + bc + ca)$

Q064 $a^3 + b^3 + c^3 = 3abc$

Q065 $a^2 + ca = b^2 + bc$

Q066 $(a + b)(b + c)(c + a) + abc = 0$

Q067 $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$

Q068 C

Q069 $\frac{ab}{a^2+b^2} = \frac{cd}{c^2+d^2}$

Q070 C

Q071 D

Q072 C

Q073 D

Q074 D

Q075 C

Q076 $\frac{ab+cd}{ab-cd} = \frac{a^2+c^2}{a^2-c^2}$

Q077 A

Q078 D

Q079 D

Q080 A

Q081 $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

Q082 B

Q083 B

Q084 A

Q085 A

Q086 $\frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$

Q087 A

Q088 $\frac{a^2+x^2}{a^2-x^2} = \frac{b^2+y^2}{b^2-y^2}$

Q089 $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

Q090 $\frac{x^2+y^2+z^2}{a^2+b^2+c^2} = \frac{xy+yz+zx}{ab+bc+ca}$

Q091 $\frac{x+y+z}{a+b+c} = \frac{x}{a}$

Q092 $\frac{x+2y+3z}{a+2b+3c} = \frac{x+y+z}{a+b+c}$

Q093 $\frac{57}{38}$

Q094 $\frac{26}{29}$

Q095 $-\frac{5}{2}$

Q096 $-\frac{29}{10}$

Q097 $\frac{94}{79}$

Q098 $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$

Quiz MCQ 1 A**Quiz MCQ 2** A**Quiz MCQ 3** A**Quiz MCQ 4** B**Quiz WRITTEN 5** $a = 1, b = -1, c = 3$ **Quiz WRITTEN 6** $x^4 - x^2 + 16$ **Quiz WRITTEN 7** $4b^2 + 4bc$ **Quiz WRITTEN 8** $(ax + by + cz)^2$

Concept 5 | Inequalities

English Student Edition • Focused formulas and guided practice

Formula opener

Square rule

$$(x - y)^2 \geq 0$$

Order

$$0 < a < b \Rightarrow a^2 < b^2$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab}$$

Worked Solutions

Inequalities

Q001 Written

Under the condition $a > b$, $x > y$, prove that $2(ax + by) > (a + b)(x + y)$.

Solution

1. Start with LHS-RHS: $2(ax+by)-(a+b)(x+y)$.
2. Expand and factor: $ax-ay-bx+by=(a-b)(x-y)$.
3. Both factors are positive, so $(a-b)(x-y)>0$; hence the required inequality holds.

Final answer

$$2(ax + by) > (a + b)(x + y)$$

Q002 MCQ

If $x > y$, prove that $3x + y > 2(x + y)$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + y > 2(x + y)$.

Correct option: A

Final answer

$$x - y$$

Worked Solutions

Inequalities

Q003 MCQ

If $x > y$, prove that $3x + 2y > 2x + 3y$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + 2y > 2x + 3y$.

Correct option: C

Final answer

$$x - y$$

Q004 MCQ

If $x > y$, prove that $3(x + y) > 2(x + 2y)$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3(x + y) > 2(x + 2y)$.

Correct option: D

Final answer

$$x - y$$

Worked Solutions

Inequalities

Q005 MCQ

If $x > y$, prove that $3x + 4y > 2x + 5y$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + 4y > 2x + 5y$.

Correct option: D

Final answer

$$x - y$$

Q006 MCQ

If $x > y$, prove that $3x + 5y > 2(x + 3y)$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + 5y > 2(x + 3y)$.

Correct option: B

Final answer

$$x - y$$

Worked Solutions

Inequalities

Q007 Written

Under the condition $a > b$, $c > d$, prove that $ac + bd > ad + bc$.

Solution

1. Subtract the right-hand side: $ac + bd - ad - bc$.
2. Factor: $a(c - d) - b(c - d) = (a - b)(c - d)$.
3. Both factors are positive, so the difference is positive; therefore $ac + bd > ad + bc$.

Final answer

$$ac + bd > ad + bc$$

Q008 Written

Under the condition $a > c$, $b > d$, prove that $ab + cd > ad + bc$.

Solution

1. Subtract the right-hand side: $ab + cd - ad - bc$.
2. Factor: $a(b - d) - c(b - d) = (a - c)(b - d)$.
3. Both factors are positive, so the difference is positive; hence $ab + cd > ad + bc$.

Final answer

$$ab + cd > ad + bc$$

Worked Solutions

Inequalities

Q009 MCQ

If $a > b$ and $c > d$, prove that $ac + bd > ad + bc$.

If $a > b$ and $c > d$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $(a - b)(c - d)$.
2. Factor the difference as $(a - b)(c - d)$.
3. The condition $a > b$ and $c > d$ makes each factor positive.
4. Hence $ac + bd > ad + bc$.

Correct option: D

Final answer

$$(a - b)(c - d)$$

Q010 MCQ

If $a > 2b$ and $c > d$, prove that $ac + 2bd > ad + 2bc$.

If $a > 2b$ and $c > d$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $(a - 2b)(c - d)$.
2. Factor the difference as $(a - 2b)(c - d)$.
3. The condition $a > 2b$ and $c > d$ makes each factor positive.
4. Hence $ac + 2bd > ad + 2bc$.

Correct option: C

Final answer

$$(a - 2b)(c - d)$$

Worked Solutions

Inequalities

Q011 MCQ

If $a > 3b$ and $c > d$, prove that $ac + 3bd > ad + 3bc$.

If $a > 3b$ and $c > d$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $(a - 3b)(c - d)$.
2. Factor the difference as $(a - 3b)(c - d)$.
3. The condition $a > 3b$ and $c > d$ makes each factor positive.
4. Hence $ac + 3bd > ad + 3bc$.

Correct option: A

Final answer

$$(a - 3b)(c - d)$$

Q012 Written

Under the condition $x > y$, prove that $3x - 4y > 2x - 3y$.

Solution

1. Subtract the right-hand side: $(3x - 4y) - (2x - 3y)$.
2. Simplify to $x - y$.
3. Since $x > y$, $x - y > 0$; therefore $3x - 4y > 2x - 3y$.

Final answer

$$3x - 4y > 2x - 3y$$

Worked Solutions

Inequalities

Q013 Written

Under the condition $x > 3$, $y > -5$, prove that $xy + 5x > 3y + 15$.

Solution

1. Start with LHS-RHS: $xy+5x-3y-15$.
2. Factor by grouping: $x(y+5) - 3(y+5) = (x-3)(y+5)$.
3. Since $x-3>0$ and $y+5>0$, the difference is positive; hence the inequality holds.

Final answer

$$xy + 5x > 3y + 15$$

Q014 Written

Under the condition $x > 2$, $y > 1$, prove that $xy + 2 > x + 2y$.

Solution

1. Start with LHS-RHS: $xy+2-x-2y$.
2. Factor: $x(y-1) - 2(y-1) = (x-2)(y-1)$.
3. Both factors are positive, so the difference is positive; hence $xy+2>x+2y$.

Final answer

$$xy + 2 > x + 2y$$

Worked Solutions

Inequalities

Q015 Written

Under the condition $a > 1$, $b > 1$, prove that $ab + 1 > a + b$.

Solution

1. Start with LHS-RHS: $ab+1-a-b$.
2. Factor: $ab-a-b+1=(a-1)(b-1)$.
3. Since $a-1>0$ and $b-1>0$, the difference is positive; hence $ab+1>a+b$.

Final answer

$$ab + 1 > a + b$$

Q016 Written

Under the condition $a > b > x > y$, prove that $ax + by > ay + bx$.

Solution

1. Subtract the right-hand side: $ax+by-ay-bx$.
2. Factor: $a(x - y) - b(x - y) = (a - b)(x - y)$.
3. Both factors are positive because $a>b$ and $x>y$; therefore the difference is positive.

Final answer

$$ax + by > ay + bx$$

Worked Solutions

Inequalities

Q017 Written

Under the condition $a > b > c > 0$, $x > y > z > 0$, prove that $ax + by + cz > ay + bz + cx$.

Solution

1. Subtract the right-hand side and regroup: $ax+by+cz-ay-bz-cx$.
2. Rewrite it as $(a-b)(x-y)+(b-c)(x-z)$.
3. Every factor in both products is positive, so the sum is positive; hence $ax+by+cz>ay+bz+cx$.

Final answer

$$ax + by + cz > ay + bz + cx$$

Worked Solutions

Inequalities

Q018 Challenge

If $a > b$ and $x > y$, prove that $2(ax + by) > 2(ay + bx)$.

If $a > b$ and $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $2(a - b)(x - y)$.
2. Factor the difference as $2(a - b)(x - y)$.
3. The condition $a > b$ and $x > y$ makes each factor positive.
4. Hence $2(ax + by) > 2(ay + bx)$.

Correct option: D

Final answer

$$2(a - b)(x - y)$$

Worked Solutions

Inequalities

Q019 Written

Prove that $a^2 + 4b^2 \geq 4ab$.

Solution

1. Start with LHS - RHS: $a^2 + 4b^2 - 4ab$.
2. Complete the square: $a^2 + 4b^2 - 4ab = (a - 2b)^2 \geq 0$.
3. Therefore $a^2 + 4b^2 \geq 4ab$.
4. Equality holds when $a = 2b$.

Final answer

$$a^2 + 4b^2 \geq 4ab$$

Q020 MCQ

Under a,b real, prove that $a^2 + 4b^2 \geq 4ab$ **select the correct factorisation of LHS - RHS:**

Solution

1. Start with LHS - RHS: $a^2 + 4b^2 - 4ab$.
2. Factor/complete the square to obtain $(a - 2b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 4b^2 \geq 4ab$.

Correct option: D

Final answer

$$(a - 2b)^2$$

Worked Solutions

Inequalities

Q021 MCQ

Under a, b real, prove that $a^2 + 9b^2 \geq 6ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 9b^2 - 6ab$.
2. Factor/complete the square to obtain $(a - 3b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 9b^2 \geq 6ab$.

Correct option: C**Final answer**

$$(a - 3b)^2$$

Q022 MCQ

Under a, b real, prove that $a^2 + 16b^2 \geq 8ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 16b^2 - 8ab$.
2. Factor/complete the square to obtain $(a - 4b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 16b^2 \geq 8ab$.

Correct option: B**Final answer**

$$(a - 4b)^2$$

Worked Solutions

Inequalities

Q023 MCQ

Under a, b real, prove that $a^2 + 25b^2 \geq 10ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 25b^2 - 10ab$.
2. Factor/complete the square to obtain $(a - 5b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 25b^2 \geq 10ab$.

Correct option: A**Final answer**

$$(a - 5b)^2$$

Q024 MCQ

Under a, b real, prove that $a^2 + 36b^2 \geq 12ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 36b^2 - 12ab$.
2. Factor/complete the square to obtain $(a - 6b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 36b^2 \geq 12ab$.

Correct option: C**Final answer**

$$(a - 6b)^2$$

Worked Solutions

Inequalities

Q025 MCQ

Under a, b real, prove that $a^2 + 49b^2 \geq 14ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 49b^2 - 14ab$.
2. Factor/complete the square to obtain $(a - 7b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 49b^2 \geq 14ab$.

Correct option: B**Final answer**

$$(a - 7b)^2$$

Q026 Written

Prove that $a^2 + 5b^2 \geq 4ab$.

Solution

1. Start with LHS - RHS: $a^2 + 5b^2 - 4ab$.
2. Complete the square: $a^2 + 5b^2 - 4ab = (a - 2b)^2 + b^2 \geq 0$.
3. Therefore $a^2 + 5b^2 \geq 4ab$.
4. Equality holds when $a = b = 0$.

Final answer

$$a^2 + 5b^2 \geq 4ab$$

Worked Solutions

Inequalities

Q027 Written

Prove that $9x^2 \geq y(6x - y)$.**Solution**

1. Subtract the right-hand side: $9x^2 - y(6x - y)$.
2. Complete the square: $9x^2 - y(6x - y) = (3x - y)^2 \geq 0$.
3. Therefore $9x^2 \geq y(6x - y)$.
4. Equality holds when $y = 3x$.

Final answer

$$9x^2 \geq y(6x - y)$$

Q028 Written

Prove that $2(x^2 + y^2) \geq (x + y)^2$.**Solution**

1. Subtract the right-hand side: $2(x^2 + y^2) - (x + y)^2$.
2. This is $(x - y)^2 \geq 0$.
3. Therefore the inequality holds.
4. Equality holds when $x = y$.

Final answer

$$2(x^2 + y^2) \geq (x + y)^2$$

Worked Solutions

Inequalities

Q029 Written

Prove that $(2x - y)^2 \geq 3(x^2 - y^2)$.**Solution**

1. Subtract the right-hand side: $(2x - y)^2 - 3(x^2 - y^2)$.
2. Simplify to $(x - 2y)^2 \geq 0$.
3. Therefore the inequality holds.
4. Equality holds when $x = 2y$.

Final answer

$$(2x - y)^2 \geq 3(x^2 - y^2)$$

Q030 Written

Prove that $(2a + 3b)^2 \geq 12ab$.**Solution**

1. Subtract the right-hand side: $(2a + 3b)^2 - 12ab$.
2. The cross terms cancel, giving $4a^2 + 9b^2 \geq 0$.
3. Therefore the inequality holds.
4. Equality holds when $a = b = 0$.

Final answer

$$(2a + 3b)^2 \geq 12ab$$

Worked Solutions

Inequalities

Q031 MCQ

Under x, y real, prove that $4x^2 \geq y(4x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $4x^2 - y(4x - y)$.
2. Factor/complete the square to obtain $(2x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $4x^2 \geq y(4x - y)$.

Correct option: A

Final answer

$$(2x - y)^2$$

Q032 MCQ

Under x, y real, prove that $9x^2 \geq y(6x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $9x^2 - y(6x - y)$.
2. Factor/complete the square to obtain $(3x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $9x^2 \geq y(6x - y)$.

Correct option: C

Final answer

$$(3x - y)^2$$

Worked Solutions

Inequalities

Q033 MCQ

Under x, y real, prove that $16x^2 \geq y(8x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $16x^2 - y(8x - y)$.
2. Factor/complete the square to obtain $(4x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $16x^2 \geq y(8x - y)$.

Correct option: D**Final answer**

$$(4x - y)^2$$

Q034 MCQ

Under x, y real, prove that $25x^2 \geq y(10x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $25x^2 - y(10x - y)$.
2. Factor/complete the square to obtain $(5x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $25x^2 \geq y(10x - y)$.

Correct option: D**Final answer**

$$(5x - y)^2$$

Worked Solutions

Inequalities

Q035 MCQ

Under x, y real, prove that $36x^2 \geq y(12x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $36x^2 - y(12x - y)$.
2. Factor/complete the square to obtain $(6x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $36x^2 \geq y(12x - y)$.

Correct option: A**Final answer**

$$(6x - y)^2$$

Q036 MCQ

Under x, y real, prove that $x^2 + y^2 \geq 2xy$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $x^2 + y^2 - 2xy$.
2. Factor/complete the square to obtain $(x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $x^2 + y^2 \geq 2xy$.

Correct option: C**Final answer**

$$(x - y)^2$$

Worked Solutions

Inequalities

Q037 Written

Prove that $x^2 + y^2 \geq 2(x - y - 1)$.**Solution**

1. Start with LHS - RHS: $x^2 + y^2 - 2(x - y - 1)$.
2. Complete the squares: $x^2 + y^2 - 2(x - y - 1) = (x - 1)^2 + (y + 1)^2 \geq 0$.
3. Therefore $x^2 + y^2 \geq 2(x - y - 1)$.
4. Equality holds when $x = 1$, $y = -1$.

Final answer

$$x^2 + y^2 \geq 2(x - y - 1)$$

Q038 Written

Prove that $x^2 + 2 > 2x$.**Solution**

1. Subtract the right-hand side: $x^2 + 2 - 2x$.
2. Complete the square: $x^2 + 2 - 2x = (x - 1)^2 + 1 > 0$.
3. Therefore $x^2 + 2 > 2x$.
4. There is no equality case.

Final answer

$$x^2 + 2 > 2x$$

Worked Solutions

Inequalities

Q039 Written

Prove that $x^2 + y^2 - 6x + 8y + 25 \geq 0$.**Solution**

1. Complete the squares: $x^2 + y^2 - 6x + 8y + 25 = (x - 3)^2 + (y + 4)^2 \geq 0$.
2. Therefore the inequality holds.
3. Equality holds when $x = 3$, $y = -4$.

Final answer

$$x^2 + y^2 - 6x + 8y + 25 \geq 0$$

Q040 Written

Prove that $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.**Solution**

1. Complete the squares: $x^2 + 2y^2 - 10x + 12y + 43 = (x - 5)^2 + 2(y + 3)^2 \geq 0$.
2. Therefore the expression is never negative.
3. Equality holds when $x = 5$, $y = -3$.

Final answer

$$x^2 + 2y^2 - 10x + 12y + 43 \geq 0$$

Worked Solutions

Inequalities

Q041 Written

Prove that $(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$.**Solution**

1. Subtract the right-hand side: $(a^2 + b^2)(x^2 + y^2) - (ax + by)^2$.
2. Use $(a^2 + b^2)(x^2 + y^2) - (ax + by)^2 = (ay - bx)^2 \geq 0$.
3. Therefore the required inequality holds.
4. Equality holds when $ay = bx$.

Final answer

$$(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$$

Worked Solutions

Inequalities

Q042 Challenge

For real x, y , prove that $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.

Solution

1. Complete both squares: $x^2 + 2y^2 - 10x + 12y + 43 = (x - 5)^2 + 2(y + 3)^2 \geq 0$.
2. Therefore $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.
3. Equality holds when $x = 5$, $y = -3$.

Final answer

0

Worked Solutions

Inequalities

Q043 Written

Prove that $\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}$.

Solution

1. Both sides are positive. Compare their squares.

$$2. \left(\sqrt{\frac{a^2+b^2}{2}} \right)^2 - \left(\frac{a+b}{2} \right)^2 = \frac{(a-b)^2}{4} \geq 0.$$

3. Hence the required inequality holds.

4. Equality holds when $a = b$.

Final answer

$$\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}$$

Worked Solutions

Inequalities

Q044 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{b^2}{2}} \geq \frac{a}{2} + \frac{b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + \frac{b^2}{2}} \geq \frac{a}{2} + \frac{b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-b)^2}{4}$$

Worked Solutions

Inequalities

Q045 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 2b^2} \geq \frac{a}{2} + b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

- Both sides are positive under the stated condition; compare their squares.
- The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-2b)^2}{4}$.
- This expression is non-negative under the stated conditions.
- Therefore $\sqrt{\frac{a^2}{2} + 2b^2} \geq \frac{a}{2} + b$.
- Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-2b)^2}{4}$$

Worked Solutions

Inequalities

Q046 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{9b^2}{2}} \geq \frac{a}{2} + \frac{3b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-3b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + \frac{9b^2}{2}} \geq \frac{a}{2} + \frac{3b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: C

Final answer

$$\frac{(a-3b)^2}{4}$$

Worked Solutions

Inequalities

Q047 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 8b^2} \geq \frac{a}{2} + 2b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

- Both sides are positive under the stated condition; compare their squares.
- The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-4b)^2}{4}$.
- This expression is non-negative under the stated conditions.
- Therefore $\sqrt{\frac{a^2}{2} + 8b^2} \geq \frac{a}{2} + 2b$.
- Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-4b)^2}{4}$$

Worked Solutions

Inequalities

Q048 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{25b^2}{2}} \geq \frac{a}{2} + \frac{5b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-5b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + \frac{25b^2}{2}} \geq \frac{a}{2} + \frac{5b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-5b)^2}{4}$$

Worked Solutions

Inequalities

Q049 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 18b^2} \geq \frac{a}{2} + 3b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-6b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + 18b^2} \geq \frac{a}{2} + 3b$.
5. Equality occurs when the displayed square is zero.

Correct option: B

Final answer

$$\frac{(a-6b)^2}{4}$$

Worked Solutions

Inequalities

Q050 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-7b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-7b)^2}{4}$$

Worked Solutions

Inequalities

Q051 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 32b^2} \geq \frac{a}{2} + 4b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

- Both sides are positive under the stated condition; compare their squares.
- The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-8b)^2}{4}$.
- This expression is non-negative under the stated conditions.
- Therefore $\sqrt{\frac{a^2}{2} + 32b^2} \geq \frac{a}{2} + 4b$.
- Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-8b)^2}{4}$$

Worked Solutions

Inequalities

Q052 Written

Prove that $\sqrt{2(a+b)} \geq \sqrt{a} + \sqrt{b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(\sqrt{2(a+b)})^2 - (\sqrt{a} + \sqrt{b})^2 = (\sqrt{a} - \sqrt{b})^2 \geq 0$.
3. Hence the required inequality holds.
4. Equality holds when $a = b$.

Final answer

$$\sqrt{2(a+b)} \geq \sqrt{a} + \sqrt{b}$$

Q053 Written

Prove that $a + 2 \geq \sqrt{3 + 6a}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(a + 2)^2 - (\sqrt{3 + 6a})^2 = (a - 1)^2 \geq 0$.
3. Hence the required inequality holds.
4. Equality holds when $a = 1$.

Final answer

$$a + 2 \geq \sqrt{3 + 6a}$$

Worked Solutions

Inequalities

Q054 Written

Prove that $\sqrt{2(a^2 + b^2)} \geq a + b$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(\sqrt{2(a^2 + b^2)})^2 - (a + b)^2 = (a - b)^2 \geq 0$.
3. Hence the required inequality holds.
4. Equality holds when $a = b$.

Final answer

$$\sqrt{2(a^2 + b^2)} \geq a + b$$

Q055 Written

Prove that $\sqrt{\frac{a+b}{2}} \geq \frac{\sqrt{a} + \sqrt{b}}{2}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $\left(\sqrt{\frac{a+b}{2}}\right)^2 - \left(\frac{\sqrt{a} + \sqrt{b}}{2}\right)^2 = \frac{(\sqrt{a} - \sqrt{b})^2}{4} \geq 0$.
3. Hence the required inequality holds.
4. Equality holds when $a = b$.

Final answer

$$\sqrt{\frac{a+b}{2}} \geq \frac{\sqrt{a} + \sqrt{b}}{2}$$

Worked Solutions

Inequalities

Q056 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + \sqrt{b} > \sqrt{a + b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 2\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + \sqrt{b} > \sqrt{a + b}$.

Correct option: B

Final answer

$$2\sqrt{a}\sqrt{b}$$

Q057 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 2\sqrt{b} > \sqrt{a + 4b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 4\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 2\sqrt{b} > \sqrt{a + 4b}$.

Correct option: A

Final answer

$$4\sqrt{a}\sqrt{b}$$

Worked Solutions

Inequalities

Q058 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 3\sqrt{b} > \sqrt{a + 9b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 6\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 3\sqrt{b} > \sqrt{a + 9b}$.

Correct option: C

Final answer

$$6\sqrt{a}\sqrt{b}$$

Q059 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 8\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Correct option: B

Final answer

$$8\sqrt{a}\sqrt{b}$$

Worked Solutions

Inequalities

Q060 Written

Prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(\sqrt{a} + 4\sqrt{b})^2 - (\sqrt{a + 16b})^2 = 8\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$$

Q061 Written

Prove that $2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(2\sqrt{a} + \sqrt{b})^2 - (\sqrt{4a + b})^2 = 4\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$$

Worked Solutions

Inequalities

Q062 Written

Prove that $\sqrt{a} + \sqrt{b} > \sqrt{a+b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(\sqrt{a} + \sqrt{b})^2 - (\sqrt{a+b})^2 = 2\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$\sqrt{a} + \sqrt{b} > \sqrt{a+b}$$

Q063 Written

Prove that $2\sqrt{a} + 3\sqrt{b} > \sqrt{4a+9b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(2\sqrt{a} + 3\sqrt{b})^2 - (\sqrt{4a+9b})^2 = 12\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$2\sqrt{a} + 3\sqrt{b} > \sqrt{4a+9b}$$

Worked Solutions

Inequalities

Q064 **Written****Prove that** $\sqrt{a-b} > \sqrt{a} - \sqrt{b}$.**Solution**

1. Both sides are positive because $a > b > 0$. Compare their squares.
2. $(\sqrt{a-b})^2 - (\sqrt{a} - \sqrt{b})^2 = 2\sqrt{b}(\sqrt{a} - \sqrt{b}) > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$\sqrt{a-b} > \sqrt{a} - \sqrt{b}$$

Q065 **Written****If** $a > b > 0$, **prove that** $\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $\left(\sqrt{a} - \frac{b}{2\sqrt{a}}\right)^2 - (\sqrt{a-b})^2 = \frac{b^2}{4a} > 0$.
3. Therefore $\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$.
4. There is no equality case.

Final answer

$$\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$$

Worked Solutions

Inequalities

Q066 Challenge

For $a, b > 0$, prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Solution

1. Both sides are positive under $a, b > 0$; compare their squares.
2. $(\sqrt{a} + 4\sqrt{b})^2 - (\sqrt{a + 16b})^2 = 8\sqrt{ab} > 0$.
3. Therefore $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Final answer

$$\sqrt{a + 16b}$$

Worked Solutions

Inequalities

Q067 Written

Prove that $\frac{b}{a} + \frac{a}{b} \geq 2$.**Solution**

1. By AM–GM, $\frac{b}{a} + \frac{a}{b} \geq 2\sqrt{\frac{b}{a} \cdot \frac{a}{b}} = 2$.
2. Equality holds when $a = b$.

Final answer

$$\frac{b}{a} + \frac{a}{b} \geq 2$$

Q068 Written

Prove that $\frac{b}{3a} + \frac{12a}{b} \geq 4$.**Solution**

1. Apply AM–GM to the two positive terms: $\frac{b}{3a} + \frac{12a}{b} \geq 2\sqrt{4} = 4$.
2. Equality holds when $b = 6a$.

Final answer

$$\frac{b}{3a} + \frac{12a}{b} \geq 4$$

Worked Solutions

Inequalities

Q069 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{a^2+b^2}{ab} \geq 2$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{a^2+b^2}{ab} - 2$.
2. Factor/complete the square to obtain $\frac{(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{a^2+b^2}{ab} \geq 2$.

Correct option: A**Final answer**

$$\frac{(a-b)^2}{ab}$$

Q070 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{4a^2 + b^2}{2ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{4a^2 + b^2}{2ab} - 2$.
2. Factor/complete the square to obtain $\frac{(2a - b)^2}{2ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{4a^2 + b^2}{2ab} \geq 2$.

Correct option: C

Final answer

$$\frac{(2a - b)^2}{2ab}$$

Worked Solutions

Inequalities

Q071 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{9a^2 + b^2}{3ab} \geq 2$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{9a^2 + b^2}{3ab} - 2$.
2. Factor/complete the square to obtain $\frac{(3a - b)^2}{3ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{9a^2 + b^2}{3ab} \geq 2$.

Correct option: B**Final answer**

$$\frac{(3a - b)^2}{3ab}$$

Q072 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{16a^2 + b^2}{4ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{16a^2 + b^2}{4ab} - 2$.
2. Factor/complete the square to obtain $\frac{(4a - b)^2}{4ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{16a^2 + b^2}{4ab} \geq 2$.

Correct option: A

Final answer

$$\frac{(4a - b)^2}{4ab}$$

Worked Solutions

Inequalities

Q073 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{25a^2 + b^2}{5ab} \geq 2$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{25a^2 + b^2}{5ab} - 2$.
2. Factor/complete the square to obtain $\frac{(5a - b)^2}{5ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{25a^2 + b^2}{5ab} \geq 2$.

Correct option: D**Final answer**

$$\frac{(5a - b)^2}{5ab}$$

Q074 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{36a^2 + b^2}{6ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{36a^2 + b^2}{6ab} - 2$.
2. Factor/complete the square to obtain $\frac{(6a - b)^2}{6ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{36a^2 + b^2}{6ab} \geq 2$.

Correct option: D

Final answer

$$\frac{(6a - b)^2}{6ab}$$

Worked Solutions

Inequalities

Q075 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{49a^2 + b^2}{7ab} \geq 2$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{49a^2 + b^2}{7ab} - 2$.
2. Factor/complete the square to obtain $\frac{(7a - b)^2}{7ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{49a^2 + b^2}{7ab} \geq 2$.

Correct option: C**Final answer**

$$\frac{(7a - b)^2}{7ab}$$

Q076 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{64a^2 + b^2}{8ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{64a^2 + b^2}{8ab} - 2$.
2. Factor/complete the square to obtain $\frac{(8a - b)^2}{8ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{64a^2 + b^2}{8ab} \geq 2$.

Correct option: D

Final answer

$$\frac{(8a - b)^2}{8ab}$$

Worked Solutions

Inequalities

Q077 Written

Prove that $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$.**Solution**

1. Expand: $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) = \frac{a}{b} + \frac{b}{a} + 2$.
2. Use $\frac{a}{b} + \frac{b}{a} \geq 2$ to obtain the lower bound 4.
3. Equality holds when $a = b$.

Final answer

$$(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$$

Q078 Written

Prove that $\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$.**Solution**

1. Expand to $x^2 + 5 + \frac{4}{x^2}$.
2. Since $x^2 + \frac{4}{x^2} \geq 4$, the product is at least 9.
3. Equality holds when $x = \sqrt{2}$.

Final answer

$$\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$$

Worked Solutions

Inequalities

Q079 Written

Prove that $(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$.**Solution**

1. Expand to $10 + \frac{a}{b} + \frac{16b}{a}$.
2. AM–GM gives $\frac{a}{b} + \frac{16b}{a} \geq 8$, so the product is at least 18.
3. Equality holds when $a = 4b$.

Final answer

$$(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$$

Q080 Written

Prove that $\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$.**Solution**

1. Expand to $ab + 6 + \frac{9}{ab}$.
2. AM–GM gives $ab + \frac{9}{ab} \geq 6$, so the product is at least 12.
3. Equality holds when $ab = 3$.

Final answer

$$\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$$

Worked Solutions

Inequalities

Q081 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+b)^2}{ab} \geq 4$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+b)^2}{ab} - 4$.
2. Factor/complete the square to obtain $\frac{(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+b)^2}{ab} \geq 4$.

Correct option: D**Final answer**

$$\frac{(a-b)^2}{ab}$$

Q082 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+4b)(4a+b)}{ab} \geq 25$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+4b)(4a+b)}{ab} - 25$.
2. Factor/complete the square to obtain $\frac{4(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+4b)(4a+b)}{ab} \geq 25$.

Correct option: B

Final answer

$$\frac{4(a-b)^2}{ab}$$

Worked Solutions

Inequalities

Q083 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+9b)(9a+b)}{ab} \geq 100$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+9b)(9a+b)}{ab} - 100$.
2. Factor/complete the square to obtain $\frac{9(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+9b)(9a+b)}{ab} \geq 100$.

Correct option: C**Final answer**

$$\frac{9(a-b)^2}{ab}$$

Q084 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+16b)(16a+b)}{ab} \geq 289$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+16b)(16a+b)}{ab} - 289$.
2. Factor/complete the square to obtain $\frac{16(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+16b)(16a+b)}{ab} \geq 289$.

Correct option: D

Final answer

$$\frac{16(a-b)^2}{ab}$$

Worked Solutions

Inequalities

Q085 Written

Prove that $\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$.**Solution**

1. Expand to $ab + 10 + \frac{9}{ab}$.
2. AM–GM gives $ab + \frac{9}{ab} \geq 6$, so the product is at least 16.
3. Equality holds when $ab = 3$.

Final answer

$$\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$$

Q086 Written

Prove that $a + \frac{4}{a} \geq 4$.**Solution**

1. Apply AM–GM: $a + \frac{4}{a} \geq 2\sqrt{4} = 4$.
2. Equality holds when $a = 2$.

Final answer

$$a + \frac{4}{a} \geq 4$$

Worked Solutions

Inequalities

Q087 Written

Prove that $3ab + \frac{4}{ab} \geq 4\sqrt{3}$.

Solution

1. Apply AM–GM to $3ab$ and $\frac{4}{ab}$: the sum is at least $2\sqrt{12} = 4\sqrt{3}$.
2. Equality holds when $ab = \frac{2}{\sqrt{3}}$.

Final answer

$$3ab + \frac{4}{ab} \geq 4\sqrt{3}$$

Q088 Written

Prove that $a + b + \frac{9}{a+b} \geq 6$.

Solution

1. Let $t = a + b > 0$. AM–GM gives $t + \frac{9}{t} \geq 6$.
2. Equality holds when $a + b = 3$.

Final answer

$$a + b + \frac{9}{a+b} \geq 6$$

Worked Solutions

Inequalities

Q089 Written

A rectangle has perimeter 200 cm. Prove that its area $LW \leq 2500 \text{ cm}^2$.

Solution

1. From the perimeter, $L + W = 100$.
2. AM–GM gives $\frac{L+W}{2} \geq \sqrt{LW}$, so $50 \geq \sqrt{LW}$.
3. Therefore $LW \leq 2500$.
4. Equality holds when $L = W = 50$.

Final answer

$$LW \leq 2500$$

Worked Solutions

Inequalities

Q090 Challenge

For $a, b > 0$, prove that $\frac{b}{a} + \frac{a}{b} \geq 2$.

Solution

1. Put the terms over a positive denominator: $\frac{b}{a} + \frac{a}{b} - 2 = \frac{(a-b)^2}{ab} \geq 0$.
2. Hence $\frac{b}{a} + \frac{a}{b} \geq 2$.
3. Equality holds when $a = b$.

Final answer

2

Worked Solutions

Inequalities

Quiz MCQ 1 Concept Quiz · MCQ

If $x > y$, prove that $3x + y > 2(x + y)$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + y > 2(x + y)$.

Correct option: A

Final answer

$$x - y$$

Quiz MCQ 2 Concept Quiz · MCQ

Under a, b real, prove that $a^2 + 25b^2 \geq 10ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 25b^2 - 10ab$.
2. Factor/complete the square to obtain $(a - 5b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 25b^2 \geq 10ab$.

Correct option: A

Final answer

$$(a - 5b)^2$$

Worked Solutions

Inequalities

Quiz MCQ 3 Concept Quiz · MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

- Both sides are positive under the stated condition; compare their squares.
- The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-7b)^2}{4}$.
- This expression is non-negative under the stated conditions.
- Therefore $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.
- Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-7b)^2}{4}$$

Worked Solutions

Inequalities

Quiz MCQ 4 Concept Quiz · MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+4b)(4a+b)}{ab} \geq 25$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+4b)(4a+b)}{ab} - 25$.
2. Factor/complete the square to obtain $\frac{4(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+4b)(4a+b)}{ab} \geq 25$.

Correct option: B

Final answer

$$\frac{4(a-b)^2}{ab}$$

Quiz WRITTEN 5 Concept Quiz · Written

If $x > y$, prove that $3x + 4y > 2x + 5y$.

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + 4y > 2x + 5y$.

Final answer

$$2x + 5y$$

Worked Solutions

Inequalities

Quiz WRITTEN 6 Concept Quiz · Written

Under x, y real, prove that $x^2 + y^2 \geq 2xy$.

Solution

1. Start with LHS - RHS: $x^2 + y^2 - 2xy$.
2. Factor/complete the square to obtain $(x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $x^2 + y^2 \geq 2xy$.

Final answer

$$2xy$$

Quiz WRITTEN 7 Concept Quiz · Written

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 6\sqrt{b} > \sqrt{a + 36b}$.

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 12\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 6\sqrt{b} > \sqrt{a + 36b}$.

Final answer

$$\sqrt{a + 36b}$$

Worked Solutions

Inequalities

Quiz WRITTEN 8 Concept Quiz · Written

Under $t > 0$, prove that $\frac{t^2+1}{t} \geq 2$.

Solution

1. Start with LHS - RHS: $\frac{t^2+1}{t} - 2$.
2. Factor/complete the square to obtain $\frac{(t-1)^2}{t}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{t^2+1}{t} \geq 2$.

Final answer

2

Answer key

Use the question number shown on each page.

Q001 $2(ax + by) > (a + b)(x + y)$

Q002 A

Q003 C

Q004 D

Q005 D

Q006 B

Q007 $ac + bd > ad + bc$

Q008 $ab + cd > ad + bc$

Q009 D

Q010 C

Q011 A

Q012 $3x - 4y > 2x - 3y$

Q013 $xy + 5x > 3y + 15$

Q014 $xy + 2 > x + 2y$

Q015 $ab + 1 > a + b$

Q016 $ax + by > ay + bx$

Q017 $ax + by + cz > ay + bz + cx$

Q018 D

Q019 $a^2 + 4b^2 \geq 4ab$

Q020 D

Q021 C

Q022 B

Q023 A

Q024 C

Q025 B

Q026 $a^2 + 5b^2 \geq 4ab$

Q027 $9x^2 \geq y(6x - y)$

Q028 $2(x^2 + y^2) \geq (x + y)^2$

Q029 $(2x - y)^2 \geq 3(x^2 - y^2)$

Q030 $(2a + 3b)^2 \geq 12ab$

Q031 A

Q032 C

Q033 D

Q034 D

Q035 A

Q036 C

Q037 $x^2 + y^2 \geq 2(x - y - 1)$

Q038 $x^2 + 2 > 2x$

Q039 $x^2 + y^2 - 6x + 8y + 25 \geq 0$

Q040 $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$

Q041 $(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$

Q042 0

Q043 $\sqrt{\frac{a^2 + b^2}{2}} \geq \frac{a + b}{2}$

Q044 A

Q045 A

Q046 C

Q047 A

Q048 A

Q049 B

Q050 A

Q051 A

Q052 $\sqrt{2(a + b)} \geq \sqrt{a} + \sqrt{b}$

Q053 $a + 2 \geq \sqrt{3 + 6a}$

Q054 $\sqrt{2(a^2 + b^2)} \geq a + b$

Q055 $\sqrt{\frac{a + b}{2}} \geq \frac{\sqrt{a} + \sqrt{b}}{2}$

Q056 B

Q057 A

Q058 C

Q059 B

Q060 $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$

Q061 $2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$

Q062 $\sqrt{a} + \sqrt{b} > \sqrt{a + b}$

Q063 $2\sqrt{a} + 3\sqrt{b} > \sqrt{4a + 9b}$

Q064 $\sqrt{a - b} > \sqrt{a} - \sqrt{b}$

Q065 $\sqrt{a - b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$

Q066 $\sqrt{a + 16b}$

Q067 $\frac{b}{a} + \frac{a}{b} \geq 2$

Q068 $\frac{b}{3a} + \frac{12a}{b} \geq 4$

Q069 A

Q070 C

Q071 B

Q072 A

Q073 D

Q074 D

Q075 C

Q076 D

Q077 $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$

Q078 $\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$

Q079 $(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$

Q080 $\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$

Q081 D

Q082 B

Q083 C

Q084 D

Q085 $\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$

Q086 $a + \frac{4}{a} \geq 4$

Q087 $3ab + \frac{4}{ab} \geq 4\sqrt{3}$

Q088 $a + b + \frac{9}{a + b} \geq 6$

Q089 $LW \leq 2500$

Q090 2

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $2x + 5y$

Quiz WRITTEN 6 $2xy$

Quiz WRITTEN 7 $\sqrt{a + 36b}$

Quiz WRITTEN 8 2

Concept 1 | Arithmetic Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Test

$$a_{n+1} - a_n = d$$

Worked Solutions

Arithmetic Sequences

Q001 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 2n - 3$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -1$, $a_2 = 1$, $a_3 = 3$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -1, a_2 = 1, a_3 = 3$$

Q002 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 - 7n$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -6$, $a_2 = -10$, $a_3 = -12$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -6, a_2 = -10, a_3 = -12$$

Worked Solutions

Arithmetic Sequences

Q003 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 5n - 8$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -3$, $a_2 = 2$, $a_3 = 7$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -3, a_2 = 2, a_3 = 7$$

Q004 Written

When the general term of the sequence $\{a_n\}$ is $a_n = -3n + 1$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -2$, $a_2 = -5$, $a_3 = -8$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -2, a_2 = -5, a_3 = -8$$

Worked Solutions

Arithmetic Sequences

Q005 Written

When the general term of the sequence $\{a_n\}$ is $a_n = -2n - 4$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -6$, $a_2 = -8$, $a_3 = -10$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -6, a_2 = -8, a_3 = -10$$

Q006 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 - 1$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 0$, $a_2 = 3$, $a_3 = 8$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 0, a_2 = 3, a_3 = 8$$

Worked Solutions

Arithmetic Sequences

Q007 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 + n + 1$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 7$, $a_3 = 13$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 3, a_2 = 7, a_3 = 13$$

Q008 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 + 2n + 4$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 7$, $a_2 = 12$, $a_3 = 19$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 7, a_2 = 12, a_3 = 19$$

Worked Solutions

Arithmetic Sequences

Q009 MCQ

For $a_n = 3n - 5$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -2$, $a_2 = 1$, $a_3 = 4$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

$(-2, 1, 4)$

Q010 MCQ

For $a_n = 4n + 7$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 11$, $a_2 = 15$, $a_3 = 19$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$(11, 15, 19)$

Worked Solutions

Arithmetic Sequences

Q011 MCQ

For $a_n = n^2 - 4n + 6$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 2$, $a_3 = 3$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

(3, 2, 3)

Q012 MCQ

For $a_n = 2n^2 + 3$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 5$, $a_2 = 11$, $a_3 = 21$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

(5, 11, 21)

Worked Solutions

Arithmetic Sequences

Q013 MCQ

For $a_n = n^2 + 5n - 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 4$, $a_2 = 10$, $a_3 = 16$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

$(4, 10, 16)$

Q014 MCQ

For $a_n = 5 - 2n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 1$, $a_3 = -1$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$(3, 1, -1)$

Worked Solutions

Arithmetic Sequences

Q015 MCQ

For $a_n = 2n^2 - 7n + 9$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 4$, $a_2 = 3$, $a_3 = 6$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

(4, 3, 6)

Q016 MCQ

For $a_n = n^2 - 3n - 1$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -3$, $a_2 = -3$, $a_3 = -1$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

(-3, -3, -1)

Worked Solutions

Arithmetic Sequences

Q017 MCQ

For $a_n = 3n^2 - 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = 10$, $a_3 = 25$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

(1, 10, 25)

Q018 MCQ

For $a_n = n^2 + 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 6$, $a_3 = 11$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

(3, 6, 11)

Worked Solutions

Arithmetic Sequences

Q019 MCQ

For $a_n = 7n - 10$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -3$, $a_2 = 4$, $a_3 = 11$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

$(-3, 4, 11)$

Q020 MCQ

For $a_n = 2n^2 + n - 1$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 9$, $a_3 = 20$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

$(2, 9, 20)$

Worked Solutions

Arithmetic Sequences

Q021 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 2 \times 3^n$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 6$, $a_2 = 18$, $a_3 = 54$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 6, a_2 = 18, a_3 = 54$$

Q022 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 3^n - 1$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 8$, $a_3 = 26$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 2, a_2 = 8, a_3 = 26$$

Worked Solutions

Arithmetic Sequences

Q023 Written

When the general term of the sequence $\{a_n\}$ is $a_n = (-3)^n$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -3$, $a_2 = 9$, $a_3 = -27$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -3, a_2 = 9, a_3 = -27$$

Q024 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 3 \times (-2)^n$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -6$, $a_2 = 12$, $a_3 = -24$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -6, a_2 = 12, a_3 = -24$$

Worked Solutions

Arithmetic Sequences

Q025 MCQ

For $a_n = 2^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 4$, $a_3 = 8$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

$(2, 4, 8)$

Q026 MCQ

For $a_n = (-2)^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -2$, $a_2 = 4$, $a_3 = -8$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

$(-2, 4, -8)$

Worked Solutions

Arithmetic Sequences

Q027 MCQ

For $a_n = 2 \times 3^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 6$, $a_2 = 18$, $a_3 = 54$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

(6, 18, 54)

Q028 MCQ

For $a_n = 5 \times (-3)^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -15$, $a_2 = 45$, $a_3 = -135$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

(-15, 45, -135)

Worked Solutions

Arithmetic Sequences

Q029 MCQ

For $a_n = 4^n - 1$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 15$, $a_3 = 63$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

(3, 15, 63)

Q030 MCQ

For $a_n = 2 \times 3^{n-1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 6$, $a_3 = 18$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

(2, 6, 18)

Worked Solutions

Arithmetic Sequences

Q031 MCQ

For $a_n = (-1)^{n-1}5^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 5$, $a_2 = -25$, $a_3 = 125$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$(5, -25, 125)$

Q032 MCQ

For $a_n = (-3)^n + 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -1$, $a_2 = 11$, $a_3 = -25$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

$(-1, 11, -25)$

Worked Solutions

Arithmetic Sequences

Q033 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n}{n+1}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{2}$, $a_2 = \frac{2}{3}$, $a_3 = \frac{3}{4}$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = \frac{1}{2}, a_2 = \frac{2}{3}, a_3 = \frac{3}{4}$$

Q034 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n}{2n+1}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{3}$, $a_2 = \frac{2}{5}$, $a_3 = \frac{3}{7}$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = \frac{1}{3}, a_2 = \frac{2}{5}, a_3 = \frac{3}{7}$$

Worked Solutions

Arithmetic Sequences

Q035 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{1}{3n-2}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = \frac{1}{4}$, $a_3 = \frac{1}{7}$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 1, a_2 = \frac{1}{4}, a_3 = \frac{1}{7}$$

Q036 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{2n}{n+3}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{2}$, $a_2 = \frac{4}{5}$, $a_3 = 1$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = \frac{1}{2}, a_2 = \frac{4}{5}, a_3 = 1$$

Worked Solutions

Arithmetic Sequences

Q037 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n+1}{2n-1}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 1$, $a_3 = \frac{4}{5}$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 2, a_2 = 1, a_3 = \frac{4}{5}$$

Q038 MCQ

For $a_n = \frac{n+1}{n+2}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{2}{3}$, $a_2 = \frac{3}{4}$, $a_3 = \frac{4}{5}$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$$\left(\frac{2}{3}, \frac{3}{4}, \frac{4}{5}\right)$$

Worked Solutions

Arithmetic Sequences

Q039 MCQ

For $a_n = \frac{2n-1}{3n+1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{4}$, $a_2 = \frac{3}{7}$, $a_3 = \frac{1}{2}$.
3. Check each value by putting its index back into the formula.

Correct option: D**Final answer**

$$\left(\frac{1}{4}, \frac{3}{7}, \frac{1}{2}\right)$$

Q040 MCQ

For $a_n = \frac{n^2+1}{n+1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = \frac{5}{3}$, $a_3 = \frac{5}{2}$.
3. Check each value by putting its index back into the formula.

Correct option: A**Final answer**

$$\left(1, \frac{5}{3}, \frac{5}{2}\right)$$

Worked Solutions

Arithmetic Sequences

Q041 MCQ

For $a_n = \frac{3n}{2n-1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 2$, $a_3 = \frac{9}{5}$.
3. Check each value by putting its index back into the formula.

Correct option: B**Final answer** $(3, 2, \frac{9}{5})$

Q042 MCQ

For $a_n = \frac{2n+3}{n+4}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = \frac{7}{6}$, $a_3 = \frac{9}{7}$.
3. Check each value by putting its index back into the formula.

Correct option: C**Final answer** $(1, \frac{7}{6}, \frac{9}{7})$

Worked Solutions

Arithmetic Sequences

Q043 MCQ

For $a_n = \frac{n}{3n-2}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = \frac{1}{4}$, $a_3 = \frac{3}{7}$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

$$(1, \frac{1}{4}, \frac{3}{7})$$

Q044 Written

For the sequence $\{a_n\} : 0, 1, 4, 9, 16, \dots$, deduce the general term.

Solution

1. The terms are (term number $- 1$)², so $a_n = (n - 1)^2$.
2. Checking $n = 1, 2, 3$ gives 0,1,4.

Final answer

$$a_n = (n - 1)^2$$

Worked Solutions

Arithmetic Sequences

Q045 **Written****For the sequence $\{a_n\} : 0, 1, 8, 27, 64, \dots$, deduce the general term.****Solution**

1. The terms are successive cubes starting with 0: $(n-1)^3$.
2. The formula gives 0,1,8,27,64 for $n = 1, 2, 3, 4, 5$.

Final answer

$$a_n = (n-1)^3$$

Q046 **Written****For the sequence $\{a_n\} : 1, 4, 9, 16, 25, \dots$, deduce the general term.****Solution**

1. These are the squares $1^2, 2^2, 3^2, \dots$, so $a_n = n^2$.

Final answer

$$a_n = n^2$$

Worked Solutions

Arithmetic Sequences

Q047 **Written**

For the sequence $\{a_n\} : 7^2, 8^2, 9^2, 10^2, 11^2, \dots$, deduce the general term.

Solution

1. The bases are $n + 6$, so $a_n = (n + 6)^2$.

Final answer

$$a_n = (n + 6)^2$$

Q048 **MCQ**

Which formula gives the general term of $\{a_n\} : 1, 8, 27, 64, \dots$?:

Solution

1. Separate the visible pattern: the rule is $a_n = n^3$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: B

Final answer

$$n^3$$

Worked Solutions

Arithmetic Sequences

Q049 MCQ

Which formula gives the general term of $\{a_n\}$: 9, 16, 25, 36, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (n + 2)^2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: C**Final answer**

$$(n + 2)^2$$

Q050 MCQ

Which formula gives the general term of $\{a_n\}$: -1, 0, 1, 8, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (n - 2)^3$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: D**Final answer**

$$(n - 2)^3$$

Worked Solutions

Arithmetic Sequences

Q051 MCQ

Which formula gives the general term of $\{a_n\}$: 36, 49, 64, 81, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (n + 5)^2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: A**Final answer**

$$(n + 5)^2$$

Q052 MCQ

Which formula gives the general term of $\{a_n\}$: 1, 9, 25, 49, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (2n - 1)^2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: B**Final answer**

$$(2n - 1)^2$$

Worked Solutions

Arithmetic Sequences

Q053 MCQ

Which formula gives the general term of $\{a_n\} : 0, 1, 16, 81, \dots$?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (n - 1)^4$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: C**Final answer**

$$(n - 1)^4$$

Q054 Written

For the sequence $\{a_n\} : 2, 5, 8, 11, 14, \dots$, deduce the general term.**Solution**

1. The common increase is 3 and the first term is 2, so $a_n = 2 + 3(n - 1) = 3n - 1$.

Final answer

$$a_n = 3n - 1$$

Worked Solutions

Arithmetic Sequences

Q055 Written

For the sequence $\{a_n\} : 0, 2, 4, 6, 8, \dots$, deduce the general term.

Solution

1. The common difference is 2 and $a_1=0$, so $a_n=2(n-1)=2n-2$.

Final answer

$$a_n = 2n - 2$$

Q056 Written

For the sequence $\{a_n\} : 0, 3, 6, 9, 12, \dots$, deduce the general term.

Solution

1. The common difference is 3 and $a_1=0$, so $a_n=3(n-1)=3n-3$.

Final answer

$$a_n = 3n - 3$$

Worked Solutions

Arithmetic Sequences

Q057 **Written****For the sequence $\{a_n\}$: 4, 9, 14, 19, 24, ..., deduce the general term.****Solution**

1. The common difference is 5 and $a_1=4$, so $a_n=4+5(n-1)=5n-1$.

Final answer

$$a_n = 5n - 1$$

Q058 **Written****For the sequence $\{a_n\}$: 6, 13, 20, 27, 34, ..., deduce the general term.****Solution**

1. The common difference is 7 and $a_1=6$, so $a_n=6+7(n-1)=7n-1$.

Final answer

$$a_n = 7n - 1$$

Worked Solutions

Arithmetic Sequences

Q059 MCQ

Which formula gives the general term of $\{a_n\}$: 7, 12, 17, 22, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = 5n + 2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: A**Final answer**

$$5n + 2$$

Q060 MCQ

Which formula gives the general term of $\{a_n\}$: -4, -1, 2, 5, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = 3n - 7$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: B**Final answer**

$$3n - 7$$

Worked Solutions

Arithmetic Sequences

Q061 MCQ

Which formula gives the general term of $\{a_n\}$: 10, 6, 2, -2 , ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = 14 - 4n$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: C**Final answer**

$$14 - 4n$$

Q062 MCQ

Which formula gives the general term of $\{a_n\}$: $\frac{1}{2}$, 1, $\frac{3}{2}$, 2, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = \frac{n}{2}$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: D**Final answer**

$$\frac{n}{2}$$

Worked Solutions

Arithmetic Sequences

Q063 Written

For the sequence $\{a_n\} : 2, -4, 6, -8, 10, \dots$, deduce the general term.

Solution

1. The magnitude is $2n$ and the first sign is positive with alternation, so the sign factor is $(-1)^{n-1}$.
2. Therefore $a_n = (-1)^{n-1} 2n$.

Final answer

$$a_n = (-1)^{n-1} 2n$$

Q064 Written

For the sequence $\{a_n\} : -3, 6, -9, 12, -15, \dots$, deduce the general term.

Solution

1. The magnitude is $3n$; the first term is negative and signs alternate, giving $(-1)^n$.
2. Therefore $a_n = (-1)^n 3n$.

Final answer

$$a_n = (-1)^n 3n$$

Worked Solutions

Arithmetic Sequences

Q065 Written

For the sequence $\{a_n\} : -5, 10, -15, 20, -25, \dots$, deduce the general term.

Solution

1. The magnitude is $5n$ and the first sign is negative, so $a_n = (-1)^n 5n$.

Final answer

$$a_n = (-1)^n 5n$$

Q066 Written

For the sequence $\{a_n\} : -1, 4, -7, 10, -13, \dots$, deduce the general term.

Solution

1. The magnitudes 1, 4, 7, 10, 13 are $3n - 2$; the first sign is negative, so $a_n = (-1)^n (3n - 2)$.

Final answer

$$a_n = (-1)^n (3n - 2)$$

Worked Solutions

Arithmetic Sequences

Q067 Written

For the sequence $\{a_n\} : 1, \frac{2}{3}, \frac{3}{5}, \frac{4}{7}, \frac{5}{9}, \dots$, deduce the general term.

Solution

1. The numerator is n and the denominator is the odd number $2n-1$.
2. Therefore $a_n = n/(2n-1)$.

Final answer

$$a_n = \frac{n}{2n-1}$$

Q068 Written

For the sequence $\{a_n\} : 1, \frac{3}{4}, \frac{5}{9}, \frac{7}{16}, \frac{9}{25}, \dots$, deduce the general term.

Solution

1. The numerator is the odd number $2n-1$ and the denominator is n^2 .
2. Therefore $a_n = (2n-1)/n^2$.

Final answer

$$a_n = \frac{2n-1}{n^2}$$

Worked Solutions

Arithmetic Sequences

Q069 Written

For the sequence $\{a_n\} : 1, -\frac{1}{3}, \frac{1}{5}, -\frac{1}{7}, \frac{1}{9}, \dots$, deduce the general term.

Solution

1. The denominator is $2n - 1$ and the signs alternate from a positive first term, so $a_n = (-1)^{n-1}/(2n - 1)$.

Final answer

$$a_n = \frac{(-1)^{n-1}}{2n - 1}$$

Q070 Written

For the sequence $\{a_n\} : \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots$, deduce the general term.

Solution

1. The numerator is n and the denominator is $n+1$, so $a_n = n/(n+1)$.

Final answer

$$a_n = \frac{n}{n + 1}$$

Worked Solutions

Arithmetic Sequences

Q071 Written

For the sequence $\{a_n\} : 1 \times 3, 2 \times 9, 3 \times 27, 4 \times 81, 5 \times 243, \dots$, deduce the general term.

Solution

1. The first factor is n and the second factor is 3^n , so $a_n = n3^n$.

Final answer

$$a_n = n 3^n$$

Q072 Written

For the sequence $\{a_n\} : 1 \times \frac{1}{2}, 2 \times \frac{1}{3}, 3 \times \frac{1}{4}, 4 \times \frac{1}{5}, 5 \times \frac{1}{6}, \dots$, deduce the general term then find the 20th term.

Solution

1. The first factor is the term number n and the second factor is $1/(n+1)$.
2. Hence the general term is $a_n = n/(n+1)$.
3. Substitute $n = 20$: $a_{20} = 20/(20+1) = 20/21$.

Final answer

$$a_n = \frac{n}{n+1}; \quad a_{20} = \frac{20}{21}$$

Worked Solutions

Arithmetic Sequences

Q073 Written

Find the first term and the common difference of the arithmetic sequences: (a) 2, 5, 8, 11, ...; (b) 3, -2, -7, -12, -17,

Solution

1. For (a), $5 - 2 = 8 - 5 = 11 - 8 = 3$, so $a = 2, d = 3$.
2. For (b), each successive difference is -5 , so $a = 3, d = -5$.

Final answer

$$(a) \quad a = 2, d = 3 \quad (b) \quad a = 3, d = -5$$

Q074 Written

Find the first term and common difference of: (a) 1, 5, 9, 13, 17, ...; (b) 23, 19, 15, 11, 7,

Solution

1. (a) The constant increase is (4), so $a = 1, d = 4$.
2. (b) The constant change is -4 , so $a = 23, d = -4$.

Final answer

$$(a) \quad a = 1, d = 4 \quad (b) \quad a = 23, d = -4$$

Worked Solutions

Arithmetic Sequences

Q075 MCQ

The listed terms $\{a_n\} = (4, 9, 14, 19, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = 4$.
2. Subtract consecutive terms: $d = 5$.

Correct option: B

Final answer

$$a = 4, d = 5$$

Q076 MCQ

The listed terms $\{a_n\} = (-7, -3, 1, 5, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = -7$.
2. Subtract consecutive terms: $d = 4$.

Correct option: A

Final answer

$$a = -7, d = 4$$

Worked Solutions

Arithmetic Sequences

Q077 MCQ

The listed terms $\{a_n\} = (18, 13, 8, 3, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = 18$.
2. Subtract consecutive terms: $d = -5$.

Correct option: C**Final answer**

$$a = 18, d = -5$$

Q078 MCQ

The listed terms $\{a_n\} = (12, 8, 4, 0, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = 12$.
2. Subtract consecutive terms: $d = -4$.

Correct option: D**Final answer**

$$a = 12, d = -4$$

Worked Solutions

Arithmetic Sequences

Q079 Written

Find the first term and common difference of: (a) 3, 5, 7, 9, 11, ...; (b) -13, -9, -5, -1, 3, ...; (c) -2, 1, 4, 7, 10, ...; (d) 10, 7, 4, 1, -2, ...; (e) $6, \frac{16}{3}, \frac{14}{3}, 4, \frac{10}{3}, \dots$; (f) $-4, -\frac{9}{2}, -5, -\frac{11}{2}, -6, \dots$

Solution

1. Subtract consecutive terms in each list. The six constant differences are 2, 4, 3, -3, $-\frac{2}{3}$, $-\frac{1}{2}$, respectively; the first terms are the first entries.

Final answer

(a)(3, 2); (b)(-13, 4); (c)(-2, 3); (d)(10, -3); (e)(6, $-\frac{2}{3}$); (f)(-4, $-\frac{1}{2}$).

Worked Solutions

Arithmetic Sequences

Q080 MCQ

The listed terms $\{a_n\} = (\frac{1}{2}, \frac{5}{6}, \frac{7}{6}, \frac{3}{2}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = \frac{1}{2}$.
2. Subtract consecutive terms: $d = \frac{1}{3}$.

Correct option: B**Final answer**

$$a = \frac{1}{2}, d = \frac{1}{3}$$

Q081 MCQ

The listed terms $\{a_n\} = (-\frac{3}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{3}{4}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = -\frac{3}{4}$.
2. Subtract consecutive terms: $d = \frac{1}{2}$.

Correct option: A**Final answer**

$$a = -\frac{3}{4}, d = \frac{1}{2}$$

Worked Solutions

Arithmetic Sequences

Q082 MCQ

The listed terms $\{a_n\} = (\frac{5}{3}, 2, \frac{7}{3}, \frac{8}{3}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = \frac{5}{3}$.
2. Subtract consecutive terms: $d = \frac{1}{3}$.

Correct option: C

Final answer

$$a = \frac{5}{3}, d = \frac{1}{3}$$

Q083 MCQ

The listed terms $\{a_n\} = (-\frac{5}{2}, -1, \frac{1}{2}, 2, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = -\frac{5}{2}$.
2. Subtract consecutive terms: $d = \frac{3}{2}$.

Correct option: B

Final answer

$$a = -\frac{5}{2}, d = \frac{3}{2}$$

Worked Solutions

Arithmetic Sequences

Q084 Written

The sequences are arithmetic. Find the common difference and the numbers that fit in (A) and (B): (a) $A, 2, -1, B, \dots$; (b) $13, A, B, -2, \dots$

Solution

- (a) $d = -1 - 2 = -3$; hence $A = 2 - d = 5$ and $B = -1 + d = -4$.
- (b) The change from 13 to -2 covers three equal steps: $3d = -15$, so $d = -5$; therefore $A = 8, B = 3$.

Final answer

$$(a) \quad d = -3, A = 5, B = -4 \quad (b) \quad d = -5, A = 8, B = 3$$

Q085 Written

The sequences are arithmetic. Find the common difference and the missing numbers: (a) $2, 9, A, B, \dots$; (b) $A, 17, B, 5, \dots$

Solution

- (a) $d = 9 - 2 = 7$, so $A = 16, B = 23$.
- (b) $5 - 17 = 2d$, hence $d = -6$; then $A = 23, B = 11$.

Final answer

$$(a) \quad d = 7, A = 16, B = 23 \quad (b) \quad d = -6, A = 23, B = 11$$

Worked Solutions

Arithmetic Sequences

Q086 MCQ

The sequence $\{a_n\} = (6, A, 14, B, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Solution

1. Set the equal-step difference (d).
2. Substitute successively to obtain $d = 4, A = 10, B = 18$.

Correct option: A

Final answer

$$d = 4, A = 10, B = 18$$

Q087 MCQ

The sequence $\{a_n\} = (A, 5, 9, B, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Solution

1. Set the equal-step difference (d).
2. Substitute successively to obtain $d = 4, A = 1, B = 13$.

Correct option: C

Final answer

$$d = 4, A = 1, B = 13$$

Worked Solutions

Arithmetic Sequences

Q088 MCQ

The sequence $\{a_n\} = (11, A, B, 23, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Solution

1. Set the equal-step difference (d).
2. Substitute successively to obtain $d = 4, A = 15, B = 19$.

Correct option: B**Final answer**

$$d = 4, A = 15, B = 19$$

Worked Solutions

Arithmetic Sequences

Q089 Written

Find the common difference and missing terms: (a) $8, 15, A, B, \dots$; (b) $7, 13, A, B, \dots$; (c) $A, 9, 4, B, \dots$; (d) $A, 15, B, 9, \dots$; (e) $11, A, B, 20, \dots$; (f) $17, A, B, 5, C, \dots$

Solution

1. Use the equal-step condition. For example, in (d), $9 - 15 = 2d$, so $d = -2$, giving $A = 17, B = 13$. Applying the same calculation to (a)–(f) gives the listed values.

Final answer

(a) $d = 7, A = 22, B = 29$; (b) $d = 6, A = 19, B = 25$; (c) $d = -5, A = 14, B = -1$; (d) $d = -2, A = 17, B = 13$; (e) $d = 3, A = 14, B = 17$; (f) $d = -4, A = 13, B = 9, C = 1$.

Worked Solutions

Arithmetic Sequences

Q090 Written

For each arithmetic sequence, find the general term and the 7th term: (a) first term (4), common difference (3); (b) first term (2), common difference -3 .

Solution

1. (a) $a_n = 4 + 3(n - 1) = 3n + 1$; $a_7 = 22$.
2. (b) $a_n = 2 - 3(n - 1) = 5 - 3n$; $a_7 = -16$.

Final answer

$$(a) \quad a_n = 3n + 1, a_7 = 22 \quad (b) \quad a_n = 5 - 3n, a_7 = -16$$

Q091 Written

For each arithmetic sequence, find the general term and the 7th term: (a) $a = 5, d = 4$; (b) $a = -2, d = 3$; (c) $a = 10, d = -3$; (d) $a = -\frac{1}{2}, d = -\frac{1}{2}$.

Solution

1. Substitute each pair into $a_n = a + (n - 1)d$, then put $n = 7$. The results are $4n + 1, 3n - 5, 13 - 3n, -\frac{n}{2}$, with 7th terms 29, 16, $-8, -\frac{7}{2}$.

Final answer

$$(a) a_n = 4n + 1, a_7 = 29; (b) a_n = 3n - 5, a_7 = 16; (c) a_n = 13 - 3n, a_7 = -8; (d) a_n = -\frac{n}{2}, a_7 = -\frac{7}{2}.$$

Worked Solutions

Arithmetic Sequences

Q092 Written

For the arithmetic sequence a_n with first term (8) and common difference -6 , find the general term and the 20th term.

Solution

1. Use $a_n = a + (n - 1)d$: $a_n = 8 - 6(n - 1) = 14 - 6n$.
2. Substitute $n = 20$: $a_{20} = 14 - 120 = -106$.

Final answer

$$a_n = 8 + (n - 1)(-6) = 14 - 6n; \quad a_{20} = -106$$

Q093 Written

For each arithmetic sequence, find the general term and the 10th term: (a) $a = -7, d = 3$; (b) $a = -2, d = -3$; (c) $a = 7, d = \frac{1}{2}$; (d) $a = \frac{2}{3}, d = -\frac{1}{3}$.

Solution

1. Use $a_n = a + (n - 1)d$ and substitute $n = 10$. Simplification gives the four displayed formulas and terms.

Final answer

$$(a)a_n = 3n - 10, a_{10} = 20; (b)a_n = 1 - 3n, a_{10} = -29; (c)a_n = \frac{n + 13}{2}, a_{10} = \frac{23}{2}; (d)a_n = \frac{3 - n}{3}, a_{10} = -\frac{7}{3}.$$

Worked Solutions

Arithmetic Sequences

Q094 Written

In an arithmetic sequence, $a_3 + a_7 = 38$ and $\frac{a_4}{a_{12}} = \frac{2}{5}$. Find the first term and common difference.

Solution

1. $a_3 + a_7 = (a + 2d) + (a + 6d) = 2a + 8d = 38$, so $a + 4d = 19$.
2. The ratio gives $5(a + 3d) = 2(a + 11d)$, hence $3a = 7d$.
3. Solving yields $d = 3$ and $a = 7$.

Final answer

$$a = 7, d = 3.$$

Q095 Written

Find the general term of the arithmetic sequence whose first term is -5 and whose 17th term is 75.

Solution

1. $75 = -5 + 16d$, so $d = 5$.
2. $a_n = -5 + 5(n - 1) = 5n - 10$.

Final answer

$$a_n = 5n - 10$$

Worked Solutions

Arithmetic Sequences

Q096 Written

The first term is 6, and the 4th term is 18. Find a_n .

Solution

1. $18 = 6 + 3d$, so $d = 4$.
2. $a_n = 6 + 4(n - 1) = 4n + 2$.

Final answer

$$a_n = 4n + 2$$

Q097 Written

The first term is 20, and the 8th term is -1 . Find a_n .

Solution

1. $-1 = 20 + 7d$, so $d = -3$.
2. $a_n = 20 - 3(n - 1) = 23 - 3n$.

Final answer

$$a_n = 23 - 3n$$

Worked Solutions

Arithmetic Sequences

Q098 **Written****The first term is -8 , and the 8th term is 55 . Find a_n .****Solution**

1. $55 = -8 + 7d$, so $d = 9$.
2. $a_n = -8 + 9(n - 1) = 9n - 17$.

Final answer

$$a_n = 9n - 17$$

Q099 **Written****The first term is -18 , and the 10th term is -63 . Find a_n .****Solution**

1. $-63 = -18 + 9d$, so $d = -5$.
2. $a_n = -18 - 5(n - 1) = -5n - 13$.

Final answer

$$a_n = -5n - 13$$

Worked Solutions

Arithmetic Sequences

Q100 MCQ

An arithmetic sequence has $a = 4$ and $a_7 = 22$. Select the correct pair (a_n, d) :

Solution

1. $22 = 4 + 6d$, so $d = 3$.
2. $a_n = 4 + 3(n - 1) = 3n + 1$.

Correct option: D

Final answer

$$(a_n, d) = (3n + 1, 3)$$

Q101 MCQ

An arithmetic sequence has $a = -3$ and $a_9 = 29$. Select the correct pair (a_n, d) :

Solution

1. $29 = -3 + 8d$, so $d = 4$.
2. $a_n = -3 + 4(n - 1) = 4n - 7$.

Correct option: B

Final answer

$$(a_n, d) = (4n - 7, 4)$$

Worked Solutions

Arithmetic Sequences

Q102 MCQ

An arithmetic sequence has $a = 12$ and $a_5 = -4$. Select the correct pair (a_n, d) :

Solution

1. $-4 = 12 + 4d$, so $d = -4$.
2. $a_n = 12 - 4(n - 1) = 16 - 4n$.

Correct option: A

Final answer

$$(a_n, d) = (16 - 4n, -4)$$

Q103 MCQ

An arithmetic sequence has $a = 5$ and $a_{11} = 45$. Select the correct pair (a_n, d) :

Solution

1. $45 = 5 + 10d$, so $d = 4$.
2. $a_n = 5 + 4(n - 1) = 4n + 1$.

Correct option: B

Final answer

$$(a_n, d) = (4n + 1, 4)$$

Worked Solutions

Arithmetic Sequences

Q104 Written

Find the general term of the arithmetic sequence whose common difference is 3 and whose 16th term is 81.

Solution

1. $81 = a + 15(3)$, so $a = 36$.
2. $a_n = 36 + 3(n - 1) = 3n + 33$.

Final answer

$$a_n = 3n + 33$$

Q105 Written

The common difference is 2, and the 8th term is 4. Find a_n .

Solution

1. $4 = a + 7(2)$, so $a = -10$.
2. $a_n = -10 + 2(n - 1) = 2n - 12$.

Final answer

$$a_n = 2n - 12$$

Worked Solutions

Arithmetic Sequences

Q106 Written

The common difference is -4 , and the 5th term is -7 . Find a_n .

Solution

1. $-7 = a + 4(-4)$, so $a = 9$.
2. $a_n = 9 - 4(n - 1) = 13 - 4n$.

Final answer

$$a_n = 13 - 4n$$

Q107 Written

The common difference is -7 , and the 6th term is -24 . Find a_n .

Solution

1. $-24 = a + 5(-7)$, so $a = 11$.
2. $a_n = 11 - 7(n - 1) = 18 - 7n$.

Final answer

$$a_n = 18 - 7n$$

Worked Solutions

Arithmetic Sequences

Q108 MCQ

An arithmetic sequence has common difference -2 and $a_4 = 1$. Select the correct pair (a_n, a) :

Solution

1. $1 = a + 3(-2)$, so $a = 7$.
2. $a_n = 7 - 2(n - 1) = 9 - 2n$.

Correct option: A

Final answer

$$(a_n, a) = (9 - 2n, 7)$$

Q109 MCQ

An arithmetic sequence has common difference 5 and $a_6 = 17$. Select the correct pair (a_n, a) :

Solution

1. $17 = a + 5(5)$, so $a = -8$.
2. $a_n = -8 + 5(n - 1) = 5n - 13$.

Correct option: D

Final answer

$$(a_n, a) = (5n - 13, -8)$$

Worked Solutions

Arithmetic Sequences

Q110 MCQ

An arithmetic sequence has common difference 4 and $a_{10} = 38$. Select the correct pair (a_n, a) :

Solution

1. $38 = a + 9(4)$, so $a = 2$.
2. $a_n = 2 + 4(n - 1) = 4n - 2$.

Correct option: C

Final answer

$$(a_n, a) = (4n - 2, 2)$$

Q111 MCQ

An arithmetic sequence has common difference -3 and $a_8 = -18$. Select the correct pair (a_n, a) :

Solution

1. $-18 = a + 7(-3)$, so $a = 3$.
2. $a_n = 3 - 3(n - 1) = 6 - 3n$.

Correct option: D

Final answer

$$(a_n, a) = (6 - 3n, 3)$$

Worked Solutions

Arithmetic Sequences

Q112 Written

Find the general term of the arithmetic sequence $\{a_n\}$ whose 3rd term is 10 and the 8th term is 50.

Solution

1. Use $a_n = a + (n - 1)d$. The data give $10 = a + 2d$ and $50 = a + 7d$.
2. Subtracting gives $5d = 40$, so $d = 8$ and $a = -6$.
3. Therefore $a_n = -6 + 8(n - 1) = 8n - 14$.

Final answer

$$a_n = 8n - 14$$

Q113 Written

Find the general term of the arithmetic sequence whose 6th term is 2 and whose 15th term is -25 .

Solution

1. $2 = a + 5d$ and $-25 = a + 14d$, so $d = -3$, $a = 17$.
2. $a_n = 17 - 3(n - 1) = 20 - 3n$.

Final answer

$$a_n = 20 - 3n$$

Worked Solutions

Arithmetic Sequences

Q114 **Written****The 4th term is 14, and the 10th term is 62. Find a_n .****Solution**

1. $14 = a + 3d$ and $62 = a + 9d$, so $d = 8$, $a = -10$.
2. $a_n = 8n - 18$.

Final answer

$$a_n = 8n - 18$$

Q115 **Written****The 3rd term is 30, and the 10th term is 2. Find a_n .****Solution**

1. $30 = a + 2d$ and $2 = a + 9d$, so $d = -4$, $a = 38$.
2. $a_n = 42 - 4n$.

Final answer

$$a_n = 42 - 4n$$

Worked Solutions

Arithmetic Sequences

Q116 Written

The 10th term is 49, and the 23rd term is 10. Find a_n .

Solution

1. $49 = a + 9d$ and $10 = a + 22d$, so $d = -3$, $a = 76$.
2. $a_n = 79 - 3n$.

Final answer

$$a_n = 79 - 3n$$

Q117 Written

The 3rd term is 6, and the 10th term is 62. Find a_n .

Solution

1. $6 = a + 2d$ and $62 = a + 9d$, so $d = 8$, $a = -10$.
2. $a_n = 8n - 18$.

Final answer

$$a_n = 8n - 18$$

Worked Solutions

Arithmetic Sequences

Q118 MCQ

An arithmetic sequence has $a_3 = 5$ and $a_9 = 17$. Select the correct pair (a_n, a) :

Solution

1. $5 = a + 2d$ and $17 = a + 8d$, so $d = 2$, $a = 1$.
2. $a_n = 1 + 2(n - 1) = 2n - 1$.

Correct option: C

Final answer

$$(a_n, a) = (2n - 1, 1)$$

Q119 MCQ

An arithmetic sequence has $a_4 = 7$ and $a_{10} = -5$. Select the correct pair (a_n, a) :

Solution

1. $7 = a + 3d$ and $-5 = a + 9d$, so $d = -2$, $a = 13$.
2. $a_n = 13 - 2(n - 1) = 15 - 2n$.

Correct option: C

Final answer

$$(a_n, a) = (15 - 2n, 13)$$

Worked Solutions

Arithmetic Sequences

Q120 MCQ

An arithmetic sequence has $a_7 = 31$ and $a_{12} = 56$. Select the correct pair (a_n, a) :

Solution

1. $31 = a + 6d$ and $56 = a + 11d$, so $d = 5$, $a = 1$.

2. $a_n = 1 + 5(n - 1) = 5n - 4$.

Correct option: B

Final answer

$$(a_n, a) = (5n - 4, 1)$$

Q121 MCQ

An arithmetic sequence has $a_2 = -4$ and $a_8 = 14$. Select the correct pair (a_n, a) :

Solution

1. $-4 = a + d$ and $14 = a + 7d$, so $d = 3$, $a = -7$.

2. $a_n = -7 + 3(n - 1) = 3n - 10$.

Correct option: A

Final answer

$$(a_n, a) = (3n - 10, -7)$$

Worked Solutions

Arithmetic Sequences

Q122 MCQ

An arithmetic sequence has $a_5 = 3$ and $a_{11} = 9$. Select the correct pair (a_n, a) :

Solution

1. $3 = a + 4d$ and $9 = a + 10d$, so $d = \frac{1}{2}$, $a = -\frac{1}{2}$.

2. $a_n = -\frac{1}{2} + \frac{1}{2}(n-1) = \frac{n-2}{2}$.

Correct option: A

Final answer

$$\left(a_n = \frac{n-2}{2}, -\frac{1}{2}\right)$$

Q123 MCQ

An arithmetic sequence has $a_2 = 6$ and $a_9 = 20$. Select the correct pair (a_n, a) :

Solution

1. $6 = a + d$ and $20 = a + 8d$, so $d = 2$, $a = 4$.

2. $a_n = 4 + 2(n-1) = 2n + 2$.

Correct option: B

Final answer

$$(a_n = 2n + 2, 4)$$

Worked Solutions

Arithmetic Sequences

Q124 Written

The common difference is $\frac{2}{5}$, and the 30th term is $\frac{23}{5}$. Find a_n .

Solution

1. $\frac{23}{5} = a + 29(\frac{2}{5})$, so $a = -7$.

2. $a_n = -7 + \frac{2}{5}(n - 1) = \frac{2n - 37}{5}$.

Final answer

$$a_n = \frac{2n - 37}{5}$$

Q125 MCQ

An arithmetic sequence has $d = -\frac{3}{2}$ and $a_5 = -9$. Select the correct pair (a_n, a) :

Solution

1. $-9 = a + 4(-\frac{3}{2})$, so $a = -3$.

2. $a_n = -3 - \frac{3}{2}(n - 1) = -\frac{3(n+1)}{2}$.

Correct option: A

Final answer

$$\left(a_n = -\frac{3(n+1)}{2}, -3 \right)$$

Worked Solutions

Arithmetic Sequences

Q126 MCQ

An arithmetic sequence has $d = \frac{2}{3}$ and $a_{12} = 10$. Select the correct pair (a_n, a) :

Solution

1. $10 = a + 11(\frac{2}{3})$, so $a = \frac{8}{3}$.
2. $a_n = \frac{8}{3} + \frac{2}{3}(n - 1) = \frac{2n+6}{3}$.

Correct option: D

Final answer

$$\left(a_n = \frac{2n+6}{3}, \frac{8}{3}\right)$$

Q127 Written

For the arithmetic sequence $\{a_n\}$ with common difference 4 and $a_8 = 33$, which term is 113?

Solution

1. First find a : $33 = a + 7(4)$, so $a = 5$.
2. Then $a_n = 5 + 4(n - 1) = 4n + 1$.
3. Solve $4n + 1 = 113$: $n = 28$.

Final answer

$n = 28$, the 28th term

Worked Solutions

Arithmetic Sequences

Q128 Written

For the arithmetic sequence with first term 1 and common difference 3, determine which term is 40.

Solution

1. $a_n = 1 + 3(n - 1) = 3n - 2$.
2. $3n - 2 = 40$ gives $n = 14$.

Final answer

$n = 14$, the 14th term

Q129 Written

For the arithmetic sequence with common difference -5 and 13th term -7 , determine which term is -52 .

Solution

1. $-7 = a + 12(-5)$, so $a = 53$ and $a_n = 58 - 5n$.
2. $58 - 5n = -52$ gives $n = 22$.

Final answer

$n = 22$, the 22nd term

Worked Solutions

Arithmetic Sequences

Q130 Written

For the arithmetic sequence whose 10th term is -20 and 20th term is -50 , determine which term is -110 .

Solution

1. $-20 = a + 9d$ and $-50 = a + 19d$, so $d = -3$, $a = 7$.
2. $a_n = 10 - 3n$, and $10 - 3n = -110$ gives $n = 40$.

Final answer

$n = 40$, the 40th term

Q131 MCQ

For $a = 3$, $d = 4$, which term of the arithmetic sequence is 59?:

Solution

1. $a_n = 3 + 4(n - 1) = 4n - 1$.
2. $4n - 1 = 59$ gives $n = 15$.

Correct option: C

Final answer

$n = 15$

Worked Solutions

Arithmetic Sequences

Q132 MCQ

For $a = -5$, $d = 6$, which term of the arithmetic sequence is 55?:

Solution

1. $a_n = -5 + 6(n - 1) = 6n - 11$.
2. $6n - 11 = 55$ gives $n = 11$.

Correct option: D

Final answer

$$n = 11$$

Q133 MCQ

For $a = 20$, $d = -3$, which term of the arithmetic sequence is -16 ?:

Solution

1. $a_n = 20 - 3(n - 1) = 23 - 3n$.
2. $23 - 3n = -16$ gives $n = 13$.

Correct option: A

Final answer

$$n = 13$$

Worked Solutions

Arithmetic Sequences

Q134 MCQ

For $a = 1$, $d = 5$, which term of the arithmetic sequence is 86?:

Solution

1. $a_n = 1 + 5(n - 1) = 5n - 4$.
2. $5n - 4 = 86$ gives $n = 18$.

Correct option: B

Final answer

$$n = 18$$

Q135 Written

For the arithmetic sequence $\{a_n\}$ with common difference 4 and $a_8 = 33$, find the first term greater than 500.

Solution

1. From the given data, $a_n = 4n + 1$.
2. Require $4n + 1 > 500$, hence $n > 124.75$.
3. The least natural index is $n = 125$.

Final answer

$n = 125$, the 125th term

Worked Solutions

Arithmetic Sequences

Q136 Written

For the arithmetic sequence with first term 1 and common difference 3, find the first term greater than 100.

Solution

1. $a_n = 3n - 2$.
2. $3n - 2 > 100$ gives $n > 34$.
3. The first valid index is $n = 35$.

Final answer

$n = 35$, the 35th term

Q137 MCQ

For $a = 2, d = 7$, find the first term greater than 100:

Solution

1. $a_n = 7n - 5$.
2. $7n - 5 > 100$ gives $n > 15$, so $n = 16$.

Correct option: A

Final answer

$n = 16$

Worked Solutions

Arithmetic Sequences

Q138 MCQ

For $a = -9, d = 4$, find the first term greater than 50:**Solution**

1. $a_n = 4n - 13$.
2. $4n - 13 > 50$ gives $n > 15.75$, so $n = 16$.

Correct option: D**Final answer**

$$n = 16$$

Q139 MCQ

For $a = 6, d = 3$, find the first term greater than 200:**Solution**

1. $a_n = 3n + 3$.
2. $3n + 3 > 200$ gives $n > 65\frac{2}{3}$, so $n = 66$.

Correct option: D**Final answer**

$$n = 66$$

Worked Solutions

Arithmetic Sequences

Q140 MCQ

For $a = -1$, $d = 9$, find the first term greater than 300:

Solution

1. $a_n = 9n - 10$.
2. $9n - 10 > 300$ gives $n > 34\frac{4}{9}$, so $n = 35$.

Correct option: B

Final answer

$$n = 35$$

Q141 Written

For the arithmetic sequence with common difference -5 and 13th term -7 , find the first term smaller than -300 .

Solution

1. $a_n = 58 - 5n$.
2. $58 - 5n < -300$ gives $n > 71.6$.
3. The first valid index is $n = 72$.

Final answer

$n = 72$, the 72nd term

Worked Solutions

Arithmetic Sequences

Q142 Written

For the arithmetic sequence whose 10th term is -20 and 20th term is -50 , find the first term smaller than -500 .

Solution

1. $a_n = 10 - 3n$.
2. $10 - 3n < -500$ gives $n > 170$.
3. The first valid index is $n = 171$.

Final answer

$n = 171$, the 171st term

Q143 MCQ

For $a = 10$, $d = -4$, find the first term smaller than -100 :

Solution

1. $a_n = 14 - 4n$.
2. $14 - 4n < -100$ gives $n > 28.5$, so $n = 29$.

Correct option: B

Final answer

$n = 29$

Worked Solutions

Arithmetic Sequences

Q144 MCQ

For $a = 25, d = -6$, find the first term smaller than -250 :**Solution**

1. $a_n = 25 - 6n$.
2. $25 - 6n < -250$ gives $n > 46\frac{5}{6}$, so $n = 47$.

Correct option: B**Final answer**

$$n = 47$$

Q145 MCQ

For $a = -3, d = -5$, find the first term smaller than -200 :**Solution**

1. $a_n = -3 - 5n$.
2. $-3 - 5n < -200$ gives $n > 40.4$, so $n = 41$.

Correct option: C**Final answer**

$$n = 41$$

Worked Solutions

Arithmetic Sequences

Q146 MCQ

For $a = 4$, $d = 6$, find the first term that is at least 100:

Solution

1. $a_n = 6n - 2$.
2. $6n - 2 \geq 100$ gives $n \geq 17$, so $n = 17$.

Correct option: A

Final answer

$$n = 17$$

Q147 MCQ

For the expression For $a = 5$, $d = 4$, does the sequence contain the term 42?, the required value is:

Solution

1. $a_n = 4n + 1$.
2. $4n + 1 = 42$ gives $n = \frac{41}{4}$, not a natural index.
3. Therefore 42 is not a term.

Correct option: B

Final answer

No term

Worked Solutions

Arithmetic Sequences

Q148 **Written**

When $a, 6, 2a$ form an arithmetic sequence in this order, find a .

Solution

1. $2(6) = a + 2a$.
2. $3a = 12$, so $a = 4$.

Final answer

$$a = 4$$

Q149 **Written**

When the three numbers $x, 6, x^2$ form an arithmetic sequence in this order, find x .

Solution

1. Use $2b = a + c$: $2(6) = x + x^2$.
2. Thus $x^2 + x - 12 = 0 = (x + 4)(x - 3)$.
3. Therefore $x = -4$ or $x = 3$.

Final answer

$$x = -4 \text{ or } x = 3$$

Worked Solutions

Arithmetic Sequences

Q150 **Written****When $x^2, x, -8$ form an arithmetic sequence in this order, find x .****Solution**

1. $2x = x^2 - 8$.
2. $x^2 - 2x - 8 = (x - 4)(x + 2) = 0$.
3. $x = -2$ or $x = 4$.

Final answer

$$x = -2 \text{ or } x = 4$$

Q151 **Written****When $-3, x, x^2$ form an arithmetic sequence in this order, find x .****Solution**

1. $2x = -3 + x^2$.
2. $x^2 - 2x - 3 = (x - 3)(x + 1) = 0$.
3. $x = -1$ or $x = 3$.

Final answer

$$x = -1 \text{ or } x = 3$$

Worked Solutions

Arithmetic Sequences

Q152 Challenge

When $2, x, y, z$ form an arithmetic sequence in this order, and $2x, y + 1, z - y$ also form an arithmetic sequence in this order, find x, y, z .

Solution

1. Let the common difference of $2, x, y, z$ be d : $x = 2 + d$, $y = 2 + 2d$, $z = 2 + 3d$.
2. $2(y + 1) = 2x + (z - y)$ gives $6 + 4d = 4 + 3d$, hence $d = -2$.
3. $(x, y, z) = (0, -2, -4)$.

Final answer

$$x = 0, \quad y = -2, \quad z = -4$$

Worked Solutions

Arithmetic Sequences

Q153 Written

Find the sum (S_7) of a sequence with first term (8), last term (22), and (7) terms.

Solution

1. Use ($S_n = \frac{n}{2}(a + l)$).
2. ($S_7 = \frac{7}{2}(8 + 22) = 105$).

Final answer

105

Q154 Written

Find (S_{10}) for a sequence with first term (7), last term (13), and (10) terms.

Solution

1. ($S_{10} = \frac{10}{2}(7 + 13) = 100$).

Final answer

100

Worked Solutions

Arithmetic Sequences

Q155 **Written****Find (S_{10}) for a sequence with first term (3), last term (21), and (10) terms.****Solution**

1. $(S_{10} = \frac{10}{2}(3 + 21) = 120).$

Final answer

120

Q156 **Written****Find (S_8) for a sequence with first term (5), last term (33), and (8) terms.****Solution**

1. $(S_8 = \frac{8}{2}(5 + 33) = 152).$

Final answer

152

Worked Solutions

Arithmetic Sequences

Q157 **Written****Find (S_{10}) for a sequence with first term (2), last term (29), and (10) terms.****Solution**

1. $(S_{10} = \frac{10}{2}(2 + 29) = 155).$

Final answer

155

Q158 **MCQ****A sequence has first term 6, last term 30, and 7 terms. Find its sum:****Solution**

1. Use $S_7 = \frac{7}{2}(6 + 30).$
2. Thus $S_7 = 126.$

Correct option: B**Final answer**

126

Worked Solutions

Arithmetic Sequences

Q159 MCQ

A sequence has first term -4 , last term 20 , and 9 terms. Find its sum:

Solution

1. $S_9 = \frac{9}{2}(-4 + 20) = 72.$

Correct option: B

Final answer

72

Q160 MCQ

Find the sum of 10 terms whose first term is 15 and last term is 3 :

Solution

1. $S_{10} = \frac{10}{2}(15 + 3) = 90.$

Correct option: B

Final answer

90

Worked Solutions

Arithmetic Sequences

Q161 MCQ

The first term is 7, the last term is 42, and there are 8 terms. Find S_8 :

Solution

1. $S_8 = \frac{8}{2}(7 + 42) = 196.$

Correct option: B

Final answer

196

Q162 MCQ

Find the sum of 6 terms with first term -12 and last term 8:

Solution

1. $S_6 = \frac{6}{2}(-12 + 8) = -12.$

Correct option: B

Final answer

-12

Worked Solutions

Arithmetic Sequences

Q163 MCQ

A sequence has $S_9 = 72$ and first term -4 . Find its last term:

Solution

1. From $72 = \frac{9}{2}(-4 + l)$, $l = 20$.

Correct option: C

Final answer

20

Q164 Written

Find (S_{13}) when the first term is (20) , the common difference is (-5) , and the number of terms is (13) .

Solution

1. Use $(S_n = \frac{n}{2}\{2a + (n - 1)d\})$.

2. $(S_{13} = \frac{13}{2}\{40 + 12(-5)\} = -130)$.

Final answer

-130

Worked Solutions

Arithmetic Sequences

Q165 **Written****Find (S_{12}) when the first term is (-6) , the common difference is (3) , and there are (12) terms.****Solution**

$$1. (S_{12} = \frac{12}{2}\{-12 + 11(3)\} = 144).$$

Final answer

144

Q166 **Written****Find (S_{15}) when the first term is (10) , the common difference is (-3) , and there are (15) terms.****Solution**

$$1. (S_{15} = \frac{15}{2}\{20 + 14(-3)\} = -165).$$

Final answer

-165

Worked Solutions

Arithmetic Sequences

Q167 Written

Find (S_{11}) when the first term is (-4) , the common difference is (4) , and there are (11) terms.

Solution

1. $(S_{11} = \frac{11}{2}\{-8 + 10(4)\} = 176).$

Final answer

176

Q168 Written

Find (S_{18}) when the first term is (3) , the common difference is (2) , and there are (18) terms.

Solution

1. $(S_{18} = \frac{18}{2}\{6 + 17(2)\} = 360).$

Final answer

360

Worked Solutions

Arithmetic Sequences

Q169 MCQ

For $a = 4, d = 3, n = 12$, find S_{12} :**Solution**

$$1. S_{12} = \frac{12}{2}\{8 + 11(3)\} = 246.$$

Correct option: C**Final answer**

246

Q170 MCQ

For $a = 9, d = -2, n = 10$, find S_{10} :**Solution**

$$1. S_{10} = 5\{18 + 9(-2)\} = 0.$$

Correct option: B**Final answer**

0

Worked Solutions

Arithmetic Sequences

Q171 MCQ

For $a = -5, d = 4, n = 9$, find S_9 :**Solution**

$$1. S_9 = \frac{9}{2}\{-10 + 8(4)\} = 88.$$

Correct option: C**Final answer**

88

Q172 MCQ

For $a = 2, d = -3, n = 15$, find S_{15} :**Solution**

$$1. S_{15} = \frac{15}{2}\{4 + 14(-3)\} = -285.$$

Correct option: B**Final answer**

-285

Worked Solutions

Arithmetic Sequences

Q173 MCQ

For $a = 11, d = 0, n = 8$, find S_8 :**Solution**1. All terms equal 11, so $S_8 = 88$.**Correct option:** C**Final answer**

88

Q174 MCQ

For $a = -3, d = -2, n = 7$, find S_7 :**Solution**1. $S_7 = \frac{7}{2}\{-6 + 6(-2)\} = -54$.**Correct option:** B**Final answer**

-54

Worked Solutions

Arithmetic Sequences

Q175 Written

Find the sum (S) of a sequence with first term (2), common difference (3), and last term (47).**Solution**

1. Find (n) from $(47 = 2 + (n - 1)3)$, giving ($n = 16$).
2. Then $(S = \frac{16}{2}(2 + 47) = 392)$.

Final answer

392

Q176 Written

Find (S) for a sequence with first term (5), common difference (3), and last term (44).**Solution**

1. Solve $(44 = 5 + (n - 1)3)$, so ($n = 14$).
2. $(S = \frac{14}{2}(5 + 44) = 343)$.

Final answer

343

Worked Solutions

Arithmetic Sequences

Q177 Written

Find (S) for a sequence with first term (3), common difference (4), and last term (75).**Solution**

1. $(75 = 3 + (n - 1)4)$ gives $(n = 19)$.
2. $(S = \frac{19}{2}(3 + 75) = 741)$.

Final answer

741

Q178 Written

Find (S) for a sequence with first term (2), common difference (5), and last term (57).**Solution**

1. $(57 = 2 + (n - 1)5)$ gives $(n = 12)$.
2. $(S = \frac{12}{2}(2 + 57) = 354)$.

Final answer

354

Worked Solutions

Arithmetic Sequences

Q179 Written

Find (S) for a sequence with first term (4), common difference (3), and last term (100).

Solution

1. $100 = 4 + (n - 1)3$ gives $(n = 33)$.
2. $S = \frac{33}{2}(4 + 100) = 1716$.

Final answer

1716

Q180 MCQ

A sequence has $a = 5, d = 4, l = 41$. Find its sum:

Solution

1. $41 = 5 + (n - 1)4$ gives $n = 10$.
2. $S = \frac{10}{2}(5 + 41) = 230$.

Correct option: C

Final answer

230

Worked Solutions

Arithmetic Sequences

Q181 MCQ

A sequence has $a = -7, d = 5, l = 48$. Find its sum:

Solution

1. $48 = -7 + (n - 1)5$ gives $n = 12$.
2. $S = 6(-7 + 48) = 246$.

Correct option: C

Final answer

246

Q182 MCQ

A sequence has $a = 12, d = 2, l = 50$. Find its sum:

Solution

1. $50 = 12 + (n - 1)2$ gives $n = 20$.
2. $S = 10(12 + 50) = 620$.

Correct option: C

Final answer

620

Worked Solutions

Arithmetic Sequences

Q183 MCQ

A sequence has $a = -10, d = 6, l = 50$. Find its sum:**Solution**

1. $50 = -10 + (n - 1)6$ gives $n = 11$.

2. $S = \frac{11}{2}(-10 + 50) = 220$.

Correct option: C**Final answer**

220

Q184 MCQ

A sequence has $a = 18, d = -3, l = -9$. Find its sum:**Solution**

1. Solve $-9 = 18 + (n - 1)(-3)$, giving $n = 10$.

2. $S = \frac{10}{2}(18 - 9) = 45$.

Correct option: C**Final answer**

45

Worked Solutions

Arithmetic Sequences

Q185 MCQ

A sequence has $a = 40, d = -4, l = -12$. Find its sum:

Solution

1. $n = 14$ from $-12 = 40 + (n - 1)(-4)$.
2. $S = 7(40 - 12) = 196$.

Correct option: B

Final answer

196

Q186 Written

For the arithmetic sequence with first term (48) and common difference (-2) , find the sum from the 20th term to the 30th term.

Solution

1. Use $(S_{30} - S_{19})$, since the range is (a_{20}) through (a_{30}) .
2. $(a_{20} = 10, a_{30} = -10)$, so $(11 \times \frac{10 + (-10)}{2} = 0)$.

Final answer

0

Worked Solutions

Arithmetic Sequences

Q187 Written

For the arithmetic sequence with first term (-53) and common difference (4) , find the sum from the 10th term to the 25th term.

Solution

1. $(a_{10} = -53 + 9(4) = -17)$ and $(a_{25} = -53 + 24(4) = 43)$.
2. There are (16) terms, so the sum is $(16 \times \frac{-17+43}{2} = 208)$.

Final answer

208

Q188 MCQ

For $a = 3, d = 4$, find the sum from a_5 to a_{12} :

Solution

1. $a_5 = 19, a_{12} = 47$, with 8 terms.
2. Sum = $8(19 + 47)/2 = 264$.

Correct option: B

Final answer

264

Worked Solutions

Arithmetic Sequences

Q189 MCQ

For $a = 20$, $d = -3$, find the sum from a_4 to a_{10} :**Solution**

1. $a_4 = 11$, $a_{10} = -7$, with 7 terms.
2. Sum = $7(11 - 7)/2 = 14$.

Correct option: C**Final answer**

14

Q190 MCQ

For $a = 60$, $d = -3$, find the sum from a_8 to a_{16} :**Solution**

1. $a_8 = 39$, $a_{16} = 15$, with 9 terms.
2. Sum = $9(39 + 15)/2 = 243$.

Correct option: C**Final answer**

243

Worked Solutions

Arithmetic Sequences

Q191 MCQ

For $a = 1, d = 7$, find the sum from a_3 to a_9 :**Solution**

1. $a_3 = 15, a_9 = 57$, with 7 terms.
2. $\text{Sum} = 7(15 + 57)/2 = 252$.

Correct option: C**Final answer**

252

Q192 MCQ

For $a = 60, d = -3$, find the sum from a_9 to a_{17} :**Solution**

1. $a_9 = 36, a_{17} = 12$, with 9 terms.
2. $\text{Sum} = 9(36 + 12)/2 = 216$.

Correct option: B**Final answer**

216

Worked Solutions

Arithmetic Sequences

Q193 MCQ

For $a = 12$, $d = 5$, find the sum from a_2 to a_8 :

Solution

1. $a_2 = 17$, $a_8 = 47$, with 7 terms.
2. Sum $= 7(17 + 47)/2 = 224$.

Correct option: C

Final answer

224

Q194 Written

For the arithmetic sequence $(90, 84, 78, \dots)$, find the sum from the 10th term to the 20th term.

Solution

1. Here $(a = 90, d = -6)$, so the required sum is $(S_{20} - S_9)$.
2. $(S_{20} = 660)$ and $(S_9 = 594)$, hence the sum is $(660 - 594 = 66)$.

Final answer

66

Worked Solutions

Arithmetic Sequences

Q195 Written

For the arithmetic sequence $(-9, -3, 3, \dots)$, find the sum from the 11th term to the 20th term.

Solution

1. Here $a = -9$, $d = 6$, so use $S_{20} - S_{10}$.
2. $a_{10} = 45$ and $a_{20} = 105$; there are 10 terms.
3. Required sum = $\frac{10(45+105)}{2} = 750$.

Final answer

750

Q196 Written

For the arithmetic sequence $(74, 69, 64, \dots)$, find the sum from the 8th term to the 18th term.

Solution

1. Here $(a = 74, d = -5)$, so $(a_8 = 39)$ and $(a_{18} = -11)$.
2. There are (11) terms; the sum is $(11 \times \frac{39-11}{2} = 154)$.

Final answer

154

Worked Solutions

Arithmetic Sequences

Q197 MCQ

For $a = 4, d = 5$, find the sum from a_6 to a_{13} :**Solution**

1. $a_6 = 29, a_{13} = 64$, with 8 terms.
2. $\text{Sum} = 8(29 + 64)/2 = 372$.

Correct option: C**Final answer**

372

Q198 MCQ

For $a = -25, d = 6$, find the sum from a_5 to a_{11} :**Solution**

1. $a_5 = -1, a_{11} = 35$, with 7 terms.
2. $\text{Sum} = 7(-1 + 35)/2 = 119$.

Correct option: C**Final answer**

119

Worked Solutions

Arithmetic Sequences

Q199 Written

Find (S_n) for a sequence with first term (1) and common difference (4).**Solution**

1. Substitute $(a = 1, d = 4)$ into $(S_n = \frac{n}{2}\{2a + (n - 1)d\})$.
2. $(S_n = \frac{n}{2}(2 + 4n - 4) = 2n^2 - n)$.

Final answer

$$2n^2 - n$$

Q200 Written

Find (S_n) for a sequence with first term (2) and common difference (-3) .**Solution**

1. $(S_n = \frac{n}{2}\{4 + (n - 1)(-3)\})$.
2. Therefore $(S_n = \frac{n(7-3n)}{2})$.

Final answer

$$\frac{n(7 - 3n)}{2}$$

Worked Solutions

Arithmetic Sequences

Q201 **Written****Find (S_n) for a sequence with first term (0) and common difference (5).****Solution**

1. $(S_n = \frac{n}{2}\{0 + 5(n - 1)\})$.
2. Hence $(S_n = \frac{5n(n-1)}{2})$.

Final answer

$$\frac{5n(n-1)}{2}$$

Q202 **Written****Find (S_n) for a sequence with first term (3) and common difference (-4) .****Solution**

1. $(S_n = \frac{n}{2}\{6 - 4(n - 1)\})$.
2. Therefore $(S_n = 5n - 2n^2)$.

Final answer

$$5n - 2n^2$$

Worked Solutions

Arithmetic Sequences

Q203 Written

Find (S_n) for a sequence with first term (-4) and common difference (-2) .**Solution**

1. $(S_n = \frac{n}{2}\{-8 - 2(n - 1)\})$.
2. Therefore $(S_n = -n^2 - 3n)$.

Final answer

$$-n^2 - 3n$$

Worked Solutions

Arithmetic Sequences

Q204 Challenge

For the arithmetic sequence $(94, 89, 84, \dots)$, find the sum of the odd-numbered terms from the first term to the 11th term.

Solution

1. The odd-numbered terms are $(a_1, a_3, a_5, a_7, a_9, a_{11})$, a new arithmetic sequence with first term (94), common difference $(2(-5) = -10)$, and (6) terms.
2. Its last term is (44), so the sum is $(\frac{6}{2}(94 + 44) = 414)$.

Final answer

414

Worked Solutions

Arithmetic Sequences

Q205 Written

Find the sum of the natural numbers from 1 to 100.**Solution**

1. There are $n = 100$ terms, with $a = 1$ and $l = 100$.
2. $S_{100} = \frac{100}{2}(1 + 100) = 5050$.

Final answer

5050

Q206 Written

Find the sum of the natural numbers from 1 to 200.**Solution**

1. Use $n = 200$, $a = 1$, $l = 200$.
2. $S = \frac{200}{2}(1 + 200) = 20100$.

Final answer

20100

Worked Solutions

Arithmetic Sequences

Q207 **Written****Find the sum of the natural numbers from 1 to 150.****Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 150$.
2. *Therefore* $S = 11325$.

Final answer

11325

Q208 **Written****Find the sum of the natural numbers from 1 to 70.****Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 70$.
2. *Therefore* $S = 2485$.

Final answer

2485

Worked Solutions

Arithmetic Sequences

Q209 Written

Find the sum of the natural numbers from 1 to 100.**Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 100$.
2. *Therefore* $S = 5050$.

Final answer

5050

Q210 Written

Find the sum of the natural numbers from 1 to 400.**Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 400$.
2. *Therefore* $S = 80200$.

Final answer

80200

Worked Solutions

Arithmetic Sequences

Q211 Written

Find the sum of the natural numbers from 1 to 250.**Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 250$.
2. *Therefore* $S = 31375$.

Final answer

31375

Q212 Written

Find the sum of the natural numbers from 1 to 81.**Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 81$.
2. *Therefore* $S = 8085$.

Final answer

8085

Worked Solutions

Arithmetic Sequences

Q213 Written

Find the sum of the natural numbers from 1 to 301.

Solution

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 301$.
2. Therefore $S = 24535$.

Final answer

24535

Q214 Written

Find the sum of the natural numbers from 1 to 99.

Solution

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 99$.
2. Therefore $S = 4950$.

Final answer

4950

Worked Solutions

Arithmetic Sequences

Q215 MCQ

Find the sum of the natural numbers from 1 to 30:**Solution**

$$1. S = \frac{30}{2}(1 + 30) = 465.$$

Correct option: B**Final answer**

465

Q216 MCQ

Find the sum of the natural numbers from 1 to 45:**Solution**

$$1. S = \frac{45}{2}(1 + 45) = 1035.$$

Correct option: B**Final answer**

1035

Worked Solutions

Arithmetic Sequences

Q217 MCQ

Find the sum of the natural numbers from 1 to 60:**Solution**

$$1. S = \frac{60}{2}(1 + 60) = 1830.$$

Correct option: B

Final answer

1830

Q218 MCQ

Find the sum of the natural numbers from 1 to 75:**Solution**

$$1. S = \frac{75}{2}(1 + 75) = 2850.$$

Correct option: B

Final answer

2850

Worked Solutions

Arithmetic Sequences

Q219 MCQ

Find the sum of the natural numbers from 1 to 90:**Solution**

$$1. S = \frac{90}{2}(1 + 90) = 4095.$$

Correct option: B**Final answer**

4095

Q220 MCQ

Find the sum of the natural numbers from 1 to 125:**Solution**

$$1. S = \frac{125}{2}(1 + 125) = 7875.$$

Correct option: B**Final answer**

7875

Worked Solutions

Arithmetic Sequences

Q221 MCQ

Find the sum of the natural numbers from 1 to 180:**Solution**

$$1. S = \frac{180}{2}(1 + 180) = 16290.$$

Correct option: B

Final answer

16290

Q222 MCQ

Find the sum of the natural numbers from 1 to 220:**Solution**

$$1. S = \frac{220}{2}(1 + 220) = 24310.$$

Correct option: B

Final answer

24310

Worked Solutions

Arithmetic Sequences

Q223 MCQ

Find the sum of the natural numbers from 1 to 275:**Solution**

$$1. S = \frac{275}{2}(1 + 275) = 37950.$$

Correct option: B**Final answer**

37950

Q224 MCQ

Find the sum of the natural numbers from 1 to 350:**Solution**

$$1. S = \frac{350}{2}(1 + 350) = 61425.$$

Correct option: B**Final answer**

61425

Worked Solutions

Arithmetic Sequences

Q225 **Written****Find the sum of the natural numbers from 101 to 200.****Solution**

1. There are $n = 100$ terms, with $a = 101$ and $l = 200$.
2. $S = \frac{100}{2}(101 + 200) = 15050$.

Final answer

15050

Q226 **Written****Find the sum of the natural numbers from 201 to 300.****Solution**

1. There are 100 terms from 201 to 300.
2. $S = \frac{100}{2}(201 + 300) = 25050$.

Final answer

25050

Worked Solutions

Arithmetic Sequences

Q227 **Written****Find the sum of all two-digit natural numbers.****Solution**

1. Use $10 + 11 + \cdots + 99 : n = 90$.
2. $S = \frac{90}{2}(10 + 99) = 4905$.

Final answer

4905

Q228 **MCQ****A sequence has first term 5, last term 35, and 7 terms. Select its sum:****Solution**

1. $S = \frac{7}{2}(5 + 35) = 140$.

Correct option: B

Final answer

140

Worked Solutions

Arithmetic Sequences

Q229 MCQ

A sequence has first term 12, last term 68, and 9 terms. Select its sum:

Solution

$$1. S = \frac{9}{2}(12 + 68) = 360.$$

Correct option: B

Final answer

360

Q230 MCQ

A sequence has first term -4, last term 20, and 9 terms. Select its sum:

Solution

$$1. S = \frac{9}{2}(-4 + 20) = 72.$$

Correct option: B

Final answer

72

Worked Solutions

Arithmetic Sequences

Q231 MCQ

A sequence has first term 7, last term 42, and 8 terms. Select its sum:

Solution

1. $S = \frac{8}{2}(7 + 42) = 196.$

Correct option: B

Final answer

196

Q232 MCQ

A sequence has first term 15, last term 3, and 10 terms. Select its sum:

Solution

1. $S = \frac{10}{2}(15 + 3) = 90.$

Correct option: B

Final answer

90

Worked Solutions

Arithmetic Sequences

Q233 MCQ

A sequence has first term 10, last term 70, and 13 terms. Select its sum:

Solution

1. $S = \frac{13}{2}(10 + 70) = 520.$

Correct option: B

Final answer

520

Q234 MCQ

A sequence has first term 2, last term 57, and 12 terms. Select its sum:

Solution

1. $S = \frac{12}{2}(2 + 57) = 354.$

Correct option: B

Final answer

354

Worked Solutions

Arithmetic Sequences

Q235 MCQ

A sequence has first term 25, last term 100, and 16 terms. Select its sum:

Solution

$$1. S = \frac{16}{2}(25 + 100) = 1000.$$

Correct option: B

Final answer

1000

Q236 MCQ

A sequence has first term 101, last term 200, and 100 terms. Select its sum:

Solution

$$1. S = \frac{100}{2}(101 + 200) = 15050.$$

Correct option: B

Final answer

15050

Worked Solutions

Arithmetic Sequences

Q237 MCQ

A sequence has first term 31, last term 91, and 16 terms. Select its sum:

Solution

1. $S = \frac{16}{2}(31 + 91) = 976.$

Correct option: B

Final answer

976

Q238 Written

Find the sum of the multiples of 3 among the two-digit natural numbers.

Solution

1. The sequence is 12, 15, 18, ..., 99, so $a = 12, d = 3, l = 99.$

2. $99 = 12 + (n - 1)3$ gives $n = 30.$

3. $S = \frac{30}{2}(12 + 99) = 1665.$

Final answer

1665

Worked Solutions

Arithmetic Sequences

Q239 Written

Find the sum of all two-digit natural numbers that are multiples of 5.**Solution**

1. The sequence is 10, 15, ..., 95; $n = (95 - 10)/5 + 1 = 18$.
2. $S = \frac{18}{2}(10 + 95) = 945$.

Final answer

945

Q240 Written

Find the sum of the multiples of 4 from 1 to 100.**Solution**

1. The sequence is 4, 8, ..., 100; $n = 25$.
2. $S = \frac{25}{2}(4 + 100) = 1300$.

Final answer

1300

Worked Solutions

Arithmetic Sequences

Q241 **Written****Find the sum of all multiples of 6 from 10 to 100.****Solution**

1. The sequence is 12, 18, ..., 96; $n = 15$.
2. $S = \frac{15}{2}(12 + 96) = 810$.

Final answer

810

Q242 **MCQ****Select the sum of the multiples of 3 from 3 to 99:****Solution**

1. $n = (99 - 3)/3 + 1 = 33$; $S = \frac{33}{2}(3 + 99) = 1683$.

Correct option: B**Final answer**

1683

Worked Solutions

Arithmetic Sequences

Q243 MCQ

Select the sum of the multiples of 4 from 4 to 100:**Solution**

$$1. \ n = (100 - 4)/4 + 1 = 25; S = \frac{25}{2}(4 + 100) = 1300.$$

Correct option: B**Final answer**

1300

Q244 Written

Find the sum of the natural numbers from 1 to 100 that leave a remainder of 1 when divided by 4.**Solution**

$$1. \ \text{The sequence is } 1, 5, \dots, 97; n = 25.$$

$$2. \ S = \frac{25}{2}(1 + 97) = 1225.$$

Final answer

1225

Worked Solutions

Arithmetic Sequences

Q245 Written

Find the sum of all two-digit natural numbers that leave a remainder of 3 when divided by 5.

Solution

1. The sequence is 13, 18, ..., 98; $n = (98 - 13)/5 + 1 = 18$.
2. $S = \frac{18}{2}(13 + 98) = 999$.

Final answer

999

Q246 Written

Find the sum of all two-digit odd natural numbers.

Solution

1. The sequence is 11, 13, ..., 99; $n = 45$.
2. $S = \frac{45}{2}(11 + 99) = 2475$.

Final answer

2475

Worked Solutions**Arithmetic Sequences****Q247** Written**Find the sum of all natural numbers from 10 to 100 that are not divisible by 6.****Solution**

1. Total 10 to 100 is 5005; multiples of 6 are 12 to 96 with sum 810.
2. Required sum = $5005 - 810 = 4195$.

Final answer

4195

Worked Solutions

Arithmetic Sequences

Q248 Challenge

Find the sum of the numbers from 1 to 100 which are not divisible by 4.

Solution

1. Total $S_{100} = 5050$. Multiples of 4 have sum 1300.
2. Required sum = $5050 - 1300 = 3750$.

Final answer

3750

Worked Solutions

Arithmetic Sequences

Q249 **Written**

In an arithmetic sequence with first term 4 and common difference 3, find how many terms from the first term have sum 50.

Solution

1. $50 = \frac{n}{2}\{2(4) + (n-1)3\} \Rightarrow 3n^2 + 5n - 100 = 0.$
2. The natural root is $n = 5$; $S_5 = 50.$

Final answer

5

Q250 **Written**

Find n when $a = 3$, $d = 2$ and $S_n = 63$.

Solution

1. $S_n = n(n+2) = 63 \Rightarrow n = 7.$

Final answer

7

Worked Solutions

Arithmetic Sequences

Q251 MCQ

Select the correct option: Find n if $S_n = 48$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 48 \Rightarrow n = 6$.

Correct option: B

Final answer

6

Q252 MCQ

Select the correct option: Find n if $S_n = 80$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 80 \Rightarrow n = 8$.

Correct option: B

Final answer

8

Worked Solutions

Arithmetic Sequences

Q253 MCQ

Select the correct option: Find n if $S_n = 120$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 120 \Rightarrow n = 10.$$

Correct option: B**Final answer**

10

Q254 MCQ

Select the correct option: Find n if $S_n = 168$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 168 \Rightarrow n = 12.$$

Correct option: B**Final answer**

12

Worked Solutions

Arithmetic Sequences

Q255 MCQ

Select the correct option: Find n if $S_n = 255$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 255 \Rightarrow n = 15.$$

Correct option: B**Final answer**

15

Q256 MCQ

Select the correct option: Find n if $S_n = 360$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 360 \Rightarrow n = 18.$$

Correct option: B**Final answer**

18

Worked Solutions

Arithmetic Sequences

Q257 MCQ

Select the correct option: Find n if $S_n = 440$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 440 \Rightarrow n = 20.$$

Correct option: B**Final answer**

20

Q258 MCQ

Select the correct option: Find n if $S_n = 624$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 624 \Rightarrow n = 24.$$

Correct option: B**Final answer**

24

Worked Solutions

Arithmetic Sequences

Q259 MCQ

Select the correct option: Find n if $S_n = 840$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 840 \Rightarrow n = 28.$$

Correct option: B

Final answer

28

Q260 MCQ

Select the correct option: Find n if $S_n = 1088$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 1088 \Rightarrow n = 32.$$

Correct option: B

Final answer

32

Worked Solutions

Arithmetic Sequences

Q261 MCQ

Select the correct option: Find n if $S_n = 1368$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 1368 \Rightarrow n = 36.$

Correct option: B

Final answer

36

Q262 MCQ

Select the correct option: Find n if $S_n = 1680$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 1680 \Rightarrow n = 40.$

Correct option: B

Final answer

40

Worked Solutions

Arithmetic Sequences

Q263 Written

For first term 17 and common difference -2 , find how many terms have sum -63 .

Solution

1. $S_n = n(18 - n)$.
2. $n(18 - n) = -63 \Rightarrow n^2 - 18n - 63 = 0$; hence $n = 21$.

Final answer

21

Q264 Written

Find n when $a = 24$, $d = -4$ and $S_n = -28$.

Solution

1. $S_n = n(26 - 2n) = -28 \Rightarrow n^2 - 13n - 14 = 0$; $n = 14$.

Final answer

14

Worked Solutions

Arithmetic Sequences

Q265 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 80$:**Solution**

1. $S_n = \frac{n}{2}(17 + 3n) = 80$; positive root $n = 5$.

Correct option: C**Final answer**

5

Q266 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 133$:**Solution**

1. $S_n = \frac{n}{2}(17 + 3n) = 133$; positive root $n = 7$.

Correct option: C**Final answer**

7

Worked Solutions

Arithmetic Sequences

Q267 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 198$:**Solution**

1. $S_n = \frac{n}{2}(17 + 3n) = 198$; positive root $n = 9$.

Correct option: C

Final answer

9

Q268 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 275$:**Solution**

1. $S_n = \frac{n}{2}(17 + 3n) = 275$; positive root $n = 11$.

Correct option: C

Final answer

11

Worked Solutions

Arithmetic Sequences

Q269 Written

Find the general term of $\{a_n\}$ with first term -60 and sum of the first 15 terms -60 .

Solution

1. $-60 = \frac{15}{2}\{2(-60) + 14d\} \Rightarrow d = 8.$
2. $a_n = -60 + 8(n - 1) = 8n - 68.$

Final answer

$$a_n = 8n - 68$$

Q270 Written

Find the general term with first term 8 and sum of the first 13 terms -130 .

Solution

1. $-130 = \frac{13}{2}(16 + 12d) \Rightarrow d = -3.$
2. $a_n = 8 - 3(n - 1) = 11 - 3n.$

Final answer

$$a_n = 11 - 3n$$

Worked Solutions

Arithmetic Sequences

Q271 Written

Find the general term with first term 4 and $S_6 = 114$.**Solution**

$$1. \frac{6}{2}(8 + 5d) = 114 \Rightarrow d = 6.$$

$$2. a_n = 4 + 6(n - 1) = 6n - 2.$$

Final answer

$$a_n = 6n - 2$$

Q272 Written

Find the general term when $a_3 = 10$ and $S_{10} = 175$.**Solution**

$$1. a + 2d = 10, 2a + 9d = 35 \Rightarrow d = 3, a = 4.$$

$$2. a_n = 4 + 3(n - 1) = 3n + 1.$$

Final answer

$$a_n = 3n + 1$$

Worked Solutions

Arithmetic Sequences

Q273 Written

For first term 65 and common difference -4 , find the number of terms giving the maximum sum and the sum.

Solution

1. $a_n = 65 - 4n$; $a_n > 0 \Rightarrow n < 17.25$.
2. The last positive term is $a_{17} = 1$; $S_{17} = \frac{17}{2}(65 + 1) = 561$.

Final answer

$$n = 17, S = 561$$

Q274 Written

For first term 70 and common difference -4 , find the number of terms giving the maximum sum and the sum.

Solution

1. $a_n = 70 - 4n$; $a_{18} = 2$ is the last positive term.
2. $S_{18} = \frac{18}{2}(70 + 2) = 648$.

Final answer

$$n = 18, S = 648$$

Worked Solutions

Arithmetic Sequences

Q275 Written

For $a = 20, d = -6$, find n giving the maximum sum and the sum.

Solution

1. $a_n = 26 - 6n$; the last positive term is $a_4 = 2$.
2. $S_4 = \frac{4}{2}(20 + 2) = 44$.

Final answer

$$n = 4, S = 44$$

Q276 Written

For $a = 126, d = -13$, find n giving the maximum sum and the sum.

Solution

1. $a_n = 139 - 13n$; the last positive term is $a_{10} = 9$.
2. $S_{10} = \frac{10}{2}(126 + 9) = 675$.

Final answer

$$n = 10, S = 675$$

Worked Solutions

Arithmetic Sequences

Q277 Written

Find the general term where $S_{10} = 10$ and $S_{20} = 40$.**Solution**

1. $2a + 9d = 2, 2a + 19d = 4 \Rightarrow d = \frac{1}{5}, a = \frac{1}{10}.$
2. $a_n = \frac{1}{10} + (n-1)\frac{1}{5} = \frac{2n-1}{10}.$

Final answer

$$a_n = \frac{2n-1}{10}$$

Q278 Written

Find the general term where $S_6 = 3$ and $S_{10} = 65$.**Solution**

1. $2a + 5d = 1, 2a + 9d = 13 \Rightarrow d = 3, a = -7.$
2. $a_n = -7 + 3(n-1) = 3n - 10.$

Final answer

$$a_n = 3n - 10$$

Worked Solutions

Arithmetic Sequences

Q279 **Written****Find the general term where $S_{20} = 400$ and $S_{30} = 900$.****Solution**

$$1. \ 2a + 19d = 40, \ 2a + 29d = 60 \Rightarrow d = 2, \ a = 1.$$

$$2. \ a_n = 1 + 2(n - 1) = 2n - 1.$$

Final answer

$$a_n = 2n - 1$$

Worked Solutions

Arithmetic Sequences

Q280 Challenge

The 15th term is 33 and the 50th term is 103. Find the sum of the first 79 terms.

Solution

1. $35d = 70 \Rightarrow d = 2, a = 5.$
2. $S_{79} = \frac{79}{2}\{2(5) + 78(2)\} = 6557.$

Final answer

6557

Worked Solutions

Arithmetic Sequences

Quiz MCQ 1 **Concept Quiz · MCQ**

For $a_n = \frac{2n+1}{n+2}$, what is a_5 ?:

Solution

1. Substitute $n = 5$: $a_5 = (2(5)+1)/(5+2) = 11/7$.

Correct option: A

Final answer

$$\frac{11}{7}$$

Quiz MCQ 2 **Concept Quiz · MCQ**

An arithmetic sequence has $a_3 = 10$ and $a_8 = 30$. Select its general term:

Solution

1. $10 = a + 2d$ and $30 = a + 7d$, so $d = 4$, $a = -2$.

2. $a_n = 4n - 2$.

Correct option: B

Final answer

$$a_n = 4n - 2$$

Worked Solutions

Arithmetic Sequences

Quiz MCQ 3 **Concept Quiz · MCQ**

For $a = 12$, $d = -2$, select the sum of odd-numbered terms through a_9 :

Solution

1. Odd terms are 12, 8, 4, 0, -4 ; their sum is 20.

Correct option: C

Final answer

20

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: If $S_{10} = 10$ and $S_{20} = 40$, find d :

Solution

1. $d = \frac{1}{5}$.

Correct option: B

Final answer

$\frac{1}{5}$

Worked Solutions

Arithmetic Sequences

Quiz WRITTEN 5 Concept Quiz · Written

If $a_n = 3n^2 - 2n + 1$, find a_1, a_2, a_3 .

Solution

1. Substitution gives $a_1=2$, $a_2=9$, and $a_3=22$.

Final answer

$$a_1 = 2, a_2 = 9, a_3 = 22$$

Quiz WRITTEN 6 Concept Quiz · Written

Find a_n for an arithmetic sequence with $a_4 = 14$ and $a_{10} = 62$.

Solution

1. $14 = a + 3d$ and $62 = a + 9d$, so $d = 8$, $a = -10$.

2. $a_n = 8n - 18$.

Final answer

$$a_n = 8n - 18$$

Worked Solutions

Arithmetic Sequences

Quiz WRITTEN 7 Concept Quiz · Written

For $a = 9, d = 4$, find the sum of odd-numbered terms from a_1 to a_{15} .

Solution

1. Selected terms have 8 terms, first 9, last 65.
2. $\text{Sum} = 8(9 + 65)/2 = 296$.

Final answer

296

Quiz WRITTEN 8 Concept Quiz · Written

Find the general term when $S_{20} = 400$ and $S_{30} = 900$.

Solution

1. $d = 2, a = 1$; therefore $a_n = 2n - 1$.

Final answer

$a_n = 2n - 1$

Answer key

Use the question number shown on each page.

Q001 $a_1 = -1, a_2 = 1, a_3 = 3$

Q002 $a_1 = -6, a_2 = -10, a_3 = -12$

Q003 $a_1 = -3, a_2 = 2, a_3 = 7$

Q004 $a_1 = -2, a_2 = -5, a_3 = -8$

Q005 $a_1 = -6, a_2 = -8, a_3 = -10$

Q006 $a_1 = 0, a_2 = 3, a_3 = 8$

Q007 $a_1 = 3, a_2 = 7, a_3 = 13$

Q008 $a_1 = 7, a_2 = 12, a_3 = 19$

Q009 B

Q010 C

Q011 D

Q012 A

Q013 B

Q014 C

Q015 D

Q016 A

Q017 B

Q018 C

Q019 D

Q020 A

Q021 $a_1 = 6, a_2 = 18, a_3 = 54$

Q022 $a_1 = 2, a_2 = 8, a_3 = 26$

Q023 $a_1 = -3, a_2 = 9, a_3 = -27$

Q024 $a_1 = -6, a_2 = 12, a_3 = -24$

Q025 A

Q026 B

Q027 C

Q028 D

Q029 A

Q030 B

Q031 C

Q032 D

Q033 $a_1 = \frac{1}{2}, a_2 = \frac{2}{3}, a_3 = \frac{3}{4}$

Q034 $a_1 = \frac{1}{3}, a_2 = \frac{2}{5}, a_3 = \frac{3}{7}$

Q035 $a_1 = 1, a_2 = \frac{1}{4}, a_3 = \frac{1}{7}$

Q036 $a_1 = \frac{1}{2}, a_2 = \frac{4}{5}, a_3 = 1$

Q037 $a_1 = 2, a_2 = 1, a_3 = \frac{4}{5}$

Q038 C

Q039 D

Q040 A

Q041 B

Q042 C

Q043 D

Q044 $a_n = (n-1)^2$

Q045 $a_n = (n-1)^3$

Q046 $a_n = n^2$

Q047 $a_n = (n+6)^2$

Q048 B

Q049 C

Q050 D

Q051 A

Q052 B

Q053 C

Q054 $a_n = 3n - 1$

Q055 $a_n = 2n - 2$

Q056 $a_n = 3n - 3$

Q057 $a_n = 5n - 1$

Q058 $a_n = 7n - 1$

Q059 A

Q060 B

Q061 C

Q062 D

Q063 $a_n = (-1)^{n-1} 2n$

Q064 $a_n = (-1)^n 3n$

Q065 $a_n = (-1)^n 5n$

Q066 $a_n = (-1)^n (3n - 2)$

Q067 $a_n = \frac{n}{2n-1}$

Q068 $a_n = \frac{2n-1}{n^2}$

Q069 $a_n = \frac{(-1)^{n-1}}{2n-1}$

Q070 $a_n = \frac{n}{n+1}$

Q071 $a_n = n 3^n$

Q072 $a_n = \frac{n}{n+1}; a_{20} = \frac{20}{21}$

Q073 (a) $a = 2, d = 3(b) a = 3, d = -5$

Q074 (a) $a = 1, d = 4(b) a = 23, d = -4$

Q075 B

Q076 A

Q077 C

Q078 D

Q079 (a)(3, 2); (b)(-13, 4); (c)(-2, 3); (d)(10, -3); (e)(6, $-\frac{2}{3}$); (f)(-4, $-\frac{1}{2}$).

Q080 B

Q081 A

Q082 C

Q083 B

Q084 (a) $d = -3, A = 5, B = -4(b) d = -5, A = 8, B = 3$

Q085 (a) $d = 7, A = 16, B = 23(b) d = -6, A = 23, B = 11$

Q086 A

Q087 C

Q088 B

Q089 (a)(d) = 7, A = 22, B = 20; (b)(d) = 6, A = 19, B = 21; (c)(d) = -5, A = 14, B = -1; (d)(d) = -2, A = 17, B = 13; (e)(d) = 3, A = 14, B = 17; (f)(d) = -4, A = 13, B = 9, C = 1.

Q090 (a) $a_n = 3n + 1, a_7 = 22(b) a_n = 5 - 3n, a_7 = -16$

Q091 (a) $a_n = 4n + 1, a_7 = 29$; (b) $a_n = 3n - 5, a_7 = 16$; (c) $a_n = 13 - 3n, a_7 = -8$; (d) $a_n = -\frac{n}{2}, a_7 = -\frac{7}{2}$.

Q092 $a_n = 8 + (n-1)(-6) = 14 - 6n; a_{20} = -106$

Q093 (a) $a_n = 3n - 10, a_{10} = 20$; (b) $a_n = 1 - 3n, a_{10} = -29$; (c) $a_n = \frac{n+13}{2}, a_{10} = \frac{23}{2}$; (d) $a_n = \frac{3-n}{3}, a_{10} = -\frac{7}{3}$.

Q094 $a = 7, d = 3$.

Q095 $a_n = 5n - 10$

Q096 $a_n = 4n + 2$

Q097 $a_n = 23 - 3n$

Q098 $a_n = 9n - 17$

Q099 $a_n = -5n - 13$

Q100 D

Q101 B	Q133 A	Q165 144	Q197 C
Q102 A	Q134 B	Q166 -165	Q198 C
Q103 B	Q135 $n = 125$, the 125th term	Q167 176	Q199 $2n^2 - n$
Q104 $a_n = 3n + 33$	Q136 $n = 35$, the 35th term	Q168 360	Q200 $\frac{n(7-3n)}{2}$
Q105 $a_n = 2n - 12$	Q137 A	Q169 C	Q201 $\frac{5n(n-1)}{2}$
Q106 $a_n = 13 - 4n$	Q138 D	Q170 B	Q202 $5n - 2n^2$
Q107 $a_n = 18 - 7n$	Q139 D	Q171 C	Q203 $-n^2 - 3n$
Q108 A	Q140 B	Q172 B	Q204 414
Q109 D	Q141 $n = 72$, the 72nd term	Q173 C	Q205 5050
Q110 C	Q142 $n = 171$, the 171st term	Q174 B	Q206 20100
Q111 D	Q143 B	Q175 392	Q207 11325
Q112 $a_n = 8n - 14$	Q144 B	Q176 343	Q208 2485
Q113 $a_n = 20 - 3n$	Q145 C	Q177 741	Q209 5050
Q114 $a_n = 8n - 18$	Q146 A	Q178 354	Q210 80200
Q115 $a_n = 42 - 4n$	Q147 B	Q179 1716	Q211 31375
Q116 $a_n = 79 - 3n$	Q148 $a = 4$	Q180 C	Q212 8085
Q117 $a_n = 8n - 18$	Q149 $x = -4$ or $x = 3$	Q181 C	Q213 24535
Q118 C	Q150 $x = -2$ or $x = 4$	Q182 C	Q214 4950
Q119 C	Q151 $x = -1$ or $x = 3$	Q183 C	Q215 B
Q120 B	Q152 $x = 0$, $y = -2$, $z = -4$	Q184 C	Q216 B
Q121 A	Q153 105	Q185 B	Q217 B
Q122 A	Q154 100	Q186 0	Q218 B
Q123 B	Q155 120	Q187 208	Q219 B
Q124 $a_n = \frac{2n-37}{5}$	Q156 152	Q188 B	Q220 B
Q125 A	Q157 155	Q189 C	Q221 B
Q126 D	Q158 B	Q190 C	Q222 B
Q127 $n = 28$, the 28th term	Q159 B	Q191 C	Q223 B
Q128 $n = 14$, the 14th term	Q160 B	Q192 B	Q224 B
Q129 $n = 22$, the 22nd term	Q161 B	Q193 C	Q225 15050
Q130 $n = 40$, the 40th term	Q162 B	Q194 66	Q226 25050
Q131 C	Q163 C	Q195 750	Q227 4905
Q132 D	Q164 -130	Q196 154	

Q228 B	Q244 1225	Q260 B	Q276 $n = 10, S = 675$
Q229 B	Q245 999	Q261 B	Q277 $a_n = \frac{2n-1}{10}$
Q230 B	Q246 2475	Q262 B	Q278 $a_n = 3n - 10$
Q231 B	Q247 4195	Q263 21	Q279 $a_n = 2n - 1$
Q232 B	Q248 3750	Q264 14	Q280 6557
Q233 B	Q249 5	Q265 C	Quiz MCQ 1 A
Q234 B	Q250 7	Q266 C	Quiz MCQ 2 B
Q235 B	Q251 B	Q267 C	Quiz MCQ 3 C
Q236 B	Q252 B	Q268 C	Quiz MCQ 4 B
Q237 B	Q253 B	Q269 $a_n = 8n - 68$	Quiz WRITTEN 5 $a_1 = 2, a_2 = 9, a_3 = 22$
Q238 1665	Q254 B	Q270 $a_n = 11 - 3n$	Quiz WRITTEN 6 $a_n = 8n - 18$
Q239 945	Q255 B	Q271 $a_n = 6n - 2$	Quiz WRITTEN 7 296
Q240 1300	Q256 B	Q272 $a_n = 3n + 1$	Quiz WRITTEN 8 $a_n = 2n - 1$
Q241 810	Q257 B	Q273 $n = 17, S = 561$	
Q242 B	Q258 B	Q274 $n = 18, S = 648$	
Q243 B	Q259 B	Q275 $n = 4, S = 44$	

Concept 2 | Geometric Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1}$$

Infinite sum

$$S_{\infty} = \frac{a}{1-r} \quad (|r| < 1)$$

Worked Solutions

Geometric Sequences

Q001 Written

Find the general term of a geometric sequence whose first term is 2 and common ratio is 3.

Solution

1. Here $a = 2$ and $r = 3$.
2. $a_n = 2 \times 3^{n-1}$

Final answer

$$2 \times 3^{n-1}$$

Q002 Written

Find the general term when the first term is 3 and the common ratio is 4.

Solution

1. $a_n = 3 \times 4^{n-1}$

Final answer

$$3 \times 4^{n-1}$$

Worked Solutions

Geometric Sequences

Q003 Written

Find the general term of the geometric sequence in part a. The first term is 5, common ratio 2.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $5 \times 2^{n-1}$.

Final answer

$$5 \times 2^{n-1}$$

Q004 Written

Find the general term of the geometric sequence in part b. The first term is 2, common ratio 5.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $2 \times 5^{n-1}$.

Final answer

$$2 \times 5^{n-1}$$

Worked Solutions

Geometric Sequences

Q005 **Written**

Find the general term of the geometric sequence in part c. The first term is 7, common ratio 4.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $7 \times 4^{n-1}$.

Final answer

$$7 \times 4^{n-1}$$

Q006 **MCQ**

Select the correct option: For first term $a = 2$, common ratio $r = 3$, and $n = 5$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 2$ and $r = 3$.

Correct option: A

Final answer

$$a_n = 2(3)^4$$

Worked Solutions

Geometric Sequences

Q007 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = 2$, and $n = 6$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 3$ and $r = 2$.

Correct option: A

Final answer

$$a_n = 3(2)^5$$

Q008 MCQ

Select the correct option: For first term $a = 5$, common ratio $r = 4$, and $n = 4$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 5$ and $r = 4$.

Correct option: A

Final answer

$$a_n = 5(4)^3$$

Worked Solutions

Geometric Sequences

Q009 MCQ

Select the correct option: For first term $a = 4$, common ratio $r = 5$, and $n = 3$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 4$ and $r = 5$.

Correct option: A

Final answer

$$a_n = 4(5)^2$$

Q010 MCQ

Select the correct option: For first term $a = 5$, common ratio $r = 2$, and $n = 7$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 5$ and $r = 2$.

Correct option: A

Final answer

$$a_n = 5(2)^6$$

Worked Solutions

Geometric Sequences

Q011 MCQ

Select the correct option: For first term $a = 7$, common ratio $r = 4$, and $n = 3$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 7$ and $r = 4$.

Correct option: A

Final answer

$$a_n = 7(4)^2$$

Q012 Written

Find the general term when the first term is 5 and the common ratio is -2 .

Solution

1. $a_n = 5(-2)^{n-1}$

Final answer

$$5(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q013 Written

Find the general term of the geometric sequence in part d. The first term is 2, common ratio -3 .

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $2(-3)^{n-1}$.

Final answer

$$2(-3)^{n-1}$$

Q014 Written

Find the general term of the geometric sequence in part e. The first term is 3, common ratio -5 .

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3(-5)^{n-1}$.

Final answer

$$3(-5)^{n-1}$$

Worked Solutions

Geometric Sequences

Q015 Written

Find the general term of the geometric sequence in part f. The first term is 4, common ratio -2 .

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $4(-2)^{n-1}$.

Final answer

$$4(-2)^{n-1}$$

Q016 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = -3$, and $n = 5$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 2$ and $r = -3$.

Correct option: A

Final answer

$$a_n = 2(-3)^4$$

Worked Solutions

Geometric Sequences

Q017 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = -2$, and $n = 6$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 3$ and $r = -2$.

Correct option: A

Final answer

$$a_n = 3(-2)^5$$

Q018 MCQ

Select the correct option: For first term $a = 6$, common ratio $r = -1$, and $n = 4$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 6$ and $r = -1$.

Correct option: A

Final answer

$$a_n = 6(-1)^3$$

Worked Solutions

Geometric Sequences

Q019 MCQ

Select the correct option: For first term $a = 8$, common ratio $r = -2$, and $n = 3$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 8$ and $r = -2$.

Correct option: A

Final answer

$$a_n = 8(-2)^2$$

Q020 MCQ

Select the correct option: For first term $a = 9$, common ratio $r = -2$, and $n = 5$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 9$ and $r = -2$.

Correct option: A

Final answer

$$a_n = 9(-2)^4$$

Worked Solutions

Geometric Sequences

Q021 Written

Find the general term when the first term is 8 and the common ratio is $-\frac{1}{3}$.

Solution

1. $a_n = 8\left(-\frac{1}{3}\right)^{n-1}$

Final answer

$$8\left(-\frac{1}{3}\right)^{n-1}$$

Q022 Written

Find the general term of the geometric sequence in part g. The first term is 4, common ratio $\frac{1}{3}$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $4\left(\frac{1}{3}\right)^{n-1}$.

Final answer

$$4\left(\frac{1}{3}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q023 Written

Find the general term of the geometric sequence in part h. The first term is 3, common ratio $-\frac{1}{2}$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3\left(-\frac{1}{2}\right)^{n-1}$.

Final answer

$$3\left(-\frac{1}{2}\right)^{n-1}$$

Q024 Written

Find the general term of the geometric sequence in part i. The first term is 6, common ratio $\frac{1}{2}$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $6\left(\frac{1}{2}\right)^{n-1}$.

Final answer

$$6\left(\frac{1}{2}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q025 MCQ

Select the correct option: For first term $a = 4$, common ratio $r = \frac{1}{3}$, and $n = 5$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 4$ and $r = \frac{1}{3}$.

Correct option: A

Final answer

$$a_n = 4 \left(\frac{1}{3} \right)^4$$

Q026 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = -\frac{1}{2}$, and $n = 6$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 3$ and $r = -\frac{1}{2}$.

Correct option: A

Final answer

$$a_n = 3 \left(-\frac{1}{2} \right)^5$$

Worked Solutions

Geometric Sequences

Q027 MCQ

Select the correct option: For first term $a = 6$, common ratio $r = \frac{1}{2}$, and $n = 4$, which expression gives a_n ?

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 6$ and $r = \frac{1}{2}$.

Correct option: A

Final answer

$$a_n = 6 \left(\frac{1}{2} \right)^3$$

Q028 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = \frac{3}{2}$, and $n = 5$, which expression gives a_n ?

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 2$ and $r = \frac{3}{2}$.

Correct option: A

Final answer

$$a_n = 2 \left(\frac{3}{2} \right)^4$$

Worked Solutions

Geometric Sequences

Q029 MCQ

Select the correct option: For first term $a = 8$, common ratio $r = -\frac{3}{2}$, and $n = 4$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 8$ and $r = -\frac{3}{2}$.

Correct option: A

Final answer

$$a_n = 8 \left(-\frac{3}{2}\right)^3$$

Q030 Written

Find the general term of 5, 10, 20, 40,

Solution

1. The common ratio is $r = 10/5 = 2$.
2. $a_n = 5 \times 2^{n-1}$

Final answer

$$5 \times 2^{n-1}$$

Worked Solutions

Geometric Sequences

Q031 Written

Find the general term of 2, 6, 18, 54,**Solution**

1. The common ratio is 3.
2. $a_n = 2 \times 3^{n-1}$

Final answer

$$2 \times 3^{n-1}$$

Q032 Written

Find the general term of the geometric sequence in part j. The sequence is 3, 6, 12, 24,**Solution**

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3 \times 2^{n-1}$.

Final answer

$$3 \times 2^{n-1}$$

Worked Solutions

Geometric Sequences

Q033 MCQ

Select the correct option: Find the general term of the geometric sequence 3, 6, 12, 24, ...:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$3 \times 2^{n-1}$$

Q034 MCQ

Select the correct option: Find the general term of the geometric sequence 2, 10, 50, 250, ...:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$2 \times 5^{n-1}$$

Worked Solutions

Geometric Sequences

Q035 MCQ

Select the correct option: Find the general term of the geometric sequence 4, 8, 16, 32, ...:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$4 \times 2^{n-1}$$

Q036 MCQ

Select the correct option: Find the general term of the geometric sequence 7, 21, 63, 189, ...:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$7 \times 3^{n-1}$$

Worked Solutions

Geometric Sequences

Q037 MCQ

Select the correct option: Find the general term of the geometric sequence 1, 4, 16, 64, ...:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$4^{n-1}$$

Q038 Written

Find the general term of the geometric sequence in part k. The sequence is 2, -6, 18, -54,

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $2(-3)^{n-1}$.

Final answer

$$2(-3)^{n-1}$$

Worked Solutions

Geometric Sequences

Q039 Written

Find the general term of the geometric sequence in part I. The sequence is 5, -10, 20, -40, ...

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $5(-2)^{n-1}$.

Final answer

$$5(-2)^{n-1}$$

Q040 MCQ

Select the correct option: Find the general term of the geometric sequence 5, -10, 20, -40, ...:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$5(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q041 MCQ

Select the correct option: Find the general term of the geometric sequence $2, -6, 18, -54, \dots$:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$2(-3)^{n-1}$$

Q042 MCQ

Select the correct option: Find the general term of the geometric sequence $-4, 8, -16, 32, \dots$:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$-4(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q043 MCQ

Select the correct option: Find the general term of the geometric sequence 6, −18, 54, −162, ...:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$6(-3)^{n-1}$$

Q044 Written

Find the general term of $-4, 6, -9, \frac{27}{2}, \dots$ **Solution**1. The common ratio is $r = 6/(-4) = -3/2$.

2. $a_n = -4\left(-\frac{3}{2}\right)^{n-1}$

Final answer

$$-4\left(-\frac{3}{2}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q045 Written

Find the general term of $-8, 12, -18, 27, \dots$ **Solution**

1. The common ratio is $-3/2$.
2. $a_n = -8\left(-\frac{3}{2}\right)^{n-1}$

Final answer

$$-8\left(-\frac{3}{2}\right)^{n-1}$$

Q046 Written

Find the general term of $2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \dots$ **Solution**

1. The common ratio is $1/3$.
2. $a_n = 2\left(\frac{1}{3}\right)^{n-1}$

Final answer

$$2\left(\frac{1}{3}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q047 Written

Find the general term of the geometric sequence in part m. The sequence is 48, 24, 12, 6,

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $48\left(\frac{1}{2}\right)^{n-1}$.

Final answer

$$48\left(\frac{1}{2}\right)^{n-1}$$

Q048 Written

Find the general term of the geometric sequence in part n. The sequence is $\frac{1}{6}, \frac{1}{4}, \frac{3}{8}, \frac{9}{16}, \dots$

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$.

Final answer

$$\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q049 Written

Find the general term of the geometric sequence in part o. The sequence is $8, 16\sqrt{2}, 64, 128\sqrt{2}, \dots$

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $8(2\sqrt{2})^{n-1}$.

Final answer

$$8(2\sqrt{2})^{n-1}$$

Q050 Written

Find the general term of the geometric sequence in part p. The sequence is $3, -\frac{15}{2}, \frac{75}{4}, -\frac{375}{8}, \dots$

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3\left(-\frac{5}{2}\right)^{n-1}$.

Final answer

$$3\left(-\frac{5}{2}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q051 Written

Find the general term of the geometric sequence in part q. The sequence is $9, -3\sqrt{3}, 3, -\sqrt{3}, \dots$

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$.

Final answer

$$9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$$

Q052 Written

Find the general term of the geometric sequence in part r. The sequence is $5, -\frac{10}{3}, \frac{20}{9}, -\frac{40}{27}, \dots$

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $5\left(-\frac{2}{3}\right)^{n-1}$.

Final answer

$$5\left(-\frac{2}{3}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q053 MCQ

Select the correct option: Find the general term of the geometric sequence $-8, 12, -18, 27, \dots$:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$-8\left(-\frac{3}{2}\right)^{n-1}$$

Q054 MCQ

Select the correct option: Find the general term of the geometric sequence $48, 24, 12, 6, \dots$:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$48\left(\frac{1}{2}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q055 MCQ

Select the correct option: Find the general term of the geometric sequence $\frac{1}{6}, \frac{1}{4}, \frac{3}{8}, \dots$:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$\frac{1}{6} \left(\frac{3}{2} \right)^{n-1}$$

Q056 MCQ

Select the correct option: Find the general term of the geometric sequence $8, 16\sqrt{2}, 64, \dots$:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$8 \left(2\sqrt{2} \right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q057 MCQ

Select the correct option: Find the general term of the geometric sequence $9, -3\sqrt{3}, 3, \dots$:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$$

Q058 MCQ

Select the correct option: Find the general term of the geometric sequence $3, -\frac{3}{2}, \frac{3}{4}, \dots$:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$3\left(-\frac{1}{2}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q059 MCQ

Select the correct option: Find the general term of the geometric sequence $10, 5, \frac{5}{2}, \dots$:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$10\left(\frac{1}{2}\right)^{n-1}$$

Q060 Written

Find the common ratio and missing values in $A, -18, 6, B, \dots$

Solution

1. $r = 6/(-18) = -\frac{1}{3}$

2. $A = 54, \quad B = 6\left(-\frac{1}{3}\right) = -2$

Final answer

$$r = -\frac{1}{3}, A = 54, B = -2$$

Worked Solutions

Geometric Sequences

Q061 MCQ

Select the correct option: The sequence $A, 12, 6, B, \dots$ is geometric. Find r, A, B :

Solution

1. Divide adjacent known terms to find r , then extend one step in each direction.

Correct option: A

Final answer

$$r = \frac{1}{2}, A = 24, B = 3$$

Q062 Written

Find the common ratio and the missing values in $A, 24, 12, B, \dots$

Solution

1. $24r = 12 \Rightarrow r = \frac{1}{2}$

2. $A\left(\frac{1}{2}\right) = 24 \Rightarrow A = 48, \quad B = 12\left(\frac{1}{2}\right) = 6$

Final answer

$$r = \frac{1}{2}, A = 48, B = 6$$

Worked Solutions

Geometric Sequences

Q063 Written

Find the common ratio and missing values in $A, -3, 9, B$.**Solution**

$$1. \ r = -3, \ A = 1, \ B = -27$$

Final answer

$$r = -3, \ A = 1, \ B = -27$$

Q064 Written

Find the common ratio and missing values in $A, \frac{1}{3}, \frac{1}{9}, B$.**Solution**

$$1. \ r = \frac{1}{3}, \ A = 1, \ B = \frac{1}{27}$$

Final answer

$$r = \frac{1}{3}, \ A = 1, \ B = \frac{1}{27}$$

Worked Solutions

Geometric Sequences

Q065 Written

Find the common ratio and missing values in $A, -\frac{1}{4}, \frac{1}{8}, B$.

Solution

1. $r = -\frac{1}{2}, \quad A = \frac{1}{2}, \quad B = -\frac{1}{16}$

Final answer

$$r = -\frac{1}{2}, A = \frac{1}{2}, B = -\frac{1}{16}$$

Worked Solutions

Geometric Sequences

Q066 Challenge

In an increasing geometric sequence, if $a_3 + a_5 = 80$ and $a_4 = 24$, find a_7 .

Solution

1. $a_3 = \frac{24}{r}$, $a_5 = 24r$, $24\left(r + \frac{1}{r}\right) = 80$

2. $3r^2 - 10r + 3 = 0 \Rightarrow r = 3$ or $\frac{1}{3}$

3. The sequence is increasing, so $r = 3$; hence $a_7 = 24(3)^3 = 648$.

Final answer

$$a_7 = 648$$

Worked Solutions

Geometric Sequences

Q067 Written

Find the general term of a geometric sequence whose 4th term is 24 and 6th term is 96.

Solution

1. $r^2 = 96/24 = 4 \Rightarrow r = \pm 2$
2. $(a, r) = (3, 2)$ or $(-3, -2)$
3. Therefore both real branches are retained.

Final answer

$$a_n = 3(2)^{n-1} \text{ or } -3(-2)^{n-1}$$

Q068 Written

Find the general term when the 3rd term is 18 and the 5th term is 162.

Solution

1. $r^2 = 162/18 = 9 \Rightarrow r = \pm 3$
2. $a = 18/r^2 = 2$

Final answer

$$a_n = 2(3)^{n-1} \text{ or } 2(-3)^{n-1}$$

Worked Solutions

Geometric Sequences

Q069 **Written****Find the requested values. The 2nd term is 6 and the 4th term is 54.****Solution**

1. $r^2 = 54/6 = 9 \Rightarrow r = \pm 3$

2. $a = 6/r$

Final answer

$$a_n = 2(3)^{n-1} \text{ or } -2(-3)^{n-1}$$

Q070 **Written****Find the requested values. The 3rd term is 27 and the 5th term is 243.****Solution**

1. $r^2 = 243/27 = 9 \Rightarrow r = \pm 3$

2. $a = 3$

Final answer

$$a_n = 3(3)^{n-1} \text{ or } 3(-3)^{n-1}$$

Worked Solutions

Geometric Sequences

Q071 **Written****Find the requested values. The 2nd term is 14 and the 5th term is 112.****Solution**

1. $r^3 = 112/14 = 8 \Rightarrow r = 2$
2. $a = 7$

Final answer

$$a_n = 7(2)^{n-1}$$

Q072 **Written****Find the general term when the 3rd term is 12 and the 6th term is -96 .****Solution**

1. $r^3 = -96/12 = -8 \Rightarrow r = -2$
2. $a = 12/r^2 = 3$
3. $a_n = 3(-2)^{n-1}$

Final answer

$$a_n = 3(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q073 Written

Find the general term when the 2nd term is -6 and the 5th term is 48 .

Solution

$$1. r^3 = 48/(-6) = -8 \Rightarrow r = -2$$

$$2. a = -6/r = 3$$

Final answer

$$a_n = 3(-2)^{n-1}$$

Q074 Written

Find the general term when the 3rd term is 12 and the 7th term is 192 .

Solution

$$1. r^4 = 192/12 = 16 \Rightarrow r = \pm 2$$

$$2. a = 12/r^2 = 3$$

Final answer

$$a_n = 3(2)^{n-1} \text{ or } 3(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q075 Written

Find the requested values. The 3rd term is -28 and the 5th term is -112 .

Solution

1. $r^2 = 4 \Rightarrow r = \pm 2$
2. $a = -28/4 = -7$

Final answer

$$a_n = -7(2)^{n-1} \text{ or } -7(-2)^{n-1}$$

Q076 Written

Find the requested values. The 4th term is 16 and the 7th term is -128 .

Solution

1. $r^3 = -8 \Rightarrow r = -2$
2. $a = -2$

Final answer

$$a_n = -2(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q077 Written

Find the requested values. The 3rd term is -20 and the 6th term is 160 .

Solution

1. $r^3 = -8 \Rightarrow r = -2$
2. $a = -5$

Final answer

$$a_n = -5(-2)^{n-1}$$

Q078 Written

Find the requested values. The 2nd term is 3 and the 6th term is 768 .

Solution

1. $r^4 = 768/3 = 256 \Rightarrow r = \pm 4$
2. $a = 3/r$

Final answer

$$a_n = \frac{3}{4}(4)^{n-1} \text{ or } -\frac{3}{4}(-4)^{n-1}$$

Worked Solutions

Geometric Sequences

Q079 Written

Find the requested values. The 3rd term is 9 and the 7th term is $729/16$.

Solution

1. $r^4 = 81/16 \Rightarrow r = \pm \frac{3}{2}$
2. $a = 4$

Final answer

$$a_n = 4\left(\frac{3}{2}\right)^{n-1} \text{ or } 4\left(-\frac{3}{2}\right)^{n-1}$$

Q080 Written

Find the requested values. The 4th term is $2/9$ and the 8th term is 18.

Solution

1. $r^4 = 81 \Rightarrow r = \pm 3$
2. $a = \frac{2/9}{r^3} = \pm \frac{2}{243}$

Final answer

$$a_n = \frac{2}{243}(3)^{n-1} \text{ or } -\frac{2}{243}(-3)^{n-1}$$

Worked Solutions

Geometric Sequences

Q081 MCQ

Select the correct option: For $a = 2$, $r = 3$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 2(3)^{n-1}$$

Q082 MCQ

Select the correct option: For $a = 3$, $r = 2$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 3(2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q083 MCQ

Select the correct option: For $a = 5$, $r = 4$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 5(4)^{n-1}$$

Q084 MCQ

Select the correct option: For $a = 4$, $r = 5$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 4(5)^{n-1}$$

Worked Solutions

Geometric Sequences

Q085 MCQ

Select the correct option: For $a = 2$, $r = -3$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 2(-3)^{n-1}$$

Q086 MCQ

Select the correct option: For $a = 3$, $r = -2$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 3(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q087 MCQ

Select the correct option: For $a = 6$, $r = -1$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 6(-1)^{n-1}$$

Q088 MCQ

Select the correct option: For $a = 8$, $r = -2$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 8(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q089 MCQ

Select the correct option: For $a = 4$, $r = \frac{1}{3}$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 4 \left(\frac{1}{3} \right)^{n-1}$$

Q090 MCQ

Select the correct option: For $a = 3$, $r = -\frac{1}{2}$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 3 \left(-\frac{1}{2} \right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q091 MCQ

Select the correct option: For $a = 6$, $r = \frac{1}{2}$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 6 \left(\frac{1}{2} \right)^{n-1}$$

Q092 MCQ

Select the correct option: For $a = 2$, $r = \frac{3}{2}$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 2 \left(\frac{3}{2} \right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q093 MCQ

Select the correct option: For $a = 8$, $r = -\frac{3}{2}$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 8 \left(-\frac{3}{2} \right)^{n-1}$$

Q094 MCQ

Select the correct option: For $a = 5$, $r = 2$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 5 (2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q095 MCQ

Select the correct option: For $a = 7, r = 4$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 7(4)^{n-1}$$

Q096 MCQ

Select the correct option: For $a = 9, r = -2$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 9(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q097 **Written**

When 2, x , 8 form a geometric sequence in this order, find x .

Solution

1. $x^2 = 2 \times 8 = 16$

2. $x = \pm 4$

Final answer

$x = \pm 4$

Q098 **Written**

Find the requested values. When 1, x , 25 form a geometric sequence, find x .

Solution

1. $x^2 = 25 \Rightarrow x = \pm 5$

Final answer

$x = \pm 5$

Worked Solutions

Geometric Sequences

Q099 Written

Find the requested values. When $1/2, x, 8$ form a geometric sequence, find x .

Solution

$$1. x^2 = \frac{1}{2} \times 8 = 4 \Rightarrow x = \pm 2$$

Final answer

$$x = \pm 2$$

Q100 Written

When $1, x, 12$ form a geometric sequence in this order, find x .

Solution

$$1. x^2 = 1 \times 12 = 12$$

$$2. x = \pm 2\sqrt{3}$$

Final answer

$$x = \pm 2\sqrt{3}$$

Worked Solutions

Geometric Sequences

Q101 MCQ

Select the correct option: If x is the middle term of the geometric triple $2, x, 8$, find x :

Solution

1. Use $x^2 = (2)(8)$.

Correct option: A

Final answer

± 4

Q102 MCQ

Select the correct option: If x is the middle term of the geometric triple $1, x, 25$, find x :

Solution

1. Use $x^2 = (1)(25)$.

Correct option: A

Final answer

± 5

Worked Solutions

Geometric Sequences

Q103 MCQ

Select the correct option: If x is the middle term of the geometric triple $\frac{1}{2}, x, 8$, find x :

Solution

1. Use $x^2 = (\frac{1}{2})(8)$.

Correct option: A

Final answer

± 2

Q104 MCQ

Select the correct option: If x is the middle term of the geometric triple $3, x, 12$, find x :

Solution

1. Use $x^2 = (3)(12)$.

Correct option: A

Final answer

± 6

Worked Solutions

Geometric Sequences

Q105 MCQ

Select the correct option: If x is the middle term of the geometric triple 4, x , 9, find x :

Solution

1. Use $x^2 = (4)(9)$.

Correct option: A

Final answer

± 6

Q106 MCQ

Select the correct option: If x is the middle term of the geometric triple 2, x , 18, find x :

Solution

1. Use $x^2 = (2)(18)$.

Correct option: A

Final answer

± 6

Worked Solutions

Geometric Sequences

Q107 Written

The numbers $6, a, b$ form an arithmetic sequence and $a, b, 16$ form a geometric sequence. Find a, b .

Solution

1. $2a = 6 + b \Rightarrow b = 2a - 6$
2. $b^2 = 16a \Rightarrow 4a^2 - 40a + 36 = 0$
3. $(a, b) = (9, 12)$ or $(1, -4)$

Final answer

$$(a, b) = (9, 12) \text{ or } (1, -4)$$

Q108 Written

Find the requested values. The numbers $a, b, 2$ form an arithmetic sequence and $a, 2, b$ form a geometric sequence. Find a, b .

Solution

1. $2b = a + 2$
2. $b^2 = 2a$
3. $(a, b) = (2, 2)$ or $(-4, -1)$

Final answer

$$(a, b) = (2, 2) \text{ or } (-4, -1)$$

Worked Solutions

Geometric Sequences

Q109 Written

The numbers $-2, a, b$ form an arithmetic sequence and $a, b, 18$ form a geometric sequence. Find a, b .

Solution

1. $b = 2a + 2$

2. $b^2 = 18a \Rightarrow (a, b) = (2, 6) \text{ or } (\frac{1}{2}, 3)$

Final answer

$$(a, b) = (2, 6) \text{ or } (\frac{1}{2}, 3)$$

Worked Solutions

Geometric Sequences

Q110 Challenge

A planning model gives $-5, a, b$ as an arithmetic sequence, while $a, b, 45$ form a geometric sequence. Find a, b .

Solution

1. $b = 2a + 5$
2. $b^2 = 45a \Rightarrow 4a^2 - 25a + 25 = 0$
3. $(a, b) = (5, 15)$ or $(\frac{5}{4}, \frac{15}{2})$

Final answer

$$(a, b) = (5, 15) \text{ or } (\frac{5}{4}, \frac{15}{2})$$

Worked Solutions

Geometric Sequences

Q111 Written

Find the sum S of a geometric sequence with first term 4, common ratio 3, and 5 terms.

Solution

1. Use $S_5 = \frac{4(3^5 - 1)}{3 - 1} = 2(3^5 - 1)$.
2. Therefore $S_5 = 484$.

Final answer

484

Q112 Written

Find the sum S of a geometric sequence with first term 3, common ratio 2, and 5 terms.

Solution

1. $S_5 = \frac{3(2^5 - 1)}{2 - 1} = 3(31)$.
2. Therefore $S_5 = 93$.

Final answer

93

Worked Solutions

Geometric Sequences

Q113 Written

Find the sum S of a geometric sequence with first term 5, common ratio 3, and 4 terms.

Solution

1. $S_4 = \frac{5(3^4 - 1)}{3 - 1} = 5(40).$
2. Therefore $S_4 = 200.$

Final answer

200

Q114 Written

Find the sum S of a geometric sequence with first term 3, common ratio 2, and 10 terms.

Solution

1. $S_{10} = \frac{3(2^{10} - 1)}{2 - 1} = 3(1023).$
2. Therefore $S_{10} = 3069.$

Final answer

3069

Worked Solutions

Geometric Sequences

Q115 MCQ

For a geometric sequence with $a = 4$, $r = 3$, and $n = 4$, calculate S_n :

Solution

1. $S_n = \frac{4(1 - (3)^4)}{1 - (3)}$.
2. Therefore $S_4 = 160$.

Correct option: D

Final answer

160

Q116 MCQ

For a geometric sequence with $a = 5$, $r = 2$, and $n = 6$, calculate S_n :

Solution

1. $S_n = \frac{5(1 - (2)^6)}{1 - (2)}$.
2. Therefore $S_6 = 315$.

Correct option: C

Final answer

315

Worked Solutions

Geometric Sequences

Q117 MCQ

For a geometric sequence with $a = 2$, $r = 4$, and $n = 5$, calculate S_n :

Solution

1. $S_n = \frac{2(1 - (4)^5)}{1 - (4)}$.

2. Therefore $S_5 = 682$.

Correct option: B

Final answer

682

Q118 MCQ

For a geometric sequence with $a = 3$, $r = 5$, and $n = 4$, calculate S_n :

Solution

1. $S_n = \frac{3(1 - (5)^4)}{1 - (5)}$.

2. Therefore $S_4 = 468$.

Correct option: A

Final answer

468

Worked Solutions

Geometric Sequences

Q119 MCQ

For a geometric sequence with $a = 7$, $r = 2$, and $n = 5$, calculate S_n :

Solution

1. $S_n = \frac{7(1 - (2)^5)}{1 - (2)}$.
2. Therefore $S_5 = 217$.

Correct option: D

Final answer

217

Q120 MCQ

For a geometric sequence with $a = 6$, $r = 3$, and $n = 3$, calculate S_n :

Solution

1. $S_n = \frac{6(1 - (3)^3)}{1 - (3)}$.
2. Therefore $S_3 = 78$.

Correct option: C

Final answer

78

Worked Solutions

Geometric Sequences

Q121 Written

Find the sum S of a geometric sequence with first term 8, common ratio $\frac{1}{2}$, and 6 terms.

Solution

1. Use $S_6 = \frac{8(1 - (1/2)^6)}{1 - 1/2} = 16\{1 - (1/2)^6\}$.
2. Therefore $S_6 = \frac{63}{4}$.

Final answer

$$\frac{63}{4}$$

Q122 Written

Find the sum S of a geometric sequence with first term 6, common ratio $\frac{1}{2}$, and 6 terms.

Solution

1. $S_6 = \frac{6(1 - (1/2)^6)}{1 - 1/2} = 12\{1 - (1/2)^6\}$.
2. Therefore $S_6 = \frac{189}{16}$.

Final answer

$$\frac{189}{16}$$

Worked Solutions

Geometric Sequences

Q123 Written

Find the sum S of a geometric sequence with first term 1, common ratio $\frac{1}{3}$, and 3 terms.

Solution

$$1. S_3 = \frac{1 - (1/3)^3}{1 - 1/3} = \frac{26}{27} \cdot \frac{3}{2}.$$

$$2. \text{ Therefore } S_3 = \frac{13}{9}.$$

Final answer

$$\frac{13}{9}$$

Q124 Written

Find the sum S of a geometric sequence with first term 1, common ratio $\frac{1}{2}$, and 8 terms.

Solution

$$1. S_8 = \frac{1 - (1/2)^8}{1 - 1/2} = 2\{1 - (1/2)^8\}.$$

$$2. \text{ Therefore } S_8 = \frac{255}{128}.$$

Final answer

$$\frac{255}{128}$$

Worked Solutions

Geometric Sequences

Q125 MCQ

For a geometric sequence with $a = 8$, $r = \frac{1}{2}$, and $n = 4$, calculate S_n :

Solution

1. $S_n = \frac{8(1 - (\frac{1}{2})^4)}{1 - (\frac{1}{2})}$.
2. Therefore $S_4 = 15$.

Correct option: B

Final answer

15

Q126 MCQ

For a geometric sequence with $a = 9$, $r = \frac{1}{3}$, and $n = 3$, calculate S_n :

Solution

1. $S_n = \frac{9(1 - (\frac{1}{3})^3)}{1 - (\frac{1}{3})}$.
2. Therefore $S_3 = 13$.

Correct option: A

Final answer

13

Worked Solutions

Geometric Sequences

Q127 MCQ

For a geometric sequence with $a = 12$, $r = \frac{1}{2}$, and $n = 5$, calculate S_n :

Solution

$$1. S_n = \frac{12(1 - (\frac{1}{2})^5)}{1 - (\frac{1}{2})}.$$

$$2. \text{ Therefore } S_5 = \frac{93}{4}.$$

Correct option: D

Final answer

$$\frac{93}{4}$$

Q128 MCQ

For a geometric sequence with $a = 5$, $r = \frac{1}{4}$, and $n = 3$, calculate S_n :

Solution

$$1. S_n = \frac{5(1 - (\frac{1}{4})^3)}{1 - (\frac{1}{4})}.$$

$$2. \text{ Therefore } S_3 = \frac{105}{16}.$$

Correct option: C

Final answer

$$\frac{105}{16}$$

Worked Solutions

Geometric Sequences

Q129 MCQ

For a geometric sequence with $a = 10$, $r = \frac{1}{2}$, and $n = 6$, calculate S_n :

Solution

$$1. S_n = \frac{10(1 - (\frac{1}{2})^6)}{1 - (\frac{1}{2})}.$$

$$2. \text{ Therefore } S_6 = \frac{315}{16}.$$

Correct option: B

Final answer

$$\frac{315}{16}$$

Q130 MCQ

For a geometric sequence with $a = 4$, $r = \frac{1}{3}$, and $n = 4$, calculate S_n :

Solution

$$1. S_n = \frac{4(1 - (\frac{1}{3})^4)}{1 - (\frac{1}{3})}.$$

$$2. \text{ Therefore } S_4 = \frac{160}{27}.$$

Correct option: A

Final answer

$$\frac{160}{27}$$

Worked Solutions

Geometric Sequences

Q131 Written

Find the sum S of a geometric sequence with first term 6, common ratio -2 , and 4 terms.

Solution

1. $S_4 = \frac{6\{1 - (-2)^4\}}{1 - (-2)} = \frac{6(-15)}{3}$.
2. Therefore $S_4 = -30$.

Final answer

-30

Q132 Written

Find the sum S of a geometric sequence with first term 1, common ratio -3 , and 4 terms.

Solution

1. $S_4 = \frac{1 - (-3)^4}{1 - (-3)} = \frac{-80}{4}$.
2. Therefore $S_4 = -20$.

Final answer

-20

Worked Solutions

Geometric Sequences

Q133 Written

Find the sum S of a geometric sequence with first term 2, common ratio -3 , and 6 terms.

Solution

1. $S_6 = \frac{2\{1 - (-3)^6\}}{1 - (-3)} = \frac{2(-728)}{4}.$
2. Therefore $S_6 = -364.$

Final answer -364

Q134 MCQ

For a geometric sequence with $a = 4$, $r = -2$, and $n = 5$, calculate S_n :

Solution

1. $S_n = \frac{4(1 - (-2)^5)}{1 - (-2)}.$
2. Therefore $S_5 = 44.$

Correct option: D

Final answer 44

Worked Solutions

Geometric Sequences

Q135 MCQ

For a geometric sequence with $a = 3$, $r = -3$, and $n = 4$, calculate S_n :

Solution

1. $S_n = \frac{3(1 - (-3)^4)}{1 - (-3)}$.
2. Therefore $S_4 = -60$.

Correct option: C

Final answer

-60

Q136 MCQ

For a geometric sequence with $a = 2$, $r = -1$, and $n = 6$, calculate S_n :

Solution

1. $S_n = \frac{2(1 - (-1)^6)}{1 - (-1)}$.
2. Therefore $S_6 = 0$.

Correct option: B

Final answer

0

Worked Solutions

Geometric Sequences

Q137 MCQ

For a geometric sequence with $a = 5$, $r = -2$, and $n = 6$, calculate S_n :

Solution

1. $S_n = \frac{5(1 - (-2)^6)}{1 - (-2)}$.
2. Therefore $S_6 = -105$.

Correct option: A

Final answer

-105

Q138 MCQ

For a geometric sequence with $a = 7$, $r = -1$, and $n = 5$, calculate S_n :

Solution

1. $S_n = \frac{7(1 - (-1)^5)}{1 - (-1)}$.
2. Therefore $S_5 = 7$.

Correct option: D

Final answer

7

Worked Solutions

Geometric Sequences

Q139 MCQ

For a geometric sequence with $a = 6$, $r = -3$, and $n = 3$, calculate S_n :

Solution

$$1. S_n = \frac{6(1 - (-3)^3)}{1 - (-3)}.$$

$$2. \text{ Therefore } S_3 = 42.$$

Correct option: C

Final answer

42

Q140 Written

Find S_n from the first term to the n -th term when the first term is 3 and the common ratio is -2 .

Solution

$$1. \text{ Here } a = 3 \text{ and } r = -2.$$

$$2. S_n = \frac{3\{1 - (-2)^n\}}{1 - (-2)} = 1 - (-2)^n.$$

Final answer $1 - (-2)^n$

Worked Solutions

Geometric Sequences

Q141 Written

Find S_n from the first term to the n -th term when the first term is 7 and the common ratio is 2.

Solution

1. $S_n = \frac{7(2^n - 1)}{2 - 1}$.
2. Hence $S_n = 7(2^n - 1)$.

Final answer

$$7(2^n - 1)$$

Q142 Written

Find S_n from the first term to the n -th term when the first term is 2 and the common ratio is $\frac{1}{3}$.

Solution

1. $S_n = \frac{2\{1 - (1/3)^n\}}{1 - 1/3}$.
2. Hence $S_n = 3\{1 - (1/3)^n\}$.

Final answer

$$3\{1 - (\frac{1}{3})^n\}$$

Worked Solutions

Geometric Sequences

Q143 Written

Find S_n from the first term to the n -th term when the first term is 4 and the common ratio is 2.

Solution

1. $S_n = \frac{4(2^n - 1)}{2 - 1}$.
2. Hence $S_n = 4(2^n - 1)$.

Final answer

$$4(2^n - 1)$$

Q144 MCQ

For the geometric sequence with $a = 3, r = 2$, find the sum S_n from the first term to the n -th term:

Solution

1. Identify a and r from the stated parameters.
2. Substitute into $S_n = \frac{a(1 - r^n)}{1 - r}$ to obtain $S_n = 3(2^n - 1)$.

Correct option: C

Final answer

$$3(2^n - 1)$$

Worked Solutions

Geometric Sequences

Q145 MCQ

For the geometric sequence with $a = 5, r = \frac{1}{2}$, find the sum S_n from the first term to the n -th term:

Solution

1. Identify a and r from the stated parameters.
2. Substitute into $S_n = \frac{a(1 - r^n)}{1 - r}$ to obtain $S_n = 10\{1 - (\frac{1}{2})^n\}$.

Correct option: D**Final answer**

$$10\{1 - (\frac{1}{2})^n\}$$

Q146 MCQ

For the geometric sequence with $a = 4, r = -2$, find the sum S_n from the first term to the n -th term:

Solution

1. Identify a and r from the stated parameters.
2. Substitute into $S_n = \frac{a(1 - r^n)}{1 - r}$ to obtain $S_n = \frac{4\{1 - (-2)^n\}}{3}$.

Correct option: A**Final answer**

$$\frac{4\{1 - (-2)^n\}}{3}$$

Worked Solutions

Geometric Sequences

Q147 **Written****Find S_n from the first term to the n -th term of 2, 6, 18, 54,****Solution**

1. The sequence has $a = 2$ and $r = 3$.
2. $S_n = \frac{2(3^n - 1)}{3 - 1} = 3^n - 1$.

Final answer

$$3^n - 1$$

Q148 **Written****Find S_n from the first term to the n -th term of 2, -6, 18, -54,****Solution**

1. The sequence has $a = 2$ and $r = -3$.
2. $S_n = \frac{2\{1 - (-3)^n\}}{1 - (-3)} = \frac{1 - (-3)^n}{2}$.

Final answer

$$\frac{1 - (-3)^n}{2}$$

Worked Solutions

Geometric Sequences

Q149 Written

Find S_n from the first term to the n -th term of $8, -4, 2, -1, \dots$

Solution

1. The sequence has $a = 8$ and $r = -\frac{1}{2}$.
2. $S_n = \frac{8\{1 - (-1/2)^n\}}{1 + 1/2} = \frac{16}{3}\{1 - (-1/2)^n\}$.

Final answer

$$\frac{16}{3}\{1 - (-\frac{1}{2})^n\}$$

Q150 Written

Find S_n from the first term to the n -th term of $1, -3, 9, -27, \dots$

Solution

1. The sequence has $a = 1$ and $r = -3$.
2. $S_n = \frac{1 - (-3)^n}{1 + 3} = \frac{1 - (-3)^n}{4}$.

Final answer

$$\frac{1 - (-3)^n}{4}$$

Worked Solutions

Geometric Sequences

Q151 Written

Find S_n from the first term to the n -th term when the first term is 16 and the common ratio is $-\frac{1}{2}$.

Solution

$$1. S_n = \frac{16\{1 - (-1/2)^n\}}{1 + 1/2}.$$

$$2. \text{Hence } S_n = \frac{32}{3}\{1 - (-1/2)^n\}.$$

Final answer

$$\frac{32}{3}\{1 - (-\frac{1}{2})^n\}$$

Q152 Written

Find S_n from the first term to the n -th term of 1, -2, 4, -8,

Solution

1. The sequence has $a = 1$ and $r = -2$.

$$2. S_n = \frac{1 - (-2)^n}{1 + 2} = \frac{1 - (-2)^n}{3}.$$

Final answer

$$\frac{1 - (-2)^n}{3}$$

Worked Solutions

Geometric Sequences

Q153 MCQ

A culture grows by 3, 9, 27, 81 units. Find the four-stage total:

Solution

1. Recognise the listed stages as a finite geometric sequence and identify its first term and ratio.
2. Apply the finite-GP formula to obtain $S_n = 120$.

Correct option: A

Final answer

120

Worked Solutions

Geometric Sequences

Q154 Challenge

A filtration scale is geometric: $3, 1, \frac{1}{3}, \frac{1}{9}, \dots$ mg/L removed per stage. Calculate the total removed in 4 stages.

Solution

1. Here $a = 3$, $r = \frac{1}{3}$, and $n = 4$.
2. $S_4 = \frac{3\{1 - (1/3)^4\}}{1 - 1/3} = \frac{40}{9}$ mg/L.

Final answer

$$\frac{40}{9}$$

Worked Solutions

Geometric Sequences

Q155 MCQ

Given $S_3 = 7$ and the sum from the 3rd to the 5th terms is 28, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 28$, so $r^2 = 4$; then $a(1 + r + r^2) = 7$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 1$ or $a = \frac{7}{3}$, respectively.

Correct option: D

Final answer

$$(a, r) = (1, 2) \text{ or } (a, r) = \left(\frac{7}{3}, -2\right)$$

Worked Solutions

Geometric Sequences

Q156 MCQ

Given $S_3 = 26$ and the sum from the 3rd to the 5th terms is 234, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2S_3 = 234$, so $r^2 = 9$; then $a(1 + r + r^2) = 26$.
2. Solving in the real numbers gives $r = 3$ or $r = -3$.
3. Back-substitution gives $a = 2$ or $a = \frac{26}{7}$, respectively.

Correct option: C

Final answer

$$(a, r) = (2, 3) \text{ or } (a, r) = \left(\frac{26}{7}, -3\right)$$

Worked Solutions

Geometric Sequences

Q157 MCQ

Given $S_3 = 21$ and the sum from the 3rd to the 5th terms is 84, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 84$, so $r^2 = 4$; then $a(1 + r + r^2) = 21$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 3$ or $a = 7$, respectively.

Correct option: B

Final answer

$$(a, r) = (3, 2) \text{ or } (a, r) = (7, -2)$$

Worked Solutions

Geometric Sequences

Q158 MCQ

Given $S_3 = 28$ and the sum from the 3rd to the 5th terms is 112, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 112$, so $r^2 = 4$; then $a(1 + r + r^2) = 28$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 4$ or $a = \frac{28}{3}$, respectively.

Correct option: A

Final answer

$$(a, r) = (4, 2) \text{ or } (a, r) = \left(\frac{28}{3}, -2\right)$$

Worked Solutions

Geometric Sequences

Q159 MCQ

Given $S_3 = 13$ and the sum from the 3rd to the 5th terms is 117, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 117$, so $r^2 = 9$; then $a(1 + r + r^2) = 13$.
2. Solving in the real numbers gives $r = 3$ or $r = -3$.
3. Back-substitution gives $a = 1$ or $a = \frac{13}{7}$, respectively.

Correct option: D

Final answer

$$(a, r) = (1, 3) \text{ or } (a, r) = \left(\frac{13}{7}, -3\right)$$

Worked Solutions

Geometric Sequences

Q160 MCQ

Given $S_3 = 35$ and the sum from the 3rd to the 5th terms is 140, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 140$, so $r^2 = 4$; then $a(1 + r + r^2) = 35$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 5$ or $a = \frac{35}{3}$, respectively.

Correct option: C**Final answer**

$$(a, r) = (5, 2) \text{ or } (a, r) = \left(\frac{35}{3}, -2\right)$$

Worked Solutions

Geometric Sequences

Q161 MCQ

Given $S_3 = \frac{19}{2}$ and the sum from the 3rd to the 5th terms is $\frac{171}{8}$, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = \frac{171}{8}$, so $r^2 = \frac{9}{4}$; then $a(1 + r + r^2) = \frac{19}{2}$.
2. Solving in the real numbers gives $r = \frac{3}{2}$ or $r = -\frac{3}{2}$.
3. Back-substitution gives $a = 2$ or $a = \frac{38}{7}$, respectively.

Correct option: D**Final answer**

$$(a, r) = \left(2, \frac{3}{2}\right) \text{ or } (a, r) = \left(\frac{38}{7}, -\frac{3}{2}\right)$$

Worked Solutions

Geometric Sequences

Q162 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 9$ and the sum from the 4th to the 6th term is -72 .

Solution

1. The 4th+6th block is $r^3 S_3 = -72$, so $r^3 = -8$ and $r = -2$.
2. Then $a(1 - 2 + 4) = 9$, giving $a = 3$.

Final answer

$$(a, r) = (3, -2)$$

Worked Solutions

Geometric Sequences

Q163 MCQ

Given $S_3 = 7$ and the sum from the 3rd to the 5th terms is $\frac{7}{4}$, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = \frac{7}{4}$, so $r^2 = \frac{1}{4}$; then $a(1 + r + r^2) = 7$.
2. Solving in the real numbers gives $r = \frac{1}{2}$ or $r = -\frac{1}{2}$.
3. Back-substitution gives $a = 4$ or $a = \frac{28}{3}$, respectively.

Correct option: C

Final answer

$$(a, r) = \left(4, \frac{1}{2}\right) \text{ or } (a, r) = \left(\frac{28}{3}, -\frac{1}{2}\right)$$

Worked Solutions

Geometric Sequences

Q164 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 35$ and the sum from the 3rd to the 5th term is 140.

Solution

1. Write $S_3 = a(1 + r + r^2) = 35$. The shifted block is $r^2 S_3 = 140$, so $r^2 = 4$ and $r = \pm 2$.
2. For $r = 2$, $7a = 35$, giving $a = 5$. For $r = -2$, $3a = 35$, giving $a = \frac{35}{3}$.

Final answer

$$(a, r) = (5, 2) \text{ or } (a, r) = \left(\frac{35}{3}, -2\right)$$

Q165 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 21$ and the sum from the 3rd to the 5th term is 84.

Solution

1. The shifted block is $r^2 S_3 = 84$, so $r^2 = 4$ and $r = \pm 2$.
2. Using $a(1 + r + r^2) = 21$ gives $a = 3$ for $r = 2$ and $a = 7$ for $r = -2$.

Final answer

$$(a, r) = (3, 2) \text{ or } (a, r) = (7, -2)$$

Worked Solutions

Geometric Sequences

Q166 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 9$ and the sum from the 3rd to the 5th term is 36.

Solution

1. The shifted block gives $r^2 S_3 = 36$, hence $r = \pm 2$.
2. Substitution into $a(1 + r + r^2) = 9$ gives $a = \frac{9}{7}$ for $r = 2$, or $a = 3$ for $r = -2$.

Final answer

$$(a, r) = \left(\frac{9}{7}, 2\right) \text{ or } (a, r) = (3, -2)$$

Q167 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 78$ and the sum from the 3rd to the 5th term is 702.

Solution

1. The shifted block gives $r^2 S_3 = 702$, so $r^2 = 9$ and $r = \pm 3$.
2. Using $a(1 + r + r^2) = 78$ gives $a = 6$ for $r = 3$, or $a = \frac{78}{7}$ for $r = -3$.

Final answer

$$(a, r) = (6, 3) \text{ or } (a, r) = \left(\frac{78}{7}, -3\right)$$

Worked Solutions

Geometric Sequences

Q168 MCQ

Given $S_3 = 6$ and the sum from the 3rd to the 5th terms is 24, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2S_3 = 24$, so $r^2 = 4$; then $a(1 + r + r^2) = 6$.
2. Solving in the real numbers gives $r = -2$ or $r = 2$.
3. Back-substitution gives $a = 2$ or $a = \frac{6}{7}$, respectively.

Correct option: B

Final answer

$$(a, r) = (2, -2) \text{ or } (a, r) = \left(\frac{6}{7}, 2\right)$$

Worked Solutions

Geometric Sequences

Q169 MCQ

Given $S_3 = 9$ and the sum from the 3rd to the 5th terms is 36, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 36$, so $r^2 = 4$; then $a(1 + r + r^2) = 9$.
2. Solving in the real numbers gives $r = -2$ or $r = 2$.
3. Back-substitution gives $a = 3$ or $a = \frac{9}{7}$, respectively.

Correct option: A

Final answer

$$(a, r) = (3, -2) \text{ or } (a, r) = \left(\frac{9}{7}, 2\right)$$

Worked Solutions

Geometric Sequences

Q170 Written

Find the first term and common ratio of a geometric sequence if the 2nd term is 3 and $S_3 = 13$.

Solution

1. From $ar = 3$, write $a = 3/r$. Then $3/r + 3 + 3r = 13$, so $3r^2 - 10r + 3 = 0$.
2. Hence $r = 3$ or $r = \frac{1}{3}$, giving $a = 1$ or $a = 9$, respectively.

Final answer

$$(a, r) = (1, 3) \text{ or } (a, r) = \left(9, \frac{1}{3}\right)$$

Q171 Written

Find the first term and common ratio of a geometric sequence if the 2nd term is 6, and $S_3 = 21$.

Solution

1. From $ar = 6$, $a = 6/r$. Substitution into $S_3 = 21$ gives $6/r + 6 + 6r = 21$, hence $2r^2 - 5r + 2 = 0$.
2. Therefore $r = 2$ or $r = \frac{1}{2}$, giving $a = 3$ or $a = 12$, respectively.

Final answer

$$(a, r) = (3, 2) \text{ or } (a, r) = \left(12, \frac{1}{2}\right)$$

Worked Solutions

Geometric Sequences

Q172 MCQ

The second term is 6 and $S_3 = 21$. Find all real pairs (a, r) :**Solution**

1. Use $ar = 6$ and $a(1 + r + r^2) = 21$; after substitution, solve the quadratic in r .
2. Solving in the real numbers gives $r = 2$ or $r = \frac{1}{2}$.
3. Back-substitution gives $a = 3$ or $a = 12$, respectively.

Correct option: C**Final answer**

$$(a, r) = (3, 2) \text{ or } (a, r) = \left(12, \frac{1}{2}\right)$$

Worked Solutions

Geometric Sequences

Q173 MCQ

The second term is 6 and $S_3 = 26$. Find all real pairs (a, r) :

Solution

1. Use $ar = 6$ and $a(1 + r + r^2) = 26$; after substitution, solve the quadratic in r .
2. Solving in the real numbers gives $r = 3$ or $r = \frac{1}{3}$.
3. Back-substitution gives $a = 2$ or $a = 18$, respectively.

Correct option: D

Final answer

$$(a, r) = (2, 3) \text{ or } (a, r) = \left(18, \frac{1}{3}\right)$$

Worked Solutions

Geometric Sequences

Q174 MCQ

The second term is 8 and $S_3 = 28$. Find all real pairs (a, r) :**Solution**

1. Use $ar = 8$ and $a(1 + r + r^2) = 28$; after substitution, solve the quadratic in r .
2. Solving in the real numbers gives $r = 2$ or $r = \frac{1}{2}$.
3. Back-substitution gives $a = 4$ or $a = 16$, respectively.

Correct option: B**Final answer**

$$(a, r) = (4, 2) \text{ or } (a, r) = \left(16, \frac{1}{2}\right)$$

Worked Solutions

Geometric Sequences

Q175 MCQ

The second term is 3 and $S_3 = 13$. Find all real pairs (a, r) :

Solution

1. Use $ar = 3$ and $a(1 + r + r^2) = 13$; after substitution, solve the quadratic in r .
2. Solving in the real numbers gives $r = 3$ or $r = \frac{1}{3}$.
3. Back-substitution gives $a = 1$ or $a = 9$, respectively.

Correct option: C

Final answer

$$(a, r) = (1, 3) \text{ or } (a, r) = \left(9, \frac{1}{3}\right)$$

Worked Solutions

Geometric Sequences

Q176 MCQ

The second term is $\frac{5}{2}$ and $S_3 = \frac{35}{4}$. Find all real pairs (a, r) :

Solution

1. Use $ar = \frac{5}{2}$ and $a(1 + r + r^2) = \frac{35}{4}$; after substitution, solve the quadratic in r .
2. Solving in the real numbers gives $r = \frac{1}{2}$ or $r = 2$.
3. Back-substitution gives $a = 5$ or $a = \frac{5}{4}$, respectively.

Correct option: D**Final answer**

$$(a, r) = \left(5, \frac{1}{2}\right) \text{ or } (a, r) = \left(\frac{5}{4}, 2\right)$$

Worked Solutions

Geometric Sequences

Q177 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 4 and $S_3 = 7$.

Solution

1. The data give $ar^2 = 4$ and $a + ar + ar^2 = 7$. Substitute $a = 4/r^2$: $4 + 4r + 4r^2 = 7r^2$, so $3r^2 - 4r - 4 = 0$.
2. Thus $r = 2$ or $r = -\frac{2}{3}$. Back-substitution gives $a = 1$ or $a = 9$, respectively.

Final answer

$$(a, r) = (1, 2) \text{ or } (a, r) = \left(9, -\frac{2}{3}\right)$$

Q178 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 12, and $S_3 = 9$.

Solution

1. Use $ar^2 = 12$ and $a + ar + ar^2 = 9$. After multiplying by r^2 , $r^2 + 4r + 4 = 0$, so $r = -2$.
2. Then $a = 12/(-2)^2 = 3$.

Final answer

$$(a, r) = (3, -2)$$

Worked Solutions

Geometric Sequences

Q179 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 20, and $S_3 = 35$.

Solution

1. Use $ar^2 = 20$ and $a + ar + ar^2 = 35$. Multiplication by r^2 gives $3r^2 - 4r - 4 = 0$.
2. Thus $r = 2$ or $r = -\frac{2}{3}$, and $a = 5$ or $a = 45$, respectively.

Final answer

$$(a, r) = (5, 2) \text{ or } (a, r) = \left(45, -\frac{2}{3}\right)$$

Q180 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 21$ and $S_6 = 189$.

Solution

1. Use $S_6 = S_3 + r^3 S_3$: $189 = 21(1 + r^3)$, hence $r^3 = 8$ and $r = 2$ in the real numbers.
2. Then $a(1 + 2 + 4) = 21$, so $a = 3$.

Final answer

$$(a, r) = (3, 2)$$

Worked Solutions

Geometric Sequences

Q181 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 26$ and $S_6 = 728$.

Solution

1. Since $S_6 = S_3 + r^3 S_3$, $728 = 26(1 + r^3)$, so $r^3 = 27$ and $r = 3$.
2. Then $a(1 + 3 + 9) = 26$, giving $a = 2$.

Final answer

$$(a, r) = (2, 3)$$

Q182 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 65$ and $S_6 = 1820$.

Solution

1. Use $S_6 = S_3(1 + r^3)$: $1820 = 65(1 + r^3)$, so $r^3 = 27$ and $r = 3$.
2. Then $13a = 65$, giving $a = 5$.

Final answer

$$(a, r) = (5, 3)$$

Worked Solutions

Geometric Sequences

Q183 Written

Find the first term and common ratio of a geometric sequence if the sum of the first and second terms is 3, and $S_4 = 51$.

Solution

1. Let $a + ar = 3$. Factor $S_4 = (a + ar)(1 + r^2)$, so $3(1 + r^2) = 51$ and $r = \pm 4$.
2. Then $a = 3/(1 + r)$, giving $a = \frac{3}{5}$ for $r = 4$, or $a = -1$ for $r = -4$.

Final answer

$$(a, r) = \left(\frac{3}{5}, 4\right) \text{ or } (a, r) = (-1, -4)$$

Q184 Written

Find the first term and common ratio of a geometric sequence if the sum of the first and second terms is -8 , and $S_4 = -80$.

Solution

1. Factor $S_4 = (a + ar)(1 + r^2)$. Thus $-8(1 + r^2) = -80$, so $r = \pm 3$.
2. Using $a(1 + r) = -8$ gives $a = -2$ for $r = 3$, or $a = 4$ for $r = -3$.

Final answer

$$(a, r) = (-2, 3) \text{ or } (a, r) = (4, -3)$$

Worked Solutions

Geometric Sequences

Q185 MCQ

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} :

Solution

1. $a_7 = S_7 - S_6 = 64$ and $a_8 = S_8 - S_7 = 128$.
2. Thus $r = a_8/a_7 = 2$, and $a_{10} = a_8 r^2 = 512$.

Correct option: D

Final answer

512

Worked Solutions

Geometric Sequences

Q186 Challenge

If S_n is the sum of the first n terms of a geometric sequence, $S_8 = 255$, $S_6 = 63$, and $a_8 - a_7 = 64$, find a_{10} .

Solution

1. $S_8 - S_6 = a_7 + a_8 = 255 - 63 = 192$. Together with $a_8 - a_7 = 64$, this gives $a_7 = 64$ and $a_8 = 128$.
2. Hence $r = a_8/a_7 = 2$, and $a_{10} = a_8 r^2 = 128(4) = 512$.

Final answer

$$a_7 = 64, a_8 = 128, r = 2, a_{10} = 512$$

Worked Solutions

Geometric Sequences

Quiz MCQ 1 [Concept Quiz · MCQ](#)

Select the correct option: For $a = 3, r = 2$, find a_6 :

Solution

1. $a_6 = 3(2)^5 = 96$

Correct option: B

Final answer

96

Quiz MCQ 2 [Concept Quiz · MCQ](#)

Select the correct option: If 2, x , 8 are consecutive geometric terms, find x :

Solution

1. $x^2 = 16 \Rightarrow x = \pm 4$

Correct option: C

Final answer

± 4

Worked Solutions

Geometric Sequences

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

For 1, 3, 9, ..., find S_n :

Solution

1. $a = 1, r = 3$, so $S_n = (3^n - 1)/2$.

Correct option: B

Final answer

$$\frac{3^n - 1}{2}$$

Quiz MCQ 4 [Concept Quiz](#) · [MCQ](#)

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} :

Solution

1. $a_7 = S_7 - S_6 = 64$, $a_8 = S_8 - S_7 = 128$, so $r = 2$.

2. Therefore $a_{10} = 128(2^2) = 512$.

Correct option: D

Final answer

512

Worked Solutions

Geometric Sequences

Quiz WRITTEN 5 Concept Quiz · Written

Find the general term of $-4, 8, -16, 32, \dots$

Solution

1. The ratio is -2 .
2. $a_n = -4(-2)^{n-1}$

Final answer

$$-4(-2)^{n-1}$$

Quiz WRITTEN 6 Concept Quiz · Written

If $1, x, 25$ are consecutive geometric terms, find x .

Solution

1. $x^2 = 25 \Rightarrow x = \pm 5$

Final answer

$$x = \pm 5$$

Worked Solutions

Geometric Sequences

Quiz WRITTEN 7 Concept Quiz · Written

A four-stage process has outputs 5, 15, 45, 135. Find the total output.

Solution

1. $S_4 = 5(1 + 3 + 9 + 27) = 200.$

Final answer

200

Quiz WRITTEN 8 Concept Quiz · Written

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} .

Solution

1. $a_7 = 64$, $a_8 = 128$, hence $r = 2$.

2. $a_{10} = a_8 r^2 = 512.$

Final answer

512

Answer key

Use the question number shown on each page.

Q001 $2 \times 3^{n-1}$

Q002 $3 \times 4^{n-1}$

Q003 $5 \times 2^{n-1}$

Q004 $2 \times 5^{n-1}$

Q005 $7 \times 4^{n-1}$

Q006 A

Q007 A

Q008 A

Q009 A

Q010 A

Q011 A

Q012 $5(-2)^{n-1}$

Q013 $2(-3)^{n-1}$

Q014 $3(-5)^{n-1}$

Q015 $4(-2)^{n-1}$

Q016 A

Q017 A

Q018 A

Q019 A

Q020 A

Q021 $8\left(-\frac{1}{3}\right)^{n-1}$

Q022 $4\left(\frac{1}{3}\right)^{n-1}$

Q023 $3\left(-\frac{1}{2}\right)^{n-1}$

Q024 $6\left(\frac{1}{2}\right)^{n-1}$

Q025 A

Q026 A

Q027 A

Q028 A

Q029 A

Q030 $5 \times 2^{n-1}$

Q031 $2 \times 3^{n-1}$

Q032 $3 \times 2^{n-1}$

Q033 A

Q034 A

Q035 A

Q036 A

Q037 A

Q038 $2(-3)^{n-1}$

Q039 $5(-2)^{n-1}$

Q040 A

Q041 A

Q042 A

Q043 A

Q044 $-4\left(-\frac{3}{2}\right)^{n-1}$

Q045 $-8\left(-\frac{3}{2}\right)^{n-1}$

Q046 $2\left(\frac{1}{3}\right)^{n-1}$

Q047 $48\left(\frac{1}{2}\right)^{n-1}$

Q048 $\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$

Q049 $8(2\sqrt{2})^{n-1}$

Q050 $3\left(-\frac{5}{2}\right)^{n-1}$

Q051 $9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$

Q052 $5\left(-\frac{2}{3}\right)^{n-1}$

Q053 A

Q054 A

Q055 A

Q056 A

Q057 A

Q058 A

Q059 A

Q060 $r = -\frac{1}{3}, A = 54, B = -2$

Q061 A

Q062 $r = \frac{1}{2}, A = 48, B = 6$

Q063 $r = -3, A = 1, B = -27$

Q064 $r = \frac{1}{3}, A = 1, B = \frac{1}{27}$

Q065 $r = -\frac{1}{2}, A = \frac{1}{2}, B = -\frac{1}{16}$

Q066 $a_7 = 648$

Q067 $a_n = 3(2)^{n-1}$ or $-3(-2)^{n-1}$

Q068 $a_n = 2(3)^{n-1}$ or $2(-3)^{n-1}$

Q069 $a_n = 2(3)^{n-1}$ or $-2(-3)^{n-1}$

Q070 $a_n = 3(3)^{n-1}$ or $3(-3)^{n-1}$

Q071 $a_n = 7(2)^{n-1}$

Q072 $a_n = 3(-2)^{n-1}$

Q073 $a_n = 3(-2)^{n-1}$

Q074 $a_n = 3(2)^{n-1}$ or $3(-2)^{n-1}$

Q075 $a_n = -7(2)^{n-1}$ or $-7(-2)^{n-1}$

Q076 $a_n = -2(-2)^{n-1}$

Q077 $a_n = -5(-2)^{n-1}$

Q078 $a_n = \frac{3}{4}(4)^{n-1}$ or $-\frac{3}{4}(-4)^{n-1}$

Q079 $a_n = 4\left(\frac{3}{2}\right)^{n-1}$ or $4\left(-\frac{3}{2}\right)^{n-1}$

Q080 $a_n = \frac{2}{243}(3)^{n-1}$ or $-\frac{2}{243}(-3)^{n-1}$

Q081 A

Q082 A

Q083 A

Q084 A

Q085 A

Q086 A

Q087 A

Q088 A

Q089 A

Q090 A

Q091 A

Q092 A

Q093 A

Q094 A

Q095 A	Q122 $\frac{189}{16}$	Q148 $\frac{1 - (-3)^n}{2}$	Q171 $(a, r) = (3, 2)$ or $(a, r) = \left(12, \frac{1}{2}\right)$
Q096 A	Q123 $\frac{13}{9}$	Q149 $\frac{16}{3}\{1 - (-\frac{1}{2})^n\}$	Q172 C
Q097 $x = \pm 4$	Q124 $\frac{255}{128}$	Q150 $\frac{1 - (-3)^n}{4}$	Q173 D
Q098 $x = \pm 5$	Q125 B	Q151 $\frac{32}{3}\{1 - (-\frac{1}{2})^n\}$	Q174 B
Q099 $x = \pm 2$	Q126 A	Q152 $\frac{1 - (-2)^n}{3}$	Q175 C
Q100 $x = \pm 2\sqrt{3}$	Q127 D	Q153 A	Q176 D
Q101 A	Q128 C	Q154 $\frac{40}{9}$	Q177 $(a, r) = (1, 2)$ or $(a, r) = \left(9, -\frac{2}{3}\right)$
Q102 A	Q129 B	Q155 D	Q178 $(a, r) = (3, -2)$
Q103 A	Q130 A	Q156 C	Q179 $(a, r) = (5, 2)$ or $(a, r) = \left(45, -\frac{2}{3}\right)$
Q104 A	Q131 -30	Q157 B	Q180 $(a, r) = (3, 2)$
Q105 A	Q132 -20	Q158 A	Q181 $(a, r) = (2, 3)$
Q106 A	Q133 -364	Q159 D	Q182 $(a, r) = (5, 3)$
Q107 $(a, b) = (9, 12)$ or $(1, -4)$	Q134 D	Q160 C	Q183 $(a, r) = \left(\frac{3}{5}, 4\right)$ or $(a, r) = (-1, -4)$
Q108 $(a, b) = (2, 2)$ or $(-4, -1)$	Q135 C	Q161 D	Q184 $(a, r) = (-2, 3)$ or $(a, r) = (4, -3)$
Q109 $(a, b) = (2, 6)$ or $(\frac{1}{2}, 3)$	Q136 B	Q162 $(a, r) = (3, -2)$	Q185 D
Q110 $(a, b) = (5, 15)$ or $(\frac{5}{4}, \frac{15}{2})$	Q137 A	Q163 C	Q186 $a_7 = 64, a_8 = 128, r = 2, a_{10} = 512$
Q111 484	Q138 D	Q164 $(a, r) = (5, 2)$ or $(a, r) = \left(\frac{35}{3}, -2\right)$	Quiz MCQ 1 B
Q112 93	Q139 C	Q165 $(a, r) = (3, 2)$ or $(a, r) = (7, -2)$	Quiz MCQ 2 C
Q113 200	Q140 $1 - (-2)^n$	Q166 $(a, r) = \left(\frac{9}{7}, 2\right)$ or $(a, r) = (3, -2)$	Quiz MCQ 3 B
Q114 3069	Q141 $7(2^n - 1)$	Q167 $(a, r) = (6, 3)$ or $(a, r) = \left(\frac{78}{7}, -3\right)$	Quiz MCQ 4 D
Q115 D	Q142 $3\{1 - (\frac{1}{3})^n\}$	Q168 B	Quiz WRITTEN 5 $-4(-2)^{n-1}$
Q116 C	Q143 $4(2^n - 1)$	Q169 A	Quiz WRITTEN 6 $x = \pm 5$
Q117 B	Q144 C	Q170 $(a, r) = (1, 3)$ or $(a, r) = \left(9, \frac{1}{3}\right)$	Quiz WRITTEN 7 200
Q118 A	Q145 D		Quiz WRITTEN 8 512
Q119 D	Q146 A		
Q120 C	Q147 $3^n - 1$		
Q121 $\frac{63}{4}$			

Concept 3 | Summation

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$\sum (au_k + bv_k) = a \sum u_k + b \sum v_k$$

Standard sums

$$\sum k = \frac{n(n+1)}{2}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Worked Solutions

Summation

Q001 Written

Write $\sum_{k=1}^5 2k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^5 2k = 2 + 4 + 6 + 8 + 10$

Final answer

$$2 + 4 + 6 + 8 + 10$$

Q002 Written

Write $\sum_{k=1}^5 (2k + 1)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^5 (2k + 1) = 3 + 5 + 7 + 9 + 11$

Final answer

$$3 + 5 + 7 + 9 + 11$$

Worked Solutions

Summation

Q003 Written

Write $\sum_{k=1}^5 3k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^5 3k = 3 + 6 + 9 + 12 + 15$

Final answer

$$3 + 6 + 9 + 12 + 15$$

Q004 Written

Write $\sum_{k=1}^4 (4k - 2)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 (4k - 2) = 2 + 6 + 10 + 14$

Final answer

$$2 + 6 + 10 + 14$$

Worked Solutions

Summation

Q005 Written

Write $\sum_{k=1}^5 (2k - 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^5 (2k - 3) = -1 + 1 + 3 + 5 + 7$

Final answer

$$-1 + 1 + 3 + 5 + 7$$

Q006 MCQ

Select the correct option: Expand $\sum_{k=1}^3 4k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$4 + 8 + 12$$

Worked Solutions

Summation

Q007 MCQ

Select the correct option: Expand $\sum_{k=1}^4 2^k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$2 + 4 + 8 + 16$$

Q008 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (5^k - 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$4 + 24 + 124$$

Worked Solutions

Summation

Q009 MCQ

Select the correct option: Expand $\sum_{k=1}^5 (k + 4)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$5 + 6 + 7 + 8 + 9$$

Q010 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (2k + 3)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$7 + 9 + 11 + 13$$

Worked Solutions

Summation

Q011 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (2^k + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$3 + 5 + 9$$

Q012 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (k^2 + 3k)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$4 + 10 + 18$$

Worked Solutions

Summation

Q013 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (k^2 + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$5 + 10 + 17 + 26$$

Q014 Written

Write $\sum_{k=1}^3 (k^2 - 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 (k^2 - 3) = -2 + 1 + 6$

Final answer

$$-2 + 1 + 6$$

Worked Solutions

Summation

Q015 Written

Write $\sum_{k=1}^3 (3k^2 + 2)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 (3k^2 + 2) = 5 + 14 + 29$

Final answer

$$5 + 14 + 29$$

Q016 Written

Write $\sum_{k=1}^4 k^2$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 k^2 = 1 + 4 + 9 + 16$

Final answer

$$1 + 4 + 9 + 16$$

Worked Solutions

Summation

Q017 Written

Write $\sum_{k=1}^3 (2k^2 + 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 (2k^2 + 3) = 5 + 11 + 21$

Final answer

$$5 + 11 + 21$$

Q018 Written

Write $\sum_{k=1}^4 (k^2 - 2k)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 (k^2 - 2k) = -1 + 0 + 3 + 8$

Final answer

$$-1 + 0 + 3 + 8$$

Worked Solutions

Summation

Q019 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (2k + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$3 + 5 + 7 + 9$$

Q020 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (3k - 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$5 + 8 + 11 + 14$$

Worked Solutions

Summation

Q021 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (2k^2 - 3)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$-1 + 5 + 15 + 29$$

Q022 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (3k^2 + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$4 + 13 + 28 + 49$$

Worked Solutions

Summation

Q023 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (k^2 - k)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$0 + 2 + 6 + 12$

Q024 MCQ

Select the correct option: Expand $\sum_{k=2}^4 5k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$10 + 15 + 20$

Worked Solutions

Summation

Q025 MCQ

Select the correct option: Expand $\sum_{k=1}^4 3$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$3 + 3 + 3 + 3$$

Q026 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (5k + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$6 + 11 + 16 + 21$$

Worked Solutions

Summation

Q027 Written

Write $\sum_{k=1}^4 3^k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 3^k = 3 + 9 + 27 + 81$

Final answer

$$3 + 9 + 27 + 81$$

Q028 Written

Write $\sum_{k=1}^3 2$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^3 2 = 2 + 2 + 2$

Final answer

$$2 + 2 + 2$$

Worked Solutions

Summation

Q029 Written

Write $\sum_{k=1}^4 2^k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 2^k = 2 + 4 + 8 + 16$

Final answer

$$2 + 4 + 8 + 16$$

Q030 Written

Write $\sum_{k=1}^6 3$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^6 3 = 3 + 3 + 3 + 3 + 3 + 3$

Final answer

$$3 + 3 + 3 + 3 + 3 + 3$$

Worked Solutions

Summation

Q031 Written

Write $\sum_{k=1}^3 3^k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 3^k = 3 + 9 + 27$

Final answer

$$3 + 9 + 27$$

Q032 Written

Write $\sum_{k=1}^3 (5^k + 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 (5^k + 3) = 8 + 28 + 128$

Final answer

$$8 + 28 + 128$$

Worked Solutions

Summation

Q033 Written

Write $\sum_{k=1}^4 (2^k - 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 (2^k - 3) = -1 + 1 + 5 + 13$

Final answer

$$-1 + 1 + 5 + 13$$

Q034 Written

Write $\sum_{k=1}^4 5$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^4 5 = 5 + 5 + 5 + 5$

Final answer

$$5 + 5 + 5 + 5$$

Worked Solutions

Summation

Q035 Written

Write $\sum_{k=1}^3 3$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^3 3 = 3 + 3 + 3$

Final answer

$$3 + 3 + 3$$

Q036 Written

Write $\sum_{k=1}^5 4$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^5 4 = 4 + 4 + 4 + 4 + 4$

Final answer

$$4 + 4 + 4 + 4 + 4$$

Worked Solutions

Summation

Q037 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (k^2 + 2)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$3 + 6 + 11$$

Q038 MCQ

Select the correct option: Expand $\sum_{k=1}^5 7$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$7 + 7 + 7 + 7 + 7$$

Worked Solutions

Summation

Q039 MCQ

Select the correct option: Expand $\sum_{k=3}^6 k^2$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$9 + 16 + 25 + 36$$

Q040 MCQ

Select the correct option: Expand $\sum_{k=1}^3 6^k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$6 + 36 + 216$$

Worked Solutions

Summation

Q041 MCQ

Select the correct option: Expand $\sum_{k=1}^5 (4k - 3)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$1 + 5 + 9 + 13 + 17$$

Q042 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (3k - 5)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$-2 + 1 + 4 + 7$$

Worked Solutions

Summation

Q043 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (4^k - 2)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$2 + 14 + 62$$

Q044 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (2k^2 - k)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$1 + 6 + 15$$

Worked Solutions

Summation

Q045 Written

Express $3 + 7 + 11 + \dots + (4n - 1)$ using the summation symbol Σ .**Solution**

1. The general term is $4k - 1$.
2. $\sum_{k=1}^n (4k - 1)$

Final answer

$$\sum_{k=1}^n (4k - 1)$$

Q046 Written

Express $5 + 8 + 11 + \dots + (3n + 2)$ using Σ .**Solution**

1. The general term is $3k + 2$.
2. $\sum_{k=1}^n (3k + 2)$

Final answer

$$\sum_{k=1}^n (3k + 2)$$

Worked Solutions

Summation

Q047 Written

Write the sum of the first n terms of the sequence $a_n = 2n + 1$ using Σ .

Solution

1. Change the term index from n to k .
2. $\sum_{k=1}^n (2k + 1)$

Final answer

$$\sum_{k=1}^n (2k + 1)$$

Q048 Written

Express $1 + 3 + 5 + \dots + (2n - 1)$ using Σ .

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (2k - 1)$

Final answer

$$\sum_{k=1}^n (2k - 1)$$

Worked Solutions

Summation

Q049 Written

Write the sum of the first n terms of $a_n = 7n - 1$ using Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (7k - 1)$

Final answer

$$\sum_{k=1}^n (7k - 1)$$

Q050 Written

Write the sum of the first n terms of $a_n = 4n + 5$ using Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (4k + 5)$

Final answer

$$\sum_{k=1}^n (4k + 5)$$

Worked Solutions

Summation

Q051 Written

Write the sum of the first n terms of $a_n = 3n - 2$ using Σ .

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (3k - 2)$

Final answer

$$\sum_{k=1}^n (3k - 2)$$

Q052 MCQ

Select the correct option: Express $2 + 5 + 8 + \cdots + (3n - 1)$ using the summation symbol Σ :

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^n (3k - 1)$$

Worked Solutions

Summation

Q053 MCQ

Select the correct option: Express $7 + 11 + 15 + \dots + 43$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=1}^{10} (4k + 3)$$

Q054 MCQ

Select the correct option: Express $6 + 10 + 14 + \dots + (4n + 2)$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=1}^n (4k + 2)$$

Worked Solutions

Summation

Q055 MCQ

Select the correct option: Express $8 + 13 + 18 + \dots + 103$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^{20} (5k + 3)$$

Q056 MCQ

Select the correct option: Express $11^2 + 12^2 + \dots + 19^2$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=11}^{19} k^2$$

Worked Solutions

Summation

Q057 MCQ

Select the correct option: Express $2^2 + 3^2 + \dots + 10^2$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=2}^{10} k^2$$

Q058 MCQ

Select the correct option: Express $1 + 3 + 9 + \dots + 3^{n-1}$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=0}^{n-1} 3^k$$

Worked Solutions

Summation

Q059 MCQ

Select the correct option: Express $2 + 7 + 12 + \dots + (5n - 3)$ using the summation symbol Σ :

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^n (5k - 3)$$

Q060 Written

Express $6^2 + 7^2 + 8^2 + \dots + 15^2$ using Σ .

Solution

1. The index starts at 6 and ends at 15.

2. $\sum_{k=6}^{15} k^2$

Final answer

$$\sum_{k=6}^{15} k^2$$

Worked Solutions

Summation

Q061 Written

Express $4^2 + 5^2 + 6^2 + \dots + 13^2$ using Σ .**Solution**

1. The index starts at 4 and ends at 13.
2. $\sum_{k=4}^{13} k^2$

Final answer

$$\sum_{k=4}^{13} k^2$$

Q062 Written

Express $1 + 7 + 25 + 79 + \dots + (3^n - 2)$ using Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (3^k - 2)$

Final answer

$$\sum_{k=1}^n (3^k - 2)$$

Worked Solutions

Summation

Q063 Written

Express $1 + 2 + 4 + 8 + \dots + 2^{n-1}$ using Σ .

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n 2^{k-1}$

Final answer

$$\sum_{k=1}^n 2^{k-1}$$

Q064 Written

Express $7^2 + 8^2 + \dots + 21^2$ using Σ .

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=7}^{21} k^2$

Final answer

$$\sum_{k=7}^{21} k^2$$

Worked Solutions

Summation

Q065 Written

Express $12^2 + 13^2 + \dots + 18^2$ using Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=12}^{18} k^2$

Final answer

$$\sum_{k=12}^{18} k^2$$

Q066 Written

Express $10^2 + 11^2 + \dots + 20^2$ using Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=10}^{20} k^2$

Final answer

$$\sum_{k=10}^{20} k^2$$

Worked Solutions

Summation

Q067 MCQ

Select the correct option: Express $4^2 + 5^2 + \dots + 12^2$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=4}^{12} k^2$$

Q068 MCQ

Select the correct option: Express $3^1 + 3^2 + \dots + 3^n$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^n 3^k$$

Worked Solutions

Summation

Q069 MCQ

Select the correct option: Express $2 + 4 + 8 + \dots + 2^n$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=1}^n 2^k$$

Q070 MCQ

Select the correct option: Express $1 + 2^1 + 2^2 + \dots + 2^{n-1}$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=0}^{n-1} 2^k$$

Worked Solutions

Summation

Q071 MCQ

Select the correct option: Express $7 + 14 + 21 + \dots + 7n$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^n 7k$$

Q072 MCQ

Select the correct option: Express $10 + 15 + 20 + \dots + 5n$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=2}^n 5k$$

Worked Solutions

Summation

Q073 MCQ

Select the correct option: Express $6^2 + 7^2 + \dots + 16^2$ using the summation symbol Σ :

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=6}^{16} k^2$$

Q074 Written

Express $2 + 5 + 8 + \dots + 29$ using Σ .

Solution

1. The general term is $3k - 1$, and $29 = 3(10) - 1$.
2. $\sum_{k=1}^{10} (3k - 1)$

Final answer

$$\sum_{k=1}^{10} (3k - 1)$$

Worked Solutions

Summation

Q075 Written

Express $8 + 16 + 24 + \cdots + 56$ **using** Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^7 8k$

Final answer

$$\sum_{k=1}^7 8k$$

Q076 Written

Express $7 + 13 + 19 + \cdots + 73$ **using** Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^{12} (6k + 1)$

Final answer

$$\sum_{k=1}^{12} (6k + 1)$$

Worked Solutions

Summation

Q077 Written

Express $4 + 9 + 14 + \dots + 119$ using Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^{24} (5k - 1)$

Final answer

$$\sum_{k=1}^{24} (5k - 1)$$

Worked Solutions

Summation

Q078 Challenge

A green-tech startup installs 1 solar panel in Week 1, 4 in Week 2, 7 in Week 3, and $3n - 2$ in Week n . Write one expression using Σ for the total panels installed from Week 1 through Week n .

Solution

1. The weekly pattern is arithmetic with general term $3k - 2$.
2. Total = $\sum_{k=1}^n (3k - 2)$

Final answer

$$\sum_{k=1}^n (3k - 2)$$

Worked Solutions

Summation

Q079 Written

Find $\sum_{k=1}^n (-3)$.

Solution

1. There are n copies of -3 .
2. $\sum_{k=1}^n (-3) = -3n$

Final answer

$$-3n$$

Q080 Written

Find $\sum_{k=1}^n 3$.

Solution

1. There are n copies of 3.
2. $\sum_{k=1}^n 3 = 3n$

Final answer

$$3n$$

Worked Solutions

Summation

Q081 Written

Find $\sum_{k=1}^n 6$.

Solution

1. $\sum_{k=1}^n 6 = 6n$

Final answer

$6n$

Q082 Written

Find $\sum_{k=1}^n (-4)$.

Solution

1. $\sum_{k=1}^n (-4) = -4n$

Final answer

$-4n$

Worked Solutions

Summation

Q083 Written

Find $\sum_{k=1}^n 1$.

Solution

1. $\sum_{k=1}^n 1 = n$

Final answer

$$n$$

Q084 MCQ

Select the correct option: Find $\sum_{k=1}^n 9$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$9n$$

Worked Solutions

Summation

Q085 MCQ

Select the correct option: Find $\sum_{k=1}^n (-2)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer $-2n$

Q086 MCQ

Select the correct option: Find $\sum_{k=1}^7 4$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

28

Worked Solutions

Summation

Q087 MCQ

Select the correct option: Find $\sum_{k=1}^{12} (-5)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

−60

Q088 MCQ

Select the correct option: Find $\sum_{k=1}^n k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$\frac{n(n+1)}{2}$

Worked Solutions

Summation

Q089 MCQ

Select the correct option: Find $\sum_{k=1}^n 4k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$2n(n + 1)$$

Q090 MCQ

Select the correct option: Find $\sum_{k=1}^{10} k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$55$$

Worked Solutions

Summation

Q091 MCQ

Select the correct option: Find $\sum_{k=1}^8 3k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

108

Q092 MCQ

Select the correct option: Find $\sum_{k=1}^6 5k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

105

Worked Solutions

Summation

Q093 MCQ

Select the correct option: Find $\sum_{k=1}^{20} k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

210

Q094 MCQ

Select the correct option: Find $\sum_{k=1}^n k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer
$$\frac{n(n+1)(2n+1)}{6}$$

Worked Solutions

Summation

Q095 MCQ

Select the correct option: Find $\sum_{k=1}^n 6k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$n(n+1)(2n+1)$$

Q096 MCQ

Select the correct option: Find $\sum_{k=1}^5 k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$55$$

Worked Solutions

Summation

Q097 MCQ

Select the correct option: Find $\sum_{k=1}^4 2k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

60

Q098 MCQ

Select the correct option: Find $\sum_{k=1}^3 4k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

56

Worked Solutions

Summation

Q099 MCQ

Select the correct option: Find $\sum_{k=1}^7 k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

140

Q100 Written

Find $\sum_{k=1}^n (5k + 4)$.

Solution

1. $\sum (5k + 4) = 5 \sum k + \sum 4$

2. $= \frac{5n(n+1)}{2} + 4n = \frac{n(5n+13)}{2}$

Final answer

$$\frac{n(5n + 13)}{2}$$

Worked Solutions

Summation

Q101 Written

Find $\sum_{k=1}^n (2k - 3)$.

Solution

$$1. \ 2 \sum k - \sum 3 = n(n+1) - 3n$$

$$2. \ = n(n-2)$$

Final answer

$$n(n-2)$$

Q102 Written

Find $\sum_{k=1}^n (4k - 1)$.

Solution

$$1. \ 4 \sum k - \sum 1 = 2n(n+1) - n$$

$$2. \ = n(2n+1)$$

Final answer

$$n(2n+1)$$

Worked Solutions

Summation

Q103 Written

Find $\sum_{k=1}^n (3k + 2)$.

Solution

$$1. \quad 3 \sum k + \sum 2 = \frac{3n(n+1)}{2} + 2n$$

$$2. \quad = \frac{n(3n+7)}{2}$$

Final answer

$$\frac{n(3n+7)}{2}$$

Q104 Written

Find $\sum_{k=1}^n (6k - 3)$.

Solution

$$1. \quad 6 \sum k - \sum 3 = 3n(n+1) - 3n$$

$$2. \quad = 3n^2$$

Final answer

$$3n^2$$

Worked Solutions

Summation

Q105 MCQ

Select the correct option: Find $\sum_{k=1}^n (3k - 1)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(3n + 1)}{2}$$

Q106 MCQ

Select the correct option: Find $\sum_{k=1}^n (2k + 5)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$n(n + 6)$$

Worked Solutions

Summation

Q107 MCQ

Select the correct option: Find $\sum_{k=1}^n (8k - 4)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$4n^2$$

Q108 MCQ

Select the correct option: Find $\sum_{k=1}^n (5k + 3)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(5n + 11)}{2}$$

Worked Solutions

Summation

Q109 Written

Find $\sum_{k=1}^n (k-1)^2$.

Solution

$$1. \sum (k^2 - 2k + 1) = \sum k^2 - 2 \sum k + \sum 1$$

$$2. = \frac{n(n-1)(2n-1)}{6}$$

Final answer

$$\frac{n(n-1)(2n-1)}{6}$$

Q110 MCQ

Select the correct option: Find $\sum_{k=1}^n (k^2 + k)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Summation

Q111 MCQ

Select the correct option: Find $\sum_{k=1}^n (2k^2 - k)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(n+1)(4n-1)}{6}$$

Q112 MCQ

Select the correct option: Find $\sum_{k=1}^n (k^2 - 1)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(2n^2 + 3n - 5)}{6}$$

Worked Solutions

Summation

Q113 MCQ

Select the correct option: Find $\sum_{k=1}^n (3k^2 + 2k)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(n+1)(2n+3)}{2}$$

Q114 MCQ

Select the correct option: Find $\sum_{k=1}^n (4k^2 - 3k + 2)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(8n^2 + 3n + 7)}{6}$$

Worked Solutions

Summation

Q115 Written

Find $1 \cdot 3 + 2 \cdot 8 + 3 \cdot 13 + \cdots + n(5n - 2)$.

Solution

$$\begin{aligned} 1. \quad & \sum_{k=1}^n k(5k - 2) = 5 \sum k^2 - 2 \sum k \\ 2. \quad & = \frac{5n(n+1)(2n+1)}{6} - n(n+1) = \frac{n(n+1)(10n-1)}{6} \end{aligned}$$

Final answer

$$\frac{n(n+1)(10n-1)}{6}$$

Q116 Written

Find $\sum_{k=1}^n (k-1)(3k-1)$.

Solution

$$\begin{aligned} 1. \quad & (k-1)(3k-1) = 3k^2 - 4k + 1 \\ 2. \quad & 3 \sum k^2 - 4 \sum k + \sum 1 = \frac{n(n-1)(2n+1)}{2} \end{aligned}$$

Final answer

$$\frac{n(n-1)(2n+1)}{2}$$

Worked Solutions

Summation

Q117 Written

Find $1 \cdot 3 + 2 \cdot 5 + 3 \cdot 7 + \cdots + n(2n + 1)$.

Solution

$$1. \sum_{k=1}^n k(2k + 1) = 2 \sum k^2 + \sum k$$

$$2. = \frac{n(n+1)(4n+5)}{6}$$

Final answer

$$\frac{n(n+1)(4n+5)}{6}$$

Q118 Written

Find $\sum_{k=1}^n k(k - 2)$.

Solution

$$1. \sum (k^2 - 2k) = \sum k^2 - 2 \sum k$$

$$2. = \frac{n(n+1)(2n-5)}{6}$$

Final answer

$$\frac{n(n+1)(2n-5)}{6}$$

Worked Solutions

Summation

Q119 Written

Find $\sum_{k=1}^n k(2k - 2)$.

Solution

$$\begin{aligned} 1. & 2 \sum k^2 - 2 \sum k \\ 2. & = \frac{2n(n+1)(n-1)}{3} \end{aligned}$$

Final answer

$$\frac{2n(n+1)(n-1)}{3}$$

Q120 Written

Find $1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 + \cdots + n(n+3)$.

Solution

$$\begin{aligned} 1. & \sum_{k=1}^n k(k+3) = \sum k^2 + 3 \sum k \\ 2. & = \frac{n(n+1)(n+5)}{3} \end{aligned}$$

Final answer

$$\frac{n(n+1)(n+5)}{3}$$

Worked Solutions

Summation

Q121 Written

Find $1 \cdot 5 + 2 \cdot 8 + 3 \cdot 11 + \cdots + n(3n + 2)$.**Solution**

$$1. \sum_{k=1}^n k(3k + 2) = 3 \sum k^2 + 2 \sum k$$

$$2. = \frac{n(n+1)(2n+3)}{2}$$

Final answer

$$\frac{n(n+1)(2n+3)}{2}$$

Q122 Written

Find $2 \cdot 5 + 3 \cdot 6 + 4 \cdot 7 + \cdots + (n+1)(n+4)$.**Solution**

$$1. \sum_{k=1}^n (k+1)(k+4) = \sum (k^2 + 5k + 4)$$

$$2. = \frac{n(n+4)(n+5)}{3}$$

Final answer

$$\frac{n(n+4)(n+5)}{3}$$

Worked Solutions

Summation

Q123 Written

Find $\sum_{k=1}^4 (2k - 1)$.

Solution

1. $2 \sum_{k=1}^4 k - \sum_{k=1}^4 1 = 2(10) - 4$
2. $= 16$

Final answer

16

Q124 Written

Find $\sum_{k=1}^5 (3k - 2)$.

Solution

1. $3 \sum_{k=1}^5 k - 2 \sum_{k=1}^5 1 = 3(15) - 10$
2. $= 35$

Final answer

35

Worked Solutions

Summation

Q125 Written

Find $\sum_{k=1}^{20} (2k - 7)$.

Solution

1. $2 \sum_{k=1}^{20} k - 7 \sum_{k=1}^{20} 1 = 2(210) - 140$

2. $= 280$

Final answer

280

Q126 Written

Find $\sum_{k=1}^8 (4k + 5)$.

Solution

1. $4 \sum_{k=1}^8 k + 5 \sum_{k=1}^8 1 = 4(36) + 40$

2. $= 184$

Final answer

184

Worked Solutions

Summation

Q127 Written

Find $\sum_{k=1}^{10} (2k - 1)$.

Solution

$$\begin{aligned} 1. & \ 2 \sum_{k=1}^{10} k - \sum_{k=1}^{10} 1 = 2(55) - 10 \\ 2. & \ = 100 \end{aligned}$$

Final answer

100

Worked Solutions

Summation

Q128 Challenge

Find $\sum_{r=4}^8 (r^2 + r + 5)$.

Solution

1. $\sum_{r=4}^8 (r^2 + r + 5) = \sum_{r=1}^8 (r^2 + r + 5) - \sum_{r=1}^3 (r^2 + r + 5)$
2. $= (204 + 36 + 40) - (14 + 6 + 15) = 245$

Final answer

245

Worked Solutions

Summation

Q129 Written

Find the sum $\sum_{k=1}^n k^2(k+1)$.

Solution

1. Expand: $k^2(k+1) = k^3 + k^2$.
2. Use $\sum k^3 = \frac{n^2(n+1)^2}{4}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $\sum_{k=1}^n k^2(k+1) = \frac{1}{12}n(n+1)(n+2)(3n+1)$

Final answer

$$\frac{n(n+1)(n+2)(3n+1)}{12}$$

Q130 Written

Find the sum $\sum_{k=1}^n k^2(k-2)$.

Solution

1. Expand: $k^2(k-2) = k^3 - 2k^2$.
2. Substitute the formulae for $\sum k^3$ and $\sum k^2$, then factor.
3. $\sum_{k=1}^n k^2(k-2) = \frac{n(n+1)(3n^2-5n-4)}{12}$

Final answer

$$\frac{n(n+1)(3n^2-5n-4)}{12}$$

Worked Solutions

Summation

Q131 Written

Find the sum $\sum_{k=1}^n k^2(k+2)$.

Solution

1. Expand: $k^2(k+2) = k^3 + 2k^2$.
2. Substitute the standard sums and factor the common $n(n+1)$.
3. $\sum_{k=1}^n k^2(k+2) = \frac{n(n+1)(3n^2 + 11n + 4)}{12}$

Final answer

$$\frac{n(n+1)(3n^2 + 11n + 4)}{12}$$

Q132 Written

Find the sum $\sum_{k=1}^n k^2(k-3)$.

Solution

1. Expand: $k^2(k-3) = k^3 - 3k^2$.
2. Substitute the standard sums and simplify the common factor.
3. $\sum_{k=1}^n k^2(k-3) = \frac{n(n+1)(n^2-3n-2)}{4}$

Final answer

$$\frac{n(n+1)(n^2-3n-2)}{4}$$

Worked Solutions

Summation

Q133 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+1)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n+2)(3n+1)}{12}$$

Q134 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-2)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2-5n-4)}{12}$$

Worked Solutions

Summation

Q135 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+2)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+11n+4)}{12}$$

Q136 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-3)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2-3n-2)}{4}$$

Worked Solutions

Summation

Q137 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+3)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(n^2+5n+2)}{4}$$

Q138 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-4)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(3n^2-13n-8)}{12}$$

Worked Solutions

Summation

Q139 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+4)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+19n+8)}{12}$$

Q140 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-1)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2-5n-4)}{12}$$

Worked Solutions

Summation

Q141 Written

Find the sum $\sum_{k=1}^n k(k^2 - 1)$.

Solution

1. Expand: $k(k^2 - 1) = k^3 - k$.
2. Use $\sum k^3$ and $\sum k$, then factor $n^2 + n - 2 = (n - 1)(n + 2)$.
3. $\sum_{k=1}^n k(k^2 - 1) = \frac{n(n + 1)(n - 1)(n + 2)}{4}$

Final answer

$$\frac{n(n + 1)(n - 1)(n + 2)}{4}$$

Q142 Written

Find the sum $\sum_{k=1}^n k(k^2 + 2)$.

Solution

1. Expand: $k(k^2 + 2) = k^3 + 2k$.
2. Use $\sum k^3$ and $\sum k$, then factor.
3. $\sum_{k=1}^n k(k^2 + 2) = \frac{n(n+1)(n^2 + n + 4)}{4}$

Final answer

$$\frac{n(n+1)(n^2 + n + 4)}{4}$$

Worked Solutions

Summation

Q143 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 1)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(n-1)(n+2)}{4}$$

Q144 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 2)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(n^2 + n + 4)}{4}$$

Worked Solutions

Summation

Q145 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 3)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n+6)}{4}$$

Q146 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 2)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n-4)}{4}$$

Worked Solutions

Summation

Q147 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 4)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n+8)}{4}$$

Q148 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 3)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n-6)}{4}$$

Worked Solutions

Summation

Q149 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+5)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+23n+10)}{12}$$

Q150 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-5)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2-17n-10)}{12}$$

Worked Solutions

Summation

Q151 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 5)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(n^2+n+10)}{4}$$

Q152 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 5)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(n^2+n-10)}{4}$$

Worked Solutions

Summation

Q153 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+6)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+27n+12)}{12}$$

Q154 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2+6)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n+12)}{4}$$

Worked Solutions

Summation

Q155 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k - 6)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(3n^2 - 25n - 12)}{12}$$

Q156 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 6)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(n^2 + n - 12)}{4}$$

Worked Solutions

Summation

Q157 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+7)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+31n+14)}{12}$$

Q158 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+8)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+35n+16)}{12}$$

Worked Solutions

Summation

Q159 Written

Find the sum $\sum_{k=1}^n 6 \times 3^{k-1}$.

Solution

1. Use $a = 6, r = 3$.
2. $S_n = \frac{6(3^n - 1)}{3 - 1} = 3(3^n - 1)$

Final answer

$$3(3^n - 1)$$

Q160 Written

Find the sum $\sum_{k=1}^n 4 \times 2^{k-1}$.

Solution

1. Use $a = 4, r = 2$.
2. $S_n = \frac{4(2^n - 1)}{2 - 1} = 4(2^n - 1)$

Final answer

$$4(2^n - 1)$$

Worked Solutions

Summation

Q161 Written

Find the sum $\sum_{k=1}^n 2^k$.

Solution

1. Rewrite as $(2 \times 2^{k-1})$.
2. $S_n = \frac{2(2^n - 1)}{2 - 1} = 2(2^n - 1)$

Final answer

$$2(2^n - 1)$$

Q162 Written

Find the sum $\sum_{k=1}^n 2 \times (-3)^{k-1}$.

Solution

1. Here $a = 2$ and $r = -3$.
2. $S_n = \frac{2(1 - (-3)^n)}{1 - (-3)} = \frac{1 - (-3)^n}{2}$

Final answer

$$\frac{1 - (-3)^n}{2}$$

Worked Solutions

Summation

Q163 Written

Find the sum $\sum_{k=1}^n 3 \times (-2)^{k-1}$.

Solution

1. Use $a = 3, r = -2$.
2. $S_n = \frac{3(1-(-2)^n)}{1-(-2)} = 1 - (-2)^n$

Final answer

$$1 - (-2)^n$$

Q164 Written

Find the sum $\sum_{k=1}^n (-5)^k$.

Solution

1. Rewrite as $(-5) \times (-5)^{k-1}$, so $a = -5, r = -5$.
2. $S_n = \frac{-5(1-(-5)^n)}{1-(-5)} = \frac{5((-5)^n - 1)}{6}$

Final answer

$$\frac{5((-5)^n - 1)}{6}$$

Worked Solutions

Summation

Q165 Written

Find the sum $\sum_{k=1}^n 2 \times (-1)^{k-1}$.

Solution

1. Use $a = 2, r = -1$.
2. $S_n = \frac{2(1-(-1)^n)}{2} = 1 - (-1)^n$

Final answer

$$1 - (-1)^n$$

Q166 Written

Find the sum $\sum_{k=1}^n 8 \times (-3)^{k-1}$.

Solution

1. Use $a = 8, r = -3$.
2. $S_n = \frac{8(1-(-3)^n)}{4} = 2(1 - (-3)^n)$

Final answer

$$2(1 - (-3)^n)$$

Worked Solutions

Summation

Q167 Written

Find the sum $\sum_{k=1}^n (-2)^{k-1}$.

Solution

1. Use $(a = 1, r = -2)$.
2. $S_n = \frac{1 - (-2)^n}{1 - (-2)} = \frac{1 - (-2)^n}{3}$

Final answer

$$\frac{1 - (-2)^n}{3}$$

Q168 Written

Find the sum $\sum_{k=1}^n (-3)^k$.

Solution

1. Rewrite as $((-3)(-3)^{k-1})$.
2. $S_n = \frac{-3(1 - (-3)^n)}{4} = \frac{3((-3)^n - 1)}{4}$

Final answer

$$\frac{3((-3)^n - 1)}{4}$$

Worked Solutions

Summation

Q169 Written

Find the sum $\sum_{k=1}^n 4^k$.

Solution

1. Rewrite $4^k = 4 \times 4^{k-1}$, so $a = 4, r = 4$.

2. $S_n = \frac{4(4^n - 1)}{4 - 1} = \frac{4(4^n - 1)}{3}$

Final answer

$$\frac{4(4^n - 1)}{3}$$

Q170 Written

Find the sum $\sum_{k=1}^n 4^{k-1}$.

Solution

1. The first term is 1 and the common ratio is 4.

2. $S_n = \frac{4^n - 1}{4 - 1} = \frac{4^n - 1}{3}$

Final answer

$$\frac{4^n - 1}{3}$$

Worked Solutions

Summation

Q171 Written

Find the sum $\sum_{k=1}^n 3^{k-1}$.

Solution

1. The first term is 1 and the ratio is 3.
2. $S_n = \frac{3^n - 1}{3 - 1} = \frac{3^n - 1}{2}$

Final answer

$$\frac{3^n - 1}{2}$$

Q172 Written

Find the sum $\sum_{k=1}^4 (-2)^{k-1}$.

Solution

1. This is a finite geometric sum with $(a = 1, r = -2, n = 4)$.
2. $S_4 = 1 - 2 + 4 - 8 = -5$

Final answer

$$-5$$

Worked Solutions

Summation

Q173 Written

Find the sum $\sum_{k=1}^3 3^k$.

Solution

1. Use $(a = 3, r = 3, n = 3)$.
2. $S_3 = 3 + 9 + 27 = 39$

Final answer

39

Q174 Written

Find the sum $\sum_{k=1}^5 2^k$.

Solution

1. Use $(a = 2, r = 2, n = 5)$.
2. $S_5 = 2 + 4 + 8 + 16 + 32 = 62$

Final answer

62

Worked Solutions

Summation

Q175 Written

Find the sum $\sum_{k=1}^4 (-3)^k$.

Solution

1. Use $(a = -3, r = -3, n = 4)$.
2. $S_4 = -3 + 9 - 27 + 81 = 60$

Final answer

60

Worked Solutions

Summation

Q176 Challenge

Find the finite sum $\sum_{r=5}^9 (2r + 7)$.

Solution

1. Substitute ($r = 5, 6, 7, 8, 9$): ($17+19+21+23+25$).
2. $17 + 19 + 21 + 23 + 25 = 105$

Final answer

105

Worked Solutions

Summation

Q177 Written

Find $\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \cdots + \frac{1}{(n+1)(n+2)}.$

Solution

1. Split each term as $\frac{1}{j} - \frac{1}{j+1}.$
2. $\frac{1}{2} - \frac{1}{n+2} = \frac{n}{2(n+2)}$

Final answer

$$\frac{n}{2(n+2)}$$

Q178 Written

Find $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \cdots + \frac{1}{n(n+1)}.$

Solution

1. Use $\frac{1}{j(j+1)} = \frac{1}{j} - \frac{1}{j+1}.$
2. $1 - \frac{1}{n+1} = \frac{n}{n+1}$

Final answer

$$\frac{n}{n+1}$$

Worked Solutions

Summation

Q179 Written

Find $\frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \frac{1}{5 \times 6} + \cdots + \frac{1}{(n+2)(n+3)}$.

Solution

1. The gap is 1 and the first term starts at 3.

2. $\frac{1}{3} - \frac{1}{n+3} = \frac{n}{3(n+3)}$

Final answer

$$\frac{n}{3(n+3)}$$

Q180 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 2} + \cdots + \frac{1}{n(n+1)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{n+1}$$

Worked Solutions

Summation

Q181 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 3} + \cdots + \frac{1}{(n+1)(n+2)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2(n+2)}$$

Q182 MCQ

Select the correct option: Find the sum $\frac{1}{3 \times 4} + \cdots + \frac{1}{(n+2)(n+3)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{3(n+3)}$$

Worked Solutions

Summation

Q183 MCQ

Select the correct option: Find the sum $\frac{1}{4 \times 5} + \cdots + \frac{1}{(n+3)(n+4)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{4(n+4)}$$

Q184 MCQ

Select the correct option: Find the sum $\frac{1}{6 \times 7} + \cdots + \frac{1}{(n+5)(n+6)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{6(n+6)}$$

Worked Solutions

Summation

Q185 MCQ

Select the correct option: Find the sum $\frac{1}{7 \times 8} + \cdots + \frac{1}{(n+6)(n+7)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{7(n+7)}$$

Q186 MCQ

Select the correct option: Find the sum $\frac{1}{9 \times 10} + \cdots + \frac{1}{(n+8)(n+9)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{9(n+9)}$$

Worked Solutions

Summation

Q187 MCQ

Select the correct option: Find the sum $\frac{1}{10 \times 11} + \cdots + \frac{1}{(n+9)(n+10)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{10(n+10)}$$

Q188 MCQ

Select the correct option: Find the sum $\frac{1}{11 \times 12} + \cdots + \frac{1}{(n+10)(n+11)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{11(n+11)}$$

Worked Solutions

Summation

Q189 Written

Find the following sum: $\frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \cdots + \frac{1}{(3n-1)(3n+2)}$.

Solution

1. Use $\frac{1}{ab} = \frac{1}{b-a} \left(\frac{1}{a} - \frac{1}{b} \right)$ with gap 3.
2. $\frac{1}{3} \left(\frac{1}{2} - \frac{1}{3n+2} \right) = \frac{n}{2(3n+2)}$

Final answer

$$\frac{n}{2(3n+2)}$$

Q190 Written

Find $\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \cdots + \frac{1}{(2n-1)(2n+1)}$.

Solution

1. The gap is 2, so factor out $\frac{1}{2}$.
2. $\frac{1}{2} \left(1 - \frac{1}{2n+1} \right) = \frac{n}{2n+1}$

Final answer

$$\frac{n}{2n+1}$$

Worked Solutions

Summation

Q191 Written

Find $\frac{1}{1 \times 3} + \frac{1}{2 \times 4} + \frac{1}{3 \times 5} + \cdots + \frac{1}{n(n+2)}.$

Solution

1. Use the constant gap 2 decomposition; two end terms remain.
2. $\frac{1}{2}(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2})$

Final answer

$$\frac{1}{2}(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2})$$

Q192 Written

Find $\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \frac{1}{7 \times 10} + \cdots + \frac{1}{(3n-2)(3n+1)}.$

Solution

1. Use gap 3 and the first denominator 1.
2. $\frac{1}{3}(1 - \frac{1}{3n+1}) = \frac{n}{3n+1}$

Final answer

$$\frac{n}{3n+1}$$

Worked Solutions

Summation

Q193 Written

Find $\frac{1}{1 \times 7} + \frac{1}{4 \times 10} + \frac{1}{7 \times 13} + \cdots + \frac{1}{(3n-2)(3n+4)}$.

Solution

1. The two factors differ by 6, while the first factors advance by 3; two terms remain at each end.
2. $\frac{1}{6}(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4})$

Final answer

$$\frac{1}{6}(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4})$$

Q194 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 3} + \cdots + \frac{1}{(2n-1)(2n+1)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2n+1}$$

Worked Solutions

Summation

Q195 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 4} + \cdots + \frac{1}{(3n-2)(3n+1)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{3n+1}$$

Q196 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 7} + \cdots + \frac{1}{(3n-2)(3n+4)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{1}{6} \left(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4} \right)$$

Worked Solutions

Summation

Q197 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 6} + \cdots + \frac{1}{(4n-2)(4n+2)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2(4n+2)}$$

Q198 MCQ

Select the correct option: Find the sum $\frac{1}{3 \times 8} + \cdots + \frac{1}{(5n-2)(5n+3)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{3(5n+3)}$$

Worked Solutions

Summation

Q199 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 5} + \cdots + \frac{1}{(4n-3)(4n+1)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{4n+1}$$

Q200 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 7} + \cdots + \frac{1}{(5n-3)(5n+2)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2(5n+2)}$$

Worked Solutions

Summation

Q201 MCQ

Select the correct option: Find the sum $\frac{1}{4 \times 10} + \cdots + \frac{1}{(6n-2)(6n+4)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{4(6n+4)}$$

Q202 Written

Find $\frac{1}{4 \times 6} + \frac{1}{5 \times 7} + \frac{1}{6 \times 8} + \cdots + \frac{1}{(n+3)(n+5)}$.

Solution

1. Write the general term as $\frac{1}{(j+3)(j+5)}$, gap 2.

2. $\frac{1}{2} \left(\frac{1}{4} + \frac{1}{5} - \frac{1}{n+4} - \frac{1}{n+5} \right)$

Final answer

$$\frac{1}{2} \left(\frac{1}{4} + \frac{1}{5} - \frac{1}{n+4} - \frac{1}{n+5} \right)$$

Worked Solutions

Summation

Q203 Written

Find $\frac{1}{2 \times 4} + \frac{1}{4 \times 6} + \frac{1}{6 \times 8} + \cdots + \frac{1}{2n(2n+2)}$.

Solution

1. Factor each denominator as $4j(j+1)$.
2. $\frac{1}{4}\left(1 - \frac{1}{n+1}\right) = \frac{n}{4(n+1)}$

Final answer

$$\frac{n}{4(n+1)}$$

Q204 Written

Find the following sum: $\sum_{k=1}^n \frac{1}{\sqrt{k+2} + \sqrt{k+1}}$.

Solution

1. Multiply by the conjugate.
2. $\frac{\sqrt{k+2}-\sqrt{k+1}}{(k+2)-(k+1)} = \sqrt{k+2} - \sqrt{k+1}$
3. $\sum_{k=1}^n (\sqrt{k+2} - \sqrt{k+1}) = \sqrt{n+2} - \sqrt{2}$

Final answer

$$\sqrt{n+2} - \sqrt{2}$$

Worked Solutions

Summation

Q205 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{k+1} + \sqrt{k}}.$

Solution

1. Rationalise with $\sqrt{k+1} - \sqrt{k}.$
2. $\sum_{k=1}^n (\sqrt{k+1} - \sqrt{k}) = \sqrt{n+1} - 1$

Final answer

$$\sqrt{n+1} - 1$$

Q206 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{k} + \sqrt{k-1}}.$

Solution

1. Rationalise: the denominator difference is 1.
2. $\sum_{k=1}^n (\sqrt{k} - \sqrt{k-1}) = \sqrt{n}$

Final answer

$$\sqrt{n}$$

Worked Solutions

Summation

Q207 Written

Find $\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \dots + \frac{1}{\sqrt{n} + \sqrt{n+1}}.$

Solution

1. Use the conjugate to obtain $\sqrt{j+1} - \sqrt{j}.$
2. $\sum_{j=1}^n (\sqrt{j+1} - \sqrt{j}) = \sqrt{n+1} - 1$

Final answer

$$\sqrt{n+1} - 1$$

Q208 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{2k-1} + \sqrt{2k+1}}.$

Solution

1. Rationalise; the square-difference gap is 2.
2. $\frac{1}{2} \sum_{k=1}^n (\sqrt{2k+1} - \sqrt{2k-1}) = \frac{\sqrt{2n+1}-1}{2}$

Final answer

$$\frac{\sqrt{2n+1} - 1}{2}$$

Worked Solutions

Summation

Q209 Written

Find $\frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \cdots + \frac{1}{\sqrt{n+1} + \sqrt{n+2}}.$

Solution

1. Multiply by each conjugate.
2. $\sum_{j=1}^n (\sqrt{j+2} - \sqrt{j+1}) = \sqrt{n+2} - \sqrt{2}$

Final answer

$$\sqrt{n+2} - \sqrt{2}$$

Worked Solutions

Summation

Q210 Challenge

Find the sum, showing your steps: $\sum_{r=1}^{30} \frac{1}{r(r+2)}$.

Solution

1. Use $\frac{1}{r(r+2)} = \frac{1}{2} \left(\frac{1}{r} - \frac{1}{r+2} \right)$.
2. $S = \frac{1}{2} \left(1 + \frac{1}{2} - \frac{1}{31} - \frac{1}{32} \right)$
3. $S = \frac{1425}{1984}$

Final answer

$$\frac{1425}{1984}$$

Worked Solutions

Summation

Q211 Written

Find the following sum $S = 3 \times 1 + 5 \times 3 + 7 \times 3^2 + 9 \times 3^3 + \cdots + (2n + 1) \times 3^{n-1}$.

Solution

1. The arithmetic coefficient is $2k + 1$ and the geometric ratio is $r = 3$.
2. Write S and $3S$, subtract with aligned powers, and simplify: $-2S = -2n \times 3^n$.
3. $S = n \times 3^n$

Final answer

$$n \times 3^n$$

Q212 Written

Find the sum $\sum_{k=1}^n (2k + 1) \times (3)^{k-1}$.

Solution

1. Write the first sum and multiply by the common ratio.
2. Subtract and keep the final shifted term with its correct sign.
3. $3^n n$

Final answer

$$3^n n$$

Worked Solutions

Summation

Q213 Written

Find the sum $\sum_{k=1}^4 (2k + 1) \times (3)^{k-1}$.

Solution

1. Identify the arithmetic coefficient and geometric ratio.
2. Apply (S-rS) over the finite limit ($n = 4$).
3. 324

Final answer

324

Q214 Written

Find the sum $\sum_{k=1}^n (2k - 1) \times (4)^{k-1}$.

Solution

1. Write the first sum and multiply by the common ratio.
2. Subtract and keep the final shifted term with its correct sign.
3. $\frac{6 \cdot 2^{2n}n - 5 \cdot 2^{2n} + 5}{9}$

Final answer

$$\frac{6 \cdot 2^{2n}n - 5 \cdot 2^{2n} + 5}{9}$$

Worked Solutions

Summation

Q215 Written

Find the sum $\sum_{k=1}^6 (2k - 1) \times (4)^{k-1}$.

Solution

1. Identify the arithmetic coefficient and geometric ratio.
2. Apply (S-rS) over the finite limit ($n = 6$).
3. 14109

Final answer

14109

Q216 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k + 1) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer $3^n n$

Worked Solutions

Summation

Q217 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k - 1) \times (4)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{6 \cdot 2^{2n}n - 5 \cdot 2^{2n} + 5}{9}$$

Q218 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k + 4) \times (2)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$2^n n + 3 \cdot 2^n - 3$$

Worked Solutions

Summation

Q219 Written

Find the following sum $S = 2 \times 1 + 3 \times 3 + 4 \times 3^2 + 5 \times 3^3 + \dots + (n + 1) \times 3^{n-1}$.

Solution

- Here the coefficient is $k + 1$ and $r = 3$.
- Subtract $3S$ from S ; the interior coefficients reduce to a geometric sum of 1s.
- $-2S = \frac{1 - (2n+1) \times 3^n}{2}$
- $S = \frac{(2n+1) \times 3^{n+1} - 1}{4}$

Final answer

$$\frac{(2n + 1) \times 3^{n+1} - 1}{4}$$

Q220 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k + 1) \times (3)^{k-1}$:

Solution

- Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{2 \times 3^{n+1} + 3^n - 1}{4}$$

Worked Solutions

Summation

Q221 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k-2) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}$$

Q222 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k-3) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$3^n n - 2 \cdot 3^n + 2$$

Worked Solutions

Summation

Q223 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k + 1) \times (-2)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$-\frac{6(-2)^n n + 5(-2)^n - 5}{9}$$

Worked Solutions

Summation

Q224 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k+2) \times (-3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$-\frac{4(-3)^n n + 9(-3)^n - 9}{16}$$

Worked Solutions

Summation

Q225 Written

Find the following sum $S = 1 \times 1 + 2 \times 2 + 3 \times 2^2 + 4 \times 2^3 + \cdots + n \times 2^{n-1}$.

Solution

1. Use $a_k = k$ and $r = 2$.
2. Applying $S - rS$ gives $-S = 1 - (n-1) \times 2^n$.
3. $S = (n-1) \times 2^n + 1$

Final answer

$$(n-1) \times 2^n + 1$$

Q226 Written

Find the following sum $S = 1 \times 3 + 3 \times 3^2 + 5 \times 3^3 + 7 \times 3^4 + \cdots + (2n-1) \times 3^n$.

Solution

1. Factor out the leading 3, leaving $\sum_{k=1}^n (2k-1)3^{k-1}$.
2. The $S - rS$ subtraction gives $\sum_{k=1}^n (2k-1)3^{k-1} = (n-1)3^n + 1$.
3. $S = (n-1) \times 3^{n+1} + 3$

Final answer

$$(n-1) \times 3^{n+1} + 3$$

Worked Solutions

Summation

Q227 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (3k) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}$$

Worked Solutions

Summation

Q228 Challenge

Find the sum of the partial sums $5 + (5 + 9) + (5 + 9 + 13) + \cdots + [5 + 9 + \cdots + (4k + 1)]$.

Solution

1. The j -th inner arithmetic sum is $5 + 9 + \cdots + (4j + 1) = j(2j + 3)$.
2. Sum the partial sums: $\sum_{j=1}^k (2j^2 + 3j)$.
3. $\frac{k(k+1)(4k+11)}{6}$

Final answer

$$\frac{k(k+1)(4k+11)}{6}$$

Worked Solutions

Summation

Q229 Written

For the sequence 3, 3 + 5, 3 + 5 + 7, 3 + 5 + 7 + 9, ..., find the k -th term.

Solution

1. The inner terms form an arithmetic sequence with first term 3 and common difference 2.
2. $u_i = 2i + 1$, $a_k = \sum_{i=1}^k u_i$
3. $a_k = \frac{k}{2} [2(3) + (k - 1)(2)] = k(k + 2)$

Final answer

$$a_k = k(k + 2)$$

Q230 Written

For the sequence 1, 1 + 3, 1 + 3 + 5, 1 + 3 + 5 + 7, ..., find the k -th term.

Solution

1. The inner terms form an arithmetic sequence with first term 1 and common difference 2.
2. $u_i = 2i - 1$, $a_k = \sum_{i=1}^k u_i$
3. $a_k = \frac{k}{2} [2(1) + (k - 1)(2)] = k^2$

Final answer

$$a_k = k^2$$

Worked Solutions

Summation

Q231 Written

For the sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, **find the k -th term.**

Solution

1. The inner terms form an arithmetic sequence with first term 2 and common difference 2.
2. $u_i = 2i, \quad a_k = \sum_{i=1}^k u_i$
3. $a_k = \frac{k}{2} [2(2) + (k-1)(2)] = k(k+1)$

Final answer

$$a_k = k(k+1)$$

Q232 Written

For the sequence $1, 1 + 5, 1 + 5 + 9, 1 + 5 + 9 + 13, \dots$, **find the k -th term.**

Solution

1. The inner terms form an arithmetic sequence with first term 1 and common difference 4.
2. $u_i = 4i - 3, \quad a_k = \sum_{i=1}^k u_i$
3. $a_k = \frac{k}{2} [2(1) + (k-1)(4)] = k(2k-1)$

Final answer

$$a_k = k(2k-1)$$

Worked Solutions

Summation

Q233 MCQ

Select the correct option: For the cumulative sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, what is a_k :**Solution**

1. The inner terms form an arithmetic sequence with first term 2 and common difference 2.

2. $u_i = 2i, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(2) + (k-1)(2)] = k(k+1)$

Correct option: A**Final answer**

$$k(k+1)$$

Q234 MCQ

Select the correct option: For the cumulative sequence $3, 3 + 6, 3 + 6 + 9, 3 + 6 + 9 + 12, \dots$, what is a_k :**Solution**

1. The inner terms form an arithmetic sequence with first term 3 and common difference 3.

2. $u_i = 3i, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(3) + (k-1)(3)] = \frac{3k(k+1)}{2}$

Correct option: A**Final answer**

$$\frac{3k(k+1)}{2}$$

Worked Solutions

Summation

Q235 MCQ

Select the correct option: For the cumulative sequence 4, 4 + 6, 4 + 6 + 8, 4 + 6 + 8 + 10, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 4 and common difference 2.

2. $u_i = 2i + 2, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(4) + (k-1)(2)] = k(k+3)$

Correct option: A

Final answer

$$k(k+3)$$

Q236 MCQ

Select the correct option: For the cumulative sequence 2, 2 + 6, 2 + 6 + 10, 2 + 6 + 10 + 14, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 2 and common difference 4.

2. $u_i = 4i - 2, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(2) + (k-1)(4)] = 2k^2$

Correct option: A

Final answer

$$2k^2$$

Worked Solutions

Summation

Q237 MCQ

Select the correct option: For the cumulative sequence 5, 5 + 8, 5 + 8 + 11, 5 + 8 + 11 + 14, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 5 and common difference 3.

$$2. u_i = 3i + 2, \quad a_k = \sum_{i=1}^k u_i$$

$$3. a_k = \frac{k}{2} [2(5) + (k-1)(3)] = \frac{k(3k+7)}{2}$$

Correct option: A

Final answer

$$\frac{k(3k+7)}{2}$$

Q238 MCQ

Select the correct option: For the cumulative sequence 3, 3 + 8, 3 + 8 + 13, 3 + 8 + 13 + 18, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 3 and common difference 5.

$$2. u_i = 5i - 2, \quad a_k = \sum_{i=1}^k u_i$$

$$3. a_k = \frac{k}{2} [2(3) + (k-1)(5)] = \frac{k(5k+1)}{2}$$

Correct option: A

Final answer

$$\frac{k(5k+1)}{2}$$

Worked Solutions

Summation

Q239 MCQ

Select the correct option: For the cumulative sequence 6, $6 + 8$, $6 + 8 + 10$, $6 + 8 + 10 + 12$, ..., **what is** a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 6 and common difference 2.

2. $u_i = 2i + 4, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(6) + (k-1)(2)] = k(k+5)$

Correct option: A

Final answer

$$k(k+5)$$

Q240 MCQ

Select the correct option: For the cumulative sequence 4, $4 + 8$, $4 + 8 + 12$, $4 + 8 + 12 + 16$, ..., **what is** a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 4 and common difference 4.

2. $u_i = 4i, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(4) + (k-1)(4)] = 2k(k+1)$

Correct option: A

Final answer

$$2k(k+1)$$

Worked Solutions

Summation

Q241 Written

For the sequence 3, 3 + 5, 3 + 5 + 7, 3 + 5 + 7 + 9, ..., find the sum from the first term to the n -th term.

Solution

1. First collapse the inner arithmetic sum: $a_k = k(k + 2)$.
2. $S_n = \sum_{k=1}^n a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(2n+7)}{6}$

Final answer

$$S_n = \frac{n(n+1)(2n+7)}{6}$$

Q242 Written

For the sequence 1, 1 + 3, 1 + 3 + 5, 1 + 3 + 5 + 7, ..., find the sum from the first term to the n -th term.

Solution

1. First collapse the inner arithmetic sum: $a_k = k^2$.
2. $S_n = \sum_{k=1}^n a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(2n+1)}{6}$

Final answer

$$S_n = \frac{n(n+1)(2n+1)}{6}$$

Worked Solutions

Summation

Q243 Written

For the sequence $1, 1 + 5, 1 + 5 + 9, 1 + 5 + 9 + 13, \dots$, **find the sum from the first term to the n -th term.**

Solution

1. First collapse the inner arithmetic sum: $a_k = k(2k - 1)$.
2. $S_n = \sum_{k=1}^n a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(4n-1)}{6}$

Final answer

$$S_n = \frac{n(n+1)(4n-1)}{6}$$

Q244 Written

For the sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, **find the sum from the first term to the n -th term.**

Solution

1. First collapse the inner arithmetic sum: $a_k = k(k + 1)$.
2. $S_n = \sum_{k=1}^n a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(n+2)}{3}$

Final answer

$$S_n = \frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Summation

Q245 MCQ

Select the correct option: For the cumulative sequence $2, 2 + 5, 2 + 5 + 8, 2 + 5 + 8 + 11, \dots$, find a_5 :

Solution

1. The inner terms form an arithmetic sequence with first term 2 and common difference 3.

2. $u_i = 3i - 1, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(2) + (k - 1)(3)] = \frac{k(3k+1)}{2}$

4. $a_5 = 40$

Correct option: A

Final answer

40

Worked Solutions

Summation

Q246 Challenge

The sum of the first n terms of $\{a_n\}$ is $S_n = 2n^2 + 5n$. Find the general term.

Solution

1. $a_1 = S_1 = 7$
2. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
3. $a_n = (2n^2 + 5n) - [2(n-1)^2 + 5(n-1)] = 4n + 3$
4. The formula also gives $a_1 = 7$, so it is valid for every positive integer n .

Final answer

$$a_n = 4n + 3$$

Worked Solutions

Summation

Q247 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(n - 3)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n - 4)(n - 1)$
3. $a_n = 2(n - 2)$
4. $a_1 = S_1 = -2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2(n - 2)$$

Worked Solutions

Summation

Q248 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(2n + 5)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(2n+3)$
3. $a_n = 4n + 3$
4. $a_1 = S_1 = 7$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 4n + 3$$

Worked Solutions

Summation

Q249 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^2$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)^2$
3. $a_n = 2n - 1$
4. $a_1 = S_1 = 1$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2n - 1$$

Worked Solutions

Summation

Q250 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(n - 4)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n - 5)(n - 1)$
3. $a_n = 2n - 5$
4. $a_1 = S_1 = -3$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2n - 5$$

Worked Solutions

Summation

Q251 MCQ

Select the correct option: If $S_n = n(n + 2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n - 1)(n + 1)$
3. $a_n = 2n + 1$
4. $a_1 = S_1 = 3$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: D

Final answer

$$a_n = 2n + 1$$

Worked Solutions

Summation

Q252 MCQ

Select the correct option: If $S_n = 2n(n + 2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 2(n-1)(n+1)$
3. $a_n = 2(2n+1)$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: C

Final answer

$$a_n = 2(2n + 1)$$

Worked Solutions

Summation

Q253 MCQ

Select the correct option: If $S_n = 3n(n - 1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 3(n - 2)(n - 1)$
3. $a_n = 6(n - 1)$
4. $a_1 = S_1 = 0$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: B

Final answer

$$a_n = 6(n - 1)$$

Worked Solutions

Summation

Q254 MCQ

Select the correct option: If $S_n = 2n(2n - 1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 2(n-1)(2n-3)$
3. $a_n = 2(4n-3)$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: A

Final answer

$$a_n = 2(4n - 3)$$

Worked Solutions

Summation

Q255 MCQ

Select the correct option: If $S_n = 5n(n + 1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 5n(n - 1)$
3. $a_n = 10n$
4. $a_1 = S_1 = 10$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: D

Final answer

$$a_n = 10n$$

Worked Solutions

Summation

Q256 MCQ

Select the correct option: If $S_n = n(2n + 3)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(2n+1)$
3. $a_n = 4n + 1$
4. $a_1 = S_1 = 5$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: C

Final answer

$$a_n = 4n + 1$$

Worked Solutions

Summation

Q257 MCQ

Select the correct option: If $S_n = n(3n + 1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(3n-2)$
3. $a_n = 2(3n-1)$
4. $a_1 = S_1 = 4$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: C

Final answer

$$a_n = 2(3n - 1)$$

Worked Solutions

Summation

Q258 MCQ

Select the correct option: If $S_n = n(n^2 + 2n + 3)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(n^2 + 2)$
3. $a_n = 3n^2 + n + 2$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: C

Final answer

$$a_n = 3n^2 + n + 2$$

Worked Solutions

Summation

Q259 MCQ

Select the correct option: If $S_n = n(n+1)(2n-1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n(n-1)(2n-3)$
3. $a_n = 2n(3n-2)$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: B

Final answer

$$a_n = 2n(3n-2)$$

Worked Solutions

Summation

Q260 MCQ

Select the correct option: If $S_n = n(3n^2 - 2n + 4)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(3n^2 - 8n + 9)$
3. $a_n = 9n^2 - 13n + 9$
4. $a_1 = S_1 = 5$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: A

Final answer

$$a_n = 9n^2 - 13n + 9$$

Worked Solutions

Summation

Q261 MCQ

Select the correct option: If $S_n = n(n^2 - n + 2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(n^2 - 3n + 4)$
3. $a_n = 3n^2 - 5n + 4$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: D

Final answer

$$a_n = 3n^2 - 5n + 4$$

Worked Solutions

Summation

Q262 MCQ

Select the correct option: If $S_n = n(n+1)(2n+1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n(n-1)(2n-1)$
3. $a_n = 6n^2$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: D

Final answer

$$a_n = 6n^2$$

Worked Solutions

Summation

Q263 MCQ

Select the correct option: If $S_n = n(n+1)(n+2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n(n-1)(n+1)$
3. $a_n = 3n(n+1)$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: B

Final answer

$$a_n = 3n(n+1)$$

Worked Solutions

Summation

Q264 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^3 + 2n + 6$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n^3 - 3n^2 + 5n + 3$
3. $a_n = 3(n^2 - n + 1)$
4. $a_1 = S_1 = 9$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 9, \quad a_n = 3(n^2 - n + 1) \quad (n \geq 2)$$

Worked Solutions

Summation

Q265 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^2 + 2n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n^2 - 2$
3. $a_n = 2n + 1$
4. $a_1 = S_1 = 2$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 2, \quad a_n = 2n + 1 \quad (n \geq 2)$$

Worked Solutions

Summation

Q266 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = (n-1)(n^2 + n + 1)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-2)(n^2 - n + 1)$
3. $a_n = 3n^2 - 3n + 1$
4. $a_1 = S_1 = 0$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 0, \quad a_n = 3n^2 - 3n + 1 \quad (n \geq 2)$$

Worked Solutions

Summation

Q267 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 5^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{5^n - 5}{5}$
3. $a_n = 4 \cdot 5^{n-1}$
4. $a_1 = S_1 = 4$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 4 \cdot 5^{n-1}$$

Worked Solutions

Summation

Q268 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 3^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{3^n - 3}{3}$
3. $a_n = 2 \cdot 3^{n-1}$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2 \cdot 3^{n-1}$$

Worked Solutions

Summation

Q269 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 2^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{2^n - 2}{2}$
3. $a_n = 2^{n-1}$
4. $a_1 = S_1 = 1$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2^{n-1}$$

Worked Solutions

Summation

Q270 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 7^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{7^n - 7}{7}$
3. $a_n = 6 \cdot 7^{n-1}$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 6 \cdot 7^{n-1}$$

Worked Solutions

Summation

Q271 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 5^n - 2$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{5^n - 10}{5}$
3. $a_n = 4 \cdot 5^{n-1}$
4. $a_1 = S_1 = 3$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 3, \quad a_n = 4 \cdot 5^{n-1} \quad (n \geq 2)$$

Worked Solutions

Summation

Q272 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 4 \cdot 2^n - 3$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 4 \cdot 2^{n-1} - 3$
3. $a_n = 2^{n+1}$
4. $a_1 = S_1 = 5$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 5, \quad a_n = 2^{n+1} \quad (n \geq 2)$$

Worked Solutions

Summation

Q273 MCQ

Select the correct option: If $S_n = n(n + 4)$, find a_5 :**Solution**

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n - 1)(n + 3)$
3. $a_n = 2n + 3$
4. $a_1 = S_1 = 5$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .
6. $a_5 = 13$

Correct option: B**Final answer**

13

Worked Solutions

Summation

Q274 Challenge

Find the general term of the sequence $\{a_n\}$ where S_n is given by $S_n = 3 \cdot 3^n - 2n + 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 3^n - 2n + 3$
3. $a_n = 2(3^n - 1)$
4. $a_1 = S_1 = 8$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 8, \quad a_n = 2(3^n - 1) \quad (n \geq 2)$$

Worked Solutions

Summation

Quiz MCQ 1 **Concept Quiz · MCQ**

Select the correct option: Expand $\sum_{k=1}^4 (3k - 1)$ into a sum of terms:

Solution

1. Substitute $k = 1, 2, 3, 4$.

Correct option: A

Final answer

$$2 + 5 + 8 + 11$$

Quiz MCQ 2 **Concept Quiz · MCQ**

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 1)$:

Solution

1. Expand $k(k^2 - 1) = k^3 - k$.

Correct option: A

Final answer

$$\frac{n(n+1)(n-1)(n+2)}{4}$$

Worked Solutions

Summation

Quiz MCQ 3 **Concept Quiz · MCQ**

Select the correct option: Find the sum $\sum_{k=1}^n k \times 2^{k-1}$:

Solution

1. Write S and rS, subtract, and isolate S.

Correct option: A

Final answer

$$(n - 1) \times 2^n + 1$$

Worked Solutions

Summation

Quiz MCQ 4 Concept Quiz · MCQ

Select the correct option: If $S_n = 2^n - 3$, find a_6 :

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{2^n - 6}{2}$
3. $a_n = 2^{n-1}$
4. $a_1 = S_1 = -1$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.
6. $a_6 = 32$

Correct option: C

Final answer

32

Worked Solutions

Summation

Quiz WRITTEN 5 Concept Quiz · Written

Expand $\sum_{k=1}^5 (k^2 - 2)$ into a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.

Final answer

$$-1 + 2 + 7 + 14 + 23$$

Quiz WRITTEN 6 Concept Quiz · Written

Find the sum $\sum_{k=1}^n k^2(k + 5)$.

Solution

1. Expand into $(k^3 + 5k^2)$.
2. Use the standard power sums and factor.

Final answer

$$\frac{n(n+1)(3n^2 + 23n + 10)}{12}$$

Worked Solutions

Summation

Quiz WRITTEN 7 Concept Quiz · Written

Find the sum $\sum_{k=1}^n (4k - 1) \times 3^k$.

Solution

1. Show the S-rS alignment (or the inner arithmetic sum for the challenge).
2. $\frac{3(4 \cdot 3^n n - 3 \cdot 3^n + 3)}{2}$

Final answer

$$\frac{3(4 \cdot 3^n n - 3 \cdot 3^n + 3)}{2}$$

Worked Solutions

Summation

Quiz WRITTEN 8 Concept Quiz · Written

Find the general term of the sequence $\{a_n\}$ when S_n is $2 \cdot 2^n - 2n + 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 2^n - 2n + 3$
3. $a_n = 2^n - 2$
4. $a_1 = S_1 = 3$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 3, \quad a_n = 2^n - 2 \quad (n \geq 2)$$

Answer key

Use the question number shown on each page.

Q001 $2 + 4 + 6 + 8 + 10$

Q002 $3 + 5 + 7 + 9 + 11$

Q003 $3 + 6 + 9 + 12 + 15$

Q004 $2 + 6 + 10 + 14$

Q005 $-1 + 1 + 3 + 5 + 7$

Q006 A

Q007 A

Q008 A

Q009 A

Q010 A

Q011 A

Q012 A

Q013 A

Q014 $-2 + 1 + 6$

Q015 $5 + 14 + 29$

Q016 $1 + 4 + 9 + 16$

Q017 $5 + 11 + 21$

Q018 $-1 + 0 + 3 + 8$

Q019 A

Q020 A

Q021 A

Q022 A

Q023 A

Q024 A

Q025 A

Q026 A

Q027 $3 + 9 + 27 + 81$

Q028 $2 + 2 + 2$

Q029 $2 + 4 + 8 + 16$

Q030 $3 + 3 + 3 + 3 + 3 + 3$

Q031 $3 + 9 + 27$

Q032 $8 + 28 + 128$

Q033 $-1 + 1 + 5 + 13$

Q034 $5 + 5 + 5 + 5$

Q035 $3 + 3 + 3$

Q036 $4 + 4 + 4 + 4 + 4$

Q037 A

Q038 A

Q039 A

Q040 A

Q041 A

Q042 A

Q043 A

Q044 A

Q045 $\sum_{k=1}^n (4k - 1)$

Q046 $\sum_{k=1}^n (3k + 2)$

Q047 $\sum_{k=1}^n (2k + 1)$

Q048 $\sum_{k=1}^n (2k - 1)$

Q049 $\sum_{k=1}^n (7k - 1)$

Q050 $\sum_{k=1}^n (4k + 5)$

Q051 $\sum_{k=1}^n (3k - 2)$

Q052 A

Q053 A

Q054 A

Q055 A

Q056 A

Q057 A

Q058 A

Q059 A

Q060 $\sum_{k=6}^{15} k^2$

Q061 $\sum_{k=4}^{13} k^2$

Q062 $\sum_{k=1}^n (3^k - 2)$

Q063 $\sum_{k=1}^n 2^{k-1}$

Q064 $\sum_{k=7}^{21} k^2$

Q065 $\sum_{k=12}^{18} k^2$

Q066 $\sum_{k=10}^{20} k^2$

Q067 A

Q068 A

Q069 A

Q070 A

Q071 A

Q072 A

Q073 A

Q074 $\sum_{k=1}^{10} (3k - 1)$

Q075 $\sum_{k=1}^7 8k$

Q076 $\sum_{k=1}^{12} (6k + 1)$

Q077 $\sum_{k=1}^{24} (5k - 1)$

Q078 $\sum_{k=1}^n (3k - 2)$

Q079 $-3n$

Q080 $3n$

Q081 $6n$

Q082 $-4n$

Q083 n

Q084 A

Q085 A

Q086 A

Q087 A

Q088 A

Q089 A	Q118 $\frac{n(n+1)(2n-5)}{6}$	Q145 A	Q174 62
Q090 A	Q119 $\frac{2n(n+1)(n-1)}{3}$	Q146 A	Q175 60
Q091 A	Q120 $\frac{n(n+1)(n+5)}{3}$	Q147 A	Q176 105
Q092 A	Q121 $\frac{n(n+1)(2n+3)}{2}$	Q148 A	Q177 $\frac{n}{2(n+2)}$
Q093 A	Q122 $\frac{n(n+4)(n+5)}{3}$	Q149 A	Q178 $\frac{n}{n+1}$
Q094 A	Q123 16	Q150 A	Q179 $\frac{n}{3(n+3)}$
Q095 A	Q124 35	Q151 A	Q180 A
Q096 A	Q125 280	Q152 A	Q181 A
Q097 A	Q126 184	Q153 A	Q182 A
Q098 A	Q127 100	Q154 A	Q183 A
Q099 A	Q128 245	Q155 A	Q184 A
Q100 $\frac{n(5n+13)}{2}$	Q129 $\frac{n(n+1)(n+2)(3n+1)}{12}$	Q156 A	Q185 A
Q101 $n(n-2)$	Q130 $\frac{n(n+1)(3n^2-5n-4)}{12}$	Q157 A	Q186 A
Q102 $n(2n+1)$	Q131 $\frac{n(n+1)(3n^2+11n+4)}{12}$	Q158 A	Q187 A
Q103 $\frac{n(3n+7)}{2}$	Q132 $\frac{n(n+1)(n^2-3n-2)}{4}$	Q159 $3(3^n-1)$	Q188 A
Q104 $3n^2$	Q133 A	Q160 $4(2^n-1)$	Q189 $\frac{n}{2(3n+2)}$
Q105 A	Q134 A	Q161 $2(2^n-1)$	Q190 $\frac{n}{2n+1}$
Q106 A	Q135 A	Q162 $\frac{1-(-3)^n}{2}$	Q191 $\frac{1}{2}(1+\frac{1}{2}-\frac{1}{n+1}-\frac{1}{n+2})$
Q107 A	Q136 A	Q163 $1-(-2)^n$	Q192 $\frac{n}{3n+1}$
Q108 A	Q137 A	Q164 $\frac{5((-5)^n-1)}{6}$	Q193 $\frac{1}{6}(1+\frac{1}{4}-\frac{1}{3n+1}-\frac{1}{3n+4})$
Q109 $\frac{n(n-1)(2n-1)}{6}$	Q138 A	Q165 $1-(-1)^n$	Q194 A
Q110 A	Q139 A	Q166 $2(1-(-3)^n)$	Q195 A
Q111 A	Q140 A	Q167 $\frac{1-(-2)^n}{3}$	Q196 A
Q112 A	Q141 $\frac{n(n+1)(n-1)(n+2)}{4}$	Q168 $\frac{3((-3)^n-1)}{4}$	Q197 A
Q113 A	Q142 $\frac{n(n+1)(n^2+n+4)}{4}$	Q169 $\frac{4(4^n-1)}{3}$	Q198 A
Q114 A	Q143 A	Q170 $\frac{4^n-1}{3}$	Q199 A
Q115 $\frac{n(n+1)(10n-1)}{6}$	Q144 A	Q171 $\frac{3^n-1}{2}$	Q200 A
Q116 $\frac{n(n-1)(2n+1)}{2}$		Q172 -5	Q201 A
Q117 $\frac{n(n+1)(4n+5)}{6}$		Q173 39	Q202 $\frac{1}{2}(\frac{1}{4}+\frac{1}{5}-\frac{1}{n+4}-\frac{1}{n+5})$

$$\text{Q203 } \frac{n}{4(n+1)}$$

$$\text{Q204 } \sqrt{n+2} - \sqrt{2}$$

$$\text{Q205 } \sqrt{n+1} - 1$$

$$\text{Q206 } \sqrt{n}$$

$$\text{Q207 } \sqrt{n+1} - 1$$

$$\text{Q208 } \frac{\sqrt{2n+1} - 1}{2}$$

$$\text{Q209 } \sqrt{n+2} - \sqrt{2}$$

$$\text{Q210 } \frac{1425}{1984}$$

$$\text{Q211 } n \times 3^n$$

$$\text{Q212 } 3^n n$$

$$\text{Q213 } 324$$

$$\text{Q214 } \frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n} + 5}{9}$$

$$\text{Q215 } 14109$$

$$\text{Q216 } A$$

$$\text{Q217 } A$$

$$\text{Q218 } A$$

$$\text{Q219 } \frac{(2n+1) \times 3^n - 1}{4}$$

$$\text{Q220 } A$$

$$\text{Q221 } A$$

$$\text{Q222 } A$$

$$\text{Q223 } A$$

$$\text{Q224 } A$$

$$\text{Q225 } (n-1) \times 2^n + 1$$

$$\text{Q226 } (n-1) \times 3^{n+1} + 3$$

$$\text{Q227 } A$$

$$\text{Q228 } \frac{k(k+1)(4k+11)}{6}$$

$$\text{Q229 } a_k = k(k+2)$$

$$\text{Q230 } a_k = k^2$$

$$\text{Q231 } a_k = k(k+1)$$

$$\text{Q232 } a_k = k(2k-1)$$

$$\text{Q233 } A$$

$$\text{Q234 } A$$

$$\text{Q235 } A$$

$$\text{Q236 } A$$

$$\text{Q237 } A$$

$$\text{Q238 } A$$

$$\text{Q239 } A$$

$$\text{Q240 } A$$

$$\text{Q241 } S_n = \frac{n(n+1)(2n+7)}{6}$$

$$\text{Q242 } S_n = \frac{n(n+1)(2n+1)}{6}$$

$$\text{Q243 } S_n = \frac{n(n+1)(4n-1)}{6}$$

$$\text{Q244 } S_n = \frac{n(n+1)(n+2)}{3}$$

$$\text{Q245 } A$$

$$\text{Q246 } a_n = 4n + 3$$

$$\text{Q247 } a_n = 2(n-2)$$

$$\text{Q248 } a_n = 4n + 3$$

$$\text{Q249 } a_n = 2n - 1$$

$$\text{Q250 } a_n = 2n - 5$$

$$\text{Q251 } D$$

$$\text{Q252 } C$$

$$\text{Q253 } B$$

$$\text{Q254 } A$$

$$\text{Q255 } D$$

$$\text{Q256 } C$$

$$\text{Q257 } C$$

$$\text{Q258 } C$$

$$\text{Q259 } B$$

$$\text{Q260 } A$$

$$\text{Q261 } D$$

$$\text{Q262 } D$$

$$\text{Q263 } B$$

$$\text{Q264 } a_1 = 9, \quad a_n = 3(n^2 - n + 1) \quad (n \geq 2)$$

$$\text{Q265 } a_1 = 2, \quad a_n = 2n + 1 \quad (n \geq 2)$$

$$\text{Q266 } a_1 = 0, \quad a_n = 3n^2 - 3n + 1 \quad (n \geq 2)$$

$$\text{Q267 } a_n = 4 \cdot 5^{n-1}$$

$$\text{Q268 } a_n = 2 \cdot 3^{n-1}$$

$$\text{Q269 } a_n = 2^{n-1}$$

$$\text{Q270 } a_n = 6 \cdot 7^{n-1}$$

$$\text{Q271 } a_1 = 3, \quad a_n = 4 \cdot 5^{n-1} \quad (n \geq 2)$$

$$\text{Q272 } a_1 = 5, \quad a_n = 2^{n+1} \quad (n \geq 2)$$

$$\text{Q273 } B$$

$$\text{Q274 } a_1 = 8, \quad a_n = 2(3^n - 1) \quad (n \geq 2)$$

$$\text{Quiz MCQ 1 } A$$

$$\text{Quiz MCQ 2 } A$$

$$\text{Quiz MCQ 3 } A$$

$$\text{Quiz MCQ 4 } C$$

$$\text{Quiz WRITTEN 5 } -1 + 2 + 7 + 14 + 23$$

$$\text{Quiz WRITTEN 6 } \frac{n(n+1)(3n^2 + 23n + 10)}{12}$$

$$\text{Quiz WRITTEN 7 } \frac{3(4 \cdot 3^n n - 3 \cdot 3^n + 3)}{2}$$

$$\text{Quiz WRITTEN 8 } a_1 = 3, \quad a_n = 2^n - 2 \quad (n \geq 2)$$

Concept 4 | Advanced Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Differences

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping

odd/even index $\rightarrow$ separate rules

Worked Solutions

Advanced Sequences

Q001 Written

Find the general term of the sequence $\{2, 5, 12, 23, 38, 57, \dots\}$.

Solution

1. The first differences are 3, 7, 11, 15, 19,
2. Hence $b_k = 4k - 1$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1} (4k - 1) = 2n^2 - 3n + 3$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = 2n^2 - 3n + 3$$

Worked Solutions

Advanced Sequences

Q002 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = 2k + 1$, then a_n is:

Solution

1. The first differences are represented by $b_k = 2k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2k + 1) = n^2$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Correct option: C

Final answer

$$a_n = n^2$$

Q003 MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k - 2$, then a_n is:

Solution

1. The first differences are represented by $b_k = 3k - 2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k - 2) = \frac{3n^2 - 7n + 8}{2}$.
4. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Correct option: B

Final answer

$$a_n = \frac{3n^2 - 7n + 8}{2}$$

Worked Solutions

Advanced Sequences

Q004 MCQ

Select the correct option: If $a_1 = 7$ and $b_k = 6k - 5$, then a_n is:

Solution

1. The first differences are represented by $b_k = 6k - 5$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 7 + \sum_{k=1}^{n-1} (6k - 5) = 3n^2 - 8n + 12$.
4. Substituting $n = 1$ gives $a_1 = 7$, matching the given first term.

Correct option: A

Final answer

$$a_n = 3n^2 - 8n + 12$$

Worked Solutions

Advanced Sequences

Q005 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = 2k + 1$, find a_5 :**Solution**

1. The first differences are represented by $b_k = 2k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2k + 1) = n^2$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.
5. $a_5 = 25$

Correct option: B**Final answer**

25

Worked Solutions

Advanced Sequences

Q006 MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k - 2$, find a_6 :**Solution**

1. The first differences are represented by $b_k = 3k - 2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k - 2) = \frac{3n^2 - 7n + 8}{2}$.
4. Substituting $n = 1$ gives $a_1 = \frac{2}{2}$, matching the given first term.
5. $a_6 = 37$

Correct option: A**Final answer**

37

Worked Solutions

Advanced Sequences

Q007 MCQ

Select the correct option: If $a_1 = 7$ and $b_k = 6k - 5$, find a_6 :**Solution**

1. The first differences are represented by $b_k = 6k - 5$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 7 + \sum_{k=1}^{n-1} (6k - 5) = 3n^2 - 8n + 12$.
4. Substituting $n = 1$ gives $a_1 = 7$, matching the given first term.
5. $a_6 = 72$

Correct option: D**Final answer**

72

Worked Solutions

Advanced Sequences

Q008 Written

Find the general term of the sequence $\{2, 10, 24, 44, 70, 102, \dots\}$.

Solution

1. The first differences are 8, 14, 20, 26, 32,
2. Hence $b_k = 2(3k + 1)$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1} (2(3k + 1)) = n(3n - 1)$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = n(3n - 1)$$

Worked Solutions

Advanced Sequences

Q009 Written

Find the general term of the sequence $\{1, 6, 15, 28, 45, 66, \dots\}$.

Solution

1. The first differences are 5, 9, 13, 17, 21,
2. Hence $b_k = 4k + 1$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 1 + \sum_{k=1}^{n-1} (4k + 1) = n(2n - 1)$.
5. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Final answer

$$a_n = n(2n - 1)$$

Worked Solutions

Advanced Sequences

Q010 MCQ

Select the correct option: If $a_1 = 3$ and $b_k = 5k + 4$, then a_n is:

Solution

1. The first differences are represented by $b_k = 5k + 4$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 3 + \sum_{k=1}^{n-1} (5k + 4) = \frac{(n+1)(5n-2)}{2}$.
4. Substituting $n = 1$ gives $a_1 = 3$, matching the given first term.

Correct option: A

Final answer

$$a_n = \frac{(n+1)(5n-2)}{2}$$

Q011 MCQ

Select the correct option: If $a_1 = 4$ and $b_k = 7k - 3$, then a_n is:

Solution

1. The first differences are represented by $b_k = 7k - 3$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 4 + \sum_{k=1}^{n-1} (7k - 3) = \frac{7n^2 - 13n + 14}{2}$.
4. Substituting $n = 1$ gives $a_1 = 4$, matching the given first term.

Correct option: D

Final answer

$$a_n = \frac{7n^2 - 13n + 14}{2}$$

Worked Solutions

Advanced Sequences

Q012 MCQ

Select the correct option: If $a_1 = 6$ and $b_k = 2(2k + 3)$, then a_n is:

Solution

1. The first differences are represented by $b_k = 2(2k + 3)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 6 + \sum_{k=1}^{n-1} (2(2k + 3)) = 2n(n + 2)$.
4. Substituting $n = 1$ gives $a_1 = 6$, matching the given first term.

Correct option: B

Final answer

$$a_n = 2n(n + 2)$$

Worked Solutions

Advanced Sequences

Q013 Written

Find the general term of the sequence $\{5, 8, 9, 8, 5, 0, \dots\}$.

Solution

1. The first differences are $3, 1, -1, -3, -5, \dots$
2. Hence $b_k = 5 - 2k$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 5 + \sum_{k=1}^{n-1} (5 - 2k) = -n(n - 6)$.
5. Substituting $n = 1$ gives $a_1 = 5$, matching the given first term.

Final answer

$$a_n = -n(n - 6)$$

Worked Solutions

Advanced Sequences

Q014 Written

Find the general term of the sequence $\{2, 3, 6, 15, 42, 123, \dots\}$.

Solution

1. The first differences are 1, 3, 9, 27, 81,
2. Hence $b_k = 3^{k-1}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1} (3^{k-1}) = \frac{3^n + 9}{6}$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = \frac{3^n + 9}{6}$$

Worked Solutions

Advanced Sequences

Q015 Written

Find the general term of the sequence $\{1, 2, 5, 14, 41, 122, \dots\}$.

Solution

1. The first differences are $1, 3, 9, 27, 81, \dots$
2. Hence $b_k = 3^{k-1}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 1 + \sum_{k=1}^{n-1} (3^{k-1}) = \frac{3^n + 3}{6}$.
5. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Final answer

$$a_n = \frac{3^n + 3}{6}$$

Worked Solutions

Advanced Sequences

Q016 Written

Find the general term of the sequence $\{5, 7, 11, 19, 35, \dots\}$.

Solution

1. The first differences are 2, 4, 8, 16,
2. Hence $b_k = 2^k$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 5 + \sum_{k=1}^{n-1} (2^k) = 2^n + 3$.
5. Substituting $n = 1$ gives $a_1 = 5$, matching the given first term.

Final answer

$$a_n = 2^n + 3$$

Worked Solutions

Advanced Sequences

Q017 Written

Find the general term of the sequence $\{2, 3, 1, 5, -3, 13, \dots\}$.

Solution

1. The first differences are $1, -2, 4, -8, 16, \dots$
2. Hence $b_k = (-2)^{k-1}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1} ((-2)^{k-1}) = \frac{(-2)^n + 14}{6}$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = \frac{(-2)^n + 14}{6}$$

Worked Solutions

Advanced Sequences

Q018 Written

Find the general term of the sequence $\{5, 4, 7, -2, 25, -56, \dots\}$.

Solution

1. The first differences are $-1, 3, -9, 27, -81, \dots$
2. Hence $b_k = \frac{(-3)^k}{3}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 5 + \sum_{k=1}^{n-1} \left(\frac{(-3)^k}{3}\right) = -\frac{(-3)^n - 57}{12}$.
5. Substituting $n = 1$ gives $a_1 = 5$, matching the given first term.

Final answer

$$a_n = -\frac{(-3)^n - 57}{12}$$

Worked Solutions

Advanced Sequences

Q019 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = k^2$, then a_n is:

Solution

1. The first differences are represented by $b_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (k^2) = \frac{(n+1)(2n^2-5n+6)}{6}$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Correct option: A

Final answer

$$a_n = \frac{(n+1)(2n^2-5n+6)}{6}$$

Worked Solutions

Advanced Sequences

Q020 Challenge

An oncology team records tissue volumes 1, 3, 7, 13, 21, 31, ... in mm^3 . Find the general formula a_n and calculate a_{12} to decide whether the 130 mm^3 threshold is exceeded.

Solution

1. The first differences are represented by $b_k = 2k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2k) = n^2 - n + 1$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.
5. $a_{12} = 133$; this is greater than 130, so immediate surgery is required.

Final answer

$$a_n = n^2 - n + 1, \quad a_{12} = 133; \text{ surgery required}$$

Worked Solutions

Advanced Sequences

Q021 Written

For groups with $a_j = 2 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , find the first term of group 10.

Solution

1. $P_9 = 81$ terms precede the group.
2. First position = $P_9 + 1 = 82$
3. First value = 164

Final answer

164

Q022 Written

For groups with $a_j = 1 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , find the first term of group 10.

Solution

1. $P_9 = 81$ terms precede the group.
2. First position = $P_9 + 1 = 82$
3. First value = 163

Final answer

163

Worked Solutions

Advanced Sequences

Q023 Written

For groups with $a_j = 1 + (j - 1)1$ and $c_j = n$ terms in group j , find the first term of group 10.

Solution

1. $P_9 = 45$ terms precede the group.
2. First position $= P_9 + 1 = 46$
3. First value $= 46$

Final answer

46

Q024 MCQ

Select the correct option: find the first term of group 3:

Solution

1. $P_2 = 3$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 4.

Correct option: B

Final answer

4

Worked Solutions

Advanced Sequences

Q025 MCQ

Select the correct option: find the first term of group 6:**Solution**

1. $P_5 = 25$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 78.

Correct option: C**Final answer**

78

Q026 MCQ

Select the correct option: find the first term of group 9:**Solution**

1. $P_8 = 36$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 37.

Correct option: D**Final answer**

37

Worked Solutions

Advanced Sequences

Q027 MCQ

Select the correct option: find the first term of group 4:**Solution**

1. $P_3 = 9$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 30.

Correct option: B**Final answer**

30

Q028 MCQ

Select the correct option: find the first term of group 7:**Solution**

1. $P_6 = 21$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 22.

Correct option: C**Final answer**

22

Worked Solutions

Advanced Sequences

Q029 MCQ

Select the correct option: find the first term of group 10:**Solution**

1. $P_9 = 81$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 246.

Correct option: D**Final answer**

246

Q030 MCQ

Select the correct option: find the first term of group 4:**Solution**

1. $P_3 = 9$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 20.

Correct option: A**Final answer**

20

Worked Solutions

Advanced Sequences

Q031 MCQ

Select the correct option: find the first term of group 7:**Solution**

1. $P_6 = 27$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 110.

Correct option: B**Final answer**

110

Q032 MCQ

Select the correct option: find the first term of group 10:**Solution**

1. $P_9 = 81$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 164.

Correct option: C**Final answer**

164

Worked Solutions

Advanced Sequences

Q033 MCQ

Select the correct option: find the first term of group 5:**Solution**

1. $P_4 = 14$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 58.

Correct option: A**Final answer**

58

Q034 MCQ

Select the correct option: find the first term of group 8:**Solution**

1. $P_7 = 49$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 100.

Correct option: B**Final answer**

100

Worked Solutions

Advanced Sequences

Q035 **Written**

For groups with $a_j = 1 + (j - 1)3$ and $c_j = 3n - 1$ terms in group j , find the first term of group 10.

Solution

1. $P_9 = 126$ terms precede the group.
2. First position = $P_9 + 1 = 127$
3. First value = 379

Final answer

379

Q036 **MCQ**

Select the correct option: find the first term of group 5:

Solution

1. $P_4 = 14$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 29.

Correct option: D**Final answer**

29

Worked Solutions

Advanced Sequences

Q037 MCQ

Select the correct option: find the first term of group 8:**Solution**

1. $P_7 = 28$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 89.

Correct option: A**Final answer**

89

Q038 Written

For groups with $a_j = 2 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , derive the sum of the terms in group n .**Solution**

1. First value $= 2(n^2 - 2n + 2)$
2. Last value $= 2n^2$
3. Number of terms $= 2n - 1$
4. $S_n = \frac{c_n}{2}(\text{first} + \text{last}) = 2(2n - 1)(n^2 - n + 1)$

Final answer $2(2n - 1)(n^2 - n + 1)$

Worked Solutions

Advanced Sequences

Q039 Written

For the same grouped sequence, find the sum of the terms in group 10.

Solution

1. First value = 163
2. Last value = 199
3. Number of terms = 19
4. $S_{10} = 3439$

Final answer

3439

Q040 Written

For the same grouped sequence, find the sum of the terms in group 10.

Solution

1. First value = 46
2. Last value = 55
3. Number of terms = 10
4. $S_{10} = 505$

Final answer

505

Worked Solutions

Advanced Sequences

Q041 Written

For the same grouped sequence, find the sum of the terms in group 10.**Solution**

1. First value = 379
2. Last value = 463
3. Number of terms = 29
4. $S_{10} = 12209$

Final answer

12209

Q042 Written

In the grouped arithmetic sequence, locate the value 88: state its group and its position within that group.**Solution**

1. Sequence position = $(88 - 2)/2 + 1 = 44$
2. $P_6 = 36 < 44 \leq P_7 = 49$
3. Therefore the value is in Group 7, position 8.

Final answer

Group 7, position 8

Worked Solutions

Advanced Sequences

Q043 Written

In the grouped arithmetic sequence, locate the value 101: state its group and its position within that group.

Solution

1. Sequence position $= (101 - 1)/2 + 1 = 51$
2. $P_7 = 49 < 51 \leq P_8 = 64$
3. Therefore the value is in Group 8, position 2.

Final answer

Group 8, position 2

Q044 Written

In the grouped arithmetic sequence, locate the value 124: state its group and its position within that group.

Solution

1. Sequence position $= (124 - 1)/1 + 1 = 124$
2. $P_{15} = 120 < 124 \leq P_{16} = 136$
3. Therefore the value is in Group 16, position 4.

Final answer

Group 16, position 4

Worked Solutions

Advanced Sequences

Q045 **Written**

In the grouped arithmetic sequence, locate the value 100: state its group and its position within that group.

Solution

1. Sequence position $= (100 - 1)/3 + 1 = 34$
2. $P_4 = 26 < 34 \leq P_5 = 40$
3. Therefore the value is in Group 5, position 8.

Final answer

Group 5, position 8

Worked Solutions

Advanced Sequences

Q046 Challenge

For the grouped sequence $2, 6 \mid 10, 14, 18 \mid 22, 26, 30, 34 \mid \dots$, solve: (a) the first term of Group 12; (b) the sum in Group n ; (c) the group and position of 202; (d) the total through Group 12.

Solution

1. $c_n = n + 1, \quad P_n = \frac{n(n+3)}{2}$
2. (a) first term = 310
3. (b) group sum = $2(n+1)(n^2 + 2n - 1)$
4. (c) 202 is in Group 9, position 7.
5. (d) total = 16200

Final answer

(a) 310; (b) $2(n+1)(n^2 + 2n - 1)$; (c) Group 9, position 7; (d) 16200

Worked Solutions

Advanced Sequences

Quiz MCQ 1 Concept Quiz · MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k + 1$, then a_n is:

Solution

1. The first differences are represented by $b_k = 3k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k + 1) = \frac{3n^2 - n + 2}{2}$.
4. Substituting $n = 1$ gives $a_1 = \frac{3(1)^2 - 1 + 2}{2} = 2$, matching the given first term.

Correct option: A

Final answer

$$a_n = \frac{3n^2 - n + 2}{2}$$

Quiz MCQ 2 Concept Quiz · MCQ

Select the correct option: If $a_1 = 3$ and $b_k = (-2)^{k-1}$, then a_n is:

Solution

1. The first differences are represented by $b_k = (-2)^{k-1}$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 3 + \sum_{k=1}^{n-1} ((-2)^{k-1}) = \frac{(-2)^n + 20}{6}$.
4. Substituting $n = 1$ gives $a_1 = 3$, matching the given first term.

Correct option: C

Final answer

$$a_n = \frac{(-2)^n + 20}{6}$$

Worked Solutions

Advanced Sequences

Quiz MCQ 3 **Concept Quiz · MCQ**

Select the correct option: find the first term of group 9:

Solution

1. $P_8 = 64$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 195.

Correct option: A

Final answer

195

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: the value 29 is in:

Solution

1. The value is sequence position 15.
2. It lies in Group 5, position 1.

Correct option: B

Final answer

Group 5, position 1

Worked Solutions

Advanced Sequences

Quiz WRITTEN 5 Concept Quiz · Written

A sequence has $a_1 = 1$ and difference sequence $b_k = 4k - 1$. Find the required general term a_n .

Solution

1. The first differences are represented by $b_k = 4k - 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (4k - 1) = 2n^2 - 3n + 2$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Final answer

$$a_n = 2n^2 - 3n + 2$$

Quiz WRITTEN 6 Concept Quiz · Written

A sequence has $a_1 = 2$ and difference sequence $b_k = (-2)^{k-1}$. Find the required general term a_n .

Solution

1. The first differences are represented by $b_k = (-2)^{k-1}$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1} ((-2)^{k-1}) = \frac{(-2)^n + 14}{6}$.
4. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = \frac{(-2)^n + 14}{6}$$

Worked Solutions

Advanced Sequences

Quiz WRITTEN 7 Concept Quiz · Written

For groups with $a_j = 5 + (j - 1)3$ and $c_j = n$ terms in group j , find the first term of group 12.

Solution

1. $P_{11} = 66$ terms precede the group.
2. First position = $P_{11} + 1 = 67$
3. First value = 203

Final answer

203

Quiz WRITTEN 8 Concept Quiz · Written

In the grouped arithmetic sequence, locate the value 66: state its group and its position within that group.

Solution

1. Sequence position = $(66 - 2)/4 + 1 = 17$
2. $P_4 = 14 < 17 \leq P_5 = 20$
3. Therefore the value is in Group 5, position 3.

Final answer

Group 5, position 3

Answer key

Use the question number shown on each page.

Q001 $a_n = 2n^2 - 3n + 3$

Q002 C

Q003 B

Q004 A

Q005 B

Q006 A

Q007 D

Q008 $a_n = n(3n - 1)$

Q009 $a_n = n(2n - 1)$

Q010 A

Q011 D

Q012 B

Q013 $a_n = -n(n - 6)$

Q014 $a_n = \frac{3^n + 9}{6}$

Q015 $a_n = \frac{3^n + 3}{6}$

Q016 $a_n = 2^n + 3$

Q017 $a_n = \frac{(-2)^n + 14}{6}$

Q018 $a_n = -\frac{(-3)^n - 57}{12}$

Q019 A

Q020 $a_n = n^2 - n + 1$, $a_{12} = 133$; surgery required

Q021 164

Q022 163

Q023 46

Q024 B

Q025 C

Q026 D

Q027 B

Q028 C

Q029 D

Q030 A

Q031 B

Q032 C

Q033 A

Q034 B

Q035 379

Q036 D

Q037 A

Q038 $2(2n - 1)(n^2 - n + 1)$

Q039 3439

Q040 505

Q041 12209

Q042 Group 7, position 8

Q043 Group 8, position 2

Q044 Group 16, position 4

Q045 Group 5, position 8

Q046 (a) 310; (b) $2(n + 1)(n^2 + 2n - 1)$; (c) Group 9, position 7; (d) 16200

Quiz MCQ 1 A

Quiz MCQ 2 C

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $a_n = 2n^2 - 3n + 2$

Quiz WRITTEN 6 $a_n = \frac{(-2)^n + 14}{6}$

Quiz WRITTEN 7 203

Quiz WRITTEN 8 Group 5, position 3

Concept 5 | Recurrence and Induction

English Student Edition • Focused formulas and guided practice

Formula opener

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1), P(k) \Rightarrow P(k + 1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k + 1) = mq'$$

Worked Solutions

Recurrence and Induction

Q001 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (2)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 8$.
2. $a_3 = 26$.
3. $a_4 = 80$.

Final answer

$$a_2 = 8, a_3 = 26, a_4 = 80$$

Q002 Written

For the sequence $\{a_n\}$ with $a_1 = -6$ and recurrence $a_{n+1} = 2a_n + (3)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = -9$.
2. $a_3 = -15$.
3. $a_4 = -27$.

Final answer

$$a_2 = -9, a_3 = -15, a_4 = -27$$

Worked Solutions

Recurrence and Induction

Q003 Written

For the sequence $\{a_n\}$ with $a_1 = -6$ and recurrence $a_{n+1} = 1a_n + (n(n-2))$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = -7$.
2. $a_3 = -7$.
3. $a_4 = -4$.

Final answer

$$a_2 = -7, a_3 = -7, a_4 = -4$$

Q004 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (-n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = -3$.
2. $a_3 = -5$.
3. $a_4 = -8$.

Final answer

$$a_2 = -3, a_3 = -5, a_4 = -8$$

Worked Solutions

Recurrence and Induction

Q005 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (2n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 4$.
2. $a_3 = 8$.
3. $a_4 = 14$.

Final answer

$$a_2 = 4, a_3 = 8, a_4 = 14$$

Q006 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (3^n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 1$.
2. $a_3 = 10$.
3. $a_4 = 37$.

Final answer

$$a_2 = 1, a_3 = 10, a_4 = 37$$

Worked Solutions

Recurrence and Induction

Q007 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (0)$. The fourth term is:

Solution

1. $a_2 = 9$.
2. $a_3 = 27$.
3. $a_4 = 81$.

Correct option: B

Final answer

81

Q008 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 4a_n + (0)$. The fourth term is:

Solution

1. $a_2 = 8$.
2. $a_3 = 32$.
3. $a_4 = 128$.

Correct option: D

Final answer

128

Worked Solutions

Recurrence and Induction

Q009 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 2a_n + (n + 1)$. The fourth term is:

Solution

1. $a_2 = 6$.
2. $a_3 = 15$.
3. $a_4 = 34$.

Correct option: A

Final answer

34

Q010 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 3a_n + (2n)$. The fourth term is:

Solution

1. $a_2 = 17$.
2. $a_3 = 55$.
3. $a_4 = 171$.

Correct option: B

Final answer

171

Worked Solutions

Recurrence and Induction

Q011 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (4)$. The fourth term is:

Solution

1. $a_2 = 5$.
2. $a_3 = 9$.
3. $a_4 = 13$.

Correct option: D

Final answer

13

Q012 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = -2a_n + (5)$. The fourth term is:

Solution

1. $a_2 = -3$.
2. $a_3 = 11$.
3. $a_4 = -17$.

Correct option: A

Final answer

-17

Worked Solutions

Recurrence and Induction

Q013 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -3a_n + (2)$. The fourth term is:

Solution

1. $a_2 = -1$.
2. $a_3 = 5$.
3. $a_4 = -13$.

Correct option: C

Final answer

-13

Q014 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (n^2)$. The fourth term is:

Solution

1. $a_2 = 5$.
2. $a_3 = 9$.
3. $a_4 = 18$.

Correct option: D

Final answer

18

Worked Solutions

Recurrence and Induction

Q015 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 2a_n + (1)$. The fourth term is:

Solution

1. $a_2 = 5$.
2. $a_3 = 11$.
3. $a_4 = 23$.

Correct option: C

Final answer

23

Q016 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (-3)$. The fourth term is:

Solution

1. $a_2 = 2$.
2. $a_3 = -1$.
3. $a_4 = -4$.

Correct option: D

Final answer

-4

Worked Solutions

Recurrence and Induction

Q017 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (n)$. The fourth term is:

Solution

1. $a_2 = 4$.
2. $a_3 = 6$.
3. $a_4 = 9$.

Correct option: B

Final answer

9

Q018 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 2a_n + (-n)$. The fourth term is:

Solution

1. $a_2 = 1$.
2. $a_3 = 0$.
3. $a_4 = -3$.

Correct option: C

Final answer

-3

Worked Solutions

Recurrence and Induction

Q019 Written

For the sequence $\{a_n\}$ with $a_1 = -1$ and recurrence $a_{n+1} = 1a_n + (5)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 5n - 6$.

Final answer

$$a_n = 5n - 6$$

Q020 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (3)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 3n - 2$.

Final answer

$$a_n = 3n - 2$$

Worked Solutions

Recurrence and Induction

Q021 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (-2)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = -2(n - 3)$.

Final answer

$$a_n = -2(n - 3)$$

Q022 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (6)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 6n - 5$.

Final answer

$$a_n = 6n - 5$$

Worked Solutions

Recurrence and Induction

Q023 Written

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (-5)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 8 - 5n$.

Final answer

$$a_n = 8 - 5n$$

Q024 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (3)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 3n - 5$.

Final answer

$$a_n = 3n - 5$$

Worked Solutions

Recurrence and Induction

Q025 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (-4)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = -4(n - 2)$.

Final answer

$$a_n = -4(n - 2)$$

Q026 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 3^n$.

Correct option: D

Final answer

$$a_n = 3^n$$

Worked Solutions

Recurrence and Induction

Q027 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 4a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2^{2n-1}$.

Correct option: A**Final answer**

$$a_n = 2^{2n-1}$$

Q028 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 5a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 4 \cdot 5^{n-1}$.

Correct option: A**Final answer**

$$a_n = 4 \cdot 5^{n-1}$$

Worked Solutions

Recurrence and Induction

Q029 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 6$ and recurrence $a_{n+1} = 2a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 3 \cdot 2^n$.

Correct option: A

Final answer

$$a_n = 3 \cdot 2^n$$

Q030 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2 \cdot 3^{n-1}$.

Final answer

$$a_n = 2 \cdot 3^{n-1}$$

Worked Solutions

Recurrence and Induction

Q031 Written

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 5a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 3 \cdot 5^{n-1}$.

Final answer

$$a_n = 3 \cdot 5^{n-1}$$

Q032 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 3a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 4 \cdot 3^{n-1}$.

Final answer

$$a_n = 4 \cdot 3^{n-1}$$

Worked Solutions

Recurrence and Induction

Q033 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 4a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2^{2n-2}$.

Final answer

$$a_n = 2^{2n-2}$$

Q034 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (4)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 4n - 3$.

Correct option: B

Final answer

$$a_n = 4n - 3$$

Worked Solutions

Recurrence and Induction

Q035 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (n^2)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.

2. $a_n = \frac{2n^3 - 3n^2 + n + 24}{6}$.

Correct option: B

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 24}{6}$$

Q036 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -4a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.

2. $a_n = (-4)^{n-1}$.

Correct option: D

Final answer

$$a_n = (-4)^{n-1}$$

Worked Solutions

Recurrence and Induction

Q037 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (3^n)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = \frac{3^{n+7}}{2}$.

Correct option: A

Final answer

$$a_n = \frac{3^n + 7}{2}$$

Q038 Written

For the sequence $\{a_n\}$ with $a_1 = -3$ and recurrence $a_{n+1} = -2a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = \frac{3(-2)^n}{2}$.

Final answer

$$a_n = \frac{3(-2)^n}{2}$$

Worked Solutions

Recurrence and Induction

Q039 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = -5a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2(-5)^{n-1}$.

Final answer

$$a_n = 2(-5)^{n-1}$$

Q040 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = -4a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = \frac{(-4)^n}{2}$.

Final answer

$$a_n = \frac{(-4)^n}{2}$$

Worked Solutions

Recurrence and Induction

Q041 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (-3)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 8 - 3n$.

Correct option: B**Final answer**

$$a_n = 8 - 3n$$

Q042 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (n)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = \frac{n^2 - n + 6}{2}$.

Correct option: C**Final answer**

$$a_n = \frac{n^2 - n + 6}{2}$$

Worked Solutions

Recurrence and Induction

Q043 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (-5)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 8 - 5n$.

Correct option: B**Final answer**

$$a_n = 8 - 5n$$

Q044 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (6)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 2(3n - 2)$.

Correct option: C**Final answer**

$$a_n = 2(3n - 2)$$

Worked Solutions

Recurrence and Induction

Q045 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (-n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 5$.
2. $a_3 = 13$.
3. $a_4 = 36$.

Final answer

$$a_2 = 5, a_3 = 13, a_4 = 36$$

Q046 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -2a_n + (5)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 3$.
2. $a_3 = -1$.
3. $a_4 = 7$.

Final answer

$$a_2 = 3, a_3 = -1, a_4 = 7$$

Worked Solutions

Recurrence and Induction

Q047 **Written**

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (-n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 8$.
2. $a_3 = 22$.
3. $a_4 = 63$.

Final answer

$$a_2 = 8, a_3 = 22, a_4 = 63$$

Worked Solutions

Recurrence and Induction

Q048 Challenge

Solve the third-order recurrence $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ for $n \geq 3$ with $a_0 = 0, a_1 = 1, a_2 = 6$.

Solution

1. $r^3 - 6r^2 + 11r - 6 = (r - 1)(r - 2)(r - 3)$

2. $a_n = A + B2^n + C3^n$

3. $A = \frac{1}{2}, B = -2, C = \frac{3}{2}$

4. $a_n = -\frac{4 \cdot 2^n - 3 \cdot 3^n - 1}{2}$

Final answer

$$a_n = -\frac{4 \cdot 2^n - 3 \cdot 3^n - 1}{2}$$

Worked Solutions

Recurrence and Induction

Q049 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (2n - 3)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2k - 3$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2k - 3) = (n - 2)^2$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = (n - 2)^2$$

Q050 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (3n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (3k) = \frac{3n^2 - 3n + 2}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3n^2 - 3n + 2}{2}$$

Worked Solutions

Recurrence and Induction

Q051 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (2n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (2k) = n^2 - n + 2$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = n^2 - n + 2$$

Q052 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k^2) = \frac{2n^3 - 3n^2 + n + 4}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$$

Worked Solutions

Recurrence and Induction

Q053 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (3n + 1)$.**Solution**

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k + 1) = \frac{3n^2 - n + 2}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3n^2 - n + 2}{2}$$

Q054 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (n(2n + 1))$.**Solution**

1. The recurrence gives the difference term $a_{k+1} - a_k = k(2k + 1)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (k(2k + 1)) = \frac{4n^3 - 3n^2 - n + 12}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{4n^3 - 3n^2 - n + 12}{6}$$

Worked Solutions

Recurrence and Induction

Q055 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = a_n + (n^2)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 3 + \sum_{k=1}^{n-1} (k^2) = \frac{2n^3 - 3n^2 + n + 18}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 18}{6}$$

Q056 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k^2) = \frac{2n^3 - 3n^2 + n + 4}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$$

Worked Solutions

Recurrence and Induction

Q057 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (n^2 + 1)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2 + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (k^2 + 1) = \frac{2n^3 - 3n^2 + 7n + 6}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = \frac{2n^3 - 3n^2 + 7n + 6}{6}$$

Q058 MCQ

Select the correct option: If $a_1 = 5$ and $a_{n+1} = a_n + (n(4n - 1))$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k(4k - 1)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 5 + \sum_{k=1}^{n-1} (k(4k - 1)) = \frac{(n+1)(8n^2 - 23n + 30)}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = \frac{(n+1)(8n^2 - 23n + 30)}{6}$$

Worked Solutions

Recurrence and Induction

Q059 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (8n)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 8k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (8k) = 2(2n^2 - 2n + 1)$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: C**Final answer**

$$a_n = 2(2n^2 - 2n + 1)$$

Q060 MCQ

Select the correct option: find a_4 when $a_1 = 3$ and $a_{n+1} = a_n + (n^2)$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 3 + \sum_{k=1}^{n-1} (k^2) = \frac{2n^3 - 3n^2 + n + 18}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A**Final answer**

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Worked Solutions

Recurrence and Induction

Q061 MCQ

Select the correct option: find a_5 when $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$:**Solution**

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k^2) = \frac{2n^3 - 3n^2 + n + 4}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: D**Final answer**

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Q062 MCQ

Select the correct option: find a_3 when $a_1 = 2$ and $a_{n+1} = a_n + (n^2 + 1)$:**Solution**

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2 + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (k^2 + 1) = \frac{2n^3 - 3n^2 + 7n + 6}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A**Final answer**

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Worked Solutions

Recurrence and Induction

Q063 MCQ

Select the correct option: find a_6 when $a_1 = 5$ and $a_{n+1} = a_n + (n(4n - 1))$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k(4k - 1)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 5 + \sum_{k=1}^{n-1} (k(4k - 1)) = \frac{(n+1)(8n^2 - 23n + 30)}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

210

Q064 MCQ

Select the correct option: find a_2 when $a_1 = 2$ and $a_{n+1} = a_n + (8n)$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 8k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (8k) = 2(2n^2 - 2n + 1)$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

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Worked Solutions

Recurrence and Induction

Q065 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (n^2)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (k^2) = \frac{2n^3 - 3n^2 + n + 12}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 12}{6}$$

Q066 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = a_n + (2n + 1)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2k + 1) = n^2$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

$$a_n = n^2$$

Worked Solutions

Recurrence and Induction

Q067 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (2^n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2^k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2^k) = 2^n - 1$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = 2^n - 1$$

Q068 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (-3^n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = -3^k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (-3^k) = -\frac{3^n - 7}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{3^n - 7}{2}$$

Worked Solutions

Recurrence and Induction

Q069 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 4$ and $a_{n+1} = a_n + (-4^n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = -4^k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 4 + \sum_{k=1}^{n-1} (-4^k) = -\frac{4^n - 16}{3}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{4^n - 16}{3}$$

Worked Solutions

Recurrence and Induction

Q070 Challenge

A technology startup has a reserve of 400,000 LE, represented by $a_1 = 4$, and burns $0.04n$ hundred-thousand LE each quarter: $a_{n+1} = a_n - 0.04n$. Find an explicit formula for a_n .

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = -\frac{k}{25}$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 4 + \sum_{k=1}^{n-1} \left(-\frac{k}{25}\right) = -\frac{n^2 - n - 200}{50}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{n^2 - n - 200}{50}$$

Worked Solutions

Recurrence and Induction

Q071 Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 3a_n + (4)$.**Solution**

1. $c = -2$ from $c = 3c + (4)$.
2. $a_{n+1} - c = 3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 3^n - 2$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = 3^n - 2$$

Worked Solutions

Recurrence and Induction

Q072 **Written****Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 5a_n + (8)$.****Solution**

1. $c = -2$ from $c = 5c + (8)$.
2. $a_{n+1} - c = 5(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3 \cdot 5^n - 10}{5}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3 \cdot 5^n - 10}{5}$$

Worked Solutions

Recurrence and Induction

Q073 Written

Find the general term of $\{a_n\}$ given $a_1 = 2$ and $a_{n+1} = 2a_n + (-1)$.

Solution

1. $c = 1$ from $c = 2c + (-1)$.
2. $a_{n+1} - c = 2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2^n + 2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2^n + 2}{2}$$

Worked Solutions

Recurrence and Induction

Q074 **Written****Find the general term of $\{a_n\}$ given $a_1 = -3$ and $a_{n+1} = 4a_n + (3)$.****Solution**

1. $c = -1$ from $c = 4c + (3)$.
2. $a_{n+1} - c = 4(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{4^n + 2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{4^n + 2}{2}$$

Worked Solutions

Recurrence and Induction

Q075 Written

Find the general term of $\{a_n\}$ given $a_1 = 3$ and $a_{n+1} = 3a_n + (-2)$.

Solution

1. $c = 1$ from $c = 3c + (-2)$.
2. $a_{n+1} - c = 3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2 \cdot 3^n + 3}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2 \cdot 3^n + 3}{3}$$

Worked Solutions

Recurrence and Induction

Q076 **Written****Find the general term of $\{a_n\}$ given $a_1 = -2$ and $a_{n+1} = -3a_n + (4)$.****Solution**

1. $c = 1$ from $c = -3c + (4)$.
2. $a_{n+1} - c = -3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = (-3)^n + 1$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = (-3)^n + 1$$

Worked Solutions

Recurrence and Induction

Q077 Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = \frac{1}{3}a_n + (2)$.

Solution

1. $c = 3$ from $c = \frac{1}{3}c + (2)$.
2. $a_{n+1} - c = \frac{1}{3}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 3 \cdot 3^{-n} (3^n - 2)$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = 3 \cdot 3^{-n} (3^n - 2)$$

Worked Solutions

Recurrence and Induction

Q078 **Written**

Find the general term of $\{a_n\}$ given $a_1 = 2$ and $a_{n+1} = \frac{2}{3}a_n + (-\frac{1}{2})$.

Solution

1. $c = -\frac{3}{2}$ from $c = \frac{2}{3}c + (-\frac{1}{2})$.
2. $a_{n+1} - c = \frac{2}{3}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3 \cdot 3^{-n}(7 \cdot 2^n - 2 \cdot 3^n)}{4}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3 \cdot 3^{-n}(7 \cdot 2^n - 2 \cdot 3^n)}{4}$$

Worked Solutions

Recurrence and Induction

Q079 Written

Find the general term of $\{a_n\}$ given $a_1 = 4$ and $a_{n+1} = -2a_n + (3)$.

Solution

1. $c = 1$ from $c = -2c + (3)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{3(-2)^n - 2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{3(-2)^n - 2}{2}$$

Worked Solutions

Recurrence and Induction

Q080 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = -2a_n + (5)$, the general term is:

Solution

1. $c = \frac{5}{3}$ from $c = -2c + (5)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{2(-2)^n - 5}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

$$a_n = -\frac{2(-2)^n - 5}{3}$$

Worked Solutions

Recurrence and Induction

Q081 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = \frac{2}{3}a_n + (-2)$, the general term is:

Solution

1. $c = -6$ from $c = \frac{2}{3}c + (-2)$.
2. $a_{n+1} - c = \frac{2}{3}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3 \cdot 3^{-n}(7 \cdot 2^n - 4 \cdot 3^n)}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = \frac{3 \cdot 3^{-n}(7 \cdot 2^n - 4 \cdot 3^n)}{2}$$

Worked Solutions

Recurrence and Induction

Q082 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = 7a_n + (-4)$, the general term is:

Solution

1. $c = \frac{2}{3}$ from $c = 7c + (-4)$.
2. $a_{n+1} - c = 7(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2(2 \cdot 7^n + 7)}{21}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = \frac{2(2 \cdot 7^n + 7)}{21}$$

Worked Solutions

Recurrence and Induction

Q083 MCQ

Select the correct option: If $a_1 = 6$ and $a_{n+1} = 2a_n + (-5)$, the general term is:

Solution

1. $c = 5$ from $c = 2c + (-5)$.
2. $a_{n+1} - c = 2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2^n + 10}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

$$a_n = \frac{2^n + 10}{2}$$

Worked Solutions

Recurrence and Induction

Q084 MCQ

Select the correct option: If $a_1 = 5$ and $a_{n+1} = -2a_n + (7)$, the general term is:

Solution

1. $c = \frac{7}{3}$ from $c = -2c + (7)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{4(-2)^n - 7}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = -\frac{4(-2)^n - 7}{3}$$

Worked Solutions

Recurrence and Induction

Q085 MCQ

Select the correct option: If $a_1 = 4$ and $a_{n+1} = -5a_n + (-2)$, the general term is:

Solution

1. $c = -\frac{1}{3}$ from $c = -5c + (-2)$.
2. $a_{n+1} - c = -5(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{13(-5)^n + 5}{15}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = -\frac{13(-5)^n + 5}{15}$$

Worked Solutions

Recurrence and Induction

Q086 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = -3a_n + (5)$, the general term is:

Solution

1. $c = \frac{5}{4}$ from $c = -3c + (5)$.
2. $a_{n+1} - c = -3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{(-3)^n - 5}{4}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

$$a_n = -\frac{(-3)^n - 5}{4}$$

Worked Solutions

Recurrence and Induction

Q087 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = \frac{1}{2}a_n + (7)$, the general term is:

Solution

1. $c = 14$ from $c = \frac{1}{2}c + (7)$.
2. $a_{n+1} - c = \frac{1}{2}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 2 \cdot 2^{-n} (7 \cdot 2^n - 13)$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = 2 \cdot 2^{-n} (7 \cdot 2^n - 13)$$

Worked Solutions

Recurrence and Induction

Q088 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = 2a_n + (-1)$, the general term is:

Solution

1. $c = 1$ from $c = 2c + (-1)$.
2. $a_{n+1} - c = 2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 1$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = 1$$

Worked Solutions

Recurrence and Induction

Q089 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = -2a_n + (5)$, the general term is:

Solution

1. $c = \frac{5}{3}$ from $c = -2c + (5)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{2(-2)^{n-5}}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = -\frac{2(-2)^n - 5}{3}$$

Worked Solutions

Recurrence and Induction

Q090 Challenge

A sequence satisfies $a_{n+1} = 4a_n - 9$ with $a_1 = 2$. Find the equilibrium value, the explicit formula, and a_5 .

Solution

1. $c = 3$ from $c = 4c + (-9)$.
2. $a_{n+1} - c = 4(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{2^{2n}-12}{4}$.
5. Substitute $n = 1$ to verify the given first term.
6. $a_5 = -253$

Final answer

$$c = 3, \quad a_n = -\frac{2^{2n} - 12}{4}, \quad a_5 = -253$$

Worked Solutions

Recurrence and Induction

Q091 Written

Use mathematical induction to prove $\sum_{i=1}^n i = \frac{n(n+1)}{2}$.

Solution

1. [1] Base case $n = 1$: both sides equal 1.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k i = \frac{k(k+1)}{2}$.
3. For $n = k + 1$, add the next term n to the left side and use the hypothesis.
4. $\frac{k(k+1)}{2} + n = \frac{(k+1)(k+2)}{2}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n+1)}{2}$$

Worked Solutions

Recurrence and Induction

Q092 Written

Use mathematical induction to prove $\sum_{i=1}^n (2i - 1)^2 = \frac{n(2n-1)(2n+1)}{3}$.

Solution

1. [1] Base case $n = 1$: both sides equal 1.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k (2i - 1)^2 = \frac{k(2k-1)(2k+1)}{3}$.
3. For $n = k + 1$, add the next term $(2k + 1)^2$ to the left side and use the hypothesis.
4. $\frac{k(2k-1)(2k+1)}{3} + (2k + 1)^2 = \frac{(k+1)(2k+1)(2k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(2n-1)(2n+1)}{3}$$

Worked Solutions

Recurrence and Induction

Q093 Written

Use mathematical induction to prove $\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$.

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k i(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k + 1$, add the next term $(k+1)(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Recurrence and Induction

Q094 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n i(i+1)$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k i(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k + 1$, add the next term $(k+1)(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: A

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Recurrence and Induction

Q095 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3 \cdot 4^{i-1}$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 3 \cdot 4^{i-1} = 4^k - 1$.
3. For $n = k + 1$, add the next term $3 \cdot 4^k$ to the left side and use the hypothesis.
4. $4^k - 1 + 3 \cdot 4^k = (2 \cdot 2^k - 1)(2 \cdot 2^k + 1)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: B

Final answer

$$4^n - 1$$

Worked Solutions

Recurrence and Induction

Q096 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i + 3$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 2i + 3 = k(k + 4)$.
3. For $n = k + 1$, add the next term $2k + 5$ to the left side and use the hypothesis.
4. $k(k + 4) + 2k + 5 = (k + 1)(k + 5)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: C

Final answer

$$n(n + 4)$$

Worked Solutions

Recurrence and Induction

Q097 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i^2 + 3i$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 2i^2 + 3i = \frac{k(k+1)(4k+11)}{6}$.
3. For $n = k + 1$, add the next term $(k + 1)(2k + 5)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(4k+11)}{6} + (k + 1)(2k + 5) = \frac{(k+1)(k+2)(4k+15)}{6}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: B

Final answer

$$\frac{n(n+1)(4n+11)}{6}$$

Worked Solutions

Recurrence and Induction

Q098 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3i^2 + 2i$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 3i^2 + 2i = \frac{k(k+1)(2k+3)}{2}$.
3. For $n = k + 1$, add the next term $(k + 1)(3k + 5)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(2k+3)}{2} + (k + 1)(3k + 5) = \frac{(k+1)(k+2)(2k+5)}{2}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: C

Final answer

$$\frac{n(n+1)(2n+3)}{2}$$

Worked Solutions

Recurrence and Induction

Q099 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 7 \cdot 2^{i-1}$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 7.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 7 \cdot 2^{i-1} = 7(2^k - 1)$.
3. For $n = k + 1$, add the next term $7 \cdot 2^k$ to the left side and use the hypothesis.
4. $7(2^k - 1) + 7 \cdot 2^k = 7(2 \cdot 2^k - 1)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: A

Final answer

$$7(2^n - 1)$$

Worked Solutions

Recurrence and Induction

Q100 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n i(i+1)$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k i(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k + 1$, add the next term $(k+1)(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: C

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Recurrence and Induction

Q101 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3 \cdot 4^{i-1}$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 3 \cdot 4^{i-1} = 4^k - 1$.
3. For $n = k + 1$, add the next term $3 \cdot 4^k$ to the left side and use the hypothesis.
4. $4^k - 1 + 3 \cdot 4^k = (2 \cdot 2^k - 1)(2 \cdot 2^k + 1)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: A

Final answer

$$4^n - 1$$

Worked Solutions

Recurrence and Induction

Q102 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i + 3$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 2i + 3 = k(k + 4)$.
3. For $n = k + 1$, add the next term $2k + 5$ to the left side and use the hypothesis.
4. $k(k + 4) + 2k + 5 = (k + 1)(k + 5)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: B

Final answer

$$n(n + 4)$$

Worked Solutions

Recurrence and Induction

Q103 Written

Use mathematical induction to prove $\sum_{i=1}^n (2i - 1) = n^2$.

Solution

1. [1] Base case $n = 1$: both sides equal 1.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k (2i - 1) = k^2$.
3. For $n = k + 1$, add the next term $2k + 1$ to the left side and use the hypothesis.
4. $k^2 + 2k + 1 = (k + 1)^2$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$n^2$$

Worked Solutions

Recurrence and Induction

Q104 Written

Use mathematical induction to prove $\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$.

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k i(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k + 1$, add the next term $(k+1)(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Recurrence and Induction

Q105 Written

Use mathematical induction to prove $\sum_{i=1}^n i(i+2) = \frac{n(n+1)(2n+7)}{6}$.

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k i(i+2) = \frac{k(k+1)(2k+7)}{6}$.
3. For $n = k + 1$, add the next term $(k+1)(k+3)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(2k+7)}{6} + (k+1)(k+3) = \frac{(k+1)(k+2)(2k+9)}{6}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n+1)(2n+7)}{6}$$

Worked Solutions

Recurrence and Induction

Q106 Written

Use mathematical induction to prove $\sum_{i=1}^n 3 \cdot 4^{i-1} = 4^n - 1$.

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 3 \cdot 4^{i-1} = 4^k - 1$.
3. For $n = k + 1$, add the next term $3 \cdot 4^k$ to the left side and use the hypothesis.
4. $4^k - 1 + 3 \cdot 4^k = (2 \cdot 2^k - 1)(2 \cdot 2^k + 1)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$4^n - 1$$

Worked Solutions

Recurrence and Induction

Q107 Written

Use mathematical induction to prove $\sum_{i=0}^n 10^i = \frac{10 \cdot 10^n - 1}{9}$.

Solution

1. [1] Base case $n = 1$: both sides equal 11.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=0}^k 10^i = \frac{10 \cdot 10^k - 1}{9}$.
3. For $n = k + 1$, add the next term 10^{k+1} to the left side and use the hypothesis.
4. $\frac{10 \cdot 10^k - 1}{9} + 10^{k+1} = \frac{100 \cdot 10^k - 1}{9}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{10 \cdot 10^n - 1}{9}$$

Worked Solutions

Recurrence and Induction

Q108 Challenge

Use mathematical induction to prove $\sum_{i=1}^n (i^2 + i + 1) = \frac{n(n^2 + 3n + 5)}{3}$.

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k (i^2 + i + 1) = \frac{k(k^2 + 3k + 5)}{3}$.
3. For $n = k + 1$, add the next term $k^2 + 3k + 3$ to the left side and use the hypothesis.
4. $\frac{k(k^2 + 3k + 5)}{3} + k^2 + 3k + 3 = \frac{(k+1)(k^2 + 5k + 9)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n^2 + 3n + 5)}{3}$$

Worked Solutions

Recurrence and Induction

Q109 Written

Prove by mathematical induction that $2^n > 3n$ for $n \geq 4$.

Solution

1. $n = 4$: $2^4 = 16 > 12 = 3 \cdot 4$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^{k+1} > 2(3k + 0)$ after multiplying by the positive base.
4. $2(3k + 0) - (3(k + 1) + 0) = 3(k - 1)$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$2^n > 3n$$

Worked Solutions

Recurrence and Induction

Q110 Written

Prove by mathematical induction that $3^n > 5n + 1$ for $n \geq 3$.

Solution

1. $n = 3$: $3^3 = 27 > 16 = 5 \cdot 3 + 1$.
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $3^k + 1 > 3(5k + 1)$ after multiplying by the positive base.
4. $3(5k + 1) - (5(k + 1) + 1) = 10k - 3$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$3^n > 5n + 1$$

Worked Solutions

Recurrence and Induction

Q111 Written

Prove by mathematical induction that $3^n > 2n$ for $n \geq 1$.

Solution

1. $n = 1$: $3^1 = 3 > 2 = 2$.
2. Assume the inequality holds for $n = k$ with $k \geq 1$.
3. $3^k + 1 > 3(2k + 0)$ after multiplying by the positive base.
4. $3(2k + 0) - (2(k + 1) + 0) = 2(2k - 1)$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$3^n > 2n$$

Worked Solutions

Recurrence and Induction

Q112 Written

Prove by mathematical induction that $2^n > 1n + 2$ for $n \geq 3$.

Solution

1. $n = 3$: $2^3 = 8 > 5 = 1 \cdot 3 + 2$.
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $2^k + 1 > 2(1k + 2)$ after multiplying by the positive base.
4. $2(1k + 2) - (1(k + 1) + 2) = k + 1$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$2^n > 1n + 2$$

Worked Solutions

Recurrence and Induction

Q113 Written

Prove by mathematical induction that $3^n > 5n + 1$ for $n \geq 4$.

Solution

1. $n = 4$: $3^4 = 81 > 21 = 5 \cdot 4 + 1$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 1)$ after multiplying by the positive base.
4. $3(5k + 1) - (5(k + 1) + 1) = 10k - 3$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$3^n > 5n + 1$$

Worked Solutions

Recurrence and Induction

Q114 Written

Prove by mathematical induction that $3^n > 5n + 2$ for $n \geq 4$.

Solution

1. $n = 4$: $3^4 = 81 > 22 = 5 \cdot 4 + 2$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 2)$ after multiplying by the positive base.
4. $3(5k + 2) - (5(k + 1) + 2) = 10k - 1$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$3^n > 5n + 2$$

Worked Solutions

Recurrence and Induction

Q115 Written

Prove by mathematical induction that $4^n > 2n + 3$ for $n \geq 4$.

Solution

1. $n = 4$: $4^4 = 256 > 11 = 11$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $4^k + 1 > 4(2k + 3)$ after multiplying by the positive base.
4. $4(2k + 3) - (2(k + 1) + 3) = 6k + 7$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$4^n > 2n + 3$$

Worked Solutions

Recurrence and Induction

Q116 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:

Solution

1. $n = 3$: $2^3 = 8 > 5 = 5$.
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $2^k + 1 > 2(1k + 2)$ after multiplying by the positive base.
4. $2(1k + 2) - (1(k + 1) + 2) = k + 1$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: D

Final answer

$$2^n > 1n + 2$$

Worked Solutions

Recurrence and Induction

Q117 MCQ

Select the correct option: which inequality is proved for $n \geq 2$?

Solution

1. $n = 2$: $3^2 = 9 > 4 = 4$.
2. Assume the inequality holds for $n = k$ with $k \geq 2$.
3. $3^k + 1 > 3(2k + 0)$ after multiplying by the positive base.
4. $3(2k + 0) - (2(k + 1) + 0) = 2(2k - 1)$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: A

Final answer

$$3^n > 2n$$

Worked Solutions

Recurrence and Induction

Q118 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:**Solution**

1. $n = 4$: $3^4 = 81 > 21 = 21$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 1)$ after multiplying by the positive base.
4. $3(5k + 1) - (5(k + 1) + 1) = 10k - 3$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: B**Final answer**

$$3^n > 5n + 1$$

Worked Solutions

Recurrence and Induction

Q119 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?

Solution

1. $n = 3$: $3^3 = 27 > 9 = 3n$.
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $3^k + 1 > 3(3k + 0)$ after multiplying by the positive base.
4. $3(3k + 0) - (3(k + 1) + 0) = 3(2k - 1)$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: D

Final answer

$$3^n > 3n$$

Worked Solutions

Recurrence and Induction

Q120 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:

Solution

1. $n = 4$: $3^4 = 81 > 22 = 22$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 2)$ after multiplying by the positive base.
4. $3(5k + 2) - (5(k + 1) + 2) = 10k - 1$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: A

Final answer

$$3^n > 5n + 2$$

Worked Solutions

Recurrence and Induction

Q121 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?

Solution

1. $n = 3$: $4^3 = 64 > 10 = 10$.
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $4^k + 1 > 4(3k + 1)$ after multiplying by the positive base.
4. $4(3k + 1) - (3(k + 1) + 1) = 9k$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: B

Final answer

$$4^n > 3n + 1$$

Worked Solutions

Recurrence and Induction

Q122 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:

Solution

1. $n = 4$: $4^4 = 256 > 11 = 11$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $4^k + 1 > 4(2k + 3)$ after multiplying by the positive base.
4. $4(2k + 3) - (2(k + 1) + 3) = 6k + 7$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: D

Final answer

$$4^n > 2n + 3$$

Worked Solutions

Recurrence and Induction

Q123 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?

Solution

1. $n = 4$: $4^4 = 256 > 17 = 17$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $4^k + 1 > 4(4k + 1)$ after multiplying by the positive base.
4. $4(4k + 1) - (4(k + 1) + 1) = 12k - 1$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: A

Final answer

$$4^n > 4n + 1$$

Worked Solutions

Recurrence and Induction

Q124 Written

Prove by mathematical induction that $2^n > 3n + 1$ for $n \geq 4$.

Solution

1. $n = 4$: $2^4 = 16 > 13 = 3 \cdot 4 + 1$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^k + 1 > 2(3k + 1)$ after multiplying by the positive base.
4. $2(3k + 1) - (3(k + 1) + 1) = 3k - 2$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$2^n > 3n + 1$$

Worked Solutions

Recurrence and Induction

Q125 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?

Solution

1. $n = 3$: $4^3 = 64 > 9 = 9$.
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $4^k + 1 > 4(3k + 0)$ after multiplying by the positive base.
4. $4(3k + 0) - (3(k + 1) + 0) = 3(3k - 1)$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: B

Final answer

$$4^n > 3n$$

Worked Solutions

Recurrence and Induction

Q126 Challenge

When $h > 0$ and $n \geq 3$, prove $(1 + h)^n > 1 + nh^2$.

Solution

1. $n = 3$: $(1 + h)^3 = 1 + 3h + 3h^2 + h^3 > 1 + 3h^2$ because $h > 0$ and $3h + h^3 > 0$.
2. Assume $(1 + h)^k > 1 + kh^2$ for $k \geq 3$.
3. Multiplying by the positive factor $1 + h$ gives a lower bound $(1 + h)(1 + kh^2)$.
4. $(1 + h)(1 + kh^2) - [1 + (k + 1)h^2] = h(1 - h + kh^2)$. The quadratic $1 - h + kh^2$ is positive for $k \geq 3$ because its discriminant is $1 - 4k < 0$.
5. Therefore the inequality holds for every $n \geq 3$ and $h > 0$.

Final answer

$$(1 + h)^n > 1 + nh^2$$

Worked Solutions

Recurrence and Induction

Q127 Written

Prove that $n^3 + (n + 1)^3 + (n + 2)^3$ is a multiple of 9 for every natural number n , using mathematical induction.

Solution

1. [1] Base case: $n = 1$, $f(1) = 36 = 9(4)$, so the statement holds.
2. [2] Assume $f(k) = 9m$ for an integer m .
3. [3] The induction difference is $f(k + 1) - f(k) = 9(k^2 + 3k + 3) = 9(k^2 + 3k + 3)$.
4. [4] Therefore $f(k + 1) = 9[m + k^2 + 3k + 3]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$$9 \mid [n^3 + (n + 1)^3 + (n + 2)^3]$$

Worked Solutions

Recurrence and Induction

Q128 Written

Prove that $n^3 + 2n$ is a multiple of 3 for every natural number n , using mathematical induction.

Solution

1. [1] Base case: $n = 1$, $f(1) = 3 = 3(1)$, so the statement holds.
2. [2] Assume $f(k) = 3m$ for an integer m .
3. [3] The induction difference is $f(k+1) - f(k) = 3(k^2 + k + 1) = 3(k^2 + k + 1)$.
4. [4] Therefore $f(k+1) = 3[m + k^2 + k + 1]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$$3 \mid (n^3 + 2n)$$

Worked Solutions

Recurrence and Induction

Q129 Written

Prove by mathematical induction that $n^2 + n$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1$, $f(1) = 2 = 2(1)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+0+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$$2 \mid (n^2 + n)$$

Worked Solutions

Recurrence and Induction

Q130 Written

Prove by mathematical induction that $n^2 + 5n + 8$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1$, $f(1) = 14 = 2(7)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+2+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$$2 \mid (n^2 + 5n + 8)$$

Worked Solutions

Recurrence and Induction

Q131 Written

Prove by mathematical induction that $n^2 + 7n + 12$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1$, $f(1) = 20 = 2(10)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+3+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$$2 \mid (n^2 + 7n + 12)$$

Worked Solutions

Recurrence and Induction

Q132 Written

Prove by mathematical induction that $n^2 + 11n + 32$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1$, $f(1) = 44 = 2(22)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+5+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$$2 \mid (n^2 + 11n + 32)$$

Worked Solutions

Recurrence and Induction

Q133 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + n$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 2 = 2(1)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k + 1) - f(k) = 2 \prod_{j=1}^1 (k + 0 + j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k + 1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: A

Final answer

2

Worked Solutions

Recurrence and Induction

Q134 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 5n + 8$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 14 = 2(7)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+2+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

2

Worked Solutions

Recurrence and Induction

Q135 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 7n + 12$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 20 = 2(10)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k + 1) - f(k) = 2 \prod_{j=1}^1 (k + 3 + j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k + 1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: C

Final answer

2

Worked Solutions

Recurrence and Induction

Q136 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 11n + 32$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 44 = 2(22)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+5+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: C

Final answer

2

Worked Solutions

Recurrence and Induction

Q137 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 3n^2 + 2n$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 6 = 6(1)$.
2. [2] Assume $f(k) = 6m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3 \prod_{j=1}^2 (k+0+j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: B

Final answer

6

Worked Solutions

Recurrence and Induction

Q138 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 9n^2 + 26n + 30$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 66 = 6(11)$.
2. [2] Assume $f(k) = 6m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3 \prod_{j=1}^2 (k+2+j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: A

Final answer

6

Worked Solutions

Recurrence and Induction

Q139 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 12n^2 + 47n + 60$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 120 = 6(20)$.
2. [2] Assume $f(k) = 6m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3 \prod_{j=1}^2 (k+3+j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

6

Worked Solutions

Recurrence and Induction

Q140 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 18n^2 + 107n + 216$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 342 = 6(57)$.
2. [2] Assume $f(k) = 6m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3 \prod_{j=1}^2 (k+5+j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

6

Worked Solutions

Recurrence and Induction

Q141 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + n + 2$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 4 = 2(2)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k + 1) - f(k) = 2 \prod_{j=1}^1 (k + 0 + j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k + 1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

2

Worked Solutions

Recurrence and Induction

Q142 Written

Prove by mathematical induction that $2n^3 - 3n^2 + n$ is a multiple of 6.

Solution

1. [1] Base case: $n = 1$, $f(1) = 0 = 6(0)$, so the statement holds.
2. [2] Assume $f(k) = 6m$ for an integer m .
3. [3] The induction difference is $f(k+1) - f(k) = 6k^2 = 6(k^2)$.
4. [4] Therefore $f(k+1) = 6[m + k^2]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$$6 \mid (2n^3 - 3n^2 + n)$$

Worked Solutions

Recurrence and Induction

Q143 Written

Prove by mathematical induction that $4n^3 - n$ is a multiple of 3.

Solution

1. [1] Base case: $n = 1$, $f(1) = 3 = 3(1)$, so the statement holds.
2. [2] Assume $f(k) = 3m$ for an integer m .
3. [3] The induction difference is $f(k+1) - f(k) = 3(2k+1)^2 = 3((2k+1)^2)$.
4. [4] Therefore $f(k+1) = 3[m + (2k+1)^2]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$$3 \mid (4n^3 - n)$$

Worked Solutions

Recurrence and Induction

Q144 Challenge

For any integer $n \geq 1$, prove that $(n + 1)^n - 1$ is a multiple of n^2 .

Solution

1. Use the binomial theorem on $(n + 1)^n = (1 + n)^n$.
2. $(1 + n)^n - 1 = \binom{n}{1}n + \binom{n}{2}n^2 + \cdots + n^n$.
3. $\binom{n}{1}n = n^2$, and every remaining term also contains n^2 .
4. $(n + 1)^n - 1 = n^2[1 + \binom{n}{2} + \binom{n}{3}n + \cdots + n^{n-2}]$. The bracket is an integer for every integer $n \geq 1$.
5. Hence the expression is a multiple of n^2 .

Final answer

$$n^2 \mid [(n + 1)^n - 1]$$

Worked Solutions

Recurrence and Induction

Q145 Written

For the sequence determined by $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 - 1}{n+1}$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = 4, \quad a_3 = 5, \quad a_4 = 6.$
2. The pattern suggests $a_n = n + 2.$

Final answer

$$a_2 = 4, \quad a_3 = 5, \quad a_4 = 6; \quad a_n = n + 2$$

Q146 Written

For the sequence determined by $a_1 = -1$ and $a_{n+1} = a_n^2 + 2na_n - 2$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = -3, \quad a_3 = -5, \quad a_4 = -7.$
2. The pattern suggests $a_n = 1 - 2n.$

Final answer

$$a_2 = -3, \quad a_3 = -5, \quad a_4 = -7; \quad a_n = 1 - 2n$$

Worked Solutions

Recurrence and Induction

Q147 Written

For the sequence determined by $a_1 = 1$ and $a_{n+1} = a_n^2 - n^2 a_n + (n+1)^2$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = 4, \quad a_3 = 9, \quad a_4 = 16.$
2. The pattern suggests $a_n = n^2.$

Final answer

$$a_2 = 4, \quad a_3 = 9, \quad a_4 = 16; \quad a_n = n^2$$

Q148 Written

For the sequence determined by $a_1 = 1$ and $a_{n+1} = \frac{a_n - 4}{a_n - 3}$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = \frac{3}{2}, \quad a_3 = \frac{5}{3}, \quad a_4 = \frac{7}{4}.$
2. The pattern suggests $a_n = \frac{2n-1}{n}.$

Final answer

$$a_2 = \frac{3}{2}, \quad a_3 = \frac{5}{3}, \quad a_4 = \frac{7}{4}; \quad a_n = \frac{2n-1}{n}$$

Worked Solutions

Recurrence and Induction

Q149 Written

For a sequence with $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 - 1}{n+1}$, prove by mathematical induction that $a_n = n + 2$.

Solution

1. [1] Base case: $a_1 = 3 = 3$.
2. [2] Assume $a_k = k + 2$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = \frac{a_k^2 - 1}{k+1}$: $a_{k+1} = k + 3$.
4. [4] This is $k + 3$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$$a_n = n + 2$$

Worked Solutions

Recurrence and Induction

Q150 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = 2k - 1$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2k + 1$.
4. [4] This is $2k + 1$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: D

Final answer

$$a_n = 2n - 1$$

Worked Solutions

Recurrence and Induction

Q151 MCQ

Select the correct option: for $a_1 = 2$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 2 = 2$.
2. [2] Assume $a_k = 2k$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2(k + 1)$.
4. [4] This is $2(k + 1)$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$$a_n = 2n$$

Worked Solutions

Recurrence and Induction

Q152 MCQ

Select the correct option: for $a_1 = 3$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 3 = 3$.
2. [2] Assume $a_k = 2k + 1$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2k + 3$.
4. [4] This is $2k + 3$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: B

Final answer

$$a_n = 2n + 1$$

Worked Solutions

Recurrence and Induction

Q153 MCQ

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 4 = 4$.
2. [2] Assume $a_k = 2(k + 1)$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2(k + 2)$.
4. [4] This is $2(k + 2)$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: A

Final answer

$$a_n = 2(n + 1)$$

Worked Solutions

Recurrence and Induction

Q154 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = 3k - 2$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3k + 1$.
4. [4] This is $3k + 1$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: D

Final answer

$$a_n = 3n - 2$$

Worked Solutions

Recurrence and Induction

Q155 MCQ

Select the correct option: for $a_1 = 2$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 2 = 2$.
2. [2] Assume $a_k = 3k - 1$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3k + 2$.
4. [4] This is $3k + 2$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$$a_n = 3n - 1$$

Worked Solutions

Recurrence and Induction

Q156 MCQ

Select the correct option: for $a_1 = 3$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 3 = 3$.
2. [2] Assume $a_k = 3k$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3(k + 1)$.
4. [4] This is $3(k + 1)$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: B

Final answer

$$a_n = 3n$$

Worked Solutions

Recurrence and Induction

Q157 MCQ

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 4 = 4$.
2. [2] Assume $a_k = 3k + 1$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3k + 4$.
4. [4] This is $3k + 4$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: A

Final answer

$$a_n = 3n + 1$$

Worked Solutions

Recurrence and Induction

Q158 Written

For a sequence with $a_1 = 1$ and $a_{n+1} = a_n^2 - n^2 a_n + (n+1)^2$, prove by mathematical induction that $a_n = n^2$.

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = k^2$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k^2 - k^2 a_k + (k+1)^2$: $a_{k+1} = (k+1)^2$.
4. [4] This is $(k+1)^2$, the proposed formula with $n = k+1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$$a_n = n^2$$

Worked Solutions

Recurrence and Induction

Q159 Written

For a sequence with $a_1 = -1$ and $a_{n+1} = a_n^2 + 2na_n - 2$, prove by mathematical induction that $a_n = 1 - 2n$.

Solution

1. [1] Base case: $a_1 = -1 = -1$.
2. [2] Assume $a_k = 1 - 2k$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k^2 + 2ka_k - 2$: $a_{k+1} = -2k - 1$.
4. [4] This is $-2k - 1$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$$a_n = 1 - 2n$$

Worked Solutions

Recurrence and Induction

Q160 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = 2a_n + 0$, the general term is:

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = 2^{k-1}$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = 2a_k + 0$: $a_{k+1} = 2^k$.
4. [4] This is 2^k , the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$$a_n = 2^{n-1}$$

Worked Solutions

Recurrence and Induction

Q161 Written

For a sequence with $a_1 = 1$ and $a_{n+1} = \frac{a_n - 4}{a_n - 3}$, prove by mathematical induction that $a_n = \frac{2n-1}{n}$.

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = \frac{2k-1}{k}$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = \frac{a_k - 4}{a_k - 3}$: $a_{k+1} = \frac{2k+1}{k+1}$.
4. [4] This is $\frac{2k+1}{k+1}$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$$a_n = \frac{2n-1}{n}$$

Worked Solutions

Recurrence and Induction

Q162 Challenge

Three numbers A, B, C in that order form a geometric sequence with common ratio r . If $A, 2B, C$ in that order form an arithmetic sequence, determine the possible values of r .

Solution

1. Let the geometric sequence be A, Ar, Ar^2 .
2. Since $A, 2B, C$ is arithmetic, its middle term is the average of the outer terms: $4B = A + C$.
3. Substitute $B = Ar, C = Ar^2$ and cancel the non-zero scale factor A : $r^2 - 4r + 1 = 0$.
4. $r = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3}$.

Final answer

$$r = 2 \pm \sqrt{3}$$

Worked Solutions

Recurrence and Induction

Quiz MCQ 1 **Concept Quiz · MCQ**

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (4)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 2(2n - 1)$.

Correct option: D

Final answer

$$a_n = 2(2n - 1)$$

Worked Solutions

Recurrence and Induction

Quiz MCQ 2 Concept Quiz · MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = -2a_n + (3)$, the general term is:

Solution

1. $c = 1$ from $c = -2c + (3)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{(-2)^n - 2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

$$a_n = -\frac{(-2)^n - 2}{2}$$

Worked Solutions

Recurrence and Induction

Quiz MCQ 3 Concept Quiz · MCQ

Select the correct option: which inequality is proved for $n \geq 4$?

Solution

1. $n = 4$: $2^4 = 16 > 13 = 13$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^k + 1 > 2(3k + 1)$ after multiplying by the positive base.
4. $2(3k + 1) - (3(k + 1) + 1) = 3k - 2$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: B

Final answer

$$2^n > 3n + 1$$

Worked Solutions

Recurrence and Induction

Quiz MCQ 4 Concept Quiz · MCQ

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 4 = 4$.
2. [2] Assume $a_k = 2(k + 1)$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2(k + 2)$.
4. [4] This is $2(k + 2)$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$$a_n = 2(n + 1)$$

Worked Solutions

Recurrence and Induction

Quiz WRITTEN 5 Concept Quiz · Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (5)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 5n - 3$.

Final answer

$$a_n = 5n - 3$$

Worked Solutions

Recurrence and Induction

Quiz WRITTEN 6 Concept Quiz · Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 5a_n + (8)$.

Solution

1. $c = -2$ from $c = 5c + (8)$.
2. $a_{n+1} - c = 5(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3 \cdot 5^n - 10}{5}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3 \cdot 5^n - 10}{5}$$

Worked Solutions

Recurrence and Induction

Quiz WRITTEN 7 Concept Quiz · Written

Prove by mathematical induction that $2^n > 4n + 1$ for $n \geq 4$.

Solution

1. $n = 4$: $2^4 = 16 > 17 = 4 \cdot 4 + 1$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^k + 1 > 2(4k + 1)$ after multiplying by the positive base.
4. $2(4k + 1) - (4(k + 1) + 1) = 4k - 3$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$2^n > 4n + 1$$

Worked Solutions

Recurrence and Induction

Quiz WRITTEN 8 Concept Quiz · Written

For a sequence with $a_1 = 5$ and $a_{n+1} = a_n + 1(2n + 1) + 2$, prove by mathematical induction that $a_n = n^2 + 2n + 2$.

Solution

1. [1] Base case: $a_1 = 5 = 5$.
2. [2] Assume $a_k = k^2 + 2k + 2$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 1(2k + 1) + 2$: $a_{k+1} = k^2 + 4k + 5$.
4. [4] This is $k^2 + 4k + 5$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$$a_n = n^2 + 2n + 2$$

Answer key

Use the question number shown on each page.

Q001 $a_2 = 8, a_3 = 26, a_4 = 80$

Q002 $a_2 = -9, a_3 = -15, a_4 = -27$

Q003 $a_2 = -7, a_3 = -7, a_4 = -4$

Q004 $a_2 = -3, a_3 = -5, a_4 = -8$

Q005 $a_2 = 4, a_3 = 8, a_4 = 14$

Q006 $a_2 = 1, a_3 = 10, a_4 = 37$

Q007 B

Q008 D

Q009 A

Q010 B

Q011 D

Q012 A

Q013 C

Q014 D

Q015 C

Q016 D

Q017 B

Q018 C

Q019 $a_n = 5n - 6$

Q020 $a_n = 3n - 2$

Q021 $a_n = -2(n - 3)$

Q022 $a_n = 6n - 5$

Q023 $a_n = 8 - 5n$

Q024 $a_n = 3n - 5$

Q025 $a_n = -4(n - 2)$

Q026 D

Q027 A

Q028 A

Q029 A

Q030 $a_n = 2 \cdot 3^{n-1}$

Q031 $a_n = 3 \cdot 5^{n-1}$

Q032 $a_n = 4 \cdot 3^{n-1}$

Q033 $a_n = 2^{2n-2}$

Q034 B

Q035 B

Q036 D

Q037 A

Q038 $a_n = \frac{3(-2)^n}{2}$

Q039 $a_n = 2(-5)^{n-1}$

Q040 $a_n = \frac{(-4)^n}{2}$

Q041 B

Q042 C

Q043 B

Q044 C

Q045 $a_2 = 5, a_3 = 13, a_4 = 36$

Q046 $a_2 = 3, a_3 = -1, a_4 = 7$

Q047 $a_2 = 8, a_3 = 22, a_4 = 63$

Q048 $a_n = -\frac{4 \cdot 2^n - 3 \cdot 3^n - 1}{2}$

Q049 $a_n = (n - 2)^2$

Q050 $a_n = \frac{3n^2 - 3n + 2}{2}$

Q051 $a_n = n^2 - n + 2$

Q052 $a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$

Q053 $a_n = \frac{3n^2 - n + 2}{2}$

Q054 $a_n = \frac{4n^3 - 3n^2 - n + 12}{6}$

Q055 C

Q056 A

Q057 C

Q058 D

Q059 C

Q060 A

Q061 D

Q062 A

Q063 B

Q064 A

Q065 $a_n = \frac{2n^3 - 3n^2 + n + 12}{6}$

Q066 A

Q067 $a_n = 2^n - 1$

Q068 $a_n = -\frac{3^n - 7}{2}$

Q069 $a_n = -\frac{4^n - 16}{3}$

Q070 $a_n = -\frac{n^2 - n - 200}{50}$

Q071 $a_n = 3^n - 2$

Q072 $a_n = \frac{3 \cdot 5^n - 10}{5}$

Q073 $a_n = \frac{2^n + 2}{2}$

Q074 $a_n = -\frac{4^n + 2}{2}$

Q075 $a_n = \frac{2 \cdot 3^n + 3}{3}$

Q076 $a_n = (-3)^n + 1$

Q077 $a_n = 3 \cdot 3^{-n} (3^n - 2)$

Q078 $a_n = \frac{3 \cdot 3^{-n} (7 \cdot 2^n - 2 \cdot 3^n)}{4}$

Q079 $a_n = -\frac{3(-2)^n - 2}{2}$

Q080 B

Q081 C

Q082 D

Q083 B

Q084 C

Q085 D

Q086 B

Q087 C

Q088 D

Q089 D

Q090 $c = 3, a_n = -\frac{2^{2n} - 12}{4}, a_5 = -253$

Q091 $\frac{n(n+1)}{2}$

Q092 $\frac{n(2n-1)(2n+1)}{3}$

Q093 $\frac{n(n+1)(n+2)}{3}$

Q094 A

Q095 B	Q113 $3^n > 5n + 1$	Q133 A	Q152 B
Q096 C	Q114 $3^n > 5n + 2$	Q134 D	Q153 A
Q097 B	Q115 $4^n > 2n + 3$	Q135 C	Q154 D
Q098 C	Q116 D	Q136 C	Q155 C
Q099 A	Q117 A	Q137 B	Q156 B
Q100 C	Q118 B	Q138 A	Q157 A
Q101 A	Q119 D	Q139 D	Q158 $a_n = n^2$
Q102 B	Q120 A	Q140 D	Q159 $a_n = 1 - 2n$
Q103 n^2	Q121 B	Q141 D	Q160 C
Q104 $\frac{n(n+1)(n+2)}{3}$	Q122 D	Q142 $6 \mid (2n^3 - 3n^2 + n)$	Q161 $a_n = \frac{2n-1}{n}$
Q105 $\frac{n(n+1)(2n+7)}{6}$	Q123 A	Q143 $3 \mid (4n^3 - n)$	Q162 $r = 2 \pm \sqrt{3}$
Q106 $4^n - 1$	Q124 $2^n > 3n + 1$	Q144 $n^2 \mid [(n+1)^n - 1]$	Quiz MCQ 1 D
Q107 $\frac{10 \cdot 10^n - 1}{9}$	Q125 B	Q145 $a_2 = 4, a_3 = 5, a_4 = 6; a_n = n + 2$	Quiz MCQ 2 A
Q108 $\frac{n(n^2 + 3n + 5)}{3}$	Q126 $(1+h)^n > 1 + nh^2$	Q146 $a_2 = -3, a_3 = -5, a_4 = -7; a_n = 1 - 2n$	Quiz MCQ 3 B
Q109 $2^n > 3n$	Q127 $9 \mid [n^3 + (n+1)^3 + (n+2)^3]$	Q147 $a_2 = 4, a_3 = 9, a_4 = 16; a_n = n^2$	Quiz MCQ 4 C
Q110 $3^n > 5n + 1$	Q128 $3 \mid (n^3 + 2n)$	Q148 $a_2 = \frac{3}{2}, a_3 = \frac{5}{3}, a_4 = \frac{7}{4}; a_n = \frac{2n-1}{n}$	Quiz WRITTEN 5 $a_n = 5n - 3$
Q111 $3^n > 2n$	Q129 $2 \mid (n^2 + n)$	Q149 $a_n = n + 2$	Quiz WRITTEN 6 $a_n = \frac{3 \cdot 5^n - 10}{5}$
Q112 $2^n > 1n + 2$	Q130 $2 \mid (n^2 + 5n + 8)$	Q150 D	Quiz WRITTEN 7 $2^n > 4n + 1$
	Q131 $2 \mid (n^2 + 7n + 12)$	Q151 C	Quiz WRITTEN 8 $a_n = n^2 + 2n + 2$
	Q132 $2 \mid (n^2 + 11n + 32)$		

Concept 1 | Complex Numbers

English Student Edition • Focused formulas and guided practice

Formula opener

Standard form

$$z = a + bi$$

Conjugate

$$\overline{a + bi} = a - bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Worked Solutions

Complex Numbers

Q001 Written

State the real and imaginary parts of $3 - 2i$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(3, -2)$.

Final answer $(3, -2)$

Q002 Written

State the real and imaginary parts of $(4 - \sqrt{5}i)/3$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(4/3, -\sqrt{5}/3)$.

Final answer $(4/3, -\sqrt{5}/3)$

Worked Solutions

Complex Numbers

Q003 Written

State the real and imaginary parts of -8 .

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-8, 0)$.

Final answer

$(-8, 0)$

Q004 Written

State the real and imaginary parts of $2i$.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0, 2)$.

Final answer

$(0, 2)$

Worked Solutions

Complex Numbers

Q005 Written

Find the real values of x and y satisfying $(2 + i)x + (1 + i)y = 2 + 3i$, x, y real.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = -1, y = 4$.

Final answer

$$x = -1, y = 4$$

Q006 Written

State the real and imaginary parts of $-3 + 5i$.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-3, 5)$.

Final answer

$$(-3, 5)$$

Worked Solutions

Complex Numbers

Q007 Written

State the real and imaginary parts of $(-1 - \sqrt{3}i)/2$.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-1/2, -\sqrt{3}/2)$.

Final answer $(-1/2, -\sqrt{3}/2)$

Q008 Written

State the real and imaginary parts of 1.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(1, 0)$.

Final answer $(1, 0)$

Worked Solutions

Complex Numbers

Q009 Written

State the real and imaginary parts of $3i$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0, 3)$.

Final answer $(0, 3)$

Q010 Written

Find the real values of x and y satisfying $(x - 6) + (x + 3y)i = 0$, x, y real.**Solution**

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 6, y = -2$.

Final answer $x = 6, y = -2$

Worked Solutions

Complex Numbers

Q011 Written

Find the real values of x and y satisfying $(3 + 2i)x + (4 - i)y = 6 - 7i$, x, y real.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = -2, y = 3$.

Final answer

$$x = -2, y = 3$$

Q012 Written

State the real and imaginary parts of $2 + 5i$.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(2, 5)$.

Final answer

$$(2, 5)$$

Worked Solutions

Complex Numbers

Q013 Written

State the real and imaginary parts of $-25 + 13i$.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-25, 13)$.

Final answer

$(-25, 13)$

Q014 Written

State the real and imaginary parts of $(2 - \sqrt{5}i)/3$.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(2/3, -\sqrt{5}/3)$.

Final answer

$(2/3, -\sqrt{5}/3)$

Worked Solutions

Complex Numbers

Q015 **Written****State the real and imaginary parts of $(7 - \sqrt{2}i)/2$.****Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(7/2, -\sqrt{2}/2)$.

Final answer $(7/2, -\sqrt{2}/2)$ Q016 **Written****State the real and imaginary parts of -4 .****Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-4, 0)$.

Final answer $(-4, 0)$

Worked Solutions

Complex Numbers

Q017 Written

State the real and imaginary parts of $1/4$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(1/4, 0)$.

Final answer $(1/4, 0)$

Q018 Written

State the real and imaginary parts of $\sqrt{3}i$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0, \sqrt{3})$.

Final answer $(0, \sqrt{3})$

Worked Solutions

Complex Numbers

Q019 Written

State the real and imaginary parts of $-i$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0, -1)$.

Final answer $(0, -1)$

Q020 Written

Find the real values of x and y satisfying $(2x + y) + (3x - y)i = 8 + 7i$.**Solution**

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 3, y = 2$.

Final answer $x = 3, y = 2$

Worked Solutions

Complex Numbers

Q021 Written

Find the real values of x and y satisfying $(x + 1) - 5i = 4 + (2y + 1)i$.**Solution**

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 3, y = -3$.

Final answer

$$x = 3, y = -3$$

Q022 Written

Find the real values of x and y satisfying $(x - 2) + (x + y)i = 0$.**Solution**

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 2, y = -2$.

Final answer

$$x = 2, y = -2$$

Worked Solutions

Complex Numbers

Q023 Written

Find the real values of x and y satisfying $(3 + 2i)x + (1 - i)y = 7 + 3i$.**Solution**

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 2, y = 1$.

Final answer

$$x = 2, y = 1$$

Q024 Written

Find the real values of x and y satisfying $(1 - i)(x + yi) = 2 + i$.**Solution**

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 1/2, y = 3/2$.

Final answer

$$x = 1/2, y = 3/2$$

Worked Solutions

Complex Numbers

Q025 Written

Solve the complex-number equation and find the exact real values of x and y : $(1 + 2i)(x + i) = x(y + i)$.

Solution

1. Expand: $((1 + 2i)(x + i) = x - 2 + (1 + 2x)i)$.
2. Equate coefficients with $(xy + xi)$: $(x - 2 = xy)$ and $(1 + 2x = x)$.
3. Solving gives $x = -1, y = 3$.

Final answer

$$x = -1, y = 3$$

Q026 Written

Calculate $(3 - 2i) - (2 + 5i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $1 - 7i$.

Final answer

$$1 - 7i$$

Worked Solutions

Complex Numbers

Q027 **Written****Calculate $(i - 2)^2$ and give the answer in standard form.****Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $3 - 4i$.

Final answer

$$3 - 4i$$

Q028 **Written****Calculate $i^6 + i^{27} + i^{17}$ and give the answer in standard form.****Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: -1 .

Final answer

$$-1$$

Worked Solutions

Complex Numbers

Q029 Written

Calculate $(4 + 6i) + (-3 + 2i)$ and give the answer in standard form.**Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $1 + 8i$.

Final answer

$$1 + 8i$$

Q030 Written

Calculate $(2 + 3i)(1 - 2i)$ and give the answer in standard form.**Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $8 - i$.

Final answer

$$8 - i$$

Worked Solutions

Complex Numbers

Q031 **Written****Calculate $i + i^2 + i^3 + i^4$ and give the answer in standard form.****Solution**

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: 0.

Final answer

0

Q032 **Written****Calculate $(7 + 4i) - (3 + 5i)$ and give the answer in standard form.****Solution**

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $4 - i$.

Final answer $4 - i$

Worked Solutions

Complex Numbers

Q033 Written

Calculate $(-4 + 3i) + (5 - 6i)$ and give the answer in standard form.**Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $1 - 3i$.

Final answer

$$1 - 3i$$

Q034 Written

Calculate $(3 - 3i) - (5 - 3i)$ and give the answer in standard form.**Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: -2 .

Final answer

$$-2$$

Worked Solutions

Complex Numbers

Q035 **Written****Calculate $(\sqrt{2} + 3i)(\sqrt{2} + i)$ and give the answer in standard form.****Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $-1 + 4\sqrt{2}i$.

Final answer

$$-1 + 4\sqrt{2}i$$

Q036 **Written****Calculate $(2 + 3i) + (5 + 8i)$ and give the answer in standard form.****Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $7 + 11i$.

Final answer

$$7 + 11i$$

Worked Solutions

Complex Numbers

Q037 Written

Calculate $(2/5 - 1/2i) - (1/3 + 3/7i)$ and give the answer in standard form.**Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $1/15 - 13/14i$.

Final answer

$$1/15 - 13/14i$$

Q038 Written

Calculate $(2i - 1)^2$ and give the answer in standard form.**Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $-3 - 4i$.

Final answer

$$-3 - 4i$$

Worked Solutions

Complex Numbers

Q039 Written

Calculate $(2/i + i)(2/i - i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: -3 .

Final answer -3

Q040 Written

Calculate $(2 + i)^3 + (2 - i)^3$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: 4.

Final answer 4

Worked Solutions

Complex Numbers

Q041 **Written****Calculate $i^5 + i^3 + i$ and give the answer in standard form.****Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: i .

Final answer i Q042 **Written****Calculate $i^9 + i^{14} + i^{11}$ and give the answer in standard form.****Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: -1 .

Final answer -1

Worked Solutions

Complex Numbers

Q043 Written

Calculate $i^{12} + i^{18} + i^{15}$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $-i$.

Final answer $-i$

Q044 Written

Given x is real, solve $(7x - 7i)(x^2 + xi) = 14x^2i^{40}$.

Solution

1. Use $(i^{40} = 1)$ and $((x - i)(x + i) = x^2 + 1)$.
2. The equation becomes $(7x(x^2 + 1) = 14x^2)$, so $(7x(x - 1)^2 = 0)$.
3. Therefore $x = 0$ or $x = 1$.

Final answer $x = 0$ or $x = 1$

Worked Solutions

Complex Numbers

Q045 Written

State the complex conjugate of $2 + 3i$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $2 - 3i$.

Final answer

$$2 - 3i$$

Q046 Written

State the complex conjugate of $7 - i\sqrt{11}$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $7 + i\sqrt{11}$.

Final answer

$$7 + i\sqrt{11}$$

Worked Solutions

Complex Numbers

Q047 **Written****State the complex conjugate of 6.****Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is 6.

Final answer

6

Q048 **Written****State the complex conjugate of $-5i$.****Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $5i$.

Final answer $5i$

Worked Solutions

Complex Numbers

Q049 Written

Calculate $(2 + i)/(2 - i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $3/5 + 4/5i$.

Final answer

$$3/5 + 4/5i$$

Q050 Written

Calculate $(3 - i)/(2i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $-1/2 - 3/2i$.

Final answer

$$-1/2 - 3/2i$$

Worked Solutions

Complex Numbers

Q051 **Written****State the complex conjugate of $1 - 4i$.****Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $1 + 4i$.

Final answer

$$1 + 4i$$

Q052 **Written****State the complex conjugate of $\sqrt{2} + i\sqrt{10}$.****Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $\sqrt{2} - i\sqrt{10}$.

Final answer

$$\sqrt{2} - i\sqrt{10}$$

Worked Solutions

Complex Numbers

Q053 Written

State the complex conjugate of 2.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is 2.

Final answer

2

Q054 Written

State the complex conjugate of $-8i$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $8i$.

Final answer $8i$

Worked Solutions

Complex Numbers

Q055 Written

Calculate $(3 + i)/(1 + 2i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $1 - i$.

Final answer

$$1 - i$$

Q056 Written

Calculate $(1 - i)/i$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $-1 - i$.

Final answer

$$-1 - i$$

Worked Solutions

Complex Numbers

Q057 **Written****State the complex conjugate of $3 + 4i$.****Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $3 - 4i$.

Final answer

$$3 - 4i$$

Q058 **Written****State the complex conjugate of $9 + 8i$.****Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $9 - 8i$.

Final answer

$$9 - 8i$$

Worked Solutions

Complex Numbers

Q059 Written

State the complex conjugate of $8 + i\sqrt{3}$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $8 - i\sqrt{3}$.

Final answer

$$8 - i\sqrt{3}$$

Q060 Written

State the complex conjugate of 5.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is 5.

Final answer

$$5$$

Worked Solutions

Complex Numbers

Q061 Written

State the complex conjugate of -15 .**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is -15 .

Final answer -15

Q062 Written

State the complex conjugate of $\sqrt{7}$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $\sqrt{7}$.

Final answer $\sqrt{7}$

Worked Solutions

Complex Numbers

Q063 Written

State the complex conjugate of i .**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $-i$.

Final answer $-i$

Q064 Written

State the complex conjugate of $-50i$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $50i$.

Final answer $50i$

Worked Solutions

Complex Numbers

Q065 Written

State the complex conjugate of $i\sqrt{6}$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $-i\sqrt{6}$.

Final answer

$$-i\sqrt{6}$$

Q066 Written

Calculate $(1 + i)/(1 - i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is i .

Final answer

$$i$$

Worked Solutions

Complex Numbers

Q067 Written

Calculate $(3 + i)/(2 + i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $7/5 - 1/5i$.

Final answer

$$7/5 - 1/5i$$

Q068 Written

Calculate $(\sqrt{5} + i)/(\sqrt{5} - i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $2/3 + \sqrt{5}/3i$.

Final answer

$$2/3 + \sqrt{5}/3i$$

Worked Solutions

Complex Numbers

Q069 **Written****Calculate $5i/(1 + 2i)$ and give the answer in the form $a+bi$.****Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $2 + i$.

Final answer

$$2 + i$$

Q070 **Written****Calculate $5i/(4 + 3i)$ and give the answer in the form $a+bi$.****Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $3/5 + 4/5i$.

Final answer

$$3/5 + 4/5i$$

Worked Solutions

Complex Numbers

Q071 Written

Calculate $(1 - 3i)/(3 - 2i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $9/13 - 7/13i$.

Final answer

$$9/13 - 7/13i$$

Q072 Written

Calculate $1/i$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $-i$.

Final answer

$$-i$$

Worked Solutions

Complex Numbers

Q073 Written

Calculate $(2 + i)/(3i)$ and give the answer in the form $a+bi$.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $1/3 - 2/3i$.

Final answer

$$1/3 - 2/3i$$

Q074 Written

Simplify exactly: $\left(\frac{i(10-12i)}{(2+i)(-1+4i)}\right)^{2027}$.

Solution

1. Rationalise the base using the conjugate of the denominator.
2. The base simplifies exactly to $(-2 - 144i)/85$.
3. Hence the exact result is $((-2 - 144i)/85)^{2027}$; no expansion of the 2027th power is required.

Final answer

$$((-2 - 144i)/85)^{2027}$$

Worked Solutions

Complex Numbers

Q075 Written

Express the square roots of -6 in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $\pm i\sqrt{6}$.

Final answer

$$\pm i\sqrt{6}$$

Q076 Written

Express $\sqrt{-12}\sqrt{-6}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-6\sqrt{2}$.

Final answer

$$-6\sqrt{2}$$

Worked Solutions

Complex Numbers

Q077 Written

Express $\sqrt{8}/\sqrt{-18}$ in terms of i .**Solution**

1. Use ($\sqrt{a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $-(2/3)i$.

Final answer

$$-(2/3)i$$

Q078 Written

Express $\sqrt{-7}$ in terms of i .**Solution**

1. Use ($\sqrt{a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $i\sqrt{7}$.

Final answer

$$i\sqrt{7}$$

Worked Solutions

Complex Numbers

Q079 Written

Express $\sqrt{-27}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $3i\sqrt{3}$.

Final answer

$$3i\sqrt{3}$$

Q080 Written

Express $-\sqrt{-12}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-2i\sqrt{3}$.

Final answer

$$-2i\sqrt{3}$$

Worked Solutions

Complex Numbers

Q081 Written

Express the square roots of -12 in terms of i .**Solution**

1. Use ($\sqrt{-a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $\pm 2i\sqrt{3}$.

Final answer

$$\pm 2i\sqrt{3}$$

Q082 Written

Express $\sqrt{-3}\sqrt{-6}$ in terms of i .**Solution**

1. Use ($\sqrt{-a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $-3\sqrt{2}$.

Final answer

$$-3\sqrt{2}$$

Worked Solutions

Complex Numbers

Q083 Written

Express $\sqrt{27}/\sqrt{-3}$ in terms of i .**Solution**

1. Use $(\sqrt{a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-3i$.

Final answer $-3i$

Q084 Written

Express $(3 + \sqrt{-2})^2$ in terms of i .**Solution**

1. Use $(\sqrt{a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $7 + 6\sqrt{2}i$.

Final answer $7 + 6\sqrt{2}i$

Worked Solutions

Complex Numbers

Q085 Written

Express $\sqrt{-2}$ in terms of i .**Solution**

1. Use ($\sqrt{a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $i\sqrt{2}$.

Final answer

$$i\sqrt{2}$$

Q086 Written

Express $\sqrt{-9}$ in terms of i .**Solution**

1. Use ($\sqrt{a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $3i$.

Final answer

$$3i$$

Worked Solutions

Complex Numbers

Q087 Written

Express $\sqrt{-45}$ in terms of i .

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $3i\sqrt{5}$.

Final answer

$$3i\sqrt{5}$$

Q088 Written

Express $-\sqrt{-24}$ in terms of i .

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-2i\sqrt{6}$.

Final answer

$$-2i\sqrt{6}$$

Worked Solutions

Complex Numbers

Q089 Written

Express $-\sqrt{-36}$ in terms of i .**Solution**

1. Use $(\sqrt{a})^2 = a$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-6i$.

Final answer $-6i$

Q090 Written

Express the square roots of -2 in terms of i .**Solution**

1. Use $(\sqrt{a})^2 = a$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $\pm i\sqrt{2}$.

Final answer $\pm i\sqrt{2}$

Worked Solutions

Complex Numbers

Q091 Written

Express the square roots of -18 in terms of i .**Solution**

1. Use ($\sqrt{-a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $\pm 3i\sqrt{2}$.

Final answer

$$\pm 3i\sqrt{2}$$

Q092 Written

Express the square roots of -16 in terms of i .**Solution**

1. Use ($\sqrt{-a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $\pm 4i$.

Final answer

$$\pm 4i$$

Worked Solutions

Complex Numbers

Q093 Written

Express $\sqrt{-2}\sqrt{-6}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-2\sqrt{3}$.

Final answer

$$-2\sqrt{3}$$

Q094 Written

Express $\sqrt{-28}\sqrt{-35}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-14\sqrt{5}$.

Final answer

$$-14\sqrt{5}$$

Worked Solutions

Complex Numbers

Q095 Written

Express $\sqrt{8}/\sqrt{-2}$ in terms of i .**Solution**

1. Use $(\sqrt{a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-2i$.

Final answer $-2i$

Q096 Written

Express $\sqrt{27}/\sqrt{-12}$ in terms of i .**Solution**

1. Use $(\sqrt{a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-(3/2)i$.

Final answer $-(3/2)i$

Worked Solutions

Complex Numbers

Q097 Written

Express $\sqrt{-32} + \sqrt{-98} + \sqrt{-8}$ in terms of i .**Solution**

1. Use ($\sqrt{a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $13i\sqrt{2}$.

Final answer

$$13i\sqrt{2}$$

Q098 Written

Express $\sqrt{-98} - \sqrt{-72} + \sqrt{-50}$ in terms of i .**Solution**

1. Use ($\sqrt{a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $6i\sqrt{2}$.

Final answer

$$6i\sqrt{2}$$

Worked Solutions

Complex Numbers

Q099 Written

Express $(\sqrt{3} + \sqrt{-5})^2$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-2 + 2i\sqrt{15}$.

Final answer

$$-2 + 2i\sqrt{15}$$

Q100 Written

Express $(2 + \sqrt{-3})(2 - \sqrt{-3})$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore 7.

Final answer

$$7$$

Worked Solutions

Complex Numbers

Q101 Written

Solve over the complex numbers: $(2x - 8)^2 = -10$.

Solution

1. Take square roots: $(2x - 8 = \pm \sqrt{10})$.
2. Divide by 2 to obtain $x = 4 \pm (\sqrt{10}/2)i$.

Final answer

$$x = 4 \pm (\sqrt{10}/2)i$$

Q102 MCQ

For $2 + 3i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $2 + 3i$ using the relevant family rule.
2. The correct value is (2, 3).

Correct option: C

Final answer

(2, 3)

Worked Solutions

Complex Numbers

Q103 MCQ

For $-4 + 5i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-4 + 5i$ using the relevant family rule.
2. The correct value is $(-4, 5)$.

Correct option: D

Final answer

$(-4, 5)$

Q104 MCQ

For $7 - 2i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $7 - 2i$ using the relevant family rule.
2. The correct value is $(7, -2)$.

Correct option: D

Final answer

$(7, -2)$

Worked Solutions

Complex Numbers

Q105 MCQ

For $(1 - \sqrt{2}i)/3$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $(1 - \sqrt{2}i)/3$ using the relevant family rule.

2. The correct value is $\left(\frac{1}{3}, -\frac{\sqrt{2}}{3}\right)$.

Correct option: B

Final answer

$$\left(\frac{1}{3}, -\frac{\sqrt{2}}{3}\right)$$

Q106 MCQ

For -6 , select the ordered pair (real part, imaginary part):

Solution

1. Simplify -6 using the relevant family rule.

2. The correct value is $(-6, 0)$.

Correct option: D

Final answer

$$(-6, 0)$$

Worked Solutions

Complex Numbers

Q107 MCQ

For $4i$, select the ordered pair (real part, imaginary part):**Solution**

1. Simplify $4i$ using the relevant family rule.
2. The correct value is $(0, 4)$.

Correct option: A**Final answer** $(0, 4)$

Q108 MCQ

For $-i$, select the ordered pair (real part, imaginary part):**Solution**

1. Simplify $-i$ using the relevant family rule.
2. The correct value is $(0, -1)$.

Correct option: B**Final answer** $(0, -1)$

Worked Solutions

Complex Numbers

Q109 MCQ

For $\sqrt{5}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $\sqrt{5}i$ using the relevant family rule.
2. The correct value is $(0, \sqrt{5})$.

Correct option: C**Final answer** $(0, \sqrt{5})$

Q110 MCQ

For $3/2 + \sqrt{3}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $3/2 + \sqrt{3}i$ using the relevant family rule.
2. The correct value is $(\frac{3}{2}, \sqrt{3})$.

Correct option: B**Final answer** $(\frac{3}{2}, \sqrt{3})$

Worked Solutions

Complex Numbers

Q111 MCQ

For $-1/4 - 2i$, select the ordered pair (real part, imaginary part):**Solution**

1. Simplify $-1/4 - 2i$ using the relevant family rule.
2. The correct value is $\left(-\frac{1}{4}, -2\right)$.

Correct option: D**Final answer**

$$\left(-\frac{1}{4}, -2\right)$$

Q112 MCQ

For $5/3$, select the ordered pair (real part, imaginary part):**Solution**

1. Simplify $5/3$ using the relevant family rule.
2. The correct value is $\left(\frac{5}{3}, 0\right)$.

Correct option: D**Final answer**

$$\left(\frac{5}{3}, 0\right)$$

Worked Solutions

Complex Numbers

Q113 MCQ

For $-\sqrt{7}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-\sqrt{7}i$ using the relevant family rule.
2. The correct value is $(0, -\sqrt{7})$.

Correct option: C

Final answer

$(0, -\sqrt{7})$

Q114 MCQ

For $-9 + \sqrt{2}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-9 + \sqrt{2}i$ using the relevant family rule.
2. The correct value is $(-9, \sqrt{2})$.

Correct option: D

Final answer

$(-9, \sqrt{2})$

Worked Solutions

Complex Numbers

Q115 MCQ

For $2/5 - 3/4i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $2/5 - 3/4i$ using the relevant family rule.
2. The correct value is $\left(\frac{2}{5}, -\frac{3}{4}\right)$.

Correct option: B

Final answer

$$\left(\frac{2}{5}, -\frac{3}{4}\right)$$

Q116 MCQ

For $-\sqrt{11}$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-\sqrt{11}$ using the relevant family rule.
2. The correct value is $(-\sqrt{11}, 0)$.

Correct option: B

Final answer

$$(-\sqrt{11}, 0)$$

Worked Solutions

Complex Numbers

Q117 MCQ

For $\sqrt{13}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $\sqrt{13}i$ using the relevant family rule.
2. The correct value is $(0, \sqrt{13})$.

Correct option: C**Final answer** $(0, \sqrt{13})$

Q118 MCQ

For $(2 + i)x + (1 - i)y = 3 + 3i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(2, -1)$.

Correct option: D**Final answer** $(2, -1)$

Worked Solutions

Complex Numbers

Q119 MCQ

For $(1 + 2i)x + (3 - i)y = 5 - 4i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(-1, 2)$.

Correct option: A

Final answer $(-1, 2)$

Q120 MCQ

For $(3 - 2i)x + (1 + i)y = 6 + i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(1, 3)$.

Correct option: B

Final answer $(1, 3)$

Worked Solutions

Complex Numbers

Q121 MCQ

For $(2 - i)x + (4 + 2i)y = 4i$, select the ordered pair (x, y) :

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(-2, 1)$.

Correct option: B

Final answer

$(-2, 1)$

Q122 MCQ

For $(1 + i)x + (2 + 3i)y = 1$, select the ordered pair (x, y) :

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(3, -1)$.

Correct option: C

Final answer

$(3, -1)$

Worked Solutions

Complex Numbers

Q123 MCQ

For $(4 + i)x + (1 - 2i)y = 2 + 5i$, select the ordered pair (x, y) :

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(1, -2)$.

Correct option: D

Final answer $(1, -2)$

Q124 MCQ

For $(2 + 3i)x + (3 + i)y = -8 - 5i$, select the ordered pair (x, y) :

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(-1, -2)$.

Correct option: D

Final answer $(-1, -2)$

Worked Solutions

Complex Numbers

Q125 MCQ

For $(1 - i)x + (2 - i)y = 6 - 4i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives (2, 2).

Correct option: C

Final answer

(2, 2)

Q126 MCQ

For $(3 + 4i) - (1 - 2i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $2 + 6i$.

Correct option: C

Final answer

$2 + 6i$

Worked Solutions

Complex Numbers

Q127 MCQ

For $(-2 + i) + (5 - 3i)$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $3 - 2i$.

Correct option: C**Final answer**

$$3 - 2i$$

Q128 MCQ

For $(1 + 2i)(3 - i)$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $5 + 5i$.

Correct option: D**Final answer**

$$5 + 5i$$

Worked Solutions

Complex Numbers

Q129 MCQ

For $(\sqrt{3} + i)(\sqrt{3} - i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is 4.

Correct option: A

Final answer

4

Q130 MCQ

For $(2 - i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(2 - i)^2$.

Correct option: C

Final answer $(2 - i)^2$

Worked Solutions

Complex Numbers

Q131 MCQ

For $(1 + 3i)^2$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-8 + 6i$.

Correct option: A**Final answer** $-8 + 6i$

Q132 MCQ

For i^{37} , select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is i .

Correct option: B**Final answer** i

Worked Solutions

Complex Numbers

Q133 MCQ

For $i^{22} + i^9$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-1 + i$.

Correct option: B**Final answer**

$$-1 + i$$

Q134 MCQ

For $i^{15} - i^6$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $1 - i$.

Correct option: B**Final answer**

$$1 - i$$

Worked Solutions

Complex Numbers

Q135 MCQ

For $(4 - 2i) + (1 + 5i)$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $5 + 3i$.

Correct option: C**Final answer** $5 + 3i$

Q136 MCQ

For $(2 + i)(2 - i)$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is 5.

Correct option: A**Final answer**

5

Worked Solutions

Complex Numbers

Q137 MCQ

For $(\sqrt{2} + 2i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(\sqrt{2} + 2i)^2$.

Correct option: C**Final answer**

$$(\sqrt{2} + 2i)^2$$

Q138 MCQ

For $i^8 + i^{11} + i^{14}$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-i$.

Correct option: C**Final answer**

$$-i$$

Worked Solutions

Complex Numbers

Q139 MCQ

For $(3 - i)^3$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(3 - i)^3$.

Correct option: D**Final answer**

$$(3 - i)^3$$

Q140 MCQ

For $(1 - 2i)(-2 + i)$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $5i$.

Correct option: D**Final answer**

$$5i$$

Worked Solutions

Complex Numbers

Q141 MCQ

For $i^5 + i^{10}$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-1 + i$.

Correct option: D**Final answer**

$$-1 + i$$

Q142 MCQ

For $(5 + 2i) - (3 - 4i)$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $2 + 6i$.

Correct option: D**Final answer**

$$2 + 6i$$

Worked Solutions

Complex Numbers

Q143 MCQ

For $(2 - 3i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(2 - 3i)^2$.

Correct option: C

Final answer

$$(2 - 3i)^2$$

Q144 MCQ

For the expression State the complex conjugate of $3 - 2i$;, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $3 + 2i$.

Correct option: C

Final answer

$$3 + 2i$$

Worked Solutions

Complex Numbers

Q145 MCQ

For the expression State the complex conjugate of $-4 + \sqrt{3}i$; the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-4 - \sqrt{3}i$.

Correct option: A

Final answer

$$-4 - \sqrt{3}i$$

Q146 MCQ

For the expression State the complex conjugate of $7i$; the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-7i$.

Correct option: C

Final answer

$$-7i$$

Worked Solutions

Complex Numbers

Q147 MCQ

For the expression State the complex conjugate of $-\sqrt{5}$, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\sqrt{5}$.

Correct option: C**Final answer**

$$-\sqrt{5}$$

Q148 MCQ

For the expression State the complex conjugate of $1/2 - 3i$, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{2} + 3i$.

Correct option: B**Final answer**

$$\frac{1}{2} + 3i$$

Worked Solutions

Complex Numbers

Q149 MCQ

For $(2 + i)/(1 - i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{2} + \frac{3i}{2}$.

Correct option: A**Final answer**

$$\frac{1}{2} + \frac{3i}{2}$$

Q150 MCQ

For $(4 - 3i)/(2 + i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $1 - 2i$.

Correct option: A**Final answer**

$$1 - 2i$$

Worked Solutions

Complex Numbers

Q151 MCQ

For $(1 + 2i)/(3 - i)$, select the value in the form $a+bi$:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{10} + \frac{7i}{10}$.

Correct option: B

Final answer

$$\frac{1}{10} + \frac{7i}{10}$$

Q152 MCQ

For $5i/(2 - i)$, select the value in the form $a+bi$:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $i(2 + i)$.

Correct option: C

Final answer

$$i(2 + i)$$

Worked Solutions

Complex Numbers

Q153 MCQ

For $(3 - i)/(1 + 3i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-i$.

Correct option: D**Final answer** $-i$

Q154 MCQ

For $1/i$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-i$.

Correct option: D**Final answer** $-i$

Worked Solutions

Complex Numbers

Q155 MCQ

For $(2 + 3i)/(2 - 3i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{5}{13} + \frac{12i}{13}$.

Correct option: B**Final answer**

$$-\frac{5}{13} + \frac{12i}{13}$$

Q156 MCQ

For $(\sqrt{2} + i)/(\sqrt{2} - i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{\sqrt{2} + i}{\sqrt{2} - i}$.

Correct option: B**Final answer**

$$\frac{\sqrt{2} + i}{\sqrt{2} - i}$$

Worked Solutions

Complex Numbers

Q157 MCQ

For $(1 - 4i)/(3 + 2i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{5}{13} - \frac{14i}{13}$.

Correct option: D**Final answer**

$$-\frac{5}{13} - \frac{14i}{13}$$

Q158 MCQ

For $7i/(1 - i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{7}{2} + \frac{7i}{2}$.

Correct option: C**Final answer**

$$-\frac{7}{2} + \frac{7i}{2}$$

Worked Solutions

Complex Numbers

Q159 MCQ

For the expression State the complex conjugate of $-2 + 5i$., the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-2 - 5i$.

Correct option: B

Final answer

$$-2 - 5i$$

Q160 MCQ

For the expression State the complex conjugate of $6 - \sqrt{7}i$., the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $6 + \sqrt{7}i$.

Correct option: D

Final answer

$$6 + \sqrt{7}i$$

Worked Solutions

Complex Numbers

Q161 MCQ

For the expression State the complex conjugate of $-i\sqrt{2}$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\sqrt{2}i$.

Correct option: B**Final answer**

$$\sqrt{2}i$$

Q162 MCQ

For the expression State the complex conjugate of 9:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is 9.

Correct option: C**Final answer**

$$9$$

Worked Solutions

Complex Numbers

Q163 MCQ

For the expression State the complex conjugate of $-3/4 + 2i$., the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{3}{4} - 2i$.

Correct option: A**Final answer**

$$-\frac{3}{4} - 2i$$

Q164 MCQ

For $(5 + 2i)/(1 - 2i)$, select the value in the form $a+bi$:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{5} + \frac{12i}{5}$.

Correct option: D**Final answer**

$$\frac{1}{5} + \frac{12i}{5}$$

Worked Solutions

Complex Numbers

Q165 MCQ

For $(2 - i)/(4 + i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{7}{17} - \frac{6i}{17}$.

Correct option: A**Final answer**

$$\frac{7}{17} - \frac{6i}{17}$$

Q166 MCQ

For $3i/(2 + 3i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{9}{13} + \frac{6i}{13}$.

Correct option: A**Final answer**

$$\frac{9}{13} + \frac{6i}{13}$$

Worked Solutions

Complex Numbers

Q167 MCQ

For $(\sqrt{3} - i)/(\sqrt{3} + i)$, select the value in the form $a+bi$:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{\sqrt{3} - i}{\sqrt{3} + i}$.

Correct option: C

Final answer

$$\frac{\sqrt{3} - i}{\sqrt{3} + i}$$

Q168 MCQ

For $(1 + i)/(2i)$, select the value in the form $a+bi$:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{2} - \frac{i}{2}$.

Correct option: A

Final answer

$$\frac{1}{2} - \frac{i}{2}$$

Worked Solutions

Complex Numbers

Q169 MCQ

For the expression State the complex conjugate of $4 - 7i$., the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $4 + 7i$.

Correct option: D

Final answer

$$4 + 7i$$

Q170 MCQ

For the expression State the complex conjugate of $\sqrt{10} + 2i$., the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\sqrt{10} - 2i$.

Correct option: A

Final answer

$$\sqrt{10} - 2i$$

Worked Solutions

Complex Numbers

Q171 MCQ

For the expression State the complex conjugate of $-8i$, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $8i$.

Correct option: B

Final answer

$8i$

Q172 MCQ

For $(3 + 2i)/(3 - 2i)$, select the value in the form $a+bi$:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{5}{13} + \frac{12i}{13}$.

Correct option: B

Final answer

$\frac{5}{13} + \frac{12i}{13}$

Worked Solutions

Complex Numbers

Q173 MCQ

For $\sqrt{-5}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $i\sqrt{5}$.

Correct option: B**Final answer**

$$i\sqrt{5}$$

Q174 MCQ

For $\sqrt{-20}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $2i\sqrt{5}$.

Correct option: A**Final answer**

$$2i\sqrt{5}$$

Worked Solutions

Complex Numbers

Q175 MCQ

For $-\sqrt{-18}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $-3i\sqrt{2}$.

Correct option: A**Final answer**

$$-3i\sqrt{2}$$

Q176 MCQ

For $\sqrt[4]{-8}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt[4]{-a} = i\sqrt[4]{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $\pm 2i\sqrt{2}$.

Correct option: C**Final answer**

$$\pm 2i\sqrt{2}$$

Worked Solutions

Complex Numbers

Q177 MCQ

For $\sqrt{-45}/\sqrt{-5}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is 3.

Correct option: A

Final answer

3

Q178 MCQ

For $\sqrt{-2}\sqrt{-8}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is -4 .

Correct option: B

Final answer -4

Worked Solutions

Complex Numbers

Q179 MCQ

For $\sqrt{12}/\sqrt{-3}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{a} = \sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $-2i$.

Correct option: C

Final answer $-2i$

Q180 MCQ

For $(2 + \sqrt{-3})^2$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{a} = \sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $1 + 4i\sqrt{3}$.

Correct option: D

Final answer $1 + 4i\sqrt{3}$

Worked Solutions

Complex Numbers

Q181 MCQ

For $\sqrt{-72}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $6i\sqrt{2}$.

Correct option: D

Final answer

$$6i\sqrt{2}$$

Q182 MCQ

For $\sqrt{-50}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $\pm 5i\sqrt{2}$.

Correct option: D

Final answer

$$\pm 5i\sqrt{2}$$

Worked Solutions

Complex Numbers

Q183 MCQ

For $\sqrt{-3}\sqrt{-12}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is -6 .

Correct option: A

Final answer -6

Q184 MCQ

For $\sqrt{8}/\sqrt{-2}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $-2i$.

Correct option: D

Final answer $-2i$

Worked Solutions

Complex Numbers

Q185 MCQ

For $\sqrt{-32} + \sqrt{-8}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $6i\sqrt{2}$.

Correct option: B

Final answer

$$6i\sqrt{2}$$

Q186 MCQ

For $\sqrt{-98} - \sqrt{-18}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $4i\sqrt{2}$.

Correct option: B

Final answer

$$4i\sqrt{2}$$

Worked Solutions

Complex Numbers

Q187 MCQ

For $(\sqrt{5} + \sqrt{-2})^2$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{a} + \sqrt{b})^2 = a + b + 2\sqrt{ab}$, then simplify the positive radical and the sign.
2. Therefore the correct value is $3 + 2i\sqrt{10}$.

Correct option: A**Final answer**

$$3 + 2i\sqrt{10}$$

Q188 MCQ

For $(3 + \sqrt{-5})(3 - \sqrt{-5})$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$, then simplify the positive radical and the sign.
2. Therefore the correct value is 14.

Correct option: D**Final answer**

$$14$$

Worked Solutions

Complex Numbers

Q189 MCQ

For $\sqrt{-7}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $i\sqrt{7}$.

Correct option: C**Final answer**

$$i\sqrt{7}$$

Q190 MCQ

For $\sqrt{-28}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $2i\sqrt{7}$.

Correct option: A**Final answer**

$$2i\sqrt{7}$$

Worked Solutions

Complex Numbers

Q191 MCQ

For $\sqrt{-12}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $\pm 2i\sqrt{3}$.

Correct option: A

Final answer

$$\pm 2i\sqrt{3}$$

Q192 MCQ

For $\sqrt{27}/\sqrt{-3}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $-3i$.

Correct option: A

Final answer

$$-3i$$

Worked Solutions

Complex Numbers

Q193 MCQ

For $\sqrt{-2}\sqrt{-18}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is -6 .

Correct option: D

Final answer -6

Q194 MCQ

For $\sqrt{-48}/\sqrt{-3}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-4i$.

Correct option: C

Final answer $-4i$

Worked Solutions

Complex Numbers

Q195 MCQ

For $\sqrt{-8} + \sqrt{-18}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $5i\sqrt{2}$.

Correct option: C**Final answer**

$$5i\sqrt{2}$$

Q196 MCQ

For $(1 + \sqrt{-2})^2$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-1 + 2i\sqrt{2}$.

Correct option: A**Final answer**

$$-1 + 2i\sqrt{2}$$

Worked Solutions

Complex Numbers

Q197 MCQ

For $(4 + \sqrt{-3})(4 - \sqrt{-3})$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is 19.

Correct option: B

Final answer

19

Q198 MCQ

For $\sqrt{-75}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $5i\sqrt{3}$.

Correct option: D

Final answer $5i\sqrt{3}$

Worked Solutions

Complex Numbers

Q199 MCQ

Solve for the real values x and y : $(1 + 3i)(x + i) = x(y + i)$:**Solution**

1. Expand and equate real and imaginary parts.
2. The system gives $(x = -1/2)$ and $(y = 7)$.

Correct option: B**Final answer**

$$\left(-\frac{1}{2}, 7\right)$$

Q200 MCQ

Given x is real, solve: $(4x - 4i)(x^2 + xi) = 10x^2i^{32}$:**Solution**

1. Use $(i^{32} = 1)$ and $((x - i)(x + i) = x^2 + 1)$.
2. The real equation is $(4x(x^2 + 1) = 10x^2)$, giving $x = 0, 1/2$, or 2 .

Correct option: D**Final answer**

$$\left\{0, 2, \frac{1}{2}\right\}$$

Worked Solutions

Complex Numbers

Q201 MCQ

Simplify exactly: $(i(8 - 6i)/((1 + 2i)(3 - i)))^{2027}$:**Solution**

1. Rationalise the base using the conjugate of the denominator.
2. The exact base simplifies to $((7+i)/5)$. Keep the 2027th power unexpanded.

Correct option: B**Final answer**

$$((7 + i)/5)^{2027}$$

Q202 MCQ

Solve over the complex numbers: $(3x - 6)^2 = -8$:**Solution**

1. Take square roots: $(3x - 6 = \pm 2i\sqrt{2})$.
2. Divide by 3 to obtain $(x = 2 \pm \frac{2}{3}i\sqrt{2})$.

Correct option: B**Final answer**

$$x = 2 \pm (2/3)i\sqrt{2}$$

Worked Solutions

Complex Numbers

Quiz MCQ 1 [Concept Quiz · MCQ](#)

For $4-3i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $4 - 3i$ using the relevant family rule.
2. The correct value is $(4, -3)$.

Correct option: A

Final answer

$(4, -3)$

Quiz MCQ 2 [Concept Quiz · MCQ](#)

For $(1+2i)(2-i)$, select the simplified value:

Solution

1. Simplify $(1 + 2i)(2 - i)$ using the relevant family rule.
2. The correct value is $4 + 3i$.

Correct option: A

Final answer

$4 + 3i$

Worked Solutions

Complex Numbers

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

For $(3+i)/(1-i)$, select the value in the form $a+bi$:

Solution

1. Simplify $(3+i)/(1-i)$ using the relevant family rule.
2. The correct value is $1+2i$.

Correct option: D

Final answer

$$1 + 2i$$

Quiz MCQ 4 [Concept Quiz](#) · [MCQ](#)

For $\sqrt{-27}$, select the expression in terms of i :

Solution

1. Simplify $\sqrt{-27}$ using the relevant family rule.
2. The correct value is $3i\sqrt{3}$.

Correct option: A

Final answer

$$3i\sqrt{3}$$

Worked Solutions

Complex Numbers

Quiz WRITTEN 5 Concept Quiz · Written

State the real and imaginary parts of $\frac{5-\sqrt{5}i}{2}$.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: $(5/2, -\sqrt{5}/2)$

Final answer

$(5/2, -\sqrt{5}/2)$

Quiz WRITTEN 6 Concept Quiz · Written

Solve for real x, y : $(2 + i)x + (1 - i)y = 3 + i$.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: $x = 4/3, y = 1/3$

Final answer

$x = 4/3, y = 1/3$

Worked Solutions

Complex Numbers

Quiz WRITTEN 7 Concept Quiz · Written

Calculate $i^{18} + i^7 - i^3$ in standard form.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: -1

Final answer

 -1

Quiz WRITTEN 8 Concept Quiz · Written

Solve over the complex numbers: $(x - 3)^2 = -12$.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: $x = 3 \pm 2i\sqrt{3}$

Final answer

 $x = 3 \pm 2i\sqrt{3}$

Answer key

Use the question number shown on each page.

Q001 $(3, -2)$	Q027 $3 - 4i$	Q053 2	Q079 $3i\sqrt{3}$
Q002 $(4/3, -\sqrt{5}/3)$	Q028 -1	Q054 $8i$	Q080 $-2i\sqrt{3}$
Q003 $(-8, 0)$	Q029 $1 + 8i$	Q055 $1 - i$	Q081 $\pm 2i\sqrt{3}$
Q004 $(0, 2)$	Q030 $8 - i$	Q056 $-1 - i$	Q082 $-3\sqrt{2}$
Q005 $x = -1, y = 4$	Q031 0	Q057 $3 - 4i$	Q083 $-3i$
Q006 $(-3, 5)$	Q032 $4 - i$	Q058 $9 - 8i$	Q084 $7 + 6\sqrt{2}i$
Q007 $(-1/2, -\sqrt{3}/2)$	Q033 $1 - 3i$	Q059 $8 - i\sqrt{3}$	Q085 $i\sqrt{2}$
Q008 $(1, 0)$	Q034 -2	Q060 5	Q086 $3i$
Q009 $(0, 3)$	Q035 $-1 + 4\sqrt{2}i$	Q061 -15	Q087 $3i\sqrt{5}$
Q010 $x = 6, y = -2$	Q036 $7 + 11i$	Q062 $\sqrt{7}$	Q088 $-2i\sqrt{6}$
Q011 $x = -2, y = 3$	Q037 $1/15 - 13/14i$	Q063 $-i$	Q089 $-6i$
Q012 $(2, 5)$	Q038 $-3 - 4i$	Q064 $50i$	Q090 $\pm i\sqrt{2}$
Q013 $(-25, 13)$	Q039 -3	Q065 $-i\sqrt{6}$	Q091 $\pm 3i\sqrt{2}$
Q014 $(2/3, -\sqrt{5}/3)$	Q040 4	Q066 i	Q092 $\pm 4i$
Q015 $(7/2, -\sqrt{2}/2)$	Q041 i	Q067 $7/5 - 1/5i$	Q093 $-2\sqrt{3}$
Q016 $(-4, 0)$	Q042 -1	Q068 $2/3 + \sqrt{5}/3i$	Q094 $-14\sqrt{5}$
Q017 $(1/4, 0)$	Q043 $-i$	Q069 $2 + i$	Q095 $-2i$
Q018 $(0, \sqrt{3})$	Q044 $x = 0$ or $x = 1$	Q070 $3/5 + 4/5i$	Q096 $-(3/2)i$
Q019 $(0, -1)$	Q045 $2 - 3i$	Q071 $9/13 - 7/13i$	Q097 $13i\sqrt{2}$
Q020 $x = 3, y = 2$	Q046 $7 + i\sqrt{11}$	Q072 $-i$	Q098 $6i\sqrt{2}$
Q021 $x = 3, y = -3$	Q047 6	Q073 $1/3 - 2/3i$	Q099 $-2 + 2i\sqrt{15}$
Q022 $x = 2, y = -2$	Q048 $5i$	Q074 $((-2 - 144i)/85)^{2027}$	Q100 7
Q023 $x = 2, y = 1$	Q049 $3/5 + 4/5i$	Q075 $\pm i\sqrt{6}$	Q101 $x = 4 \pm (\sqrt{10}/2)i$
Q024 $x = 1/2, y = 3/2$	Q050 $-1/2 - 3/2i$	Q076 $-6\sqrt{2}$	Q102 C
Q025 $x = -1, y = 3$	Q051 $1 + 4i$	Q077 $-(2/3)i$	Q103 D
Q026 $1 - 7i$	Q052 $\sqrt{2} - i\sqrt{10}$	Q078 $i\sqrt{7}$	Q104 D

Q105 B	Q132 B	Q159 B	Q186 B
Q106 D	Q133 B	Q160 D	Q187 A
Q107 A	Q134 B	Q161 B	Q188 D
Q108 B	Q135 C	Q162 C	Q189 C
Q109 C	Q136 A	Q163 A	Q190 A
Q110 B	Q137 C	Q164 D	Q191 A
Q111 D	Q138 C	Q165 A	Q192 A
Q112 D	Q139 D	Q166 A	Q193 D
Q113 C	Q140 D	Q167 C	Q194 C
Q114 D	Q141 D	Q168 A	Q195 C
Q115 B	Q142 D	Q169 D	Q196 A
Q116 B	Q143 C	Q170 A	Q197 B
Q117 C	Q144 C	Q171 B	Q198 D
Q118 D	Q145 A	Q172 B	Q199 B
Q119 A	Q146 C	Q173 B	Q200 D
Q120 B	Q147 C	Q174 A	Q201 B
Q121 B	Q148 B	Q175 A	Q202 B
Q122 C	Q149 A	Q176 C	Quiz MCQ 1 A
Q123 D	Q150 A	Q177 A	Quiz MCQ 2 A
Q124 D	Q151 B	Q178 B	Quiz MCQ 3 D
Q125 C	Q152 C	Q179 C	Quiz MCQ 4 A
Q126 C	Q153 D	Q180 D	Quiz WRITTEN 5 $(5/2, -\sqrt{5}/2)$
Q127 C	Q154 D	Q181 D	Quiz WRITTEN 6 $x = 4/3, y = 1/3$
Q128 D	Q155 B	Q182 D	Quiz WRITTEN 7 -1
Q129 A	Q156 B	Q183 A	Quiz WRITTEN 8 $x = 3 \pm 2i\sqrt{3}$
Q130 C	Q157 D	Q184 D	
Q131 A	Q158 C	Q185 B	

Concept 2 | Quadratic Equations

English Student Edition • Focused formulas and guided practice

Formula opener

Roots

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum/product

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$$

Discriminant

$$\Delta = b^2 - 4ac$$

Worked Solutions

Quadratic Equations

Q001 Written

Solve $x^2 + 8 = 0$.**Solution**

1. $x^2 = -8$
2. $x = \pm\sqrt{-8} = \pm i\sqrt{8} = \pm 2i\sqrt{2}$
3. Therefore the final answer is $x = \pm 2i\sqrt{2}$.

Final answer

$$x = \pm 2i\sqrt{2}$$

Q002 Written

Solve $x^2 + 4 = 0$.**Solution**

1. $x^2 = -4$
2. $x = \pm 2i$
3. Therefore the final answer is $x = \pm 2i$.

Final answer

$$x = \pm 2i$$

Worked Solutions

Quadratic Equations

Q003 Written

Solve $x^2 + 49 = 0$.

Solution

1. $x^2 = -49$
2. Therefore the final answer is $x = \pm 7i$.

Final answer

$$x = \pm 7i$$

Q004 Written

Solve $9x^2 - 4 = 0$.

Solution

1. $(3x - 2)(3x + 2) = 0$
2. Therefore the final answer is $x = \pm \frac{2}{3}$.

Final answer

$$x = \pm \frac{2}{3}$$

Worked Solutions

Quadratic Equations

Q005 Written

Solve $x^2 - 12 = 0$.**Solution**

1. $x = \pm\sqrt{12}$
2. Therefore the final answer is $x = \pm 2\sqrt{3}$.

Final answer

$$x = \pm 2\sqrt{3}$$

Q006 Written

Solve $4x^2 + 3 = 0$.**Solution**

1. $x^2 = -3/4$
2. Therefore the final answer is $x = \pm \frac{i\sqrt{3}}{2}$.

Final answer

$$x = \pm \frac{i\sqrt{3}}{2}$$

Worked Solutions

Quadratic Equations

Q007 MCQ

Solve $x^2 + 8 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm 2i\sqrt{2}$.

Correct option: A**Final answer**

$$x = \pm 2i\sqrt{2}$$

Q008 MCQ

Solve $x^2 + 4 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm 2i$.

Correct option: A**Final answer**

$$x = \pm 2i$$

Worked Solutions

Quadratic Equations

Q009 MCQ

Solve $x^2 + 49 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm 7i$.

Correct option: C**Final answer**

$$x = \pm 7i$$

Q010 MCQ

Solve $9x^2 - 4 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm \frac{2}{3}$.

Correct option: C**Final answer**

$$x = \pm \frac{2}{3}$$

Worked Solutions

Quadratic Equations

Q011 MCQ

Solve $x^2 - 12 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm 2\sqrt{3}$.

Correct option: B**Final answer**

$$x = \pm 2\sqrt{3}$$

Q012 MCQ

Solve $4x^2 + 3 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm \frac{i\sqrt{3}}{2}$.

Correct option: B**Final answer**

$$x = \pm \frac{i\sqrt{3}}{2}$$

Worked Solutions

Quadratic Equations

Q013 Written

Solve $(x - 3)^2 = -8$.**Solution**

1. $x - 3 = \pm i\sqrt{8} = \pm 2i\sqrt{2}$
2. Therefore the final answer is $x = 3 \pm 2i\sqrt{2}$.

Final answer

$$x = 3 \pm 2i\sqrt{2}$$

Q014 Written

Solve $(x - 2)^2 = 5$.**Solution**

1. $x - 2 = \pm\sqrt{5}$
2. Therefore the final answer is $x = 2 \pm \sqrt{5}$.

Final answer

$$x = 2 \pm \sqrt{5}$$

Worked Solutions

Quadratic Equations

Q015 Written

Solve $(x + 6)^2 = -6$.**Solution**

1. $x + 6 = \pm i\sqrt{6}$
2. Therefore the final answer is $x = -6 \pm i\sqrt{6}$.

Final answer

$$x = -6 \pm i\sqrt{6}$$

Q016 Written

Solve $(x + 4)^2 = -18$.**Solution**

1. $x + 4 = \pm i\sqrt{18} = \pm 3i\sqrt{2}$
2. Therefore the final answer is $x = -4 \pm 3i\sqrt{2}$.

Final answer

$$x = -4 \pm 3i\sqrt{2}$$

Worked Solutions

Quadratic Equations

Q017 **Written****Solve** $(2x - 3)^2 = 7$.**Solution**

1. $2x - 3 = \pm\sqrt{7}$

2. Therefore the final answer is $x = \frac{3 \pm \sqrt{7}}{2}$.

Final answer

$$x = \frac{3 \pm \sqrt{7}}{2}$$

Q018 **MCQ****Solve** $(x - 3)^2 = -8$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = 3 \pm 2i\sqrt{2}$.**Correct option:** A**Final answer**

$$x = 3 \pm 2i\sqrt{2}$$

Worked Solutions

Quadratic Equations

Q019 MCQ

Solve $(x - 2)^2 = 5$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = 2 \pm \sqrt{5}$.

Correct option: B**Final answer**

$$x = 2 \pm \sqrt{5}$$

Q020 MCQ

Solve $(x + 6)^2 = -6$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = -6 \pm i\sqrt{6}$.

Correct option: C**Final answer**

$$x = -6 \pm i\sqrt{6}$$

Worked Solutions

Quadratic Equations

Q021 MCQ

Solve $(x + 4)^2 = -18$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = -4 \pm 3i\sqrt{2}$.

Correct option: C**Final answer**

$$x = -4 \pm 3i\sqrt{2}$$

Q022 MCQ

Solve $(2x - 3)^2 = 7$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \frac{3 \pm \sqrt{7}}{2}$.

Correct option: D**Final answer**

$$x = \frac{3 \pm \sqrt{7}}{2}$$

Worked Solutions

Quadratic Equations

Q023 Written

Solve $2x^2 - 7x - 9 = 0$.**Solution**

1. $(x + 1)(2x - 9) = 0$
2. $x = -1$ or $x = 9/2$
3. Therefore the final answer is $x = -1, \frac{9}{2}$.

Final answer

$$x = -1, \frac{9}{2}$$

Q024 Written

Solve $2x^2 + 5x - 3 = 0$.**Solution**

1. $(2x - 1)(x + 3) = 0$
2. Therefore the final answer is $x = \frac{1}{2}, -3$.

Final answer

$$x = \frac{1}{2}, -3$$

Worked Solutions

Quadratic Equations

Q025 Written

Solve $2x^2 + 7x = 0$.**Solution**

1. $x(2x + 7) = 0$
2. Therefore the final answer is $x = 0, -\frac{7}{2}$.

Final answer

$$x = 0, -\frac{7}{2}$$

Q026 Written

Solve $2x^2 + 5x - 12 = 0$.**Solution**

1. $(2x - 3)(x + 4) = 0$
2. Therefore the final answer is $x = \frac{3}{2}, -4$.

Final answer

$$x = \frac{3}{2}, -4$$

Worked Solutions

Quadratic Equations

Q027 **Written****Solve** $x^2 - 8x + 15 = 0$.**Solution**

1. $(x - 3)(x - 5) = 0$
2. Therefore the final answer is $x = 3, 5$.

Final answer $x = 3, 5$ Q028 **MCQ****Solve** $2x^2 - 7x - 9 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = -1, \frac{9}{2}$.

Correct option: A**Final answer** $x = -1, \frac{9}{2}$

Worked Solutions

Quadratic Equations

Q029 MCQ

Solve $2x^2 + 5x - 3 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \frac{1}{2}, -3$.

Correct option: D**Final answer**

$$x = \frac{1}{2}, -3$$

Q030 MCQ

Solve $2x^2 + 7x = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = 0, -\frac{7}{2}$.

Correct option: D**Final answer**

$$x = 0, -\frac{7}{2}$$

Worked Solutions

Quadratic Equations

Q031 MCQ

Solve $2x^2 + 5x - 12 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \frac{3}{2}, -4$.

Correct option: A**Final answer**

$$x = \frac{3}{2}, -4$$

Q032 MCQ

Solve $x^2 - 8x + 15 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = 3, 5$.

Correct option: A**Final answer**

$$x = 3, 5$$

Worked Solutions

Quadratic Equations

Q033 Written

Solve $3x^2 - 4x - 1 = 0$.**Solution**

1. $D = 16 + 12 = 28$
2. $x = (4 \pm 2\sqrt{7})/6 = (2 \pm \sqrt{7})/3$
3. Therefore the final answer is $x = \frac{2 \pm \sqrt{7}}{3}$.

Final answer

$$x = \frac{2 \pm \sqrt{7}}{3}$$

Q034 MCQ

Solve $3x^2 - 4x - 1 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \frac{2 \pm \sqrt{7}}{3}$.

Correct option: B**Final answer**

$$x = \frac{2 \pm \sqrt{7}}{3}$$

Worked Solutions

Quadratic Equations

Q035 Written

Solve $2x^2 + 2\sqrt{6}x + 11 = 0$.**Solution**

1. $D = (2\sqrt{6})^2 - 4(2)(11) = -64$

2. $x = (-2\sqrt{6} \pm \sqrt{(-64)})/4 = (-\sqrt{6} \pm 4i)/2$

3. Therefore the final answer is $x = \frac{-\sqrt{6} \pm 4i}{2}$.

Final answer

$$x = \frac{-\sqrt{6} \pm 4i}{2}$$

Q036 Written

Solve $2x^2 + 2\sqrt{3}x + 5 = 0$.**Solution**

1. $D = 12 - 40 = -28$

2. $x = (-2\sqrt{3} \pm i\sqrt{28})/4 = (-\sqrt{3} \pm i\sqrt{7})/2$

3. Therefore the final answer is $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$.

Final answer

$$x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$$

Worked Solutions

Quadratic Equations

Q037 Written

Solve $x^2 + x + 1 = 0$.**Solution**

1. $D = 1 - 4 = -3$
2. $x = (-1 \pm i\sqrt{3})/2$
3. Therefore the final answer is $x = \frac{-1 \pm i\sqrt{3}}{2}$.

Final answer

$$x = \frac{-1 \pm i\sqrt{3}}{2}$$

Q038 Written

Solve $3x^2 - 5x + 3 = 0$.**Solution**

1. $D = 25 - 36 = -11$
2. $x = (5 \pm i\sqrt{11})/6$
3. Therefore the final answer is $x = \frac{5 \pm i\sqrt{11}}{6}$.

Final answer

$$x = \frac{5 \pm i\sqrt{11}}{6}$$

Worked Solutions

Quadratic Equations

Q039 Written

Solve $x^2 - 2\sqrt{3}x + 5 = 0$.**Solution**

1. $D = 12 - 20 = -8$
2. $x = (2\sqrt{3} \pm i\sqrt{8})/2 = \sqrt{3} \pm i\sqrt{2}$
3. Therefore the final answer is $x = \sqrt{3} \pm i\sqrt{2}$.

Final answer

$$x = \sqrt{3} \pm i\sqrt{2}$$

Q040 Written

Solve $3x^2 + 2\sqrt{2}x + 1 = 0$.**Solution**

1. $D = 8 - 12 = -4$
2. $x = (-2\sqrt{2} \pm 2i)/6 = (-\sqrt{2} \pm i)/3$
3. Therefore the final answer is $x = \frac{-\sqrt{2} \pm i}{3}$.

Final answer

$$x = \frac{-\sqrt{2} \pm i}{3}$$

Worked Solutions

Quadratic Equations

Q041 Written

Solve $x^2 - x + 2 = 0$.**Solution**

1. $D = 1 - 8 = -7$
2. $x = (1 \pm i\sqrt{7})/2$
3. Therefore the final answer is $x = \frac{1 \pm i\sqrt{7}}{2}$.

Final answer

$$x = \frac{1 \pm i\sqrt{7}}{2}$$

Q042 Written

Solve $3x^2 - \sqrt{5}x + 1 = 0$.**Solution**

1. $D = 5 - 12 = -7$
2. $x = (\sqrt{5} \pm i\sqrt{7})/6$
3. Therefore the final answer is $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$.

Final answer

$$x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$$

Worked Solutions

Quadratic Equations

Q043 **Written****Solve** $6x^2 - 12x + 15 = 0$.**Solution**

1. $D = 144 - 360 = -216$
2. $x = (12 \pm i6\sqrt{6})/12 = 1 \pm i\sqrt{6}/2$
3. Therefore the final answer is $x = 1 \pm \frac{i\sqrt{6}}{2}$.

Final answer

$$x = 1 \pm \frac{i\sqrt{6}}{2}$$

Q044 **MCQ****Solve** $2x^2 + 2\sqrt{6}x + 11 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \frac{-\sqrt{6} \pm 4i}{2}$.

Correct option: B**Final answer**

$$x = \frac{-\sqrt{6} \pm 4i}{2}$$

Worked Solutions

Quadratic Equations

Q045 MCQ

Solve $2x^2 + 2\sqrt{3}x + 5 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$.**Correct option:** C**Final answer**

$$x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$$

Q046 MCQ

Solve $x^2 + x + 1 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \frac{-1 \pm i\sqrt{3}}{2}$.**Correct option:** D**Final answer**

$$x = \frac{-1 \pm i\sqrt{3}}{2}$$

Worked Solutions

Quadratic Equations

Q047 MCQ

Solve $3x^2 - 5x + 3 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \frac{5 \pm i\sqrt{11}}{6}$.**Correct option:** B**Final answer**

$$x = \frac{5 \pm i\sqrt{11}}{6}$$

Q048 MCQ

Solve $x^2 - 2\sqrt{3}x + 5 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \sqrt{3} \pm i\sqrt{2}$.**Correct option:** D**Final answer**

$$x = \sqrt{3} \pm i\sqrt{2}$$

Worked Solutions

Quadratic Equations

Q049 MCQ

Solve $3x^2 + 2\sqrt{2}x + 1 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \frac{-\sqrt{2} \pm i}{3}$.**Correct option:** B**Final answer**

$$x = \frac{-\sqrt{2} \pm i}{3}$$

Q050 MCQ

Solve $x^2 - x + 2 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \frac{1 \pm i\sqrt{7}}{2}$.**Correct option:** C**Final answer**

$$x = \frac{1 \pm i\sqrt{7}}{2}$$

Worked Solutions

Quadratic Equations

Q051 MCQ

Solve $3x^2 - \sqrt{5}x + 1 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$.**Correct option:** C**Final answer**

$$x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$$

Q052 MCQ

Solve $6x^2 - 12x + 15 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = 1 \pm \frac{i\sqrt{6}}{2}$.**Correct option:** B**Final answer**

$$x = 1 \pm \frac{i\sqrt{6}}{2}$$

Worked Solutions

Quadratic Equations

Q053 Written

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$.**Solution**

1. $D = 4^2 - 4(3)(7) = -68 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Q054 Written

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions.**Solution**

1. $D = (-m)^2 - 4(m^2 - 3m - 9) = -3m^2 + 12m + 36$
2. $D \geq 0 \rightarrow m^2 - 4m - 12 \leq 0 \rightarrow (m + 2)(m - 6) \leq 0$
3. Therefore the final answer is $-2 \leq m \leq 6$.

Final answer $-2 \leq m \leq 6$

Worked Solutions

Quadratic Equations

Q055 Written

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies.

Solution

1. $D = k^2 - 4(3 - k) = (k - 2)(k + 6)$
2. Use $D > 0, D = 0, D < 0$ intervals
3. Therefore the final answer is $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary.

Final answer

$k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

Q056 Written

For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions.

Solution

1. $D = 25 - 48 = -23 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q057 **Written****For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions.****Solution**

1. $D = 49 + 32 = 81 > 0$
2. Therefore the final answer is Two distinct real solutions.

Final answer

Two distinct real solutions

Q058 **Written****For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions.****Solution**

1. $D = 100 - 100 = 0$
2. Therefore the final answer is One repeated real solution.

Final answer

One repeated real solution

Worked Solutions

Quadratic Equations

Q059 Written

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$.**Solution**

1. $D = 25 - 12 = 13 > 0$
2. Therefore the final answer is Two distinct real solutions.

Final answer

Two distinct real solutions

Q060 Written

Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$.**Solution**

1. $D = 144 - 144 = 0$
2. Therefore the final answer is One repeated real solution.

Final answer

One repeated real solution

Worked Solutions

Quadratic Equations

Q061 Written

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$.**Solution**

1. $D = 64 - 80 = -16 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Q062 Written

Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$.**Solution**

1. $D = 16 - 16 = 0$
2. Therefore the final answer is One repeated real solution.

Final answer

One repeated real solution

Worked Solutions

Quadratic Equations

Q063 Written

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$.

Solution

1. $D = 9 - 20 = -11 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Q064 Written

Determine the nature of the solutions of $x^2 - x - 2 = 0$.

Solution

1. $D = 1 + 8 = 9 > 0$
2. Therefore the final answer is Two distinct real solutions.

Final answer

Two distinct real solutions

Worked Solutions

Quadratic Equations

Q065 MCQ

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: A**Final answer**

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q066 MCQ

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-2 \leq m \leq 6$.

Correct option: D

Final answer

$$-2 \leq m \leq 6$$

Worked Solutions

Quadratic Equations

Q067 MCQ

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary.

Correct option: C

Final answer

$k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

Worked Solutions

Quadratic Equations

Q068 MCQ

For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: C

Final answer

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q069 MCQ

For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct real solutions.

Correct option: A

Final answer

Two distinct real solutions

Worked Solutions

Quadratic Equations

Q070 MCQ

For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is One repeated real solution.

Correct option: A

Final answer

One repeated real solution

Worked Solutions

Quadratic Equations

Q071 MCQ

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct real solutions.

Correct option: C**Final answer**

Two distinct real solutions

Worked Solutions

Quadratic Equations

Q072 MCQ

Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is One repeated real solution.

Correct option: C**Final answer**

One repeated real solution

Worked Solutions

Quadratic Equations

Q073 MCQ

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: B**Final answer**

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q074 MCQ

Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is One repeated real solution.

Correct option: A**Final answer**

One repeated real solution

Worked Solutions

Quadratic Equations

Q075 MCQ

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: A**Final answer**

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q076 MCQ

Determine the nature of the solutions of $x^2 - x - 2 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct real solutions.

Correct option: A

Final answer

Two distinct real solutions

Worked Solutions

Quadratic Equations

Q077 Written

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions.

Solution

1. $D = 4m^2 - 4(m + 2)$
2. $D < 0 \rightarrow m^2 - m - 2 < 0 \rightarrow (m - 2)(m + 1) < 0$
3. Therefore the final answer is $-1 < m < 2$.

Final answer

$$-1 < m < 2$$

Q078 Written

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies.

Solution

1. $D = a^2 - 4(2a - 3) = (a - 2)(a - 6)$
2. Classify by sign of D
3. Therefore the final answer is $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary.

Final answer $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Worked Solutions

Quadratic Equations

Q079 Written

Find the values of a for which $x^2 - (a + 1)x + 4 = 0$ has a repeated solution.

Solution

1. $D = (a + 1)^2 - 16 = 0$
2. $a + 1 = \pm 4$
3. Therefore the final answer is $a = 3$ or $a = -5$.

Final answer $a = 3$ or $a = -5$

Q080 Written

Find the range of m for which $x^2 + (m + 6)x - 2m = 0$ has two distinct imaginary solutions.

Solution

1. $D = (m + 6)^2 + 8m = m^2 + 20m + 36 = (m + 2)(m + 18)$
2. $D < 0$
3. Therefore the final answer is $-18 < m < -2$.

Final answer $-18 < m < -2$

Worked Solutions

Quadratic Equations

Q081 Written

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions.

Solution

1. $D = 4m^2 - 4(2m^2 - 3m) = 4m(3 - m)$
2. $D \geq 0$
3. Therefore the final answer is $0 \leq m \leq 3$.

Final answer

$$0 \leq m \leq 3$$

Q082 MCQ

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-1 < m < 2$.

Correct option: C

Final answer

$$-1 < m < 2$$

Worked Solutions

Quadratic Equations

Q083 MCQ

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary.

Correct option: C

Final answer

$a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Worked Solutions

Quadratic Equations

Q084 MCQ

Find the values of a for which $x^2 - (a + 1)x + 4 = 0$ has a repeated solution Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $a = 3$ or $a = -5$.

Correct option: A**Final answer** $a = 3$ or $a = -5$

Q085 MCQ

Find the range of m for which $x^2 + (m + 6)x - 2m = 0$ has two distinct imaginary solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-18 < m < -2$.

Correct option: D**Final answer** $-18 < m < -2$

Worked Solutions

Quadratic Equations

Q086 MCQ

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $0 \leq m \leq 3$.

Correct option: C

Final answer

$$0 \leq m \leq 3$$

Q087 Written

Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$.

Solution

1. Use $S = -b/a = 6/3 = 2$ and $P = c/a = -5/3$
2. Therefore the final answer is $\alpha + \beta = 2$, $\alpha\beta = -\frac{5}{3}$.

Final answer

$$\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$$

Worked Solutions

Quadratic Equations

Q088 Written

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$.**Solution**

1. $S = -2/4 = -1/2; P = -7/4$
2. Therefore the final answer is $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$.

Final answer

$$\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$$

Q089 Written

Find the sum and product of the roots of $x^2 + x + 2 = 0$.**Solution**

1. $S = -1; P = 2$
2. Therefore the final answer is $S = -1, \quad P = 2$.

Final answer

$$S = -1, \quad P = 2$$

Worked Solutions

Quadratic Equations

Q090 Written

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$.**Solution**

1. $S = 7/3; P = -2$
2. Therefore the final answer is $S = \frac{7}{3}, P = -2$.

Final answer

$$S = \frac{7}{3}, P = -2$$

Q091 MCQ

Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $\alpha + \beta = 2, \alpha\beta = -\frac{5}{3}$.

Correct option: C**Final answer**

$$\alpha + \beta = 2, \alpha\beta = -\frac{5}{3}$$

Worked Solutions

Quadratic Equations

Q092 MCQ

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $\alpha + \beta = -\frac{1}{2}$, $\alpha\beta = -\frac{7}{4}$.

Correct option: A**Final answer**

$$\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$$

Q093 MCQ

Find the sum and product of the roots of $x^2 + x + 2 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $S = -1$, $P = 2$.

Correct option: B**Final answer**

$$S = -1, \quad P = 2$$

Worked Solutions

Quadratic Equations

Q094 MCQ

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $S = \frac{7}{3}$, $P = -2$.

Correct option: A**Final answer**

$$S = \frac{7}{3}, \quad P = -2$$

Q095 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$.**Solution**

1. $S = 3, P = -3/2$
2. $S^2 - 2P = 9 + 3 = 12$
3. Therefore the final answer is 12.

Final answer

12

Worked Solutions

Quadratic Equations

Q096 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 3, P = -3/2$
2. $S^3 - 3PS = 27 + 27/2 = 81/2$
3. Therefore the final answer is $\frac{81}{2}$.

Final answer

$$\frac{81}{2}$$

Q097 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = 2, P = 3/2$
2. $S^2 - 2P = 4 - 3 = 1$
3. Therefore the final answer is 1.

Final answer

$$1$$

Worked Solutions

Quadratic Equations

Q098 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 2, P = 3/2$
2. $S^3 - 3PS = 8 - 9 = -1$
3. Therefore the final answer is -1 .

Final answer -1

Q099 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = 2, P = 3$
2. $S^2 - 2P = 4 - 6 = -2$
3. Therefore the final answer is -2 .

Final answer -2

Worked Solutions

Quadratic Equations

Q100 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 2, P = 3$
2. $S^3 - 3PS = 8 - 18 = -10$
3. Therefore the final answer is -10 .

Final answer

-10

Q101 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = -2, P = -1/3$
2. $S^2 - 2P = 4 + 2/3 = 14/3$
3. Therefore the final answer is $\frac{14}{3}$.

Final answer

$\frac{14}{3}$

Worked Solutions

Quadratic Equations

Q102 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = -2, P = -1/3$
2. $S^3 - 3PS = -8 - 2 = -10$
3. Therefore the final answer is -10 .

Final answer

-10

Q103 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 12.

Correct option: A

Final answer

12

Worked Solutions

Quadratic Equations

Q104 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{81}{2}$.

Correct option: B**Final answer**

$$\frac{81}{2}$$

Q105 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 1.

Correct option: A**Final answer**

$$1$$

Worked Solutions

Quadratic Equations

Q106 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is -1 .

Correct option: B**Final answer** -1

Q107 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is -2 .

Correct option: D**Final answer** -2

Worked Solutions

Quadratic Equations

Q108 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is -10 .

Correct option: B**Final answer** -10

Q109 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{14}{3}$.

Correct option: B**Final answer** $\frac{14}{3}$

Worked Solutions

Quadratic Equations

Q110 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is -10 .

Correct option: D

Final answer -10

Q111 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$.

Solution

1. $2(\alpha + \beta)/(\alpha\beta) = 2(3)/(-3/2) = -4$
2. Therefore the final answer is -4 .

Final answer -4

Worked Solutions

Quadratic Equations

Q112 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$.

Solution

1. $(\alpha^2 + \beta^2)/(\alpha\beta) = 1/(3/2) = 2/3$
2. Therefore the final answer is $\frac{2}{3}$.

Final answer

$$\frac{2}{3}$$

Q113 Written

Find the sum and product of the roots of $5x^2 - 3 = 0$.

Solution

1. $S = 0; P = -3/5$
2. Therefore the final answer is $S = 0, P = -\frac{3}{5}$.

Final answer

$$S = 0, P = -\frac{3}{5}$$

Worked Solutions

Quadratic Equations

Q114 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Solution

1. $S/P = 2/3$
2. Therefore the final answer is $\frac{2}{3}$.

Final answer

$$\frac{2}{3}$$

Q115 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Solution

1. $S/P = (-2)/(-1/3) = 6$
2. Therefore the final answer is 6.

Final answer

$$6$$

Worked Solutions

Quadratic Equations

Q116 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is -4 .

Correct option: C

Final answer

-4

Q117 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{2}{3}$.

Correct option: D

Final answer

$\frac{2}{3}$

Worked Solutions

Quadratic Equations

Q118 MCQ

Find the sum and product of the roots of $5x^2 - 3 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $S = 0$, $P = -\frac{3}{5}$.

Correct option: D**Final answer**

$$S = 0, \quad P = -\frac{3}{5}$$

Q119 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{2}{3}$.

Correct option: D**Final answer**

$$\frac{2}{3}$$

Worked Solutions

Quadratic Equations

Q120 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 6.

Correct option: D**Final answer**

6

Q121 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$.

Solution

1. $P + S + 1 = -3/2 + 3 + 1 = 5/2$
2. Therefore the final answer is $\frac{5}{2}$.

Final answer $\frac{5}{2}$

Worked Solutions

Quadratic Equations

Q122 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$.

Solution

1. $P - S + 1 = 3/2 - 2 + 1 = 1/2$
2. Therefore the final answer is $\frac{1}{2}$.

Final answer

$$\frac{1}{2}$$

Q123 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$.

Solution

1. $P + 2S + 4 = 3 + 4 + 4 = 11$
2. Therefore the final answer is 11.

Final answer

$$11$$

Worked Solutions

Quadratic Equations

Q124 **Written**

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$.

Solution

1. $P - 2S + 4 = -1/3 + 4 + 4 = 23/3$
2. Therefore the final answer is $\frac{23}{3}$.

Final answer

$$\frac{23}{3}$$

Q125 **MCQ**

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{5}{2}$.

Correct option: A**Final answer**

$$\frac{5}{2}$$

Worked Solutions

Quadratic Equations

Q126 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{1}{2}$.

Correct option: C

Final answer

$$\frac{1}{2}$$

Q127 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 11.

Correct option: D

Final answer

$$11$$

Worked Solutions

Quadratic Equations

Q128 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{23}{3}$.

Correct option: D

Final answer

$$\frac{23}{3}$$

Q129 Written

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots.

Solution

1. Let roots $\alpha, 3\alpha$. $4\alpha = 8 \Rightarrow \alpha = 2$.
2. $k = \alpha(3\alpha) = 6$
3. Therefore the final answer is $k = 6, \quad x = 2, 6$.

Final answer

$$k = 6, \quad x = 2, 6$$

Worked Solutions

Quadratic Equations

Q130 Written

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots.

Solution

1. Roots $\alpha, 2\alpha$; $3\alpha = -6 \Rightarrow \alpha = -2$; $m = 2\alpha^2 = 8$
2. Therefore the final answer is $m = 8$, $x = -2, -4$.

Final answer

$$m = 8, \quad x = -2, -4$$

Q131 Written

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots.

Solution

1. Roots $\alpha, 2\alpha$; $2\alpha^2 = 1/8 \Rightarrow \alpha = \pm 1/4$.
2. $m = 8(3\alpha) = \pm 6$.
3. Therefore the final answer is $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$.

Final answer

$$(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2}) \text{ or } (-6, -\frac{1}{4}, -\frac{1}{2})$$

Worked Solutions

Quadratic Equations

Q132 MCQ

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $k = 6$, $x = 2, 6$.

Correct option: A**Final answer**

$$k = 6, \quad x = 2, 6$$

Q133 MCQ

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $m = 8$, $x = -2, -4$.

Correct option: A**Final answer**

$$m = 8, \quad x = -2, -4$$

Worked Solutions

Quadratic Equations

Q134 MCQ

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$.

Correct option: D

Final answer

$$(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2}) \text{ or } (-6, -\frac{1}{4}, -\frac{1}{2})$$

Worked Solutions

Quadratic Equations

Q135 Written

If the difference between the two roots of $x^2 - (m - 1)x - 6 = 0$ is 5, find m and the roots.

Solution

1. Let roots be α and $\alpha + 5$.
2. $\alpha(\alpha + 5) = -6 \Rightarrow \alpha = -2$ or -3 .
3. $S = 2\alpha + 5 = m - 1$ gives $m = 2$ or 0 ; obtain the paired roots.
4. Therefore the final answer is $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$.

Final answer

$$(m, x_1, x_2) = (2, -2, 3) \text{ or } (0, -3, 2)$$

Q136 Written

If the difference between the roots of $x^2 - (m + 1)x + 2 = 0$ is 1, find m and the roots.

Solution

1. Let roots be α and $\alpha + 1$. $\alpha(\alpha + 1) = 2 \Rightarrow \alpha = 1$ or -2 .
2. $m + 1 = 2\alpha + 1 \Rightarrow m = 2$ or -4 .
3. Therefore the final answer is $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$.

Final answer

$$(m, x_1, x_2) = (2, 1, 2) \text{ or } (-4, -2, -1)$$

Worked Solutions

Quadratic Equations

Q137 Written

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots.

Solution

1. Roots $\alpha, \alpha + 4$; $2\alpha + 4 = -6 \rightarrow \alpha = -5$; $m = \alpha(\alpha + 4) = 5$
2. Therefore the final answer is $m = 5$, $x = -5, -1$.

Final answer

$$m = 5, \quad x = -5, -1$$

Q138 Written

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots.

Solution

1. Roots $\alpha, \alpha + 1$; $\alpha(\alpha + 1) = 3/4 \rightarrow \alpha = 1/2$ or $-3/2$.
2. $-m/4 = 2\alpha + 1 \rightarrow m = -8$ or 8 .
3. Therefore the final answer is $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$.

Final answer

$$(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2}) \text{ or } (8, -\frac{3}{2}, -\frac{1}{2})$$

Worked Solutions

Quadratic Equations

Q139 MCQ

If the difference between the two roots of $x^2 - (m - 1)x - 6 = 0$ is 5, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$.

Correct option: D

Final answer

$$(m, x_1, x_2) = (2, -2, 3) \text{ or } (0, -3, 2)$$

Worked Solutions

Quadratic Equations

Q140 MCQ

If the difference between the roots of $x^2 - (m + 1)x + 2 = 0$ is 1, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$.

Correct option: A

Final answer

$(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$

Worked Solutions

Quadratic Equations

Q141 MCQ

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $m = 5$, $x = -5, -1$.

Correct option: B

Final answer

$$m = 5, \quad x = -5, -1$$

Worked Solutions

Quadratic Equations

Q142 MCQ

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$.

Correct option: B

Final answer

$$(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2}) \text{ or } (8, -\frac{3}{2}, -\frac{1}{2})$$

Worked Solutions

Quadratic Equations

Q143 Written

If the ratio of the roots of $x^2 - 2x + m = 0$ is $2 : 3$, find m and the roots.

Solution

- Let roots $2t, 3t$; $5t = 2 \rightarrow t = 2/5$; $m = 6t^2 = 24/25$
- Therefore the final answer is $m = \frac{24}{25}$, $x = \frac{4}{5}, \frac{6}{5}$.

Final answer

$$m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$$

Q144 MCQ

If the ratio of the roots of $x^2 - 2x + m = 0$ is $2 : 3$, find m and the roots Select the correct answer:

Solution

- Apply the same quadratic method as the given demand.
- The correct value is $m = \frac{24}{25}$, $x = \frac{4}{5}, \frac{6}{5}$.

Correct option: A

Final answer

$$m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$$

Worked Solutions

Quadratic Equations

Q145 Written

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots.

Solution

1. Roots α, α^2 ; $\alpha + \alpha^2 = 6 \Rightarrow \alpha = 2$ or -3 .
2. $k = \alpha^3 = 8$ or -27 .
3. Therefore the final answer is $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$.

Final answer

$$(k, x_1, x_2) = (8, 2, 4) \text{ or } (-27, -3, 9)$$

Worked Solutions

Quadratic Equations

Q146 MCQ

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$.

Correct option: B

Final answer

$$(k, x_1, x_2) = (8, 2, 4) \text{ or } (-27, -3, 9)$$

Worked Solutions

Quadratic Equations

Q147 Written

Factorise $2x^2 - 3x + 2$ over the complex numbers.

Solution

1. $D = 9 - 16 = -7$; roots $(3 \pm i\sqrt{7})/4$
2. Keep leading coefficient 2
3. Therefore the final answer is $2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$.

Final answer

$$2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$$

Q148 Written

Factorise $x^2 + 2x + 5$ over the complex numbers.

Solution

1. Roots $-1 \pm 2i$
2. Therefore the final answer is $(x + 1 - 2i)(x + 1 + 2i)$.

Final answer

$$(x + 1 - 2i)(x + 1 + 2i)$$

Worked Solutions

Quadratic Equations

Q149 **Written****Factorise $2x^2 - 6x + 1$ over the complex numbers.****Solution**

1. $D = 28$; roots $(3 \pm \sqrt{7})/2$
2. Keep leading coefficient 2
3. Therefore the final answer is $2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$.

Final answer

$$2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$$

Q150 **Written****Factorise $x^2 + 4x + 5$ over the complex numbers.****Solution**

1. $D = 16 - 20 = -4$; roots $-2 \pm i$
2. Therefore the final answer is $(x + 2 - i)(x + 2 + i)$.

Final answer

$$(x + 2 - i)(x + 2 + i)$$

Worked Solutions

Quadratic Equations

Q151 Written

Factorise $x^2 + 4x + 2$ over the complex numbers.

Solution

1. *Roots* $-2 \pm \sqrt{2}$
2. Therefore the final answer is $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$.

Final answer

$$(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$$

Q152 Written

Factorise $x^2 - 4x + 7$ over the complex numbers.

Solution

1. $D = 16 - 28 = -12$; *roots* $2 \pm i\sqrt{3}$
2. Therefore the final answer is $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$.

Final answer

$$(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$$

Worked Solutions

Quadratic Equations

Q153 **Written****Factorise $2x^2 - x + 1$ over the complex numbers.****Solution**

1. $D = 1 - 8 = -7$; roots $(1 \pm i\sqrt{7})/4$

2. Therefore the final answer is $2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$.

Final answer

$$2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$$

Q154 **Written****Factorise $3x^2 - 5x + 3$ over the complex numbers.****Solution**

1. $D = 25 - 36 = -11$; roots $(5 \pm i\sqrt{11})/6$

2. Therefore the final answer is $3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$.

Final answer

$$3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$$

Worked Solutions

Quadratic Equations

Q155 **Written****Factorise $2x^2 - 2x - 1$ over the complex numbers.****Solution**

1. $D = 4 + 8 = 12$; roots $(1 \pm \sqrt{3})/2$

2. Therefore the final answer is $2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$.

Final answer

$$2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$$

Worked Solutions

Quadratic Equations

Q156 MCQ

Factorise $2x^2 - 3x + 2$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$.

Correct option: C**Final answer**

$$2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$$

Worked Solutions

Quadratic Equations

Q157 MCQ

Factorise $x^2 + 2x + 5$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x + 1 - 2i)(x + 1 + 2i)$.

Correct option: A**Final answer**

$$(x + 1 - 2i)(x + 1 + 2i)$$

Worked Solutions

Quadratic Equations

Q158 MCQ

Factorise $2x^2 - 6x + 1$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$.

Correct option: D**Final answer**

$$2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$$

Worked Solutions

Quadratic Equations

Q159 MCQ

Factorise $x^2 + 4x + 5$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x + 2 - i)(x + 2 + i)$.

Correct option: D**Final answer**

$$(x + 2 - i)(x + 2 + i)$$

Q160 MCQ

Factorise $x^2 + 4x + 2$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$.

Correct option: B**Final answer**

$$(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$$

Worked Solutions

Quadratic Equations

Q161 MCQ

Factorise $x^2 - 4x + 7$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$.

Correct option: A**Final answer**

$$(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$$

Worked Solutions

Quadratic Equations

Q162 MCQ

Factorise $2x^2 - x + 1$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$.**Correct option: C****Final answer**

$$2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$$

Worked Solutions

Quadratic Equations

Q163 MCQ

Factorise $3x^2 - 5x + 3$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$.

Correct option: A**Final answer**

$$3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$$

Worked Solutions

Quadratic Equations

Q164 MCQ

Factorise $2x^2 - 2x - 1$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$.

Correct option: B**Final answer**

$$2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$$

Worked Solutions

Quadratic Equations

Q165 **Written****Factorise $x^4 + 2x^2 - 3$ over the complex numbers.****Solution**

1. $x^4 + 2x^2 - 3 = (x^2 - 1)(x^2 + 3)$
2. Factor each quadratic
3. Therefore the final answer is $(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$.

Final answer

$$(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$$

Q166 **Written****Factorise $x^4 - x^2 - 6$ over the complex numbers.****Solution**

1. Let $y = x^2$: $y^2 - y - 6 = (y - 3)(y + 2)$
2. Factor $x^2 - 3$ and $x^2 + 2$
3. Therefore the final answer is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$.

Final answer

$$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$$

Worked Solutions

Quadratic Equations

Q167 Written

Factorise $x^4 - 9$ over the complex numbers.

Solution

1. $x^4 - 9 = (x^2 - 3)(x^2 + 3)$
2. Factor each quadratic
3. Therefore the final answer is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$.

Final answer

$$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$$

Q168 Written

Factorise $x^4 - 3x^2 + 2$ over the complex numbers.

Solution

1. Let $y = x^2$: $y^2 - 3y + 2 = (y - 1)(y - 2)$
2. Factor $x^2 - 1$ and $x^2 - 2$
3. Therefore the final answer is $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$.

Final answer

$$(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$$

Worked Solutions

Quadratic Equations

Q169 MCQ

Factorise $x^4 + 2x^2 - 3$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$.

Correct option: B**Final answer**

$$(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$$

Q170 MCQ

Factorise $x^4 - x^2 - 6$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$.

Correct option: B**Final answer**

$$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$$

Worked Solutions

Quadratic Equations

Q171 MCQ

Factorise $x^4 - 9$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$.

Correct option: D**Final answer**

$$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$$

Q172 MCQ

Factorise $x^4 - 3x^2 + 2$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$.

Correct option: C**Final answer**

$$(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$$

Worked Solutions

Quadratic Equations

Q173 Written

Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots.

Solution

1. $S = 1, P = 1/2$
2. $x^2 - x + 1/2 = 0$; multiply by 2
3. Therefore the final answer is $2x^2 - 2x + 1 = 0$.

Final answer

$$2x^2 - 2x + 1 = 0$$

Q174 Written

Find two numbers whose sum is -1 and product is 2 .

Solution

1. They are roots of $x^2 + x + 2 = 0$
2. $D = -7$
3. Therefore the final answer is $\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$.

Final answer

$$\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$$

Worked Solutions

Quadratic Equations

Q175 Written

Create a quadratic equation with roots $3 + 2i$ and $3 - 2i$.

Solution

1. $S = 6, P = 13$
2. $x^2 - Sx + P = 0$
3. Therefore the final answer is $x^2 - 6x + 13 = 0$.

Final answer

$$x^2 - 6x + 13 = 0$$

Q176 Written

Find two numbers whose sum and product are both 1.

Solution

1. Roots of $x^2 - x + 1 = 0$
2. $D = -3$
3. Therefore the final answer is $\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$.

Final answer

$$\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$$

Worked Solutions

Quadratic Equations

Q177 Written

Create a quadratic equation with roots -1 and 2 .**Solution**

1. $S = 1, P = -2$
2. Therefore the final answer is $x^2 - x - 2 = 0$.

Final answer

$$x^2 - x - 2 = 0$$

Q178 Written

Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$.**Solution**

1. $S = 2, P = 17$
2. Therefore the final answer is $x^2 - 2x + 17 = 0$.

Final answer

$$x^2 - 2x + 17 = 0$$

Worked Solutions

Quadratic Equations

Q179 Written

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$.

Solution

1. $S = 10/3, P = 2$
2. $x^2 - (10/3)x + 2 = 0$; multiply by 3
3. Therefore the final answer is $3x^2 - 10x + 6 = 0$.

Final answer

$$3x^2 - 10x + 6 = 0$$

Q180 Written

Find two numbers whose sum is 2 and product is 5.

Solution

1. Roots of $x^2 - 2x + 5 = 0$
2. $D = -16$
3. Therefore the final answer is $1 + 2i, 1 - 2i$.

Final answer

$$1 + 2i, 1 - 2i$$

Worked Solutions

Quadratic Equations

Q181 **Written****Find two numbers whose sum is 4 and product is 2.****Solution**

1. Roots of $x^2 - 4x + 2 = 0$
2. Therefore the final answer is $2 + \sqrt{2}$, $2 - \sqrt{2}$.

Final answer

$$2 + \sqrt{2}, 2 - \sqrt{2}$$

Q182 **MCQ****Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $2x^2 - 2x + 1 = 0$.

Correct option: B**Final answer**

$$2x^2 - 2x + 1 = 0$$

Worked Solutions

Quadratic Equations

Q183 MCQ

Find two numbers whose sum is -1 and product is 2 Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$.**Correct option: B****Final answer**

$$\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$$

Q184 MCQ

Create a quadratic equation with roots $3 + 2i$ and $3 - 2i$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x^2 - 6x + 13 = 0$.**Correct option: B****Final answer**

$$x^2 - 6x + 13 = 0$$

Worked Solutions

Quadratic Equations

Q185 MCQ

Find two numbers whose sum and product are both 1 Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$.**Correct option: A****Final answer**

$$\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$$

Q186 MCQ

Create a quadratic equation with roots -1 and 2 Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x^2 - x - 2 = 0$.**Correct option: A****Final answer**

$$x^2 - x - 2 = 0$$

Worked Solutions

Quadratic Equations

Q187 MCQ

Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x^2 - 2x + 17 = 0$.

Correct option: D

Final answer

$$x^2 - 2x + 17 = 0$$

Q188 MCQ

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $3x^2 - 10x + 6 = 0$.

Correct option: A

Final answer

$$3x^2 - 10x + 6 = 0$$

Worked Solutions

Quadratic Equations

Q189 MCQ

Find two numbers whose sum is 2 and product is 5 Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $1 + 2i$, $1 - 2i$.

Correct option: B**Final answer** $1 + 2i, 1 - 2i$

Q190 MCQ

Find two numbers whose sum is 4 and product is 2 Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $2 + \sqrt{2}$, $2 - \sqrt{2}$.

Correct option: B**Final answer** $2 + \sqrt{2}, 2 - \sqrt{2}$

Worked Solutions

Quadratic Equations

Q191 Written

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$.

Solution

1. $S = -2, P = -4$
2. $\text{Newsum} = S + 4 = 2; \text{newproduct} = P + 2S + 4 = -4$
3. $\text{Equation } x^2 - 2x - 4 = 0$
4. Therefore the final answer is $x^2 - 2x - 4 = 0$.

Final answer

$$x^2 - 2x - 4 = 0$$

Q192 MCQ

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x^2 - 2x - 4 = 0$.

Correct option: C

Final answer

$$x^2 - 2x - 4 = 0$$

Worked Solutions

Quadratic Equations

Q193 Written

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2 .

Solution

1. $S = -1, P = -3$
2. $\text{Newsum} S^2 - 2P = 1 + 6 = 7$; $\text{product} P^2 = 9$
3. Therefore the final answer is $x^2 - 7x + 9 = 0$.

Final answer

$$x^2 - 7x + 9 = 0$$

Q194 Written

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2 .

Solution

1. $S = 2, P = 7$
2. $\text{Newsum} = S^2 - 2P = -10$; $\text{product} = P^2 = 49$
3. Therefore the final answer is $x^2 + 10x + 49 = 0$.

Final answer

$$x^2 + 10x + 49 = 0$$

Worked Solutions

Quadratic Equations

Q195 MCQ

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2 . Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x^2 - 7x + 9 = 0$.

Correct option: D**Final answer**

$$x^2 - 7x + 9 = 0$$

Q196 MCQ

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2 . Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x^2 + 10x + 49 = 0$.

Correct option: C**Final answer**

$$x^2 + 10x + 49 = 0$$

Worked Solutions

Quadratic Equations

Q197 Written

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$.

Solution

1. $S = -3/2, P = 2$
2. $\text{Newsum} = S/P = -3/4$; $\text{product} = 1/P = 1/2$
3. $x^2 + (3/4)x + 1/2 = 0$; multiply by 4
4. Therefore the final answer is $4x^2 + 3x + 2 = 0$.

Final answer

$$4x^2 + 3x + 2 = 0$$

Q198 MCQ

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $4x^2 + 3x + 2 = 0$.

Correct option: A

Final answer

$$4x^2 + 3x + 2 = 0$$

Worked Solutions

Quadratic Equations

Q199 Written

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$.

Solution

1. $S = -2, P = 3$
2. $\text{Newsum} = 4S = -8$
3. $\text{Newproduct} = 3(\alpha^2 + \beta^2) + 10P = 3(S^2 - 2P) + 10P = 24$
4. Therefore the final answer is $x^2 + 8x + 24 = 0$.

Final answer

$$x^2 + 8x + 24 = 0$$

Q200 MCQ

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x^2 + 8x + 24 = 0$.

Correct option: D

Final answer

$$x^2 + 8x + 24 = 0$$

Worked Solutions

Quadratic Equations

Q201 Written

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m .

Solution

1. $D > 0 \rightarrow (m - 2)(m + 1) > 0 \rightarrow m < -1 \text{ or } m > 2$
2. $S = -2m > 0 \rightarrow m < 0$
3. $P = m + 2 > 0 \rightarrow m > -2$
4. Intersection : $-2 < m < -1$
5. Therefore the final answer is $-2 < m < -1$.

Final answer

$$-2 < m < -1$$

Worked Solutions

Quadratic Equations

Q202 Written

If $x^2 + 2(m - 1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m .

Solution

1. $D = 4(4 - m)(m + 1) > 0 \rightarrow -1 < m < 4$
2. $S = 2(1 - m) > 0 \rightarrow m < 1$
3. $P = (2m + 1)(m - 3) > 0 \rightarrow m < -1/2 \text{ or } m > 3$
4. Intersection : $-1 < m < -1/2$
5. Therefore the final answer is $-1 < m < -\frac{1}{2}$.

Final answer

$$-1 < m < -\frac{1}{2}$$

Worked Solutions

Quadratic Equations

Q203 MCQ

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-2 < m < -1$.

Correct option: A**Final answer**

$$-2 < m < -1$$

Q204 MCQ

If $x^2 + 2(m - 1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-1 < m < -\frac{1}{2}$.

Correct option: B**Final answer**

$$-1 < m < -\frac{1}{2}$$

Worked Solutions

Quadratic Equations

Q205 **Written**

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m .

Solution

1. $D > 0 : 4m^2 - 4(m + 6) > 0 \rightarrow (m - 3)(m + 2) > 0$
2. $S = 2m < 0 \rightarrow m < 0$
3. $P = m + 6 > 0 \rightarrow m > -6$
4. *Intersection* : $-6 < m < -2$
5. Therefore the final answer is $-6 < m < -2$.

Final answer

$$-6 < m < -2$$

Worked Solutions

Quadratic Equations

Q206 **Written**

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k .

Solution

1. $D = (k - 10)(k + 2) > 0 \rightarrow k < -2 \text{ or } k > 10$
2. $S = -k < 0 \rightarrow k > 0$
3. $P = 2k + 5 > 0 \rightarrow k > -5/2$
4. *Intersection* $k > 10$
5. Therefore the final answer is $k > 10$.

Final answer $k > 10$

Worked Solutions

Quadratic Equations

Q207 MCQ

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-6 < m < -2$.

Correct option: C**Final answer**

$$-6 < m < -2$$

Q208 MCQ

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $k > 10$.

Correct option: B**Final answer**

$$k > 10$$

Worked Solutions

Quadratic Equations

Q209 Written

If $x^2 - (m - 3)x - m - 4 = 0$ has roots of different signs, find the range of m .

Solution

1. Opposite signs if $P = -m - 4 < 0$
2. Therefore the final answer is $m > -4$.

Final answer

$$m > -4$$

Q210 Written

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k .

Solution

1. $P = -k - 3 < 0$
2. Therefore the final answer is $k > -3$.

Final answer

$$k > -3$$

Worked Solutions

Quadratic Equations

Q211 MCQ

If $x^2 - (m - 3)x - m - 4 = 0$ has roots of different signs, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $m > -4$.

Correct option: A**Final answer**

$$m > -4$$

Q212 MCQ

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $k > -3$.

Correct option: D**Final answer**

$$k > -3$$

Worked Solutions

Quadratic Equations

Q213 Challenge

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies.

Solution

1. $D = a^2 + 4a = a(a + 4)$
2. Classify sign intervals at -4 and 0
3. Therefore the final answer is $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary.

Final answer

$a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary

Worked Solutions

Quadratic Equations

Q214 Challenge

For $x^2 - (k + 2)x + k^2 = 0$, determine the nature of the solutions as k varies.

Solution

1. $D = (k + 2)^2 - 4k^2 = -3k^2 + 4k + 4 = -3(k - 2)(k + 2/3)$

2. Classify D sign

3. Therefore the final answer is $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary.

Final answer

$-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary

Worked Solutions

Quadratic Equations

Q215 Challenge

For $x^2 - 2(a - 1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies.

Solution

1. $D = 4(a - 1)^2 - 4(a^2 - 3a + 4) = 4(a - 3)$
2. Classify by sign of $a - 3$
3. Therefore the final answer is $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary.

Final answer

$a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary

Worked Solutions

Quadratic Equations

Q216 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$.

Solution

1. $\alpha\beta(\alpha + \beta) = PS = (3/2)(3) = 9/2$
2. Therefore the final answer is $\frac{9}{2}$.

Final answer

$$\frac{9}{2}$$

Worked Solutions

Quadratic Equations

Q217 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$.

Solution

1. $S^2 - 4P = 9 - 6 = 3$
2. Therefore the final answer is 3.

Final answer

3

Worked Solutions

Quadratic Equations

Q218 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 3, P = 3/2$
2. $S^3 - 3PS = 27 - 27/2 = 27/2$
3. Therefore the final answer is $\frac{27}{2}$.

Final answer

$$\frac{27}{2}$$

Worked Solutions

Quadratic Equations

Q219 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$.

Solution

1. $(\alpha^3 + \beta^3)/(\alpha\beta) = (27/2)/(3/2) = 9$
2. Therefore the final answer is 9.

Final answer

9

Worked Solutions

Quadratic Equations

Q220 Challenge

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$.

Solution

1. *Subtract equations* : $(a - b)(x - c) = 0$, so *common root* $r = c$.
2. *Substitute* c : $c(c + a + b) = 0$; $c \neq 0 \rightarrow a + b + c = 0$.
3. *The other roots are* $bc/c = b$ and $ca/c = a$. *Their equation is* $x^2 - (a + b)x + ab = x^2 + cx + ab$.
4. Therefore the final answer is Common root c ; other roots a and b .

Final answer

Common root c ; other roots a and b .

Worked Solutions

Quadratic Equations

Q221 Challenge

For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model.

Solution

1. Let $y = x^2$: $6y^2 - 7y - 3 = (2y - 3)(3y + 1)$
2. $x^2 = 3/2$ or $-1/3$; hence $x = \pm\sqrt{6}/2$ or $\pm i\sqrt{3}/3$
3. For real displacement x , only $\pm\sqrt{6}/2$ are admissible
4. Therefore the final answer is $x = \pm\frac{\sqrt{6}}{2}$, $x = \pm\frac{i\sqrt{3}}{3}$; real: $\pm\sqrt{6}/2$.

Final answer

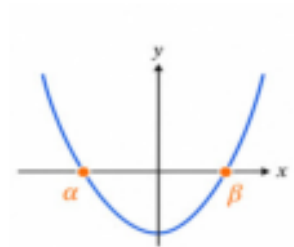
$$x = \pm\frac{\sqrt{6}}{2}, \quad x = \pm\frac{i\sqrt{3}}{3}; \text{ real: } \pm\sqrt{6}/2$$

Worked Solutions

Quadratic Equations

Q222 Challenge

In the opposite figure, α, β are the two roots. If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$.



Solution

1. $\text{Newsum} = S/P; \text{newproduct} = 1/P$
2. Equation $x^2 - (S/P)x + 1/P = 0$; multiply by P
3. Therefore the final answer is For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$.

Final answer

For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$

Worked Solutions

Quadratic Equations

Q223 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive.

Solution

1. $D = (m + 1)^2 - 4(m + 4) > 0 \rightarrow m < -3 \text{ or } m > 5$
2. $S = -(m + 1) > 0 \rightarrow m < -1$
3. $P = m + 4 > 0 \rightarrow m > -4$
4. Intersection $(-4, -3)$
5. Therefore the final answer is $-4 < m < -3$.

Final answer

$$-4 < m < -3$$

Worked Solutions

Quadratic Equations

Q224 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative.

Solution

1. $D > 0 \rightarrow m < -3$ or $m > 5$
2. $S < 0 \rightarrow m > -1$
3. $P > 0 \rightarrow m > -4$
4. Intersection $m > 5$
5. Therefore the final answer is $m > 5$.

Final answer

$$m > 5$$

Worked Solutions

Quadratic Equations

Q225 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two roots of opposite signs.

Solution

1. Opposite signs if $P = m + 4 < 0$
2. Therefore the final answer is $m < -4$.

Final answer

$$m < -4$$

Worked Solutions

Quadratic Equations

Q226 Challenge

A model has $R = 3x^2 + 2\sqrt{3}x + 2$. Determine the values of x for which $R = 0$, and interpret the result.

Solution

1. Set $R = 0$. $D = 12 - 24 = -12$.
2. $x = (-2\sqrt{3} \pm i2\sqrt{3})/6 = (-\sqrt{3} \pm i\sqrt{3})/3$.
3. Because x is real, the complex pair is not admissible; no real equilibrium exists.
4. Therefore the final answer is $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium.

Final answer

$$x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}; \text{ no real equilibrium}$$

Worked Solutions

Quadratic Equations

Q227 Challenge

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary.

Correct option: B

Final answer

$a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary

Worked Solutions

Quadratic Equations

Q228 Challenge

For $x^2 - (k + 2)x + k^2 = 0$, determine the nature of the solutions as k varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary.

Correct option: B

Final answer

$-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary

Worked Solutions

Quadratic Equations

Q229 Challenge

For $x^2 - 2(a - 1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary.

Correct option: B

Final answer

$a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary

Worked Solutions

Quadratic Equations

Q230 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{9}{2}$.

Correct option: B

Final answer

$$\frac{9}{2}$$

Worked Solutions

Quadratic Equations

Q231 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 3.

Correct option: D

Final answer

3

Worked Solutions

Quadratic Equations

Q232 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{27}{2}$.

Correct option: B

Final answer

$$\frac{27}{2}$$

Worked Solutions

Quadratic Equations

Q233 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 9.

Correct option: A

Final answer

9

Worked Solutions

Quadratic Equations

Q234 Challenge

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is Common root c ; other roots a and b .

Correct option: C

Final answer

Common root c ; other roots a and b .

Worked Solutions

Quadratic Equations

Q235 Challenge

For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm \frac{\sqrt{6}}{2}$, $x = \pm \frac{i\sqrt{3}}{3}$; real: $\pm \sqrt{6}/2$.

Correct option: A

Final answer

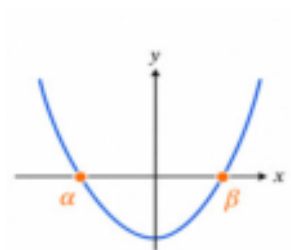
$$x = \pm \frac{\sqrt{6}}{2}, \quad x = \pm \frac{i\sqrt{3}}{3}; \text{ real: } \pm \sqrt{6}/2$$

Worked Solutions

Quadratic Equations

Q236 Challenge

If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$. Select the correct answer:



Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$.

Correct option: A

Final answer

For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$

Worked Solutions

Quadratic Equations

Q237 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-4 < m < -3$.

Correct option: C

Final answer

$$-4 < m < -3$$

Worked Solutions

Quadratic Equations

Q238 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $m > 5$.

Correct option: D

Final answer

$$m > 5$$

Worked Solutions

Quadratic Equations

Q239 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two roots of opposite signs. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $m < -4$.

Correct option: D

Final answer

$$m < -4$$

Worked Solutions

Quadratic Equations

Q240 Challenge

A model has $R = 3x^2 + 2\sqrt{3}x + 2$. Determine the values of x for which $R = 0$, and interpret the result. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium.

Correct option: D

Final answer

$$x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}; \text{ no real equilibrium}$$

Worked Solutions

Quadratic Equations

Quiz MCQ 1 Concept Quiz · MCQ

Solve $x^2 - 5x + 6 = 0$ **Select the correct answer:**

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 2, 3.

Correct option: B

Final answer

2, 3

Worked Solutions

Quadratic Equations

Quiz MCQ 2 Concept Quiz · MCQ

For $x^2 - 4x + 7 = 0$, determine the nature of the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct non-real complex roots.

Correct option: A

Final answer

Two distinct non-real complex roots

Worked Solutions

Quadratic Equations

Quiz MCQ 3 **Concept Quiz · MCQ**

If α, β are roots of $x^2 - 3x + 1 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 7.

Correct option: C

Final answer

7

Quiz MCQ 4 **Concept Quiz · MCQ**

Create a monic quadratic with roots $2 + i$ and $2 - i$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x^2 - 4x + 5 = 0$.

Correct option: A

Final answer

$x^2 - 4x + 5 = 0$

Worked Solutions

Quadratic Equations

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $2x^2 + 3x - 2 = 0$.**Solution**

1. Apply the relevant quadratic identity.
2. Final answer: $x = \frac{1}{2}, -2$

Final answer

$$x = \frac{1}{2}, -2$$

Quiz WRITTEN 6 Concept Quiz · Written**Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary roots.****Solution**

1. Apply the relevant quadratic identity.
2. Final answer: $-1 < m < 2$

Final answer

$$-1 < m < 2$$

Worked Solutions

Quadratic Equations

Quiz WRITTEN 7 Concept Quiz · Written

If α, β are roots of $x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. Apply the relevant quadratic identity.
2. Final answer: 28

Final answer

28

Quiz WRITTEN 8 Concept Quiz · Written

If α, β are roots of $x^2 - 2x + 5 = 0$, form the quadratic with roots α^2, β^2 .

Solution

1. Apply the relevant quadratic identity.
2. Final answer: $x^2 + 6x + 25 = 0$

Final answer

$x^2 + 6x + 25 = 0$

Answer key

Use the question number shown on each page.

Q001 $x = \pm 2i\sqrt{2}$

Q002 $x = \pm 2i$

Q003 $x = \pm 7i$

Q004 $x = \pm \frac{2}{3}$

Q005 $x = \pm 2\sqrt{3}$

Q006 $x = \pm \frac{i\sqrt{3}}{2}$

Q007 A

Q008 A

Q009 C

Q010 C

Q011 B

Q012 B

Q013 $x = 3 \pm 2i\sqrt{2}$

Q014 $x = 2 \pm \sqrt{5}$

Q015 $x = -6 \pm i\sqrt{6}$

Q016 $x = -4 \pm 3i\sqrt{2}$

Q017 $x = \frac{3 \pm \sqrt{7}}{2}$

Q018 A

Q019 B

Q020 C

Q021 C

Q022 D

Q023 $x = -1, \frac{9}{2}$

Q024 $x = \frac{1}{2}, -3$

Q025 $x = 0, -\frac{7}{2}$

Q026 $x = \frac{3}{2}, -4$

Q027 $x = 3, 5$

Q028 A

Q029 D

Q030 D

Q031 A

Q032 A

Q033 $x = \frac{2 \pm \sqrt{7}}{3}$

Q034 B

Q035 $x = \frac{-\sqrt{6} \pm 4i}{2}$

Q036 $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$

Q037 $x = \frac{-1 \pm i\sqrt{3}}{2}$

Q038 $x = \frac{5 \pm i\sqrt{11}}{6}$

Q039 $x = \sqrt{3} \pm i\sqrt{2}$

Q040 $x = \frac{-\sqrt{2} \pm i}{3}$

Q041 $x = \frac{1 \pm i\sqrt{7}}{2}$

Q042 $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$

Q043 $x = 1 \pm \frac{i\sqrt{6}}{2}$

Q044 B

Q045 C

Q046 D

Q047 B

Q048 D

Q049 B

Q050 C

Q051 C

Q052 B

Q053 Two distinct imaginary (non-real complex) solutions

Q054 $-2 \leq m \leq 6$

Q055 $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

Q056 Two distinct imaginary (non-real complex) solutions

Q057 Two distinct real solutions

Q058 One repeated real solution

Q059 Two distinct real solutions

Q060 One repeated real solution

Q061 Two distinct imaginary (non-real complex) solutions

Q062 One repeated real solution

Q063 Two distinct imaginary (non-real complex) solutions

Q064 Two distinct real solutions

Q065 A

Q066 D

Q067 C

Q068 C

Q069 A

Q070 A

Q071 C

Q072 C

Q073 B

Q074 A

Q075 A

Q076 A

Q077 $-1 < m < 2$

Q078 $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Q079 $a = 3$ or $a = -5$

Q080 $-18 < m < -2$

Q081 $0 \leq m \leq 3$

Q082 C

Q083 C

Q084 A

Q085 D

Q086 C

Q087 $\alpha + \beta = 2, \alpha\beta = -\frac{5}{3}$

Q088 $\alpha + \beta = -\frac{1}{2}, \alpha\beta = -\frac{7}{4}$

Q089 $S = -1, P = 2$

Q090 $S = \frac{7}{3}, P = -2$

Q091 C

Q092 A

Q093 B

Q094 A

Q095 12

Q096 $\frac{81}{2}$
 Q097 1
 Q098 -1
 Q099 -2
 Q100 -10
 Q101 $\frac{14}{3}$
 Q102 -10
 Q103 A
 Q104 B
 Q105 A
 Q106 B
 Q107 D
 Q108 B
 Q109 B
 Q110 D
 Q111 -4
 Q112 $\frac{2}{3}$
 Q113 $S = 0, \quad P = -\frac{3}{5}$
 Q114 $\frac{2}{3}$
 Q115 6
 Q116 C
 Q117 D
 Q118 D
 Q119 D
 Q120 D
 Q121 $\frac{5}{2}$
 Q122 $\frac{1}{2}$
 Q123 11
 Q124 $\frac{23}{3}$
 Q125 A

Q126 C
 Q127 D
 Q128 D
 Q129 $k = 6, \quad x = 2, 6$
 Q130 $m = 8, \quad x = -2, -4$
 Q131 $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$
 Q132 A
 Q133 A
 Q134 D
 Q135 $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$
 Q136 $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$
 Q137 $m = 5, \quad x = -5, -1$
 Q138 $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$
 Q139 D
 Q140 A
 Q141 B
 Q142 B
 Q143 $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$
 Q144 A
 Q145 $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$
 Q146 B
 Q147 $2\left(x - \frac{3+i\sqrt{7}}{4}\right)\left(x - \frac{3-i\sqrt{7}}{4}\right)$
 Q148 $(x+1-2i)(x+1+2i)$
 Q149 $2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right)$
 Q150 $(x+2-i)(x+2+i)$
 Q151 $(x+2-\sqrt{2})(x+2+\sqrt{2})$
 Q152 $(x-2-i\sqrt{3})(x-2+i\sqrt{3})$
 Q153 $2\left(x - \frac{1+i\sqrt{7}}{4}\right)\left(x - \frac{1-i\sqrt{7}}{4}\right)$
 Q154 $3\left(x - \frac{5+i\sqrt{11}}{6}\right)\left(x - \frac{5-i\sqrt{11}}{6}\right)$

Q155 $2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right)$
 Q156 C
 Q157 A
 Q158 D
 Q159 D
 Q160 B
 Q161 A
 Q162 C
 Q163 A
 Q164 B
 Q165 $(x-1)(x+1)(x-i\sqrt{3})(x+i\sqrt{3})$
 Q166 $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{2})(x+i\sqrt{2})$
 Q167 $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{3})(x+i\sqrt{3})$
 Q168 $(x-1)(x+1)(x-\sqrt{2})(x+\sqrt{2})$
 Q169 B
 Q170 B
 Q171 D
 Q172 C
 Q173 $2x^2 - 2x + 1 = 0$
 Q174 $\frac{-1+i\sqrt{7}}{2}, \frac{-1-i\sqrt{7}}{2}$
 Q175 $x^2 - 6x + 13 = 0$
 Q176 $\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$
 Q177 $x^2 - x - 2 = 0$
 Q178 $x^2 - 2x + 17 = 0$
 Q179 $3x^2 - 10x + 6 = 0$
 Q180 $1+2i, 1-2i$
 Q181 $2+\sqrt{2}, 2-\sqrt{2}$
 Q182 B
 Q183 B
 Q184 B

Q185 A
 Q186 A
 Q187 D
 Q188 A
 Q189 B
 Q190 B
 Q191 $x^2 - 2x - 4 = 0$
 Q192 C
 Q193 $x^2 - 7x + 9 = 0$
 Q194 $x^2 + 10x + 49 = 0$
 Q195 D
 Q196 C
 Q197 $4x^2 + 3x + 2 = 0$
 Q198 A
 Q199 $x^2 + 8x + 24 = 0$
 Q200 D
 Q201 $-2 < m < -1$
 Q202 $-1 < m < -\frac{1}{2}$
 Q203 A
 Q204 B
 Q205 $-6 < m < -2$
 Q206 $k > 10$
 Q207 C
 Q208 B
 Q209 $m > -4$
 Q210 $k > -3$
 Q211 A
 Q212 D
 Q213 $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary
 Q214 $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary
 Q215 $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary
 Q216 $\frac{9}{2}$

Q217 3

Q218 $\frac{27}{2}$

Q219 9

Q220 Common root c ; other roots a and b .Q221 $x = \pm \frac{\sqrt{6}}{2}$, $x = \pm \frac{i\sqrt{3}}{3}$; real: $\pm \sqrt{6}/2$ Q222 For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$ Q223 $-4 < m < -3$ Q224 $m > 5$ Q225 $m < -4$ Q226 $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium

Q227 B

Q228 B

Q229 B

Q230 B

Q231 D

Q232 B

Q233 A

Q234 C

Q235 A

Q236 A

Q237 C

Q238 D

Q239 D

Q240 D

Quiz MCQ 1 B

Quiz MCQ 2 A

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $x = \frac{1}{2}, -2$ Quiz WRITTEN 6 $-1 < m < 2$

Quiz WRITTEN 7 28

Quiz WRITTEN 8 $x^2 + 6x + 25 = 0$

Concept 3 | Polynomial Theorems

English Student Edition • Focused formulas and guided practice

Formula opener

Remainder

$$P(a) = R \text{ when divisor} = x - a$$

Factor

$$P(a) = 0 \Rightarrow (x - a) \mid P(x)$$

Roots

$$P(x) = a \prod (x - r_i)$$

Worked Solutions

Polynomial Theorems

Q001 **Written****Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$.****Solution**

1. $U_{\text{sex}} = -2$
2. $P(-2) = 2$
3. Therefore the final answer is 2.

Final answer

2

Q002 **Written****Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$.****Solution**

1. $U_{\text{sex}} = 1$
2. $P(1) = 6$
3. Therefore the final answer is 6.

Final answer

6

Worked Solutions

Polynomial Theorems

Q003 MCQ

Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 2.

Correct option: C

Final answer

2

Q004 MCQ

Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 6.

Correct option: C

Final answer

6

Worked Solutions

Polynomial Theorems

Q005 Written

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$.**Solution**

1. Use $x = 2/3$
2. Evaluate $P(2/3) = 8$
3. Therefore the final answer is 8.

Final answer

8

Q006 Written

Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$.**Solution**

1. Use $x = 1/2$
2. $P(1/2) = -1/8$
3. Therefore the final answer is $-\frac{1}{8}$.

Final answer $-\frac{1}{8}$

Worked Solutions

Polynomial Theorems

Q007 **Written****Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$.****Solution**

1. Use $x = 2/3$
2. $P(2/3) = 0$
3. Therefore the final answer is 0.

Final answer

0

Q008 **Written****Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$.****Solution**

1. Use $x = -1/2$
2. $P(-1/2) = 4$
3. Therefore the final answer is 4.

Final answer

4

Worked Solutions

Polynomial Theorems

Q009 MCQ

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 8.

Correct option: D**Final answer**

8

Q010 MCQ

Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $-\frac{1}{8}$.

Correct option: A**Final answer** $-\frac{1}{8}$

Worked Solutions

Polynomial Theorems

Q011 MCQ

Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 0.

Correct option: D**Final answer**

0

Q012 MCQ

Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 4.

Correct option: A**Final answer**

4

Worked Solutions

Polynomial Theorems

Q013 Written

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a .

Solution

1. $P(-3) = 4$
2. $9a - 35 = 4$, hence $a = 13/3$
3. Therefore the final answer is $a = 13/3$.

Final answer

$$a = 13/3$$

Q014 Written

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a .

Solution

1. $P(3) = 1$
2. $9a + 37 = 1$, hence $a = -4$
3. Therefore the final answer is $a = -4$.

Final answer

$$a = -4$$

Worked Solutions

Polynomial Theorems

Q015 Written

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2 , find a .

Solution

1. $P(-1) = -2$
2. $a = -4$
3. Therefore the final answer is $a = -4$.

Final answer

$$a = -4$$

Q016 Written

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8 , find a .

Solution

1. $P(-3) = 8$
2. $a = -16$
3. Therefore the final answer is $a = -16$.

Final answer

$$a = -16$$

Worked Solutions

Polynomial Theorems

Q017 MCQ

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $a = 13/3$.

Correct option: B**Final answer**

$$a = 13/3$$

Q018 MCQ

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $a = -4$.

Correct option: B**Final answer**

$$a = -4$$

Worked Solutions

Polynomial Theorems

Q019 MCQ

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2 , find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $a = -4$.

Correct option: C**Final answer**

$$a = -4$$

Q020 MCQ

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8 , find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $a = -16$.

Correct option: A**Final answer**

$$a = -16$$

Worked Solutions

Polynomial Theorems

Q021 **Written**

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a .

Solution

1. $P(2) = 0$
2. $a = -4$
3. Therefore the final answer is $a = -4$.

Final answer

$$a = -4$$

Q022 **MCQ**

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $a = -4$.

Correct option: A

Final answer

$$a = -4$$

Worked Solutions

Polynomial Theorems

Q023 Written

A polynomial $P(x)$ leaves remainders 3 on division by $x - 1$ and -5 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$.

Solution

1. Let $R = ax + b$
2. $a + b = 3$ and $-3a + b = -5$
3. $R = 2x + 1$
4. Therefore the final answer is $2x + 1$.

Final answer

$$2x + 1$$

Q024 Written

A polynomial $P(x)$ leaves remainders 3 on division by $x - 2$ and -7 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$.

Solution

1. Let $R = ax + b$
2. $2a + b = 3$ and $-3a + b = -7$
3. $R = -10x + 13$
4. Therefore the final answer is $-10x + 13$.

Final answer

$$-10x + 13$$

Worked Solutions

Polynomial Theorems

Q025 Written

A polynomial $P(x)$ leaves remainders 5 on division by $x - 1$ and -1 on division by $x + 2$. Find the remainder on division by $(x - 1)(x + 2)$.

Solution

1. Let $R = ax + b$
2. $a + b = 5$ and $-2a + b = -1$
3. $R = 2x + 3$
4. Therefore the final answer is $2x + 3$.

Final answer

$$2x + 3$$

Q026 Written

A polynomial $P(x)$ leaves remainders 4 on division by $x - 2$ and -11 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$.

Solution

1. Let $R = ax + b$
2. $2a + b = 4$ and $-3a + b = -11$
3. $R = -15x + 19$
4. Therefore the final answer is $-15x + 19$.

Final answer

$$-15x + 19$$

Worked Solutions

Polynomial Theorems

Q027 MCQ

A polynomial $P(x)$ leaves remainders 3 on division by $x - 1$ and -5 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $2x + 1$.

Correct option: A

Final answer $2x + 1$

Q028 MCQ

A polynomial $P(x)$ leaves remainders 3 on division by $x - 2$ and -7 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $-10x + 13$.

Correct option: A

Final answer $-10x + 13$

Worked Solutions

Polynomial Theorems

Q029 MCQ

A polynomial $P(x)$ leaves remainders 5 on division by $x - 1$ and -1 on division by $x + 2$. Find the remainder on division by $(x - 1)(x + 2)$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $2x + 3$.

Correct option: A

Final answer

$$2x + 3$$

Q030 MCQ

A polynomial $P(x)$ leaves remainders 4 on division by $x - 2$ and -11 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $-15x + 19$.

Correct option: A

Final answer

$$-15x + 19$$

Worked Solutions

Polynomial Theorems

Q031 Written

A polynomial $P(x)$ leaves remainder $7x + 6$ on division by $x^2 - x - 6$, and $4x - 3$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 + x - 2$.

Solution

1. Let $R = ax + b$
2. Use shared roots -2 and 1 : $-2a + b = -8$ and $a + b = 1$
3. $R = 3x - 2$
4. Therefore the final answer is $3x - 2$.

Final answer

$$3x - 2$$

Q032 Written

A polynomial $P(x)$ leaves remainder $3x + 2$ on division by $x^2 - x - 2$, and $7x - 6$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 - 1$.

Solution

1. Let $R = ax + b$
2. At -1 and 1 : $-a + b = -1$ and $a + b = 1$
3. $R = x$
4. Therefore the final answer is x .

Final answer

$$x$$

Worked Solutions

Polynomial Theorems

Q033 Written

A polynomial $P(x)$ leaves remainder $2x - 2$ on division by $x^2 - 2x - 3$, and $-x + 7$ on division by $x^2 - 4$. Find the remainder on division by $x^2 - x - 2$.

Solution

1. Use shared roots -1 and 2
2. $-a + b = -4$ and $2a + b = 5$, so $R = 3x - 1$
3. Therefore the final answer is $3x - 1$.

Final answer

$$3x - 1$$

Q034 Written

A polynomial $P(x)$ leaves remainder $-x + 4$ on division by $x^2 - 3x + 2$, and $3x$ on division by $x^2 - 4x + 3$. Find the remainder on division by $x^2 - 5x + 6$.

Solution

1. Use shared roots 2 and 3
2. $2a + b = 2$ and $3a + b = 9$, so $R = 7x - 12$
3. Therefore the final answer is $7x - 12$.

Final answer

$$7x - 12$$

Worked Solutions

Polynomial Theorems

Q035 MCQ

A polynomial $P(x)$ leaves remainder $7x + 6$ on division by $x^2 - x - 6$, and $4x - 3$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 + x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $3x - 2$.

Correct option: B

Final answer $3x - 2$

Q036 MCQ

A polynomial $P(x)$ leaves remainder $3x + 2$ on division by $x^2 - x - 2$, and $7x - 6$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is x .

Correct option: B

Final answer x

Worked Solutions

Polynomial Theorems

Q037 MCQ

A polynomial $P(x)$ leaves remainder $2x - 2$ on division by $x^2 - 2x - 3$, and $-x + 7$ on division by $x^2 - 4$. Find the remainder on division by $x^2 - x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $3x - 1$.

Correct option: A

Final answer $3x - 1$

Q038 MCQ

A polynomial $P(x)$ leaves remainder $-x + 4$ on division by $x^2 - 3x + 2$, and $3x$ on division by $x^2 - 4x + 3$. Find the remainder on division by $x^2 - 5x + 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $7x - 12$.

Correct option: D

Final answer $7x - 12$

Worked Solutions

Polynomial Theorems

Q039 Written

A polynomial $P(x)$ leaves remainder $x - 5$ on division by $(x + 1)^2$, and 6 on division by $x - 2$. Find the remainder on division by $(x + 1)^2(x - 2)$.

Solution

1. Let $R = ax^2 + bx + c$
2. $R(-1) = -6, R'(-1) = 1, R(2) = 6$
3. Solve to get $R = x^2 + 3x - 4$
4. Therefore the final answer is $x^2 + 3x - 4$.

Final answer

$$x^2 + 3x - 4$$

Q040 MCQ

A polynomial $P(x)$ leaves remainder $x - 5$ on division by $(x + 1)^2$, and 6 on division by $x - 2$. Find the remainder on division by $(x + 1)^2(x - 2)$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $x^2 + 3x - 4$.

Correct option: A

Final answer

$$x^2 + 3x - 4$$

Worked Solutions

Polynomial Theorems

Q041 **Written**

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$.

Solution

1. Use -2 and coefficients 1,0,5,11
2. Bottom row 1,-2,9,-7
3. Therefore the final answer is $Q = x^2 - 2x + 9, R = -7$.

Final answer

$$Q = x^2 - 2x + 9, R = -7$$

Q042 **MCQ**

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = x^2 - 2x + 9, R = -7$.

Correct option: A

Final answer

$$Q = x^2 - 2x + 9, R = -7$$

Worked Solutions

Polynomial Theorems

Q043 Written

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$.

Solution

1. Use 1 and coefficients 1,0,5,-6
2. Bottom row 1,1,6,0
3. Therefore the final answer is $Q = x^2 + x + 6, R = 0$.

Final answer

$$Q = x^2 + x + 6, R = 0$$

Q044 Written

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$.

Solution

1. Use 3 and coefficients 3,-5,-3,-1
2. Bottom row 3,4,9,26
3. Therefore the final answer is $Q = 3x^2 + 4x + 9, R = 26$.

Final answer

$$Q = 3x^2 + 4x + 9, R = 26$$

Worked Solutions

Polynomial Theorems

Q045 Written

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$.

Solution

1. Use 2 and coefficients 1,0,-5,6
2. Bottom row 1,2,-1,4
3. Therefore the final answer is $Q = x^2 + 2x - 1, R = 4$.

Final answer

$$Q = x^2 + 2x - 1, R = 4$$

Q046 Written

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$.

Solution

1. Use 1 and coefficients 1,2,-1,-3
2. Bottom row 1,3,2,-1
3. Therefore the final answer is $Q = x^2 + 3x + 2, R = -1$.

Final answer

$$Q = x^2 + 3x + 2, R = -1$$

Worked Solutions

Polynomial Theorems

Q047 **Written**

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$.

Solution

1. Use 1 and coefficients 4,-2,2
2. Bottom row 4,2,4
3. Therefore the final answer is $Q = 4x + 2, R = 4$.

Final answer

$$Q = 4x + 2, R = 4$$

Q048 **MCQ**

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = x^2 + x + 6, R = 0$.

Correct option: B

Final answer

$$Q = x^2 + x + 6, R = 0$$

Worked Solutions

Polynomial Theorems

Q049 MCQ

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = 3x^2 + 4x + 9, R = 26$.

Correct option: D**Final answer**

$$Q = 3x^2 + 4x + 9, R = 26$$

Q050 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = x^2 + 2x - 1, R = 4$.

Correct option: D**Final answer**

$$Q = x^2 + 2x - 1, R = 4$$

Worked Solutions

Polynomial Theorems

Q051 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = x^2 + 3x + 2, R = -1$.

Correct option: A

Final answer

$$Q = x^2 + 3x + 2, R = -1$$

Q052 MCQ

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = 4x + 2, R = 4$.

Correct option: B

Final answer

$$Q = 4x + 2, R = 4$$

Worked Solutions

Polynomial Theorems

Q053 Written

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$.

Solution

1. Use -2 and coefficients 3,5,0,-2,-12
2. Bottom row 3,-1,2,-6,0
3. Therefore the final answer is $Q = 3x^3 - x^2 + 2x - 6, R = 0$.

Final answer

$$Q = 3x^3 - x^2 + 2x - 6, R = 0$$

Q054 MCQ

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = 3x^3 - x^2 + 2x - 6, R = 0$.

Correct option: D

Final answer

$$Q = 3x^3 - x^2 + 2x - 6, R = 0$$

Worked Solutions

Polynomial Theorems

Q055 Written

Factorise $x^3 + 2x^2 - x - 2$.**Solution**

1. $P(1) = 0$; divide by $x - 1$
2. Quotient $x^2 + 3x + 2 = (x + 1)(x + 2)$
3. Therefore the final answer is $(x - 1)(x + 1)(x + 2)$.

Final answer

$$(x - 1)(x + 1)(x + 2)$$

Q056 Written

Factorise $x^3 - 2x^2 - 5x + 6$.**Solution**

1. $P(1) = 0$; quotient $x^2 - x - 6 = (x - 3)(x + 2)$
2. Completed factorisation
3. Therefore the final answer is $(x - 1)(x - 3)(x + 2)$.

Final answer

$$(x - 1)(x - 3)(x + 2)$$

Worked Solutions

Polynomial Theorems

Q057 Written

Factorise $x^3 - x^2 - 16x - 20$.**Solution**

1. $P(-2) = 0$; divide by $x + 2$ twice
2. Remaining $x - 5$
3. Therefore the final answer is $(x + 2)^2(x - 5)$.

Final answer

$$(x + 2)^2(x - 5)$$

Q058 Written

Factorise $2x^3 - x^2 - 5x - 2$.**Solution**

1. $P(2) = 0$; quotient $2x^2 + 3x + 1$
2. Therefore the final answer is $(x - 2)(2x + 1)(x + 1)$.

Final answer

$$(x - 2)(2x + 1)(x + 1)$$

Worked Solutions

Polynomial Theorems

Q059 Written

Factorise $x^3 - 7x - 6$.**Solution**

1. Test divisor 3 and divide
2. Remaining linear factors
3. Therefore the final answer is $(x - 3)(x + 1)(x + 2)$.

Final answer

$$(x - 3)(x + 1)(x + 2)$$

Q060 Written

Factorise $x^3 + 4x^2 + x - 6$.**Solution**

1. Test divisor 1 and divide
2. Remaining linear factors
3. Therefore the final answer is $(x - 1)(x + 2)(x + 3)$.

Final answer

$$(x - 1)(x + 2)(x + 3)$$

Worked Solutions

Polynomial Theorems

Q061 Written

Factorise $x^3 + 6x^2 + 11x + 6$.**Solution**

1. Test divisor -1 and divide
2. Remaining linear factors
3. Therefore the final answer is $(x + 1)(x + 2)(x + 3)$.

Final answer

$$(x + 1)(x + 2)(x + 3)$$

Q062 Written

Factorise $x^3 - 3x + 2$.**Solution**

1. Test divisor 1 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x - 1)^2(x + 2)$.

Final answer

$$(x - 1)^2(x + 2)$$

Worked Solutions

Polynomial Theorems

Q063 Written

Factorise $x^3 + x^2 - 8x - 12$.**Solution**

1. Test divisor -2 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x - 3)(x + 2)^2$.

Final answer

$$(x - 3)(x + 2)^2$$

Q064 Written

Factorise $x^3 - 3x^2 - 9x - 5$.**Solution**

1. Test divisor -1 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x - 5)(x + 1)^2$.

Final answer

$$(x - 5)(x + 1)^2$$

Worked Solutions

Polynomial Theorems

Q065 Written

Factorise $2x^3 - 3x^2 - 3x + 2$.**Solution**

1. Test divisor 2 and divide
2. *Factorquotient* $2x^2 + x - 1$
3. Therefore the final answer is $(x - 2)(x + 1)(2x - 1)$.

Final answer

$$(x - 2)(x + 1)(2x - 1)$$

Q066 Written

Factorise $3x^3 + x^2 - 8x + 4$.**Solution**

1. Test divisor 1 and divide
2. *Factorquotient* $3x^2 + 4x - 4$
3. Therefore the final answer is $(x - 1)(x + 2)(3x - 2)$.

Final answer

$$(x - 1)(x + 2)(3x - 2)$$

Worked Solutions

Polynomial Theorems

Q067 **Written****Factorise** $2x^3 + 7x^2 + 8x + 3$.**Solution**

1. Test divisor -1 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x + 1)^2(2x + 3)$.

Final answer

$$(x + 1)^2(2x + 3)$$

Q068 **MCQ****Factorise** $x^3 + 2x^2 - x - 2$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x + 1)(x + 2)$.

Correct option: D**Final answer**

$$(x - 1)(x + 1)(x + 2)$$

Worked Solutions

Polynomial Theorems

Q069 MCQ

Factorise $x^3 - 2x^2 - 5x + 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x - 3)(x + 2)$.

Correct option: D**Final answer**

$$(x - 1)(x - 3)(x + 2)$$

Q070 MCQ

Factorise $x^3 - x^2 - 16x - 20$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x + 2)^2(x - 5)$.

Correct option: C**Final answer**

$$(x + 2)^2(x - 5)$$

Worked Solutions

Polynomial Theorems

Q071 MCQ

Factorise $2x^3 - x^2 - 5x - 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 2)(2x + 1)(x + 1)$.

Correct option: B**Final answer**

$$(x - 2)(2x + 1)(x + 1)$$

Q072 MCQ

Factorise $x^3 - 7x - 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 3)(x + 1)(x + 2)$.

Correct option: B**Final answer**

$$(x - 3)(x + 1)(x + 2)$$

Worked Solutions

Polynomial Theorems

Q073 MCQ

Factorise $x^3 + 4x^2 + x - 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x + 2)(x + 3)$.

Correct option: C**Final answer**

$$(x - 1)(x + 2)(x + 3)$$

Q074 MCQ

Factorise $x^3 + 6x^2 + 11x + 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x + 1)(x + 2)(x + 3)$.

Correct option: B**Final answer**

$$(x + 1)(x + 2)(x + 3)$$

Worked Solutions

Polynomial Theorems

Q075 MCQ

Factorise $x^3 - 3x + 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)^2(x + 2)$.

Correct option: D**Final answer**

$$(x - 1)^2(x + 2)$$

Q076 MCQ

Factorise $x^3 + x^2 - 8x - 12$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 3)(x + 2)^2$.

Correct option: D**Final answer**

$$(x - 3)(x + 2)^2$$

Worked Solutions

Polynomial Theorems

Q077 MCQ

Factorise $x^3 - 3x^2 - 9x - 5$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 5)(x + 1)^2$.

Correct option: D**Final answer**

$$(x - 5)(x + 1)^2$$

Q078 MCQ

Factorise $2x^3 - 3x^2 - 3x + 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 2)(x + 1)(2x - 1)$.

Correct option: A**Final answer**

$$(x - 2)(x + 1)(2x - 1)$$

Worked Solutions

Polynomial Theorems

Q079 MCQ

Factorise $3x^3 + x^2 - 8x + 4$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x + 2)(3x - 2)$.

Correct option: C**Final answer**

$$(x - 1)(x + 2)(3x - 2)$$

Q080 MCQ

Factorise $2x^3 + 7x^2 + 8x + 3$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x + 1)^2(2x + 3)$.

Correct option: D**Final answer**

$$(x + 1)^2(2x + 3)$$

Worked Solutions

Polynomial Theorems

Q081 Written

Factorise $x^4 - 6x^3 + 8x^2 + 8x - 16$.**Solution**

1. $P(2) = 0$; divide by $x - 2$ twice
2. Remaining quadratic $x^2 - 2x - 4$
3. Therefore the final answer is $(x - 2)^2(x^2 - 2x - 4)$.

Final answer

$$(x - 2)^2(x^2 - 2x - 4)$$

Q082 Written

Factorise $x^4 + 3x^3 - 27x^2 + 13x + 42$.**Solution**

1. Test roots and divide successively
2. Four linear factors
3. Therefore the final answer is $(x - 3)(x - 2)(x + 1)(x + 7)$.

Final answer

$$(x - 3)(x - 2)(x + 1)(x + 7)$$

Worked Solutions

Polynomial Theorems

Q083 **Written****Factorise** $x^4 - 4x^3 + 8x^2 - 8x + 3$.**Solution**

1. $P(1) = 0$; *dividetwice*
2. *Remainingquadratic* $x^2 - 2x + 3$
3. Therefore the final answer is $(x - 1)^2(x^2 - 2x + 3)$.

Final answer

$$(x - 1)^2(x^2 - 2x + 3)$$

Q084 **Written****Factorise** $x^4 - 10x^3 + 35x^2 - 50x + 24$.**Solution**

1. Test roots 1,2,3,4 successively
2. Four linear factors
3. Therefore the final answer is $(x - 4)(x - 3)(x - 2)(x - 1)$.

Final answer

$$(x - 4)(x - 3)(x - 2)(x - 1)$$

Worked Solutions

Polynomial Theorems

Q085 Written

Factorise $x^4 - 5x^3 - 3x^2 + 17x - 10$.**Solution**

1. Test roots 1 and -2 successively
2. Completed factorisation
3. Therefore the final answer is $(x - 5)(x - 1)^2(x + 2)$.

Final answer

$$(x - 5)(x - 1)^2(x + 2)$$

Q086 Written

Factorise $2x^4 - 5x^3 - 20x^2 + 20x + 48$.**Solution**

1. Test roots 2, -2, 4, -3/2
2. Completed factorisation
3. Therefore the final answer is $(x - 4)(x - 2)(x + 2)(2x + 3)$.

Final answer

$$(x - 4)(x - 2)(x + 2)(2x + 3)$$

Worked Solutions

Polynomial Theorems

Q087 **Written****Factorise** $x^4 - 4x^3 - 4x^2 + 13x - 6$.**Solution**

1. Test roots 1,-2
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 1)(x + 2)(x^2 - 5x + 3)$.

Final answer

$$(x - 1)(x + 2)(x^2 - 5x + 3)$$

Q088 **Written****Factorise** $x^4 - 4x^3 + 3x^2 + 2x - 6$.**Solution**

1. Test roots 3,-1
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 3)(x + 1)(x^2 - 2x + 2)$.

Final answer

$$(x - 3)(x + 1)(x^2 - 2x + 2)$$

Worked Solutions

Polynomial Theorems

Q089 Written

Factorise $x^4 - x^3 + x^2 + x - 2$.**Solution**

1. Test roots 1,-1
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 1)(x + 1)(x^2 - x + 2)$.

Final answer

$$(x - 1)(x + 1)(x^2 - x + 2)$$

Q090 MCQ

Factorise $x^4 - 6x^3 + 8x^2 + 8x - 16$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 2)^2(x^2 - 2x - 4)$.

Correct option: D**Final answer**

$$(x - 2)^2(x^2 - 2x - 4)$$

Worked Solutions

Polynomial Theorems

Q091 MCQ

Factorise $x^4 + 3x^3 - 27x^2 + 13x + 42$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 3)(x - 2)(x + 1)(x + 7)$.

Correct option: A**Final answer**

$$(x - 3)(x - 2)(x + 1)(x + 7)$$

Q092 MCQ

Factorise $x^4 - 4x^3 + 8x^2 - 8x + 3$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)^2(x^2 - 2x + 3)$.

Correct option: D**Final answer**

$$(x - 1)^2(x^2 - 2x + 3)$$

Worked Solutions

Polynomial Theorems

Q093 MCQ

Factorise $x^4 - 10x^3 + 35x^2 - 50x + 24$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 4)(x - 3)(x - 2)(x - 1)$.

Correct option: D**Final answer**

$$(x - 4)(x - 3)(x - 2)(x - 1)$$

Q094 MCQ

Factorise $x^4 - 5x^3 - 3x^2 + 17x - 10$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 5)(x - 1)^2(x + 2)$.

Correct option: C**Final answer**

$$(x - 5)(x - 1)^2(x + 2)$$

Worked Solutions

Polynomial Theorems

Q095 MCQ

Factorise $2x^4 - 5x^3 - 20x^2 + 20x + 48$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 4)(x - 2)(x + 2)(2x + 3)$.

Correct option: D**Final answer**

$$(x - 4)(x - 2)(x + 2)(2x + 3)$$

Q096 MCQ

Factorise $x^4 - 4x^3 - 4x^2 + 13x - 6$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x + 2)(x^2 - 5x + 3)$.

Correct option: A**Final answer**

$$(x - 1)(x + 2)(x^2 - 5x + 3)$$

Worked Solutions

Polynomial Theorems

Q097 MCQ

Factorise $x^4 - 4x^3 + 3x^2 + 2x - 6$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 3)(x + 1)(x^2 - 2x + 2)$.

Correct option: B**Final answer**

$$(x - 3)(x + 1)(x^2 - 2x + 2)$$

Q098 MCQ

Factorise $x^4 - x^3 + x^2 + x - 2$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x + 1)(x^2 - x + 2)$.

Correct option: C**Final answer**

$$(x - 1)(x + 1)(x^2 - x + 2)$$

Worked Solutions

Polynomial Theorems

Q099 Written

Factorise $2x^3 - 3x^2 - x - 2$.**Solution**

1. $P(2) = 0$; quotient $2x^2 + x + 1$ has discriminant -7
2. Therefore the final answer is $(x - 2)(2x^2 + x + 1)$.

Final answer

$$(x - 2)(2x^2 + x + 1)$$

Q100 Written

Factorise $2x^3 - 7x^2 + 3x + 6$.**Solution**

1. Test divisor 2 and divide
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 2)(2x^2 - 3x - 3)$.

Final answer

$$(x - 2)(2x^2 - 3x - 3)$$

Worked Solutions

Polynomial Theorems

Q101 Written

Factorise $x^3 + 4x^2 + 2x - 4$.**Solution**

1. Test divisor -2 and divide
2. Quadratic quotient remains
3. Therefore the final answer is $(x + 2)(x^2 + 2x - 2)$.

Final answer

$$(x + 2)(x^2 + 2x - 2)$$

Q102 Written

Factorise $x^3 - 3x^2 + 2$.**Solution**

1. Test divisor 1 and divide
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 1)(x^2 - 2x - 2)$.

Final answer

$$(x - 1)(x^2 - 2x - 2)$$

Worked Solutions

Polynomial Theorems

Q103 MCQ

Factorise $2x^3 - 3x^2 - x - 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 2)(2x^2 + x + 1)$.

Correct option: C**Final answer**

$$(x - 2)(2x^2 + x + 1)$$

Q104 MCQ

Factorise $2x^3 - 7x^2 + 3x + 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 2)(2x^2 - 3x - 3)$.

Correct option: B**Final answer**

$$(x - 2)(2x^2 - 3x - 3)$$

Worked Solutions

Polynomial Theorems

Q105 MCQ

Factorise $x^3 + 4x^2 + 2x - 4$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x + 2)(x^2 + 2x - 2)$.

Correct option: B**Final answer**

$$(x + 2)(x^2 + 2x - 2)$$

Q106 MCQ

Factorise $x^3 - 3x^2 + 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x^2 - 2x - 2)$.

Correct option: D**Final answer**

$$(x - 1)(x^2 - 2x - 2)$$

Worked Solutions

Polynomial Theorems

Q107 Challenge

Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$.

Solution

1. *Substitute* $x = -1$
2. $(-1)^{57} + 57 = 56$
3. Therefore the final answer is 56.

Final answer

56

Worked Solutions

Polynomial Theorems

Q108 Challenge

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a , then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial.

Solution

1. Use the divisor zeros: $P(-1/2) = 0$ and $P(-2) = 0$.
2. Hence $a = 10$ for $2x + 1$, and $a = -11$ for $x + 2$.
3. Each linear factor corresponds to a polynomial root.

Final answer

$$a = 10 \quad \text{for } 2x + 1$$

$$a = -11 \quad \text{for } x + 2$$

linear factors $\leftrightarrow$ roots

Worked Solutions

Polynomial Theorems

Q109 Challenge

A polynomial $P(x)$ is divisible by $x - 1$ and leaves remainder -4 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$.

Solution

1. $a + b = 0$ and $-3a + b = -4$
2. $R = x - 1$
3. Therefore the final answer is $x - 1$.

Final answer $x - 1$

Worked Solutions

Polynomial Theorems

Q110 Challenge

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts.

Solution

1. $P = (2x - 1)^2(x^2 - 3)$
2. Remaining roots are $+ / - \sqrt{3}$
3. Therefore the final answer is $x = \sqrt{3}, -\sqrt{3}$.

Final answer

$$x = \sqrt{3}, -\sqrt{3}$$

Worked Solutions

Polynomial Theorems

Q111 Challenge

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b .

Solution

1. $P(1) = 0$ gives $a + b = -10$
2. $P(-2) = 0$ gives $2a - b = -26$
3. $a = -12, b = 2$
4. Therefore the final answer is $a = -12, b = 2$.

Final answer

$$a = -12, b = 2$$

Worked Solutions

Polynomial Theorems

Q112 Challenge

For the polynomial above with $a = -12, b = 2$, determine all roots of $P(x) = 0$.

Solution

1. $P = (x - 1)(x + 2)(x^2 - 4x - 6)$
2. Solve quadratic : $x = 2 \pm \sqrt{10}$
3. All four roots satisfy $P = 0$
4. Therefore the final answer is $x = 1, -2, 2 \pm \sqrt{10}$.

Final answer

$$x = 1, -2, 2 \pm \sqrt{10}$$

Worked Solutions

Polynomial Theorems

Q113 Challenge

Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 56.

Correct option: B

Final answer

56

Worked Solutions

Polynomial Theorems

Q114 Challenge

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a , then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.

$$a = 10 \quad \text{for } 2x + 1$$

2. The correct value is $a = -11$ for $x + 2$.

linear factors $\leftrightarrow$ roots

Correct option: D

Final answer

$$a = 10 \quad \text{for } 2x + 1$$

$$a = -11 \quad \text{for } x + 2$$

linear factors $\leftrightarrow$ roots

Worked Solutions

Polynomial Theorems

Q115 Challenge

A polynomial $P(x)$ is divisible by $x - 1$ and leaves remainder -4 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $x - 1$.

Correct option: C

Final answer

$x - 1$

Worked Solutions

Polynomial Theorems

Q116 Challenge

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $x = \sqrt{3}, -\sqrt{3}$.

Correct option: A

Final answer

$$x = \sqrt{3}, -\sqrt{3}$$

Worked Solutions

Polynomial Theorems

Q117 Challenge

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $a = -12, b = 2$.

Correct option: C

Final answer

$$a = -12, b = 2$$

Worked Solutions

Polynomial Theorems

Q118 Challenge

For the polynomial above with $a = -12, b = 2$, determine all roots of $P(x) = 0$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $x = 1, -2, 2 \pm \sqrt{10}$.

Correct option: A

Final answer

$$x = 1, -2, 2 \pm \sqrt{10}$$

Worked Solutions

Polynomial Theorems

Quiz MCQ 1 [Concept Quiz · MCQ](#)

Find the remainder when $P(x) = x^3 - 2x + 5$ is divided by $x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 9.

Correct option: C

Final answer

9

Quiz MCQ 2 [Concept Quiz · MCQ](#)

If $P(x)$ is divisible by $x + 1$, which statement is true? Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $P(-1) = 0$.

Correct option: C

Final answer

$P(-1) = 0$

Worked Solutions

Polynomial Theorems

Quiz MCQ 3 [Concept Quiz · MCQ](#)

The remainder on division by a quadratic divisor has the form $ax + b$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $ax + b$.

Correct option: B

Final answer

$$ax + b$$

Quiz MCQ 4 [Concept Quiz · MCQ](#)

Which method records quotient and remainder for division by $x - a$? Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is *Syntheticdivision*.

Correct option: C

Final answer

Syntheticdivision

Worked Solutions

Polynomial Theorems

Quiz WRITTEN 5 Concept Quiz · Written

Find the remainder when $2x^3 + 3x^2 - x + 4$ is divided by $2x + 1$.

Solution

1. Apply the relevant polynomial theorem.
2. Final answer: 4

Final answer

4

Quiz WRITTEN 6 Concept Quiz · Written

Find the remainder $R = ax + b$ when $P(x)$ has values $P(1) = 3$ and $P(-2) = -6$.

Solution

1. Apply the relevant polynomial theorem.
2. Final answer: $R = 3x$

Final answer

$R = 3x$

Worked Solutions

Polynomial Theorems

Quiz WRITTEN 7 Concept Quiz · Written**Factorise** $x^3 - 6x^2 + 11x - 6$.**Solution**

1. Apply the relevant polynomial theorem.
2. Final answer: $(x - 1)(x - 2)(x - 3)$

Final answer

$$(x - 1)(x - 2)(x - 3)$$

Quiz WRITTEN 8 Concept Quiz · Written**If** $x - 1$ **and** $x + 2$ **are factors of** $x^4 - 3x^3 + ax^2 + bx + 12$, **find** a, b .**Solution**

1. Apply the relevant polynomial theorem.
2. Final answer: $a = -12, b = 2$

Final answer

$$a = -12, b = 2$$

Answer key

Use the question number shown on each page.

Q001 2	Q027 A	Q053 $Q = 3x^3 - x^2 + 2x - 6, R = 0$	Q079 C
Q002 6	Q028 A	Q054 D	Q080 D
Q003 C	Q029 A	Q055 $(x - 1)(x + 1)(x + 2)$	Q081 $(x - 2)^2(x^2 - 2x - 4)$
Q004 C	Q030 A	Q056 $(x - 1)(x - 3)(x + 2)$	Q082 $(x - 3)(x - 2)(x + 1)(x + 7)$
Q005 8	Q031 $3x - 2$	Q057 $(x + 2)^2(x - 5)$	Q083 $(x - 1)^2(x^2 - 2x + 3)$
Q006 $-\frac{1}{8}$	Q032 x	Q058 $(x - 2)(2x + 1)(x + 1)$	Q084 $(x - 4)(x - 3)(x - 2)(x - 1)$
Q007 0	Q033 $3x - 1$	Q059 $(x - 3)(x + 1)(x + 2)$	Q085 $(x - 5)(x - 1)^2(x + 2)$
Q008 4	Q034 $7x - 12$	Q060 $(x - 1)(x + 2)(x + 3)$	Q086 $(x - 4)(x - 2)(x + 2)(2x + 3)$
Q009 D	Q035 B	Q061 $(x + 1)(x + 2)(x + 3)$	Q087 $(x - 1)(x + 2)(x^2 - 5x + 3)$
Q010 A	Q036 B	Q062 $(x - 1)^2(x + 2)$	Q088 $(x - 3)(x + 1)(x^2 - 2x + 2)$
Q011 D	Q037 A	Q063 $(x - 3)(x + 2)^2$	Q089 $(x - 1)(x + 1)(x^2 - x + 2)$
Q012 A	Q038 D	Q064 $(x - 5)(x + 1)^2$	Q090 D
Q013 $a = 13/3$	Q039 $x^2 + 3x - 4$	Q065 $(x - 2)(x + 1)(2x - 1)$	Q091 A
Q014 $a = -4$	Q040 A	Q066 $(x - 1)(x + 2)(3x - 2)$	Q092 D
Q015 $a = -4$	Q041 $Q = x^2 - 2x + 9, R = -7$	Q067 $(x + 1)^2(2x + 3)$	Q093 D
Q016 $a = -16$	Q042 A	Q068 D	Q094 C
Q017 B	Q043 $Q = x^2 + x + 6, R = 0$	Q069 D	Q095 D
Q018 B	Q044 $Q = 3x^2 + 4x + 9, R = 26$	Q070 C	Q096 A
Q019 C	Q045 $Q = x^2 + 2x - 1, R = 4$	Q071 B	Q097 B
Q020 A	Q046 $Q = x^2 + 3x + 2, R = -1$	Q072 B	Q098 C
Q021 $a = -4$	Q047 $Q = 4x + 2, R = 4$	Q073 C	Q099 $(x - 2)(2x^2 + x + 1)$
Q022 A	Q048 B	Q074 B	Q100 $(x - 2)(2x^2 - 3x - 3)$
Q023 $2x + 1$	Q049 D	Q075 D	Q101 $(x + 2)(x^2 + 2x - 2)$
Q024 $-10x + 13$	Q050 D	Q076 D	Q102 $(x - 1)(x^2 - 2x - 2)$
Q025 $2x + 3$	Q051 A	Q077 D	Q103 C
Q026 $-15x + 19$	Q052 B	Q078 A	Q104 B

Q105 B**Q106** D**Q107** 56

$$a = 10 \quad \text{for } 2x + 1$$

Q108 $a = -11$ for $x + 2$ linear factors $\leftrightarrow$ roots**Q109** $x - 1$

Q110 $x = \sqrt{3}, -\sqrt{3}$

Q111 $a = -12, b = 2$

Q112 $x = 1, -2, 2 \pm \sqrt{10}$

Q113 B**Q114** D**Q115** C**Q116** A**Q117** C**Q118** A**Quiz MCQ 1** C**Quiz MCQ 2** C**Quiz MCQ 3** B**Quiz MCQ 4** C**Quiz WRITTEN 5** 4**Quiz WRITTEN 6** $R = 3x$ **Quiz WRITTEN 7** $(x - 1)(x - 2)(x - 3)$ **Quiz WRITTEN 8** $a = -12, b = 2$

Concept 4 | Higher-Degree Equations

English Student Edition • Focused formulas and guided practice

Formula opener

Factor theorem

$$P(a) = 0 \Rightarrow x - a \text{ is a factor}$$

Degree

$$\deg(PQ) = \deg P + \deg Q$$

Conjugate roots

$$a + bi \Rightarrow a - bi$$

Worked Solutions

Higher-Degree Equations

Q001 Written

$$x^3 - 8 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$.

Final answer

$$x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$$

Q002 Written

$$x^3 = -1$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$.

Final answer

$$x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$$

Worked Solutions

Higher-Degree Equations

Q003 Written

$$x^3 = 27$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$.

Final answer

$$x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$$

Q004 Written

$$x^3 = -125$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$.

Final answer

$$x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$$

Worked Solutions

Higher-Degree Equations

Q005 Written

$$8x^3 = 1$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$.

Final answer

$$x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$$

Q006 MCQ

$$x^3 - 8 = 0$$
 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$.

Correct option: B

Final answer

$$x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$$

Worked Solutions

Higher-Degree Equations

Q007 MCQ

 $x^3 = -1$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$.

Correct option: D

Final answer

$$x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$$

Q008 MCQ

 $x^3 = 27$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$.

Correct option: B

Final answer

$$x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$$

Worked Solutions

Higher-Degree Equations

Q009 MCQ

 $x^3 = -125$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$.

Correct option: A

Final answer

$$x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$$

Q010 MCQ

 $8x^3 = 1$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$.

Correct option: C

Final answer

$$x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$$

Worked Solutions

Higher-Degree Equations

Q011 Written

$$x^3 - 2x^2 - 5x + 6 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, 1, 3$.

Final answer

$$x = -2, 1, 3$$

Q012 Written

$$3x^3 + x^2 - 8x + 4 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, 2/3, 1$.

Final answer

$$x = -2, 2/3, 1$$

Worked Solutions

Higher-Degree Equations

Q013 Written

$$2x^3 - 3x^2 - x - 2 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$.

Final answer

$$x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$$

Q014 Written

$$x^3 - x^2 - 16x - 20 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2(\text{repeated}), 5$.

Final answer

$$x = -2(\text{repeated}), 5$$

Worked Solutions

Higher-Degree Equations

Q015 Written

$$x^3 + 6x^2 + 11x + 6 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, -2, -1$.

Final answer

$$x = -3, -2, -1$$

Q016 Written

$$x^3 + 3x^2 - 4x - 12 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, -2, 2$.

Final answer

$$x = -3, -2, 2$$

Worked Solutions

Higher-Degree Equations

Q017 Written

$$3x^3 - 2x^2 - 3x + 2 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1, 2/3, 1$.

Final answer

$$x = -1, 2/3, 1$$

Q018 Written

$$x^3 - x^2 - 4 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$.

Final answer

$$x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$$

Worked Solutions

Higher-Degree Equations

Q019 Written

$$x^3 + 4x^2 + 2x - 4 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$.

Final answer

$$x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$$

Q020 Written

$$x^3 - 4x^2 + 9x - 10 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, 1 - 2i, 1 + 2i$.

Final answer

$$x = 2, 1 - 2i, 1 + 2i$$

Worked Solutions

Higher-Degree Equations

Q021 Written

$$x^3 + x^2 - 8x - 12 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2$ (repeated), 3.

Final answer

$$x = -2(\text{repeated}), 3$$

Q022 Written

$$2x^3 + 7x^2 + 8x + 3 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1$ (repeated), $-3/2$.

Final answer

$$x = -1(\text{repeated}), -3/2$$

Worked Solutions

Higher-Degree Equations

Q023 Written

$$x^3 + 6x^2 + 12x + 8 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2$ (triple).

Final answer

$$x = -2(\text{triple})$$

Q024 MCQ

$$x^3 - 2x^2 - 5x + 6 = 0$$
 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2, 1, 3$.

Correct option: C

Final answer

$$x = -2, 1, 3$$

Worked Solutions

Higher-Degree Equations

Q025 MCQ

 $3x^3 + x^2 - 8x + 4 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2, 2/3, 1$.

Correct option: D

Final answer

$$x = -2, 2/3, 1$$

Q026 MCQ

 $2x^3 - 3x^2 - x - 2 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$.

Correct option: D

Final answer

$$x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$$

Worked Solutions

Higher-Degree Equations

Q027 MCQ

 $x^3 - x^2 - 16x - 20 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2$ (repeated), 5.

Correct option: B

Final answer $x = -2$ (repeated), 5

Q028 MCQ

 $x^3 + 6x^2 + 11x + 6 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -3, -2, -1$.

Correct option: C

Final answer $x = -3, -2, -1$

Worked Solutions

Higher-Degree Equations

Q029 MCQ

 $x^3 + 3x^2 - 4x - 12 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -3, -2, 2$.

Correct option: D

Final answer

$$x = -3, -2, 2$$

Q030 MCQ

 $3x^3 - 2x^2 - 3x + 2 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -1, 2/3, 1$.

Correct option: D

Final answer

$$x = -1, 2/3, 1$$

Worked Solutions

Higher-Degree Equations

Q031 MCQ

 $x^3 - x^2 - 4 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$.

Correct option: A

Final answer

$$x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$$

Worked Solutions

Higher-Degree Equations

Q032 MCQ

 $x^3 + 4x^2 + 2x - 4 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$.

Correct option: D

Final answer

$$x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$$

Worked Solutions

Higher-Degree Equations

Q033 MCQ

 $x^3 - 4x^2 + 9x - 10 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 2, 1 - 2i, 1 + 2i$.

Correct option: D

Final answer

$$x = 2, 1 - 2i, 1 + 2i$$

Q034 MCQ

 $x^3 + x^2 - 8x - 12 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2(\text{repeated}), 3$.

Correct option: D

Final answer

$$x = -2(\text{repeated}), 3$$

Worked Solutions

Higher-Degree Equations

Q035 MCQ

 $2x^3 + 7x^2 + 8x + 3 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -1$ (repeated), $-3/2$.

Correct option: B**Final answer**

$$x = -1(\text{repeated}), -3/2$$

Q036 MCQ

 $x^3 + 6x^2 + 12x + 8 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2$ (triple).

Correct option: D**Final answer**

$$x = -2(\text{triple})$$

Worked Solutions

Higher-Degree Equations

Q037 Written

$$x^4 - 7x^2 - 18 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, 3, -\sqrt{2}i, \sqrt{2}i$.

Final answer

$$x = -3, 3, -\sqrt{2}i, \sqrt{2}i$$

Q038 Written

$$x^4 - 5x^2 + 4 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, -1, 1, 2$.

Final answer

$$x = -2, -1, 1, 2$$

Worked Solutions

Higher-Degree Equations

Q039 Written

$$x^4 - 5x^2 - 36 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, 3, -2i, 2i$.

Final answer

$$x = -3, 3, -2i, 2i$$

Q040 Written

$$x^4 - 6x^2 - 27 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, 3, -\sqrt{3}i, \sqrt{3}i$.

Final answer

$$x = -3, 3, -\sqrt{3}i, \sqrt{3}i$$

Worked Solutions

Higher-Degree Equations

Q041 Written

$$x^4 - x^2 - 2 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -\sqrt{2}, \sqrt{2}, -i, i$.

Final answer

$$x = -\sqrt{2}, \sqrt{2}, -i, i$$

Q042 MCQ

$$x^4 - 7x^2 - 18 = 0$$
 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -3, 3, -\sqrt{2}i, \sqrt{2}i$.

Correct option: D

Final answer

$$x = -3, 3, -\sqrt{2}i, \sqrt{2}i$$

Worked Solutions

Higher-Degree Equations

Q043 MCQ

 $x^4 - 5x^2 + 4 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2, -1, 1, 2$.

Correct option: D

Final answer

$$x = -2, -1, 1, 2$$

Q044 MCQ

 $x^4 - 5x^2 - 36 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -3, 3, -2i, 2i$.

Correct option: C

Final answer

$$x = -3, 3, -2i, 2i$$

Worked Solutions

Higher-Degree Equations

Q045 MCQ

 $x^4 - 6x^2 - 27 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -3, 3, -\sqrt{3}i, \sqrt{3}i$.

Correct option: C

Final answer

$$x = -3, 3, -\sqrt{3}i, \sqrt{3}i$$

Q046 MCQ

 $x^4 - x^2 - 2 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -\sqrt{2}, \sqrt{2}, -i, i$.

Correct option: B

Final answer

$$x = -\sqrt{2}, \sqrt{2}, -i, i$$

Worked Solutions

Higher-Degree Equations

Q047 Written

$$x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$.

Final answer

$$x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$$

Q048 Written

$$x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 1, 2, 3, 4$.

Final answer

$$x = 1, 2, 3, 4$$

Worked Solutions

Higher-Degree Equations

Q049 Written

$$x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 1, 2, 3(\text{repeated})$.

Final answer

$$x = 1, 2, 3(\text{repeated})$$

Q050 Written

$$x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1, 3, 1 - i, 1 + i$.

Final answer

$$x = -1, 3, 1 - i, 1 + i$$

Worked Solutions

Higher-Degree Equations

Q051 MCQ

 $x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 2$ (repeated), $1 - \sqrt{5}$, $1 + \sqrt{5}$.

Correct option: B

Final answer

$$x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$$

Worked Solutions

Higher-Degree Equations

Q052 MCQ

 $x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 1, 2, 3, 4$.

Correct option: A

Final answer $x = 1, 2, 3, 4$

Q053 MCQ

 $x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 1, 2, 3(\text{repeated})$.

Correct option: C

Final answer $x = 1, 2, 3(\text{repeated})$

Worked Solutions

Higher-Degree Equations

Q054 MCQ

 $x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -1, 3, 1 - i, 1 + i$.

Correct option: C

Final answer

$$x = -1, 3, 1 - i, 1 + i$$

Q055 Written

 $x^3 + 4x^2 + ax + b = 0$ has roots -3 and 1; find a,b and the other root.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = 1, b = -6$; other roots: -2 .

Final answer

$$a = 1, b = -6; \text{ other roots: } -2$$

Worked Solutions

Higher-Degree Equations

Q056 Written

$x^3 - ax^2 - x + b = 0$ has roots -1 and 2; find a, b and the other root.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = 2, b = 2$; other roots: 1.

Final answer

$a = 2, b = 2$; other roots: 1

Q057 Written

$x^3 + ax^2 + bx - 8 = 0$ has roots 1 and 2; find a, b and the other root.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -7, b = 14$; other roots: 4.

Final answer

$a = -7, b = 14$; other roots: 4

Worked Solutions

Higher-Degree Equations

Q058 MCQ

$x^3 + 4x^2 + ax + b = 0$ has roots -3 and 1; find a,b and the other root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = 1, b = -6$; other roots: -2 .

Correct option: C

Final answer

$a = 1, b = -6$; other roots: -2

Worked Solutions

Higher-Degree Equations

Q059 MCQ

$x^3 - ax^2 - x + b = 0$ has roots -1 and 2; find a, b and the other root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = 2, b = 2$; other roots: 1.

Correct option: B

Final answer

$a = 2, b = 2$; other roots: 1

Worked Solutions

Higher-Degree Equations

Q060 MCQ

$x^3 + ax^2 + bx - 8 = 0$ has roots 1 and 2; find a, b and the other root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -7, b = 14$; other roots: 4.

Correct option: C

Final answer

$a = -7, b = 14$; other roots: 4

Worked Solutions

Higher-Degree Equations

Q061 Written

$x^3 + ax^2 + bx + 5 = 0$ has root $2+i$; find real a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -3, b = 1$; other roots: $-1, 2 - i$.

Final answer

$a = -3, b = 1$; other roots: $-1, 2 - i$

Q062 Written

$x^3 + ax^2 + bx - 4 = 0$ has root $1+i$; find a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -4, b = 6$; other roots: $1 - i, 2$.

Final answer

$a = -4, b = 6$; other roots: $1 - i, 2$

Worked Solutions

Higher-Degree Equations

Q063 Written

$x^3 + ax^2 + bx - 20 = 0$ has root $1 - 3i$; find real a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -4, b = 14$; other roots: $1 + 3i, 2$.

Final answer

$a = -4, b = 14$; other roots: $1 + 3i, 2$

Q064 Written

$x^3 + ax^2 + bx + 10 = 0$ has root $1 + 2i$; find real a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = 0, b = 1$; other roots: $1 - 2i, -2$.

Final answer

$a = 0, b = 1$; other roots: $1 - 2i, -2$

Worked Solutions

Higher-Degree Equations

Q065 MCQ

$x^3 + ax^2 + bx + 5 = 0$ has root $2+i$; find real a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -3, b = 1$; other roots: $-1, 2 - i$.

Correct option: A

Final answer

$a = -3, b = 1$; other roots: $-1, 2 - i$

Q066 MCQ

$x^3 + ax^2 + bx - 4 = 0$ has root $1+i$; find a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -4, b = 6$; other roots: $1 - i, 2$.

Correct option: B

Final answer

$a = -4, b = 6$; other roots: $1 - i, 2$

Worked Solutions

Higher-Degree Equations

Q067 MCQ

$x^3 + ax^2 + bx - 20 = 0$ has root $1 - 3i$; find real a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -4, b = 14$; other roots: $1 + 3i, 2$.

Correct option: D

Final answer

$a = -4, b = 14$; other roots: $1 + 3i, 2$

Q068 MCQ

$x^3 + ax^2 + bx + 10 = 0$ has root $1 + 2i$; find real a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = 0, b = 1$; other roots: $1 - 2i, -2$.

Correct option: B

Final answer

$a = 0, b = 1$; other roots: $1 - 2i, -2$

Worked Solutions

Higher-Degree Equations

Q069 Written

$x^3 - kx^2 + (3k - 1)x - 24 = 0$ has root 2; find k and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $k = 9$; other roots: 3, 4.

Final answer

$k = 9$; other roots: 3, 4

Q070 MCQ

$x^3 - kx^2 + (3k - 1)x - 24 = 0$ has root 2; find k and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $k = 9$; other roots: 3, 4.

Correct option: B

Final answer

$k = 9$; other roots: 3, 4

Worked Solutions

Higher-Degree Equations

Q071 Written

For an imaginary cube root of unity w , evaluate w^9 .**Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Q072 Written

For an imaginary cube root of unity w , evaluate w^{2010} .**Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Worked Solutions

Higher-Degree Equations

Q073 **Written****For an imaginary cube root of unity w , evaluate w^3 .****Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Q074 **MCQ****For an imaginary cube root of unity w , evaluate w^9 Select the correct answer:****Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 1.

Correct option: A**Final answer**

1

Worked Solutions

Higher-Degree Equations

Q075 MCQ

For an imaginary cube root of unity w , evaluate w^{2010} Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 1.

Correct option: B**Final answer**

1

Q076 MCQ

For an imaginary cube root of unity w , evaluate w^3 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 1.

Correct option: D**Final answer**

1

Worked Solutions

Higher-Degree Equations

Q077 Written

For an imaginary cube root of unity w , evaluate $w^4 + w^5$.**Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is -1 .

Final answer -1

Q078 Written

For an imaginary cube root of unity w , evaluate $w^2 + w$.**Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is -1 .

Final answer -1

Worked Solutions

Higher-Degree Equations

Q079 **Written**

For an imaginary cube root of unity w , evaluate $w^2 + w^4 + w^6$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 0.

Final answer

0

Q080 **MCQ**

For an imaginary cube root of unity w , evaluate $w^4 + w^5$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is -1 .

Correct option: A

Final answer -1

Worked Solutions

Higher-Degree Equations

Q081 MCQ

For an imaginary cube root of unity w , evaluate $w^2 + w$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is -1 .

Correct option: B**Final answer** -1

Q082 MCQ

For an imaginary cube root of unity w , evaluate $w^2 + w^4 + w^6$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 0 .

Correct option: D**Final answer** 0

Worked Solutions

Higher-Degree Equations

Q083 Written

For an imaginary cube root of unity w , evaluate $w + 1/w$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is -1 .

Final answer -1

Q084 Written

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer 1

Worked Solutions

Higher-Degree Equations

Q085 Written

For an imaginary cube root of unity w , evaluate $1/w + 1/w^2 + 1$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 0.

Final answer

0

Q086 MCQ

For an imaginary cube root of unity w , evaluate $w + 1/w$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is -1 .

Correct option: D

Final answer -1

Worked Solutions

Higher-Degree Equations

Q087 MCQ

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 1.

Correct option: C**Final answer**

1

Q088 MCQ

For an imaginary cube root of unity w , evaluate $1/w + 1/w^2 + 1$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 0.

Correct option: D**Final answer**

0

Worked Solutions

Higher-Degree Equations

Q089 Written

$x^3 - 2x^2 + (a - 3)x + a = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -4$ or $9/4$.

Final answer

$a = -4$ or $9/4$

Q090 Written

$x^3 - x^2 + (a - 2)x + a = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -3$ or 1 .

Final answer

$a = -3$ or 1

Worked Solutions

Higher-Degree Equations

Q091 Written

 $x^3 - 7x^2 + (a + 6)x - a = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = 5$ or 9 .

Final answer

 $a = 5$ or 9

Q092 Written

 $x^3 - (a + 1)x^2 + (3a + 5)x - 2a - 5 = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -6, -2$, or 10 .

Final answer

 $a = -6, -2$, or 10

Worked Solutions

Higher-Degree Equations

Q093 **Written** $x^3 + (2a - 1)x^2 - 3(a - 2)x + a - 6 = 0$ Find all real parameter values a producing a repeated root.**Solution**

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -7, -3$, or 2 .

Final answer $a = -7, -3$, or 2 Q094 **MCQ** $x^3 - 2x^2 + (a - 3)x + a = 0$ Find all real parameter values a producing a repeated root Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -4$ or $9/4$.

Correct option: D

Final answer $a = -4$ or $9/4$

Worked Solutions

Higher-Degree Equations

Q095 MCQ

$x^3 - x^2 + (a - 2)x + a = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -3$ or 1 .

Correct option: D

Final answer

$a = -3$ or 1

Q096 MCQ

$x^3 - 7x^2 + (a + 6)x - a = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = 5$ or 9 .

Correct option: B

Final answer

$a = 5$ or 9

Worked Solutions

Higher-Degree Equations

Q097 MCQ

$x^3 - (a + 1)x^2 + (3a + 5)x - 2a - 5 = 0$ Find all real parameter values a producing a repeated root. Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -6, -2$, or 10 .

Correct option: A**Final answer** $a = -6, -2$, or 10

Q098 MCQ

$x^3 + (2a - 1)x^2 - 3(a - 2)x + a - 6 = 0$ Find all real parameter values a producing a repeated root. Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -7, -3$, or 2 .

Correct option: A**Final answer** $a = -7, -3$, or 2

Worked Solutions

Higher-Degree Equations

Q099 Challenge

$x^3 + ax + b = 0$ has root $1 + \sqrt{2}i$; find real a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$.

Final answer

$a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$

Worked Solutions

Higher-Degree Equations

Q100 Challenge

For an imaginary cube root of unity w , evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 4.

Final answer

4

Worked Solutions

Higher-Degree Equations

Q101 Challenge

$x^3 + (a - 1)x^2 + (a - 3)x - 2a + 3 = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = 2/3, 2$, or 6 .

Final answer

$a = 2/3, 2$, or 6

Worked Solutions

Higher-Degree Equations

Q102 Challenge

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$.

Solution

1. Set inflow equal to outflow: $(x^4 + x^2 + x = x^3 + 2)$.
2. Move all terms to one side.
3. Therefore the final answer is $x^4 - x^3 + x^2 + x - 2 = 0$.

Final answer

$$x^4 - x^3 + x^2 + x - 2 = 0$$

Worked Solutions

Higher-Degree Equations

Q103 Challenge

Solve $x^4 - x^3 + x^2 + x - 2 = 0$ **for the stable pressure levels.**

Solution

1. Factor $(x^4 - x^3 + x^2 + x - 2)$ and solve the remaining quadratic.
2. Apply the stated physical restriction ($x \geq 0$) only after listing the algebraic roots.
3. Therefore the final answer is $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$.

Final answer

$$x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$$

Worked Solutions

Higher-Degree Equations

Q104 Challenge

Classify the stable pressure solutions as real or complex.

Solution

1. Separate the real roots from the non-real conjugate pair.

Both types occur: real $x = -1, 1$

2. Therefore the final answer is non-real conjugates $x = (1 \pm \sqrt{7}i)/2$.

if $x \geq 0 : x = 1$

Final answer

Both types occur: real $x = -1, 1$

non-real conjugates $x = (1 \pm \sqrt{7}i)/2$

if $x \geq 0 : x = 1$

Worked Solutions

Higher-Degree Equations

Q105 Challenge

$x^3 + ax + b = 0$ has root $1 + \sqrt{2}i$; find real a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$.

Correct option: C

Final answer

$a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$

Worked Solutions

Higher-Degree Equations

Q106 Challenge

For an imaginary cube root of unity w , evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion. Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 4.

Correct option: D

Final answer

4

Worked Solutions

Higher-Degree Equations

Q107 Challenge

$x^3 + (a - 1)x^2 + (a - 3)x - 2a + 3 = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = 2/3, 2$, or 6 .

Correct option: D

Final answer

$a = 2/3, 2$, or 6

Worked Solutions

Higher-Degree Equations

Q108 Challenge

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x^4 - x^3 + x^2 + x - 2 = 0$.

Correct option: A

Final answer

$$x^4 - x^3 + x^2 + x - 2 = 0$$

Worked Solutions

Higher-Degree Equations

Q109 Challenge

Solve $x^4 - x^3 + x^2 + x - 2 = 0$ **for the stable pressure levels Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$.

Correct option: D

Final answer

$$x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$$

Worked Solutions

Higher-Degree Equations

Q110 Challenge

Classify the stable pressure solutions as real or complex Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.

Both types occur: real $x = -1, 1$

2. The correct value is non-real conjugates $x = (1 \pm \sqrt{7}i)/2$.

if $x \geq 0 : x = 1$

Correct option: B

Final answer

Both types occur: real $x = -1, 1$

non-real conjugates $x = (1 \pm \sqrt{7}i)/2$

if $x \geq 0 : x = 1$

Worked Solutions

Higher-Degree Equations

Quiz MCQ 1 **Concept Quiz · MCQ**

Solve $x^3 - 8 = 0$. Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$.

Correct option: D

Final answer

$$x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$$

Quiz MCQ 2 **Concept Quiz · MCQ**

Are $1 + 2i$ and $1 - 2i$ roots of $x^3 - 1 = 0$? Which root must also occur? Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $1 - 2i$.

Correct option: D

Final answer

$$1 - 2i$$

Worked Solutions

Higher-Degree Equations

Quiz MCQ 3 **Concept Quiz · MCQ**

For an imaginary cube root of unity w , evaluate $w^2 + w$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is -1 .

Correct option: B

Final answer

-1

Quiz MCQ 4 **Concept Quiz · MCQ**

For the remaining quadratic factor, a repeated root requires Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $D = 0$.

Correct option: B

Final answer

$D = 0$

Worked Solutions

Higher-Degree Equations

Quiz WRITTEN 5 Concept Quiz · Written

$$\text{Solve } x^4 - 5x^2 + 4 = 0.$$

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: $x = -2, -1, 1, 2$

Final answer

$$x = -2, -1, 1, 2$$

Quiz WRITTEN 6 Concept Quiz · Written

$$\text{For } x^3 + ax^2 + bx + 10 = 0 \text{ with root } 1 + 2i, \text{ find } a, b \text{ and the other roots.}$$

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: $a = 0, b = 1$; other roots: $1 - 2i, -2$

Final answer

$$a = 0, b = 1; \text{ other roots: } 1 - 2i, -2$$

Worked Solutions

Higher-Degree Equations

Quiz WRITTEN 7 Concept Quiz · Written

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$.

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: 1

Final answer

1

Quiz WRITTEN 8 Concept Quiz · Written

Find all real a for $x^3 - 2x^2 + (a - 3)x + a = 0$ to have a repeated root.

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: $a = -4$ or $9/4$

Final answer

$a = -4$ or $9/4$

Answer key

Use the question number shown on each page.

Q001 $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$	Q027 B	Q053 C	Q079 0
Q002 $x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$	Q028 C	Q054 C	Q080 A
Q003 $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$	Q029 D	Q055 $a = 1, b = -6$; other roots: -2	Q081 B
Q004 $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$	Q030 D	Q056 $a = 2, b = 2$; other roots: 1	Q082 D
Q005 $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$	Q031 A	Q057 $a = -7, b = 14$; other roots: 4	Q083 -1
Q006 B	Q032 D	Q058 C	Q084 1
Q007 D	Q033 D	Q059 B	Q085 0
Q008 B	Q034 D	Q060 C	Q086 D
Q009 A	Q035 B	Q061 $a = -3, b = 1$; other roots: $-1, 2 - i$	Q087 C
Q010 C	Q036 D	Q062 $a = -4, b = 6$; other roots: $1 - i, 2$	Q088 D
Q011 $x = -2, 1, 3$	Q037 $x = -3, 3, -\sqrt{2}i, \sqrt{2}i$	Q063 $a = -4, b = 14$; other roots: $1 + 3i, 2$	Q089 $a = -4$ or $9/4$
Q012 $x = -2, 2/3, 1$	Q038 $x = -2, -1, 1, 2$	Q064 $a = 0, b = 1$; other roots: $1 - 2i, -2$	Q090 $a = -3$ or 1
Q013 $x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$	Q039 $x = -3, 3, -2i, 2i$	Q065 A	Q091 $a = 5$ or 9
Q014 $x = -2$ (repeated), 5	Q040 $x = -3, 3, -\sqrt{3}i, \sqrt{3}i$	Q066 B	Q092 $a = -6, -2$, or 10
Q015 $x = -3, -2, -1$	Q041 $x = -\sqrt{2}, \sqrt{2}, -i, i$	Q067 D	Q093 $a = -7, -3$, or 2
Q016 $x = -3, -2, 2$	Q042 D	Q068 B	Q094 D
Q017 $x = -1, 2/3, 1$	Q043 D	Q069 $k = 9$; other roots: $3, 4$	Q095 D
Q018 $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$	Q044 C	Q070 B	Q096 B
Q019 $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$	Q045 C	Q071 1	Q097 A
Q020 $x = 2, 1 - 2i, 1 + 2i$	Q046 B	Q072 1	Q098 A
Q021 $x = -2$ (repeated), 3	Q047 $x = 2$ (repeated), $1 - \sqrt{5}, 1 + \sqrt{5}$	Q073 1	Q099 $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$
Q022 $x = -1$ (repeated), $-3/2$	Q048 $x = 1, 2, 3, 4$	Q074 A	Q100 4
Q023 $x = -2$ (triple)	Q049 $x = 1, 2, 3$ (repeated)	Q075 B	Q101 $a = 2/3, 2$, or 6
Q024 C	Q050 $x = -1, 3, 1 - i, 1 + i$	Q076 D	Q102 $x^4 - x^3 + x^2 + x - 2 = 0$
Q025 D	Q051 B	Q077 -1	Q103 $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$
Q026 D	Q052 A	Q078 -1	

Both types occur: real $x = -1, 1$

Q104 non-real conjugates $x = (1 \pm \sqrt{7}i)/2$
if $x \geq 0 : x = 1$

Q105 C

Q106 D

Q107 D

Q108 A

Q109 D

Q110 B

Quiz MCQ 1 D

Quiz MCQ 2 D

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $x = -2, -1, 1, 2$

Quiz WRITTEN 6 $a = 0, b = 1$; other roots: $1 - 2i, -2$

Quiz WRITTEN 7 1

Quiz WRITTEN 8 $a = -4$ or $9/4$

Concept 1 | Coordinate Geometry Basics

English Student Edition • Focused formulas and guided practice

Formula opener

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Gradient

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Worked Solutions

Coordinate Geometry Basics

Q001 **Written****A (−4), B (8): distance AB****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 12.
3. Final answer: 12.

Final answer

12

Q002 **Written****A (−5), B (7): distance AB****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 12.
3. Final answer: 12.

Final answer

12

Worked Solutions

Coordinate Geometry Basics

Q003 Written

For two points A (-2) and B (6) , find the distance AB.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 8.
3. Final answer: 8.

Final answer

8

Q004 Written

For two points A (4) and B (10) , find the distance AB.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 6.
3. Final answer: 6.

Final answer

6

Worked Solutions

Coordinate Geometry Basics

Q005 MCQ

Find a (-4) , B (8) : distance AB:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 12.

Correct option: C**Final answer**

12

Q006 MCQ

Find a (-5) , B (7) : distance AB:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 12.

Correct option: C**Final answer**

12

Worked Solutions

Coordinate Geometry Basics

Q007 MCQ

For two points A (-2) and B (6) , find the distance AB:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 8.

Correct option: C**Final answer**

8

Q008 MCQ

For two points A (4) and B (10) , find the distance AB:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 6.

Correct option: C**Final answer**

6

Worked Solutions

Coordinate Geometry Basics

Q009 **Written****A (−4), B (8): internal point C in ratio 2:1****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 4.
3. Final answer: 4.

Final answer

4

Q010 **Written****A (−5), B (7): internal point C in ratio 2:1****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 3.
3. Final answer: 3.

Final answer

3

Worked Solutions

Coordinate Geometry Basics

Q011 Written

For two points A (-2) and B (6) , find the coordinate of point C dividing segment AB internally in the ratio 3:1.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 4.
3. Final answer: 4.

Final answer

4

Q012 Written

For two points A (4) and B (10) , find the coordinate of point C dividing segment AB internally in the ratio 2:1.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 8.
3. Final answer: 8.

Final answer

8

Worked Solutions

Coordinate Geometry Basics

Q013 MCQ

Find a (-4) , B (8) : internal point C in ratio 2:1:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 4.

Correct option: B**Final answer**

4

Q014 MCQ

Find a (-5) , B (7) : internal point C in ratio 2:1:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 3.

Correct option: C**Final answer**

3

Worked Solutions

Coordinate Geometry Basics

Q015 MCQ

For two points A (-2) and B (6) , find the coordinate of point C dividing segment AB internally in the ratio 3:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 4.

Correct option: B

Final answer

4

Q016 MCQ

For two points A (4) and B (10) , find the coordinate of point C dividing segment AB internally in the ratio 2:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 8.

Correct option: B

Final answer

8

Worked Solutions

Coordinate Geometry Basics

Q017 **Written****A (−4), B (8): external point D in ratio 2:3****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives -28 .
3. Final answer: -28 .

Final answer -28 Q018 **Written****A (−5), B (7): external point D in ratio 2:5****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives -13 .
3. Final answer: -13 .

Final answer -13

Worked Solutions

Coordinate Geometry Basics

Q019 **Written**

For two points A (-2) and B (6) , find the coordinate of point D dividing segment AB externally in the ratio 5:3.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 18.
3. Final answer: 18.

Final answer

18

Q020 **Written**

For two points A (4) and B (10) , find the coordinate of point D dividing segment AB externally in the ratio 1:3.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 1.
3. Final answer: 1.

Final answer

1

Worked Solutions

Coordinate Geometry Basics

Q021 MCQ

Find a (-4) , B (8) : external point D in ratio 2:3:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -28 .

Correct option: A**Final answer** -28

Q022 MCQ

Find a (-5) , B (7) : external point D in ratio 2:5:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -13 .

Correct option: B**Final answer** -13

Worked Solutions

Coordinate Geometry Basics

Q023 MCQ

For two points A (-2) and B (6) , find the coordinate of point D dividing segment AB externally in the ratio 5:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 18.

Correct option: A**Final answer**

18

Q024 MCQ

For two points A (4) and B (10) , find the coordinate of point D dividing segment AB externally in the ratio 1:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 1.

Correct option: B**Final answer**

1

Worked Solutions

Coordinate Geometry Basics

Q025 **Written****A (−4), B (8): midpoint M****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 2.
3. Final answer: 2.

Final answer

2

Q026 **Written****A (−5), B (7): midpoint M****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 1.
3. Final answer: 1.

Final answer

1

Worked Solutions

Coordinate Geometry Basics

Q027 **Written****For two points A (-2) and B (6) , find the coordinate of midpoint M of segment AB.****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 2.
3. Final answer: 2.

Final answer

2

Q028 **Written****For two points A (4) and B (10) , find the coordinate of midpoint M of segment AB.****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 7.
3. Final answer: 7.

Final answer

7

Worked Solutions

Coordinate Geometry Basics

Q029 MCQ

Find a (-4) , B (8) : midpoint M:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 2.

Correct option: C**Final answer**

2

Q030 MCQ

Find a (-5) , B (7) : midpoint M:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 1.

Correct option: A**Final answer**

1

Worked Solutions

Coordinate Geometry Basics

Q031 MCQ

For two points A (-2) and B (6) , find the coordinate of midpoint M of segment AB:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 2.

Correct option: D**Final answer**

2

Q032 MCQ

For two points A (4) and B (10) , find the coordinate of midpoint M of segment AB:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 7.

Correct option: A**Final answer**

7

Worked Solutions**Coordinate Geometry Basics****Q033 Challenge****Pods A (-8) and B (-3) : fibre-optic tether length****Solution**

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is 5.

Final answer

5

Worked Solutions

Coordinate Geometry Basics

Q034 Challenge**Pods A (-8) , B (-3) : internal sensor point in ratio 1:4****Solution**

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is -7 .

Final answer -7

Worked Solutions

Coordinate Geometry Basics

Q035 Challenge**Pods A (-8) , B (-3) : external outpost in ratio 3:4****Solution**

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is -23 .

Final answer -23

Worked Solutions**Coordinate Geometry Basics****Q036 Challenge****Pods A (−8), B (−3): midpoint supply hub****Solution**

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $-11/2$.

Final answer $-11/2$

Worked Solutions

Coordinate Geometry Basics

Q037 Challenge

Find pods A (-8) and B (-3) : fibre-optic tether length:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: A

Final answer

5

Worked Solutions

Coordinate Geometry Basics

Q038 Challenge

Find pods A (-8) , B (-3) : internal sensor point in ratio 1:4:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -7 .

Correct option: C

Final answer

-7

Worked Solutions

Coordinate Geometry Basics

Q039 Challenge

Find pods A (-8) , B (-3) : external outpost in ratio 3:4:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -23 .

Correct option: B

Final answer

-23

Worked Solutions

Coordinate Geometry Basics

Q040 Challenge**Find pods A (-8) , B (-3) : midpoint supply hub:****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $-11/2$.

Correct option: C**Final answer** $-11/2$

Worked Solutions

Coordinate Geometry Basics

Q041 **Written****Point a has coordinates (2,3): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *I*.

Final answer*I*Q042 **Written****Point b has coordinates (2,-3): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *IV*.

Final answer*IV*

Worked Solutions

Coordinate Geometry Basics

Q043 **Written****Point c has coordinates $(-2,3)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *II*.

Final answer*II*Q044 **Written****Point d has coordinates $(-2,-3)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *III*.

Final answer*III*

Worked Solutions

Coordinate Geometry Basics

Q045 **Written****Point a has coordinates $(-4,2)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *II*.

Final answer*II*Q046 **Written****Point b has coordinates $(1,2)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *I*.

Final answer*I*

Worked Solutions

Coordinate Geometry Basics

Q047 **Written****Point c has coordinates (3,-2): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *IV*.

Final answer*IV*Q048 **Written****Point d has coordinates (-3,-1): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *III*.

Final answer*III*

Worked Solutions

Coordinate Geometry Basics

Q049 **Written****Point a has coordinates (-3,1): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *II*.

Final answer*II*Q050 **Written****Point b has coordinates (2,-8): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *IV*.

Final answer*IV*

Worked Solutions

Coordinate Geometry Basics

Q051 **Written****Point c has coordinates (4,4): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *I*.

Final answer*I*Q052 **Written****Point d has coordinates (-6,-4): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *III*.

Final answer*III*

Worked Solutions

Coordinate Geometry Basics

Q053 Written

Point e has coordinates $(-2,6)$: determine its quadrant**Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *II*.

Final answer*II*

Q054 Written

Point f has coordinates $(3,-3)$: determine its quadrant**Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *IV*.

Final answer*IV*

Worked Solutions

Coordinate Geometry Basics

Q055 **Written****Point g has coordinates $(-5,-7)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *III*.

Final answer*III*Q056 **Written****Point h has coordinates $(1,4)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *I*.

Final answer*I*

Worked Solutions

Coordinate Geometry Basics

Q057 MCQ

Point a has coordinates (2,3): determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *I*.

Correct option: C**Final answer***I*

Q058 MCQ

Point b has coordinates (2,-3): determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *IV*.

Correct option: B**Final answer***IV*

Worked Solutions

Coordinate Geometry Basics

Q059 MCQ

Point c has coordinates $(-2,3)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *II*.

Correct option: A**Final answer***II*

Q060 MCQ

Point d has coordinates $(-2,-3)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *III*.

Correct option: C**Final answer***III*

Worked Solutions

Coordinate Geometry Basics

Q061 MCQ

Point a has coordinates $(-4,2)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *II*.

Correct option: A**Final answer***II*

Q062 MCQ

Point b has coordinates $(1,2)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *I*.

Correct option: C**Final answer***I*

Worked Solutions

Coordinate Geometry Basics

Q063 MCQ

Point c has coordinates (3,-2): determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *IV*.

Correct option: B**Final answer***IV*

Q064 MCQ

Point d has coordinates (-3,-1): determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *III*.

Correct option: A**Final answer***III*

Worked Solutions

Coordinate Geometry Basics

Q065 MCQ

Point a has coordinates $(-3,1)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *II*.

Correct option: B**Final answer***II*

Q066 MCQ

Point b has coordinates $(2,-8)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *IV*.

Correct option: A**Final answer***IV*

Worked Solutions

Coordinate Geometry Basics

Q067 MCQ

Point c has coordinates (4,4): determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *I*.

Correct option: A**Final answer***I*

Q068 MCQ

Point d has coordinates (-6,-4): determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *III*.

Correct option: C**Final answer***III*

Worked Solutions

Coordinate Geometry Basics

Q069 MCQ

Point e has coordinates $(-2,6)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *II*.

Correct option: B**Final answer***II*

Q070 MCQ

Point f has coordinates $(3,-3)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *IV*.

Correct option: C**Final answer***IV*

Worked Solutions

Coordinate Geometry Basics

Q071 MCQ

Point g has coordinates $(-5, -7)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *III*.

Correct option: C**Final answer***III*

Q072 MCQ

Point h has coordinates $(1, 4)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *I*.

Correct option: D**Final answer***I*

Worked Solutions

Coordinate Geometry Basics

Q073 Written

For point A $(-3, 2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-3, -2)$.

Final answer

$(-3, -2)$

Q074 Written

For point A $(-1, 2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-1, -2)$.

Final answer

$(-1, -2)$

Worked Solutions

Coordinate Geometry Basics

Q075 Written

For point A $(-4, -2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-4, 2)$.

Final answer

$(-4, 2)$

Q076 Written

For point A $(1, -2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(1, 2)$.

Final answer

$(1, 2)$

Worked Solutions

Coordinate Geometry Basics

Q077 Written

For point A $(-3, 4)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-3, -4)$.

Final answer

$(-3, -4)$

Q078 MCQ

For point A $(-3, 2)$, find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-3, -2)$.

Correct option: C

Final answer

$(-3, -2)$

Worked Solutions

Coordinate Geometry Basics

Q079 MCQ

For point A $(-1, 2)$, find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-1, -2)$.

Correct option: D

Final answer

$(-1, -2)$

Q080 MCQ

For point A $(-4, -2)$, find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-4, 2)$.

Correct option: C

Final answer

$(-4, 2)$

Worked Solutions

Coordinate Geometry Basics

Q081 MCQ

For point A (1, -2), find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (1, 2).

Correct option: D

Final answer

(1, 2)

Q082 MCQ

For point A (-3, 4), find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (-3, -4).

Correct option: C

Final answer

(-3, -4)

Worked Solutions

Coordinate Geometry Basics

Q083 Written

For point A $(-3, 2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3, 2)$.

Final answer

$(3, 2)$

Q084 Written

For point A $(-1, 2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(1, 2)$.

Final answer

$(1, 2)$

Worked Solutions

Coordinate Geometry Basics

Q085 Written

For point A $(-4, -2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(4, -2)$.

Final answer

$(4, -2)$

Q086 Written

For point A $(1, -2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-1, -2)$.

Final answer

$(-1, -2)$

Worked Solutions

Coordinate Geometry Basics

Q087 Written

For point A $(-3, 4)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3, 4)$.

Final answer

$(3, 4)$

Q088 MCQ

For point A $(-3, 2)$, find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, 2)$.

Correct option: C

Final answer

$(3, 2)$

Worked Solutions

Coordinate Geometry Basics

Q089 MCQ

For point A $(-1, 2)$, find the coordinates of point Q symmetric with respect to the y-axis:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(1, 2)$.

Correct option: C**Final answer** $(1, 2)$

Q090 MCQ

For point A $(-4, -2)$, find the coordinates of point Q symmetric with respect to the y-axis:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4, -2)$.

Correct option: A**Final answer** $(4, -2)$

Worked Solutions

Coordinate Geometry Basics

Q091 MCQ

For point A $(1, -2)$, find the coordinates of point Q symmetric with respect to the y-axis:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-1, -2)$.

Correct option: A**Final answer** $(-1, -2)$

Q092 MCQ

For point A $(-3, 4)$, find the coordinates of point Q symmetric with respect to the y-axis:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, 4)$.

Correct option: A**Final answer** $(3, 4)$

Worked Solutions

Coordinate Geometry Basics

Q093 **Written****For point A $(-3, 2)$, find the coordinates of point R symmetric with respect to the origin.****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3, -2)$.

Final answer $(3, -2)$ Q094 **Written****For point A $(-1, 2)$, find the coordinates of point R symmetric with respect to the origin.****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(1, -2)$.

Final answer $(1, -2)$

Worked Solutions

Coordinate Geometry Basics

Q095 Written

For point A $(-4, -2)$, find the coordinates of point R symmetric with respect to the origin.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(4, 2)$.

Final answer

$(4, 2)$

Q096 Written

For point A $(1, -2)$, find the coordinates of point R symmetric with respect to the origin.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-1, 2)$.

Final answer

$(-1, 2)$

Worked Solutions

Coordinate Geometry Basics

Q097 **Written****For point A $(-3, 4)$, find the coordinates of point R symmetric with respect to the origin.****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3, -4)$.

Final answer $(3, -4)$ Q098 **MCQ****For point A $(-3, 2)$, find the coordinates of point R symmetric with respect to the origin:****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, -2)$.

Correct option: B**Final answer** $(3, -2)$

Worked Solutions

Coordinate Geometry Basics

Q099 MCQ

For point A $(-1, 2)$, find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(1, -2)$.

Correct option: B

Final answer

$(1, -2)$

Q100 MCQ

For point A $(-4, -2)$, find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4, 2)$.

Correct option: B

Final answer

$(4, 2)$

Worked Solutions

Coordinate Geometry Basics

Q101 MCQ

For point A (1, -2), find the coordinates of point R symmetric with respect to the origin:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (-1, 2).

Correct option: A**Final answer**

(-1, 2)

Q102 MCQ

For point A (-3, 4), find the coordinates of point R symmetric with respect to the origin:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (3, -4).

Correct option: D**Final answer**

(3, -4)

Worked Solutions

Coordinate Geometry Basics

Q103 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(-3, -4)$.

Final answer

$(-3, -4)$

Worked Solutions

Coordinate Geometry Basics

Q104 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(3, 4)$.

Final answer

$(3, 4)$

Worked Solutions

Coordinate Geometry Basics

Q105 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point R symmetric with respect to the origin.

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(3, -4)$.

Final answer

$(3, -4)$

Worked Solutions

Coordinate Geometry Basics

Q106 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-3, -4)$.

Correct option: A

Final answer

$(-3, -4)$

Worked Solutions

Coordinate Geometry Basics

Q107 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, 4)$.

Correct option: B

Final answer

$(3, 4)$

Worked Solutions

Coordinate Geometry Basics

Q108 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, -4)$.

Correct option: C

Final answer

$(3, -4)$

Worked Solutions

Coordinate Geometry Basics

Q109 Written

Distance between A (2, 6) and B (5, 10)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives 5.

Final answer

5

Q110 Written

Distance between A (4, 3) and B (1, 2)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $\sqrt{10}$.

Final answer $\sqrt{10}$

Worked Solutions

Coordinate Geometry Basics

Q111 Written

Distance between A($\sqrt{3}$, -1) and B (0, 1)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $\sqrt{7}$.

Final answer

$$\sqrt{7}$$

Q112 Written

Distance between A (1, 2) and B (11, 7)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $5\sqrt{5}$.

Final answer

$$5\sqrt{5}$$

Worked Solutions

Coordinate Geometry Basics

Q113 Written

Distance between A (3, -1) and B (-1, 3)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $4\sqrt{2}$.

Final answer

$$4\sqrt{2}$$

Q114 MCQ

Find distance between A (2, 6) and B (5, 10):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: C**Final answer**

$$5$$

Worked Solutions

Coordinate Geometry Basics

Q115 MCQ

Find distance between A (4, 3) and B (1, 2):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $\sqrt{10}$.

Correct option: A**Final answer**

$$\sqrt{10}$$

Q116 MCQ

Find distance between A($\sqrt{3}$, -1) and B (0, 1):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $\sqrt{7}$.

Correct option: C**Final answer**

$$\sqrt{7}$$

Worked Solutions

Coordinate Geometry Basics

Q117 MCQ

Find distance between A (1, 2) and B (11, 7):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $5\sqrt{5}$.

Correct option: B**Final answer**

$$5\sqrt{5}$$

Q118 MCQ

Find distance between A (3, -1) and B (-1, 3):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $4\sqrt{2}$.

Correct option: A**Final answer**

$$4\sqrt{2}$$

Worked Solutions

Coordinate Geometry Basics

Q119 Written

A (3, -2), **B** (x, 1), **distance 5: find x****Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -1, 7$.

Final answer

$$x = -1, 7$$

Q120 Written

A (-1, 3), **B** (x, 1), **distance $\sqrt{5}$: find x****Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -2, 0$.

Final answer

$$x = -2, 0$$

Worked Solutions

Coordinate Geometry Basics

Q121 Written

A (3, -1), B (x, 2), distance 5: find x

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -1, 7$.

Final answer

$$x = -1, 7$$

Q122 Written

A (1, -3), B (-2, y), distance $\sqrt{13}$: find y

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $y = -5, -1$.

Final answer

$$y = -5, -1$$

Worked Solutions

Coordinate Geometry Basics

Q123 **Written****A** $(-4, -2)$, **B** $(x, 3)$, **distance** $5\sqrt{5}$: **find x****Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -14, 6$.

Final answer

$$x = -14, 6$$

Q124 **MCQ****A** $(3, -2)$, **B** $(x, 1)$, **distance** 5: **find x:****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -1, 7$.

Correct option: B**Final answer**

$$x = -1, 7$$

Worked Solutions

Coordinate Geometry Basics

Q125 MCQ

A $(-1, 3)$, **B** $(x, 1)$, **distance** $\sqrt{5}$: **find x:****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -2, 0$.

Correct option: C**Final answer**

$$x = -2, 0$$

Q126 MCQ

A $(3, -1)$, **B** $(x, 2)$, **distance** 5: **find x:****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -1, 7$.

Correct option: B**Final answer**

$$x = -1, 7$$

Worked Solutions

Coordinate Geometry Basics

Q127 MCQ

A (1, -3), B (-2, y), distance $\sqrt{13}$: find y:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $y = -5, -1$.

Correct option: A

Final answer

$$y = -5, -1$$

Q128 MCQ

A (-4, -2), B (x, 3), distance $5\sqrt{5}$: find x:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -14, 6$.

Correct option: D

Final answer

$$x = -14, 6$$

Worked Solutions

Coordinate Geometry Basics

Q129 **Written****Distance from origin to A (2, -4)****Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $2\sqrt{5}$.

Final answer

$$2\sqrt{5}$$

Q130 **Written****Distance from origin to A (-3, 4)****Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives 5.

Final answer

$$5$$

Worked Solutions

Coordinate Geometry Basics

Q131 MCQ

Find distance from origin to A (2, -4):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $2\sqrt{5}$.

Correct option: C**Final answer**

$$2\sqrt{5}$$

Q132 MCQ

Find distance from origin to A (-3, 4):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: C**Final answer**

$$5$$

Worked Solutions

Coordinate Geometry Basics

Q133 Challenge

Distance between $(x, 7)$ and $(3x-1, -5)$ is 13: find x

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $x = -2, 3$.

Final answer

$$x = -2, 3$$

Worked Solutions

Coordinate Geometry Basics

Q134 Challenge

Distance between $(x, 7)$ and $(3x-1, -5)$ is 13: find x :

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -2, 3$.

Correct option: C

Final answer

$$x = -2, 3$$

Worked Solutions

Coordinate Geometry Basics

Q135 **Written****Point on the x-axis equidistant from A $(-2, 1)$ and B $(3, 4)$** **Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(2, 0)$.

Final answer $(2, 0)$ Q136 **Written****Point on the y-axis equidistant from A $(4, 2)$ and B $(1, -1)$** **Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(0, 3)$.

Final answer $(0, 3)$

Worked Solutions

Coordinate Geometry Basics

Q137 **Written****Point on the x-axis equidistant from A $(-1, 2)$ and B $(4, 3)$** **Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(2, 0)$.

Final answer $(2, 0)$ Q138 **Written****Point on the y-axis equidistant from A $(-5, 2)$ and B $(3, 5)$** **Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(0, 5/6)$.

Final answer $(0, 5/6)$

Worked Solutions

Coordinate Geometry Basics

Q139 **Written****Point on the x-axis equidistant from A $(-1, 2)$ and B $(3, 8)$** **Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(17/2, 0)$.

Final answer $(17/2, 0)$ Q140 **MCQ****Find point on the x-axis equidistant from A $(-2, 1)$ and B $(3, 4)$:****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, 0)$.

Correct option: C**Final answer** $(2, 0)$

Worked Solutions

Coordinate Geometry Basics

Q141 MCQ

Find point on the y-axis equidistant from A (4, 2) and B (1, -1):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (0, 3).

Correct option: A**Final answer**

(0, 3)

Q142 MCQ

Find point on the x-axis equidistant from A (-1, 2) and B (4, 3):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (2, 0).

Correct option: C**Final answer**

(2, 0)

Worked Solutions

Coordinate Geometry Basics

Q143 MCQ

Find point on the y-axis equidistant from A $(-5, 2)$ and B $(3, 5)$:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0, 5/6)$.

Correct option: C**Final answer** $(0, 5/6)$

Q144 MCQ

Find point on the x-axis equidistant from A $(-1, 2)$ and B $(3, 8)$:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(17/2, 0)$.

Correct option: B**Final answer** $(17/2, 0)$

Worked Solutions

Coordinate Geometry Basics

Q145 **Written****Point on $y = 2x - 1$ equidistant from A (1, 4) and B (3, 1)****Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(13/8, 9/4)$.

Final answer $(13/8, 9/4)$ Q146 **Written****Point on $y = 3x + 1$ equidistant from A (4, -5) and B (2, -3)****Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(-4, -11)$.

Final answer $(-4, -11)$

Worked Solutions

Coordinate Geometry Basics

Q147 **Written****Point on $y = 2x - 3$ equidistant from A (1, 4) and B (3, -2)****Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are (2, 1).

Final answer

(2, 1)

Q148 **Written****Point on $y = x - 1$ equidistant from A (-1, 5) and B (7, -1)****Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are (3, 2).

Final answer

(3, 2)

Worked Solutions

Coordinate Geometry Basics

Q149 **Written****Point on $y = -x + 1$ equidistant from A (1, 2) and B (−3, 6)****Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are (−2, 3).

Final answer

(−2, 3)

Q150 **MCQ****Find point on $y = 2x - 1$ equidistant from A (1, 4) and B (3, 1):****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (13/8, 9/4).

Correct option: B**Final answer**

(13/8, 9/4)

Worked Solutions

Coordinate Geometry Basics

Q151 MCQ

Find point on $y = 3x + 1$ equidistant from A (4, -5) and B (2, -3):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (-4, -11).

Correct option: C**Final answer**

(-4, -11)

Q152 MCQ

Find point on $y = 2x - 3$ equidistant from A (1, 4) and B (3, -2):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (2, 1).

Correct option: A**Final answer**

(2, 1)

Worked Solutions

Coordinate Geometry Basics

Q153 MCQ

Find point on $y = x - 1$ equidistant from A $(-1, 5)$ and B $(7, -1)$:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, 2)$.

Correct option: C**Final answer** $(3, 2)$

Q154 MCQ

Find point on $y = -x + 1$ equidistant from A $(1, 2)$ and B $(-3, 6)$:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-2, 3)$.

Correct option: B**Final answer** $(-2, 3)$

Worked Solutions

Coordinate Geometry Basics

Q155 Challenge

A (2, 1), B (4, 5), C (3, y) with CA=CB and D (x, 0) with DA=DB: find CD

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $3\sqrt{5}$.

Final answer

$$3\sqrt{5}$$

Worked Solutions

Coordinate Geometry Basics

Q156 Challenge

A (2, 1), B (4, 5), C (3, y) with CA=CB and D (x, 0) with DA=DB: find CD:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $3\sqrt{5}$.

Correct option: A

Final answer

$$3\sqrt{5}$$

Worked Solutions

Coordinate Geometry Basics

Q157 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(4/3, 11/3)$.

Final answer $(4/3, 11/3)$

Worked Solutions

Coordinate Geometry Basics

Q158 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point P dividing segment AC internally in the ratio 3:1.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-15/4, 13/4)$.

Final answer $(-15/4, 13/4)$

Worked Solutions

Coordinate Geometry Basics

Q159 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point P dividing segment AB internally in the ratio 4:1.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are (2, -9/5).

Final answer

(2, -9/5)

Worked Solutions

Coordinate Geometry Basics

Q160 Written

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point P dividing segment BC internally in the ratio 3:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(3, 3)$.

Final answer

$(3, 3)$

Worked Solutions

Coordinate Geometry Basics

Q161 Written

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-2/3, -5/3)$.

Final answer

$(-2/3, -5/3)$

Worked Solutions

Coordinate Geometry Basics

Q162 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4/3, 11/3)$.

Correct option: C

Final answer

$(4/3, 11/3)$

Worked Solutions

Coordinate Geometry Basics

Q163 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point P dividing segment AC internally in the ratio 3:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-15/4, 13/4)$.

Correct option: C

Final answer

$(-15/4, 13/4)$

Worked Solutions

Coordinate Geometry Basics

Q164 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point P dividing segment AB internally in the ratio 4:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (2, -9/5).

Correct option: C

Final answer

(2, -9/5)

Worked Solutions

Coordinate Geometry Basics

Q165 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point P dividing segment BC internally in the ratio 3:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, 3)$.

Correct option: C

Final answer

$(3, 3)$

Worked Solutions

Coordinate Geometry Basics

Q166 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-2/3, -5/3)$.

Correct option: A

Final answer

$(-2/3, -5/3)$

Worked Solutions

Coordinate Geometry Basics

Q167 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2, 19/2)$.

Final answer $(2, 19/2)$

Worked Solutions

Coordinate Geometry Basics

Q168 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(11, -18)$.

Final answer $(11, -18)$

Worked Solutions

Coordinate Geometry Basics

Q169 Written

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point Q dividing segment BC externally in the ratio 2:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are (10, -15).

Final answer

(10, -15)

Worked Solutions

Coordinate Geometry Basics

Q170 Written

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point Q dividing segment CA externally in the ratio 3:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-8, -11)$.

Final answer

$(-8, -11)$

Worked Solutions

Coordinate Geometry Basics

Q171 **Written**

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(0, -15/2)$.

Final answer $(0, -15/2)$

Worked Solutions

Coordinate Geometry Basics

Q172 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, 19/2)$.

Correct option: B

Final answer

$(2, 19/2)$

Worked Solutions

Coordinate Geometry Basics

Q173 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(11, -18)$.

Correct option: C

Final answer

$(11, -18)$

Worked Solutions

Coordinate Geometry Basics

Q174 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point Q dividing segment BC externally in the ratio 2:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (10, -15).

Correct option: B

Final answer

(10, -15)

Worked Solutions

Coordinate Geometry Basics

Q175 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point Q dividing segment CA externally in the ratio 3:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-8, -11)$.

Correct option: B

Final answer

$(-8, -11)$

Worked Solutions

Coordinate Geometry Basics

Q176 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0, -15/2)$.

Correct option: C**Final answer** $(0, -15/2)$

Worked Solutions

Coordinate Geometry Basics

Q177 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of midpoint M of segment CA.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(3/2, -1/2)$.

Final answer $(3/2, -1/2)$

Worked Solutions

Coordinate Geometry Basics

Q178 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2, 2)$.

Final answer $(2, 2)$

Worked Solutions

Coordinate Geometry Basics

Q179 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of midpoint M of segment AB.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-1, 0)$.

Final answer $(-1, 0)$

Worked Solutions

Coordinate Geometry Basics

Q180 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-2, 1)$.

Final answer $(-2, 1)$

Worked Solutions

Coordinate Geometry Basics

Q181 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of midpoint M of segment CA.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are (0, 3).

Final answer

(0, 3)

Worked Solutions

Coordinate Geometry Basics

Q182 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2/3, 1)$.

Final answer $(2/3, 1)$

Worked Solutions

Coordinate Geometry Basics

Q183 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of midpoint M of segment AB.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2, -1)$.

Final answer $(2, -1)$

Worked Solutions

Coordinate Geometry Basics

Q184 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(5/3, 5/3)$.

Final answer $(5/3, 5/3)$

Worked Solutions

Coordinate Geometry Basics

Q185 Written

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of midpoint M of segment BC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2, -5/2)$.

Final answer

$(2, -5/2)$

Worked Solutions

Coordinate Geometry Basics

Q186 **Written**

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(0, 0)$.

Final answer $(0, 0)$

Worked Solutions

Coordinate Geometry Basics

Q187 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of midpoint M of segment CA:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3/2, -1/2)$.

Correct option: A

Final answer

$(3/2, -1/2)$

Worked Solutions

Coordinate Geometry Basics

Q188 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, 2)$.

Correct option: C

Final answer

$(2, 2)$

Worked Solutions

Coordinate Geometry Basics

Q189 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of midpoint M of segment AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-1, 0)$.

Correct option: B

Final answer

$(-1, 0)$

Worked Solutions

Coordinate Geometry Basics

Q190 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-2, 1)$.

Correct option: A

Final answer

$(-2, 1)$

Worked Solutions

Coordinate Geometry Basics

Q191 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of midpoint M of segment CA:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (0, 3).

Correct option: A

Final answer

(0, 3)

Worked Solutions

Coordinate Geometry Basics

Q192 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (2/3, 1).

Correct option: D

Final answer

(2/3, 1)

Worked Solutions

Coordinate Geometry Basics

Q193 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of midpoint M of segment AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, -1)$.

Correct option: B

Final answer

$(2, -1)$

Worked Solutions

Coordinate Geometry Basics

Q194 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(5/3, 5/3)$.

Correct option: A

Final answer

$(5/3, 5/3)$

Worked Solutions

Coordinate Geometry Basics

Q195 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of midpoint M of segment BC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, -5/2)$.

Correct option: B

Final answer

$(2, -5/2)$

Worked Solutions

Coordinate Geometry Basics

Q196 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0, 0)$.

Correct option: C

Final answer

$(0, 0)$

Worked Solutions

Coordinate Geometry Basics

Q197 Challenge

A $(-2, 3)$, **B** $(6, 1)$, **C** $(0, y)$, **CA=CB: find the centroid G**

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(4/3, -2/3)$.

Final answer

$(4/3, -2/3)$

Worked Solutions

Coordinate Geometry Basics

Q198 Challenge

A $(-2, 3)$, **B** $(6, 1)$, **C** $(0, y)$, **CA=CB: find the centroid G:**

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4/3, -2/3)$.

Correct option: C

Final answer

$(4/3, -2/3)$

Worked Solutions

Coordinate Geometry Basics

Q199 **Written****Point with A $(-3, 2)$ symmetric about centre $(2, 6)$** **Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(7, 10)$.

Final answer $(7, 10)$ Q200 **Written****Point with A $(1, -2)$ symmetric about centre $(2, 3)$** **Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(3, 8)$.

Final answer $(3, 8)$

Worked Solutions

Coordinate Geometry Basics

Q201 **Written****Point with A (3, 2) symmetric about centre (-1, 5)****Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-5, 8)$.

Final answer $(-5, 8)$ Q202 **Written****Point with A $(-3, 1)$ symmetric about centre $(2, 0)$** **Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(7, -1)$.

Final answer $(7, -1)$

Worked Solutions

Coordinate Geometry Basics

Q203 **Written****Point with A (7, 4) symmetric about centre (-3, 5)****Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-13, 6)$.

Final answer $(-13, 6)$ Q204 **Written****Point with A (4, -8) symmetric about centre (-3, 2)****Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-10, 12)$.

Final answer $(-10, 12)$

Worked Solutions

Coordinate Geometry Basics

Q205 **Written****Point with A (0, -4) symmetric about centre (-4, 1)****Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-8, 6)$.

Final answer $(-8, 6)$ Q206 **Written****Point with A $(-4, 2)$ symmetric about centre $(-1, -3)$** **Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(2, -8)$.

Final answer $(2, -8)$

Worked Solutions

Coordinate Geometry Basics

Q207 MCQ

Find point with A $(-3, 2)$ symmetric about centre $(2, 6)$:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(7, 10)$.

Correct option: C**Final answer** $(7, 10)$

Q208 MCQ

Find point with A $(1, -2)$ symmetric about centre $(2, 3)$:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, 8)$.

Correct option: B**Final answer** $(3, 8)$

Worked Solutions

Coordinate Geometry Basics

Q209 MCQ

Find point with A (3, 2) symmetric about centre (-1, 5):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (-5, 8).

Correct option: A**Final answer**

(-5, 8)

Q210 MCQ

Find point with A (-3, 1) symmetric about centre (2, 0):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (7, -1).

Correct option: C**Final answer**

(7, -1)

Worked Solutions

Coordinate Geometry Basics

Q211 MCQ

Find point with A (7, 4) symmetric about centre (-3, 5):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-13, 6)$.

Correct option: C**Final answer** $(-13, 6)$

Q212 MCQ

Find point with A (4, -8) symmetric about centre (-3, 2):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-10, 12)$.

Correct option: C**Final answer** $(-10, 12)$

Worked Solutions

Coordinate Geometry Basics

Q213 MCQ

Find point with A (0, -4) symmetric about centre (-4, 1):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (-8, 6).

Correct option: B**Final answer**

(-8, 6)

Q214 MCQ

Find point with A (-4, 2) symmetric about centre (-1, -3):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (2, -8).

Correct option: A**Final answer**

(2, -8)

Worked Solutions

Coordinate Geometry Basics

Q215 Challenge

B (m , 5), **C** (4, n) symmetric about **A** (1, -1): find $m+n$

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is -9 .

Final answer

-9

Worked Solutions

Coordinate Geometry Basics

Q216 Challenge

B (m , 5), **C** (4, n) symmetric about **A** (1, -1): find $m+n$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -9 .

Correct option: C

Final answer

-9

Worked Solutions

Coordinate Geometry Basics

Quiz MCQ 1 [Concept Quiz · MCQ](#)

Point A ($-3, 4$) lies in which quadrant?:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *II*.

Correct option: A

Final answer

II

Quiz MCQ 2 [Concept Quiz · MCQ](#)

Find the distance between A ($1, 2$) and B ($4, 6$):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: B

Final answer

5

Worked Solutions

Coordinate Geometry Basics

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

Point P divides the segment from A $(-2, 3)$ to B $(6, 1)$ internally in ratio 1:3. Find P:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0, 5/2)$.

Correct option: C

Final answer

$(0, 5/2)$

Quiz MCQ 4 [Concept Quiz](#) · [MCQ](#)

Point B is symmetric to A $(2, -1)$ about P $(-3, 4)$. Find B:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-8, 9)$.

Correct option: C

Final answer

$(-8, 9)$

Worked Solutions

Coordinate Geometry Basics

Quiz WRITTEN 5 Concept Quiz · Written

Find the point on $y = 2x + 1$ equidistant from A (0, 2) and B (4, 0).

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are (1, 3).

Final answer

(1, 3)

Quiz WRITTEN 6 Concept Quiz · Written

For points A (−5) and B (7), find the point dividing AB externally in ratio 2:5.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives −13.
3. Final answer: −13.

Final answer

−13

Worked Solutions

Coordinate Geometry Basics

Quiz WRITTEN 7 Concept Quiz · Written

If the distance between A (3, -1) and B (x, 2) is 5, find x.

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -1, 7$.

Final answer

$$x = -1, 7$$

Quiz WRITTEN 8 Concept Quiz · Written

Find the centroid of A (-2, 1), B (6, -3), C (1, 7).

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are (5/3, 5/3).

Final answer

$$(5/3, 5/3)$$

Answer key

Use the question number shown on each page.

Q001 12	Q027 2	Q053 <i>II</i>	Q079 D
Q002 12	Q028 7	Q054 <i>IV</i>	Q080 C
Q003 8	Q029 C	Q055 <i>III</i>	Q081 D
Q004 6	Q030 A	Q056 <i>I</i>	Q082 C
Q005 C	Q031 D	Q057 C	Q083 (3, 2)
Q006 C	Q032 A	Q058 B	Q084 (1, 2)
Q007 C	Q033 5	Q059 A	Q085 (4, -2)
Q008 C	Q034 -7	Q060 C	Q086 (-1, -2)
Q009 4	Q035 -23	Q061 A	Q087 (3, 4)
Q010 3	Q036 -11/2	Q062 C	Q088 C
Q011 4	Q037 A	Q063 B	Q089 C
Q012 8	Q038 C	Q064 A	Q090 A
Q013 B	Q039 B	Q065 B	Q091 A
Q014 C	Q040 C	Q066 A	Q092 A
Q015 B	Q041 <i>I</i>	Q067 A	Q093 (3, -2)
Q016 B	Q042 <i>IV</i>	Q068 C	Q094 (1, -2)
Q017 -28	Q043 <i>II</i>	Q069 B	Q095 (4, 2)
Q018 -13	Q044 <i>III</i>	Q070 C	Q096 (-1, 2)
Q019 18	Q045 <i>II</i>	Q071 C	Q097 (3, -4)
Q020 1	Q046 <i>I</i>	Q072 D	Q098 B
Q021 A	Q047 <i>IV</i>	Q073 (-3, -2)	Q099 B
Q022 B	Q048 <i>III</i>	Q074 (-1, -2)	Q100 B
Q023 A	Q049 <i>II</i>	Q075 (-4, 2)	Q101 A
Q024 B	Q050 <i>IV</i>	Q076 (1, 2)	Q102 D
Q025 2	Q051 <i>I</i>	Q077 (-3, -4)	Q103 (-3, -4)
Q026 1	Q052 <i>III</i>	Q078 C	Q104 (3, 4)

Q105 (3, -4)

Q106 A

Q107 B

Q108 C

Q109 5

Q110 $\sqrt{10}$ Q111 $\sqrt{7}$ Q112 $5\sqrt{5}$ Q113 $4\sqrt{2}$

Q114 C

Q115 A

Q116 C

Q117 B

Q118 A

Q119 $x = -1, 7$ Q120 $x = -2, 0$ Q121 $x = -1, 7$ Q122 $y = -5, -1$ Q123 $x = -14, 6$

Q124 B

Q125 C

Q126 B

Q127 A

Q128 D

Q129 $2\sqrt{5}$

Q130 5

Q131 C

Q132 C

Q133 $x = -2, 3$

Q134 C

Q135 (2, 0)

Q136 (0, 3)

Q137 (2, 0)

Q138 (0, $5/6$)

Q139 (17/2, 0)

Q140 C

Q141 A

Q142 C

Q143 C

Q144 B

Q145 (13/8, 9/4)

Q146 (-4, -11)

Q147 (2, 1)

Q148 (3, 2)

Q149 (-2, 3)

Q150 B

Q151 C

Q152 A

Q153 C

Q154 B

Q155 $3\sqrt{5}$

Q156 A

Q157 (4/3, 11/3)

Q158 (-15/4, 13/4)

Q159 (2, -9/5)

Q160 (3, 3)

Q161 (-2/3, -5/3)

Q162 C

Q163 C

Q164 C

Q165 C

Q166 A

Q167 (2, 19/2)

Q168 (11, -18)

Q169 (10, -15)

Q170 (-8, -11)

Q171 (0, -15/2)

Q172 B

Q173 C

Q174 B

Q175 B

Q176 C

Q177 (3/2, -1/2)

Q178 (2, 2)

Q179 (-1, 0)

Q180 (-2, 1)

Q181 (0, 3)

Q182 (2/3, 1)

Q183 (2, -1)

Q184 (5/3, 5/3)

Q185 (2, -5/2)

Q186 (0, 0)

Q187 A

Q188 C

Q189 B

Q190 A

Q191 A

Q192 D

Q193 B

Q194 A

Q195 B

Q196 C

Q197 (4/3, -2/3)

Q198 C

Q199 (7, 10)

Q200 (3, 8)

Q201 (-5, 8)

Q202 (7, -1)

Q203 (-13, 6)

Q204 (-10, 12)

Q205 (-8, 6)

Q206 (2, -8)

Q207 C

Q208 B

Q209 A

Q210 C

Q211 C

Q212 C

Q213 B

Q214 A

Q215 -9

Q216 C

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 C

Quiz WRITTEN 5 (1, 3)

Quiz WRITTEN 6 -13

Quiz WRITTEN 7 $x = -1, 7$

Quiz WRITTEN 8 (5/3, 5/3)

Concept 2 | Triangles and Lines

English Student Edition • Focused formulas and guided practice

Formula opener

Line

$$y - y_1 = m(x - x_1)$$

Perpendicular

$$m_1 m_2 = -1$$

Area

$$A = \frac{1}{2}bh$$

Worked Solutions

Triangles and Lines

Q001 **Written**

Determine the shape of triangle ABC with A (3, 2), B (−1, 0), C $(1 + \sqrt{3}, 1 - 2\sqrt{3})$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *equilateral*.

Final answer*equilateral*

Worked Solutions

Triangles and Lines

Q002 **Written**

Determine the shape of triangle ABC with A (2, 2), B (2, 0), C $(2 + \sqrt{3}, 1)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *equilateral*.

Final answer*equilateral*

Worked Solutions

Triangles and Lines

Q003 MCQ

Determine the shape of triangle ABC with A (3, 2), B (−1, 0), C $(1 + \sqrt{3}, 1 - 2\sqrt{3})$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *equilateral*.

Correct option: D

Final answer

equilateral

Worked Solutions

Triangles and Lines

Q004 MCQ

Determine the shape of triangle ABC with A (2, 2), B (2, 0), C $(2 + \sqrt{3}, 1)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *equilateral*.

Correct option: C

Final answer*equilateral*

Q005 Written

Determine the shape of triangle ABC with A (−1, 2), B (1, 1), C (0, 4).

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *isosceles right – angled; right angle at A*.

Final answer*isosceles right – angled; right angle at A*

Worked Solutions

Triangles and Lines

Q006 Written

Determine the shape of triangle ABC with A (0, 2), B (−1, −2), C (4, 1).**Solution**

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *isosceles right – angled; right angle at A*.

Final answer*isosceles right – angled; right angle at A*

Q007 MCQ

Determine the shape of triangle ABC with A (−1, 2), B (1, 1), C (0, 4):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *isosceles right – angled; right angle at A*.

Correct option: B**Final answer***isosceles right – angled; right angle at A*

Worked Solutions

Triangles and Lines

Q008 MCQ

Determine the shape of triangle ABC with A (0, 2), B (−1, −2), C (4, 1):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *isosceles right – angled; right angle at A*.

Correct option: D**Final answer***isosceles right – angled; right angle at A*

Q009 Written

Determine the shape of triangle ABC with A (−3, 4), B (1, 7), C (3, −4).**Solution**

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *scalene right – angled; right angle at A*.

Final answer*scalene right – angled; right angle at A*

Worked Solutions

Triangles and Lines

Q010 MCQ

Determine the shape of triangle ABC with A (−3, 4), B (1, 7), C (3, −4):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *scalene right – angled; right angle at A*.

Correct option: D**Final answer***scalene right – angled; right angle at A*

Q011 Written

Determine the shape of triangle ABC with A (4, 9), B (0, 6), C (3, 2).**Solution**

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *isosceles right – angled; right angle at B*.

Final answer*isosceles right – angled; right angle at B*

Worked Solutions

Triangles and Lines

Q012 Written

Determine the shape of triangle ABC with A (1, 2), B (5, 8), C (11, 4).

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *isosceles right – angled; right angle at B.*

Final answer

isosceles right – angled; right angle at B

Q013 Written

Determine the shape of triangle ABC with A (−1, 1), B (4, 2), C (1, 4).

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *isosceles right – angled; right angle at C.*

Final answer

isosceles right – angled; right angle at C

Worked Solutions

Triangles and Lines

Q014 MCQ

Determine the shape of triangle ABC with A (4, 9), B (0, 6), C (3, 2):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *isosceles right – angled; right angle at B*.

Correct option: B**Final answer***isosceles right – angled; right angle at B*

Q015 MCQ

Determine the shape of triangle ABC with A (1, 2), B (5, 8), C (11, 4):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *isosceles right – angled; right angle at B*.

Correct option: C**Final answer***isosceles right – angled; right angle at B*

Worked Solutions

Triangles and Lines

Q016 MCQ

Determine the shape of triangle ABC with A $(-1, 1)$, B $(4, 2)$, C $(1, 4)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *isosceles right – angled; right angle at C*.

Correct option: B**Final answer***isosceles right – angled; right angle at C*

Worked Solutions

Triangles and Lines

Q017 Challenge

If A (1, 2), B (2k, k), C (4, 5) and ABC is isosceles and right-angled at B, find k.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $k = 2$.

Final answer

$$k = 2$$

Worked Solutions

Triangles and Lines

Q018 Challenge

If A (1, 2), B (2k, k), C (4, 5) and ABC is isosceles and right-angled at B, find k:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $k = 2$.

Correct option: C

Final answer

$$k = 2$$

Worked Solutions

Triangles and Lines

Q019 Written

Find the equation of the line through (3,-4) parallel to the x-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -4$.

Final answer

$$y = -4$$

Q020 Written

Find the equation of the line through (-3,8) parallel to the y-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -3$.

Final answer

$$x = -3$$

Worked Solutions

Triangles and Lines

Q021 Written

Find the equation of the line through (2,6) parallel to the x-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 6$.

Final answer

$$y = 6$$

Q022 Written

Find the equation of the line through (3,5) perpendicular to the x-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 3$.

Final answer

$$x = 3$$

Worked Solutions

Triangles and Lines

Q023 Written

Find the equation of the line through $(-4,5)$ parallel to x-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 5$.

Final answer

$$y = 5$$

Q024 Written

Find the equation of the line through $(2,1)$ parallel to x-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 1$.

Final answer

$$y = 1$$

Worked Solutions

Triangles and Lines

Q025 Written

Find the equation of the line through $(-4, -7)$ parallel to x-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -7$.

Final answer

$$y = -7$$

Q026 Written

Find the equation of the line through $(1, -2)$ perpendicular to x-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 1$.

Final answer

$$x = 1$$

Worked Solutions

Triangles and Lines

Q027 Written

Find the equation of the line through $(-3, -4)$ perpendicular to x-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -3$.

Final answer

$$x = -3$$

Q028 Written

Find the equation of the line through $(3, -5)$ perpendicular to x-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 3$.

Final answer

$$x = 3$$

Worked Solutions

Triangles and Lines

Q029 Written

Find the equation of the line through (2,-7) parallel to y-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 2$.

Final answer

$$x = 2$$

Q030 Written

Find the equation of the line through (4,3) parallel to y-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 4$.

Final answer

$$x = 4$$

Worked Solutions

Triangles and Lines

Q031 Written

Find the equation of the line through $(-6, -2)$ parallel to y-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -6$.

Final answer

$$x = -6$$

Q032 MCQ

Find the equation of the line through $(3, -4)$ parallel to the x-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -4$.

Correct option: B**Final answer**

$$y = -4$$

Worked Solutions

Triangles and Lines

Q033 MCQ

Find the equation of the line through $(-3, 8)$ parallel to the y-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -3$.

Correct option: C**Final answer**

$$x = -3$$

Q034 MCQ

Find the equation of the line through $(2, 6)$ parallel to the x-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 6$.

Correct option: D**Final answer**

$$y = 6$$

Worked Solutions

Triangles and Lines

Q035 MCQ

Find the equation of the line through (3,5) perpendicular to the x-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 3$.

Correct option: D**Final answer**

$$x = 3$$

Q036 MCQ

Find the equation of the line through (-4,5) parallel to x-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 5$.

Correct option: B**Final answer**

$$y = 5$$

Worked Solutions

Triangles and Lines

Q037 MCQ

Find the equation of the line through (2,1) parallel to x-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 1$.

Correct option: D**Final answer**

$$y = 1$$

Q038 MCQ

Find the equation of the line through (-4,-7) parallel to x-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -7$.

Correct option: B**Final answer**

$$y = -7$$

Worked Solutions

Triangles and Lines

Q039 MCQ

Find the equation of the line through (1,-2) perpendicular to x-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 1$.

Correct option: C**Final answer**

$$x = 1$$

Q040 MCQ

Find the equation of the line through (-3,-4) perpendicular to x-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -3$.

Correct option: D**Final answer**

$$x = -3$$

Worked Solutions

Triangles and Lines

Q041 MCQ

Find the equation of the line through (3,-5) perpendicular to x-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 3$.

Correct option: D**Final answer**

$$x = 3$$

Q042 MCQ

Find the equation of the line through (2,-7) parallel to y-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 2$.

Correct option: B**Final answer**

$$x = 2$$

Worked Solutions

Triangles and Lines

Q043 MCQ

Find the equation of the line through (4,3) parallel to y-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 4$.

Correct option: D**Final answer**

$$x = 4$$

Q044 MCQ

Find the equation of the line through (-6,-2) parallel to y-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -6$.

Correct option: A**Final answer**

$$x = -6$$

Worked Solutions

Triangles and Lines

Q045 **Written****Find the equation of the line through $(-3,1)$ and $(-3,2)$.****Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -3$.

Final answer

$$x = -3$$

Q046 **Written****Find the equation of the line through $(-2,3)$ and $(-5,3)$.****Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 3$.

Final answer

$$y = 3$$

Worked Solutions

Triangles and Lines

Q047 Written

Find the equation of the line through (1,-1) and (4,0).**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = x/3 - 4/3$ (equivalently $x - 3y - 4 = 0$).

Final answer

$$y = x/3 - 4/3 \text{ (equivalently } x - 3y - 4 = 0)$$

Q048 Written

Find the equation of the line through (-2,-2) and (2,-2).**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -2$.

Final answer

$$y = -2$$

Worked Solutions

Triangles and Lines

Q049 Written

Find the equation of the line through $(-4,2)$ and $(-4,8)$.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -4$.

Final answer

$$x = -4$$

Q050 Written

Find the equation of the line through $(-2,-6)$ and $(-4,1)$.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $7x + 2y + 26 = 0$.

Final answer

$$7x + 2y + 26 = 0$$

Worked Solutions

Triangles and Lines

Q051 Written

Find the equation of the line through (4,0) and (4,3).**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 4$.

Final answer

$$x = 4$$

Q052 Written

Find the equation of the line through (-1,2) and (-1,6).**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -1$.

Final answer

$$x = -1$$

Worked Solutions

Triangles and Lines

Q053 Written

Find the equation of the line through $(-5,-3)$ and $(-5,2)$.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -5$.

Final answer

$$x = -5$$

Q054 Written

Find the equation of the line through $(-2,3)$ and $(-3,3)$.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 3$.

Final answer

$$y = 3$$

Worked Solutions

Triangles and Lines

Q055 Written

Find the equation of the line through (4,-2) and (1,-2).**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -2$.

Final answer

$$y = -2$$

Q056 Written

Find the equation of the line through (-3,6) and (2,6).**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 6$.

Final answer

$$y = 6$$

Worked Solutions

Triangles and Lines

Q057 **Written****Find the equation of the line through (1,2) and (3,10).****Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 4x - 2$.

Final answer

$$y = 4x - 2$$

Q058 **Written****Find the equation of the line through (-1,-8) and (4,2).****Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 2x - 6$.

Final answer

$$y = 2x - 6$$

Worked Solutions

Triangles and Lines

Q059 MCQ

Find the equation of the line through $(-3,1)$ and $(-3,2)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -3$.

Correct option: B**Final answer**

$$x = -3$$

Q060 MCQ

Find the equation of the line through $(-2,3)$ and $(-5,3)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3$.

Correct option: C**Final answer**

$$y = 3$$

Worked Solutions

Triangles and Lines

Q061 MCQ

Find the equation of the line through (1,-1) and (4,0):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = x/3 - 4/3$ (equivalently $x - 3y - 4 = 0$).

Correct option: D**Final answer**

$$y = x/3 - 4/3 \text{ (equivalently } x - 3y - 4 = 0)$$

Q062 MCQ

Find the equation of the line through (-2,-2) and (2,-2):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -2$.

Correct option: A**Final answer**

$$y = -2$$

Worked Solutions

Triangles and Lines

Q063 MCQ

Find the equation of the line through $(-4,2)$ and $(-4,8)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -4$.

Correct option: B**Final answer**

$$x = -4$$

Q064 MCQ

Find the equation of the line through $(-2,-6)$ and $(-4,1)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $7x + 2y + 26 = 0$.

Correct option: D**Final answer**

$$7x + 2y + 26 = 0$$

Worked Solutions

Triangles and Lines

Q065 MCQ

Find the equation of the line through (4,0) and (4,3):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 4$.

Correct option: C**Final answer**

$$x = 4$$

Q066 MCQ

Find the equation of the line through (-1,2) and (-1,6):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -1$.

Correct option: A**Final answer**

$$x = -1$$

Worked Solutions

Triangles and Lines

Q067 MCQ

Find the equation of the line through $(-5,-3)$ and $(-5,2)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -5$.

Correct option: C**Final answer**

$$x = -5$$

Q068 MCQ

Find the equation of the line through $(-2,3)$ and $(-3,3)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3$.

Correct option: D**Final answer**

$$y = 3$$

Worked Solutions

Triangles and Lines

Q069 MCQ

Find the equation of the line through (4,-2) and (1,-2):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -2$.

Correct option: D**Final answer**

$$y = -2$$

Q070 MCQ

Find the equation of the line through (-3,6) and (2,6):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 6$.

Correct option: C**Final answer**

$$y = 6$$

Worked Solutions

Triangles and Lines

Q071 MCQ

Find the equation of the line through (1,2) and (3,10):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 4x - 2$.

Correct option: D**Final answer**

$$y = 4x - 2$$

Q072 MCQ

Find the equation of the line through (-1,-8) and (4,2):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 2x - 6$.

Correct option: D**Final answer**

$$y = 2x - 6$$

Worked Solutions

Triangles and Lines

Q073 Written

Find the equation of the line through (2,-1) with gradient 3.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 3x - 7$.

Final answer

$$y = 3x - 7$$

Q074 Written

Find the equation of the line through (-3,4) with gradient 2.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 2x + 10$.

Final answer

$$y = 2x + 10$$

Worked Solutions

Triangles and Lines

Q075 Written

Find the equation of the line through (-1,5) with gradient -1.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -x + 4$.

Final answer

$$y = -x + 4$$

Q076 Written

Find the equation of the line through (-3,-8) with gradient 1.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = x - 5$.

Final answer

$$y = x - 5$$

Worked Solutions

Triangles and Lines

Q077 Written

Find the equation of the line through (2,-1) with gradient -2.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -2x + 3$.

Final answer

$$y = -2x + 3$$

Q078 MCQ

Find the equation of the line through (2,-1) with gradient 3:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3x - 7$.

Correct option: D**Final answer**

$$y = 3x - 7$$

Worked Solutions

Triangles and Lines

Q079 MCQ

Find the equation of the line through (-3,4) with gradient 2:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 2x + 10$.

Correct option: C**Final answer**

$$y = 2x + 10$$

Q080 MCQ

Find the equation of the line through (-1,5) with gradient -1:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -x + 4$.

Correct option: D**Final answer**

$$y = -x + 4$$

Worked Solutions

Triangles and Lines

Q081 MCQ

Find the equation of the line through $(-3, -8)$ with gradient 1:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = x - 5$.

Correct option: B**Final answer**

$$y = x - 5$$

Q082 MCQ

Find the equation of the line through $(2, -1)$ with gradient -2:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -2x + 3$.

Correct option: D**Final answer**

$$y = -2x + 3$$

Worked Solutions

Triangles and Lines

Q083 Challenge

Sensors A $(-4, 2)$ and B $(9, 2)$ have the same height. Find the straight-line boundary through A and B.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $y = 2$.

Final answer

$$y = 2$$

Worked Solutions

Triangles and Lines

Q084 Challenge

Sensors A $(-4, 2)$ and B $(9, 2)$ have the same height. Find the straight-line boundary through A and B:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 2$.

Correct option: D

Final answer

$$y = 2$$

Worked Solutions

Triangles and Lines

Q085 **Written**

A (2, 3), B (4, -1), and C ($a - 1$, $2 - a$) are collinear. Find a .

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 7$.

Final answer

$$a = 7$$

Q086 **Written**

A (2, 5), B (4, 9), and C (-1 , a) are collinear. Find a .

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = -1$.

Final answer

$$a = -1$$

Worked Solutions

Triangles and Lines

Q087 **Written****A (1, -1), B (-2, -3), and C (4, a) are collinear. Find a.****Solution**

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 1$.

Final answer

$$a = 1$$

Q088 **Written****A (1, 2), B (3, 1), and C (a, -1) are collinear. Find a.****Solution**

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 7$.

Final answer

$$a = 7$$

Worked Solutions

Triangles and Lines

Q089 **Written****A** $(-3, -9)$, **B** $(5, 7)$, and **C** $(a + 3, 3a - 2)$ are collinear. Find a .**Solution**

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 5$.

Final answer

$$a = 5$$

Q090 **MCQ****A** $(2, 3)$, **B** $(4, -1)$, and **C** $(a - 1, 2 - a)$ are collinear. Find a :**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 7$.

Correct option: D**Final answer**

$$a = 7$$

Worked Solutions

Triangles and Lines

Q091 MCQ

A (2, 5), B (4, 9), and C (-1, a) are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = -1$.

Correct option: D**Final answer**

$$a = -1$$

Q092 MCQ

A (1, -1), B (-2, -3), and C (4, a) are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 1$.

Correct option: C**Final answer**

$$a = 1$$

Worked Solutions

Triangles and Lines

Q093 MCQ

A (1, 2), B (3, 1), and C (a, -1) are collinear. Find a:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 7$.

Correct option: D**Final answer**

$$a = 7$$

Q094 MCQ

A (-3, -9), B (5, 7), and C (a + 3, 3a - 2) are collinear. Find a:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 5$.

Correct option: D**Final answer**

$$a = 5$$

Worked Solutions

Triangles and Lines

Q095 **Written****Find the line through the intersection of $2x - 3y + 1 = 0$ and $x + 2y - 3 = 0$, passing through $(-1,3)$.****Solution**

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $x + y - 2 = 0$.

Final answer

$$x + y - 2 = 0$$

Q096 **Written****Find the line through the intersection of $x + 2y - 10 = 0$ and $2x + 3y - 7 = 0$, passing through $(5,6)$.****Solution**

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $x + 3y - 23 = 0$.

Final answer

$$x + 3y - 23 = 0$$

Worked Solutions

Triangles and Lines

Q097 Written

Find the line through the intersection of $3x + 5y - 1 = 0$ and $7x + 8y - 6 = 0$, passing through $(-1, 2)$.**Solution**

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $x + y - 1 = 0$.

Final answer

$$x + y - 1 = 0$$

Q098 Written

Find the line through the intersection of $x + 2y - 4 = 0$ and $2x - y - 3 = 0$, passing through $(-3, -1)$.**Solution**

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $2x - 5y + 1 = 0$.

Final answer

$$2x - 5y + 1 = 0$$

Worked Solutions

Triangles and Lines

Q099 MCQ

Find the line through the intersection of $2x - 3y + 1 = 0$ and $x + 2y - 3 = 0$, passing through $(-1,3)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + y - 2 = 0$.

Correct option: B**Final answer**

$$x + y - 2 = 0$$

Q100 MCQ

Find the line through the intersection of $x + 2y - 10 = 0$ and $2x + 3y - 7 = 0$, passing through $(5,6)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + 3y - 23 = 0$.

Correct option: A**Final answer**

$$x + 3y - 23 = 0$$

Worked Solutions

Triangles and Lines

Q101 MCQ

Find the line through the intersection of $3x + 5y - 1 = 0$ and $7x + 8y - 6 = 0$, passing through $(-1, 2)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + y - 1 = 0$.

Correct option: B**Final answer**

$$x + y - 1 = 0$$

Q102 MCQ

Find the line through the intersection of $x + 2y - 4 = 0$ and $2x - y - 3 = 0$, passing through $(-3, -1)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - 5y + 1 = 0$.

Correct option: D**Final answer**

$$2x - 5y + 1 = 0$$

Worked Solutions

Triangles and Lines

Q103 Challenge

The intersection of $6x + 5y + 7 = 0$ and $3x - 2y + 5 = 0$ is joined to the origin. Find the equation of that trajectory.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $3x + 13y = 0$.

Final answer

$$3x + 13y = 0$$

Worked Solutions

Triangles and Lines

Q104 Challenge

The intersection of $6x + 5y + 7 = 0$ and $3x - 2y + 5 = 0$ is joined to the origin. Find the equation of that trajectory:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x + 13y = 0$.

Correct option: C

Final answer

$$3x + 13y = 0$$

Worked Solutions

Triangles and Lines

Q105 Written

Find the gradient of line $y = 2x + 3$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 2$.

Final answer

$$m = 2$$

Q106 Written

Find the gradient of line $y = -3x + 2$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -3$.

Final answer

$$m = -3$$

Worked Solutions

Triangles and Lines

Q107 Written

Find the gradient of line $x - 3y + 1 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 1/3$.

Final answer

$$m = 1/3$$

Q108 Written

Find the gradient of line $2x - y - 1 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 2$.

Final answer

$$m = 2$$

Worked Solutions

Triangles and Lines

Q109 Written

Find the gradient of line $x + 3y + 3 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/3$.

Final answer

$$m = -1/3$$

Q110 Written

Find the gradient of line $y = 3x + 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3$.

Final answer

$$m = 3$$

Worked Solutions

Triangles and Lines

Q111 Written

Find the gradient of line $y = 4x - 1$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 4$.

Final answer

$$m = 4$$

Q112 Written

Find the gradient of line $y = -4x - 7$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -4$.

Final answer

$$m = -4$$

Worked Solutions

Triangles and Lines

Q113 Written

Find the gradient of line $6x - 2y = 5$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3$.

Final answer

$$m = 3$$

Q114 Written

Find the gradient of line $8x + 2y - 3 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -4$.

Final answer

$$m = -4$$

Worked Solutions

Triangles and Lines

Q115 Written

Find the gradient of line $x + 4y - 4 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/4$.

Final answer

$$m = -1/4$$

Q116 Written

Find the gradient of line $y = 3x - 1$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3$.

Final answer

$$m = 3$$

Worked Solutions

Triangles and Lines

Q117 Written

Find the gradient of line $y = -3x + 4$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -3$.

Final answer

$$m = -3$$

Q118 Written

Find the gradient of line $y = x - 3$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 1$.

Final answer

$$m = 1$$

Worked Solutions

Triangles and Lines

Q119 Written

Find the gradient of line $6x + 2y + 5 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -3$.

Final answer

$$m = -3$$

Q120 Written

Find the gradient of line $2x - 2y - 1 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 1$.

Final answer

$$m = 1$$

Worked Solutions

Triangles and Lines

Q121 Written

Find the gradient of line $x + 3y - 2 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/3$.

Final answer

$$m = -1/3$$

Q122 Written

Find the gradient of line $y = 2x - 1$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 2$.

Final answer

$$m = 2$$

Worked Solutions

Triangles and Lines

Q123 Written

Find the gradient of line $y = -x + 4$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1$.

Final answer

$$m = -1$$

Q124 Written

Find the gradient of line $y = 3x/2 + 6$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3/2$.

Final answer

$$m = 3/2$$

Worked Solutions

Triangles and Lines

Q125 Written

Find the gradient of line $x + y + 3 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1$.

Final answer

$$m = -1$$

Q126 Written

Find the gradient of line $2x + 3y + 8 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -2/3$.

Final answer

$$m = -2/3$$

Worked Solutions

Triangles and Lines

Q127 Written

Find the gradient of line $3x + 6y - 1 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/2$.

Final answer

$$m = -1/2$$

Q128 MCQ

Find the gradient of line $y = 2x + 3$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 2$.

Correct option: D**Final answer**

$$m = 2$$

Worked Solutions

Triangles and Lines

Q129 MCQ

Find the gradient of line $y = -3x + 2$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -3$.

Correct option: C**Final answer**

$$m = -3$$

Q130 MCQ

Find the gradient of line $x - 3y + 1 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 1/3$.

Correct option: A**Final answer**

$$m = 1/3$$

Worked Solutions

Triangles and Lines

Q131 MCQ

Find the gradient of line $2x - y - 1 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 2$.

Correct option: B**Final answer**

$$m = 2$$

Q132 MCQ

Find the gradient of line $x + 3y + 3 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/3$.

Correct option: A**Final answer**

$$m = -1/3$$

Worked Solutions

Triangles and Lines

Q133 MCQ

Find the gradient of line $y = 3x + 1$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3$.

Correct option: A**Final answer**

$$m = 3$$

Q134 MCQ

Find the gradient of line $y = 4x - 1$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 4$.

Correct option: B**Final answer**

$$m = 4$$

Worked Solutions

Triangles and Lines

Q135 MCQ

Find the gradient of line $y = -4x - 7$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -4$.

Correct option: D**Final answer**

$$m = -4$$

Q136 MCQ

Find the gradient of line $6x - 2y = 5$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3$.

Correct option: A**Final answer**

$$m = 3$$

Worked Solutions

Triangles and Lines

Q137 MCQ

Find the gradient of line $8x + 2y - 3 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -4$.

Correct option: C**Final answer**

$$m = -4$$

Q138 MCQ

Find the gradient of line $x + 4y - 4 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/4$.

Correct option: C**Final answer**

$$m = -1/4$$

Worked Solutions

Triangles and Lines

Q139 MCQ

Find the gradient of line $y = 3x - 1$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3$.

Correct option: D**Final answer**

$$m = 3$$

Q140 MCQ

Find the gradient of line $y = -3x + 4$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -3$.

Correct option: A**Final answer**

$$m = -3$$

Worked Solutions

Triangles and Lines

Q141 MCQ

Find the gradient of line $y = x - 3$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 1$.

Correct option: D**Final answer**

$$m = 1$$

Q142 MCQ

Find the gradient of line $6x + 2y + 5 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -3$.

Correct option: C**Final answer**

$$m = -3$$

Worked Solutions

Triangles and Lines

Q143 MCQ

Find the gradient of line $2x - 2y - 1 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 1$.

Correct option: C**Final answer**

$$m = 1$$

Q144 MCQ

Find the gradient of line $x + 3y - 2 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/3$.

Correct option: A**Final answer**

$$m = -1/3$$

Worked Solutions

Triangles and Lines

Q145 MCQ

Find the gradient of line $y = 2x - 1$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 2$.

Correct option: D**Final answer**

$$m = 2$$

Q146 MCQ

Find the gradient of line $y = -x + 4$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1$.

Correct option: A**Final answer**

$$m = -1$$

Worked Solutions

Triangles and Lines

Q147 MCQ

Find the gradient of line $y = 3x/2 + 6$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3/2$.

Correct option: B**Final answer**

$$m = 3/2$$

Q148 MCQ

Find the gradient of line $x + y + 3 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1$.

Correct option: D**Final answer**

$$m = -1$$

Worked Solutions

Triangles and Lines

Q149 MCQ

Find the gradient of line $2x + 3y + 8 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -2/3$.

Correct option: B**Final answer**

$$m = -2/3$$

Q150 MCQ

Find the gradient of line $3x + 6y - 1 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/2$.

Correct option: B**Final answer**

$$m = -1/2$$

Worked Solutions

Triangles and Lines

Q151 **Written****Find the line through (2,-4) parallel to $3x - 2y + 1 = 0$.****Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $3x - 2y - 14 = 0$.

Final answer

$$3x - 2y - 14 = 0$$

Q152 **Written****Find the line through (1,-3) parallel to $6x + 3y - 5 = 0$.****Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + y + 1 = 0$.

Final answer

$$2x + y + 1 = 0$$

Worked Solutions

Triangles and Lines

Q153 **Written****Find the equation of the line through (1,-2) parallel to $2x - y + 5 = 0$.****Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x - y - 4 = 0$.

Final answer

$$2x - y - 4 = 0$$

Q154 **Written****Find the equation of the line through (-1,2) parallel to $2x + 3y = 0$.****Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + 3y - 4 = 0$.

Final answer

$$2x + 3y - 4 = 0$$

Worked Solutions

Triangles and Lines

Q155 **Written****Find the equation of the line through (3,-1) parallel to $3x + 2y + 1 = 0$.****Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $3x + 2y - 7 = 0$.

Final answer

$$3x + 2y - 7 = 0$$

Q156 **MCQ****Find the line through (2,-4) parallel to $3x - 2y + 1 = 0$:****Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x - 2y - 14 = 0$.

Correct option: C**Final answer**

$$3x - 2y - 14 = 0$$

Worked Solutions

Triangles and Lines

Q157 MCQ

Find the line through (1,-3) parallel to $6x + 3y - 5 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + y + 1 = 0$.

Correct option: D**Final answer**

$$2x + y + 1 = 0$$

Q158 MCQ

Find the equation of the line through (1,-2) parallel to $2x - y + 5 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - y - 4 = 0$.

Correct option: C**Final answer**

$$2x - y - 4 = 0$$

Worked Solutions

Triangles and Lines

Q159 MCQ

Find the equation of the line through $(-1,2)$ parallel to $2x + 3y = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + 3y - 4 = 0$.

Correct option: D**Final answer**

$$2x + 3y - 4 = 0$$

Q160 MCQ

Find the equation of the line through $(3,-1)$ parallel to $3x + 2y + 1 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x + 2y - 7 = 0$.

Correct option: D**Final answer**

$$3x + 2y - 7 = 0$$

Worked Solutions

Triangles and Lines

Q161 Written

Find the line through $(-2,5)$ perpendicular to $3x + 5y + 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $5x - 3y + 25 = 0$.

Final answer

$$5x - 3y + 25 = 0$$

Q162 Written

Find the line through $(2,1)$ perpendicular to $2x + 3y + 4 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $3x - 2y - 4 = 0$.

Final answer

$$3x - 2y - 4 = 0$$

Worked Solutions

Triangles and Lines

Q163 Written

Find the equation of the line through (2,-5) perpendicular to $2x + y - 1 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $x - 2y - 12 = 0$.

Final answer

$$x - 2y - 12 = 0$$

Q164 Written

Find the equation of the line through (-1,3) perpendicular to $5x - 2y + 3 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + 5y - 13 = 0$.

Final answer

$$2x + 5y - 13 = 0$$

Worked Solutions

Triangles and Lines

Q165 **Written****Find the equation of the line through $(-2, -5)$ perpendicular to $3x - 2y - 7 = 0$.****Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + 3y + 19 = 0$.

Final answer

$$2x + 3y + 19 = 0$$

Q166 **MCQ****Find the line through $(-2, 5)$ perpendicular to $3x + 5y + 1 = 0$:****Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $5x - 3y + 25 = 0$.

Correct option: D**Final answer**

$$5x - 3y + 25 = 0$$

Worked Solutions

Triangles and Lines

Q167 MCQ

Find the line through (2,1) perpendicular to $2x + 3y + 4 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x - 2y - 4 = 0$.

Correct option: D**Final answer**

$$3x - 2y - 4 = 0$$

Q168 MCQ

Find the equation of the line through (2,-5) perpendicular to $2x + y - 1 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x - 2y - 12 = 0$.

Correct option: B**Final answer**

$$x - 2y - 12 = 0$$

Worked Solutions

Triangles and Lines

Q169 MCQ

Find the equation of the line through $(-1,3)$ perpendicular to $5x - 2y + 3 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + 5y - 13 = 0$.

Correct option: A**Final answer**

$$2x + 5y - 13 = 0$$

Q170 MCQ

Find the equation of the line through $(-2,-5)$ perpendicular to $3x - 2y - 7 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + 3y + 19 = 0$.

Correct option: D**Final answer**

$$2x + 3y + 19 = 0$$

Worked Solutions

Triangles and Lines

Q171 Challenge

If $L1 : 2x + 3y - 10 = 0$ is perpendicular to $L2 : kx - 2y + 5 = 0$ and parallel to $L3 : 4x + my - 1 = 0$, find m/k .

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $m/k = 2$ ($k = 3, m = 6$).

Final answer

$$m/k = 2 \text{ (} k = 3, m = 6 \text{)}$$

Worked Solutions

Triangles and Lines

Q172 Challenge

If $L1 : 2x + 3y - 10 = 0$ is perpendicular to $L2 : kx - 2y + 5 = 0$ and parallel to $L3 : 4x + my - 1 = 0$, find m/k :

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m/k = 2$ ($k = 3$, $m = 6$).

Correct option: D

Final answer

$$m/k = 2 \ (k = 3, \ m = 6)$$

Worked Solutions

Triangles and Lines

Q173 Written

For A (5, 11), B (1, -1), find the equation of the perpendicular bisector of AB.**Solution**

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $x + 3y - 18 = 0$.

Final answer

$$x + 3y - 18 = 0$$

Q174 Written

For A (-1, 2), B (3, -4), find the equation of the perpendicular bisector of AB.**Solution**

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $2x - 3y - 5 = 0$.

Final answer

$$2x - 3y - 5 = 0$$

Worked Solutions

Triangles and Lines

Q175 **Written****Find the perpendicular bisector of segment A $(-2, 3)$, B $(2, 1)$.****Solution**

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $2x - y + 2 = 0$.

Final answer

$$2x - y + 2 = 0$$

Q176 **Written****Find the perpendicular bisector of segment A $(-1, 2)$, B $(5, -2)$.****Solution**

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $3x - 2y - 6 = 0$.

Final answer

$$3x - 2y - 6 = 0$$

Worked Solutions

Triangles and Lines

Q177 **Written****Find the perpendicular bisector of segment A (0, 6), B (4, 4).****Solution**

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $2x - y + 1 = 0$.

Final answer

$$2x - y + 1 = 0$$

Q178 **MCQ****For A (5, 11), B (1, -1), find the equation of the perpendicular bisector of AB:****Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + 3y - 18 = 0$.

Correct option: A**Final answer**

$$x + 3y - 18 = 0$$

Worked Solutions

Triangles and Lines

Q179 MCQ

For A $(-1, 2)$, B $(3, -4)$, find the equation of the perpendicular bisector of AB:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - 3y - 5 = 0$.

Correct option: C**Final answer**

$$2x - 3y - 5 = 0$$

Q180 MCQ

Find the perpendicular bisector of segment A $(-2, 3)$, B $(2, 1)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - y + 2 = 0$.

Correct option: C**Final answer**

$$2x - y + 2 = 0$$

Worked Solutions

Triangles and Lines

Q181 MCQ

Find the perpendicular bisector of segment A $(-1, 2)$, B $(5, -2)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x - 2y - 6 = 0$.

Correct option: D**Final answer**

$$3x - 2y - 6 = 0$$

Q182 MCQ

Find the perpendicular bisector of segment A $(0, 6)$, B $(4, 4)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - y + 1 = 0$.

Correct option: A**Final answer**

$$2x - y + 1 = 0$$

Worked Solutions

Triangles and Lines

Q183 **Written****Find the point B symmetric to A $(-4, 3)$ with respect to $3x - y + 5 = 0$.****Solution**

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (2, 1)$.

Final answer

$$B = (2, 1)$$

Q184 **Written****Find the point B symmetric to A $(0, 4)$ with respect to $2x - y - 1 = 0$.****Solution**

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (4, 2)$.

Final answer

$$B = (4, 2)$$

Worked Solutions

Triangles and Lines

Q185 Written

Find the point B symmetric to A (1, 2) with respect to $x + 2y - 10 = 0$.**Solution**

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (3, 6)$.

Final answer

$$B = (3, 6)$$

Q186 Written

Find the point B symmetric to A (4, -1) with respect to $4x - 2y - 3 = 0$.**Solution**

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (-2, 2)$.

Final answer

$$B = (-2, 2)$$

Worked Solutions

Triangles and Lines

Q187 MCQ

Find the point B symmetric to A $(-4, 3)$ with respect to $3x - y + 5 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (2, 1)$.

Correct option: A**Final answer**

$$B = (2, 1)$$

Q188 MCQ

Find the point B symmetric to A $(0, 4)$ with respect to $2x - y - 1 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (4, 2)$.

Correct option: A**Final answer**

$$B = (4, 2)$$

Worked Solutions

Triangles and Lines

Q189 MCQ

Find the point B symmetric to A (1, 2) with respect to $x + 2y - 10 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (3, 6)$.

Correct option: C**Final answer**

$$B = (3, 6)$$

Q190 MCQ

Find the point B symmetric to A (4, -1) with respect to $4x - 2y - 3 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (-2, 2)$.

Correct option: C**Final answer**

$$B = (-2, 2)$$

Worked Solutions

Triangles and Lines

Q191 Challenge

Reflect A (5, 0) across the targeting line $y = 2x$. Find the receiver point B.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $B = (-3, 4)$.

Final answer

$$B = (-3, 4)$$

Worked Solutions

Triangles and Lines

Q192 Challenge

Reflect A (5, 0) across the targeting line $y = 2x$. Find the receiver point B:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (-3, 4)$.

Correct option: C

Final answer

$$B = (-3, 4)$$

Worked Solutions

Triangles and Lines

Q193 Written

Find the distance from $(-2,1)$ to $3x - 2y + 5 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $3/\sqrt{13} = 3\sqrt{13}/13$.

Final answer

$$3/\sqrt{13} = 3\sqrt{13}/13$$

Q194 Written

Find the distance from $(3,-2)$ to $x + 2y - 4 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{5}$.

Final answer

$$\sqrt{5}$$

Worked Solutions

Triangles and Lines

Q195 Written

Find the perpendicular distance from $(-3,2)$ to $2x - 3y + 6 = 0$.**Solution**

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $6/\sqrt{13} = 6\sqrt{13}/13$.

Final answer

$$6/\sqrt{13} = 6\sqrt{13}/13$$

Q196 Written

Find the perpendicular distance from $(-3,7)$ to $12x - 5y = 7$.**Solution**

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: 6.

Final answer

6

Worked Solutions

Triangles and Lines

Q197 **Written****Find the perpendicular distance from $(-8,1)$ to $y = 2x - 3$.****Solution**

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $4\sqrt{5}$.

Final answer

$$4\sqrt{5}$$

Q198 **Written****Find the perpendicular distance from the origin to $3x - 4y - 5 = 0$.****Solution**

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: 1.

Final answer

$$1$$

Worked Solutions

Triangles and Lines

Q199 Written

Find the perpendicular distance from (2,3) to $3x + 2y + 1 = 0$.**Solution**

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{13}$.

Final answer $\sqrt{13}$

Q200 Written

Find the perpendicular distance from (13,4) to $3x + 4y - 5 = 0$.**Solution**

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: 10.

Final answer

10

Worked Solutions

Triangles and Lines

Q201 Written

Find the perpendicular distance from (1,2) to $y = 3x + 1$.**Solution**

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{10}/5$.

Final answer

$$\sqrt{10}/5$$

Q202 MCQ

Find the distance from (-2,1) to $3x - 2y + 5 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3/\sqrt{13} = 3\sqrt{13}/13$.

Correct option: A**Final answer**

$$3/\sqrt{13} = 3\sqrt{13}/13$$

Worked Solutions

Triangles and Lines

Q203 MCQ

Find the distance from (3,-2) to $x + 2y - 4 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{5}$.

Correct option: D**Final answer**

$$\sqrt{5}$$

Q204 MCQ

Find the perpendicular distance from (-3,2) to $2x - 3y + 6 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $6/\sqrt{13} = 6\sqrt{13}/13$.

Correct option: D**Final answer**

$$6/\sqrt{13} = 6\sqrt{13}/13$$

Worked Solutions

Triangles and Lines

Q205 MCQ

Find the perpendicular distance from $(-3,7)$ to $12x - 5y = 7$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 6.

Correct option: B**Final answer**

6

Q206 MCQ

Find the perpendicular distance from $(-8,1)$ to $y = 2x - 3$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $4\sqrt{5}$.

Correct option: D**Final answer** $4\sqrt{5}$

Worked Solutions

Triangles and Lines

Q207 MCQ

Find the perpendicular distance from the origin to $3x - 4y - 5 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 1.

Correct option: D**Final answer**

1

Q208 MCQ

Find the perpendicular distance from (2,3) to $3x + 2y + 1 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{13}$.

Correct option: A**Final answer** $\sqrt{13}$

Worked Solutions

Triangles and Lines

Q209 MCQ

Find the perpendicular distance from (13,4) to $3x + 4y - 5 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 10.

Correct option: B**Final answer**

10

Q210 MCQ

Find the perpendicular distance from (1,2) to $y = 3x + 1$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}/5$.

Correct option: D**Final answer** $\sqrt{10}/5$

Worked Solutions

Triangles and Lines

Q211 Written

Find the perpendicular length from the origin to $3x - y = 5$.**Solution**

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{10}/2$.

Final answer

$$\sqrt{10}/2$$

Q212 Written

Find the perpendicular length from $(-1,5)$ to $y = 3x - 2$.**Solution**

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{10}$.

Final answer

$$\sqrt{10}$$

Worked Solutions

Triangles and Lines

Q213 MCQ

Find the perpendicular length from the origin to $3x - y = 5$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}/2$.

Correct option: D**Final answer**

$$\sqrt{10}/2$$

Q214 MCQ

Find the perpendicular length from $(-1,5)$ to $y = 3x - 2$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}$.

Correct option: B**Final answer**

$$\sqrt{10}$$

Worked Solutions

Triangles and Lines

Q215 Challenge

Find the shortest distance from P (−1, 5) to the radiation line $y = x/3 + 2$.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $\sqrt{10}$.

Final answer

$$\sqrt{10}$$

Worked Solutions

Triangles and Lines

Q216 Challenge

Find the shortest distance from P $(-1, 5)$ to the radiation line $y = x/3 + 2$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}$.

Correct option: A

Final answer

$\sqrt{10}$

Worked Solutions

Triangles and Lines

Q217 **Written****Find the area of triangle ABC with A (1, 1), B (3, 7), C (−3, −1).****Solution**

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2} \text{ base} \times \text{height}$.
3. Area: 10.

Final answer

10

Q218 **Written****Find the area for A (1, 5), B (−3, −3), C (3, −6).****Solution**

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2} \text{ base} \times \text{height}$.
3. Area: 30.

Final answer

30

Worked Solutions

Triangles and Lines

Q219 **Written****Find the area for A** $(-2, 1)$, **B** $(2, -1)$, **C** $(0, 4)$.**Solution**

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2} \text{ base} \times \text{height}$.
3. Area: 8.

Final answer

8

Q220 **Written****Find the area for A** $(0, -6)$, **B** $(2, -2)$, **C** $(1, 5)$.**Solution**

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2} \text{ base} \times \text{height}$.
3. Area: 9.

Final answer

9

Worked Solutions

Triangles and Lines

Q221 **Written****Find the area for A** $(-2, 4)$, **B** $(-3, -5)$, **C** $(5, -1)$.**Solution**

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2} \text{ base} \times \text{height}$.
3. Area: 34.

Final answer

34

Q222 **Written****Find the area for A** $(2, 5)$, **B** $(0, 1)$, **C** $(4, 2)$.**Solution**

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2} \text{ base} \times \text{height}$.
3. Area: 7.

Final answer

7

Worked Solutions

Triangles and Lines

Q223 MCQ

Find the area of triangle ABC with A (1, 1), B (3, 7), C (−3, −1):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 10.

Correct option: D**Final answer**

10

Q224 MCQ

Find the area for A (1, 5), B (−3, −3), C (3, −6):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 30.

Correct option: C**Final answer**

30

Worked Solutions

Triangles and Lines

Q225 MCQ

Find the area for A $(-2, 1)$, B $(2, -1)$, C $(0, 4)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 8.

Correct option: A**Final answer**

8

Q226 MCQ

Find the area for A $(0, -6)$, B $(2, -2)$, C $(1, 5)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 9.

Correct option: B**Final answer**

9

Worked Solutions

Triangles and Lines

Q227 MCQ

Find the area for A $(-2, 4)$, B $(-3, -5)$, C $(5, -1)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 34.

Correct option: B**Final answer**

34

Q228 MCQ

Find the area for A $(2, 5)$, B $(0, 1)$, C $(4, 2)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 7.

Correct option: D**Final answer**

7

Worked Solutions

Triangles and Lines

Q229 Challenge

Find the area of the wound triangle A $(-2, 1)$, **B** $(1, 5)$, **C** $(-21, -21)$.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: 5.

Final answer

5

Worked Solutions

Triangles and Lines

Q230 Challenge

Find the area of the wound triangle A $(-2, 1)$, B $(1, 5)$, C $(-21, -21)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 5.

Correct option: D

Final answer

5

Worked Solutions

Triangles and Lines

Q231 **Written**

The line $(k - 4)x + (k + 2)y + k + 5 = 0$ passes through a fixed point for every k . Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(1/2, -3/2)$.

Final answer $(1/2, -3/2)$ Q232 **Written**

The line $(2k + 1)x - (k + 2)y + 7k + 8 = 0$ passes through a fixed point for every k . Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(-2, 3)$.

Final answer $(-2, 3)$

Worked Solutions

Triangles and Lines

Q233 Written

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point for every k . Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(3, -3)$.

Final answer $(3, -3)$

Q234 Written

The line $(k + 3)x - (2k + 1)y + k - 2 = 0$ passes through a fixed point for every k . Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(1, 1)$.

Final answer $(1, 1)$

Worked Solutions

Triangles and Lines

Q235 **Written**

The line $(k + 3)x + (3k + 2)y + 7 = 0$ passes through a fixed point for every k . Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(-3, 1)$.

Final answer $(-3, 1)$ Q236 **MCQ**

The line $(k - 4)x + (k + 2)y + k + 5 = 0$ passes through a fixed point for every k . Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(1/2, -3/2)$.

Correct option: D

Final answer $(1/2, -3/2)$

Worked Solutions

Triangles and Lines

Q237 MCQ

The line $(2k + 1)x - (k + 2)y + 7k + 8 = 0$ passes through a fixed point for every k . Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(-2, 3)$.

Correct option: B**Final answer** $(-2, 3)$

Q238 MCQ

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point for every k . Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(3, -3)$.

Correct option: D**Final answer** $(3, -3)$

Worked Solutions

Triangles and Lines

Q239 MCQ

The line $(k + 3)x - (2k + 1)y + k - 2 = 0$ passes through a fixed point for every k . Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(1, 1)$.

Correct option: D**Final answer** $(1, 1)$

Q240 MCQ

The line $(k + 3)x + (3k + 2)y + 7 = 0$ passes through a fixed point for every k . Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(-3, 1)$.

Correct option: A**Final answer** $(-3, 1)$

Worked Solutions

Triangles and Lines

Q241 Challenge

The line $-(k - 2)x + (k - 3)y = -3k + 5$ passes through a fixed point for every k . Find it.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: (4, 1).

Final answer

(4, 1)

Worked Solutions

Triangles and Lines

Q242 Challenge

The line $-(k - 2)x + (k - 3)y = -3k + 5$ passes through a fixed point for every k . Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is (4, 1).

Correct option: A

Final answer

(4, 1)

Worked Solutions

Triangles and Lines

Q243 Written

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a .

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = -4, -3, \text{ or } 6$.

Final answer

$$a = -4, -3, \text{ or } 6$$

Q244 Written

For $x - 2y + 8 = 0$, $x + y - 1 = 0$, and $ax + y - 5 = 0$ not to form a triangle, find a .

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = -1, -1/2, \text{ or } 1$.

Final answer

$$a = -1, -1/2, \text{ or } 1$$

Worked Solutions

Triangles and Lines

Q245 Written

For $x - 2y + 1 = 0$, $3x + 2y = 6$, and $ax - 3y + 2 = 0$ not to form a triangle, find a .

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = 11/10$, $3/2$, or $-9/2$.

Final answer

$$a = 11/10, 3/2, \text{ or } -9/2$$

Q246 Written

For $x - y = -1$, $3x + 2y = 12$, and $ax - y = a - 1$ not to form a triangle, find a .

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = 1$, $-3/2$, or 2 .

Final answer

$$a = 1, -3/2, \text{ or } 2$$

Worked Solutions

Triangles and Lines

Q247 MCQ

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a :

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = -4, -3, \text{ or } 6$.

Correct option: A

Final answer $a = -4, -3, \text{ or } 6$

Q248 MCQ

For $x - 2y + 8 = 0$, $x + y - 1 = 0$, and $ax + y - 5 = 0$ not to form a triangle, find a :

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = -1, -1/2, \text{ or } 1$.

Correct option: A

Final answer $a = -1, -1/2, \text{ or } 1$

Worked Solutions

Triangles and Lines

Q249 MCQ

For $x - 2y + 1 = 0$, $3x + 2y = 6$, and $ax - 3y + 2 = 0$ not to form a triangle, find a :

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 11/10$, $3/2$, or $-9/2$.

Correct option: C**Final answer** $a = 11/10, 3/2, \text{ or } -9/2$

Q250 MCQ

For $x - y = -1$, $3x + 2y = 12$, and $ax - y = a - 1$ not to form a triangle, find a :

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 1$, $-3/2$, or 2 .

Correct option: D**Final answer** $a = 1, -3/2, \text{ or } 2$

Worked Solutions

Triangles and Lines

Q251 Challenge

For $x + 2y - 1 = 0$, $3x + y + 2 = 0$, and $mx + 4y - 3 = 0$ not to form a triangle, find the sum of all possible m .

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: 15.

Final answer

15

Worked Solutions

Triangles and Lines

Q252 Challenge

For $x + 2y - 1 = 0$, $3x + y + 2 = 0$, and $mx + 4y - 3 = 0$ not to form a triangle, find the sum of all possible m :

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 15.

Correct option: A

Final answer

15

Worked Solutions

Triangles and Lines

Quiz MCQ 1 [Concept Quiz · MCQ](#)

Determine the shape of triangle A (0, 0), B (3, 0), C (0, 4):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *scalene right – angled*.

Correct option: C

Final answer

scalene right – angled

Quiz MCQ 2 [Concept Quiz · MCQ](#)

Find the equation of the line through (2,-1) with gradient 3:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3x - 7$.

Correct option: B

Final answer

$y = 3x - 7$

Worked Solutions

Triangles and Lines

Quiz MCQ 3 [Concept Quiz · MCQ](#)

Find the perpendicular distance from (1,2) to $3x + 4y - 10 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 1/5.

Correct option: D

Final answer

1/5

Quiz MCQ 4 [Concept Quiz · MCQ](#)

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is (3, -3).

Correct option: C

Final answer

(3, -3)

Worked Solutions

Triangles and Lines

Quiz WRITTEN 5 Concept Quiz · Written

Find the area of triangle A (0, 0), B (4, 0), C (0, 3).

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2} \text{ base} \times \text{height}$.
3. Area: 6.

Final answer

6

Quiz WRITTEN 6 Concept Quiz · Written

Find the perpendicular bisector of the segment joining A (1, 2) and B (5, 4).

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $y = -2x + 9$.

Final answer

$y = -2x + 9$

Worked Solutions

Triangles and Lines

Quiz WRITTEN 7 Concept Quiz · Written

Find the equation of the line through (1,2) perpendicular to $y = 3x - 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $y = -x/3 + 7/3$.

Final answer

$$y = -x/3 + 7/3$$

Quiz WRITTEN 8 Concept Quiz · Written

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a .

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = -4$, -3 , or 6 .

Final answer

$$a = -4, -3, \text{ or } 6$$

Answer key

Use the question number shown on each page.

Q001 <i>equilateral</i>	Q027 $x = -3$	Q053 $x = -5$	Q079 C
Q002 <i>equilateral</i>	Q028 $x = 3$	Q054 $y = 3$	Q080 D
Q003 D	Q029 $x = 2$	Q055 $y = -2$	Q081 B
Q004 C	Q030 $x = 4$	Q056 $y = 6$	Q082 D
Q005 <i>isosceles right – angled; right angle at A</i>	Q031 $x = -6$	Q057 $y = 4x - 2$	Q083 $y = 2$
Q006 <i>isosceles right – angled; right angle at A</i>	Q032 B	Q058 $y = 2x - 6$	Q084 D
Q007 B	Q033 C	Q059 B	Q085 $a = 7$
Q008 D	Q034 D	Q060 C	Q086 $a = -1$
Q009 <i>scalene right – angled; right angle at A</i>	Q035 D	Q061 D	Q087 $a = 1$
Q010 D	Q036 B	Q062 A	Q088 $a = 7$
Q011 <i>isosceles right – angled; right angle at B</i>	Q037 D	Q063 B	Q089 $a = 5$
Q012 <i>isosceles right – angled; right angle at B</i>	Q038 B	Q064 D	Q090 D
Q013 <i>isosceles right – angled; right angle at A</i>	Q039 C	Q065 C	Q091 D
Q014 B	Q040 D	Q066 A	Q092 C
Q015 C	Q041 D	Q067 C	Q093 D
Q016 B	Q042 B	Q068 D	Q094 D
Q017 $k = 2$	Q043 D	Q069 D	Q095 $x + y - 2 = 0$
Q018 C	Q044 A	Q070 C	Q096 $x + 3y - 23 = 0$
Q019 $y = -4$	Q045 $x = -3$	Q071 D	Q097 $x + y - 1 = 0$
Q020 $x = -3$	Q046 $y = 3$	Q072 D	Q098 $2x - 5y + 1 = 0$
Q021 $y = 6$	Q047 $y = x/3 - 4/3$ (equivalently $x - 3y - 4 = 0$)	Q073 $y = 3x - 7$	Q099 B
Q022 $x = 3$	Q048 $y = -2$	Q074 $y = 2x + 10$	Q100 A
Q023 $y = 5$	Q049 $x = -4$	Q075 $y = -x + 4$	Q101 B
Q024 $y = 1$	Q050 $7x + 2y + 26 = 0$	Q076 $y = x - 5$	Q102 D
Q025 $y = -7$	Q051 $x = 4$	Q077 $y = -2x + 3$	Q103 $3x + 13y = 0$
Q026 $x = 1$	Q052 $x = -1$	Q078 D	Q104 C

Q105 $m = 2$	Q137 C	Q169 A	Q201 $\sqrt{10}/5$
Q106 $m = -3$	Q138 C	Q170 D	Q202 A
Q107 $m = 1/3$	Q139 D	Q171 $m/k = 2 (k = 3, m = 6)$	Q203 D
Q108 $m = 2$	Q140 A	Q172 D	Q204 D
Q109 $m = -1/3$	Q141 D	Q173 $x + 3y - 18 = 0$	Q205 B
Q110 $m = 3$	Q142 C	Q174 $2x - 3y - 5 = 0$	Q206 D
Q111 $m = 4$	Q143 C	Q175 $2x - y + 2 = 0$	Q207 D
Q112 $m = -4$	Q144 A	Q176 $3x - 2y - 6 = 0$	Q208 A
Q113 $m = 3$	Q145 D	Q177 $2x - y + 1 = 0$	Q209 B
Q114 $m = -4$	Q146 A	Q178 A	Q210 D
Q115 $m = -1/4$	Q147 B	Q179 C	Q211 $\sqrt{10}/2$
Q116 $m = 3$	Q148 D	Q180 C	Q212 $\sqrt{10}$
Q117 $m = -3$	Q149 B	Q181 D	Q213 D
Q118 $m = 1$	Q150 B	Q182 A	Q214 B
Q119 $m = -3$	Q151 $3x - 2y - 14 = 0$	Q183 $B = (2, 1)$	Q215 $\sqrt{10}$
Q120 $m = 1$	Q152 $2x + y + 1 = 0$	Q184 $B = (4, 2)$	Q216 A
Q121 $m = -1/3$	Q153 $2x - y - 4 = 0$	Q185 $B = (3, 6)$	Q217 10
Q122 $m = 2$	Q154 $2x + 3y - 4 = 0$	Q186 $B = (-2, 2)$	Q218 30
Q123 $m = -1$	Q155 $3x + 2y - 7 = 0$	Q187 A	Q219 8
Q124 $m = 3/2$	Q156 C	Q188 A	Q220 9
Q125 $m = -1$	Q157 D	Q189 C	Q221 34
Q126 $m = -2/3$	Q158 C	Q190 C	Q222 7
Q127 $m = -1/2$	Q159 D	Q191 $B = (-3, 4)$	Q223 D
Q128 D	Q160 D	Q192 C	Q224 C
Q129 C	Q161 $5x - 3y + 25 = 0$	Q193 $3/\sqrt{13} = 3\sqrt{13}/13$	Q225 A
Q130 A	Q162 $3x - 2y - 4 = 0$	Q194 $\sqrt{5}$	Q226 B
Q131 B	Q163 $x - 2y - 12 = 0$	Q195 $6/\sqrt{13} = 6\sqrt{13}/13$	Q227 B
Q132 A	Q164 $2x + 5y - 13 = 0$	Q196 6	Q228 D
Q133 A	Q165 $2x + 3y + 19 = 0$	Q197 $4\sqrt{5}$	Q229 5
Q134 B	Q166 D	Q198 1	Q230 D
Q135 D	Q167 D	Q199 $\sqrt{13}$	Q231 $(1/2, -3/2)$
Q136 A	Q168 B	Q200 10	Q232 $(-2, 3)$

Q233 (3, -3)

Q234 (1, 1)

Q235 (-3, 1)

Q236 D

Q237 B

Q238 D

Q239 D

Q240 A

Q241 (4, 1)

Q242 A

Q243 $a = -4, -3, \text{ or } 6$ Q244 $a = -1, -1/2, \text{ or } 1$ Q245 $a = 11/10, 3/2, \text{ or } -9/2$ Q246 $a = 1, -3/2, \text{ or } 2$

Q247 A

Q248 A

Q249 C

Q250 D

Q251 15

Q252 A

Quiz MCQ 1 C

Quiz MCQ 2 B

Quiz MCQ 3 D

Quiz MCQ 4 C

Quiz WRITTEN 5 6

Quiz WRITTEN 6 $y = -2x + 9$ Quiz WRITTEN 7 $y = -x/3 + 7/3$ Quiz WRITTEN 8 $a = -4, -3, \text{ or } 6$

Concept 3 | Circles

English Student Edition • Focused formulas and guided practice

Formula opener

Circle

$$(x - a)^2 + (y - b)^2 = r^2$$

General form

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Tangent

$$xx_1 + yy_1 = r^2$$

Worked Solutions

Circles

Q001 Written

Find the equation of the circle with centre (3,-2) and radius 5.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $(x - 3)^2 + (y + 2)^2 = 25$.

Final answer

$$(x - 3)^2 + (y + 2)^2 = 25$$

Q002 Written

Find the equation of the circle with centre (-3,1) and radius 2.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $(x + 3)^2 + (y - 1)^2 = 4$.

Final answer

$$(x + 3)^2 + (y - 1)^2 = 4$$

Worked Solutions

Circles

Q003 Written

Find the equation of the circle with centre the origin and radius 4.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $x^2 + y^2 = 16$.

Final answer

$$x^2 + y^2 = 16$$

Q004 Written

Find the equation of the circle with centre (3,4), radius 6.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $(x - 3)^2 + (y - 4)^2 = 36$.

Final answer

$$(x - 3)^2 + (y - 4)^2 = 36$$

Worked Solutions

Circles

Q005 Written

Find the equation of the circle with centre $(-2, 6)$, radius $2\sqrt{3}$.**Solution**

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $(x + 2)^2 + (y - 6)^2 = 12$.

Final answer

$$(x + 2)^2 + (y - 6)^2 = 12$$

Q006 Written

Find the equation of the circle with centre the origin, radius 5.**Solution**

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $x^2 + y^2 = 25$.

Final answer

$$x^2 + y^2 = 25$$

Worked Solutions

Circles

Q007 MCQ

Find the equation of the circle with centre (3,-2) and radius 5:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 3)^2 + (y + 2)^2 = 25$.

Correct option: A**Final answer**

$$(x - 3)^2 + (y + 2)^2 = 25$$

Q008 MCQ

Find the equation of the circle with centre (-3,1) and radius 2:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 3)^2 + (y - 1)^2 = 4$.

Correct option: A**Final answer**

$$(x + 3)^2 + (y - 1)^2 = 4$$

Worked Solutions

Circles

Q009 MCQ

Find the equation of the circle with centre the origin and radius 4:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 = 16$.

Correct option: D**Final answer**

$$x^2 + y^2 = 16$$

Q010 MCQ

Find the equation of the circle with centre (3,4), radius 6:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 3)^2 + (y - 4)^2 = 36$.

Correct option: B**Final answer**

$$(x - 3)^2 + (y - 4)^2 = 36$$

Worked Solutions

Circles

Q011 MCQ

Find the equation of the circle with centre $(-2,6)$, radius $2\sqrt{3}$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 2)^2 + (y - 6)^2 = 12$.

Correct option: B**Final answer**

$$(x + 2)^2 + (y - 6)^2 = 12$$

Q012 MCQ

Find the equation of the circle with centre the origin, radius 5:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 = 25$.

Correct option: A**Final answer**

$$x^2 + y^2 = 25$$

Worked Solutions

Circles

Q013 Written

Determine what kind of figure $x^2 + y^2 - 6x + 2y + 6 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(3, -1)$, radius 2.

Final answer

circle; centre $(3, -1)$, radius 2

Q014 Written

Determine what kind of figure $x^2 + y^2 - 4x - 6y + 4 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(2, 3)$, radius 3.

Final answer

circle; centre $(2, 3)$, radius 3

Worked Solutions

Circles

Q015 Written

Determine what kind of figure $x^2 + y^2 - 4x + 8y + 4 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(2, -4)$, radius 4.

Final answer

circle; centre $(2, -4)$, radius 4

Q016 Written

Determine what kind of figure $x^2 + y^2 - 4x = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(2, 0)$, radius 2.

Final answer

circle; centre $(2, 0)$, radius 2

Worked Solutions

Circles

Q017 Written

Determine what kind of figure $x^2 + y^2 + 2x - 4y - 20 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(-1, 2)$, radius 5.

Final answer

circle; centre $(-1, 2)$, radius 5

Q018 Written

Determine what kind of figure $x^2 + y^2 + 4x - 6y = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(-2, 3)$, radius $\sqrt{13}$.

Final answer

circle; centre $(-2, 3)$, radius $\sqrt{13}$

Worked Solutions

Circles

Q019 MCQ

Determine what kind of figure $x^2 + y^2 - 6x + 2y + 6 = 0$ represents:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre $(3, -1)$, radius 2.

Correct option: A**Final answer**circle; centre $(3, -1)$, radius 2

Worked Solutions

Circles

Q020 MCQ

Determine what kind of figure $x^2 + y^2 - 4x - 6y + 4 = 0$ represents:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre (2, 3), radius 3.

Correct option: C**Final answer**

circle; centre (2, 3), radius 3

Worked Solutions

Circles

Q021 MCQ

Determine what kind of figure $x^2 + y^2 - 4x + 8y + 4 = 0$ represents:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre $(2, -4)$, radius 4.

Correct option: D**Final answer**circle; centre $(2, -4)$, radius 4

Worked Solutions

Circles

Q022 MCQ

Determine what kind of figure $x^2 + y^2 - 4x = 0$ represents:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre $(2, 0)$, radius 2.

Correct option: C**Final answer**circle; centre $(2, 0)$, radius 2

Worked Solutions

Circles

Q023 MCQ

Determine what kind of figure $x^2 + y^2 + 2x - 4y - 20 = 0$ represents:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre $(-1, 2)$, radius 5.

Correct option: C**Final answer**circle; centre $(-1, 2)$, radius 5

Worked Solutions

Circles

Q024 MCQ

Determine what kind of figure $x^2 + y^2 + 4x - 6y = 0$ represents:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre $(-2, 3)$, radius $\sqrt{13}$.

Correct option: C**Final answer**circle; centre $(-2, 3)$, radius $\sqrt{13}$

Worked Solutions

Circles

Q025 Challenge

A circle passes through A (2, 5), B (8, 1), and has its centre on $x + y = 8$. Find its standard equation, centre and radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: centre (5, 3), radius² = 13; $(x - 5)^2 + (y - 3)^2 = 13$.

Final answer

centre (5, 3), radius² = 13; $(x - 5)^2 + (y - 3)^2 = 13$

Worked Solutions

Circles

Q026 Challenge

A circle passes through A (2, 5), B (8, 1), and has its centre on $x + y = 8$. Find its standard equation, centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is centre (5, 3), $\text{radius}^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13$.

Correct option: B

Final answer

centre (5, 3), $\text{radius}^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13$

Worked Solutions

Circles

Q027 Written

Find the equation of the circle with centre (2,3) passing through (4,6).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x - 2)^2 + (y - 3)^2 = 13$.

Final answer

$$(x - 2)^2 + (y - 3)^2 = 13$$

Q028 Written

Find the equation of the circle with centre (-3,1) passing through (4,-2).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x + 3)^2 + (y - 1)^2 = 58$.

Final answer

$$(x + 3)^2 + (y - 1)^2 = 58$$

Worked Solutions

Circles

Q029 Written

Find the equation of the circle with centre (1,-3) through (-3,0).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x - 1)^2 + (y + 3)^2 = 25$.

Final answer

$$(x - 1)^2 + (y + 3)^2 = 25$$

Q030 Written

Find the equation of the circle with centre (-4,1) through (3,-2).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x + 4)^2 + (y - 1)^2 = 58$.

Final answer

$$(x + 4)^2 + (y - 1)^2 = 58$$

Worked Solutions

Circles

Q031 Written

Find the equation of the circle with centre (2,-1) through the origin.**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x - 2)^2 + (y + 1)^2 = 5$.

Final answer

$$(x - 2)^2 + (y + 1)^2 = 5$$

Q032 Written

Find the equation of the circle with centre the origin through (4,-3).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 = 25$.

Final answer

$$x^2 + y^2 = 25$$

Worked Solutions

Circles

Q033 MCQ

Find the equation of the circle with centre (2,3) passing through (4,6):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 2)^2 + (y - 3)^2 = 13$.

Correct option: C**Final answer**

$$(x - 2)^2 + (y - 3)^2 = 13$$

Q034 MCQ

Find the equation of the circle with centre (-3,1) passing through (4,-2):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 3)^2 + (y - 1)^2 = 58$.

Correct option: A**Final answer**

$$(x + 3)^2 + (y - 1)^2 = 58$$

Worked Solutions

Circles

Q035 MCQ

Find the equation of the circle with centre (1,-3) through (-3,0):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 1)^2 + (y + 3)^2 = 25$.

Correct option: D**Final answer**

$$(x - 1)^2 + (y + 3)^2 = 25$$

Q036 MCQ

Find the equation of the circle with centre (-4,1) through (3,-2):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 4)^2 + (y - 1)^2 = 58$.

Correct option: A**Final answer**

$$(x + 4)^2 + (y - 1)^2 = 58$$

Worked Solutions

Circles

Q037 MCQ

Find the equation of the circle with centre (2,-1) through the origin:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 2)^2 + (y + 1)^2 = 5$.

Correct option: B**Final answer**

$$(x - 2)^2 + (y + 1)^2 = 5$$

Q038 MCQ

Find the equation of the circle with centre the origin through (4,-3):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 = 25$.

Correct option: A**Final answer**

$$x^2 + y^2 = 25$$

Worked Solutions

Circles

Q039 Written

Find the equation of the circle with diameter endpoints A (1, 3), B (3, -5).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x - 2)^2 + (y + 1)^2 = 17$.

Final answer

$$(x - 2)^2 + (y + 1)^2 = 17$$

Q040 Written

Find the equation of the circle with diameter endpoints (3,4), (5,-2).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x - 4)^2 + (y - 1)^2 = 10$.

Final answer

$$(x - 4)^2 + (y - 1)^2 = 10$$

Worked Solutions

Circles

Q041 Written

Find the equation of the circle with diameter endpoints (4,7), (-2,-1).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x - 1)^2 + (y - 3)^2 = 25$.

Final answer

$$(x - 1)^2 + (y - 3)^2 = 25$$

Q042 Written

Find the equation of the circle with diameter endpoints (4,-2), (-6,-2).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x + 1)^2 + (y + 2)^2 = 25$.

Final answer

$$(x + 1)^2 + (y + 2)^2 = 25$$

Worked Solutions

Circles

Q043 Written

Find the equation of the circle with diameter endpoints $(-3,0)$, $(-1,2)$.**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x + 2)^2 + (y - 1)^2 = 2$.

Final answer

$$(x + 2)^2 + (y - 1)^2 = 2$$

Q044 Written

Find the equation of the circle with diameter endpoints $(-3,6)$, $(3,-2)$.**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + (y - 2)^2 = 25$.

Final answer

$$x^2 + (y - 2)^2 = 25$$

Worked Solutions

Circles

Q045 MCQ

Find the equation of the circle with diameter endpoints A (1, 3), B (3, -5):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 2)^2 + (y + 1)^2 = 17$.

Correct option: A**Final answer**

$$(x - 2)^2 + (y + 1)^2 = 17$$

Q046 MCQ

Find the equation of the circle with diameter endpoints (3,4), (5,-2):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 4)^2 + (y - 1)^2 = 10$.

Correct option: C**Final answer**

$$(x - 4)^2 + (y - 1)^2 = 10$$

Worked Solutions

Circles

Q047 MCQ

Find the equation of the circle with diameter endpoints (4,7), (-2,-1):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 1)^2 + (y - 3)^2 = 25$.

Correct option: C**Final answer**

$$(x - 1)^2 + (y - 3)^2 = 25$$

Q048 MCQ

Find the equation of the circle with diameter endpoints (4,-2), (-6,-2):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y + 2)^2 = 25$.

Correct option: A**Final answer**

$$(x + 1)^2 + (y + 2)^2 = 25$$

Worked Solutions

Circles

Q049 MCQ

Find the equation of the circle with diameter endpoints $(-3,0)$, $(-1,2)$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 2)^2 + (y - 1)^2 = 2$.

Correct option: B**Final answer**

$$(x + 2)^2 + (y - 1)^2 = 2$$

Q050 MCQ

Find the equation of the circle with diameter endpoints $(-3,6)$, $(3,-2)$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + (y - 2)^2 = 25$.

Correct option: C**Final answer**

$$x^2 + (y - 2)^2 = 25$$

Worked Solutions

Circles

Q051 Written

Find the equation of the circle passing through (1,1), (2,-1), and (3,2).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 5x - y + 4 = 0$.

Final answer

$$x^2 + y^2 - 5x - y + 4 = 0$$

Q052 Written

Find the equation of the circle passing through (0,1), (-1,2), and (3,1).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 3x - 7y + 6 = 0$.

Final answer

$$x^2 + y^2 - 3x - 7y + 6 = 0$$

Worked Solutions

Circles

Q053 Written

Find the equation of the circle with through (0,0), (-1,1), (2,4).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 2x - 4y = 0$.

Final answer

$$x^2 + y^2 - 2x - 4y = 0$$

Q054 Written

Find the equation of the circle with through (-4,6), (0,8), (3,-1).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 6y - 16 = 0$.

Final answer

$$x^2 + y^2 - 6y - 16 = 0$$

Worked Solutions

Circles

Q055 **Written****Find the equation of the circle with through (1,1), (5,-1), (-3,-7).****Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 2x + 8y - 8 = 0$.

Final answer

$$x^2 + y^2 - 2x + 8y - 8 = 0$$

Q056 **MCQ****Find the equation of the circle passing through (1,1), (2,-1), and (3,2):****Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 5x - y + 4 = 0$.

Correct option: A**Final answer**

$$x^2 + y^2 - 5x - y + 4 = 0$$

Worked Solutions

Circles

Q057 MCQ

Find the equation of the circle passing through (0,1), (-1,2), and (3,1):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 3x - 7y + 6 = 0$.

Correct option: B**Final answer**

$$x^2 + y^2 - 3x - 7y + 6 = 0$$

Q058 MCQ

Find the equation of the circle with through (0,0), (-1,1), (2,4):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 2x - 4y = 0$.

Correct option: D**Final answer**

$$x^2 + y^2 - 2x - 4y = 0$$

Worked Solutions

Circles

Q059 MCQ

Find the equation of the circle with through $(-4,6)$, $(0,8)$, $(3,-1)$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 6y - 16 = 0$.

Correct option: D**Final answer**

$$x^2 + y^2 - 6y - 16 = 0$$

Q060 MCQ

Find the equation of the circle with through $(1,1)$, $(5,-1)$, $(-3,-7)$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 2x + 8y - 8 = 0$.

Correct option: D**Final answer**

$$x^2 + y^2 - 2x + 8y - 8 = 0$$

Worked Solutions

Circles

Q061 Challenge

Find the circle through (0,2), (2,0), and (4,6), giving its centre and radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10}$.

Final answer

$$x^2 + y^2 - 6x - 6y + 8 = 0; \text{ centre } (3, 3), \text{ radius } \sqrt{10}$$

Worked Solutions

Circles

Q062 Challenge

Find the circle through (0,2), (2,0), and (4,6), giving its centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10}$.

Correct option: A

Final answer

$$x^2 + y^2 - 6x - 6y + 8 = 0; \text{ centre } (3, 3), \text{ radius } \sqrt{10}$$

Worked Solutions

Circles

Q063 Written

Find the circle with centre (4,-2) tangent to the x-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 4)^2 + (y + 2)^2 = 4$.

Final answer

$$(x - 4)^2 + (y + 2)^2 = 4$$

Q064 Written

Find the circle with centre (-2,3) tangent to the y-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 2)^2 + (y - 3)^2 = 4$.

Final answer

$$(x + 2)^2 + (y - 3)^2 = 4$$

Worked Solutions

Circles

Q065 Written

Find the circle with centre $(-1,4)$ tangent to x-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 1)^2 + (y - 4)^2 = 16$.

Final answer

$$(x + 1)^2 + (y - 4)^2 = 16$$

Q066 Written

Find the circle with centre $(3,-4)$ tangent to x-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 3)^2 + (y + 4)^2 = 16$.

Final answer

$$(x - 3)^2 + (y + 4)^2 = 16$$

Worked Solutions

Circles

Q067 Written

Find the circle with centre (5,-4) tangent to x-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 5)^2 + (y + 4)^2 = 16$.

Final answer

$$(x - 5)^2 + (y + 4)^2 = 16$$

Q068 Written

Find the circle with centre (-1,6) tangent to y-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 1)^2 + (y - 6)^2 = 1$.

Final answer

$$(x + 1)^2 + (y - 6)^2 = 1$$

Worked Solutions

Circles

Q069 Written

Find the circle with centre (2,-5) tangent to y-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 2)^2 + (y + 5)^2 = 4$.

Final answer

$$(x - 2)^2 + (y + 5)^2 = 4$$

Q070 Written

Find the circle with centre (-3,-4) tangent to y-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 3)^2 + (y + 4)^2 = 9$.

Final answer

$$(x + 3)^2 + (y + 4)^2 = 9$$

Worked Solutions

Circles

Q071 MCQ

Find the circle with centre (4,-2) tangent to the x-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 4)^2 + (y + 2)^2 = 4$.

Correct option: B**Final answer**

$$(x - 4)^2 + (y + 2)^2 = 4$$

Worked Solutions

Circles

Q072 MCQ

Find the circle with centre $(-2,3)$ tangent to the y -axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 2)^2 + (y - 3)^2 = 4$.

Correct option: B**Final answer**

$$(x + 2)^2 + (y - 3)^2 = 4$$

Worked Solutions

Circles

Q073 MCQ

Find the circle with centre $(-1,4)$ tangent to x-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y - 4)^2 = 16$.

Correct option: B**Final answer**

$$(x + 1)^2 + (y - 4)^2 = 16$$

Worked Solutions

Circles

Q074 MCQ

Find the circle with centre (3,-4) tangent to x-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 3)^2 + (y + 4)^2 = 16$.

Correct option: D**Final answer**

$$(x - 3)^2 + (y + 4)^2 = 16$$

Worked Solutions

Circles

Q075 MCQ

Find the circle with centre (5,-4) tangent to x-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 5)^2 + (y + 4)^2 = 16$.

Correct option: D**Final answer**

$$(x - 5)^2 + (y + 4)^2 = 16$$

Worked Solutions

Circles

Q076 MCQ

Find the circle with centre $(-1,6)$ tangent to y-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y - 6)^2 = 1$.

Correct option: D**Final answer**

$$(x + 1)^2 + (y - 6)^2 = 1$$

Worked Solutions

Circles

Q077 MCQ

Find the circle with centre (2,-5) tangent to y-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 2)^2 + (y + 5)^2 = 4$.

Correct option: B**Final answer**

$$(x - 2)^2 + (y + 5)^2 = 4$$

Worked Solutions

Circles

Q078 MCQ

Find the circle with centre $(-3,-4)$ tangent to y-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 3)^2 + (y + 4)^2 = 9$.

Correct option: C**Final answer**

$$(x + 3)^2 + (y + 4)^2 = 9$$

Worked Solutions

Circles

Q079 Written

Find the circles through $(-2,1)$ tangent to both the x-axis and y-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 1)^2 + (y - 1)^2 = 1$ or $(x + 5)^2 + (y - 5)^2 = 25$.

Final answer

$$(x + 1)^2 + (y - 1)^2 = 1 \text{ or } (x + 5)^2 + (y - 5)^2 = 25$$

Q080 Written

Find the circles through $(1,2)$ tangent to both axes.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 1)^2 + (y - 1)^2 = 1$ or $(x - 5)^2 + (y - 5)^2 = 25$.

Final answer

$$(x - 1)^2 + (y - 1)^2 = 1 \text{ or } (x - 5)^2 + (y - 5)^2 = 25$$

Worked Solutions

Circles

Q081 **Written****Find the circle with through $(-1, -2)$ tangent to both axes.****Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$.

Final answer

$$(x + 1)^2 + (y + 1)^2 = 1 \text{ or } (x + 5)^2 + (y + 5)^2 = 25$$

Q082 **Written****Find the circle with through $(-2, -1)$ tangent to both axes.****Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$.

Final answer

$$(x + 1)^2 + (y + 1)^2 = 1 \text{ or } (x + 5)^2 + (y + 5)^2 = 25$$

Worked Solutions

Circles

Q083 Written

Find the circle with through (8,-1) tangent to both axes.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 5)^2 + (y + 5)^2 = 25$ or $(x - 13)^2 + (y + 13)^2 = 169$.

Final answer

$$(x - 5)^2 + (y + 5)^2 = 25 \text{ or } (x - 13)^2 + (y + 13)^2 = 169$$

Worked Solutions

Circles

Q084 MCQ

Find the circles through $(-2,1)$ tangent to both the x-axis and y-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y - 1)^2 = 1$ or $(x + 5)^2 + (y - 5)^2 = 25$.

Correct option: C**Final answer**

$$(x + 1)^2 + (y - 1)^2 = 1 \text{ or } (x + 5)^2 + (y - 5)^2 = 25$$

Worked Solutions

Circles

Q085 MCQ

Find the circles through (1,2) tangent to both axes:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 1)^2 + (y - 1)^2 = 1$ or $(x - 5)^2 + (y - 5)^2 = 25$.

Correct option: D**Final answer**

$$(x - 1)^2 + (y - 1)^2 = 1 \text{ or } (x - 5)^2 + (y - 5)^2 = 25$$

Worked Solutions

Circles

Q086 MCQ

Find the circle with through $(-1,-2)$ tangent to both axes:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$.

Correct option: C**Final answer**

$$(x + 1)^2 + (y + 1)^2 = 1 \text{ or } (x + 5)^2 + (y + 5)^2 = 25$$

Worked Solutions

Circles

Q087 MCQ

Find the circle with through $(-2,-1)$ tangent to both axes:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$.

Correct option: C**Final answer**

$$(x + 1)^2 + (y + 1)^2 = 1 \text{ or } (x + 5)^2 + (y + 5)^2 = 25$$

Worked Solutions

Circles

Q088 MCQ

Find the circle with through (8,-1) tangent to both axes:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 5)^2 + (y + 5)^2 = 25$ or $(x - 13)^2 + (y + 13)^2 = 169$.

Correct option: B**Final answer**

$$(x - 5)^2 + (y + 5)^2 = 25 \text{ or } (x - 13)^2 + (y + 13)^2 = 169$$

Worked Solutions

Circles

Q089 Written

Find the circle whose centre is on $y = -x$ and which passes through (5,2) and (-2,3).**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 1)^2 + (y + 1)^2 = 25$.

Final answer

$$(x - 1)^2 + (y + 1)^2 = 25$$

Q090 Written

Find the circle whose centre is on $y = x + 5$ and which passes through the origin and (1,2).**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 5/2)^2 + (y - 5/2)^2 = 25/2$.

Final answer

$$(x + 5/2)^2 + (y - 5/2)^2 = 25/2$$

Worked Solutions

Circles

Q091 Written

Find the circle with centre on $y = x + 1$ through (3,4) and (1,4).**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 2)^2 + (y - 3)^2 = 2$.

Final answer

$$(x - 2)^2 + (y - 3)^2 = 2$$

Q092 Written

Find the circle with centre on $y = 2x - 6$ through (3,7) and (-1,3).**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 4)^2 + (y - 2)^2 = 26$.

Final answer

$$(x - 4)^2 + (y - 2)^2 = 26$$

Worked Solutions

Circles

Q093 MCQ

Find the circle whose centre is on $y = -x$ and which passes through (5,2) and (-2,3):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 1)^2 + (y + 1)^2 = 25$.

Correct option: D**Final answer**

$$(x - 1)^2 + (y + 1)^2 = 25$$

Worked Solutions

Circles

Q094 MCQ

Find the circle whose centre is on $y = x + 5$ and which passes through the origin and (1,2):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 5/2)^2 + (y - 5/2)^2 = 25/2$.

Correct option: C**Final answer**

$$(x + 5/2)^2 + (y - 5/2)^2 = 25/2$$

Worked Solutions

Circles

Q095 MCQ

Find the circle with centre on $y = x + 1$ through (3,4) and (1,4):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 2)^2 + (y - 3)^2 = 2$.

Correct option: C**Final answer**

$$(x - 2)^2 + (y - 3)^2 = 2$$

Worked Solutions

Circles

Q096 MCQ

Find the circle with centre on $y = 2x - 6$ through (3,7) and (-1,3):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 4)^2 + (y - 2)^2 = 26$.

Correct option: D**Final answer**

$$(x - 4)^2 + (y - 2)^2 = 26$$

Worked Solutions

Circles

Q097 Challenge

A circle centre is constrained to the x-axis and passes through P1(-2,2), P2(1,1). Find its standard equation, centre and radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5}$.

Final answer

$$(x + 1)^2 + y^2 = 5; \text{ centre } (-1, 0), \text{ radius } \sqrt{5}$$

Worked Solutions

Circles

Q098 Challenge

A circle centre is constrained to the x-axis and passes through P1(-2,2), P2(1,1). Find its standard equation, centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5}$.

Correct option: C

Final answer

$$(x + 1)^2 + y^2 = 5; \text{ centre } (-1, 0), \text{ radius } \sqrt{5}$$

Worked Solutions

Circles

Q099 Written

Find the range of k for $x^2 + y^2 - 2kx + 2ky + 4k + 6 = 0$ to represent a circle.**Solution**

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < -1$ or $k > 3$.

Final answer

$$k < -1 \text{ or } k > 3$$

Q100 Written

Find the range of k for $x^2 + y^2 + 2kx - 4ky + 4k^2 + 6 = 0$ to represent a circle.**Solution**

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < -\sqrt{6}$ or $k > \sqrt{6}$.

Final answer

$$k < -\sqrt{6} \text{ or } k > \sqrt{6}$$

Worked Solutions

Circles

Q101 Written

Find the range of k for $x^2 + y^2 + 2x - y + k = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < 5/4$.

Final answer

$$k < 5/4$$

Q102 Written

Find the range of k for $x^2 + y^2 + 6kx - 2ky + 28k + 6 = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < -1/5$ or $k > 3$.

Final answer

$$k < -1/5 \text{ or } k > 3$$

Worked Solutions

Circles

Q103 Written

Find the range of k for $x^2 + y^2 + 2kx - 2(k + 3)y + 4k^2 + 1 = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $-1 < k < 4$.

Final answer

$$-1 < k < 4$$

Worked Solutions

Circles

Q104 MCQ

Find the range of k for $x^2 + y^2 - 2kx + 2ky + 4k + 6 = 0$ to represent a circle:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < -1$ or $k > 3$.

Correct option: C**Final answer**

$$k < -1 \text{ or } k > 3$$

Worked Solutions

Circles

Q105 MCQ

Find the range of k for $x^2 + y^2 + 2kx - 4ky + 4k^2 + 6 = 0$ to represent a circle:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < -\sqrt{6}$ or $k > \sqrt{6}$.

Correct option: B**Final answer**

$$k < -\sqrt{6} \text{ or } k > \sqrt{6}$$

Worked Solutions

Circles

Q106 MCQ

Find the range of k for $x^2 + y^2 + 2x - y + k = 0$ to represent a circle:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < 5/4$.

Correct option: C**Final answer**

$$k < 5/4$$

Worked Solutions

Circles

Q107 MCQ

Find the range of k for $x^2 + y^2 + 6kx - 2ky + 28k + 6 = 0$ to represent a circle:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < -1/5$ or $k > 3$.

Correct option: B**Final answer**

$$k < -1/5 \text{ or } k > 3$$

Worked Solutions

Circles

Q108 MCQ

Find the range of k for $x^2 + y^2 + 2kx - 2(k + 3)y + 4k^2 + 1 = 0$ to represent a circle:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-1 < k < 4$.

Correct option: D

Final answer

$$-1 < k < 4$$

Worked Solutions

Circles

Q109 Challenge

An acoustic-safety circle is $x^2 + y^2 + 6x - 2ky + 2k^2 - 4k + 9 = 0$. Find the exact k -range giving a positive radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $0 < k < 4$.

Final answer

$$0 < k < 4$$

Worked Solutions

Circles

Q110 Challenge

An acoustic-safety circle is $x^2 + y^2 + 6x - 2ky + 2k^2 - 4k + 9 = 0$. Find the exact k -range giving a positive radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $0 < k < 4$.

Correct option: B

Final answer

$$0 < k < 4$$

Worked Solutions

Circles

Q111 Written

Find the coordinates of the intersections of $(x + 2)^2 + y^2 = 10$ and $y = -2x + 1$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(1, -1)$ and $(-1, 3)$.

Final answer

$(1, -1)$ and $(-1, 3)$

Q112 Written

Find the coordinates of the intersections of $x^2 + y^2 = 4$ and $x + y = 2$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(0, 2)$ and $(2, 0)$.

Final answer

$(0, 2)$ and $(2, 0)$

Worked Solutions

Circles

Q113 Written

Find the coordinates of the intersections of $x^2 + y^2 = 10$ with $y = 3x - 10$.**Solution**

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(3, -1)$.

Final answer $(3, -1)$

Q114 Written

Find the coordinates of the intersections of $x^2 + y^2 = 5$ with $2x - y + 5 = 0$.**Solution**

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(-2, 1)$.

Final answer $(-2, 1)$

Worked Solutions

Circles

Q115 Written

Find the coordinates of the intersections of $(x - 1)^2 + y^2 = 10$ with $x + 3y - 1 = 0$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(-2, 1)$ and $(4, -1)$.

Final answer

$(-2, 1)$ and $(4, -1)$

Worked Solutions

Circles

Q116 MCQ

Find the coordinates of the intersections of $(x + 2)^2 + y^2 = 10$ and $y = -2x + 1$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(1, -1)$ and $(-1, 3)$.

Correct option: C**Final answer** $(1, -1)$ and $(-1, 3)$

Worked Solutions

Circles

Q117 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 4$ and $x + y = 2$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is (0, 2) and (2, 0).

Correct option: D**Final answer**

(0, 2) and (2, 0)

Worked Solutions

Circles

Q118 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 10$ with $y = 3x - 10$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(3, -1)$.

Correct option: C**Final answer** $(3, -1)$

Worked Solutions

Circles

Q119 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 5$ with $2x - y + 5 = 0$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-2, 1)$.

Correct option: D**Final answer** $(-2, 1)$

Worked Solutions

Circles

Q120 MCQ

Find the coordinates of the intersections of $(x - 1)^2 + y^2 = 10$ with $x + 3y - 1 = 0$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-2, 1)$ and $(4, -1)$.

Correct option: B

Final answer

$(-2, 1)$ and $(4, -1)$

Worked Solutions

Circles

Q121 Written

Find the number of intersections of $x^2 + y^2 = 2$ and $x - y + 3 = 0$.**Solution**

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: 0 points.

Final answer

0 points

Q122 Written

Find the number of intersections of $x^2 + y^2 = 10$ and $y = 2x - 5$.**Solution**

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: 2 points : (1, -3) and (3, 1).

Final answer

2 points : (1, -3) and (3, 1)

Worked Solutions

Circles

Q123 Written

Find the number of intersections of $x^2 + y^2 = 5$ with $y = x + 2$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: 2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$ and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$.

Final answer

2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$ and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$

Q124 Written

Find the number of intersections of $x^2 + y^2 = 2$ with $2x + 3y - 6 = 0$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: 0 points.

Final answer

0 points

Worked Solutions

Circles

Q125 **Written****Find the number of intersections of $(x + 1)^2 + (y - 2)^2 = 5$ with $x + 2y + 2 = 0$.****Solution**

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: 1 point : $(-2, 0)$.

Final answer1 point : $(-2, 0)$

Worked Solutions

Circles

Q126 MCQ

Find the number of intersections of $x^2 + y^2 = 2$ and $x - y + 3 = 0$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 0 points.

Correct option: D**Final answer**

0 points

Worked Solutions

Circles

Q127 MCQ

Find the number of intersections of $x^2 + y^2 = 10$ and $y = 2x - 5$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 2 points : $(1, -3)$ and $(3, 1)$.

Correct option: A**Final answer**2 points : $(1, -3)$ and $(3, 1)$

Worked Solutions

Circles

Q128 MCQ

Find the number of intersections of $x^2 + y^2 = 5$ with $y = x + 2$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$ and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$.

Correct option: C**Final answer**2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$ and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$

Worked Solutions

Circles

Q129 MCQ

Find the number of intersections of $x^2 + y^2 = 2$ with $2x + 3y - 6 = 0$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 0 points.

Correct option: C**Final answer**

0 points

Worked Solutions

Circles

Q130 MCQ

Find the number of intersections of $(x + 1)^2 + (y - 2)^2 = 5$ with $x + 2y + 2 = 0$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 1 point : $(-2, 0)$.

Correct option: D**Final answer**1 point : $(-2, 0)$

Worked Solutions

Circles

Q131 Written

If $x^2 + y^2 = 5$ and $y = 2x - k$ intersect at two distinct points, find the range of k .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $-5 < k < 5$.

Final answer

$$-5 < k < 5$$

Q132 Written

If $x^2 + y^2 = r^2$ and $2x + y - 5 = 0$ are tangent, find r .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $r = \sqrt{5}$.

Final answer

$$r = \sqrt{5}$$

Worked Solutions

Circles

Q133 Written

If $x^2 + y^2 = 4$ and $y = mx + 4$ are tangent, find m .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $m = \pm\sqrt{3}$.

Final answer

$$m = \pm\sqrt{3}$$

Q134 Written

If $x^2 + y^2 = r^2$ and $x - 3y - 10 = 0$ have no intersections, find the positive-radius range.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $0 < r < \sqrt{10}$.

Final answer

$$0 < r < \sqrt{10}$$

Worked Solutions

Circles

Q135 Written

If $x^2 + y^2 = 8$ and $y = x + m$ are tangent, find m .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $m = \pm 4$.

Final answer

$$m = \pm 4$$

Q136 Written

If $x^2 + y^2 = 10$ and $y = 3x + n$ have no intersections, find n .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $n < -10$ or $n > 10$.

Final answer

$$n < -10 \text{ or } n > 10$$

Worked Solutions

Circles

Q137 Written

If $x^2 + y^2 = 5$ and $y = 2x + m$ have points of intersection, find the range of m .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $-5 \leq m \leq 5$.

Final answer

$$-5 \leq m \leq 5$$

Q138 Written

If $x^2 + y^2 = r^2$ and $y = x + 2$ are tangent, find r .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $r = \sqrt{2}$.

Final answer

$$r = \sqrt{2}$$

Worked Solutions

Circles

Q139 Written

If $x^2 + y^2 = r^2$ and $4x - y + 17 = 0$ intersect at two distinct points, find the positive-radius range.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $r > \sqrt{17}$.

Final answer

$$r > \sqrt{17}$$

Worked Solutions

Circles

Q140 MCQ

If $x^2 + y^2 = 5$ and $y = 2x - k$ intersect at two distinct points, find the range of k :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-5 < k < 5$.

Correct option: D

Final answer

$$-5 < k < 5$$

Worked Solutions

Circles

Q141 MCQ

If $x^2 + y^2 = r^2$ and $2x + y - 5 = 0$ are tangent, find r :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $r = \sqrt{5}$.

Correct option: B

Final answer

$$r = \sqrt{5}$$

Worked Solutions

Circles

Q142 MCQ

If $x^2 + y^2 = 4$ and $y = mx + 4$ are tangent, find m :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $m = \pm\sqrt{3}$.

Correct option: B

Final answer

$$m = \pm\sqrt{3}$$

Worked Solutions

Circles

Q143 MCQ

If $x^2 + y^2 = r^2$ and $x - 3y - 10 = 0$ have no intersections, find the positive-radius range:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $0 < r < \sqrt{10}$.

Correct option: B

Final answer

$$0 < r < \sqrt{10}$$

Worked Solutions

Circles

Q144 MCQ

If $x^2 + y^2 = 8$ and $y = x + m$ are tangent, find m :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $m = \pm 4$.

Correct option: C

Final answer

$$m = \pm 4$$

Worked Solutions

Circles

Q145 MCQ

If $x^2 + y^2 = 10$ and $y = 3x + n$ have no intersections, find n :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $n < -10$ or $n > 10$.

Correct option: A

Final answer

$$n < -10 \text{ or } n > 10$$

Worked Solutions

Circles

Q146 MCQ

If $x^2 + y^2 = 5$ and $y = 2x + m$ have points of intersection, find the range of m :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-5 \leq m \leq 5$.

Correct option: C

Final answer

$$-5 \leq m \leq 5$$

Worked Solutions

Circles

Q147 MCQ

If $x^2 + y^2 = r^2$ and $y = x + 2$ are tangent, find r :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $r = \sqrt{2}$.

Correct option: B

Final answer

$$r = \sqrt{2}$$

Worked Solutions

Circles

Q148 MCQ

If $x^2 + y^2 = r^2$ and $4x - y + 17 = 0$ intersect at two distinct points, find the positive-radius range:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $r > \sqrt{17}$.

Correct option: D

Final answer

$$r > \sqrt{17}$$

Worked Solutions

Circles

Q149 Challenge

A radar circle $x^2 + y^2 = r^2$ must have no intersection with $3x + 4y - 10 = 0$. Find the positive range of r .

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $0 < r < 2$.

Final answer

$$0 < r < 2$$

Worked Solutions

Circles

Q150 Challenge

A radar circle $x^2 + y^2 = r^2$ must have no intersection with $3x + 4y - 10 = 0$. Find the positive range of r :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $0 < r < 2$.

Correct option: D

Final answer

$$0 < r < 2$$

Worked Solutions

Circles

Q151 Written

Find the chord length cut off by $x^2 + y^2 = 4$ on $x + y - 1 = 0$.**Solution**

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $\sqrt{14}$.

Final answer $\sqrt{14}$

Q152 Written

Find the chord length cut off by $(x - 2)^2 + (y + 1)^2 = 9$ on $x + 2y - 5 = 0$.**Solution**

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: 4.

Final answer

4

Worked Solutions

Circles

Q153 Written

Find the chord length cut off by $x^2 + y^2 = 10$ on $2x - y = 5$.**Solution**

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $2\sqrt{5}$.

Final answer

$$2\sqrt{5}$$

Q154 Written

Find the chord length cut off by $(x - 2)^2 + (y - 1)^2 = 5$ on $x + y = 4$.**Solution**

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $3\sqrt{2}$.

Final answer

$$3\sqrt{2}$$

Worked Solutions

Circles

Q155 **Written****Find the chord length cut off by $x^2 + y^2 = 3$ on $x - y = 2$.****Solution**

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: 2.

Final answer

2

Q156 **MCQ****Find the chord length cut off by $x^2 + y^2 = 4$ on $x + y - 1 = 0$:****Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $\sqrt{14}$.

Correct option: A**Final answer** $\sqrt{14}$

Worked Solutions

Circles

Q157 MCQ

Find the chord length cut off by $(x - 2)^2 + (y + 1)^2 = 9$ on $x + 2y - 5 = 0$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 4.

Correct option: C**Final answer**

4

Q158 MCQ

Find the chord length cut off by $x^2 + y^2 = 10$ on $2x - y = 5$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2\sqrt{5}$.

Correct option: D**Final answer** $2\sqrt{5}$

Worked Solutions

Circles

Q159 MCQ

Find the chord length cut off by $(x - 2)^2 + (y - 1)^2 = 5$ on $x + y = 4$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $3\sqrt{2}$.

Correct option: B**Final answer** $3\sqrt{2}$

Q160 MCQ

Find the chord length cut off by $x^2 + y^2 = 3$ on $x - y = 2$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 2.

Correct option: D**Final answer**

2

Worked Solutions

Circles

Q161 Written

For $x^2 + y^2 = 3$ and $x - y + k = 0$, find k when the chord length is 2.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = \pm 2$.

Final answer

$$k = \pm 2$$

Q162 Written

For $x^2 + y^2 = 16$ and $y = x + k$, find k when the chord length is $\sqrt{2}$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = \pm\sqrt{31}$.

Final answer

$$k = \pm\sqrt{31}$$

Worked Solutions

Circles

Q163 Written

For $x^2 + y^2 = 10$ and $y = x + k$, find k when the chord length is $4\sqrt{2}$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = \pm 2$.

Final answer

$$k = \pm 2$$

Q164 Written

For $(x - 1)^2 + y^2 = 5$ and $y = x + k$, find k when the chord length is $3\sqrt{2}$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = 0$ or $k = -2$.

Final answer

$$k = 0 \text{ or } k = -2$$

Worked Solutions

Circles

Q165 MCQ

For $x^2 + y^2 = 3$ and $x - y + k = 0$, find k when the chord length is 2:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = \pm 2$.

Correct option: C

Final answer

$$k = \pm 2$$

Q166 MCQ

For $x^2 + y^2 = 16$ and $y = x + k$, find k when the chord length is $\sqrt{2}$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = \pm\sqrt{31}$.

Correct option: D

Final answer

$$k = \pm\sqrt{31}$$

Worked Solutions

Circles

Q167 MCQ

For $x^2 + y^2 = 10$ and $y = x + k$, find k when the chord length is $4\sqrt{2}$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = \pm 2$.

Correct option: A

Final answer

$$k = \pm 2$$

Worked Solutions

Circles

Q168 MCQ

For $(x - 1)^2 + y^2 = 5$ and $y = x + k$, find k when the chord length is $3\sqrt{2}$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = 0$ or $k = -2$.

Correct option: B

Final answer

$$k = 0 \text{ or } k = -2$$

Worked Solutions

Circles

Q169 Challenge

For $x^2 + y^2 + 4x - 2y - 7 = 0$ and $y = x + k$, find k when the chord length is 4.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $k = -1$ or $k = 7$.

Final answer

$$k = -1 \text{ or } k = 7$$

Worked Solutions

Circles

Q170 Challenge

For $x^2 + y^2 + 4x - 2y - 7 = 0$ and $y = x + k$, find k when the chord length is 4:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = -1$ or $k = 7$.

Correct option: A

Final answer

$$k = -1 \text{ or } k = 7$$

Worked Solutions

Circles

Q171 Written

Find the tangent to $x^2 + y^2 = 25$ at (4,3).**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $4x + 3y = 25$.

Final answer

$$4x + 3y = 25$$

Q172 Written

Find the tangent to $x^2 + y^2 = 34$ at (5,3).**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $5x + 3y = 34$.

Final answer

$$5x + 3y = 34$$

Worked Solutions

Circles

Q173 Written

Find the tangents $x^2 + y^2 = 10$ at (3,-1).**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $3x - y = 10$.

Final answer

$$3x - y = 10$$

Q174 Written

Find the tangents $x^2 + y^2 = 29$ at (-5,-2).**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-5x - 2y = 29$.

Final answer

$$-5x - 2y = 29$$

Worked Solutions

Circles

Q175 Written

Find the tangents $x^2 + y^2 = 5$ at $(-2,1)$.**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-2x + y = 5$.

Final answer

$$-2x + y = 5$$

Worked Solutions

Circles

Q176 MCQ

Find the tangent to $x^2 + y^2 = 25$ at (4,3):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $4x + 3y = 25$.

Correct option: A**Final answer**

$$4x + 3y = 25$$

Worked Solutions

Circles

Q177 MCQ

Find the tangent to $x^2 + y^2 = 34$ at (5,3):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $5x + 3y = 34$.

Correct option: C**Final answer**

$$5x + 3y = 34$$

Worked Solutions

Circles

Q178 MCQ

Find the tangents $x^2 + y^2 = 10$ at (3,-1):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $3x - y = 10$.

Correct option: A**Final answer**

$$3x - y = 10$$

Worked Solutions

Circles

Q179 MCQ

Find the tangents $x^2 + y^2 = 29$ at $(-5, -2)$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-5x - 2y = 29$.

Correct option: C**Final answer**

$$-5x - 2y = 29$$

Worked Solutions

Circles

Q180 MCQ

Find the tangents $x^2 + y^2 = 5$ at $(-2,1)$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-2x + y = 5$.

Correct option: A**Final answer**

$$-2x + y = 5$$

Worked Solutions

Circles

Q181 Written

Find the tangents from (1,3) to $x^2 + y^2 = 5$.**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $2x + y = 5$ and $-x + 2y = 5$.

Final answer

$$2x + y = 5 \text{ and } -x + 2y = 5$$

Q182 Written

Find the tangents from (-2,4) to $x^2 + y^2 = 10$.**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-3x + y = 10$ and $x + 3y = 10$.

Final answer

$$-3x + y = 10 \text{ and } x + 3y = 10$$

Worked Solutions

Circles

Q183 **Written****Find the tangents from $(-1,3)$ to $x^2 + y^2 = 5$.****Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-2x + y = 5$ and $x + 2y = 5$.

Final answer

$$-2x + y = 5 \text{ and } x + 2y = 5$$

Q184 **Written****Find the tangents from $(2,1)$ to $x^2 + y^2 = 1$.****Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $y = 1$ and $4x - 3y = 5$.

Final answer

$$y = 1 \text{ and } 4x - 3y = 5$$

Worked Solutions

Circles

Q185 Written

Find the tangents from (2,6) to $x^2 + y^2 = 20$.**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-x + 2y = 10$ and $2x + y = 10$.

Final answer

$$-x + 2y = 10 \text{ and } 2x + y = 10$$

Worked Solutions

Circles

Q186 MCQ

Find the tangents from (1,3) to $x^2 + y^2 = 5$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2x + y = 5$ and $-x + 2y = 5$.

Correct option: D**Final answer**

$$2x + y = 5 \text{ and } -x + 2y = 5$$

Worked Solutions

Circles

Q187 MCQ

Find the tangents from $(-2,4)$ to $x^2 + y^2 = 10$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-3x + y = 10$ and $x + 3y = 10$.

Correct option: A**Final answer**

$$-3x + y = 10 \text{ and } x + 3y = 10$$

Worked Solutions

Circles

Q188 MCQ

Find the tangents from $(-1,3)$ to $x^2 + y^2 = 5$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-2x + y = 5$ and $x + 2y = 5$.

Correct option: D**Final answer**

$$-2x + y = 5 \text{ and } x + 2y = 5$$

Worked Solutions

Circles

Q189 MCQ

Find the tangents from (2,1) to $x^2 + y^2 = 1$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $y = 1$ and $4x - 3y = 5$.

Correct option: B**Final answer**

$$y = 1 \text{ and } 4x - 3y = 5$$

Worked Solutions

Circles

Q190 MCQ

Find the tangents from (2,6) to $x^2 + y^2 = 20$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-x + 2y = 10$ and $2x + y = 10$.

Correct option: C**Final answer**

$$-x + 2y = 10 \text{ and } 2x + y = 10$$

Worked Solutions

Circles

Q191 Challenge

A drone at (2,6) needs tangents to $x^2 + y^2 = 20$. Find both sightline equations and their exact contact points.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$.

Final answer

$-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$

Worked Solutions

Circles

Q192 Challenge

A drone at (2,6) needs tangents to $x^2 + y^2 = 20$. Find both sightline equations and their exact contact points:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$.

Correct option: B

Final answer

$$-x + 2y = 10 \text{ at } (-2, 4); 2x + y = 10 \text{ at } (4, 2)$$

Worked Solutions

Circles

Q193 Written

Classify $x^2 + y^2 = 4$ **and** $(x - 4)^2 + y^2 = 9$.**Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: intersect at 2 points.

Final answer

intersect at 2 points

Q194 Written

Classify $x^2 + y^2 = 9$ **and** $(x - 3)^2 + (y - 4)^2 = 4$ **from A-E**.**Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: B : touch externally.

Final answer B : touch externally

Worked Solutions

Circles

Q195 **Written****Classify** $(x - 1)^2 + y^2 = 5^2$ **and** $x^2 + y^2 + 2x - 6y = 3$ **from A-E.****Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: E : intersect at 2 points.

Final answer E : intersect at 2 pointsQ196 **Written****Classify** $x^2 + y^2 = 4$ **and** $(x - 1)^2 + y^2 = 1$ **from A-E.****Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: C : touch internally.

Final answer C : touch internally

Worked Solutions

Circles

Q197 Written

Classify $x^2 + y^2 = 18$ and $(x + 2)^2 + (y - 2)^2 = 2$ from A-E.**Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: C : touch internally.

Final answer C : touch internally

Q198 Written

Classify $(x + 3)^2 + y^2 = 16$ and $x^2 + y^2 - 2x - 8y - 8 = 0$ from A-E.**Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: E : intersect at 2 points.

Final answer E : intersect at 2 points

Worked Solutions

Circles

Q199 **Written****Classify** $x^2 + (y - 3)^2 = 45$ **and** $x^2 + y^2 + 2x - 2y = 3$ **from A-E.****Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: D : one is inside the other .

Final answer D : one is inside the other

Worked Solutions

Circles

Q200 MCQ

For the expression Classify $x^2 + y^2 = 4$ and $(x - 4)^2 + y^2 = 9$, the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is intersect at 2 points.

Correct option: C

Final answer

intersect at 2 points

Worked Solutions

Circles

Q201 MCQ

For the expression Classify $x^2 + y^2 = 9$ and $(x - 3)^2 + (y - 4)^2 = 4$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is B : touch externally.

Correct option: D

Final answer

B : touch externally

Worked Solutions

Circles

Q202 MCQ

For the expression Classify $(x - 1)^2 + y^2 = 5^2$ and $x^2 + y^2 + 2x - 6y = 3$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is E : intersect at 2 points.

Correct option: B**Final answer** E : intersect at 2 points

Worked Solutions

Circles

Q203 MCQ

For the expression Classify $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 1$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is C : touch internally.

Correct option: A

Final answer

C : touch internally

Worked Solutions

Circles

Q204 MCQ

For the expression Classify $x^2 + y^2 = 18$ and $(x + 2)^2 + (y - 2)^2 = 2$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is C : touch internally.

Correct option: D

Final answer

C : touch internally

Worked Solutions

Circles

Q205 MCQ

For the expression Classify $(x + 3)^2 + y^2 = 16$ and $x^2 + y^2 - 2x - 8y - 8 = 0$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is E : intersect at 2 points.

Correct option: B

Final answer

E : intersect at 2 points

Worked Solutions

Circles

Q206 MCQ

For the expression Classify $x^2 + (y - 3)^2 = 45$ and $x^2 + y^2 + 2x - 2y = 3$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is D : one is inside the other .

Correct option: C

Final answer

D : one is inside the other

Worked Solutions

Circles

Q207 **Written****Find the circle with centre (3,1) touching $(x - 1)^2 + (y + 3)^2 = 5$ internally.****Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: $(x - 3)^2 + (y - 1)^2 = 45$.

Final answer

$$(x - 3)^2 + (y - 1)^2 = 45$$

Q208 **Written****Find the circle with centre (-3,-9) touching $(x + 6)^2 + (y + 5)^2 = 1$ externally.****Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: $(x + 3)^2 + (y + 9)^2 = 16$.

Final answer

$$(x + 3)^2 + (y + 9)^2 = 16$$

Worked Solutions

Circles

Q209 Written

Find the circle with centre (3,-9) touching $(x - 3)^2 + y^2 = 9$ internally.**Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: $(x - 3)^2 + (y + 9)^2 = 144$.

Final answer

$$(x - 3)^2 + (y + 9)^2 = 144$$

Worked Solutions

Circles

Q210 MCQ

Find the circle with centre (3,1) touching $(x - 1)^2 + (y + 3)^2 = 5$ internally:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 3)^2 + (y - 1)^2 = 45$.

Correct option: A**Final answer**

$$(x - 3)^2 + (y - 1)^2 = 45$$

Worked Solutions

Circles

Q211 MCQ

Find the circle with centre $(-3, -9)$ touching $(x + 6)^2 + (y + 5)^2 = 1$ externally:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 3)^2 + (y + 9)^2 = 16$.

Correct option: D**Final answer**

$$(x + 3)^2 + (y + 9)^2 = 16$$

Worked Solutions

Circles

Q212 MCQ

Find the circle with centre (3,-9) touching $(x - 3)^2 + y^2 = 9$ internally:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 3)^2 + (y + 9)^2 = 144$.

Correct option: A**Final answer**

$$(x - 3)^2 + (y + 9)^2 = 144$$

Worked Solutions

Circles

Q213 Challenge

Find the circle with centre $(-1, -2)$ touching $x^2 + y^2 - 4x - 4y + 4 = 0$ externally.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $(x + 1)^2 + (y + 2)^2 = 9$.

Final answer

$$(x + 1)^2 + (y + 2)^2 = 9$$

Worked Solutions

Circles

Q214 Challenge

Find the circle with centre $(-1, -2)$ touching $x^2 + y^2 - 4x - 4y + 4 = 0$ externally:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y + 2)^2 = 9$.

Correct option: D

Final answer

$$(x + 1)^2 + (y + 2)^2 = 9$$

Worked Solutions

Circles

Q215 Written

Find the line through the intersections of $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x - 4y = 9$.**Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$).

Final answer

$$2x - 4y - 5 = 0 \text{ (equivalently } y = x/2 - 5/4)$$

Q216 Written

Find the line through the intersections of $x^2 + y^2 = 6$ and $x^2 + y^2 + 3x + 3y = 9$.**Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $x + y - 1 = 0$.

Final answer

$$x + y - 1 = 0$$

Worked Solutions

Circles

Q217 Written

Find the line through the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 - 12x + 6y = 3$.**Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $12x - 6y - 7 = 0$.

Final answer

$$12x - 6y - 7 = 0$$

Q218 Written

Find the line through the intersections of $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 4$.**Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $x = 1/2$.

Final answer

$$x = 1/2$$

Worked Solutions

Circles

Q219 MCQ

Find the line through the intersections of $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x - 4y = 9$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$).

Correct option: A**Final answer**

$$2x - 4y - 5 = 0 \text{ (equivalently } y = x/2 - 5/4)$$

Worked Solutions

Circles

Q220 MCQ

Find the line through the intersections of $x^2 + y^2 = 6$ and $x^2 + y^2 + 3x + 3y = 9$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x + y - 1 = 0$.

Correct option: C**Final answer**

$$x + y - 1 = 0$$

Worked Solutions

Circles

Q221 MCQ

Find the line through the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 - 12x + 6y = 3$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $12x - 6y - 7 = 0$.

Correct option: B**Final answer**

$$12x - 6y - 7 = 0$$

Worked Solutions

Circles

Q222 MCQ

Find the line through the intersections of $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 4$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x = 1/2$.

Correct option: D**Final answer**

$$x = 1/2$$

Worked Solutions

Circles

Q223 Written

Find the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 + 2x - y = 5$.**Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $(-3, -1)$ and $(-1, 3)$.

Final answer $(-3, -1)$ and $(-1, 3)$

Q224 Written

Find the intersections of $x^2 + y^2 = 4$ and $(x - 4)^2 + (y - 2)^2 = 16$.**Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $(0, 2)$ and $(8/5, -6/5)$.

Final answer $(0, 2)$ and $(8/5, -6/5)$

Worked Solutions

Circles

Q225 **Written****Find the intersections of $x^2 + y^2 = 5$ and $(x - 2)^2 + (y - 2)^2 = 1$.****Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: (1, 2) and (2, 1).

Final answer

(1, 2) and (2, 1)

Worked Solutions

Circles

Q226 MCQ

Find the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 + 2x - y = 5$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-3, -1)$ and $(-1, 3)$.

Correct option: B**Final answer** $(-3, -1)$ and $(-1, 3)$

Worked Solutions

Circles

Q227 MCQ

Find the intersections of $x^2 + y^2 = 4$ and $(x - 4)^2 + (y - 2)^2 = 16$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(0, 2)$ and $(8/5, -6/5)$.

Correct option: C**Final answer** $(0, 2)$ and $(8/5, -6/5)$

Worked Solutions

Circles

Q228 MCQ

Find the intersections of $x^2 + y^2 = 5$ and $(x - 2)^2 + (y - 2)^2 = 1$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is (1, 2) and (2, 1).

Correct option: B**Final answer**

(1, 2) and (2, 1)

Worked Solutions

Circles

Q229 Challenge

Two radar circles are $x^2 + y^2 = 25$ and $(x + 5)^2 + (y + 5)^2 = 65$. Find their intersection points.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $(-4, 3)$ and $(3, -4)$.

Final answer

$(-4, 3)$ and $(3, -4)$

Worked Solutions

Circles

Q230 Challenge

Two radar circles are $x^2 + y^2 = 25$ and $(x + 5)^2 + (y + 5)^2 = 65$. Find their intersection points:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-4, 3)$ and $(3, -4)$.

Correct option: C

Final answer

$(-4, 3)$ and $(3, -4)$

Worked Solutions

Circles

Q231 Written

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 5 = 0$ and $x^2 + y^2 = 5$, passing through (3,0).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 2x + 6y - 15 = 0$.

Final answer

$$x^2 + y^2 + 2x + 6y - 15 = 0$$

Q232 Written

Find the circle through the intersections of $x^2 + y^2 - 4x - 2y - 8 = 0$ and $x^2 + y^2 = 4$, passing through the origin.

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 4x + 2y = 0$.

Final answer

$$x^2 + y^2 + 4x + 2y = 0$$

Worked Solutions

Circles

Q233 Written

Find the circle through the intersections of $x^2 + y^2 - 2x - 4y + 3 = 0$ and $x^2 + y^2 = 4$, passing through (3,0).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 10x + 20y - 39 = 0$.

Final answer

$$x^2 + y^2 + 10x + 20y - 39 = 0$$

Q234 Written

Find the circle through the intersections of $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 1 = 0$, passing through (-3,2).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 24x + 32y - 5 = 0$.

Final answer

$$x^2 + y^2 + 24x + 32y - 5 = 0$$

Worked Solutions

Circles

Q235 **Written**

Find the circle through the intersections of $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$, passing through (4,0).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 - 10x - 6y + 24 = 0$.

Final answer

$$x^2 + y^2 - 10x - 6y + 24 = 0$$

Worked Solutions

Circles

Q236 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 5 = 0$ and $x^2 + y^2 = 5$, passing through (3,0):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 2x + 6y - 15 = 0$.

Correct option: D

Final answer

$$x^2 + y^2 + 2x + 6y - 15 = 0$$

Worked Solutions

Circles

Q237 MCQ

Find the circle through the intersections of $x^2 + y^2 - 4x - 2y - 8 = 0$ and $x^2 + y^2 = 4$, passing through the origin:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 4x + 2y = 0$.

Correct option: A

Final answer

$$x^2 + y^2 + 4x + 2y = 0$$

Worked Solutions

Circles

Q238 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 4y + 3 = 0$ and $x^2 + y^2 = 4$, passing through (3,0):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 10x + 20y - 39 = 0$.

Correct option: D

Final answer

$$x^2 + y^2 + 10x + 20y - 39 = 0$$

Worked Solutions

Circles

Q239 MCQ

Find the circle through the intersections of $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 1 = 0$, passing through $(-3, 2)$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 24x + 32y - 5 = 0$.

Correct option: C

Final answer

$$x^2 + y^2 + 24x + 32y - 5 = 0$$

Worked Solutions

Circles

Q240 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$, passing through (4,0):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 10x - 6y + 24 = 0$.

Correct option: B

Final answer

$$x^2 + y^2 - 10x - 6y + 24 = 0$$

Worked Solutions

Circles

Q241 Challenge

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 4x - 2y - 11 = 0$, passing through (2,2).

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$.

Final answer

$$x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$$

Worked Solutions

Circles

Q242 Challenge

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 4x - 2y - 11 = 0$, passing through (2,2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$.

Correct option: A

Final answer

$$x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$$

Worked Solutions

Circles

Quiz MCQ 1 **Concept Quiz · MCQ**

For the circle $(x - 2)^2 + (y + 3)^2 = 25$, determine its centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is centre $(2, -3)$, radius 5.

Correct option: B

Final answer

centre $(2, -3)$, radius 5

Worked Solutions

Circles

Quiz MCQ 2 **Concept Quiz · MCQ**

Determine the substitution discriminant when a line meets a circle at exactly one real point:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is 0.

Correct option: D

Final answer

0

Worked Solutions

Circles

Quiz MCQ 3 Concept Quiz · MCQ

Determine the chord length for radius r and centre-to-chord distance d :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2\sqrt{r^2 - d^2}$.

Correct option: A

Final answer

$$2\sqrt{r^2 - d^2}$$

Worked Solutions

Circles

Quiz MCQ 4 **Concept Quiz · MCQ**

Determine the relative position of two non-overlapping circles when $D = r_1 + r_2$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is externally tangent.

Correct option: D

Final answer

externally tangent

Worked Solutions

Circles

Quiz WRITTEN 5 Concept Quiz · Written

Find the circle with centre (1,-2) passing through (4,2).

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $(x - 1)^2 + (y + 2)^2 = 25$.

Final answer

$$(x - 1)^2 + (y + 2)^2 = 25$$

Quiz WRITTEN 6 Concept Quiz · Written

Find the intersections of $x^2 + y^2 = 25$ and $y = 3$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(4, 3)$, $(-4, 3)$.

Final answer

$$(4, 3), (-4, 3)$$

Worked Solutions

Circles

Quiz WRITTEN 7 Concept Quiz · Written

Find the tangent to $x^2 + y^2 = 25$ at (3,4).

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $3x + 4y = 25$.

Final answer

$$3x + 4y = 25$$

Quiz WRITTEN 8 Concept Quiz · Written

The circles $S_1 = x^2 + y^2 - 5 = 0$ and $S_2 = x^2 + y^2 - 2x - 2y - 4 = 0$ meet. Find the circle $S_1 + \lambda S_2 = 0$ passing through (1,1).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $\lambda = -1/2$; $x^2 + y^2 + 2x + 2y - 6 = 0$.

Final answer

$$\lambda = -1/2; x^2 + y^2 + 2x + 2y - 6 = 0$$

Answer key

Use the question number shown on each page.

Q001 $(x - 3)^2 + (y + 2)^2 = 25$

Q002 $(x + 3)^2 + (y - 1)^2 = 4$

Q003 $x^2 + y^2 = 16$

Q004 $(x - 3)^2 + (y - 4)^2 = 36$

Q005 $(x + 2)^2 + (y - 6)^2 = 12$

Q006 $x^2 + y^2 = 25$

Q007 A

Q008 A

Q009 D

Q010 B

Q011 B

Q012 A

Q013 circle; centre (3, -1), radius 2

Q014 circle; centre (2, 3), radius 3

Q015 circle; centre (2, -4), radius 4

Q016 circle; centre (2, 0), radius 2

Q017 circle; centre (-1, 2), radius 5

Q018 circle; centre (-2, 3), radius $\sqrt{13}$

Q019 A

Q020 C

Q021 D

Q022 C

Q023 C

Q024 C

Q025 centre (5, 3), radius² = 13; $(x - 5)^2 + (y - 3)^2 = 13$

Q026 B

Q027 $(x - 2)^2 + (y - 3)^2 = 13$

Q028 $(x + 3)^2 + (y - 1)^2 = 58$

Q029 $(x - 1)^2 + (y + 3)^2 = 25$

Q030 $(x + 4)^2 + (y - 1)^2 = 58$

Q031 $(x - 2)^2 + (y + 1)^2 = 5$

Q032 $x^2 + y^2 = 25$

Q033 C

Q034 A

Q035 D

Q036 A

Q037 B

Q038 A

Q039 $(x - 2)^2 + (y + 1)^2 = 17$

Q040 $(x - 4)^2 + (y - 1)^2 = 10$

Q041 $(x - 1)^2 + (y - 3)^2 = 25$

Q042 $(x + 1)^2 + (y + 2)^2 = 25$

Q043 $(x + 2)^2 + (y - 1)^2 = 2$

Q044 $x^2 + (y - 2)^2 = 25$

Q045 A

Q046 C

Q047 C

Q048 A

Q049 B

Q050 C

Q051 $x^2 + y^2 - 5x - y + 4 = 0$

Q052 $x^2 + y^2 - 3x - 7y + 6 = 0$

Q053 $x^2 + y^2 - 2x - 4y = 0$

Q054 $x^2 + y^2 - 6y - 16 = 0$

Q055 $x^2 + y^2 - 2x + 8y - 8 = 0$

Q056 A

Q057 B

Q058 D

Q059 D

Q060 D

Q061 $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10}$

Q062 A

Q063 $(x - 4)^2 + (y + 2)^2 = 4$

Q064 $(x + 2)^2 + (y - 3)^2 = 4$

Q065 $(x + 1)^2 + (y - 4)^2 = 16$

Q066 $(x - 3)^2 + (y + 4)^2 = 16$

Q067 $(x - 5)^2 + (y + 4)^2 = 16$

Q068 $(x + 1)^2 + (y - 6)^2 = 1$

Q069 $(x - 2)^2 + (y + 5)^2 = 4$

Q070 $(x + 3)^2 + (y + 4)^2 = 9$

Q071 B

Q072 B

Q073 B

Q074 D

Q075 D

Q076 D

Q077 B

Q078 C

Q079 $(x + 1)^2 + (y - 1)^2 = 1$ or $(x + 5)^2 + (y - 5)^2 = 25$

Q080 $(x - 1)^2 + (y - 1)^2 = 1$ or $(x - 5)^2 + (y - 5)^2 = 25$

Q081 $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$

Q082 $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$

Q083 $(x - 5)^2 + (y + 5)^2 = 25$ or $(x - 13)^2 + (y + 13)^2 = 169$

Q084 C

Q085 D

Q086 C

Q087 C

Q088 B

Q089 $(x - 1)^2 + (y + 1)^2 = 25$

Q090 $(x + 5/2)^2 + (y - 5/2)^2 = 25/2$

Q091 $(x - 2)^2 + (y - 3)^2 = 2$

Q092 $(x - 4)^2 + (y - 2)^2 = 26$

Q093 D

Q094 C

Q095 C

Q096 D

Q097 $(x + 1)^2 + y^2 = 5$; centre (-1, 0), radius $\sqrt{5}$

Q098 C

Q099 $k < -1$ or $k > 3$

Q100 $k < -\sqrt{6}$ or $k > \sqrt{6}$

Q101 $k < 5/4$

Q102 $k < -1/5$ or $k > 3$

Q103 $-1 < k < 4$

Q104 C

Q105 B	Q137 $-5 \leq m \leq 5$	Q169 $k = -1$ or $k = 7$	Q201 D
Q106 C	Q138 $r = \sqrt{2}$	Q170 A	Q202 B
Q107 B	Q139 $r > \sqrt{17}$	Q171 $4x + 3y = 25$	Q203 A
Q108 D	Q140 D	Q172 $5x + 3y = 34$	Q204 D
Q109 $0 < k < 4$	Q141 B	Q173 $3x - y = 10$	Q205 B
Q110 B	Q142 B	Q174 $-5x - 2y = 29$	Q206 C
Q111 (1, -1) and (-1, 3)	Q143 B	Q175 $-2x + y = 5$	Q207 $(x - 3)^2 + (y - 1)^2 = 45$
Q112 (0, 2) and (2, 0)	Q144 C	Q176 A	Q208 $(x + 3)^2 + (y + 9)^2 = 16$
Q113 (3, -1)	Q145 A	Q177 C	Q209 $(x - 3)^2 + (y + 9)^2 = 144$
Q114 (-2, 1)	Q146 C	Q178 A	Q210 A
Q115 (-2, 1) and (4, -1)	Q147 B	Q179 C	Q211 D
Q116 C	Q148 D	Q180 A	Q212 A
Q117 D	Q149 $0 < r < 2$	Q181 $2x + y = 5$ and $-x + 2y = 5$	Q213 $(x + 1)^2 + (y + 2)^2 = 9$
Q118 C	Q150 D	Q182 $-3x + y = 10$ and $x + 3y = 10$	Q214 D
Q119 D	Q151 $\sqrt{14}$	Q183 $-2x + y = 5$ and $x + 2y = 5$	Q215 $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$)
Q120 B	Q152 4	Q184 $y = 1$ and $4x - 3y = 5$	Q216 $x + y - 1 = 0$
Q121 0 points	Q153 $2\sqrt{5}$	Q185 $-x + 2y = 10$ and $2x + y = 10$	Q217 $12x - 6y - 7 = 0$
Q122 2 points : (1, -3) and (3, 1)	Q154 $3\sqrt{2}$	Q186 D	Q218 $x = 1/2$
Q123 2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$ and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$	Q155 2	Q187 A	Q219 A
Q124 0 points	Q156 A	Q188 D	Q220 C
Q125 1 point : (-2, 0)	Q157 C	Q189 B	Q221 B
Q126 D	Q158 D	Q190 C	Q222 D
Q127 A	Q159 B	Q191 $-x + 2y = 10$ at (-2, 4); $2x + y = 10$ at (4, 0)	Q223 (-3, -1) and (-1, 3)
Q128 C	Q160 D	Q192 B	Q224 (0, 2) and $(8/5, -6/5)$
Q129 C	Q161 $k = \pm 2$	Q193 intersect at 2 points	Q225 (1, 2) and (2, 1)
Q130 D	Q162 $k = \pm\sqrt{31}$	Q194 B : touch externally	Q226 B
Q131 $-5 < k < 5$	Q163 $k = \pm 2$	Q195 E : intersect at 2 points	Q227 C
Q132 $r = \sqrt{5}$	Q164 $k = 0$ or $k = -2$	Q196 C : touch internally	Q228 B
Q133 $m = \pm\sqrt{3}$	Q165 C	Q197 C : touch internally	Q229 (-4, 3) and (3, -4)
Q134 $0 < r < \sqrt{10}$	Q166 D	Q198 E : intersect at 2 points	Q230 C
Q135 $m = \pm 4$	Q167 A	Q199 D : one is inside the other	Q231 $x^2 + y^2 + 2x + 6y - 15 = 0$
Q136 $n < -10$ or $n > 10$	Q168 B	Q200 C	Q232 $x^2 + y^2 + 4x + 2y = 0$

Q233 $x^2 + y^2 + 10x + 20y - 39 = 0$

Q234 $x^2 + y^2 + 24x + 32y - 5 = 0$

Q235 $x^2 + y^2 - 10x - 6y + 24 = 0$

Q236 D**Q237** A**Q238** D**Q239** C**Q240** B

Q241 $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$

Q242 A**Quiz MCQ 1** B**Quiz MCQ 2** D**Quiz MCQ 3** A**Quiz MCQ 4** D

Quiz WRITTEN 5 $(x - 1)^2 + (y + 2)^2 = 25$

Quiz WRITTEN 6 $(4, 3), (-4, 3)$

Quiz WRITTEN 7 $3x + 4y = 25$

Quiz WRITTEN 8 $\lambda = -1/2; x^2 + y^2 + 2x + 2y - 6 = 0$